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(12) **United States Patent**
Schönfeld et al.(10) **Patent No.: US 12,384,843 B2**(45) **Date of Patent: *Aug. 12, 2025**(54) **ANTI-SEMAPHORIN 3A ANTIBODIES**(71) Applicant: **BAYER AKTIENGESELLSCHAFT**,
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Leverkusen (DE)(*) Notice: Subject to any disclaimer, the term of this
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U.S.C. 154(b) by 0 days.This patent is subject to a terminal dis-
claimer.(21) Appl. No.: **17/706,543**(22) Filed: **Mar. 28, 2022**(65) **Prior Publication Data**

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(51) **Int. Cl.****C07K 16/28** (2006.01)**A61K 39/395** (2006.01)**A61P 9/10** (2006.01)**A61P 13/12** (2006.01)**C07K 16/18** (2006.01)**A61K 39/00** (2006.01)(52) **U.S. Cl.**CPC **C07K 16/2803** (2013.01); **A61K 39/3955**
(2013.01); **A61P 9/10** (2018.01); **A61P 13/12**
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2039/505 (2013.01); **C07K 2317/33** (2013.01);
C07K 2317/565 (2013.01); **C07K 2317/76**
(2013.01); **C07K 2317/92** (2013.01)(58) **Field of Classification Search**

None

See application file for complete search history.

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Pierce, P.L.C.(57) **ABSTRACT**The present disclosure relates to an isolated antibody or
antigen-binding fragment thereof that binds to human Sema-
phorin 3A (Sema3A). An antibody conjugate and a pharma-
ceutical composition each comprising the isolated antibody
or antigen-binding fragment thereof that binds to human
Sema3A are also provided.

11 Claims, 15 Drawing Sheets

Specification includes a Sequence Listing.

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Fig. 1A

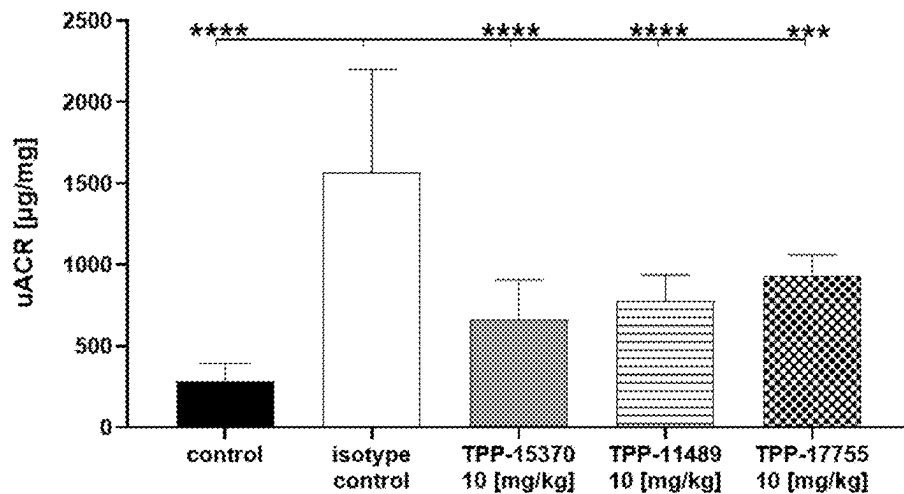


Fig.1B

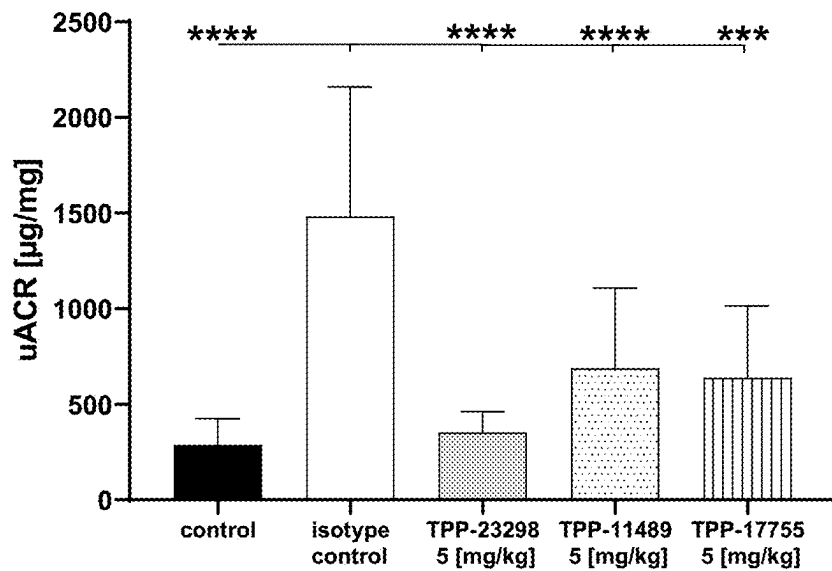


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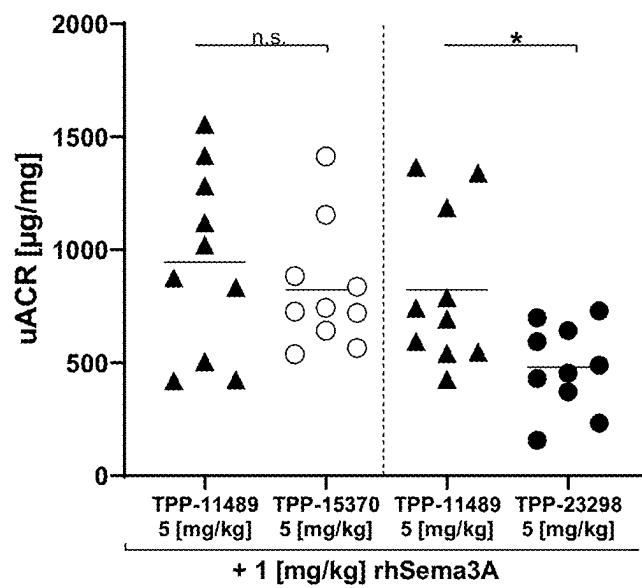


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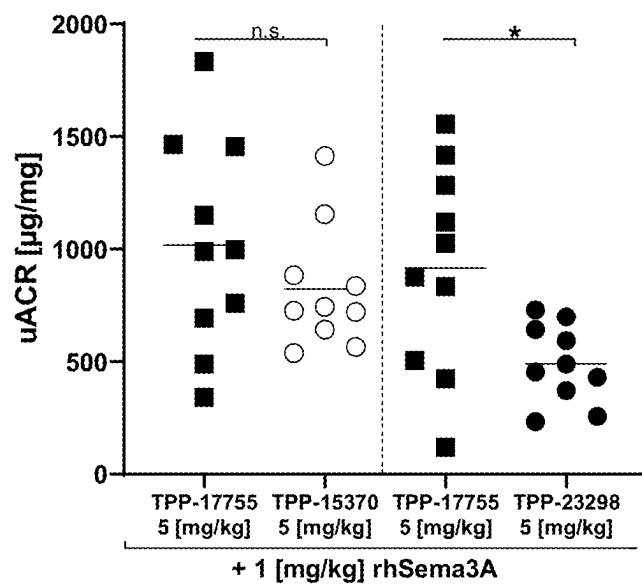


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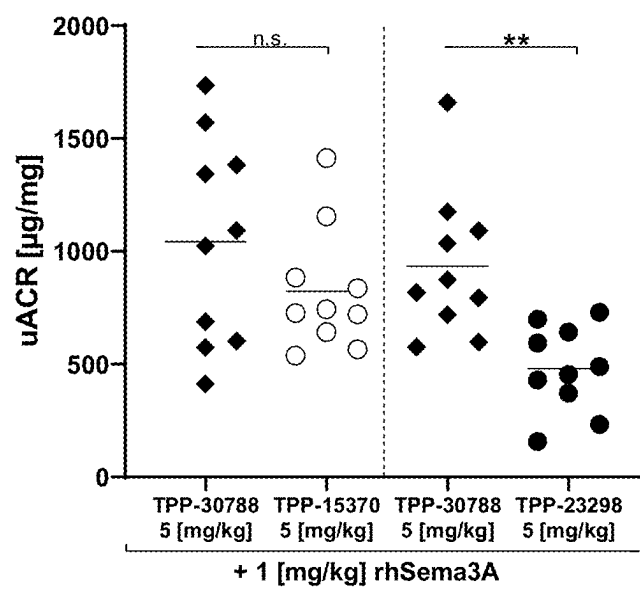


Fig. 3A

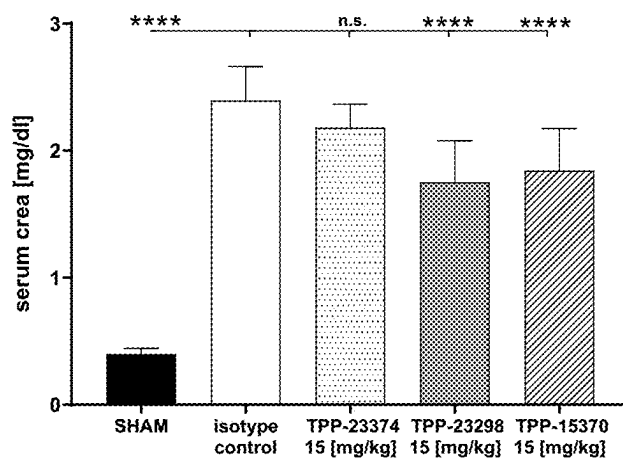


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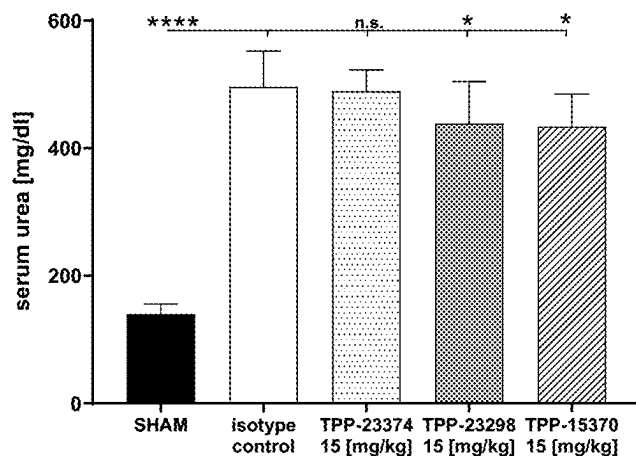


Fig. 3C

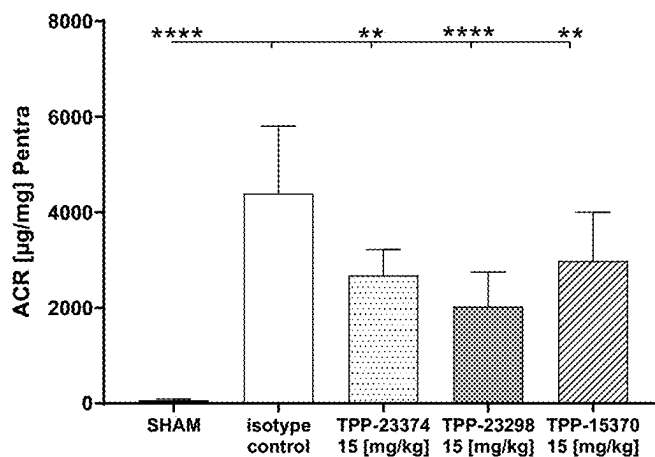


Fig. 4A

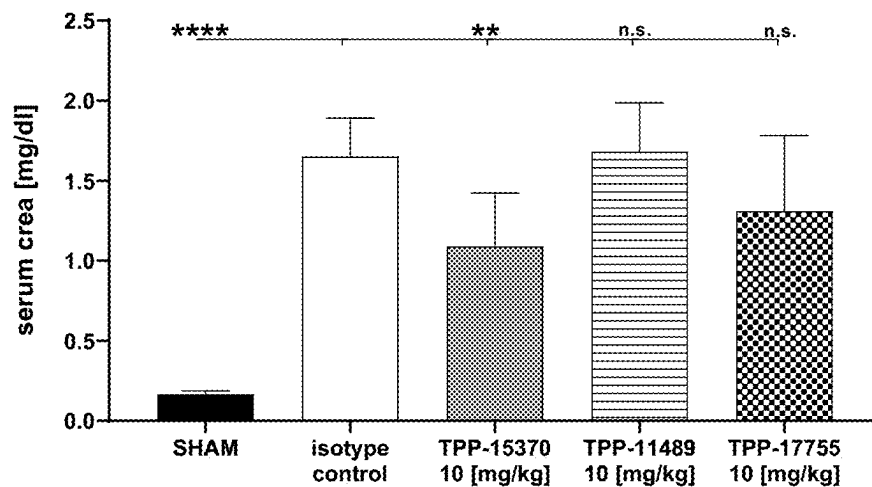


Fig. 4B

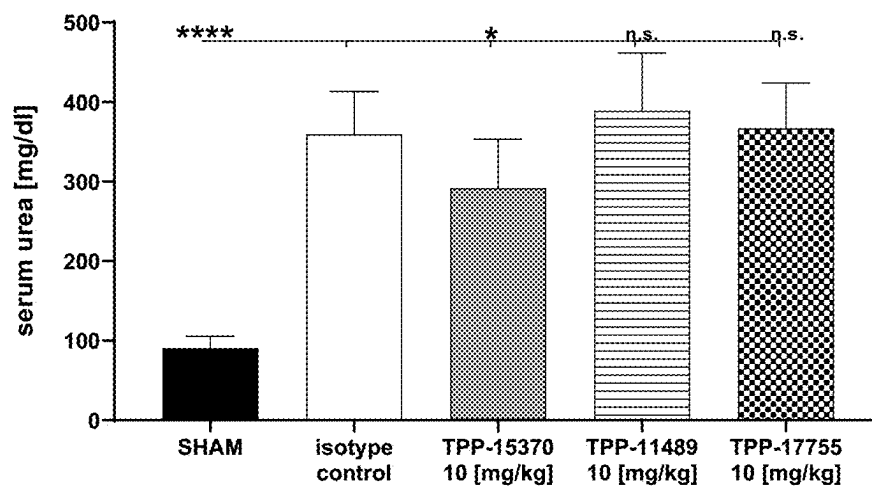


Fig. 4C

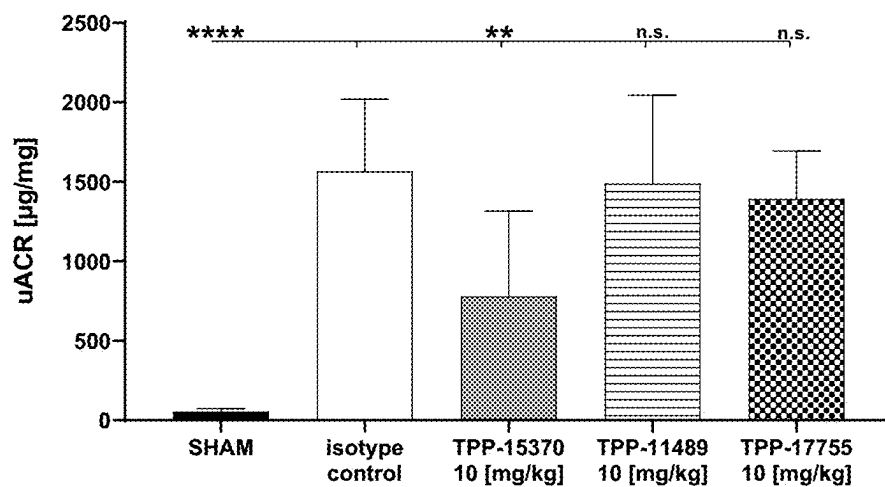


Fig. 5A

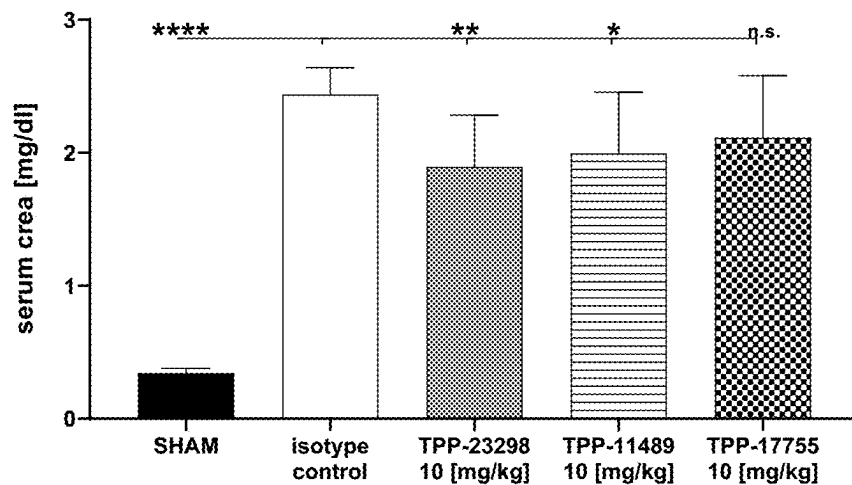


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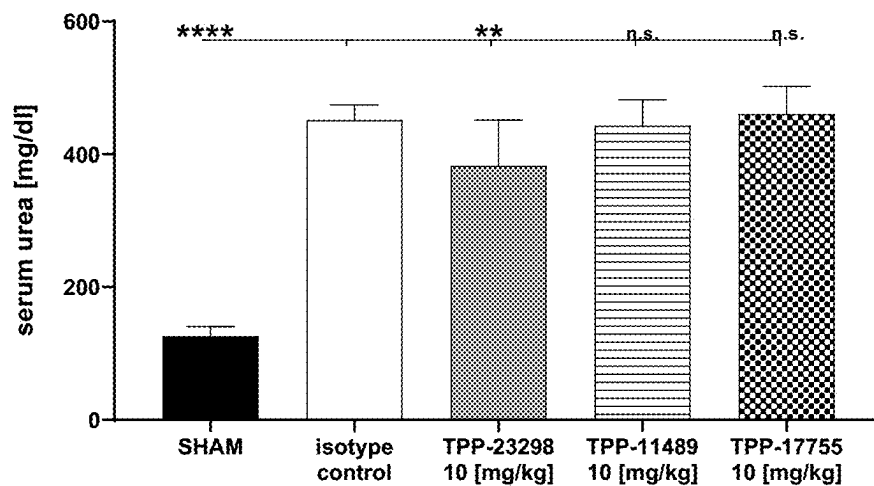


Fig. 5C

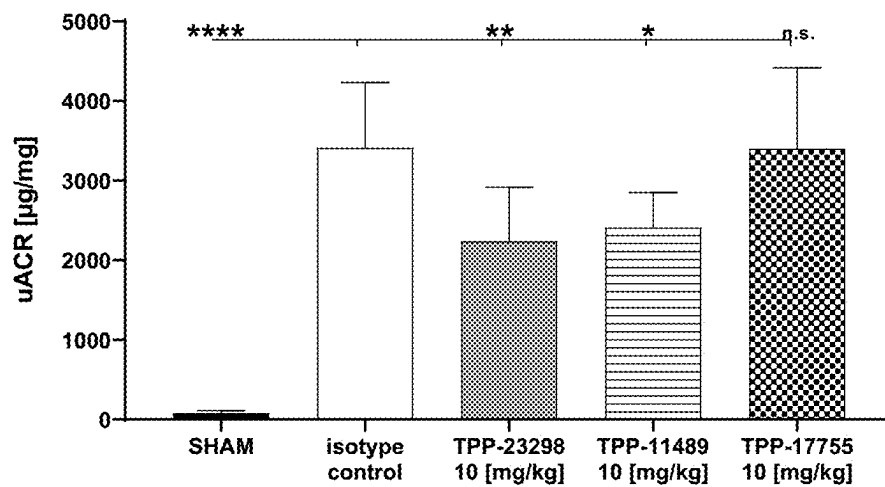


Fig. 6A

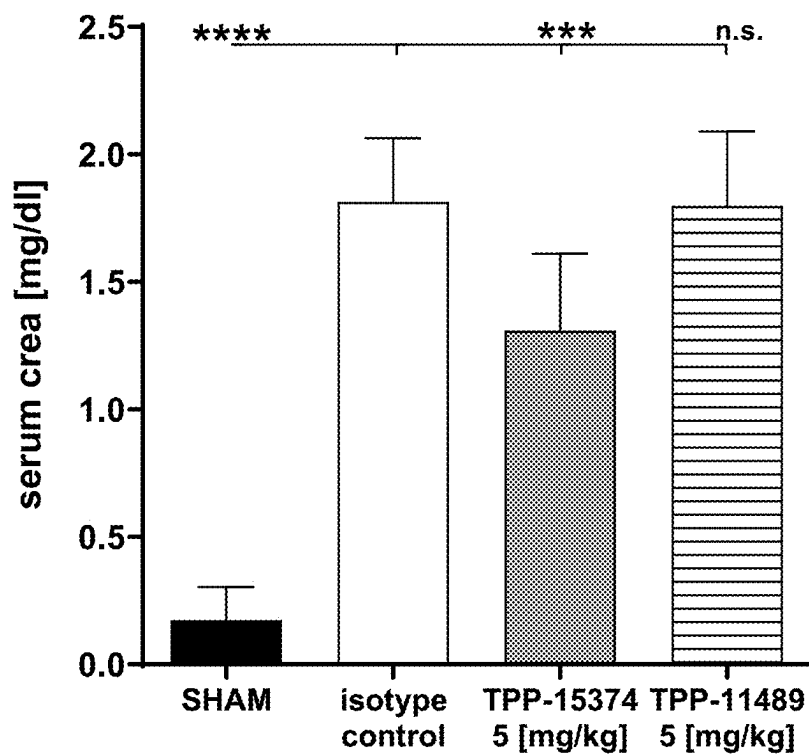


Fig. 6B

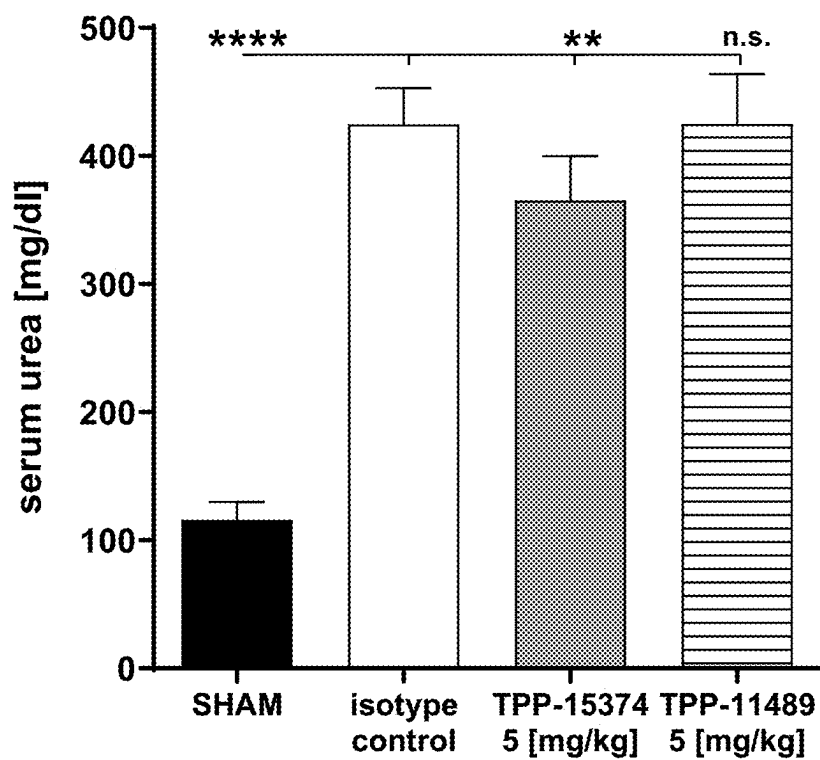


Fig. 6C

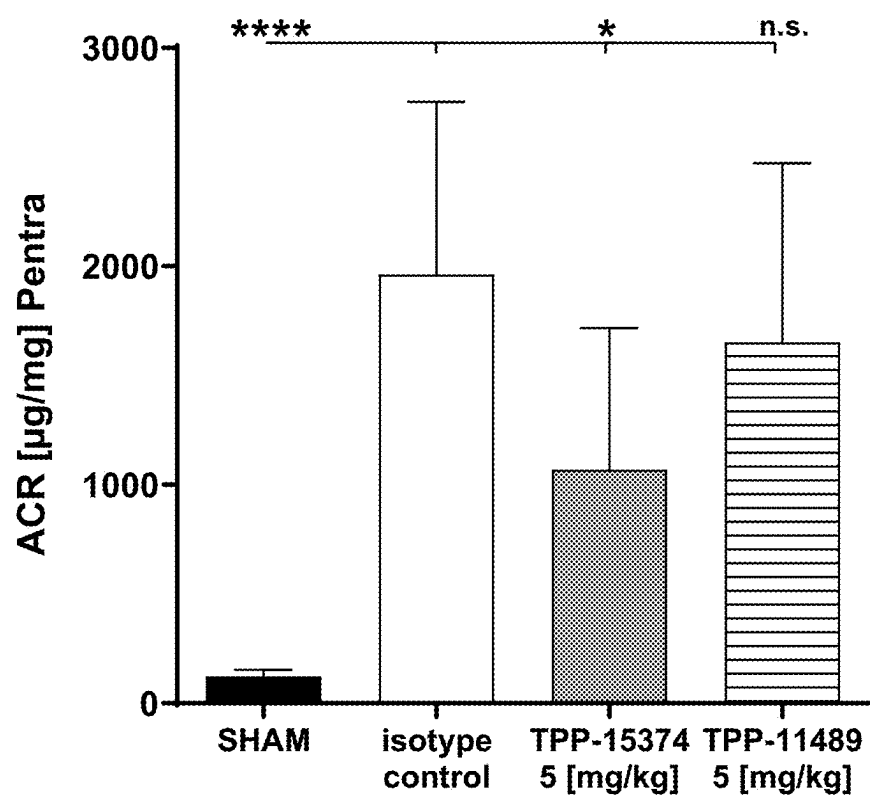


Fig. 7

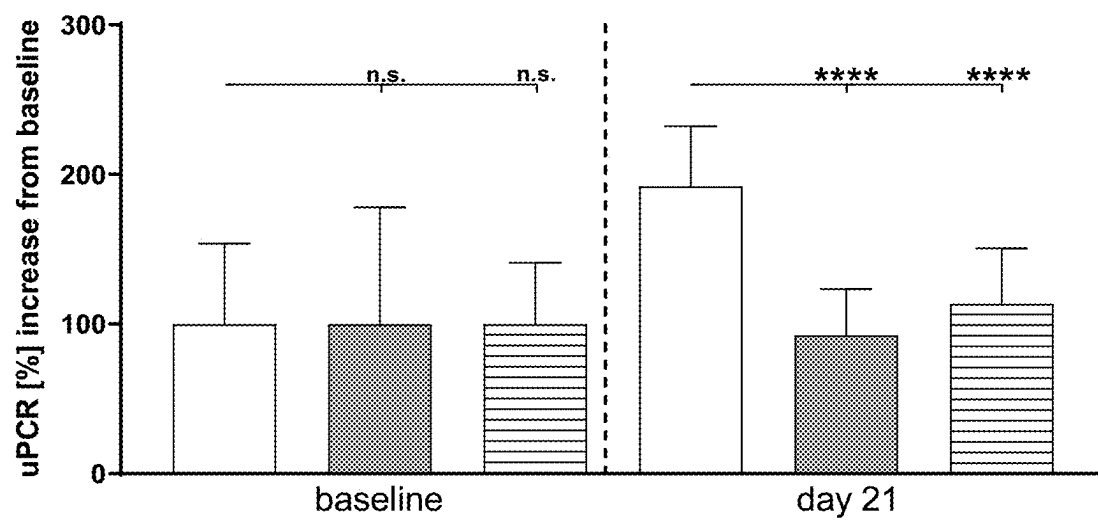


Fig. 8A

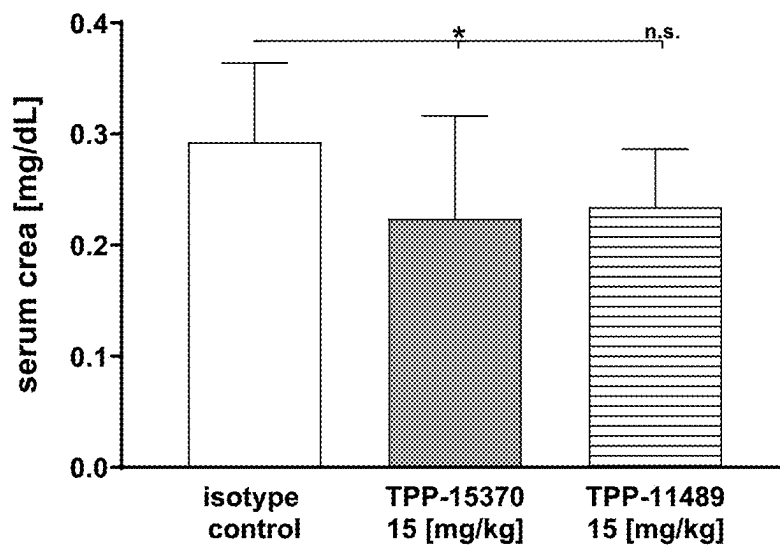


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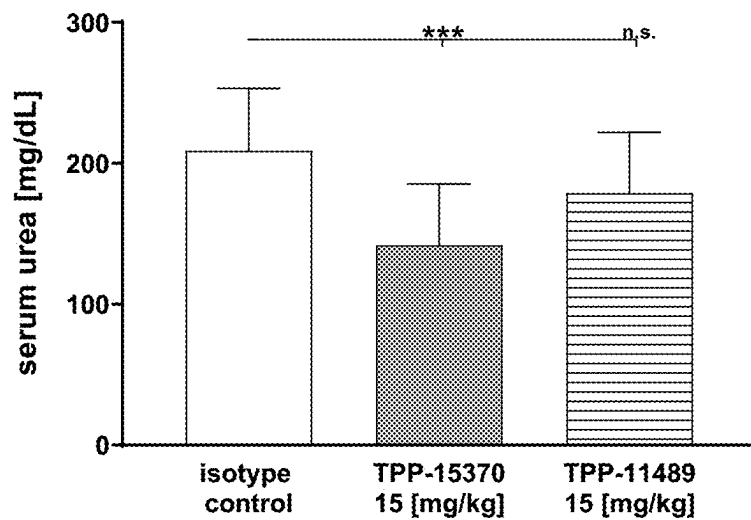


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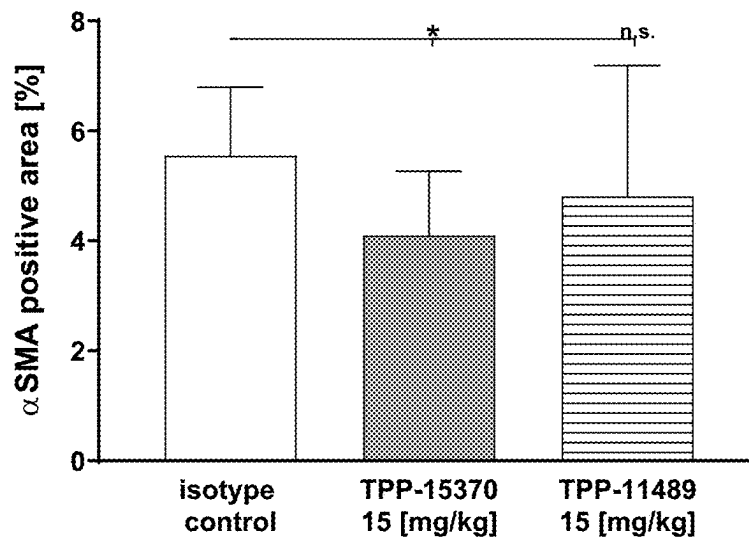


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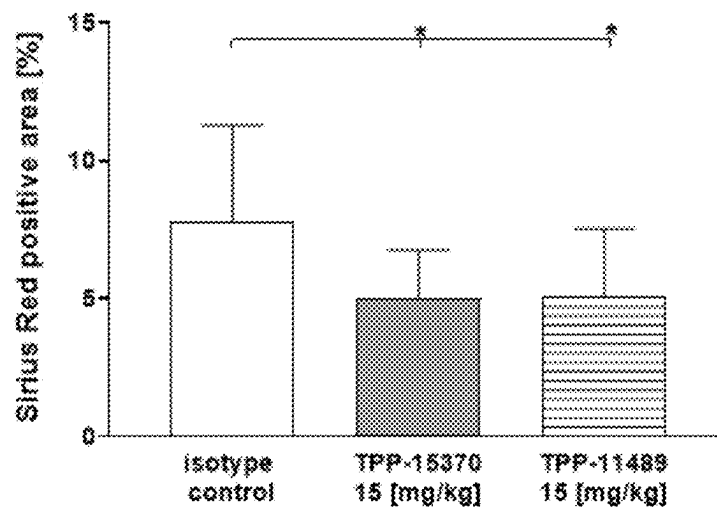


Fig. 9A

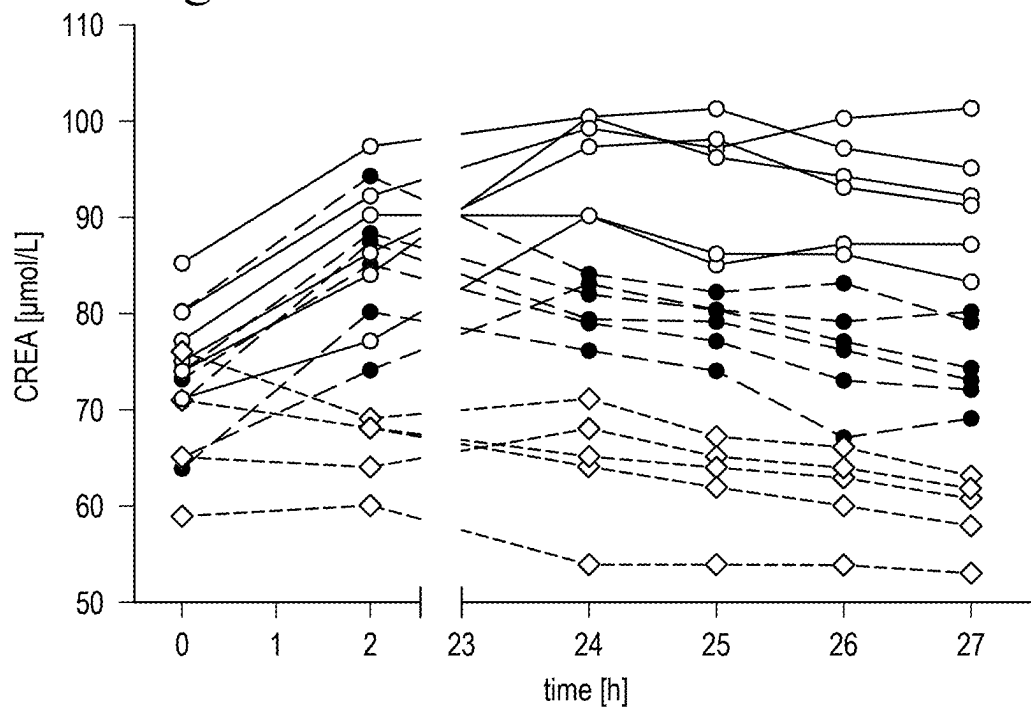


Fig. 9B

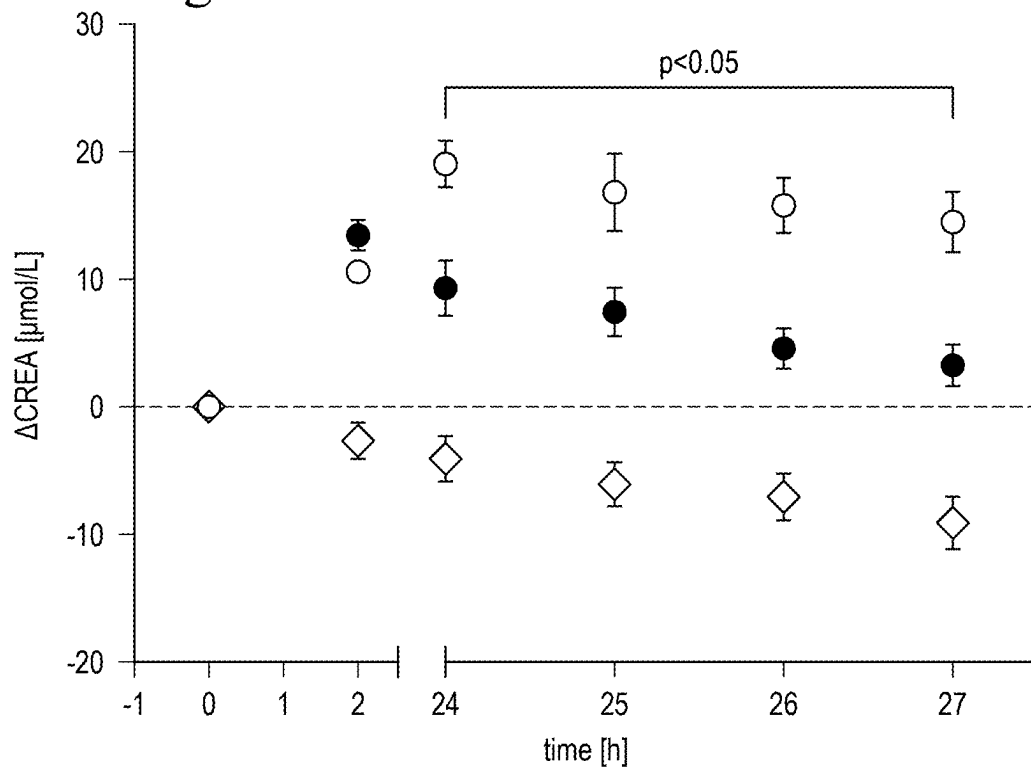


Fig. 9C

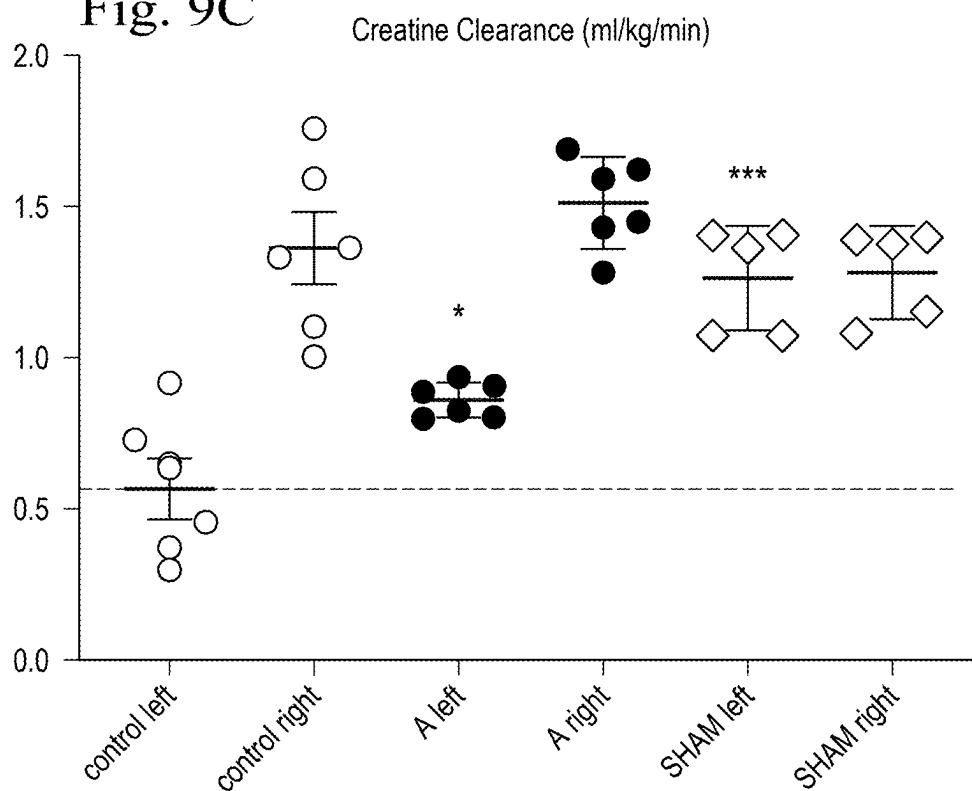
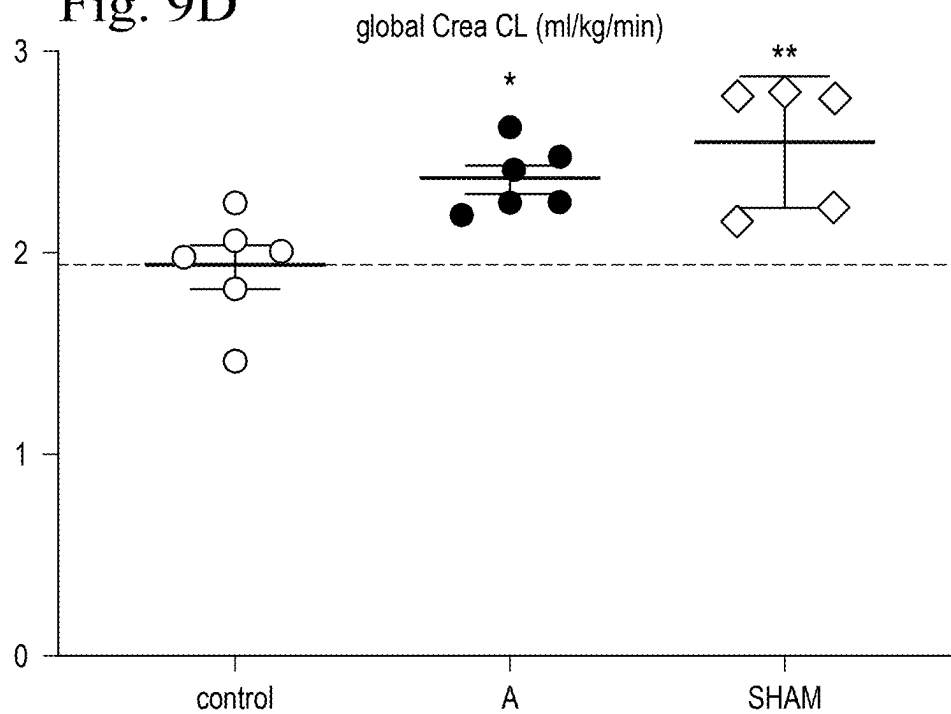


Fig. 9D



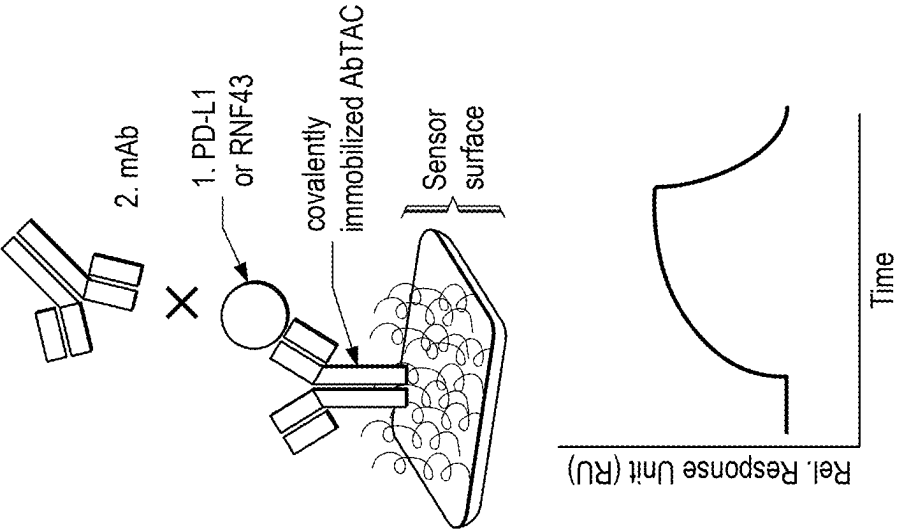


Fig. 10C

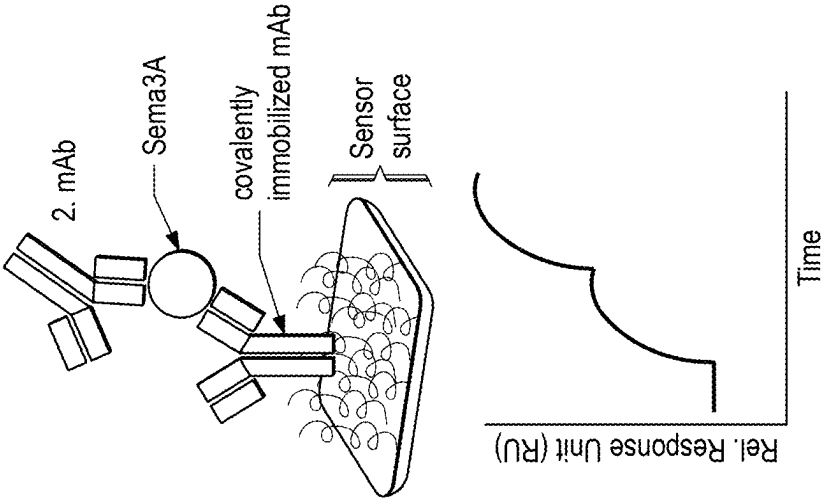


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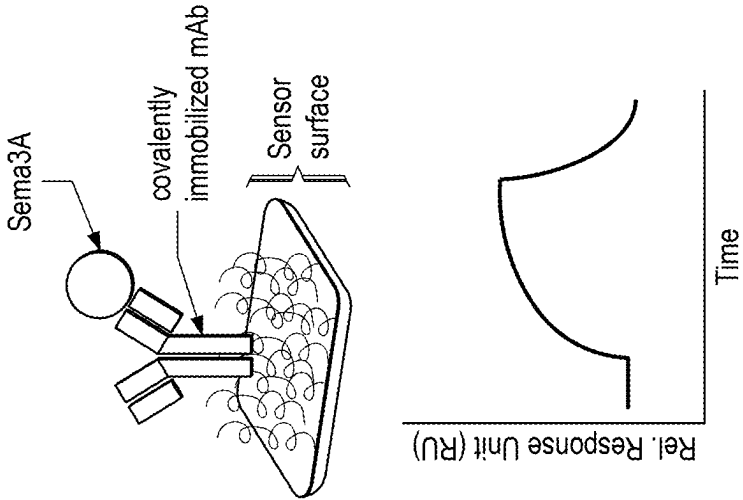


Fig. 10A

Fig. 11A

Fig. 11B

Fig. 11C

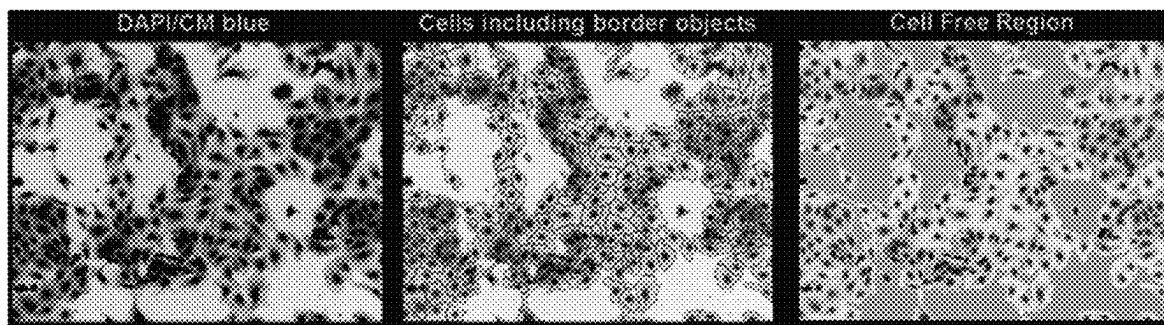
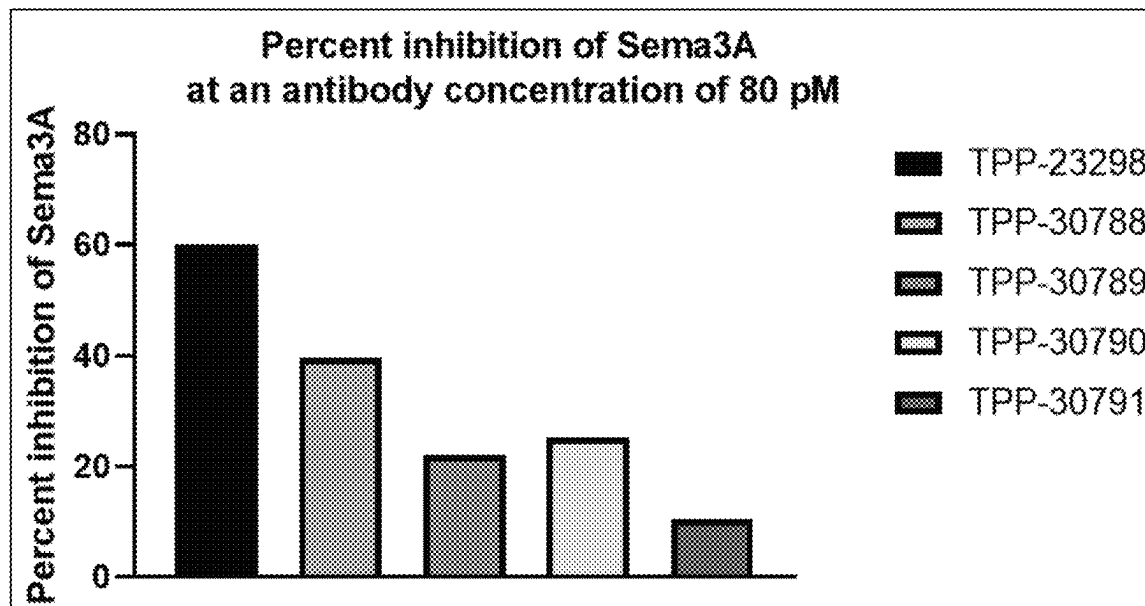


Fig. 12



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ANTI-SEMAPHORIN 3A ANTIBODIES**CROSS-REFERENCE TO RELATED APPLICATION**

This application claims the benefit of, and priority to, European Patent Application Serial No. 21165960.2 filed Mar. 30, 2021, the entire disclosure of which is hereby incorporated by reference.

SEQUENCE LISTING

This application contains references to amino acid sequences and/or nucleic acid sequences which have been submitted concurrently herewith as the sequence listing text file entitled "Seqs3_BHC201071_FC_US_ST25.txt", file size 596,189 bytes, created on Jun. 14, 2022. The aforementioned sequence listing is hereby incorporated by reference in its entirety pursuant to 37 C.F.R. § 1.52 (e) (5).

FIELD

The present disclosure provides isolated antibodies or antigen-binding fragments thereof that bind to human semaphorin 3A (Sema3A). The isolated antibody or antigen-binding fragments according to the present disclosure i) bind to human Sema3A of the sequence of SEQ ID NO: 600 with a dissociation constant (KD) ≤ 50 nM, ≤ 20 nM, ≤ 10 nM, ≤ 1 nM, or ≤ 0.1 nM; ii) cross-react with mouse, cynomolgus, rat, pig and/or dog Sema3A, particularly wherein said isolated antibodies or antigen-binding fragments thereof binds to mouse, cynomolgus, rat, pig and/or dog Sema3A with a dissociation constant (KD) ≤ 50 nM, ≤ 20 nM, ≤ 10 nM, ≤ 1 nM, or ≤ 0.1 nM; iii) bind to human Sema3A of the sequence of SEQ ID NO: 600 with a binding activity as measured by surface plasmon resonance (SPR) of $\geq 60\%$, $\geq 70\%$, $\geq 80\%$, or $\geq 90\%$; iv) inhibit the activity of human Sema3A of the sequence of SEQ ID NO: 600 in an in vitro mesangial cell migration assay with an EC50 of ≤ 10 nM, ≤ 5 nM, ≤ 2.5 nM, or ≤ 1 nM; v) inhibit the activity of human Sema3A of the sequence of SEQ ID NO: 600 in an in vitro growth cone collapse assay with an EC50 of ≤ 50 nM, ≤ 25 nM, ≤ 10 nM, or ≤ 5 nM; and/or vi) inhibit the activity of human Sema3A of the sequence of SEQ ID NO: 600 in an in vitro HUVEC repulsion assay with an EC50 of ≤ 1 nM, or ≤ 0.3 nM, ≤ 0.1 nM, ≤ 0.07 nM, ≤ 0.06 nM and/or vii) exhibiting an increased potency against cellular Sema3A, of the sequence of SEQ ID NO: 600, induced HUVEC repulsion. The present disclosure further provides isolated nucleic acid sequences encoding said antibodies or antigen-binding fragments thereof and vectors comprising same, isolated cells expressing said antibodies or antigen-binding fragments thereof, methods of producing said antibodies or antigen-binding fragments thereof and pharmaceutical compositions and kits comprising said antibodies or antigen-binding fragments thereof.

Antibodies according to the present disclosure can be used in the treatment of diseases associated with increased Sema3A levels or activity such as Alport syndrome, acute kidney injury (AKI) primary focal segmental glomerular sclerosis (FSGS), or chronic kidney disease (CKD).

BACKGROUND

Semaphorin 3A (Sema3A) is a secreted dimeric protein that acts as guidance protein. It forms a ternary complex with neuropilin-1 and different plexins which leads to the activation of different signaling pathways. It is a key regu-

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lator of cell migration, adhesion, cytoskeletal stabilization and apoptosis. Sema3A is expressed in podocyte in adult kidneys where it is induced after injury.

Excess of Sema3A interferes with the glomerular filtration barrier inducing ultrastructural changes of the filtration barrier leading to podocyte foot process effacement and albuminuria. Sema3A is also highly induced after AKI and exacerbates the injury by promoting tubular inflammation, tubular epithelial cell apoptosis and ultrastructural abnormalities of the filtration barrier. Genetic deficiency or pharmacological inhibition of Sema3A in rodents results in reduced renal damage in different animal models of kidney diseases.

Furthermore, Sema3A is expressed in retinal neurons and endothelium. It has been shown to increase vascular permeability, to promote retinal inflammation and cellular senescence and to inhibit retinal vascular regeneration in rodent models. Sema3A also plays a role in CNS disorders. Sema3A inhibition results in enhanced regeneration and/or preservation of injured axons, decreased apoptotic cell numbers and enhancement of angiogenesis, resulting in considerably better functional recovery.

WO 20141/23186 discloses an avian-mouse chimeric antibody (clone No. 4-2 strain-derived) and two humanized IgG1 variants thereof and suggests their suitability in the treatment of Alzheimer's disease.

WO 2017/074013 discloses anti-Sema3A IgG antibodies A08, C10 and F11 and suggests their suitability in the treatment of cancer.

Currently, no therapeutic option to inhibit Sema3A interaction with its receptors is available to treat patients with e.g. proteinuric kidney disease like Alport syndrome and it is presumed that monoclonal therapeutic Sema3A antibodies could be optimally suited for this. Thus, there exists a great need for novel therapeutic Sema3A antibodies useful for the treatment of diseases that are associated with elevated Sema3A levels or activity such as Alport syndrome, acute kidney injury (AKI) primary focal segmental glomerular sclerosis (FSGS), or chronic kidney disease (CKD) that has not been met so far.

OBJECTS

In view of the prior art, it is an object of the present disclosure to provide novel therapeutic Sema3A antibodies that overcome the shortcomings of Sema3A antibodies of the prior art. In particular it is an object of the present disclosure to provide novel Sema3A antibodies that are high affinity binders of human Sema3A that efficiently block Sema3A activity. Desirable Sema3A antibodies are cross-reactive to Sema3A of multiple species in order to allow for preclinical experiments. They are non-immunogenic in human therapy, i.e. they are human or humanized antibodies. Desirable Sema3A antibodies are selective to Sema3A; they do not bind to off-targets and in particular do not cross-react with other semaphorin protein family members.

Such novel Sema3A antibodies would offer major advances in the treatment of diseases associated with elevated Sema3A levels or activity such as Alport syndrome, acute kidney injury (AKI) primary focal segmental glomerular sclerosis (FSGS), or chronic kidney disease (CKD).

SUMMARY

The above-mentioned object and other objects are achieved by the teaching of the present disclosure. The present disclosure is based on the discovery of novel anti-

bodies that have a specific affinity for Sema3A and can deliver a therapeutic benefit to a subject.

Thus, in a first aspect, the present disclosure relates to an isolated antibody or antigen-binding fragment thereof that binds to human Sema3A, wherein said isolated antibody or antigen-binding fragment thereof i) binds to human Sema3A of the sequence of SEQ ID NO: 600 with a dissociation constant (KD) ≤ 50 nM, ≤ 20 nM, ≤ 10 nM, ≤ 1 nM, or ≤ 0.1 nM; ii) cross-reacts with mouse, cynomolgus, rat, pig and/or dog Sema3A, particularly wherein said isolated antibody or antigen-binding fragment thereof binds to mouse, cynomolgus, rat, pig and/or dog Sema3A with a dissociation constant (KD) ≤ 50 nM, ≤ 20 nM, ≤ 10 nM, ≤ 1 nM, or ≤ 0.1 nM; iii) binds to human Sema3A of the sequence of SEQ ID NO: 600 with a binding activity as measured by surface plasmon resonance (SPR) of $\geq 60\%$, $\geq 70\%$, $\geq 80\%$, or $\geq 90\%$; iv) inhibits the activity of human Sema3A of the sequence of SEQ ID NO: 600 in an in vitro mesangial cell migration assay with an EC50 of ≤ 10 nM, ≤ 5 nM, ≤ 2.5 nM, or ≤ 1 nM; v) inhibits the activity of human Sema3A of the sequence of SEQ ID NO: 600 in an in vitro growth cone collapse assay with an EC50 of ≤ 50 nM, ≤ 25 nM, ≤ 10 nM, or ≤ 5 nM; vi) inhibits the activity of human Sema3A of the sequence of SEQ ID NO: 600 in an in vitro HUVEC repulsion assay with an EC50 of ≤ 1 nM, or ≤ 0.3 nM, ≤ 0.1 nM, ≤ 0.07 nM, ≤ 0.06 nM and/or vii) exhibits an increased potency against cellular Sema3A, of the sequence of SEQ ID NO: 600, induced HUVEC repulsion.

The isolated antibody or antigen-binding fragment according to the present disclosure binds with high affinity to human Sema3A and inhibits its function. Thus, the isolated antibody or antigen-binding fragment according to the present disclosure may be used in the treatment of diseases associated with increased Sema3A levels or activity such as i) renal diseases, in particular acute and chronic kidney diseases, diabetic kidney diseases, Alport syndrome, acute and chronic renal failure, polycystic kidney disease (PCKD) and syndrome of inadequate ADH secretion (SIADH); ii) sequelae of renal insufficiency, in particular pulmonary edema, heart failure, uremia, anemia, electrolyte disturbances such as hyperkalemia and hyponatremia and disturbances in bone and carbohydrate metabolism; iii) vascular hyperpermeability, diabetic retinopathy, deterioration of the blood retinal barrier, macular edema, particularly age related macular edema, non-proliferative age-related macular edema and non-proliferative diabetic macular edema; iv) diseases of the central or peripheral nervous system in particular neuropathic pain, spinal cord injury, multiple sclerosis, traumatic brain injury, brain edema and neurodegenerative diseases, particularly Alzheimer's disease, Parkinson's disease, Huntington's disease, amyotrophic lateral sclerosis, progressive supranuclear paralysis, black substance degeneration, Shy-Drager syndrome, olivopontocerebellar atrophy and spinocerebellar degeneration; v) cancer, in particular intestinal cancer, colorectal cancer, lung cancer, breast cancer, brain cancer, melanoma, renal cell cancer, leukemia, lymphoma, T-cell lymphoma, stomach cancer, pancreatic cancer, cervical cancer, endometrial cancer, ovarian cancer, esophageal cancer, liver cancer, squamous cell carcinoma of the head and neck, skin cancer, urinary tract cancer, prostate cancer, choriocarcinoma, pharyngeal cancer and larynx cancer.

The isolated antibody or antigen-binding fragment according to the present disclosure may further be used in the diagnosis of Sema3A-related disorders.

In a further aspect, the present disclosure relates to an isolated nucleic acid sequence that encodes the antibody or antigen-binding fragment according to the present disclosure.

In a further aspect, the present disclosure relates to a vector comprising a nucleic acid sequence according to the present disclosure.

In a further aspect, the present disclosure relates to an isolated cell expressing the antibody or antigen-binding fragment according to the present disclosure and/or comprising the nucleic acid according to the present disclosure or the vector according to the present disclosure.

In a further aspect, the present disclosure relates to a method of producing the isolated antibody or antigen-binding fragment according to the present disclosure comprising culturing of the cell according to the present disclosure and optionally purification of the antibody or antigen-binding fragment thereof.

In a further aspect, the present disclosure relates to a pharmaceutical composition comprising the isolated antibody or antigen-binding fragment according to the present disclosure or the antibody conjugate according to the present disclosure.

In a further aspect, the present disclosure relates to a kit comprising the isolated antibody or antigen-binding fragment according to the present disclosure or the conjugate according to the present disclosure and instructions for use.

DETAILED DESCRIPTION

The present disclosure may be understood more readily by reference to the following detailed description of the disclosure and the examples included therein.

Definitions

Unless defined otherwise, all technical and scientific terms used herein have the meaning commonly understood by one of ordinary skill in the art to which this disclosure belongs. The following references, however, can provide one of skill in the art to which this disclosure pertains with a general definition of many of the terms used in this disclosure, and can be referenced and used so long as such definitions are consistent with the meaning commonly understood in the art. Such references include, but are not limited to, Singleton et al., Dictionary of Microbiology and Molecular Biology (2nd ed. 1994); The Cambridge Dictionary of Science and Technology (Walker ed., 1988); Hale & Marham, The Harper Collins Dictionary of Biology (1991); Lackie et al., The Dictionary of Cell & Molecular Biology (3d ed. 1999); and Cellular and Molecular Immunology, Eds. Abbas, Lichtman and Pober, 2nd Edition, W.B. Saunders Company. Any additional technical resource available to the person of ordinary skill in the art providing definitions of terms used herein having the meaning commonly understood in the art can be consulted. For the purposes of the present disclosure, the following terms are further defined. Additional terms are defined elsewhere in the description. As used herein and in the appended claims, the singular forms "a," and "the" include plural reference unless the context clearly dictates otherwise. Thus, for example, reference to "a gene" is a reference to one or more genes and includes equivalents thereof known to those skilled in the art, and so forth.

In the context of the present disclosure, the term "comprises" or "comprising" means "including, but not limited to". The term is intended to be open-ended, to specify the

presence of any stated features, elements, integers, steps or components, but not to preclude the presence or addition of one or more other features, elements, integers, steps, components or groups thereof. The term “comprising” thus includes the more restrictive terms “consisting of” and “essentially consisting of”. In one embodiment the term “comprising” as used throughout the application and in particular within the claims may be replaced by the term “consisting of”.

In this context, the term “about” or “approximately” means within 80% to 120%, alternatively within 90% to 110%, including within 95% to 105% of a given value or range.

The terms “polypeptide” and “protein” are used interchangeably herein to refer to a polymer of amino acid residues. The terms apply to amino acid polymers in which one or more amino acid residue is an artificial chemical mimetic of a corresponding naturally occurring amino acid, as well as to naturally occurring amino acid polymers and non-naturally occurring amino acid polymers. Unless otherwise indicated, a particular polypeptide sequence also implicitly encompasses conservatively modified variants thereof.

As used herein “Sema3A” designates “semaphorin 3A”, also known as “HH16”, “SemD”, “COLL1”, “SEMA1”, “SEMA2”, “SEMA3”, “coll-1”, “Hsema-1”, “SEMAIII”, “Hsema-III”, “collapsin 1”, “semaphorin D”, “semaphorin III”, and “semaphorin L”.

The terms “anti-Sema3A antibody” and “an antibody that binds to Sema3A” refer to an antibody that is capable of binding Sema3A with sufficient affinity such that the antibody is useful as a diagnostic and/or therapeutic agent in targeting Sema3A. In one embodiment, the extent of binding of an anti-Sema3A antibody to an unrelated, non-Sema3A protein is less than about 10%, less than about 5%, or less than about 2% of the binding of the antibody to Sema3A as measured, e.g., by standard ELISA procedure. In certain embodiments, an antibody that binds to Sema3A has a binding activity (EC50) of $\leq 1 \mu\text{M}$, $\leq 100 \text{ nM}$, $\leq 10 \text{ nM}$, $\leq 1 \text{ nM}$, $\leq 0.1 \text{ nM}$, $\leq 0.01 \text{ nM}$, or $\leq 0.001 \text{ nM}$ (e.g. 10^{-8} M or less, e.g. from 10^{-8} M to 10^{-13} M , e.g., from 10^{-9} M to 10^{-13} M). In certain embodiments, an anti-Sema3A antibody binds to an epitope of Sema3A that is conserved among Sema3A from different species.

The term “antibody”, as used herein, is intended to refer to immunoglobulin molecules. Antibodies may comprise four polypeptide chains, two heavy (H) chains (about 50-70 kDa) and two light (L) chains (about 25 kDa) which are typically inter-connected by disulfide bonds. In particular embodiments, the antibody is composed of two identical pairs of polypeptide chains. The amino-terminal portion of each chain includes a “variable” region of about 100 to 110 or more amino acids primarily responsible for antigen recognition. The heavy chain variable region is abbreviated herein as VH, the light chain variable region is abbreviated herein as VL. The carboxyl-terminal portion of each chain defines a constant region primarily responsible for effector function. The heavy chain constant region can comprise e.g. three domains CH1, CH2 and CH3. The light chain constant region is comprised of one domain (CL). The VH and VL regions can be further subdivided into regions of hypervariability, termed complementarity determining regions (CDR), interspersed with regions that are more conserved, termed framework regions (FR). Each VH and VL is typically composed of three CDRs and up to four FRs, arranged from amino-terminus to carboxy-terminus e.g., in the following order: FR1, CDR1, FR2, CDR2, FR3, CDR3, FR4.

As used herein, the term “Complementarity Determining Regions” (CDRs; e.g., CDR1, CDR2, and CDR3) refers to the amino acid residues of an antibody variable domain the presence of which are necessary for antigen binding. Each variable domain typically has three CDR regions identified as CDR1, CDR2 and CDR3. Each complementarity determining region may comprise amino acid residues from a “complementarity determining region” as defined by Kabat (Kabat et al., Sequences of Proteins of Immunological Interest, 5th Ed. Public Health Service, National Institutes of Health, Bethesda, MD. (1991)) and/or those residues from a “hypervariable loop” (Chothia and Lesk; J Mol Biol 196: 901-917 (1987)). In some instances, a complementarity determining region can include amino acids from both a CDR region defined according to Kabat and a hypervariable loop.

“Framework” or FR residues are those variable domain residues other than the hypervariable region residues.

The phrase “constant region” refers to the portion of the antibody molecule that confers effector functions.

The term “Fc region” herein is used to define a C-terminal region of an immunoglobulin heavy chain that contains at least a portion of the constant region. The term includes native sequence Fc regions and variant Fc regions. In one embodiment, a human IgG heavy chain Fc region extends from Cys226, or from Pro230, to the carboxyl-terminus of the heavy chain. However, the C-terminal lysine (Lys447) of the Fc region may or may not be present. Unless otherwise specified herein, numbering of amino acid residues in the Fc region or constant region is according to the EU numbering system, also called the EU index, as described in Kabat et al., Sequences of Proteins of Immunological Interest, 5th Ed. Public Health Service, National Institutes of Health, Bethesda, MD, 1991.

Immunoglobulins can be assigned to different classes depending on the amino acid sequence of the constant domain of their heavy chains. Heavy chains are classified as μ (μ), δ (Δ), γ (γ), α (α), and ϵ (ϵ), and define the antibody’s isotype as IgM, IgD, IgG, IgA, and IgE, respectively. In particular embodiments, the antibody according to the present disclosure is an IgG antibody. Several of these may be further divided into subclasses or isotypes, e.g. IgG1, IgG2, IgG3, IgG4, IgA1 and IgA2. In particular embodiments, the antibody according to the present disclosure is an IgG1, an IgG2, an IgG3 or an IgG4 antibody, more particularly an IgG1 or an IgG4 antibody. Different isotypes may have different effector functions. Human light chains are classified as kappa (κ) and lambda (λ) light chains. Within light and heavy chains, the variable and constant regions are joined by a “J” region of about 12 or more amino acids, with the heavy chain also including a “D” region of about 10 more amino acids. See generally, Fundamental Immunology, Ch. 7 (Paul, W., ed., 2nd ed. Raven Press, N.Y. (1989)).

A “functional fragment” or “antigen-binding antibody fragment” of an antibody/immunoglobulin hereby is defined as a fragment of an antibody/immunoglobulin (e.g., a variable region of an IgG) that retains the antigen-binding region. An “antigen-binding region” of an antibody typically is found in one or more hyper variable region(s) of an antibody, e.g., the CDR1, -2, and/or -3 regions; however, the variable “framework” regions can also play an important role in antigen binding, such as by providing a scaffold for the CDRs. Preferably, the “antigen-binding region” comprises at least amino acid residues 4 to 103 of the variable light (VL) chain and 5 to 109 of the variable heavy (VH) chain, more preferably amino acid residues 3 to 107 of VL

and 4 to 111 of VH, and particularly preferred are the complete VL and VH chains (amino acid positions 1 to 109 of VL and 1 to 113 of VH; numbering according to WO 97/08320).

Nonlimiting examples of “functional fragments” or “antigen-binding antibody fragments” include Fab, Fab', F(ab')₂, Fv fragments, domain antibodies (dAb), complementarity determining region (CDR) fragments, single-chain antibodies (scFv), single chain antibody fragments, diabodies, triabodies, tetrabodies, minibodies, linear antibodies (Zapata et al., Protein Eng., 8 (10): 1057-1062 (1995)); chelating recombinant antibodies, tribodies or bibodies, intrabodies, nanobodies, small modular immunopharmaceuticals (SMIPs), an antigen-binding-domain immunoglobulin fusion protein, a camelized antibody, a VHH containing antibody, or muteins or derivatives thereof, and polypeptides that contain at least a portion of an immunoglobulin that is sufficient to confer specific antigen binding to the polypeptide, such as a CDR sequence, as long as the antibody retains the desired biological activity; and multispecific antibodies such as bi- and tri-specific antibodies formed from antibody fragments (C. A. K. Borrebaeck, editor (1995) *Antibody Engineering (Breakthroughs in Molecular Biology)*, Oxford University Press; R. Kontermann & S. Diebel, editors (2001) *Antibody Engineering (Springer Laboratory Manual)*, Springer Verlag). An antibody other than a “bispecific” or “bifunctional” antibody is understood to have each of its binding sites identical. The F(ab')₂ or Fab may be engineered to minimize or completely remove the intermolecular disulfide interactions that occur between the C_{H1} and C_L domains. Papain digestion of antibodies produces two identical antigen-binding fragments, called “Fab” fragments, each with a single antigen-binding site, and a residual “Fc” fragment, whose name reflects its ability to crystallize readily. Pepsin treatment yields an F(ab')₂ fragment that has two “Fv” fragments. An “Fv” fragment is the minimum antibody fragment that contains a complete antigen recognition and binding site. This region consists of a dimer of one heavy- and one light-chain variable domain in tight, non-covalent association. It is in this configuration that the three CDRs of each variable domain interact to define an antigen binding site on the surface of the VH-VL dimer. Collectively, the six CDRs confer antigen-binding specificity to the antibody. However, even a single variable domain (or half of an Fv comprising only three CDRs specific for an antigen) has the ability to recognize and bind antigen.

“Single-chain Fv” or “sFv” or “scFv” antibody fragments comprise the VH and VL domains of antibody, wherein these domains are present in a single polypeptide chain.

Preferably, the Fv polypeptide further comprises a polypeptide linker between the VH and VL domains that enables the Fv to form the desired structure for antigen binding. For a review of Fvs see Pluckthun in *The Pharmacology of Monoclonal Antibodies*, vol. 113, Rosenberg and Moore eds., Springer-Verlag, New York, pp. 269-315 (1994).

The Fab fragment also contains the constant domain of the light chain and the first constant domain (CH1) of the heavy chain. Fab fragments differ from Fab' fragments by the addition of a few residues at the carboxyl terminus of the heavy chain CH1 domain including one or more cysteine residues from the antibody hinge region. Fab'-SH is the designation herein for Fab' in which the cysteine residue(s) of the constant domains bear a free thiol group. F(ab')₂ antibody fragments originally were produced as pairs of Fab' fragments which have hinge cysteine residues between them.

The term “mutein” or “variant” can be used interchangeably and refers to an antibody or antigen-binding fragment that contains at least one amino acid substitution, deletion, or insertion in the variable region or the portion equivalent to the variable region, provided that the mutein or variant retains the desired binding affinity or biological activity. Variants of the antibodies or antigen-binding antibody fragments contemplated in the disclosure are molecules in which the binding activity of the antibody or antigen-binding antibody fragment is maintained.

A “chimeric antibody” or antigen-binding fragment thereof is defined herein as one, wherein the variable domains are derived from a non-human origin and some or all constant domains are derived from a human origin.

“Humanized antibodies” contain CDR regions derived from a non-human species, such as mouse, that have, for example, been engrafted, along with any necessary framework back-mutations, into human sequence-derived V regions. Thus, for the most part, humanized antibodies are human immunoglobulins (recipient antibody) in which residues from a hypervariable region of the recipient are replaced by residues from a hypervariable region of a non-human species (donor antibody) such as mouse, rat, rabbit or non-human primate having the desired specificity, affinity, and capacity. See, for example, U.S. Pat. Nos. 5,225,539; 5,585,089; 5,693,761; 5,693,762; 5,859,205, each herein incorporated by reference. In some instances, framework residues of the human immunoglobulin are replaced by corresponding non-human residues (see, for example, U.S. Pat. Nos. 5,585,089; 5,693,761; 5,693,762, each herein incorporated by reference). Furthermore, humanized antibodies may comprise residues that are not found in the recipient antibody or in the donor antibody. These modifications are made to further refine antibody performance (e.g., to obtain desired affinity). In general, the humanized antibody will comprise substantially all of at least one, and typically two, variable domains, in which all or substantially all of the hypervariable regions correspond to those of a non-human immunoglobulin and all or substantially all of the framework regions are those of a human immunoglobulin sequence. The humanized antibody optionally also will comprise at least a portion of an immunoglobulin constant region (Fc), typically that of a human immunoglobulin. For further details see Jones et al., *Nature* 331:522-25 (1986); Riechmann et al., *Nature* 332:323-27 (1988); and Presta, *Curr. Opin. Struct. Biol.* 2:593-96 (1992), each herein incorporated by reference.

“Human antibodies” or “fully human antibodies” comprise human derived CDRs, i.e. CDRs of human origin. Fully human antibodies may comprise a low number of germline deviations compared with the closest human germline reference determined based on the IMGT database (<http://www.imgt.org>). For example, a fully human antibody according to the current disclosure may comprise up to 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 germline deviations in the CDRs compared with the closest human germline reference. Fully human antibodies can be developed from human derived B cells by cloning techniques in combination with a cell enrichment or immortalization step. The majority of fully human antibodies, however, are isolated either from immunized mice transgenic for the human IgG locus or from sophisticated combinatorial libraries by phage display (Brüggemann M., Osborn M. J., Ma B., Hayre J., Avis S., Lundstrom B. and Buelow R., *Human Antibody Production in Transgenic Animals*, *Arch Immunol Ther Exp (Warsz.)* 63 (2015), 101-108; Carter P. J., *Potent antibody therapeutics by design*, *Nat Rev Immunol* 6 (2006), 343-357; Frenzel A.,

Schirrmann T. and Hust M., Phage display-derived human antibodies in clinical development and therapy, *MAbs* 8 (2016), 1177-1194; Nelson A. L., Dhimoalea E. and Reichert J. M., Development trends for human monoclonal antibody therapeutics, *Nat Rev Drug Discov* 9 (2010), 767-774.).

Several techniques are available to generate fully human antibodies (cf. WO2008/112640 A3). Cambridge Antibody Technologies (CAT) and Dyax have obtained antibody cDNA sequences from peripheral B cells isolated from immunized humans and devised phage display libraries for the identification of human variable region sequences of a particular specificity. Briefly, the antibody variable region sequences are fused either with the Gene III or Gene VIII structure of the M13 bacteriophage. These antibody variable region sequences are expressed either as Fab or single chain Fv (scFv) structures at the tip of the phage carrying the respective sequences. Through rounds of a panning process using different levels of antigen binding conditions (stringencies), phages expressing Fab or scFv structures that are specific for the antigen of interest can be selected and isolated. The antibody variable region cDNA sequences of selected phages can then be elucidated using standard sequencing procedures. These sequences may then be used for the reconstruction of a full antibody having the desired isotype using established antibody engineering techniques. Antibodies constructed in accordance with this method are considered fully human antibodies (including the CDRs). In order to improve the immunoreactivity (antigen binding affinity and specificity) of the selected antibody, an in vitro maturation process can be introduced, including a combinatorial association of different heavy and light chains, deletion/addition/mutation at the CDR3 of the heavy and light chains (to mimic V-J, and V-D-J recombination), and random mutations (to mimic somatic hypermutation). An example of a "fully human" antibody generated by this method is the anti-tumor necrosis factor α antibody, Humira (adalimumab).

"Human Engineered™" antibodies generated by altering the parent sequence according to the methods set forth in Studnicka et al., U.S. Pat. No. 5,766,886.

An antibody of the disclosure may be derived from a recombinant antibody gene library. The development of technologies for making repertoires of recombinant human antibody genes, and the display of the encoded antibody fragments on the surface of filamentous bacteriophage, has provided a recombinant means for directly making and selecting human antibodies, which also can be applied to humanized, chimeric, murine or mutein antibodies. The antibodies produced by phage technology are produced as antigen binding fragments—usually Fv or Fab fragments—in bacteria and thus lack effector functions. Effector functions can be introduced by one of two strategies: The fragments can be engineered either into complete antibodies for expression in mammalian cells, or into bispecific antibody fragments with a second binding site capable of triggering an effector function. Typically, heavy chain VH-CH1 and light chain VL-CL of antibodies are separately cloned by PCR and recombined randomly in combinatorial phage display libraries, which can then be selected for binding to a particular antigen. The Fab fragments are expressed on the phage surface, i.e., physically linked to the genes that encode them. Thus, selection of Fab by antigen binding co-selects for the Fab encoding sequences, which can be amplified subsequently. By several rounds of antigen binding and re-amplification, a procedure termed panning, Fab specific for the antigen are enriched and finally isolated.

A variety of procedures have been described for human antibodies deriving from phage-display libraries. Such libraries may be built on a single master framework, into which diverse in vivo-formed (i. e. human-derived) CDRs are allowed to recombine as described by Carlsson and Söderlind *Exp. Rev. Mol. Diagn.* 1 (1), 102-108 (2001), Söderlin et al., *Nat. Biotech.* 18, 852-856 (2000) and U.S. Pat. No. 6,989,250. Alternatively, such an antibody library may be based on amino acid sequences that have been designed in silico and encoded by nucleic acids that are synthetically created. In silico design of an antibody sequence is achieved, for example, by analyzing a database of human sequences and devising a polypeptide sequence utilizing the data obtained therefrom. Methods for designing and obtaining in silico-created sequences are described, for example, in Knappik et al., *J. Mol. Biol.* (2000) 296:57; Krebs et al., *J. Immunol. Methods.* (2001) 254: 67; and U.S. Pat. No. 6,300,064. For a review of phage display screening (for example see Hoet R M et al, *Nat Biotechnol* 2005; 23 (3): 344-8), the well-established hybridoma technology (for example see Köhler and Milstein *Nature*. 1975 Aug. 7; 256 (5517): 495-7), or immunization of mice inter alia immunization of hMAb mice (e.g. VelocImmune Mouse®).

The term "monoclonal antibody" as used herein refers to an antibody obtained from a population of substantially homogeneous antibodies, i.e., the individual antibodies comprising the population are identical except for possible mutations, e.g., naturally occurring mutations, that may be present in minor amounts. Thus, the term "monoclonal" indicates the character of the antibody as not being a mixture of discrete antibodies. In contrast to polyclonal antibody preparations, which typically include different antibodies directed against different determinants (epitopes), each monoclonal antibody of a monoclonal antibody preparation is directed against a single determinant on an antigen. In addition to their specificity, monoclonal antibody preparations are advantageous in that they are typically uncontaminated by other immunoglobulins. The term "monoclonal" is not to be construed as to require production of the antibody by any particular method. For example, the monoclonal antibodies to be used may be made by the hybridoma method first described by Kohler et al., *Nature*, 256:495 [1975, or may be made by recombinant DNA methods (see, e.g., U.S. Pat. No. 4,816,567). The "monoclonal antibodies" may also be recombinant, chimeric, humanized, human, Human Engineered™, or antibody fragments, for example.

An "isolated" antibody is one that has been identified and separated from a component of the cell that expressed it. Contaminant components of the cell are materials that would interfere with diagnostic or therapeutic uses of the antibody, and may include enzymes, hormones, and other proteinaceous or non-proteinaceous solutes.

An "isolated" nucleic acid is one that has been identified and separated from a component of its natural environment. An isolated nucleic acid includes a nucleic acid molecule contained in cells that ordinarily contain the nucleic acid molecule, but the nucleic acid molecule is present extrachromosomally or at a chromosomal location that is different from its natural chromosomal location.

As used herein, an antibody "binds specifically to", is "specific to/for" or "specifically recognizes" an antigen of interest, e.g. Sema3A, is one that binds the antigen with sufficient affinity such that the antibody is useful as a therapeutic agent in targeting a cell or tissue expressing the antigen. The term "specifically recognizes" or "binds specifically to" or is "specific to/for" a particular polypeptide or an epitope on a particular polypeptide target as used herein

can be exhibited, for example, by an antibody, or antigen-binding fragment thereof, having a monovalent K_D for the antigen of less than about 10^{-4} M, alternatively less than about 10^{-5} M, alternatively less than about 10^{-6} M, alternatively less than about 10^{-7} M, alternatively less than about 10^{-8} M, alternatively less than about 10^{-9} M, alternatively less than about 10^{-10} M, alternatively less than about 10^{-11} M, alternatively less than about 10^{-12} M, or less.

An antibody “binds selectively to,” is “selective to/for” or “selectively recognizes” an antigen if such antibody is able to discriminate between such antigen and one or more reference antigen(s). In particular, an antibody that “binds selectively to” an antigen does not significantly cross-react with proteins other than orthologs and variants (e.g. mutant forms, splice variants, or proteolytically truncated forms) of the aforementioned antigen target. In its most general form, “selective binding,” “binds selectively to,” is “selective to/for” or “selectively recognizes” is referring to the ability of the antibody to discriminate between the antigen of interest and an unrelated antigen, as determined, for example, in accordance with one of the following methods. Such methods comprise but are not limited to surface plasmon resonance (SPR), Western blots, ELISA-, RIA-, ECL-, IRMA-tests and peptide scans. For example, a standard ELISA assay can be carried out. The scoring may be carried out by standard color development (e.g. secondary antibody with horseradish peroxidase and tetramethyl benzidine with hydrogen peroxide). The reaction in certain wells is scored by the optical density, for example, at 450 nm. Typical background (=negative reaction) may be 0.1 OD; typical positive reaction may be 1 OD. This means the difference positive/negative is more than 5-fold, 10-fold, 50-fold, and preferably more than 100-fold. Typically, determination of binding selectivity is performed by using not a single reference antigen, but a set of about three to five unrelated antigens, such as milk powder, BSA, transferrin or the like.

“Binding affinity” or “affinity” refers to the strength of the total sum of non-covalent interactions between a single binding site of a molecule and its binding partner. Unless indicated otherwise, as used herein, “binding affinity” refers to intrinsic binding affinity which reflects a 1:1 interaction between members of a binding pair (e.g. an antibody and an antigen). The dissociation constant “ K_D ” is commonly used to describe the affinity between a molecule (such as an antibody) and its binding partner (such as an antigen) i.e. how tightly a ligand binds to a particular protein. Ligand-protein affinities are influenced by non-covalent intermolecular interactions between the two molecules. Affinity can be measured by common methods known in the art, including those described herein. In one embodiment, the “ K_D ” or “ K_D value” according to this disclosure is measured by using surface plasmon resonance assays using a Biacore T200 instrument (GE Healthcare Biacore, Inc.). Other suitable devices are BIACORE T100, BIACORE (R)-2000, BIACORE 4000, a BIACORE (R)-3000 (BIAcore, Inc., Piscataway, NJ), or ProteOn XPR36 instrument (Bio-Rad Laboratories, Inc.).

As used herein, the term “epitope” includes any protein determinant capable of specific binding to an immunoglobulin or T-cell receptor. Epitopic determinants usually consist of chemically active surface groupings of molecules such as amino acids or sugar side chains, or combinations thereof and usually have specific three-dimensional structural characteristics, as well as specific charge characteristics.

An “antibody that binds to the same epitope” as a reference antibody or “an antibody which competes for binding”

to a reference antibody refers to an antibody that blocks binding of the reference antibody to its antigen in a competition assay by 10%, 20%, 30%, 40%, 50% or more, and conversely, the reference antibody blocks binding of the antibody to its antigen in a competition assay by 10%, 20%, 30%, 40%, 50% or more.

The term “maturated antibodies” or “maturated antigen-binding fragments” such as maturated Fab variants or “optimized” variants includes derivatives of an antibody or antibody fragment exhibiting stronger binding—i.e. binding with increased affinity—to a given antigen such as the extracellular domain of a target protein. Maturation is the process of identifying a small number of mutations within the six CDRs of an antibody or antibody fragment leading to this affinity increase. The maturation process is the combination of molecular biology methods for introduction of mutations into the antibody and screening for identifying the improved binders.

“Percent (%) sequence identity” with respect to a reference polynucleotide or polypeptide sequence, respectively, is defined as the percentage of nucleic acid or amino acid residues, respectively, in a candidate sequence that are identical with the nucleic acid or amino acid residues, respectively, in the reference polynucleotide or polypeptide sequence, respectively, after aligning the sequences and introducing gaps, if necessary, to achieve the maximum percent sequence identity. Conservative substitutions are not considered as part of the sequence identity. Preferred are un-gapped alignments. Alignment for purposes of determining percent amino acid sequence identity can be achieved in various ways that are within the skill in the art, for instance, using publicly available computer software such as BLAST, BLAST-2, ALIGN or Megalign (DNASTAR) software. Those skilled in the art can determine appropriate parameters for aligning sequences, including any algorithms needed to achieve maximal alignment over the full length of the sequences being compared.

“Sequence homology” indicates the percentage of amino acids that either is identical or that represent conservative amino acid substitutions.

An “antagonistic” antibody or a “blocking” antibody is one which significantly inhibits (either partially or completely) a biological activity of the antigen it binds. In particular embodiments, the antibody or antigen-binding fragment according to the present disclosure is a Sema3A blocking antibody or antigen-binding fragment thereof.

The term “antibody conjugate” refers to an antibody conjugated to one or more molecules including drugs—in which case the antibody conjugate is referred to as “antibody-drug conjugate” (“ADC”)—and high molecular weight molecules such as peptides or proteins.

The term “antibody-drug conjugate” or “ADC” refers to an antibody conjugated to one or more cytotoxic or cytostatic agents, such as a chemotherapeutic agent, a drug, a growth inhibitory agent, a toxin (e.g., a protein toxin, an enzymatically active toxin of bacterial, fungal, plant, or animal origin, or fragments thereof), or a radioactive isotope (i.e., a radioconjugate). Immunoconjugates have been used for the local delivery of cytotoxic agents, i.e., drugs that kill or inhibit the growth or proliferation of cells, in the treatment of cancer (e.g. Liu et al., Proc Natl. Acad. Sci. (1996), 93, 8618-8623)). Immunoconjugates allow for the targeted delivery of a drug moiety to a tumor, and intracellular accumulation therein, where systemic administration of unconjugated drugs may result in unacceptable levels of toxicity to normal cells and/or tissues. Toxins used in antibody-toxin conjugates include bacterial toxins such as

diphtheria toxin, plant toxins such as ricin, small molecule toxins such as geldanamycin. The toxins may exert their cytotoxic effects by mechanisms including tubulin binding, DNA binding, or topoisomerase inhibition.

Amino acids may be referred to herein by their commonly known three letter symbols or by the one-letter symbols recommended by the IUPAC-IUB Biochemical Nomenclature Commission. Nucleotides, likewise, may be referred to by their commonly accepted single-letter codes.

The term “vector”, as used herein, refers to a nucleic acid molecule capable of propagating another nucleic acid to which it is linked. The term includes the vector as a self-replicating nucleic acid structure as well as the vector incorporated into the genome of a host cell into which it has been introduced. Certain vectors are capable of directing the expression of nucleic acids to which they are operatively linked. Such vectors are referred to herein as “expression vectors.”

The terms “host cell”, “host cell line”, and “host cell culture” are used interchangeably and refer to cells into which at least one exogenous nucleic acid has been introduced, including the progeny of such cells. Host cells include “transformants” and “transformed cells”, “transfectants” and “transfected cells” and “transduced cells” which include the primary transformed/transfected/transduced cell and progeny derived therefrom without regard to the number of passages. Progeny may not be completely identical in nucleic acid content to a parent cell but may contain mutations. Mutant progeny that have the same function or biological activity as screened or selected for in the originally transformed cell are included herein.

As used herein, the phrase “therapeutically effective amount” is meant to refer to an amount of therapeutic or prophylactic antibody that would be appropriate to elicit the desired therapeutic or prophylactic effect or response, including alleviating some or all of such symptoms of disease or reducing the predisposition to the disease, when administered in accordance with the desired treatment regimen.

The term “pharmaceutical formulation”/“pharmaceutical composition” refers to a preparation which is in such form as to permit the biological activity of an active ingredient contained therein to be effective, and which contains no additional components which are unacceptably toxic to a subject to which the formulation would be administered. Antibodies According to the Present Disclosure

In one aspect the present disclosure relates to an isolated antibody or antigen-binding fragment thereof binding to human *Sema3A*, wherein said isolated antibody or antigen-binding fragment thereof binds to human *Sema3A* of the sequence of SEQ ID NO: 600 with a dissociation constant (KD) ≤ 50 nM, ≤ 20 nM, ≤ 10 nM, ≤ 1 nM, or ≤ 0.1 nM. In particular embodiments, the isolated antibody or antigen-binding fragment thereof binds to the His-tagged human *Sema3A* domain of SEQ ID NO: 582 with a dissociation constant (KD) ≤ 50 nM, ≤ 20 nM, ≤ 10 nM, ≤ 1 nM, or ≤ 0.1 nM.

In another aspect the present disclosure relates to an isolated antibody or antigen-binding fragment thereof binding to human *Sema3A*, wherein said isolated antibody or antigen-binding fragment thereof cross-reacts with mouse, cynomolgus, rat, pig and/or dog *Sema3A*, particularly wherein said isolated antibody or antigen-binding fragment thereof binds to mouse, cynomolgus, rat, pig and/or dog *Sema3A* with a dissociation constant (KD) ≤ 50 nM, ≤ 20 nM, ≤ 10 nM, ≤ 1 nM, or ≤ 0.1 nM.

In particular such embodiments, said affinities are to mouse *Sema3A* of SEQ ID NO: 601, to cynomolgus *Sema3A* of SEQ ID NO: 602, to rat *Sema3A* of SEQ ID NO: 603, to pig *Sema3A* of SEQ ID NO: 604 and to dog *Sema3A* of SEQ ID NO: 605. In particular embodiments, said affinities are to His-tagged mouse *Sema3A* domain of SEQ ID NO: 583, to His-tagged cynomolgus *Sema3A* domain of SEQ ID NO: 586, to His-tagged rat *Sema3A* domain of SEQ ID NO: 584, to His-tagged pig *Sema3A* domain of SEQ ID NO: 587 and to His-tagged dog *Sema3A* domain of SEQ ID NO: 585.

In another aspect the present disclosure relates to an isolated antibody or antigen-binding fragment thereof binding to human *Sema3A*, wherein said isolated antibody or antigen-binding fragment thereof binds to human *Sema3A* with a binding activity as measured by surface plasmon resonance (SPR) of $\geq 60\%$, $\geq 70\%$, $\geq 80\%$, or $\geq 90\%$. In particular embodiments, the isolated antibody or antigen-binding fragment thereof binds to human *Sema3A* of the sequence of SEQ ID NO: 600 with a binding activity as measured by surface plasmin resonance (SPR) of $\geq 60\%$, $\geq 70\%$, $\geq 80\%$, or $\geq 90\%$. In particular embodiments, the isolated antibody or antigen-binding fragment thereof binds to His-tagged human *Sema3A* domain of the sequence of SEQ ID NO: 582 with a binding activity as measured by surface plasmin resonance (SPR) of $\geq 60\%$, $\geq 70\%$, $\geq 80\%$, or $\geq 90\%$.

In another aspect the present disclosure relates to an isolated antibody or antigen-binding fragment thereof binding to human *Sema3A*, wherein said isolated antibody or antigen-binding fragment thereof inhibits the activity of human *Sema3A* of the sequence of SEQ ID NO: 600 in an in vitro mesangial cell migration assay with an EC50 of ≤ 10 nM, ≤ 5 nM, ≤ 2.5 nM, or ≤ 1 nM.

In particular, the isolated antibody or antigen-binding fragment according to the present disclosure inhibits the activity of human *Sema3A* of the sequence of SEQ ID NO: 600 in an in vitro scratch assay using human primary mesangial cells and described in more detail in Example 9.

In another aspect the present disclosure relates to an isolated antibody or antigen-binding fragment thereof binding to human *Sema3A*, wherein said isolated antibody or antigen-binding fragment thereof inhibits the activity of human *Sema3A* of the sequence of SEQ ID NO: 600 in an in vitro growth cone collapse assay with an EC50 of ≤ 50 nM, ≤ 25 nM, ≤ 10 nM, or ≤ 5 nM.

In particular, the isolated antibody or antigen-binding fragment according to the present disclosure inhibits *Sema3A*-induced cytoskeletal collapse in an in vitro growth cone collapse assay using mouse dorsal root ganglion (DRG) neurons as described in more detail in Example 10. The in vitro growth cone assay described in Example 10 is a modified version of the growth cone assay described in Mikule et al. (PMID: 12077190).

In another aspect the present disclosure relates to an isolated antibody or antigen-binding fragment thereof binding to human *Sema3A*, wherein said isolated antibody or antigen-binding fragment thereof inhibits the activity of human *Sema3A* of the sequence of SEQ ID NO: 600 in an in vitro HUVEC repulsion assay with an EC50 of ≤ 1 nM, or ≤ 0.3 nM, ≤ 0.1 nM, ≤ 0.07 nM, ≤ 0.06 nM.

In particular, the isolated antibody or antigen-binding fragment according to the present disclosure inhibits *Sema3A* induced cell repulsion in an in vitro repulsion assay using *Sema3A*, of the sequence of SEQ ID NO: 600,

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expressing HEK293 cells seeded on a confluent monolayer of human umbilical vein endothelial cells (HUVEC) as described in Example 11.

In a further aspect, the present disclosure relates to an isolated antibody or antigen-binding fragment thereof binding to Sema3A, of the sequence of SEQ ID NO: 600, wherein said isolated antibody or antigen-binding fragment thereof exhibits an improved potency in HUVEC repulsion assay; i) wherein said isolated antibody or antigen-binding fragment thereof exhibits an improved potency in HUVEC repulsion assay in comparison to TPP-17755 with SEQ IDs 81, 85, 97, 98, or to TPP-11489 with SEQ IDs 1, 5, 17, 18, or to TPP-30788 with SEQ IDs 800, 804, 810, 811, or to TPP-30789 with SEQ IDs 814, 818, 824, 825, or to TPP-30790 with SEQ IDs 828, 832, 838, 839, or to TPP-30791 with SEQ IDs 842, 846, 852, 853; ii) wherein said isolated antibody or antigen-binding fragment thereof exhibits preferably a >400-fold, preferably a >50-fold, preferably >5-fold, preferably >2-fold increased potency against cellular Sema3A induced HUVEC repulsion based on the EC₅₀ values, in comparison to TPP-17755 with SEQ IDs 81, 85, 97, 98, or to TPP-11489 with SEQ IDs 1, 5, 17, 18, or to TPP-30788 with SEQ IDs 800, 804, 810, 811, or to TPP-30789 with SEQ IDs 814, 818, 824, 825, or to TPP-30790 with SEQ IDs 828, 832, 838, 839, or to TPP-30791 with SEQ IDs 842, 846, 852, 853; iii) wherein said isolated antibody or antigen-binding fragment thereof exhibits at least a 30% increased percent inhibition, preferably at least 50% increased percent inhibition of Sema3A in comparison to TPP-17755, to TPP-11489, to TPP-30788, to TPP-30789, TPP-30790, or to TPP-30791, with aforementioned sequences; iv) wherein said isolated antibody or antigen-binding fragment thereof has a two-digit picomolar activity against human Sema3A in vitro HUVEC repulsion assay, while prior art antibody potencies of TPP-17755, TPP-11489, TPP-30788, TPP-30789, TPP-30790, or TPP-30791, with aforementioned sequences, are in the three-digit picomolar or even nanomolar range; v) wherein said isolated antibody or antigen-binding fragment thereof inhibits the activity of human Sema3A in an in vitro HUVEC repulsion assay with an EC₅₀ of ≤ 1 nM, or ≤ 0.3 nM, ≤ 0.1 nM, ≤ 0.07 nM, ≤ 0.06 nM, as described in Example 11.

The isolated antibody or antigen-binding fragment of the present disclosure show an improved potency in HUVEC repulsion assay compared to TPP-30788-TPP-30791 (BI clone I to IV), which might be due to a binding to a different epitope of human Sema3A.

In another aspect, the present disclosure relates to an isolated antibody or antigen-binding fragment thereof binding to Sema3A, wherein said isolated antibody or antigen-binding fragment thereof inhibits the activity of Sema3A in vivo, since the antibodies according to the present disclosure reduce Sema3A-induced urinary Albumin excretion. Thus in a further aspect, the present disclosure relates to an isolated antibody or antigen-binding fragment thereof binding to Sema3A, wherein said isolated antibody or antigen-binding fragment thereof exhibits an improved inhibitory activity of Sema3A in vivo, i) wherein said the antibodies exhibit an increased reduction of Sema3A-induced urinary Albumin excretion compared to TPP-30788 (BI clone I); ii) wherein said the antibodies exhibit an increased reduction of Sema3A-induced urinary Albumin excretion compared to TPP-17755 (Samsung); iii) wherein said the antibodies exhibit an increased reduction of Sema3A-induced urinary Albumin excretion compared to TPP-11489 (Chiome) as described in Example 12.

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The isolated antibody or antigen-binding fragment of the present disclosure show an improved efficacy in an in vivo model for induced urinary Albumin excretion compared to TPP-30788-TPP-30791 (BI clone I to IV), which might be due to a binding to a different epitope of human Sema3A.

Thus, in a further aspect, the present disclosure relates to an isolated antibody or antigen-binding fragment thereof that binds to human Sema3A, wherein said isolated antibody or antigen-binding fragment thereof i) exhibits an increased stability (e.g. increased stress-stability when diluted in PBS to 25 mg/ml and incubated at 700 rpm and 40° C. for two weeks) compared to TPP-30788 (BI clone I); ii) wherein the increased stability exhibits an increased amount of monomeric anti-Sema3A antibody compared to TPP-30788 (BI clone I) measured by SEC; iii) wherein the increased stability exhibits a decreased percentage of the LC and HC of the anti-Sema3A antibody compared to TPP-30788 (BI clone I) measured by cGE, proving a reduced rate of degradation which is measured by the presence of remaining LC and HC, iv) wherein the increased stability exhibits that the amount of monomeric anti-Sema3A antibody is maintained, e.g. Δ % monomer=1 after the incubation at 40° C., 700 rpm for two weeks; v) wherein the increased stability exhibits that the amount of LC and HC of the anti-Sema3A antibody is maintained e.g. Δ % LC+HC<1 after the incubation at 40° C., 700 rpm for two weeks.

Thus, in a further aspect, the present disclosure relates to TPP-23298, that binds to human Sema3A, wherein said isolated antibody or antigen-binding fragment thereof i) exhibits an increased stability (e.g. increased stress-stability when diluted in PBS to 25 mg/ml and incubated at 700 rpm and 40° C. for two weeks) compared to TPP-30788 (BI clone I); ii) wherein the increased stability exhibits an increased amount of monomeric anti-Sema3A antibody compared to TPP-30788 (BI clone I) measured by SEC; iii) wherein the increased stability exhibits a decreased percentage of the LC and HC of the anti-Sema3A antibody compared to TPP-30788 (BI clone I) measured by cGE, proving a reduced rate of degradation which is measured by the presence of remaining LC and HC, iv) wherein the increased stability exhibits that the amount of monomeric anti-Sema3A antibody is maintained, e.g. Δ % monomer=1 after the incubation at 40° C., 700 rpm for two weeks; v) wherein the increased stability exhibits that the amount of LC and HC of the anti-Sema3A antibody is maintained e.g. Δ % LC+HC <1 after the incubation at 40° C., 700 rpm for two weeks.

Thus, in a further aspect, the present disclosure relates to an isolated antibody or antigen-binding fragment thereof that binds to human Sema3A, wherein said isolated antibody or antigen-binding fragment thereof i) exhibits an increased solubility; ii) wherein the increased solubility is measured in mg/ml after concentration at 90% recovery; iii) wherein the solubility is increased compared to TPP-30788 (BI clone I); iv) wherein the solubility is increased ≤ 1 fold, ≤ 1.5 fold, ≤ 2 fold compared to TPP-30788 (BI clone I); v) wherein the increased solubility exhibits that the percentage of monomeric anti-Sema3A antibody is not increased after concentration e.g. Δ monomer <1 measured by SEC.

Thus, in a further aspect, the present disclosure relates to TPP-23298, that binds to human Sema3A, wherein said isolated antibody or antigen-binding fragment thereof i) exhibits an increased solubility; ii) wherein the increased solubility is measured in mg/ml after concentration at 90% recovery; iii) wherein the solubility is increased compared to TPP-30788 (BI clone I); iv) wherein the solubility is increased ≤ 1 fold, ≤ 1.5 fold, ≤ 2 fold compared to TPP-

30788 (BI clone I); v) wherein the increased solubility exhibits that the percentage of monomeric anti-Sema3A antibody is not increased after concentration e.g. Δ % monomer < 1 measured by SEC.

Thus, in a further aspect, the present disclosure relates to an isolated antibody or antigen-binding fragment thereof that binds to human Sema3A, wherein said isolated antibody or antigen-binding fragment thereof i) exhibits an increased viscosity compared to water or PBS; ii) exhibits a reduced viscosity in PBS compared to TPP-30788 (BI clone I); iii) wherein the viscosity is measured by a Viscosizer and exhibits a cP value of 5.1 (150 mg/ml).

Thus, in a further aspect, the present disclosure relates to TPP-23298, that binds to human Sema3A, wherein said isolated antibody or antigen-binding fragment thereof i) exhibits an increased viscosity compared to water or PBS; ii) exhibits a reduced viscosity in PBS compared to TPP-30788 (BI clone I); iii) wherein the viscosity is measured by a Viscosizer and exhibits a cP value of 5.1 (150 mg/ml).

In particular the isolated antibody or antigen-binding fragment according to the present disclosure shows a much higher solubility and stability, is more resistant to heat stress and is less viscous in PBS buffer than TPP-30788 as described in Example 17.

In particular TPP-23298 shows a much higher solubility and stability, is more resistant to heat stress and is less viscous in PBS buffer than TPP-30788 as described in Example 17.

In another aspect the present disclosure relates to an isolated antibody or antigen-binding fragment thereof binding to human Sema3A, which can be produced with high titers in mammalian cells; i) wherein high titer is ≤ 200 mg/L as described in Example 16.

In another aspect the present disclosure relates to an isolated antibody or antigen-binding fragment thereof binding to human Sema3A, wherein the antibody exhibits a higher binding selectivity to active Sema3A (TPP-13211) over cleaved Sema3A TPP-19068; i) wherein the antibody exhibits a higher binding selectivity to active Sema3A (TPP-13211) compared to the binding selectivity of TPP-30788-TPP-30791 to active Sema3A, as described in Example 8.

In another aspect the present disclosure relates to TPP-23298 binding to human Sema3A, wherein the antibody exhibits a higher binding selectivity to active Sema3A (TPP-13211) over cleaved Sema3A TPP-19068; i) wherein the antibody exhibits a higher binding selectivity to active Sema3A (TPP-13211) compared to the binding selectivity of TPP-30788-TPP-30791 to active Sema3A, as described in Example 8.

In another aspect the present disclosure relates to an isolated antibody or antigen-binding fragment thereof binding to human Sema3A, wherein the antibody binds a different epitope on Sema3A compared to TPP-30788; i) wherein the epitope binding is measured in SPR assay, as described in Example 5a. All antibodies binding the same epitope and competing with the binding of the isolated antibody or antigen-binding fragment according to the present disclosure are comprised by the present disclosure.

In another aspect the present disclosure relates to TPP-23298 binding to human Sema3A, wherein the antibody binds a different epitope on Sema3A compared to TPP-30788; i) wherein the epitope binding is measured in SPR assay, as described in Example 5a. All antibodies binding the same epitope and competing with the binding of the isolated antibody or antigen-binding fragment according to the present disclosure are comprised by the present disclosure.

In another aspect the present disclosure relates to an isolated antibody or antigen-binding fragment thereof that competes with the isolated antibody or antigen-binding fragment according to any one of the preceding claims for binding to Sema3A and wherein the isolated antibody or antigen-binding fragment does not compete with the binding of an antibody with the SEQ IDs NO 800, NO 804, NO 810 or NO 811 to Sema3A.

The isolated antibody or antigen-binding fragment according to the present disclosure may exhibit any combination of the above described characteristics.

The isolated antibody or antigen-binding fragment according to the present disclosure is a Sema3A blocking antibody or antigen-binding fragment thereof. In particular embodiments, the antibody binds specifically and more particularly selectively to the Sema3A domain of Semaphorin3A and interferes with the interaction of its receptor neuropilin-1.

In particular embodiments, the isolated antibody or antigen-binding fragment thereof according to the present disclosure cross-reacts with mouse, cynomolgus, rat, pig and/or dog Sema3A, particularly having an affinity to mouse, cynomolgus, rat, pig and/or dog Sema3A that is less than 100-fold, particularly less than 50-fold, more particularly less than 25-fold, even more particularly less than 10-fold and most particularly less than 5-fold different to that to human Sema3A.

In particular embodiments, the isolated antibody or antigen-binding fragment thereof according to the present disclosure does not significantly cross-react with human Sema3B, Sema3C, Sema3D, Sema3E, Sema3F and/or Sema3G. In particular, the isolated antibody or antigen-binding fragment thereof does not significantly cross-react with human Sema3G.

In particular embodiments, the isolated antibody or antigen-binding fragment thereof according to the present disclosure inhibits Sema3A-induced albuminuria and/or proteinuria.

In particular embodiments, the isolated antibody or antigen-binding fragment thereof according to the present disclosure inhibits Sema3A-induced fibrosis.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure comprises a heavy chain variable domain that is at least 90%, at least 95%, at least 98%, at least 99%, or 100% identical to SEQ ID NO: 141, and a light chain variable domain that is at least 90%, at least 95%, at least 98%, at least 99%, or 100% identical to SEQ ID NO: 145.

In particular other embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure comprises a heavy chain variable domain that is at least 90%, at least 95%, at least 98%, at least 99%, or 100% identical to SEQ ID NO: 61, and a light chain variable domain that is at least 90%, at least 95%, at least 98%, at least 99%, or 100% identical to SEQ ID NO: 65.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure comprises a heavy chain antigen-binding region that comprises an H-CDR3 comprising the sequence RDDYTSTRDAFDX (SEQ ID NO: 594), wherein X is selected from the group consisting of Y and V. Particularly, X is Y.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure comprises a light chain antigen-binding region that comprises an L-CDR3 comprising the sequence X₁AWDDSLNX₂X₃X₄V (SEQ ID NO: 598), wherein X₁ is

selected from the group consisting of A and H, wherein X_2 is selected from the group consisting of V, D, and G, in particular wherein X_2 is selected from the group consisting of V and D, wherein X_3 is selected from the group consisting of I and Y, and wherein X_4 is selected from the group consisting of P and V.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure comprises a heavy chain antigen-binding region that comprises an H-CDR3 as defined above and a light chain antigen-binding region that comprises an L-CDR3 as defined above.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure comprises a heavy chain antigen-binding region that comprises an H-CDR3 comprising the sequence SGYSSS-WFDPDFDY (SEQ ID NO: 64).

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure comprises a light chain antigen-binding region that comprises an L-CDR3 comprising the sequence X_1 SYX₂GX₃NPYVV (SEQ ID NO: 599), wherein X_1 is selected from the group consisting of S and Q; wherein X_2 is selected from the group consisting of E and A; and wherein X_3 is selected from the group consisting of P, I, and S. In particular, X_3 is selected from the group consisting of P and I.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure comprises a heavy chain antigen-binding region that comprises an H-CDR3 as defined above and a light chain antigen-binding region that comprises an L-CDR3 as defined above.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure comprises a heavy chain antigen-binding region that comprises an H-CDR1 comprising the sequence SYX₁MX₂ (SEQ ID NO: 588), wherein X_1 is selected from G and A and wherein X_2 is selected from H, S and L. Particularly, the heavy chain antigen-binding region comprises an H-CDR1 comprising the sequence SYAMX (SEQ ID NO: 589), wherein X is selected from S and L.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure comprises a heavy chain antigen-binding region that comprises an H-CDR2 comprising the sequence AIGX₁GGDTYYADSVX₂G (SEQ ID NO: 590), wherein X_1 is selected from T and Y, and wherein X_2 is selected from K and M. Particularly, the heavy chain antigen-binding region comprises an H-CDR2 comprising the sequence AIGXGGDTYYADSVKG (SEQ ID NO: 591), wherein X is selected from T and Y.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure comprises a heavy chain antigen-binding region that comprises an H-CDR3 comprising the sequence RDDYTSRDAFDX (SEQ ID NO: 594), wherein X is selected from the group consisting of Y and V. Particularly, X is Y.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure comprises a heavy chain antigen-binding region that comprises an H-CDR1, an H-CDR2 and an H-CDR3 as defined above.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure

comprises a light chain antigen-binding region that comprises an L-CDR1 comprising the sequence SGSSSNIG-SNTVN (SEQ ID NO: 46).

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure comprises a light chain antigen-binding region that comprises an L-CDR2 comprising the sequence YDDLXPS (SEQ ID NO: 596), wherein X is selected from L and R. Particularly, the light chain antigen-binding region comprises an L-CDR2 comprising the sequence YDDLRLPS (SEQ ID NO: 127).

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure comprises a light chain antigen-binding region that comprises an L-CDR3 comprising the sequence X_1 AWDDSLNX₂X₃X₄V (SEQ ID NO: 598), wherein X_1 is selected from the group consisting of A and H, wherein X_2 is selected from the group consisting of V, D, and G, in particular wherein X_2 is selected from the group consisting of V and D, wherein X_3 is selected from the group consisting of I and Y, and wherein X_4 is selected from the group consisting of P and V.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure comprises a light chain antigen-binding region that comprises an L-CDR1, and L-CDR2 and an L-CDR3 as defined above.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure comprises a heavy chain antigen-binding region that comprises an H-CDR1, an H-CDR2 and an H-CDR3 as defined above and a light chain antigen-binding region that comprises an L-CDR1, and L-CDR2 and an L-CDR3 as defined above.

In particular such embodiments, the amino acid residue directly adjacent to the H-CDR1 at its 5' end (corresponding to residue 30 of reference VII domain of SEQ ID NO: 121) is S or Y.

In particular other embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure comprises a heavy chain antigen-binding region that comprises an H-CDR1 comprising the sequence SYEMN (SEQ ID NO: 62).

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure comprises a heavy chain antigen-binding region that comprises an H-CDR2 comprising the sequence GISWNSGX₁IX₂YADSVKG (SEQ ID NO: 592), wherein X_1 is selected from W and S and X_2 is selected from G and D. Particularly, the heavy chain antigen-binding region comprises an H-CDR2 comprising the sequence GISWNSGWIXYADSVKG (SEQ ID NO: 593), wherein X is selected from G and D.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure comprises a heavy chain antigen-binding region that comprises an H-CDR3 comprising the sequence SGYSSS-WFDPDFDY (SEQ ID NO: 64).

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure comprises a heavy chain antigen-binding region that comprises an H-CDR1, an H-CDR2 and an H-CDR3 as defined above.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure comprises a light chain antigen-binding region that com-

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prises an L-CDR1 comprising the sequence TGSSSX1-GAGYDVH (SEQ ID NO: 595), wherein X is selected from N and D.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure comprises a light chain antigen-binding region that comprises an L-CDR2 comprising the sequence GXSNRPS (SEQ ID NO: 597), wherein X is selected from N and A.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure comprises a light chain antigen-binding region that comprises an L-CDR3 comprising the sequence X₁SYX₂GX₃NPYVV (SEQ ID NO: 599), wherein X₁ is selected from the group consisting of S and Q, wherein X₂ is selected from the group consisting of E and A, and wherein X₃ is selected from the group consisting of P, I, and S. Particularly, X₃ is selected from the group consisting of P and I.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure comprises a light chain antigen-binding region that comprises an L-CDR1, and L-CDR2 and an L-CDR3 as defined above.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure comprises a heavy chain antigen-binding region that comprises an H-CDR1, an H-CDR2 and an H-CDR3 as defined above and a light chain antigen-binding region that comprises an L-CDR1, and L-CDR2 and an L-CDR3 as defined above.

In particular such embodiments, the three amino acid residues directly adjacent to the H-CDR1 at its 5' end (corresponding to residues 28 to 30 of reference VH domain of SEQ ID NO: 101) are X₁FX₂, wherein X₁ is selected from T and D and X₂ is selected from S and D.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure comprises:

- i) a heavy chain antigen-binding region that comprises an H-CDR3 comprising SEQ ID NO: 44 and a light chain antigen-binding region that comprises an L-CDR3 comprising SEQ ID NO: 48; or
- ii) a heavy chain antigen-binding region that comprises an H-CDR3 comprising SEQ ID NO: 64 and a light chain antigen-binding region that comprises an L-CDR3 comprising SEQ ID NO: 68; or
- iii) a heavy chain antigen-binding region that comprises an H-CDR3 comprising SEQ ID NO: 104 and a light chain antigen-binding region that comprises an L-CDR3 comprising SEQ ID NO: 108; or
- iv) a heavy chain antigen-binding region that comprises an H-CDR3 comprising SEQ ID NO: 124 and a light chain antigen-binding region that comprises an L-CDR3 comprising SEQ ID NO: 128; or
- v) a heavy chain antigen-binding region that comprises an H-CDR3 comprising SEQ ID NO: 144 and a light chain antigen-binding region that comprises an L-CDR3 comprising SEQ ID NO: 148; or
- vi) a heavy chain antigen-binding region that comprises an H-CDR3 comprising SEQ ID NO: 164 and a light chain antigen-binding region that comprises an L-CDR3 comprising SEQ ID NO: 168; or
- vii) a heavy chain antigen-binding region that comprises an H-CDR3 comprising SEQ ID NO: 184 and a light chain antigen-binding region that comprises an L-CDR3 comprising SEQ ID NO: 188; or

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- viii) a heavy chain antigen-binding region that comprises an H-CDR3 comprising SEQ ID NO: 204 and a light chain antigen-binding region that comprises an L-CDR3 comprising SEQ ID NO: 208; or
- ix) a heavy chain antigen-binding region that comprises an H-CDR3 comprising SEQ ID NO: 224 and a light chain antigen-binding region that comprises an L-CDR3 comprising SEQ ID NO: 228; or
- x) a heavy chain antigen-binding region that comprises an H-CDR3 comprising SEQ ID NO: 244 and a light chain antigen-binding region that comprises an L-CDR3 comprising SEQ ID NO: 248; or
- xi) a heavy chain antigen-binding region that comprises an H-CDR3 comprising SEQ ID NO: 264 and a light chain antigen-binding region that comprises an L-CDR3 comprising SEQ ID NO: 268; or
- xii) a heavy chain antigen-binding region that comprises an H-CDR3 comprising SEQ ID NO: 284 and a light chain antigen-binding region that comprises an L-CDR3 comprising SEQ ID NO: 288; or
- xiii) a heavy chain antigen-binding region that comprises an H-CDR3 comprising SEQ ID NO: 304 and a light chain antigen-binding region that comprises an L-CDR3 comprising SEQ ID NO: 308; or
- xiv) a heavy chain antigen-binding region that comprises an H-CDR3 comprising SEQ ID NO: 324 and a light chain antigen-binding region that comprises an L-CDR3 comprising SEQ ID NO: 328; or
- xv) a heavy chain antigen-binding region that comprises an H-CDR3 comprising SEQ ID NO: 344 and a light chain antigen-binding region that comprises an L-CDR3 comprising SEQ ID NO: 348; or
- xvi) a heavy chain antigen-binding region that comprises an H-CDR3 comprising SEQ ID NO: 364 and a light chain antigen-binding region that comprises an L-CDR3 comprising SEQ ID NO: 368; or
- xvii) a heavy chain antigen-binding region that comprises an H-CDR3 comprising SEQ ID NO: 384 and a light chain antigen-binding region that comprises an L-CDR3 comprising SEQ ID NO: 388; or
- xviii) a heavy chain antigen-binding region that comprises an H-CDR3 comprising SEQ ID NO: 404 and a light chain antigen-binding region that comprises an L-CDR3 comprising SEQ ID NO: 408; or
- xix) a heavy chain antigen-binding region that comprises an H-CDR3 comprising SEQ ID NO: 424 and a light chain antigen-binding region that comprises an L-CDR3 comprising SEQ ID NO: 428; or
- xx) a heavy chain antigen-binding region that comprises an H-CDR3 comprising SEQ ID NO: 444 and a light chain antigen-binding region that comprises an L-CDR3 comprising SEQ ID NO: 448; or
- xxi) a heavy chain antigen-binding region that comprises an H-CDR3 comprising SEQ ID NO: 464 and a light chain antigen-binding region that comprises an L-CDR3 comprising SEQ ID NO: 468; or
- xxii) a heavy chain antigen-binding region that comprises an H-CDR3 comprising SEQ ID NO: 484 and a light chain antigen-binding region that comprises an L-CDR3 comprising SEQ ID NO: 488; or
- xxiii) a heavy chain antigen-binding region that comprises an H-CDR3 comprising SEQ ID NO: 504 and a light chain antigen-binding region that comprises an L-CDR3 comprising SEQ ID NO: 508; or
- xxiv) a heavy chain antigen-binding region that comprises an H-CDR3 comprising SEQ ID NO: 524 and a light

chain antigen-binding region that comprises an L-CDR3 comprising SEQ ID NO: 528; or
xxv) a heavy chain antigen-binding region that comprises an H-CDR3 comprising SEQ ID NO: 544 and a light chain antigen-binding region that comprises an L-CDR3 comprising SEQ ID NO: 548; or
xxvi) a heavy chain antigen-binding region that comprises an H-CDR3 comprising SEQ ID NO: 564 and a light chain antigen-binding region that comprises an L-CDR3 comprising SEQ ID NO: 568.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure comprises a heavy chain antigen-binding region that comprises an H-CDR1 comprising any one of SEQ ID NOs: 42, 62, 102, 122, 142, 162, 182, 202, 222, 242, 262, 282, 302, 322, 342, 362, 382, 402, 422, 442, 462, 482, 502, 522, 542, and 562.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure comprises a heavy chain antigen-binding region that comprises an H-CDR2 comprising any one of SEQ ID NOs: 43, 63, 103, 123, 143, 163, 183, 203, 223, 243, 263, 283, 303, 323, 343, 363, 383, 403, 423, 443, 463, 483, 503, 523, 543, and 563.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure comprises a light chain antigen-binding region that comprises an L-CDR1 comprising any one of SEQ ID NOs: 46, 66, 106, 126, 146, 166, 186, 206, 226, 246, 266, 286, 306, 326, 346, 366, 386, 406, 426, 446, 466, 486, 506, 526, 546, and 566.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure comprises a light chain antigen-binding region that comprises an L-CDR2 comprising any one of SEQ ID NOS: 47, 67, 107, 127, 147, 167, 187, 207, 227, 247, 267, 287, 307, 327, 347, 367, 387, 407, 427, 447, 467, 487, 507, 527, 547, and 567.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure comprises:

- i) a heavy chain antigen-binding region that comprises an H-CDR1 comprising SEQ ID NO: 42, an H-CDR2 comprising SEQ ID NO: 43, and an H-CDR3 comprising SEQ ID NO: 44 and a light chain antigen-binding region that comprises an L-CDR1 comprising SEQ ID NO: 46, an L-CDR2 comprising SEQ ID NO: 47, and an L-CDR3 comprising SEQ ID NO: 48; or
- ii) a heavy chain antigen-binding region that comprises an H-CDR1 comprising SEQ ID NO: 62, an H-CDR2 comprising SEQ ID NO: 63, and an H-CDR3 comprising SEQ ID NO: 64 and a light chain antigen-binding region that comprises an L-CDR1 comprising SEQ ID NO: 66, an L-CDR2 comprising SEQ ID NO: 67, and an L-CDR3 comprising SEQ ID NO: 68; or
- iii) a heavy chain antigen-binding region that comprises an H-CDR1 comprising SEQ ID NO: 102, an H-CDR2 comprising SEQ ID NO: 103, and an H-CDR3 comprising SEQ ID NO: 104 and a light chain antigen-binding region that comprises an L-CDR1 comprising SEQ ID NO: 106, an L-CDR2 comprising SEQ ID NO: 107, and an L-CDR3 comprising SEQ ID NO: 108; or
- iv) a heavy chain antigen-binding region that comprises an H-CDR1 comprising SEQ ID NO: 122, an H-CDR2 comprising SEQ ID NO: 123, and an H-CDR3 comprising SEQ ID NO: 124 and a light chain antigen-binding region that comprises an L-CDR1 comprising

- [illegible]

[illegible]

SEQ ID NO: 506, an L-CDR2 comprising SEQ ID NO: 507, and an L-CDR3 comprising SEQ ID NO: 508; or xxiv) a heavy chain antigen-binding region that comprises an H-CDR1 comprising SEQ ID NO: 522, an H-CDR2 comprising SEQ ID NO: 523, and an H-CDR3 comprising SEQ ID NO: 524 and a light chain antigen-binding region that comprises an L-CDR1 comprising SEQ ID NO: 526, an L-CDR2 comprising SEQ ID NO: 527, and an L-CDR3 comprising SEQ ID NO: 528; or xxv) a heavy chain antigen-binding region that comprises an H-CDR1 comprising SEQ ID NO: 542, an H-CDR2 comprising SEQ ID NO: 543, and an H-CDR3 comprising SEQ ID NO: 544 and a light chain antigen-binding region that comprises an L-CDR1 comprising SEQ ID NO: 546, an L-CDR2 comprising SEQ ID NO: 547, and an L-CDR3 comprising SEQ ID NO: 548; or xxvi) a heavy chain antigen-binding region that comprises an H-CDR1 comprising SEQ ID NO: 562, an H-CDR2 comprising SEQ ID NO: 563, and an H-CDR3 comprising SEQ ID NO: 564 and a light chain antigen-binding region that comprises an L-CDR1 comprising SEQ ID NO: 566, an L-CDR2 comprising SEQ ID NO: 567, and an L-CDR3 comprising SEQ ID NO: 568.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure comprises:

- i) a variable heavy chain domain comprising SEQ ID NO: 41 and a variable light chain domain comprising SEQ ID NO: 45; or
- ii) a variable heavy chain domain comprising SEQ ID NO: 61 and a variable light chain domain comprising SEQ ID NO: 65; or
- iii) a variable heavy chain domain comprising SEQ ID NO: 101 and a variable light chain domain comprising SEQ ID NO: 105; or
- iv) a variable heavy chain domain comprising SEQ ID NO: 121 and a variable light chain domain comprising SEQ ID NO: 125; or
- v) a variable heavy chain domain comprising SEQ ID NO: 141 and a variable light chain domain comprising SEQ ID NO: 145; or
- vi) a variable heavy chain domain comprising SEQ ID NO: 161 and a variable light chain domain comprising SEQ ID NO: 165; or
- vii) a variable heavy chain domain comprising SEQ ID NO: 181 and a variable light chain domain comprising SEQ ID NO: 185; or
- viii) a variable heavy chain domain comprising SEQ ID NO: 201 and a variable light chain domain comprising SEQ ID NO: 205; or
- ix) a variable heavy chain domain comprising SEQ ID NO: 221 and a variable light chain domain comprising SEQ ID NO: 225; or
- x) a variable heavy chain domain comprising SEQ ID NO: 241 and a variable light chain domain comprising SEQ ID NO: 245; or
- xi) a variable heavy chain domain comprising SEQ ID NO: 261 and a variable light chain domain comprising SEQ ID NO: 265; or
- xii) a variable heavy chain domain comprising SEQ ID NO: 281 and a variable light chain domain comprising SEQ ID NO: 285; or
- xiii) a variable heavy chain domain comprising SEQ ID NO: 301 and a variable light chain domain comprising SEQ ID NO: 305; or

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- xiv) a variable heavy chain domain comprising SEQ ID NO: 321 and a variable light chain domain comprising SEQ ID NO: 325; or
- xv) a variable heavy chain domain comprising SEQ ID NO: 341 and a variable light chain domain comprising SEQ ID NO: 345; or
- xvi) a variable heavy chain domain comprising SEQ ID NO: 361 and a variable light chain domain comprising SEQ ID NO: 365; or
- xvii) a variable heavy chain domain comprising SEQ ID NO: 381 and a variable light chain domain comprising SEQ ID NO: 385; or
- xviii) a variable heavy chain domain comprising SEQ ID NO: 401 and a variable light chain domain comprising SEQ ID NO: 405; or
- xix) a variable heavy chain domain comprising SEQ ID NO: 421 and a variable light chain domain comprising SEQ ID NO: 425; or
- xx) a variable heavy chain domain comprising SEQ ID NO: 441 and a variable light chain domain comprising SEQ ID NO: 445; or
- xxi) a variable heavy chain domain comprising SEQ ID NO: 461 and a variable light chain domain comprising SEQ ID NO: 465; or
- xxii) a variable heavy chain domain comprising SEQ ID NO: 481 and a variable light chain domain comprising SEQ ID NO: 485; or
- xxiii) a variable heavy chain domain comprising SEQ ID NO: 501 and a variable light chain domain comprising SEQ ID NO: 505; or
- xxiv) a variable heavy chain domain comprising SEQ ID NO: 521 and a variable light chain domain comprising SEQ ID NO: 525; or
- xxv) a variable heavy chain domain comprising SEQ ID NO: 541 and a variable light chain domain comprising SEQ ID NO: 545; or
- xxvi) a variable heavy chain domain comprising SEQ ID NO: 561 and a variable light chain domain comprising SEQ ID NO: 565.

In particular embodiments, the isolated antibody according to the present disclosure is an IgG antibody. In particular such embodiments, the isolated antibody according to the present disclosure is an IgG1, IgG2, IgG3 or an IgG4 antibody. Most particularly, the isolated antibody according to the present disclosure is an IgG1 or an IgG4 antibody.

In particular embodiments, the isolated antibody according to the present disclosure comprises:

- i) a heavy chain comprising SEQ ID NO: 57 and a light chain comprising SEQ ID NO: 58; or
- ii) a heavy chain comprising SEQ ID NO: 77 and a light chain comprising SEQ ID NO: 78; or
- iii) a heavy chain comprising SEQ ID NO: 117 and a light chain comprising SEQ ID NO: 118; or
- iv) a heavy chain comprising SEQ ID NO: 137 and a light chain comprising SEQ ID NO: 138; or
- v) a heavy chain comprising SEQ ID NO: 157 and a light chain comprising SEQ ID NO: 158; or
- vi) a heavy chain comprising SEQ ID NO: 177 and a light chain comprising SEQ ID NO: 178; or
- vii) a heavy chain comprising SEQ ID NO: 197 and a light chain comprising SEQ ID NO: 198; or
- viii) a heavy chain comprising SEQ ID NO: 217 and a light chain comprising SEQ ID NO: 218; or
- ix) a heavy chain comprising SEQ ID NO: 237 and a light chain comprising SEQ ID NO: 238; or
- x) a heavy chain comprising SEQ ID NO: 257 and a light chain comprising SEQ ID NO: 258; or

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- xi) a heavy chain comprising SEQ ID NO: 277 and a light chain comprising SEQ ID NO: 278; or
- xii) a heavy chain comprising SEQ ID NO: 297 and a light chain comprising SEQ ID NO: 298; or
- xiii) a heavy chain comprising SEQ ID NO: 317 and a light chain comprising SEQ ID NO: 318; or
- xiv) a heavy chain comprising SEQ ID NO: 337 and a light chain comprising SEQ ID NO: 338; or
- xv) a heavy chain comprising SEQ ID NO: 357 and a light chain comprising SEQ ID NO: 358; or
- xvi) a heavy chain comprising SEQ ID NO: 377 and a light chain comprising SEQ ID NO: 378; or
- xvii) a heavy chain comprising SEQ ID NO: 397 and a light chain comprising SEQ ID NO: 398; or
- xviii) a heavy chain comprising SEQ ID NO: 417 and a light chain comprising SEQ ID NO: 418; or
- xix) a heavy chain comprising SEQ ID NO: 437 and a light chain comprising SEQ ID NO: 438; or
- xx) a heavy chain comprising SEQ ID NO: 457 and a light chain comprising SEQ ID NO: 458; or
- xxi) a heavy chain comprising SEQ ID NO: 477 and a light chain comprising SEQ ID NO: 478; or
- xxii) a heavy chain comprising SEQ ID NO: 497 and a light chain comprising SEQ ID NO: 498; or
- xxiii) a heavy chain comprising SEQ ID NO: 517 and a light chain comprising SEQ ID NO: 518; or
- xxiv) a heavy chain comprising SEQ ID NO: 537 and a light chain comprising SEQ ID NO: 538; or
- xxv) a heavy chain comprising SEQ ID NO: 557 and a light chain comprising SEQ ID NO: 558; or
- xxvi) a heavy chain comprising SEQ ID NO: 577 and a light chain comprising SEQ ID NO: 578.

In particular embodiments, the antigen-binding fragment according to the present disclosure is an scFv, Fab, Fab' fragment or a F(ab')₂ fragment.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is a monoclonal antibody or antigen-binding fragment thereof.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is a human, humanized or chimeric antibody or antigen-binding fragment thereof, more particularly a fully human antibody or antigen-binding fragment thereof.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is a monospecific antibody. In particular other embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is a multispecific antibody that binds to Sema3A and at least one further antigen, such as a bispecific, trispecific or tetraspecific antibody.

In another aspect the present disclosure relates to an isolated antibody or antigen-binding fragment thereof that competes with the isolated antibody or antigen-binding fragment according to the present disclosure for binding to human Sema3A.

In a further aspect, the present disclosure relates to an antibody conjugate, comprising the isolated antibody or antigen binding fragment according to the present disclosure. For example, an antibody could be conjugated to a cytotoxic agent, an immunotoxin, a toxophore or a radioisotope. Also provided are anti-Sema3A antibodies conjugated to a detectable marker. Preferred markers are a radiolabel, an enzyme, a chromophore or a fluorophore. The antibody may also be conjugated to high molecular weight molecules such as peptides or proteins, such as interleukins.

The ADC according to the present disclosure comprises an anti-Sema3A antibody conjugated to one or more cyto-

toxic agents, such as chemotherapeutic agents or drugs, growth inhibitory agents, toxins (e.g., protein toxins, enzymatically active toxins of bacterial, fungal, plant, human or animal origin, or fragments thereof), or radioactive isotopes.

In one embodiment, the ADC according to the present disclosure comprises an anti-Sema3A antibody as described herein conjugated to one or more drugs, including but not limited to a maytansinoid (see U.S. Pat. Nos. 5,208,020, 5,416,064 and European Patent EP0425235); an auristatin such as monomethylauristatin drug moieties DE and DF (MMAE and MMAF) (see U.S. Pat. Nos. 5,635,483 and 5,780,588, and 7,498,298); a dolastatin; a calicheamicin or derivative thereof; an anthracycline such as daunomycin or doxorubicin; methotrexate; vindesine; a taxane such as docetaxel, paclitaxel, larotaxel, tesetaxel, and ortataxel; a trichothecene; and CC1065.

In another embodiment, the ADC according to the present disclosure comprises an anti-Sema3A antibody as described herein conjugated to an enzymatically active toxin or fragment thereof, including but not limited to diphtheria A chain, nonbinding active fragments of diphtheria toxin, exotoxin A chain (from *Pseudomonas aeruginosa*), ricin A chain, abrin A chain, modeccin A chain, alphasarcin, *Aleurites fordii* proteins, dianthin proteins, *Phytolacca americana* proteins (P API, P APII, and PAP-S), *Momordica charantia* inhibitor, curcin, crotin, *Sapaonaria officinalis* inhibitor, gelonin, mitogellin, restrictocin, phenomycin, enomycin, and the tricothecenes.

In another embodiment, the ACD according to the present disclosure comprises an anti-Sema3A antibody as described herein conjugated to a radioactive atom to form a radioconjugate. A variety of radioactive isotopes are available for the production of radioconjugates. Examples include 227Th, 225Ac, 211At, 131I, 125I, 90Y, 186Re, 188Re, 153Sm, 212Bi, 32P, 212Pb and radioactive isotopes of Lu. When the radioconjugate is used for detection, it may comprise a radioactive atom for scintigraphic studies, for example Tc99m, or a spin label for nuclear magnetic resonance (NMR) imaging, such as iodine-123 again, iodine-131, indium-111, fluorine-19, carbon-13, nitrogen-15, oxygen-17, gadolinium, manganese or iron.

Conjugates of an antibody and cytotoxic agent may be made using a variety of bifunctional protein coupling agents such as N-succinimidyl-3-(2-pyridyldithio) propionate (SPDP), succinimidyl-4-(N-maleimidomethyl) cyclohexane-1-carboxylate (SMCC), iminothiolane (IT), bifunctional derivatives of imidoesters (such as dimethyl adipimide HCl), active esters (such as disuccinimidyl suberate), aldehydes (such as glutaraldehyde), bis-azido compounds (such as bis (p-azidobenzoyl) hexanediamine), bis-diazonium derivatives (such as bis-(p-diazoniumbenzoyl)-ethylenediamine), diisocyanates (such as toluene 2,6-diisocyanate), and bis-active fluorine compounds (such as 1,5-difluoro-2,4-dinitrobenzene).

The linker may be a "cleavable linker" facilitating release of a cytotoxic drug in the cell. For example, an acid-labile linker, peptidase-sensitive linker, photolabile linker, dimethyl linker or disulfide-containing linker (Chari et al., Cancer Res. 52:12 7-131 (1992)).

The ACD according to the present disclosure includes ADCs prepared with cross-linker reagents including, but not limited to, BMPS, EMCS, GMBS, HBVS, LC-SMCC, MBS, MPBH, SBAP, SIA, SIAB, SMCC, SMPB, SMPH, sulfo-EMCS, sulfo-GMBS, sulfo-KMUS, sulfo-MBS, sulfo-SIAB, sulfo-SMCC, and sulfo-SMPB, and SVSB

(succinimidyl-(4-vinylsulfone)benzoate) which are commercially available (e.g., from Pierce Biotechnology, Inc., Rockford, IL., U.S.A)

Amino acid and nucleic acid sequences of preferred antibodies according to the present disclosure and three prior art antibodies are listed in Table 1 and Table 1A.

Peptide Variants

Antibodies or antigen-binding fragments of the disclosure are not limited to the specific peptide sequences provided herein. Rather, the disclosure also embodies variants of these polypeptides. With reference to the instant disclosure and conventionally available technologies and references, the skilled worker will be able to prepare, test and utilize functional variants of the antibodies disclosed herein, while appreciating these variants having the ability to bind to Sema3A fall within the scope of the present disclosure.

A variant can include, for example, an antibody that has at least one altered complementary determining region (CDR) (hyper-variable) and/or framework (FR) (variable) domain/position, vis-à-vis a peptide sequence disclosed herein.

By altering one or more amino acid residues in a CDR or FR region, the skilled worker routinely can generate mutated or diversified antibody sequences, which can be screened against the antigen, for new or improved properties, for example.

A further preferred embodiment of the disclosure is an antibody or antigen-binding fragment thereof in which the VH and VL sequences are selected as shown in Table 1 and Table 1A. The skilled worker can use the data in Table 1 and Table 1A to design peptide variants that are within the scope of the present disclosure. It is preferred that variants are constructed by changing amino acids within one or more CDR regions; a variant might also have one or more altered framework regions. For example, a peptide FR domain might be altered where there is a deviation in a residue compared to a germline sequence.

Alternatively, the skilled worker could make the same analysis by comparing the amino acid sequences disclosed herein to known sequences of the same class of such antibodies, using, for example, the procedure described by Knappik A., et al., JMB 2000, 296:57-86.

Furthermore, variants may be obtained by using one antibody as starting point for further optimization by diversifying one or more amino acid residues in the antibody, preferably amino acid residues in one or more CDRs, and by screening the resulting collection of antibody variants for variants with improved properties. Particularly preferred is diversification of one or more amino acid residues in CDR3 of VL and/or VH. Diversification can be done e.g. by synthesizing a collection of DNA molecules using trinucleotide mutagenesis (TRIM) technology (Virnekäs B. et al., Nucl. Acids Res. 1994, 22:5600.). Antibodies or antigen-binding fragments thereof include molecules with modifications/variations including but not limited to e.g. modifications leading to altered half-life (e.g. modification of the Fc part or attachment of further molecules such as PEG), altered binding affinity or altered ADCC or CDC activity.

Conservative Amino Acid Variants

Polypeptide variants may be made that conserve the overall molecular structure of an antibody peptide sequence described herein. Given the properties of the individual amino acids, some rational substitutions will be recognized by the skilled worker. Amino acid substitutions, i.e., "conservative substitutions," may be made, for instance, on the

basis of similarity in polarity, charge, solubility, hydrophobicity, hydrophilicity, and/or the amphipathic nature of the residues involved.

For example, (a) nonpolar (hydrophobic) amino acids include alanine, leucine, isoleucine, valine, proline, phenylalanine, tryptophane, and methionine; (b) polar neutral amino acids include glycine, serine, threonine, cysteine, tyrosine, asparagine, and glutamine; (c) positively charged (basic) amino acids include arginine, lysine, and histidine; and (d) negatively charged (acidic) amino acids include aspartic acid and glutamic acid. Substitutions typically may be made within groups (a)-(d). In addition, glycine and proline may be substituted for one another based on their ability to disrupt α -helices. Similarly, certain amino acids, such as alanine, cysteine, leucine, methionine, glutamic acid, glutamine, histidine and lysine are more commonly found in α -helices, while valine, isoleucine, phenylalanine, tyrosine, tryptophan and threonine are more commonly found in β -pleated sheets. Glycine, serine, aspartic acid, asparagine, and proline are commonly found in turns. Some preferred substitutions may be made among the following groups: (i) S and T; (ii) P and G; and (iii) A, V, L and I. Given the known genetic code, and recombinant and synthetic DNA techniques, the skilled scientist readily can construct DNAs encoding the conservative amino acid variants.

Glycosylation Variants

Where the antibody comprises an Fc region, the carbohydrate attached thereto may be altered. Native antibodies produced by mammalian cells typically comprise a branched, biantennary oligosaccharide that is generally attached by an N-linkage to Asn297 using Kabat EU numbering of the CH2 domain of the Fc region; see, e.g., Wright et al. *Trends Biotechnol.* 15:26-32 (1997).

In certain embodiments, an antibody provided herein is altered to increase or decrease the extent to which the antibody is glycosylated. Addition or deletion of glycosylation sites to an antibody may be conveniently accomplished by altering the expression system (e.g. host cell) and/or by altering the amino acid sequence such that one or more glycosylation sites is created or removed.

In one embodiment of this disclosure, aglycosyl antibodies having decreased effector function or antibody derivatives are prepared by expression in a prokaryotic host. Suitable prokaryotic hosts for include but are not limited to *E. coli*, *Bacillus subtilis*, *Salmonella typhimurium* and various species within the genera *Pseudomonas*, *Streptomyces*, and *Staphylococcus*.

In one embodiment, antibody variants are provided having decreased effector function, which are characterized by a modification at the conserved N-linked site in the CH2 domains of the Fc portion of said antibody. In one embodiment of present disclosure, the modification comprises a mutation at the heavy chain glycosylation site to prevent glycosylation at the site. Thus, in one preferred embodiment of this disclosure, the aglycosyl antibodies or antibody derivatives are prepared by mutation of the heavy chain glycosylation site, —i.e., mutation of N297 using Kabat EU numbering and expressed in an appropriate host cell.

In another embodiment of the present disclosure, aglycosyl antibodies or antibody derivatives have decreased effector function, wherein the modification at the conserved N-linked site in the CH2 domains of the Fc portion of said antibody or antibody derivative comprises the removal of the CH2 domain glycans, —i.e., deglycosylation. These aglycosyl antibodies may be generated by conventional methods and then deglycosylated enzymatically. Methods

for enzymatic deglycosylation of antibodies are well known in the art (e.g. Winkelhake & Nicolson (1976), *J Biol Chem.* 251 (4): 1074-80).

In another embodiment of this disclosure, deglycosylation may be achieved using the glycosylation inhibitor tunicamycin (Nose & Wigzell (1983), *Proc Natl Acad Sci USA*, 80 (21): 6632-6). That is, the modification is the prevention of glycosylation at the conserved N-linked site in the CH2 domains of the Fc portion of said antibody.

In one embodiment, antibody variants are provided having a carbohydrate structure that lacks fucose attached (directly or indirectly) to an Fc region. For example, the amount of fucose in such antibody may be from 1% to 80%, from 1% to 65%, from 5% to 65% or from 20% to 40%. The amount of fucose is determined by calculating the average amount of fucose within the sugar chain at Asn297, relative to the sum of all glycostructures attached to Asn 297 (e.g. complex, hybrid and high mannose structures) as measured by MALDI-TOF mass spectrometry, as described in WO 2008/077546, for example. Asn297 refers to the asparagine residue located at about position 297 in the Fc region (Eu numbering of Fc region residues); however, Asn297 may also be located about ± 3 amino acids upstream or downstream of position 297, i.e., between positions 294 and 300, due to minor sequence variations in antibodies. Such fucosylation variants may have improved ADCC function.

Examples of publications related to “defucosylated” or “fucose-deficient” antibody variants include: Okazaki et al. *J Mol. Biol.* 336:1239-1249 (2004); Yamane-Ohnuki et al. *Biotech. Bioeng.* 87:614 (2004).

Examples of cell lines capable of producing defucosylated antibodies include Lec13 CHO cells deficient in protein fucosylation (Ripka et al. *Arch. Biochem. Biophys.* 249: 533-545 (1986); and WO 2004/056312), and knockout cell lines, such as alpha-1,6-fucosyltransferase gene, FUT8, knockout CHO cells (see, e.g., Yamane-Ohnuki et al. *Biotech. Bioeng.* 87:614 (2004); Kanda, Y. et al., *Biotechnol. Bioeng.*, 94 (4): 680-688 (2006)).

Antibody variants are further provided with bisected oligosaccharides, e.g., in which a biantennary oligosaccharide attached to the Fc region of the antibody is bisected by GlcNAc. Such antibody variants may have reduced fucosylation and/or improved ADCC function. Examples of such antibody variants are described, e.g., in WO 2003/011878; U.S. Pat. No. 6,602,684; and US 2005/0123546.

Antibody variants with at least one galactose residue in the oligosaccharide attached to the Fc region are also provided. Such antibody variants may have improved CDC function. Such antibody variants are described, e.g., in WO1997/30087; WO1998/58964; and WO1999/22764.

FC Region Variants

In certain embodiments, one or more amino acid modifications (e.g. a substitution) may be introduced into the Fc region of an antibody (e.g., a human IgG1, IgG2, IgG3 or IgG4 Fc region) provided herein, thereby generating an Fc region variant.

In certain embodiments, the disclosure contemplates an antibody variant that possesses some but not all effector functions, which make it a desirable candidate for applications in which the half-life of the antibody in vivo is important yet certain effector functions (such as complement and ADCC) are unnecessary or deleterious. In vitro and/or in vivo cytotoxicity assays can be conducted to confirm the reduction/depletion of CDC and/or ADCC activities. For example, Fc receptor (FcR) binding assays can be conducted to ensure that the antibody lacks Fc γ R binding (hence likely lacking ADCC activity) but retains FcRn binding ability. In

some embodiments, alterations are made in the Fc region that result in altered (i.e., either improved or diminished) C1q binding and/or Complement Dependent Cytotoxicity (CDC).

In certain embodiments, the disclosure contemplates an antibody variant that possesses an increased or decreased half-life. Antibodies with increased half-lives and improved binding to the neonatal Fc receptor (FcRn), which is responsible for the transfer of maternal IgGs to the fetus (Guyer et al., *J Immunol.* 117:587 (1976) and Kim et al., *J Immunol.* 24:249 (1994)), are described in US2005/0014934 (Hinton et al.). Those antibodies comprise an Fc region with one or more substitutions therein which improve binding of the Fc region to FcRn.

Antibody Generation

An antibody of the disclosure may be derived from a recombinant antibody library that is based on amino acid sequences that have been isolated from the antibodies of a large number of healthy volunteers e.g. using the n-CoDeR® technology the fully human CDRs are recombined into new antibody molecules (Carlson & Söderlind, *Expert Rev Mol Diagn.* 2001 May; 1 (1): 102-8). Or alternatively for example antibody libraries as the fully human antibody phage display library described in Hoet R M et al., *Nat Biotechnol* 2005; 23 (3): 344-8) can be used to isolate Sema3A-specific antibodies. Antibodies or antibody fragments isolated from human antibody libraries are considered human antibodies or human antibody fragments herein.

Human antibodies may be further prepared by administering an immunogen to a transgenic animal that has been modified to produce intact human antibodies or intact antibodies with human variable regions in response to antigenic challenge. Such animals typically contain all or a portion of the human immunoglobulin loci, which replace the endogenous immunoglobulin loci, or which are present extrachromosomally or integrated randomly into the animal's chromosomes. For example immunization of genetically engineered mice inter alia immunization of hMAb mice (e.g. VelocImmune Mouse® or XENOMOUSE®) may be performed.

Further antibodies may be generated using the hybridoma technology (for example see Köhler and Milstein *Nature*. 1975 Aug. 7; 256 (5517): 495-7), resulting in for example murine, rat, or rabbit antibodies which can be converted into chimeric or humanized antibodies. Humanized antibodies and methods of making them are reviewed, e.g., in Almagro and Fransson, *Front. Biosci.* 13:1619-1633 (2008), and are further described, e.g., in Riechmann et al., *Nature* 332:323-329 (1988); Queen et al., *Proc. Natl Acad. Sci. USA* 86:10029-10033 (1989); U.S. Pat. Nos. 5,821,337, 7,527, 791, 6,982,321, and 7,087,409; Kashmiri et al., *Methods* 36:25-34 (2005) (describing specificity determining region (SDR) grafting); Padlan, *Mol. Immunol.* 28:489-498 (1991) (describing "resurfacing"); Dall'Acqua et al., *Methods* 36:43-60 (2005) (describing "FR shuffling"); and Osbourn et al., *Methods* 36:61-68 (2005) and Klimka et al., *Br. J. Cancer*, 83:252-260 (2000) (describing the "guided selection" approach to FR shuffling).

Examples are provided for the generation of antibodies using a recombinant antibody library.

DNA Molecules According to the Present Disclosure

The present disclosure also relates to an isolated nucleic acid sequence that encodes the antibody or antigen-binding fragment according to the present disclosure. The isolated nucleic acid sequence encoding the antibody or antigen-binding fragment according to the present disclosure can for instance be produced by techniques described in Sambrook

et al., 1989, and Ausubel et al., 1989, or alternatively, by chemical synthesis. (e.g. techniques described in *Oligonucleotide Synthesis* (1984, Gait, ed., IRL Press, Oxford)). The DNA sequences and respective SEQ IDs used for the antibodies expressed are given in Table 1 and 1A. These sequences are optimized in certain cases for mammalian expression. DNA molecules of the disclosure are not limited to the sequences disclosed herein, but also include variants thereof. DNA variants within the disclosure may be described by reference to their physical properties in hybridization. The skilled worker will recognize that DNA can be used to identify its complement and, since DNA is double stranded, its equivalent or homolog, using nucleic acid hybridization techniques. It also will be recognized that hybridization can occur with less than 100% complementarity. However, given appropriate choice of conditions, hybridization techniques can be used to differentiate among DNA sequences based on their structural relatedness to a particular probe. For guidance regarding such conditions see, Sambrook et al., 1989 *supra* and Ausubel et al., 1995 (Ausubel, F. M., Brent, R., Kingston, R. E., Moore, D. D., Sedman, J. G., Smith, J. A., & Struhl, K. eds. (1995). *Current Protocols in Molecular Biology*. New York: John Wiley and Sons).

Structural similarity between two polynucleotide sequences can be expressed as a function of "stringency" of the conditions under which the two sequences will hybridize with one another. As used herein, the term "stringency" refers to the extent that the conditions disfavor hybridization. Stringent conditions strongly disfavor hybridization, and only the most structurally related molecules will hybridize to one another under such conditions. Conversely, non-stringent conditions favor hybridization of molecules displaying a lesser degree of structural relatedness. Hybridization stringency, therefore, directly correlates with the structural relationships of two nucleic acid sequences.

Hybridization stringency is a function of many factors, including overall DNA concentration, ionic strength, temperature, probe size and the presence of agents which disrupt hydrogen bonding. Factors promoting hybridization include high DNA concentrations, high ionic strengths, low temperatures, longer probe size and the absence of agents that disrupt hydrogen bonding. Hybridization typically is performed in two phases: the "binding" phase and the "washing" phase.

Functionally Equivalent DNA Variants

Yet another class of DNA variants within the scope of the disclosure may be described with reference to the product they encode. These functionally equivalent polynucleotides are characterized by the fact that they encode the same peptide sequences due to the degeneracy of the genetic code.

It is recognized that variants of DNA molecules provided herein can be constructed in several different ways. For example, they may be constructed as completely synthetic DNAs. Methods of efficiently synthesizing oligonucleotides are widely available. See Ausubel et al., section 2.11, Supplement 21 (1993). Overlapping oligonucleotides may be synthesized and assembled in a fashion first reported by Khorana et al., *J. Mol. Biol.* 72:209-217 (1971); see also Ausubel et al., *supra*, Section 8.2. Synthetic DNAs preferably are designed with convenient restriction sites engineered at the 5' and 3' ends of the gene to facilitate cloning into an appropriate vector.

As indicated, a method of generating variants is to start with one of the DNAs disclosed herein and then to conduct site-directed mutagenesis. See Ausubel et al., *supra*, chapter 8, Supplement 37 (1997). In a typical method, a target DNA

is cloned into a single-stranded DNA bacteriophage vehicle. Single-stranded DNA is isolated and hybridized with an oligonucleotide containing the desired nucleotide alteration (s). The complementary strand is synthesized and the double stranded phage is introduced into a host. Some of the resulting progeny will contain the desired mutant, which can be confirmed using DNA sequencing. In addition, various methods are available that increase the probability that the progeny phage will be the desired mutant. These methods are well known to those in the field and kits are commercially available for generating such mutants.

Recombinant DNA Constructs and Expression

The present disclosure further provides recombinant DNA constructs comprising one or more of the nucleotide sequences according to the present disclosure. The recombinant constructs of the present disclosure can be used in connection with a vector, such as a plasmid, phagemid, phage or viral vector, into which a DNA molecule encoding an antibody of the disclosure or antigen-binding fragment thereof or variant thereof is inserted.

Thus, in one aspect, the present disclosure relates to a vector comprising a nucleic acid sequence according to the present disclosure.

An antibody, antigen binding portion, or variant thereof provided herein can be prepared by recombinant expression of nucleic acid sequences encoding light and heavy chains or portions thereof in a host cell. To express an antibody, antigen binding portion, or variant thereof recombinantly a host cell can be transfected with one or more recombinant expression vectors carrying DNA fragments encoding the light and/or heavy chains or portions thereof such that the light and heavy chains are expressed in the host cell. Standard recombinant DNA methodologies are used to prepare and/or obtain nucleic acids encoding the heavy and light chains, incorporate these nucleic acids into recombinant expression vectors and introduce the vectors into host cells, such as those described in Sambrook, Fritsch and Maniatis (eds.), *Molecular Cloning; A Laboratory Manual*, Second Edition, Cold Spring Harbor, N.Y., (1989), Ausubel, F. M. et al. (eds.) *Current Protocols in Molecular Biology*, Greene Publishing Associates, (1989) and in U.S. Pat. No. 4,816,397 by Boss et al.

In addition, the nucleic acid sequences encoding variable regions of the heavy and/or light chains can be converted, for example, to nucleic acid sequences encoding full-length antibody chains, Fab fragments, or to scFv. The VL- or VH-encoding DNA fragment can be operatively linked, (such that the amino acid sequences encoded by the two DNA fragments are in-frame) to another DNA fragment encoding, for example, an antibody constant region or a flexible linker. The sequences of human heavy chain and light chain constant regions are known in the art (see e.g., Kabat, E. A., et al. (1991) *Sequences of Proteins of Immunological Interest*, Fifth Edition, U.S. Department of Health and Human Services, NIH Publication No. 91-3242) and DNA fragments encompassing these regions can be obtained by standard PCR amplification.

To create a polynucleotide sequence that encodes a scFv, the VH- and VL-encoding nucleic acids can be operatively linked to another fragment encoding a flexible linker such that the VH and VL sequences can be expressed as a contiguous single-chain protein, with the VL and VH regions joined by the flexible linker (see e.g., Bird et al. (1988) *Science* 242:423-426; Huston et al. (1988) *Proc. Natl. Acad. Sci. USA* 85:5879-5883; McCafferty et al., *Nature* (1990) 348:552-554).

To express the antibodies, antigen binding fragments thereof or variants thereof standard recombinant DNA expression methods can be used (see, for example, Goeddel; *Gene Expression Technology. Methods in Enzymology* 185, Academic Press, San Diego, Calif. (1990)). For example, DNA encoding the desired polypeptide can be inserted into an expression vector which is then transfected into a suitable host cell. Suitable host cells are prokaryotic and eukaryotic cells. Examples for prokaryotic host cells are e.g. bacteria, examples for eukaryotic hosts cells are yeasts, insects and insect cells, plants and plant cells, transgenic animals, or mammalian cells. Introduction of the recombinant construct into the host cell can be carried out using standard techniques such as calcium phosphate transfection, DEAE dextran mediated transfection, electroporation, transduction or phage infection.

In some embodiments, the DNAs encoding the heavy and light chains are inserted into separate vectors. In other embodiments, the DNA encoding the heavy and light chains is inserted into the same vector. It is understood that the design of the expression vector, including the selection of regulatory sequences is affected by factors such as the choice of the host cell, the level of expression of protein desired and whether expression is constitutive or inducible.

Thus, in a further aspect, the present disclosure relates to an isolated cell expressing the antibody or antigen-binding fragment according to the present disclosure and/or comprising the nucleic acid according to the present disclosure or the vector according to the present disclosure.

The isolated cell can be virtually any cell for which expression vectors are available. The isolated cell can for example a higher eukaryotic host cell, such as a mammalian cell, a lower eukaryotic host cell, such as a yeast cell, and may be a prokaryotic cell, such as a bacterial cell.

In a further aspect, the present disclosure relates to a method of producing the isolated antibody or antigen-binding fragment according to the present disclosure comprising culturing of the cell according to the present disclosure. In particular embodiments, the cell according to the present disclosure is cultivated under suitable conditions for antibody expression and the antibody or antigen-binding fragment thereof is recovered. In particular embodiments, the antibody or antigen-binding fragment thereof is purified, particularly to at least 95% homogeneity by weight.

Bacterial Expression

Useful expression vectors for bacterial use are constructed by inserting a DNA sequence encoding a desired protein together with suitable translation initiation and termination signals in operable reading phase with a functional promoter. The vector will comprise one or more phenotypic selectable markers and an origin of replication to ensure maintenance of the vector and, if desirable, to provide amplification within the host. Suitable prokaryotic hosts for transformation include but are not limited to *E. coli*, *Bacillus subtilis*, *Salmonella typhimurium* and various species within the genera *Pseudomonas*, *Streptomyces*, and *Staphylococcus*.

Bacterial vectors may be, for example, bacteriophage-, plasmid- or phagemid-based. These vectors can contain a selectable marker and a bacterial origin of replication derived from commercially available plasmids typically containing elements of the well-known cloning vector pBR322 (ATCC 37017). Following transformation of a suitable host strain and growth of the host strain to an appropriate cell density, the selected promoter is de-repressed/induced by appropriate means (e.g., temperature shift or chemical induction) and cells are cultured for an

additional period. Cells are typically harvested by centrifugation, disrupted by physical or chemical means, and the resulting crude extract retained for further purification.

In bacterial systems, a number of expression vectors may be advantageously selected depending upon the use intended for the protein being expressed. For example, when a large quantity of such a protein is to be produced, for the generation of antibodies or to screen peptide libraries, for example, vectors which direct the expression of high levels of fusion protein products that are readily purified may be desirable.

Therefore, an embodiment of the present disclosure is an expression vector comprising a nucleic acid sequence encoding for the novel antibodies of the present disclosure.

Antibodies of the present disclosure or antigen-binding fragments thereof or variants thereof include naturally purified products, products of chemical synthetic procedures, and products produced by recombinant techniques from a prokaryotic host, including, for example, *E. coli*, *Bacillus subtilis*, *Salmonella typhimurium* and various species within the genera *Pseudomonas*, *Streptomyces*, and *Staphylococcus*, preferably, from *E. coli* cells.

Mammalian Expression

Preferred regulatory sequences for mammalian host cell expression include viral elements that direct high levels of protein expression in mammalian cells, such as promoters and/or enhancers derived from cytomegalovirus (CMV) (such as the CMV promoter/enhancer), Simian Virus 40 (SV40) (such as the SV40 promoter/enhancer), adenovirus, (e.g., the adenovirus major late promoter (AdMLP)) and polyoma. Expression of the antibodies may be constitutive or regulated (e.g. inducible by addition or removal of small molecule inductors such as Tetracyclin in conjunction with Tet system). For further description of viral regulatory elements, and sequences thereof, see e.g., U.S. Pat. No. 5,168,062 by Stinski, U.S. Pat. No. 4,510,245 by Bell et al. and U.S. Pat. No. 4,968,615 by Schaffner et al., The recombinant expression vectors can also include origins of replication and selectable markers (see e.g., U.S. Pat. Nos. 4,399,216, 4,634,665 and 5,179,017). Suitable selectable markers include genes that confer resistance to drugs such as G418, puromycin, hygromycin, blasticidin, zeocin/bleomycin or methotrexate or selectable marker that exploit auxotrophies such as Glutamine Synthetase (Bebington et al., Biotechnology (N Y). 1992 February; 10 (2): 169-75), on a host cell into which the vector has been introduced. For example, the dihydrofolate reductase (DHFR) gene confers resistance to methotrexate, neo gene confers resistance to G418, the *bsd* gene from *Aspergillus terreus* confers resistance to blasticidin, puromycin N-acetyl-transferase confers resistance to puromycin, the *Sh ble* gene product confers resistance to zeocin, and resistance to hygromycin is conferred by the *E. coli* hygromycin resistance gene (*hyg* or *hph*). Selectable markers like DHFR or Glutamine Synthetase are also useful for amplification techniques in conjunction with MTX and MSX.

Transfection of the expression vector into a host cell can be carried out using standard techniques such as electroporation, nucleofection, calcium-phosphate precipitation, lipofection, polycation-based transfection such as polyethylenimine (PEI)-based transfection and DEAE-dextran transfection.

Suitable mammalian host cells for expressing the antibodies, antigen binding fragments thereof or variants thereof provided herein include Chinese Hamster Ovary (CHO cells) such as CHO-K1, CHO-S, CHO-K1SV [including dhfr-CHO cells, described in Urlaub and Chasin, (1980)

Proc. Natl. Acad. Sci. USA 77:4216-4220 and Urlaub et al., Cell. 1983 June; 33 (2):405-12, used with a DHFR selectable marker, e.g., as described in R. J. Kaufman and P. A. Sharp (1982) Mol. Biol. 159:601-621; and other knockout cells exemplified in Fan et al., Biotechnol Bioeng. 2012 April; 109 (4):1007-15], NS0 myeloma cells, COS cells, HEK293 cells, HKB11 cells, BHK21 cells, CAP cells, EB66 cells, and SP2 cells.

Expression might also be transient or semi-stable in expression systems such as HEK293, HEK293T, HEK293-EBNA, HEK293E, HEK293-6E, HEK293-Freestyle, HKB11, Expi293F, 293EBNALT75, CHO Freestyle, CHO-S, CHO-K1, CHO-K1SV, CHOEBNALT85, CHOS-XE, CHO-3E7 or CAP-T cells (for instance Durocher et al., Nucleic Acids Res. 2002 Jan. 15; 30 (2): E9).

In some embodiments, the expression vector is designed such that the expressed protein is secreted into the culture medium in which the host cells are grown. The antibodies, antigen binding fragments thereof or variants thereof can be recovered from the culture medium using standard protein purification methods.

Purification

Antibodies of the disclosure or antigen-binding fragments thereof or variants thereof can be recovered and purified from recombinant cell cultures by well-known methods including, but not limited to ammonium sulfate or ethanol precipitation, acid extraction, Protein A chromatography, Protein G chromatography, anion or cation exchange chromatography, phospho-cellulose chromatography, hydrophobic interaction chromatography, affinity chromatography, hydroxylapatite chromatography and lectin chromatography. High performance liquid chromatography ("HPLC") can also be employed for purification. See, e.g., Colligan, Current Protocols in Immunology, or Current Protocols in Protein Science, John Wiley & Sons, NY, N.Y., (1997-2001), e.g., Chapters 1, 4, 6, 8, 9, 10, each entirely incorporated herein by reference.

Antibodies of the present disclosure or antigen-binding fragments thereof or variants thereof include naturally purified products, products of chemical synthetic procedures, and products produced by recombinant techniques from a eukaryotic host, including, for example, yeast, higher plant, insect and mammalian cells. Depending upon the host employed in a recombinant production procedure, the antibody of the present disclosure can be glycosylated or can be non-glycosylated. Such methods are described in many standard laboratory manuals, such as Sambrook, supra, Sections 17.37-17.42; Ausubel, supra, Chapters 10, 12, 13, 16, 18 and 20.

In preferred embodiments, the antibody is purified (1) to greater than 95% by weight of antibody as determined e.g. by the Lowry method, UV-Vis spectroscopy or by SDS-Capillary Gel electrophoresis (for example on a Caliper LabChip GXII, GX 90 or Biorad Bioanalyzer device), and in further preferred embodiments more than 99% by weight, (2) to a degree sufficient to obtain at least 15 residues of N-terminal or internal amino acid sequence, or (3) to homogeneity by SDS-PAGE under reducing or non-reducing conditions using Coomassie blue or, preferably, silver stain. Isolated naturally occurring antibody includes the antibody in situ within recombinant cells since at least one component of the antibody's natural environment will not be present. Ordinarily, however, isolated antibody will be prepared by at least one purification step.

Therapeutic Methods

Therapeutic methods involve administering to a subject in need of treatment a therapeutically effective amount of an

antibody or an antigen-binding fragment thereof or a variant thereof contemplated by the disclosure. A “therapeutically effective” amount hereby is defined as the amount of an antibody or antigen-binding fragment thereof that is of sufficient quantity to decrease Sema3A activity in a subject—either as a single dose or according to a multiple dose regimen, alone or in combination with other agents, which leads to the alleviation of an adverse condition, yet which amount is toxicologically tolerable. The subject may be a human or non-human animal (e.g., rabbit, rat, mouse, dog, monkey or other lower-order primate).

Thus, in one aspect, the present disclosure relates to the isolated antibody or antigen-binding fragment according to the present disclosure or to a conjugate comprising the isolated antibody or antigen-binding fragment according to the present disclosure or to a pharmaceutical composition comprising the isolated antibody or antigen-binding fragment according to the present disclosure for use as a medicament.

The isolated antibody or antigen-binding fragment according to the present disclosure can be used as a therapeutic or a diagnostic tool in a variety of Sema3A-associated disorders.

Thus, in a further aspect, the present disclosure relates to the isolated antibody or antigen-binding fragment according to the present disclosure or to a conjugate comprising the isolated antibody or antigen-binding fragment according to the present disclosure or to a pharmaceutical composition comprising the isolated antibody or antigen-binding fragment according to the present disclosure for use in the treatment and/or prevention of renal diseases, in particular of acute and chronic kidney diseases, diabetic kidney diseases, Alport syndrome and of acute and chronic renal failure. The general terms ‘renal disease’ or ‘kidney disease’ describes a class of conditions in which the kidneys fail to filter and remove waste products from the blood. There are two major forms of kidney disease: acute kidney disease (acute kidney injury, AKI) and chronic kidney disease (CKD). The isolated antibody or antigen-binding fragment according to the present disclosure or a conjugate or pharmaceutical composition comprising the same may further be used for the treatment and/or prevention of sequelae of acute kidney injury arising from multiple insults such as ischemia-reperfusion injury, radiocontrast administration, cardiopulmonary bypass surgery, shock and sepsis. In the context of the present disclosure, the terms renal failure and renal insufficiency comprise both acute and chronic manifestations of renal insufficiency, as well as underlying or related kidney diseases such as renal hypoperfusion, intradialytic hypotension, obstructive uropathy, glomerulopathies, IgA nephropathy, glomerulonephritis, acute glomerulonephritis, glomerulosclerosis, tubulointerstitial diseases, nephropathic diseases such as primary and congenital kidney disease, nephritis, Alport syndrome, kidney inflammation, immunological kidney diseases such as kidney transplant rejection, immune complex-induced kidney diseases, nephropathy induced by toxic substances, contrast medium-induced nephropathy; minimal change glomerulonephritis (lipoid); Membranous glomerulonephritis; focal segmental glomerulosclerosis (FSGS); hemolytic uremic syndrome (HUS), amyloidosis, Goodpasture’s syndrome, Wegener’s granulomatosis, Purpura Schönlein-Henoch, diabetic and non-diabetic nephropathy, pyelonephritis, renal cysts, nephrosclerosis, hypertensive nephrosclerosis and nephrotic syndrome, which can be characterized diagnostically, for example, by abnormally reduced creatinine and/or water excretion, abnormally increased blood concentrations of urea, nitrogen, potassium and/or creatinine, altered activity of renal enzymes such as, for example,

glutamyl synthetase, altered urine osmolality or urine volume, increased microalbuminuria, macroalbuminuria, lesions of glomeruli and arterioles, tubular dilatation, hyperphosphatemia and/or the need for dialysis. The present disclosure also relates to the isolated antibody or antigen-binding fragment according to the present disclosure or a conjugate or pharmaceutical composition comprising same for use in the treatment and/or prevention of sequelae of renal insufficiency, for example pulmonary edema, heart failure, uremia, anemia, electrolyte disturbances (e.g. hyperkalemia, hyponatremia) and disturbances in bone and carbohydrate metabolism. The compounds according to the disclosure are also suitable for the treatment and/or prevention of polycystic kidney disease (PKD) and of the syndrome of inadequate ADH secretion (SIADH).

Additionally, the isolated antibody or antigen-binding fragment according to the present disclosure or a conjugate or pharmaceutical composition comprising the same may be used for the treatment and/or prevention of vascular hyperpermeability, diabetic retinopathy, deterioration of the blood retinal barrier and consequent macular edema, preferably, age related macular edema, non-proliferative age-related macular edema and non-proliferative diabetic macular edema.

Further, the isolated antibody or antigen-binding fragment according to the present disclosure or a conjugate or pharmaceutical composition comprising same is suitable for the prevention or treatment of disease of the central or peripheral nervous system like neuropathic pain, spinal cord injury, multiple sclerosis, traumatic brain injury, brain edema or neurodegenerative diseases in which the neurodegenerative disease is Alzheimer’s disease, Parkinson’s disease, Huntington’s disease, amyotrophic lateral sclerosis, progressive supranuclear paralysis, black substance degeneration, Shy-Drager syndrome, olivopontocerebellar atrophy or spinocerebellar degeneration.

Furthermore, the isolated antibody or antigen-binding fragment according to the present disclosure or a conjugate or pharmaceutical composition comprising the same may be useful for the treatment and/or prevention of cancer, wherein the cancer is intestinal cancer, colorectal cancer, lung cancer, breast cancer, brain cancer, melanoma, renal cell cancer, leukemia, lymphoma, T-cell lymphoma, stomach cancer, pancreatic cancer, cervical cancer, endometrial cancer, ovarian cancer, esophageal cancer, liver cancer, squamous cell carcinoma of the head and neck, skin cancer, urinary tract cancer, prostate cancer, choriocarcinoma, pharyngeal cancer or larynx cancer.

The disorders mentioned above have been well characterized in humans, but also exist with a similar etiology in other animals, including mammals, and can be treated by administering pharmaceutical compositions according to the present disclosure.

The antibody or the antigen-binding fragment according to the present disclosure or a variant thereof might be co-administered with known medicaments, and in some instances the antibody or antigen-binding fragment thereof might itself be modified. For example, an antibody or an antigen-binding fragment thereof or a variant thereof could be conjugated to a drug or to another peptide or protein to potentially further increase efficacy.

Antibodies of the present disclosure or antigen-binding fragments thereof or variants thereof may be administered as the sole pharmaceutical agent or in combination with one or more additional therapeutic agents where the combination causes no unacceptable adverse effects.

Thus, in a further aspect, the present disclosure relates to the isolated antibody or antigen-binding fragment according to the present disclosure or the conjugate according to the present disclosure or the pharmaceutical composition according to the present disclosure for use in simultaneous, separate, or sequential combination with one or more further therapeutically active compounds.

Non-limiting examples of therapeutically active compounds to be used in combination with the antibody or antigen-binding fragment according to the present disclosure are:

blood pressure lowering agents, for example and preferably from the group of calcium antagonists, angiotensin II antagonists, ACE inhibitors, NEP inhibitors, vaso-peptidase inhibitors, endothelin antagonists, renin inhibitors, alpha-blockers, beta-blockers, mineralocorticoid receptor antagonists and diuretics;

antidiabetic agents (hypoglycemic or antihyperglycemic agents), such as for example and preferably insulin and derivatives, sulfonylureas, biguanides, thiazolidinediones, acarbose, DPP4 inhibitors, GLP-1 analogues, or SGLT inhibitors (gliflozins);

compounds inhibiting the signal transduction cascade, in particular tyrosine and/or serine/threonine kinase inhibitors, such as for example nintedanib, dasatinib, nilotinib, bosutinib, regorafenib, sorafenib, sunitinib, cediranib, axitinib, telatinib, imatinib, brivanib, pazopanib, vatalanib, gefitinib, erlotinib, lapatinib, canertinib, lestaurtinib, pelitinib, semaxanib or tandutinib;

anti-inflammatory drugs such as non-steroidal anti-inflammatory drugs (NSAIDs) including acetylsalicylic acid (aspirin), ibuprofen and naproxen, glucocorticoids such as for example and preferably prednisolone, prednisolone, methylprednisolone, triamcinolone, dexamethasone, beclomethasone, betamethasone, flunisolide, budesonide or fluticasone, or 5-aminosalicylic acid derivatives, leukotriene antagonists, TNF-alpha inhibitors and chemokine receptor antagonists such as CCR1, 2 and/or 5 inhibitors, NF-kB inhibitors and Nrf2 activators; anti-fibrotic drugs such as TGFbeta antagonist, or microRNA-21 inhibitors;

organic nitrates and NO-donors, for example sodium nitroprusside, nitroglycerin, isosorbide mononitrate, isosorbide dinitrate, molsidomine or SIN-1, and inhalational NO;

compounds that inhibit the degradation of cyclic guanosine monophosphate (cGMP), for example inhibitors of phosphodiesterases (PDE) 1, 2, 5 and/or 9, in particular PDE-5 inhibitors such as sildenafil, vardenafil, tadalafil, udenafil, dasantafil, avanafil, mirodenafil, lodenafil, CTP-499 or PF-00489791;

calcium sensitizers, such as for example and preferably levosimendan;

antithrombotic agents, particularly selected from the group consisting of platelet aggregation inhibitors, anticoagulants and profibrinolytic substances;

agents, that stimulate NO- and heme-dependent as well as NO- and heme-independent the synthesis of cGMP, for example and with preference soluble guanylate cyclase (sGC) modulators, for example and with preference riociguat, cinaciguat, vericiguat or BAY 1101042;

fat metabolism altering agents, for example and preferably from the group of thyroid receptor agonists, cholesterol synthesis inhibitors, such as for example and preferably HMG-CoA-reductase or squalene synthesis inhibitors, ACAT inhibitors, CETP inhibitors, MTP inhibitors, PPAR-alpha, PPAR-gamma and/or PPAR-

delta agonists, cholesterol absorption inhibitors, lipase inhibitors, polymeric bile acid adsorbers, bile acid reabsorption inhibitors and lipoprotein (a) antagonists.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with a platelet aggregation inhibitor, particularly aspirin, clopidogrel, ticlopidine or dipyridamole.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with a thrombin inhibitor, particularly ximelagatran, dabigatran, melagatran, bivalirudin or enoxaparin.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with a GPIIb/IIIa antagonist, particularly tirofiban or abciximab.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with a factor Xa inhibitor, particularly selected from rivaroxaban, apixaban, otamixaban, fidexaban, razaxaban, fondaparinux, idraparinux, DU-176b, PMD-3112, YM-150, KFA-1982, EMD-503982, MCM-17, MLN-1021, DX 9065a, DPC 906, JTV 803, SSR-126512 and SSR-128428.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with heparin or a low molecular weight (LMW) heparin derivative.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with a vitamin K antagonist, particularly selected from coumarin.

Blood pressure lowering agents are particularly selected from the group consisting of calcium antagonists, angiotensin AII antagonists, ACE inhibitors, NEP inhibitors, vaso-peptidase inhibitors, endothelin antagonists, renin inhibitors, alpha-blockers, beta-blockers, mineralocorticoid receptor antagonists and diuretics.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with a calcium antagonist, particularly selected from nifedipine, amlodipine, verapamil and diltiazem.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with an angiotensin AII receptor antagonist, particularly selected from the group consisting of losartan, candesartan, valsartan, telmisartan, irbesartan, olmesartan, eprosartan, embursartan and azilsartan.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with an ACE inhibitor, particularly selected from the group consisting of enalapril, captopril, lisinopril, ramipril, delapril, fosinopril, quinopril, perindopril, benazepril andtrandopril.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with an endothelin antagonist, particularly selected from the group consisting of bosentan, darusentan, ambrisentan, tezosentan, sitaxsentan, avosentan, macitentan and atrasentan.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with a renin inhibitor, particularly selected from the group consisting of aliskiren, SPP-600 and SPP-800.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with a mineralocorticoid receptor antagonist, particularly selected from the group consisting of finerenone, spironolactone, canrenone, potassium canrenoate, eplerenone, esaxerenone (CS-3150), or aparenone (MT-3995), CS-3150, and MT-3995.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with a diuretic, particularly selected from the group consisting of furosemide, bumetanide, piretanide, torsemide, bendroflumethiazide, chlorothiazide, hydrochlorothiazide, xipamide, indapamide, hydroflumethiazide, methyclothiazide, polythiazide, trichloromethiazide, chlorothalidone, metolazone, quinethazone, acetazolamide, dichlorophenamide, methazolamide, glycerine, isosorbide, mannitol, amiloride and triamterene.

Fat metabolism altering agents are particularly selected from the group consisting of CETP inhibitors, thyroid receptor agonists, cholesterol synthesis inhibitors such as HMG-CoA-reductase or squalene synthesis inhibitors, ACAT inhibitors, MTP inhibitors, PPAR-alpha, PPAR-gamma and/or PPAR-delta agonists, cholesterol absorption inhibitors, polymeric bile acid adsorbers, bile acid reabsorption inhibitors, lipase inhibitors and lipoprotein (a) antagonists.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with a Nrf2 activator, particularly selected from Bardoxolone methyl.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with a thyroid receptor agonist, particularly selected from the group consisting of D-thyroxine, 3,5,3'-triiodothyronine (T3), CGS 23425 and axitirome (CGS 26214).

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with an HMG-CoA-reductase inhibitor from the class of statins, particularly selected from the group consisting of lovastatin, simvastatin, pravastatin, fluvastatin, atorvastatin, rosuvastatin and pitavastatin.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with a PPAR-gamma modulator, particularly selected from pioglitazone and rosiglitazone.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with a PPAR-delta modulator, particularly selected from the group consisting of ASP1128, GW 501516 and BAY 68-5042.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with a cholesterol absorption inhibitor, particularly selected from the group consisting of ezetimibe, tiqueside and pamaqueside.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with a lipase inhibitor, particularly selected from orlistat.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with a polymeric bile acid adsorber, particularly selected from the group consisting of cholestyramine, colestipol, colesolvam, CholestaGel and colestimide.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with a bile acid reabsorption inhibitor, particularly selected from the group consisting of ASBT (IBAT) inhibitors such as AZD-7806, S-8921, AK-105, BARI-1741, SC-435 and SC-635.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with a lipoprotein (a) antagonist, particularly selected from the group consisting of gemcabene calcium (CI-1027) and nicotinic acid.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with a TGFbeta antagonist, particularly selected from pirfenidone and fresolimumab.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with anti-microRNA-21 oligonucleotides, particularly selected from Lademisen.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with HIF-PH inhibitors, particularly selected from molidustat and roxadustat.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with a CCR2 antagonist, particularly selected from CCX-140.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with a TNFalpha antagonist, particularly selected from adalimumab.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with a galectin-3 inhibitor, particularly selected from GCS-100.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with a hepatocyte growth factor mimetic, particularly selected from Refanalina.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with a p53 modulator, particularly selected from QPI-1002.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with a NOX1/4 inhibitor, particularly selected from GKT-137831.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with a medicament which affects the vitamin D metabolism, particularly selected from cholecalciferol and paracalcitol.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with a cytostatic agent, particularly selected from cyclophosphamide.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with anti-VEGF therapy, particularly selected from the group consisting of ranibizumab, bevacizumab and aflibercept.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with an immunosuppressive agent, particularly selected from ciclosporin.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is

administered in combination with a phosphate binder, particularly selected from sevelamer and lanthanum carbonate.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with a calcimimetic for therapy of hyperparathyroidism.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with agents for iron deficit therapy, particularly selected from iron products.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with agents for the therapy of hyperurikaemia, particularly selected from allopurinol and rasburicase.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with glycoprotein hormone for the therapy of anaemia, particularly selected from erythropoietin.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with biologics for immune therapy, particularly selected from the group consisting of abatacept, rituximab, eculizumab and belimumab.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with Jak inhibitors, particularly selected from the group consisting of ruxolitinib, tofacitinib, baricitinib, CYT387, GSK2586184, lestauritinib, pacritinib (SB1518) and TG101348.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with prostacyclin analogs for therapy of microthrombi.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with an alkali therapy, particularly selected from sodium bicarbonate.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with an mTOR inhibitor, particularly selected from everolimus and rapamycin.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with an NHE3 inhibitor, particularly selected from AZD1722.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with an eNOS modulator, particularly selected from sapropterin.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with a CTGF inhibitor, particularly selected from FG-3019.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with one or more additional therapeutic agents selected from the group consisting of diuretics, angiotensin AII antagonists, ACE inhibitors, beta-receptor blockers, mineralocorticoid receptor antagonists, antidiabetics, organic nitrates and NO donors, activators and stimulators of the soluble guanylate cyclase (sGC), and positive-inotropic agents.

In particular embodiments, the isolated antibody or antigen-binding fragment according to the present disclosure is administered in combination with one or more additional

therapeutic agents selected from the group consisting of diuretics, angiotensin AII antagonists, ACE inhibitors, beta-receptor blockers, mineralocorticoid receptor antagonists, antidiabetics, organic nitrates and NO donors, activators and stimulators of the soluble guanylate cyclase (sGC), positive-inotropic agents, anti-inflammatory agents, immunosuppressive agents, phosphate binders and antibodies which modulate vitamin D metabolism.

Combination therapy includes administration of a single pharmaceutical dosage formulation which comprises the antibody or antigen-binding fragment according to the present disclosure or a variant thereof and one or more additional therapeutic agents, as well as administration of the antibody or antigen-binding fragment according to the present disclosure and each additional therapeutic agent in its own separate pharmaceutical dosage formulation. For example, an antibody of the disclosure or an antigen-binding fragment thereof or a variant thereof and a therapeutic agent may be administered to the patient together in a single liquid composition, or each agent may be administered in separate dosage formulation.

Where separate dosage formulations are used, the antibody or antigen-binding fragment according to the present disclosure or the variant thereof and one or more additional therapeutic agents may be administered at essentially the same time (e.g., concurrently) or at separately staggered times (e.g., sequentially).

The antibody or the antigen-binding fragment according to the present disclosure or a variant thereof might be used in combination with surgical interventions, including but not limited to:

major cardiovascular surgeries e.g. coronary artery bypass grafting (CABG), heart valve repair or replacement, insertion of a pacemaker or an implantable cardioverter defibrillator (ICD), maze surgery, aneurysm repair, aortic artery surgery/endarterectomy and thrombectomy;

major non-cardiac surgeries e.g., thoracic, orthopedic urologic surgeries.

Diagnostic Methods

Furthermore, the antibody or antigen-binding fragment according to the present disclosure may be utilized, as such or in compositions, in research and diagnostics, or as analytical reference standards, and the like.

Anti-Sema3A antibodies or antigen-binding fragments thereof can be used for detecting the presence of Sema3A. Thus, in a further aspect, the present disclosure relates to the isolated antibody or antigen-binding fragment according to the present disclosure or the antibody conjugate according to the present disclosure for use as a diagnostic agent.

Pharmaceutical Compositions and Administration

In a further aspect, the present disclosure relates to a pharmaceutical composition comprising the isolated antibody or antigen-binding fragment according to the present disclosure or the antibody conjugate according to the present disclosure. To treat any of the foregoing disorders, pharmaceutical compositions for use in accordance with the present disclosure may be formulated in any conventional manner using one or more physiologically acceptable carriers, excipients, or auxiliaries. Further details on techniques for formulation and administration may be found in the latest edition of Remington's Pharmaceutical Sciences (Ed. Maack Publishing Co, Easton, Pa.).

The antibody or antigen-binding fragment according to the present disclosure can be administered by any suitable means, which can vary, depending on the type of disorder being treated. Possible administration routes include oral,

parenteral, and topical administration. Methods of parenteral delivery include intra-arterial, intramuscular, subcutaneous, intramedullary, intrathecal, intraventricular, intravenous, intraperitoneal, or intranasal administration. In addition, the antibody or antigen-binding fragment according to the present disclosure may be administered by pulse infusion, with, e.g., declining doses of the antibody. Preferably, administration is by injections, most preferably intravenous or subcutaneous injections, depending in part on whether the administration is brief or prolonged. The amount to be administered will depend on a variety of factors such as the clinical symptoms, weight of the individual, whether other drugs are administered, and the like. The skilled artisan will recognize that the route of administration will vary depending on the disorder or condition to be treated.

The pharmaceutical composition according to the present disclosure comprises the antibody or antigen-binding fragment according to the present disclosure alone or in combination with at least one other agent, such as a stabilizing compound. The antibody or antigen-binding fragment according to the present disclosure may be administered in any sterile, biocompatible pharmaceutical carrier, including, but not limited to, saline, buffered saline, dextrose, and water. In particular embodiments, the pharmaceutical composition according to the present disclosure may comprise one or more further pharmaceutically active compounds, in particular one or more further pharmaceutically active compounds that are suitable to treat Sema3A associated disorders. Any of these agents can be administered to a patient alone, or in combination with other agents or drugs, in pharmaceutical compositions where it is mixed with excipient(s) or pharmaceutically acceptable carriers. In particular embodiments, the pharmaceutically acceptable carrier is pharmaceutically inert.

Pharmaceutical compositions for oral administration can be formulated using pharmaceutically acceptable carriers well known in the art in dosages suitable for oral administration. Such carriers enable the pharmaceutical compositions to be formulated as tablets, pills, dragees, capsules, liquids, gels, syrups, slurries, suspensions and the like, for ingestion by the patient.

Pharmaceutical preparations for oral use can be obtained through combination of active compounds with solid excipient, optionally grinding a resulting mixture, and processing the mixture of granules, after adding suitable auxiliaries, if desired, to obtain tablets or dragee cores. Suitable excipients are carbohydrate or protein fillers such as sugars, including lactose, sucrose, mannitol, or sorbitol; starch from corn, wheat, rice, potato, or other plants; cellulose such as methylcellulose, hydroxypropylmethylcellulose, or sodium carboxymethyl cellulose; and gums including arabic and tragacanth; and proteins such as gelatin and collagen. If desired, disintegrating or solubilizing agents may be added, such as the cross-linked polyvinyl pyrrolidone, agar, alginic acid, or a salt thereof, such as sodium alginate.

Dragee cores can be provided with suitable coatings such as concentrated sugar solutions, which may also contain gum arabic, talc, polyvinyl pyrrolidone, carbopol gel, polyethylene glycol and/or titanium dioxide, lacquer solutions, and suitable organic solvents or solvent mixtures. Dyestuffs or pigments may be added to the tablets or dragee coatings for product identification or to characterize the quantity of active compound, i.e. dosage.

Pharmaceutical preparations that can be used orally include push-fit capsules made of gelatin, as well as soft, sealed capsules made of gelatin and a coating such as glycerol or sorbitol. Push-fit capsules can contain active

ingredients mixed with a filler or binders such as lactose or starches, lubricants such as talc or magnesium stearate, and optionally, stabilizers. In soft capsules, the active compounds may be dissolved or suspended in suitable liquids, such as fatty oils, liquid paraffin, or liquid polyethylene glycol with or without stabilizers.

Pharmaceutical formulations for parenteral administration include aqueous solutions of active compounds. For injection, the pharmaceutical compositions of the disclosure may be formulated in aqueous solutions, preferably in physiologically compatible buffers such as Hank's solution, Ringer's solution, or physiologically buffered saline. Aqueous injection suspensions may contain substances that increase viscosity of the suspension, such as sodium carboxymethyl cellulose, sorbitol, or dextran. Additionally, suspensions of the active compounds may be prepared as appropriate oily injection suspensions. Suitable lipophilic solvents or vehicles include fatty oils such as sesame oil, or synthetic fatty acid esters, such as ethyl oleate or triglycerides, or liposomes. Optionally, the suspension may also contain suitable stabilizers or agents which increase the solubility of the compounds to allow for the preparation of highly concentrated solutions.

For topical or nasal administration, penetrants appropriate to the particular barrier to be permeated are used in the formulation. Such penetrants are generally known in the art.

The pharmaceutical compositions of the present disclosure may be manufactured in a manner that is known in the art, e.g., by means of conventional mixing, dissolving, granulating, dragee-making, levigating, emulsifying, encapsulating, entrapping or lyophilizing processes.

The pharmaceutical composition may be provided as a salt and can be formed with acids, including but not limited to hydrochloric, sulfuric, acetic, lactic, tartaric, malic, succinic, etc. Salts tend to be more soluble in aqueous or other protic solvents that are the corresponding free base forms. In other cases, the preferred preparation may be a lyophilized powder in 1 mM-50 mM histidine or phosphate or Tris, 0.1%-2% sucrose and/or 2%-7% mannitol at a pH range of 4.5 to 7.5 optionally comprising additional substances like polysorbate that is combined with buffer prior to use.

After pharmaceutical compositions comprising a compound of the disclosure formulated in an acceptable carrier have been prepared, they can be placed in an appropriate container and labeled for treatment of an indicated condition. For administration of anti-Sema3A antibodies or antigen-binding fragment thereof, such labeling would include amount, frequency and method of administration.

Therapeutically Effective Dose

The determination of an effective dose is well within the capability of those skilled in the art. Determining a therapeutically effective amount of the novel antibody of this disclosure or an antigen-binding fragment thereof or a variant thereof, largely will depend on particular patient characteristics, route of administration, and the nature of the disorder being treated. General guidance can be found, for example, in the publications of the International Conference on Harmonization and in REMINGTON'S PHARMACEUTICAL SCIENCES, chapters 27 and 28, pp. 484-528 (18th ed., Alfonso R. Gennaro, Ed., Easton, Pa.: Mack Pub. Co., 1990). More specifically, determining a therapeutically effective amount will depend on such factors as toxicity and efficacy of the medicament. Toxicity may be determined using methods well known in the art and found in the foregoing references. Efficacy may be determined utilizing

the same guidance in conjunction with the methods described below in the Examples.

For any compound, the therapeutically effective dose can be estimated initially either in cell culture assays, or in animal models, usually mice, rabbits, dogs, pigs or monkeys. The animal model is also used to achieve a desirable concentration range and route of administration. Such information can then be used to determine useful doses and routes for administration in humans.

A therapeutically effective dose refers to that amount of antibody or antigen-binding fragment thereof, that ameliorates the symptoms or condition. Therapeutic efficacy and toxicity of such compounds can be determined by standard pharmaceutical procedures in cell cultures or experimental animals, e.g., ED_{50} (the dose therapeutically effective in 50% of the population) and LD_{50} (the dose lethal to 50% of the population). The dose ratio between therapeutic and toxic effects is the therapeutic index, and it can be expressed as the ratio, ED_{50}/LD_{50} . Pharmaceutical compositions that exhibit large therapeutic indices are preferred. The data obtained from cell culture assays and animal studies are used in formulating a range of dosage for human use. The dosage of such compounds lies preferably within a range of circulating concentrations that include the ED_{50} with little or no toxicity. The dosage varies within this range depending upon the dosage form employed, sensitivity of the patient, and the route of administration.

The exact dosage is chosen by the individual physician in view of the patient to be treated. Dosage and administration are adjusted to provide sufficient levels of the active moiety or to maintain the desired effect. Additional factors that may be taken into account include the severity of the disease state, age, weight and gender of the patient; diet, time and frequency of administration, drug combination(s), reaction sensitivities, and tolerance/response to therapy. Long acting pharmaceutical compositions might be administered for example every 3 to 4 days, every week, once every two weeks, or once every three weeks, depending on half-life and clearance rate of the particular formulation.

Normal dosage amounts may vary from 0.1 to 100,000 micrograms, up to a total dose of about 10 g, depending upon the route of administration. Guidance as to particular dosages and methods of delivery is provided in the literature. See U.S. Pat. Nos. 4,657,760; 5,206,344; or 5,225,212.

Kits

In a further aspect, the present disclosure relates to a kit comprising the isolated antibody or antigen-binding fragment according to the present disclosure or the conjugate according to the present disclosure and instructions for use. In particular embodiments, the kit comprises one or more containers filled with one or more of the ingredients of the aforementioned compositions of the disclosure. Associated with such container(s) can be a notice in the form prescribed by a governmental agency regulating the manufacture, use or sale of pharmaceuticals or biological products, reflecting approval by the agency of the manufacture, use or sale of the product for human administration.

DRAWINGS

FIG. 1A: Effects of Sema3A inhibition with TPP-15370 (grey bar), TPP-11489 (striped bar) and TPP-17755 (squared bar) on Sema3A-induced albumin excretion in mice. Shown are mean±S.D. (n=10). ***, ****: p<0.001 p<0.0001 vs. isotype control. Dunnett's post hoc test.

FIG. 1B: Effects of Sema3A inhibition with TPP-23298 (grey bars), TPP-11489 (dotted bar) and TPP-17755 (striped

bar) on Sema3A-induced albumin excretion in mice. Shown are mean±S.D. (n=10). ***, ****: p<0.001 p<0.0001 vs. isotype control. Dunnett's post hoc test.

FIG. 2A: Sema3A induced albuminuria in mice after treatment with TPP-15370 (white circles) and TPP-23298 (black circles) in comparison to TPP-11489 (black triangles). The comparisons were performed in two separate experiments. Shown are mean values. (n=10). n.s.=statistically not significant vs. TPP-15370; *p<0.05 vs. TPP-23298. Unpaired T-test.

FIG. 2B: Sema3A induced albuminuria in mice after treatment with TPP-15370 (white circles) and TPP-23298 (black circles) in comparison to TPP-17755 (black squares). The comparisons were performed in two separate experiments. Shown are mean values. (n=10). n.s.=statistically not significant vs. TPP-15370; *p<0.05 vs. TPP-23298. Unpaired T-test.

FIG. 2C: Sema3A induced albuminuria in mice after treatment with TPP-15370 (white circles) and TPP-23298 (black circles) in comparison to TPP-30788 (black rhombus). The comparisons were performed in two separate experiments. Shown are mean values. (n=10). n.s.=statistically not significant vs. TPP-15370; *p<0.05 vs. TPP-23298. Unpaired T-test.

FIGS. 3A-3C: Effects of Sema3A inhibition with TPP-23374 (dotted bars) TPP-23298 (grey bars) and TPP-15370 (striped bars) on FIG. 3A: serum creatinine levels, FIG. 3B: serum urea levels and FIG. 3C: urinary albumin excretion after I/R injury in mice. Shown are mean±S.D. (n=8-10). *, **, ****: p<0.05, p<0.01, p<0.0001 vs. isotype control. Dunnett's post hoc test.

FIGS. 4A-4C: Effects of Sema3A inhibition with TPP-15370 (grey bars), TPP-11489 (striped bars) and TPP-17755 (squared bars) on FIG. 4A: serum creatinine levels, FIG. 4B: serum urea levels and FIG. 4C: urinary albumin excretion after I/R injury in mice. Shown are mean±S.D. (n=8-10). *, **, ****: p<0.05, p<0.01, p<0.0001 vs. isotype control. Dunnett's post hoc test.

FIGS. 5A-5C: Effects of Sema3A inhibition with TPP-23298 (grey bars), TPP-11489 (striped bars) and TPP-17755 (squared bars) on FIG. 5A: serum creatinine levels, FIG. 5B: serum urea levels and FIG. 5C: urinary albumin excretion after I/R injury in mice. Shown are mean±S.D. (n=10-12). *, **, ****: p<0.05, p<0.01, p<0.0001 vs. isotype control. Dunnett's post hoc test.

FIGS. 6A-6C: Effects of Sema3A inhibition with TPP-15374 (grey bars), TPP-11489 (striped bars) on FIG. 6A: serum creatinine levels, FIG. 6B: serum urea levels and FIG. 6C: urinary albumin excretion after I/R injury in mice. Shown are mean±S.D. (n=10-12). *, **, ****: p<0.05, p<0.01, p<0.0001 vs. isotype control. Dunnett's post hoc test.

FIG. 7: Effects of Sema3A inhibition with TPP-15370 (grey bars), TPP-11489 (striped bars) on proteinuria in Alport mice. Shown are mean±S.D. (n=8-10). ****: p<0.0001 vs. isotype control. Dunnett's post hoc test.

FIGS. 8A-8D: Effects of Sema3A inhibition with TPP-15370 (grey bars), TPP-11489 (striped bars) on FIG. 8A: serum creatinine levels, FIG. 8B: serum urea levels and fibrosis FIG. 8C: myofibroblast and FIG. 8D: collagen deposition in Alport mice. Shown are mean±S.D. ***, ****: p<0.05, p<0.001, p<0.0001 vs. isotype control. Dunnett's post hoc test. *, ***, ****.

FIGS. 9A-9D: Effects of Sema3A inhibition with TPP-23298 in a single dose preventive setting in a unilateral kidney IRI model in pigs, 105 min of ischemia. TPP-23298 (FIG. 9A; black dots) or control IgG (open circles) (10

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mg/kg) were given 30 min before inflating the balloon in the left renal artery. Values from SHAM animals are indicated diamonds. Time course of plasma creatinine concentrations of individual animals (FIG. 9A), and time course of mean change of creatinine plasma concentrations versus base line values at start of experimentation (0 h) (FIG. 9B). Mean values of creatinine clearance for 24-27 h interval. Creatinine clearance side separated for left (damaged) and right (non-damaged) kidneys and kidneys from sham animals (FIG. 9C). Global creatinine clearance (FIG. 9D); means $\pm$ SEM, p-value in (FIG. 9B) from t-test, */*** in (FIG. 9C) and FIG. 9D): p<0.05/0.001, one-way ANOVA versus corresponding control followed by Dunnett's multiple comparison

FIGS. 10A-10C: Schematic representation of a sandwich-based epitope binning experiment using SPR (see also Example 5A): FIG. 10A) One antibody is immobilized to a SPR chip, Sema3A is injected, and the binding is monitored; FIG. 10B) A second (competitive) antibody is injected on to the complex of the immobilized mAb bound to Sema3A, and the binding is monitored; FIG. 10C) A second (non-competitive) antibody is injected on to the complex of the immobilized mAb bound to Sema3A, and the binding is monitored.

FIGS. 11A-11C: HRA image analysis steps: FIG. 11A) Fluorescence microscopy image of DAPI/CM cells; FIG. 11B) Identification of cells in the selected area; FIG. 11C) Calculation of cells-free region size (grey area).

FIG. 12: The percent inhibition of Sema3A in a HUVEC repulsion assay at an antibody concentration of 80 pM is shown (see Example 11). Each column represents one antibody in the following left to right order: TPP-23298 (black column), TPP-30788, TPP-TPP-30789, TPP-30790, and TPP-30791.

EXAMPLES

Example 1: Sema3A Sequences and Tool Generation

TABLE 2

Tools used in this disclosure				
TPP-No.	Protein	Boundaries [aa]	Uniprot ID	Catalog No.
TPP-13211	Human Semaphorin3A-Fc (R&D Systems)	26-771	Q14563	1250-S3
No TPP-No.	Human Semaphorin3G (Abnova)	1-782	Q9NS98	H00056920-P01
No TPP-No.	Human Semaphorin3F-Fc (R&D Systems)	19-772	Q13275	9878-S3
TPP-13357	Mouse Semaphorin3A-Fc (R&D Systems)	21-747	O08665	5926-S3
TPP-19068	Human Semaphorin3A - Sema Domain	21-569	Q14563	Produced inhouse
TPP-19069	Mouse Semaphorin3A - Sema Domain	21-569	O08665	Produced inhouse
TPP-19122	Cyno Semaphorin3A - Sema Domain	21-569	Q63548	Produced inhouse
TPP-19120	Rat Semaphorin3A - Sema Domain	21-569	E2QX94	Produced inhouse
TPP-19121	Dog Semaphorin3A - Sema Domain	21-569	A0A2K5VGJ0	Produced inhouse
TPP-20176	Pig Semaphorin3A - Sema Domain	49-658	A0A480WHT2	Produced inhouse

Sema3A domains were produced by mammalian cell culture using transiently transfected HEK293-6E cells (National Research Council Canada). All constructs were under

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the control of a CMV promoter and sequences contain a C-terminal FXa cleavage site followed by a 6 $\times$ his-tag. Cell culture was performed using F17 medium (Life Technologies) supplemented with 0.1% pluronic F68 (Life Technologies) and 4 mM Glutamax (Life Technologies). 24 h post-transfection, 1% FCS ultra-low IgG (Life Technologies) and 0.5 mM valproic acid (Sigma Aldrich) were added. Cell supernatant was sterile filtered and subsequently purified or concentrated via crossflow filtration prior to purification.

Sema3A domains were purified using a two-step purification consisting of affinity and size exclusion chromatography. In brief, cell culture supernatant was loaded on to a Ni²⁺-NTA column (GE Healthcare) connected to an Äkta Avant system (GE Healthcare). Column was equilibrated with 4 CV of 50 mM NaH₂PO₄, 300 mM NaCl, pH 8 and washed afterwards with 10 CV of running buffer until baseline was reached. Elution was carried out using 6 CV of running buffer containing 250 mM imidazole, pH 8.0. Fractions of the elution peak were unified, concentrated using a Vivaflow 200 Hydrosart membrane (cut-off 10 kDa, Sartorius) and subjected to size exclusion chromatography using a Superdex 200 column (GE Healthcare) connected to an Äkta Pure 25 system. The column was equilibrated and run in DPBS, pH 7.4. Fractions of the domain elution peak were unified and concentrated using a Vivaflow 200 Hydrosart membrane (cut-off 10 kDa, Sartorius). The final protein quality was assessed on an analytical size exclusion chromatography (Superdex 200) for purity and monodispersity as well as SDS-PAGE. Sema domains were aliquoted and snap frozen in liquid nitrogen and stored at -80° C. until further use.

Example 2: Antibody Generation from BioInvent Antibody Libraries

A fully human antibody phage display library (BioInvent n-CoDeR Fab lambda library) was used to isolate human monoclonal antibodies of the present disclosure by selection against recombinant human Sema3A (TPP-13211, R&D Systems) using the following protocol. Briefly, Immuno-

tubes (Nunc) were coated for one hour at room temperature (RT) with the 100 μ g of the target molecule (huSema3A) or an irrelevant Fc-containing off-target in 1 ml PBS (Phos-

phate Buffered Saline) with end-over-end rotation. The target and depletion antigen-coated immunotube as well as an empty immunotube were washed 4 times with PBS+ 0.05% Tween20 (PBST) and subsequently blocked using 3 ml of a 3% Milk powder in PBST solution for 1 h at RT with end-over-end rotation. An aliquot of the phage library was thawed and allowed to block in a solution of 3% milk powder in PBST for 1 h at RT with end-over-end rotation. The non-coated depletion immunotube was washed 3 times in 4 ml PBS before addition of the blocked phage library and incubation with end-over-end rotation for 30 min at RT. This step was repeated for the non-target antigen-coated depletion immunotube. The huSema3A-coated immunotube was washed 3 times in 4 ml PBS before addition of the depleted library and incubation for 90 min at room temperature with end-over-end rotation. After stringent washing (4 × with 4 ml PBST and 1 × with 4 ml PBS) Fab-expressing phages binding specifically to the coated target were eluted using 500 µl 100 nM TEA, 10 min incubation at room temperature followed by neutralization by addition of 500 µl Tris-HCl pH 7.5. 500 µl of eluted phage were used to infect *Escherichia coli* strain HB101. Subsequently the phages were amplified in *Escherichia coli* strain HB101 using M13KO7 Helper Phage (Invitrogen™). In two subsequent selection rounds the target concentration was decreased to 25 µg/ml. For a first qualitative assessment, 88 randomly picked Fab-expressing phage clones from each selection round were expressed in single wells and tested for binding to huSema3A compared to an irrelevant off-target. The clone pool from Round 3 in this example was found to contain a 60% positive hit rate and was chosen for further screening.

In a next step, the expression of soluble Fabs was enabled by bulk removal of the gene III fusion in this pool and 2208 single clones were picked for expression in *Escherichia coli* strain Top10 and evaluation of Fab-containing supernatants in a huSema3A binding ELISA. The VH and VL sequences for all 2208 clones was also determined using NGS methods. 154 distinct clones positive for binding to huSema3A were identified. These positive binding Fab fragments were tested in a confirmatory binding ELISA and were also evaluated for binding to mouse Sema3A-Fc (TPP-13357, R&D Systems) as well as specificity testing using an additional off target molecule, murine Sema3F (R&D Systems). Based on this analysis, 48 human/mouse cross-reactive Sema3A binding Fabs were prioritized. These Fab fragments were subsequently purified from 25 ml expression cultures using Capture Select CH1 matrix (LifeTechnologies), eluted using 12.5 mM Citric acid at pH 2.5 and finally buffer exchanged to PBS using a Zeba™ Spin desalting plate (ThermoFisher). A kinetic ranking was performed for all 48 purified Fab fragments by surface plasmon resonance (SPR), examining the binding to both human and mouse Sema3A and reformatted in to a full-length human IgG1 and again tested for binding in SPR (see Example 4).

Example 3: Sequence Optimization, Germlining & Affinity Maturation of Lead Antibodies TPP-15370 and TPP-15374

IgG1 antibodies TPP-15370 and TPP-15374 were subjected to lead optimization procedures aiming to (i) optimize its affinity, (ii) increase functional efficiency, (iii) reduce the risk of sequence-based immunogenicity and (iv) improve compatibility with downstream development processes.

Affinity maturation was done by a first single mutation gathering round followed by recombination of the most

affinity- and potency-increasing amino acid exchanges in a germlined and sequence optimized antibody backbone.

For mutation gathering NNK (N=A or G or C or T, K=G or T) randomizations at the following individual amino acid positions were generated by site directed mutagenesis using synthetic oligonucleotides including NNK for codon-diversification. For TPP-15370 the following regions were analyzed for their effect on affinity: GFTFSSYGMH (residues 26 to 35 of VH SEQ ID NO: 41), WVSAIGTGGDTYY-ADSVMG (residues 47 to 65 of VH SEQ ID NO: 41), ARDDYTSRDAFDV (residues 96 to 109 of VH SEQ ID NO:41), SGSSSNIGSNTVNWY (residues 23 to 37 of VL SEQ ID NO: 45), LLIYYDDLPS (residues 47 to 57 of VL SEQ ID NO: 45), and AAWDDSLNGYVV (residues 90 to 101 of VL SEQ ID NO: 45).

For TPP-15374 the following regions were analyzed for their effect on affinity: GFTFSSYEMN (residues 26 to 35 of VH SEQ ID NO: 61), WVSGISWNSGSIGYADSVKG (residues 47 to 66 of VH SEQ ID NO: 61), ARSGYSSS-WFDPDFDY (residues 97 to 112 of VH SEQ ID NO: 61), TGSSSNIGAGYDVHWY (residues 23 to 38 of VL SEQ ID NO: 65), LLIYGNSNRPS (residues 48 to 58 of VL SEQ ID NO: 65), and SSYAGSNPYV (residues 91 to 101 of VL SEQ ID NO: 65).

The resulting single NNK libraries were sequenced and about 1000 single amino acid exchange variants of TPP15370 and TPP-15374, respectively, were identified. They were expressed by transient transfection of mammalian cells and resulting expression supernatants were normalized in terms of antibody concentrations to be screened in surface plasmon resonance and competition ELISA.

For the germlining and sequence optimization process of TPP-15370 and TPP-15374 the closest germline families for light and heavy chain were selected and scrutinized for potential CMC relevant residues. Deviations from closest human germlines in CDR regions and FW regions and potential CMC relevant residues in CDR regions were adjusted by site directed mutagenesis and tested for in functional and biophysical assays (unspecific binding, temperature stability in DSC). The resulting single reversions and following combinatorial IgG variants were expressed by transient transfection of mammalian cells and resulting expression supernatants were normalized in terms of antibody concentrations to be screened in binding assays (SPR, competition ELISA) and functional assays. This led to germlined and sequence optimized molecules TPP-21565 for TPP-15370 and TPP-18533 for TPP-15374. TPP-21565 carries in comparison to TPP-15370 reversions L55R and R80Q in the light chain and G33A, H35S, M64K and V109Y in the heavy chain. TPP-18533 carries in comparison to TPP-15374 reversions A10V, T13A, S78T, R81Q, S82A in the light chain.

For the final recombination library of TPP-21565 eight single substitution variants that were shown in the NNK library screening step to exhibit improved affinity and functional efficiency were selected. Light chain mutations A90H, G98D, G98V, Y99I and V100P and heavy chain mutations S30Y, S35L and T53Y were recombined in one recombination library (continuous amino acid nomenclature, reference is TPP-21565 as defined by SEQ ID NOs: 121-VH and 125-VL).

For the final recombination library of TPP-18533 eleven single substitution variants that were shown in the NNK library screening step to exhibit improved affinity and functional efficiency were selected. Light chain mutations N28D, N53A, S91K, S91Q, A94E, S96I and S96P and heavy chain mutations T28D, S30D, S57W and G59Y were recombined

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in one recombination library (continuous amino acid nomenclature, reference is TPP-18533 as defined by SEQ ID NOs: 101-VH and 105-VL).

For TPP-18533 oligonucleotides were generated to introduce selected mutations or the corresponding wild type amino acid at each selected position. Library construction was performed using sequential rounds of overlap extension PCR. The final PCR product was ligated into a mammalian IgG4 (S228P) expression vector and variants were sequenced using massive-parallel sequencing techniques. For TPP-21565 the recombinatorial variants were designed as distinct clones and cloned into an IgG4 (S228P) containing expression plasmid.

More than 1000 unique combinatorial amino acid exchange variants of TPP-18533 and more than 100 unique combinatorial variants of TPP-21565 were generated in that way, expressed by transient transfection of mammalian cells, and resulting expression supernatants were normalized in terms of antibody concentrations to be screened in varying number in SPR, competition ELISA and functional assays. Based on the result in these assays, mutants were either categorized as 'improved' or 'non-improved'.

Table 1 and 1A lists i.a. preferred antibodies candidates according to the present disclosure that were selected in the combination library screening step as being most potent in terms of binding to Sema3A and in terms of antagonizing the Sema3A-dependent biological activity as well as the respective amino acid and nucleic acid sequences of antibodies according to the present disclosure.

Example 4: Determination of Affinity and Species Cross-Reactivity Using Surface Plasmon Resonance

To assess the binding kinetics and affinity of anti-Sema3A antibodies as well as their species cross-reactivity profile, binding assays were conducted using surface plasmon resonance (SPR). Binding assays were performed on a Biacore T200 instrument or on a Biacore 8K+ instrument (Cytiva) at 25° C. using assay buffer HBS P+, 300 mM NaCl, 0.75 mM CaCl₂, 2.5 mM MgCl₂, 1 mg/ml BSA, 0.05% NaN₃. Antibodies were captured either via anti-human Fc IgGs ("Human antibody capture kit", Order No. BR100839, Cytiva) or in case of Fc-tagged analytes by anti-human Fab IgGs ("Human Fab capture kit", Order No. 28958325, Cytiva) covalently amine coupled to a Series S CM5 sensor chip (Cytiva). The amine coupling was carried out according to the manufacturer's instructions using 1-ethyl-3-(3-dimethylaminopropyl) carbodiimide hydrochloride (EDC), N-hydroxysuccinimide (NHS) and ethanolamine HCl, pH 8.5 ("Amine Coupling Kit" BR-1000-50, Cytiva.). For phage display hits Fc-tagged human and mouse Sema3A was used as analytes in a concentration range from 1.56-200 nM. Human, mouse, cynomolgus, rat, dog and pig monovalent Sema3A domain were used as analytes in a concentration series from 0.024-3.125 nM in multi cycle kinetics mode or in 100 nM for binding analysis only. The sensor surface was regenerated with glycine pH 2.0 after each antigen injection. Obtained sensorgrams were double referenced (subtraction of reference flow cell signal and buffer injection) and were fitted to a 1:1 Langmuir binding model to derive kinetic data using the Biacore T200 Evaluation software. Results are shown in Tables 3,4 and 4a.

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TABLE 3

Affinity of anti-Sema3A IgG1 antibodies derived from phage display hits determined by SPR using TPP-13211 and TPP-13357.

	Nomenclature	Mouse K _D [M]	Human K _D [M]
5			
10	TPP-15355	4.0E-09	3.5E-09
	TPP-15356	n.b.	3.2E-09
	TPP-15357	1.0E-07	5.0E-08
	TPP-15358	3.1E-09	9.5E-10
15	TPP-15359	n.b.	1.1E-08
	TPP-15360	1.1E-07	7.2E-09
	TPP-15361	n.b.	5.5E-09
	TPP-15362	n.b.	n.b.
	TPP-15363	n.b.	2.4E-09
20	TPP-15364	n.b.	2.6E-09
	TPP-15365	2.4E-07	6.5E-08
	TPP-15366	1.4E-08	1.3E-08
	TPP-15367	5.4E-09	2.2E-09
	TPP-15368	8.2E-07	1.5E-07
25	TPP-15369	4.1E-08	3.5E-08
	TPP-15370	3.2E-09	2.8E-09
	TPP-15371	7.4E-09	4.5E-09
	TPP-15372	n.b.	3.7E-09
30	TPP-15373	2.0E-07	1.3E-07
	TPP-15374	1.8E-08	1.8E-08
	TPP-15375	5.8E-09	5.2E-09
	TPP-15376	8.4E-09	5.8E-09
	TPP-15377	3.3E-09	1.9E-09
35	TPP-15378	n.d.	1.2E-08
	TPP-15379	4.3E-07	2.1E-07
	TPP-15380	n.b.	n.b.
	TPP-15381	9.9E-09	3.3E-09
40	TPP-15382	2.5E-07	1.9E-07
	TPP-15383	5.3E-08	2.8E-08
	TPP-15384	9.6E-09	9.1E-09
	TPP-15385	8.5E-09	7.2E-09
	TPP-15386	n.b.	n.b.
45	TPP-15387	1.6E-07	1.1E-07
	TPP-15388	1.7E-07	1.3E-08
	TPP-15389	4.2E-09	2.8E-09
	TPP-15390	9.8E-08	5.7E-08
50	TPP-15391	n.b.	7.0E-09
	TPP-15392	n.d.	n.d.
	TPP-15393	5.9E-08	9.3E-09
	TPP-15394	n.d.	n.d.
	TPP-15395	1.1E-06	2.2E-07
55	TPP-15396	6.2E-09	2.1E-09
	TPP-15397	2.7E-07	9.7E-09
	TPP-15398	8.5E-09	8.4E-09
	TPP-15399	1.9E-07	1.5E-07
60	TPP-15400	4.9E-09	4.6E-09
	TPP-15401	7.6E-07	1.2E-08

n.b. = no binding.

n.d. = not determinable

The majority of phage display hits bind to human and mouse dimeric Sema3A in the lower nanomolar range.

TABLE 4

Affinity of anti-Sema3A antibodies derived from TPP-15370 and TPP-15374 determined by SPR using TPP-19068, TPP-19069, TPP-19122, TPP-19120, TPP-19121, TPP-20176 as analytes as well as prior art antibodies (TPP-30972 was purified from HEK cell expression).						
Nomenclature	Mouse K_D [M]	Pig K_D [M]	Cyno K_D [M]	Dog K_D [M]	Human K_D [M]	Rat K_D [M]
TPP-11489 (Chiome)	1.6E-07	1.0E-07	6.3E-08	5.0E-08	7.3E-08	3.3E-08
TPP-17755 (Samsung)	3.9E-09	4.0E-09	1.4E-08	7.5E-09	6.9E-09	5.4E-09
TPP-30791 (BI clone IV)	2.8E-11	2.9E-11	5.7E-11	7.8E-11	1.2E-11	1.7E-11
TPP-30790 (BI clone III)	4.0E-11	3.6E-11	7.7E-11	1.0E-10	1.5E-11	2.2E-11
TPP-30789 (BI clone II)	4.2E-11	3.9E-11	7.9E-11	1.1E-10	2.2E-11	2.6E-11
TPP-30788 (BI clone I)	4.3E-11	3.8E-11	7.8E-11	1.1E-10	1.8E-11	2.6E-11
TPP-30792 (3H4 Univ Ramot)	no binding	no binding	no binding	no binding	no binding	no binding
TPP-15370	7.2E-09	9.0E-09	4.0E-08	2.2E-08	1.0E-08	1.4E-08
TPP-23298	7.4E-11	6.7E-11	7.8E-11	7.0E-11	8.7E-11	3.0E-11
TPP-23334	6.2E-11	1.4E-11	1.5E-11	8.4E-12	2.1E-11	5.6E-11
TPP-23337	5.0E-11	1.1E-11	2.6E-11	4.5E-12	5.0E-11	1.1E-10
TPP-23338	4.5E-11	4.6E-11	4.2E-11	5.3E-11	5.4E-11	
TPP-23340	5.9E-11	6.2E-11	6.0E-11	5.8E-11	2.2E-11	
TPP-23341	9.2E-11	8.6E-11	8.7E-11	8.4E-11	9.1E-11	
TPP-23345	6.3E-11	5.5E-11	6.2E-11	4.6E-11	6.5E-11	
TPP-23346	6.4E-11	5.8E-11	6.1E-11	6.1E-11	7.2E-11	
TPP-23347	5.5E-11	5.3E-11	5.4E-11	5.1E-11	6.0E-11	
TPP-23373	8.3E-11	7.8E-11	7.2E-11	1.0E-10	1.1E-10	
TPP-23374	1.6E-11	below 3 pM	below 3 pM	7.3E-12	8.1E-12	3.3E-12
TPP-23375	4.2E-11	4.7E-11	4.5E-11	4.5E-11	5.3E-11	
TPP-15374	8.3E-09	7.2E-09	4.6E-08	1.9E-08	1.5E-08	9.8E-09
TPP-18533	8.1E-09		6.4E-09		8.7E-09	
TPP-25497					5.2E-11	
TPP-25256					4.9E-11	
TPP-25255					5.1E-11	
TPP-25257					5.3E-11	
TPP-25248					5.0E-11	
TPP-25064					4.9E-11	
TPP-26111					5.2E-11	
TPP-25224					4.9E-11	
TPP-25448					5.3E-11	
TPP-25655					4.9E-11	

All derivative antibodies of TPP-15370 and TPP-15374 have a significantly increased affinity to the Sema3A domain in the lower picomolar range compared to their parental antibodies as well as to most prior art antibodies.

TABLE 4a

Affinity of anti-Sema3A IgG1 antibodies determined by SPR using 100 nM TPP-19068 (human) in a binding experiment.	
Nomenclature	Human K_D [M]
TPP-23298	1.3E-10
TPP-17755 (Samsung)	6.2E-09
TPP-11489 (Chiome)	n.d.
TPP-30788 (BI clone I)	9.8E-11
TPP-31357 (Fab of 3H4 Univ Ramot)	3.5E-10

n.d. = not determinable due to multiphasic behaviour

In contrast to the full length 3H4 IgG1 (TPP-30792) which showed no binding in SPR to Sema3A molecules, the Fab variant of TPP-30792, TPP-31357 shows binding to human Sema3A, but with less affinity as TPP-23298.

Example 5: Determination of Binding Activity Using Surface Plasmon Resonance

To assess the binding activity of anti-Sema3A antibodies binding assays were conducted using surface plasmon resonance (SPR). Binding assays were performed on a Biacore T200 instrument (Cytiva) at 25° C. using assay buffer HBS P+, 300 mM NaCl, 0.75 mM CaCl₂, 2.5 mM MgCl₂, 1 mg/ml BSA, 0.05% NaN₃. Antibodies were captured via anti-human Fc IgGs ("Human antibody capture kit", Order No. BR100839, Cytiva) covalently amine coupled to a Series S CM5 sensor chip (Cytiva). The amine coupling was carried out according to the manufacturer's instructions using 1-ethyl-3-(3-dimethylaminopropyl) carbodiimide hydrochloride (EDC), N-hydroxysuccinimide (NHS) and ethanolamine HCl, pH 8.5 ("Amine Coupling Kit" BR-1000-50, Cytiva.). Human, mouse, cynomolgus, rat, dog and pig monovalent Sema3A domain were used as analytes in a concentration series from 0.024-3.125 nM in multi cycle kinetics mode. The sensor surface was regenerated with glycine pH 2.0 after each antigen injection. Obtained sensorgrams were double referenced (subtraction of reference flow cell signal and buffer injection) and were fitted to a 1:1 Langmuir binding model using the Biacore T200 Evaluation software obtaining the experimental fitted R_{Max} value. To

calculate the binding activity first the theoretical R_{Max} needs to be calculated according to equation 1:

$$R_{Max} = \frac{R_{Ligand} * Mr_{Analyte} * Valency_{Ligand}}{Mr_{Ligand}}$$

Equation 1: Theoretical calculation of R_{Max} . R_{Ligand} =Ligand Level in RU, Mr =molecular weight, $Valency_{Ligand}$ =number of binding sites per antibody molecule, here 2

Binding activity was determined by dividing the experimental determined R_{Max} by the theoretical calculated R_{Max} according to equation 2:

$$\text{Activity in \%} = \frac{R_{Max \text{ experimental}}}{R_{Max \text{ theoretical}}} * 100$$

Equation 2: Calculation of binding activity in %

TABLE 5

Summary of ligand levels after capture, experimental, theoretical and binding activity of tested antibodies					
Ligand	Analyte	Ligand Level [RU]	Experimental Rmax [RU]	Theoretical Rmax [RU]	Binding Activity [%]
TPP-11489 (Chiome)	Rat	798	212	681	31
TPP-15370	Sema3A	53	51	45	113
TPP-15374	domain	53	42	45	94
TPP-17755 (Samsung)		54	26	46	56
TPP-23298		46	42	39	108
TPP-30791 (BI clone IV)		46	42	39	109
TPP-30790 (BI clone III)		62	50	53	94
TPP-30789 (BI clone II)		50	44	42	104
TPP-30788 (BI clone I)		46	43	40	109
TPP-11489 (Chiome)	Dog	797	148	680	22
TPP-15370	Sema3A	53	51	45	114
TPP-15374	domain	53	44	45	98
TPP-17755 (Samsung)		54	25	46	55
TPP-23298		45	42	39	107
TPP-30791 (BI clone IV)		47	43	40	106
TPP-30790 (BI clone III)		61	48	52	92
TPP-30789 (BI clone II)		50	44	43	102
TPP-30788 (BI clone I)		47	42	40	106
TPP-11489 (Chiome)	Pig	801	525	684	77
TPP-15370	Sema3A	53	50	45	111
TPP-15374	domain	53	47	45	103
TPP-17755 (Samsung)		54	28	46	60
TPP-23298		46	42	39	107
TPP-30791 (BI clone IV)		49	44	42	105
TPP-30790 (BI clone III)		61	48	52	92
TPP-30789 (BI clone II)		51	45	43	103
TPP-30788 (BI clone I)		47	43	40	107
TPP-11489 (Chiome)	Cyno	800	85	682	13
TPP-15370	Sema3A	53	63	45	139
TPP-15374	domain	53	47	45	104
TPP-17755 (Samsung)		53	24	45	53
TPP-23298		46	41	39	106
TPP-30791 (BI clone IV)		47	43	40	107
TPP-30790 (BI clone III)		62	48	52	92
TPP-30789 (BI clone II)		50	44	42	103
TPP-30788 (BI clone I)		47	43	40	107
TPP-11489 (Chiome)	Human	798	257	681	38
TPP-15370	Sema3A	53	51	45	112
TPP-15374	domain	53	45	45	101
TPP-17755 (Samsung)		54	25	46	55
TPP-23298		46	42	39	107
TPP-30791 (BI clone IV)		48	44	41	107
TPP-30790 (BI clone III)		61	48	52	93
TPP-30789 (BI clone II)		49	45	42	106

TABLE 5-continued

Summary of ligand levels after capture, experimental, theoretical and binding activity of tested antibodies					
Ligand	Analyte	Ligand Level [RU]	Experimental Rmax [RU]	Theoretical Rmax [RU]	Binding Activity [%]
10 TPP-30788 (BI clone I)		47	43	40	107
TPP-11489 (Chiome)	Mouse	796	803	680	118
TPP-15370	Sema3A	53	50	45	111
TPP-15374	domain	53	48	45	106
TPP-17755 (Samsung)		54	26	46	57
TPP-23298		46	42	39	108
15 TPP-30791 (BI clone IV)		47	43	40	107
TPP-30790 (BI clone III)		62	49	52	93
TPP-30789 (BI clone II)		51	45	43	103
TPP-30788 (BI clone I)		47	43	40	108

The binding activity calculated in the SPR experiment is a measure of the activity of the surface-attached ligand. As can be seen from Table 5, TPP-15370, TPP-15374, TPP-23298 and TPPs 30788-30791 are able to bind to all tested Sema3A domains with around 100% activity meaning all binding regions are fully able to bind. Prior art antibody TPP-17755 only reaches an activity level of 50-60% depending on the species. Prior art antibody TPP-11489 shows an even more reduced level of below 50%, except for mouse and pig where it is higher. Strikingly, to reach such an activity level, the ligand level of TPP-11489 needs to be over 10-fold higher as compared to the other antibodies pointing in general to a much lower binding activity as compared to TPP-15370, TPP-15374 and TPP-23298.

Example 6: Competition ELISA

For screening in a competition ELISA setup, human Sema3a (TPP-13211) was coated onto 384-well plates (Greiner bio-one, 781077) with a concentration of 0.5 µg/ml in coating buffer (Carbonate-Basis pH 9.6, Candor 121125) over night at 10° C. After washing the plates 3 times with 50 µl PBS 0.05% Tween the plates were blocked with 50 µl Smart Block® (Candor 113500) for 1 h at 20° C. and washed again 3 times as described.

Subsequently, 20 µl of pre-mixed antibody solution was added to the plates and incubate for 18 h at 10° C. For the pre-mixed antibody solution, for each well, one biotinylated, parental antibody being either TPP-15370 or TPP-15374 was mixed in a ratio 1:1, 1:5 or 5:1 with an antibody containing one or more amino acid variations within its CDR regions (recombination variants) and not containing any biotin tag. As additional controls an isotype control antibody not demonstrating any binding to human Sema3A was also used as competition antibody. The total concentration of the added antibody solution was 0.25 µg/ml. During the incubation time the antibodies bound to the plates in a competitive manner as they compete for the same epitope on the human Sema3A protein.

After subsequent washing with 50 µl PBS 0.05% Tween for 3 times, 20 µl of a Streptavidin-HRP solution (R&D Systems, DY998, 1:200 in PBS 0.05% Tween 10% Smart Block) were added and incubated for 1 h and 20° C. followed by subsequent washing 3 times with 50 µl PBS 0.05% Tween and addition of 20 µl Amplex Red solution (Invitrogen A12222, 1:1000 in NaP-buffer 50 mM pH7.6 with 1:10000 of 30% H2O2). After a final incubation for 20 min at 20° C. the signal was determined using an emission

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wavelength of 595 nm and excitation of 530 nm. Due to the biotinylation of the parental antibodies TPP-15370 and TPP-15374 only the binding of these variants can be detected. Hence, competition with an antibody variant demonstrating superior binding shows a lower binding signal in comparison to e.g. competition of the parental antibody with a non-biotinylated version of itself.

In total, 103 recombination variants of TPP-15370 and 1136 recombination variants for TPP-15374 were measured. For analysis, and to allow for correction of plate-to-plate variations, the ELISA raw values were normalized to the value of the competition with the isotype control antibody TPP-9809.

Table 6 lists the values for the competition ELISA for selected recombination variants of TPP-15370 and TPP-15374. Depicted are the ratios vs. the isotype control antibody TPP-9809 in the measurement with a 1 to 5 or a 1 to 1 ratio, respectively.

TABLE 6

Values for the competition ELISA for recombination variants of TPP-15374 and TPP-15370. Depicted are the ratios vs. the isotype control antibody for selected recombination variants, respectively, when normalized to the isotype control antibody TPP-9809 in the measurement with a 1 to 5 or a 1 to 1 ratio, respectively					
TPP-15374 family			TPP-15370 family		
TPP Number	VAL norm to TPP-9809 (1 to 5 ratio)	VAL norm to TPP-9809 (1 to 1 ratio)	TPP Number	VAL norm to TPP-9809 (1 to 5 ratio)	VAL norm to TPP-9809 (1 to 1 ratio)
TPP-15374	0.41	0.69	TPP-15370	0.54	0.67
TPP-9809	1.00	1.00	TPP-9809	1.00	1.00
TPP-25497	0.26	0.39	TPP-23298	0.09	0.18
TPP-25256	0.15	0.41	TPP-23334	0.11	0.28
TPP-25255	0.17	0.37	TPP-23337	0.14	0.27
TPP-25257	0.18	0.36	TPP-23338		0.33
TPP-25248	0.20	0.36	TPP-23340		0.40
TPP-25064	0.19	0.48	TPP-23341	0.18	0.38
TPP-26111	0.18	0.49	TPP-23345	0.08	0.27
TPP-25224	0.17	0.43	TPP-23346	0.13	0.22
TPP-25448	0.19	0.47	TPP-23347	0.16	0.30
TPP-25655	0.23	0.39	TPP-23373	0.20	0.35
			TPP-23374	0.08	0.19
			TPP-23375	0.16	0.30

Example 5a: Epitope Binning Using Surface Plasmon Resonance (SPR)

An epitope binning experiment was performed to determine the epitope bins of anti-Sema3A antibodies using SPR by employing a classical sandwich approach. In this experiment, one antibody is immobilized to a SPR chip, Sema3A is injected, and the binding is monitored (FIG. 10A). After successful binding of Sema3A to the first antibody, a second antibody is injected on to the complex of the immobilized mAb bound to Sema3A and the additional binding is monitored (FIG. 10B and FIG. 10C). If the second antibody competes with the first antibody for the binding to Sema3A than no additional binding signal is detected after injection of the second antibody, showing that the two antibodies bind to the same or very adjacent Sema3A epitope (FIG. 10C). If the second antibody does not compete with the first antibody for the binding to Sema3A than an additional binding signal is detected after injection of the second antibody, showing that the two antibodies bind to different Sema3A epitopes (FIG. 10B).

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Experiments were performed on a Biacore T200 instrument (Cytiva) at 25° C. using assay buffer HBS P+, 300 mM NaCl, 0.75 mM CaCl₂, 2.5 mM MgCl₂, 1 mg/ml BSA, 0.05% NaN₃. Antibodies were covalently amine coupled to a Series S CM5 sensor chip (Cytiva). The amine coupling was carried out according to the manufacturer's instructions using 1-ethyl-3-(3-dimethylaminopropyl) carbodiimide hydrochloride (EDC), N-hydroxysuccinimide (NHS) and ethanolamine HCl, pH 8.5 ("Amine Coupling Kit" BR-1000-50, Cytiva.). Human, monovalent Sema3A domain was used as first analyte in a concentration of 200 nM followed by a second injection of the competitor antibody. This setup was performed with all possible combinations. The sensor surface was regenerated with glycine pH 2.0 after each antigen injection. Table 6a shows the binning results.

TABLE 6a

Matrix view of the epitope binning results (+ = additional binding, - = no additional binding)			
First antibody/ second antibody	TPP-30788 (BI Clone I)	TPP-23298	TPP-17755 (Samsung)
TPP-30788 (BI Clone I)		+	+
TPP-23298	+		-
TPP-17755 (Samsung)	+	-	

"+" means injection of second antibody resulted in additional binding signal showing that the two tested antibodies bind to two different Sema3A epitopes
 "-" means injection of second antibody did not result in additional binding signal showing that the two tested antibodies compete for binding to overlapping or adjacent epitopes Sema3A epitopes

The binning experiment strongly points to another epitope for TPP-23298 compared to TPP-30788 (BI clone I) meaning that both antibodies target an independent/different epitope on Sema3A, whereas TPP-23298 might have overlapping or adjacent epitopes with TPP-17755 (Samsung).

Example 7: Assessment of Binding to Off-Targets

To assess the specificity of an anti-Sema3A mAb (TPP-15370, parental mAb) an off-target screen using Retrogenix technology was conducted. For primary screening, 5484

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expression vectors, encoding both ZsGreen1 and a full-length human plasma membrane protein or a cell-surface tethered human secreted protein, were arrayed in duplicate across 16 microarray slides. Human HEK293 cells were used for reverse transfection/expression.

The test antibody was added to each slide after cell fixation giving a final concentration of 20 µg/ml. Detection of binding was performed by using the same AF647 anti-hlgG Fc detection antibody as used in the Pre-screen. Two replicate slides were screened for each of the 16 slide-sets. Hits were classified as 'strong, medium, weak or very weak', depending on the intensity of the duplicate spots.

Following a screen for binding against fixed HEK293 cells expressing 5484 human plasma membrane proteins and human secreted proteins, Retrogenix's technology identified no specific off-target interactions for test antibody TPP-15370. Binding to Sema3A—its primary target—was observed. These data indicate high specificity of TPP-15370 for its primary target.

Example 8: Selectivity of Anti-Sema3A mAbs

Semaphorin proteins can be subdivided in five classes occurring in vertebrates (class 3-7). To assess the selectivity profile of parental anti-Sema3A mAbs TPP-15370 and TPP-15374 in the Semaphorin 3 class (Sema3A-G) an ELISA assay was conducted using Sema3A, Sema3B, Sema3C, Sema3D, Sema3E, and Sema3F molecules from R&D Systems. Both antibodies showed no binding to Sema3B, Sema3C, Sema3D, Sema3E and Sema3F.

Because Sema3G has been recently identified as kidney protective (PMID: 27180624), it was important to test whether the antibodies do not bind to Sema3G. For the assessment of binding selectivity to Sema3A vs Sema3G, 1 nM recombinant human Sema3A-Fc chimera (R&D Systems) or recombinant human GST-Sema3G (Abnova) were coated on Maxisorb plates, incubated with antibodies in a dose-response curve from 0.00015-10 µg/ml, and the binding of antibodies quantified using HRP coupled anti-human antiserum and chemiluminescent substrate.

TABLE 7

Off-target ELISA values for testing of Sema3G as off-target			
Antibody	EC50 [nM]		Selectivity Score
	Coating: Sema3A	Coating: Sema3G	
TPP-23298	1.6	>66667	>41666
TPP-23334	9.2	>66667	>7220
TPP-23337	15.5	>66667	>4308
TPP-23338	9.6	>66667	>6949
TPP-23340	12.3	>66667	>5435
TPP-23341	21.3	>66667	>3133
TPP-23347	8.4	>66667	>7918
TPP-23373	17.6	>66667	>3786
TPP-23374	6.1	>66667	>10951
TPP-23375	7.7	>66667	>8651
TPP-11489 (Chiome)	Weak binding (EC50 not determinable)	>66667	n.d.
TPP-17755 (Samsung)	Slight dose-response (not determinable)	>66667	n.d.
TPP-30791 (BI clone IV)	0.08	>66667	>833337
TPP-30790 (BI clone III)	0.08	>66667	>833337
TPP-30789 (BI clone II)	0.10	>66667	>666670
TPP-30788 (BI clone I)	0.15	>66667	>444446

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All tested antibodies of the present disclosure as well as prior art antibodies do not bind to kidney protective Sema3G, as shown in Table 7.

Sema3A is a secreted protein that contains two furin cleavage sites and is present in an active and inactive cleaved form. In the in vivo situation Sema3A exists in both forms side by side. To test if anti-Sema3A antibodies are able to differentiate between the inactive and active form and to test how antibodies perform in binding to active Sema3A (resembled by full-length Sema3A (TPP-13211) in contrast to a inactive version as it only contains the Sema3A domain (resembled by cleaved Sema3A TPP-19068), an ELISA assay was performed. As readout out the ELISA signals of the tested antibody to the active Sema3A has been divided by the ELISA signals of the tested antibody to the inactive Sema3A.

TABLE 7a

Ratio for binding of anti-Sema3A antibodies to active vs. inactive Sema3A as determined by ELISA	
Antibody	Ratio ELISA binding TPP-13211/TPP-19068*
TPP-23298	0.66 ± 0.14
TPP-30788 (BI clone I)	0.19 ± 0.03
TPP-30789 (BI clone II)	0.20 ± 0.07
TPP-30790 (BI clone III)	0.19 ± 0.03
TPP-30791 (BI clone IV)	0.21 ± 0.004

*A Ratio ELISA binding TPP-13211/TPP-19068 below 1 shows a higher binding activity to active Sema3A.

A Ratio ELISA binding TPP-13211/TPP-19068 above 1 shows a higher binding activity to inactive Sema3A.

The binding analysis as shown in Table 7a clearly showed that the antibody of the present disclosure (TPP-23298) shows increased binding to active Sema3A than TPP-30788-TPP-30791 (BI clones) presumably since they target a different epitope indicating a higher selectivity for active Sema3A.

Example 9: In Vitro Efficacy in a Mesangial Cell Migration Assay

A confluent monolayer of human primary mesangial cells was generated by seeding cells in serum-containing culture medium into image lock plates for 24 hours. After switching to serum-free culture medium, scratch wounds were created using the WoundMaker tool, after which the cells were treated with 1 nM recombinant human Sema3A-Fc chimera (R&D Systems) in the absence or presence of inhibitory antibodies. The cells were imaged in the Incucyte and after 24 hrs the extent of wound closure was assessed using the Incucyte Integrated Cell Migration Analysis software module.

TABLE 8

EC50 values for phage display hits and recombination variants in the MCM assay	
Antibody	EC50 [nM]
TPP-15051 (Chiome)	42.87
TPP-15354	31.87
TPP-15355	>200
TPP-15356	>200
TPP-15357	158.13
TPP-15358	>200
TPP-15359	>200

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TABLE 8-continued

EC50 values for phage display hits and recombination variants in the MCM assay	
Antibody	EC50 [nM]
TPP-15360	37.47
TPP-15361	118.67
TPP-15362	>200
TPP-15363	>200
TPP-15364	>200
TPP-15365	>200
TPP-15366	2.27
TPP-15367	148.07
TPP-15368	>200
TPP-15369	45.47
TPP-15370	4.13
TPP-15371	>200
TPP-15372	86.87
TPP-15373	123.53
TPP-15374	5.07
TPP-15375	>200
TPP-15376	>200
TPP-15377	>200
TPP-15378	67.00
TPP-15379	>200
TPP-15380	125.53
TPP-15381	>200
TPP-15382	199.87
TPP-15384	1.60
TPP-15385	1.20
TPP-15386	>200
TPP-15387	>200
TPP-15388	>200
TPP-15389	103.60
TPP-15390	>200
TPP-15391	>200
TPP-15392	62.53
TPP-15393	131.93
TPP-15394	>200
TPP-15395	>200
TPP-15396	82.67
TPP-15397	>200
TPP-15398	6.00
TPP-15399	197.13
TPP-15400	4.73
TPP-15401	>200
TPP-17755 (Samsung)	11.33
TPP-23298	0.40
TPP-23334	0.67
TPP-23337	0.33
TPP-23338	0.60
TPP-23340	0.87
TPP-23341	0.90
TPP-23345	0.93
TPP-23346	1.27
TPP-23347	0.67
TPP-23373	0.63
TPP-23374	0.30
TPP-23375	1.03
TPP-30788 (BI clone I)	1.43

We identified antibodies with potencies in the three-digit picomolar range in the human Mesangial Cell Migration Assay, which is considerably more potent than the prior art antibodies, as shown in Table 8.

Example 10: In Vitro Efficacy in a Growth Cone Collapse Assay

In the direction of determining the potency of the antibodies against Sema3A induced cytoskeletal collapse, a growth cone collapse assay was used similarly as described (PMID: 12077190) with a few modifications. In brief, mouse dorsal root ganglion (DRG) neurons were isolated from E13 C57B1/6J mouse embryos, cultured on poly-L-lysine and laminin-coated 96-wells with Neurobasal medium+100 ng/ml NGF+B-27+10% FCS. After 20 hours,

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the cells were treated for 1 hour with 10 nM recombinant human Sema3A-Fc chimera (RnD Systems) in the absence or presence of inhibitory antibodies followed by PFA fixation and staining with Alexa488-phalloidin. The extent of growth cone collapse was assessed using immunofluorescence microscopy via actin growth cone area/shape/texture for more than 100 growth cones per well.

TABLE 9

EC50 values for phage display hits and recombination variants in the GCC assay	
Antibody	EC50 (nM)
TPP-15051 (Chiome)	243.40
TPP-15354	67.73
TPP-15355	>200
TPP-15356	>200
TPP-15357	50.73
TPP-15358	>200
TPP-15359	>200
TPP-15360	31.07
TPP-15361	>200
TPP-15362	>200
TPP-15363	>200
TPP-15364	>200
TPP-15365	142.87
TPP-15366	4.13
TPP-15367	170.87
TPP-15368	>200
TPP-15369	76.60
TPP-15370	4.33
TPP-15371	>200
TPP-15372	109.47
TPP-15373	>200
TPP-15374	8.13
TPP-15375	>200
TPP-15376	>200
TPP-15377	>200
TPP-15378	138.60
TPP-15379	>200
TPP-15380	135.40
TPP-15381	>200
TPP-15382	>200
TPP-15384	18.80
TPP-15385	6.00
TPP-15386	>200
TPP-15387	>200
TPP-15388	>200
TPP-15389	160.67
TPP-15390	>200
TPP-15391	>200
TPP-15392	>200
TPP-15393	>200
TPP-15394	>200
TPP-15395	66.47
TPP-15396	180.80
TPP-15397	>200
TPP-15398	12.00
TPP-15399	>200
TPP-15400	25.73
TPP-15401	>200
TPP17755 (Samsung)	52.67
TPP-23298	2.40
TPP-23334	2.24
TPP-23337	2.12
TPP-23374	2.19

The identified antibodies also show potencies in the single digit nanomolar range in the murine Growth Cone Collapse Assay, again considerably more potent than the tested prior art antibodies (two- to three-digit nanomolar potency), as shown in Table 9.

Example 11: In Vitro Efficacy in a HUVEC
Repulsion Assay

Recombinant human Sema3A-Fc chimera (R&D Systems) is not identical to Sema3A in human biofluids because it contains several mutated amino acids and an extra protein fragment at its carboxy-terminus. Furthermore, the above described assays (human Mesangial Cell Migration Assay and murine Growth Cone Collapse Assay) use Sema3A in homogenous distribution, which is in contrast to the gradient distribution described for Sema3A in tissues. We hypothesized that these differences could result in a different potency of the antibodies towards recombinant versus endogenous protein. Therefore, we adapted an assay using a gradient of human wild-type Sema3A as agonist (PMID: 17569671). In brief, in this HUVEC repulsion assay, human embryonic kidney 293 cells (HEK293) cells expressing human Sema3A of the sequence of SEQ ID NO: 600, were seeded on a confluent monolayer of human umbilical vein endothelial cells (HUVEC) in EGM-2 medium in the absence or presence of inhibitory antibodies, cultured for 72 hours, fixed, stained with DAPI and the extent of cell repulsion assessed by immunofluorescence microscopy (measurement of cell free areas). Consequently, the substrate human Sema3A exists in excess.

Based on immunofluorescence microscopy images of the DAPI/CM stained cells (CM=HCS CellMask™ Stain, stains the whole cell in order to define the total cell area) data analysis is performed as follows: Cells are identified based on the DAPI/CM signals (FIG. 11B). The cell area for analysis is defined and selected. In this area the cell-free region is calculated (FIG. 11C). Percent inhibition is calculated based on the “cell free-region” that is induced by Sema3A in the antibody-treated wells in comparison to the isotype-treated wells. Percent inhibition is plotted over antibody concentration and EC-50 values of the respective antibodies are calculated.

In detail the following steps are performed for the data analysis:

1. Four fields are imaged per well which corresponds to 80% of the well area. All of these fields stitched together are used for the detection of the cells via the DAPI/CM fluorescence.
2. The “cell area” is calculated based on the DAPI/CM area.
3. The “cell-free region” is calculated based on the “total area” subtracted by the “cell area”.
4. Percent inhibition is calculated based on the “cell free-region” that is induced by Sema3A in the antibody-treated wells vs the isotype-treated wells.
5. The software GraphPad Prism is used to determine the EC50 values using nonlinear regression (Variable slope model=four-parameter dose-response curve).

TABLE 10

EC50 values for selected antibodies in the repulsion assay, first experiment	
Antibody	EC50 (pM)
TPP-15370	800
TPP-23298	80
TPP-23334	120
TPP-23337	170
TPP-23340	180
TPP-23341	113
TPP-23373	180

TABLE 10-continued

EC50 values for selected antibodies in the repulsion assay, first experiment	
Antibody	EC50 (pM)
TPP-23374	77
TPP-23375	123

TABLE 10a

EC50 values for TPP-23298 in the repulsion assay in a second experiment to compare to prior art antibodies	
Antibody	EC50 (pM)
TPP-23298	54
TPP-30788 (BI clone I)	104
TPP-30789 (BI clone II)	165
TPP-30790 (BI clone III)	121
TPP-30791 (BI clone IV)	221
TPP-17755 (Samsung)	2794
TPP-11489 (Chiome)	>20000

The potency distinction to the prior art antibodies in the human Mesangial Cell Migration Assay and murine Growth Cone Collapse Assay above, is even more pronounced in this HUVEC Repulsion Assay that uses a gradient of native wt Sema3A (mixture of processed inactive and undigested active Sema3A) as shown in Table 10 and 10a. The improved potency in HUVEC repulsion assay in comparison to TPP-17755, to TPP-11489, to TPP-30788, to TPP-30789, TPP-30790, or to TPP-30791 is quantified measuring the picomolar activity as shown by the corresponding EC-50 values. While TPP-23298 shows two-digit picomolar activities, prior art antibody potencies of TPP-17755, TPP-11489, TPP-30788, TPP-30789, TPP-30790, or TPP-30791, are in the three-digit picomolar or even nanomolar range.

As an alternative illustration of the results, the improved potency in HUVEC repulsion assay is quantified by measuring the cell-free region at a specified concentration of 80 pM of the respective antibodies. TPP-23298 shows a higher percent inhibition of Sema3A than to TPP-30788, to TPP-30789, TPP-30790, or to TPP-30791 (FIG. 12).

Analyzing the data from both assays displayed in table 10 and 10a TPP-23298 shows the highest potency against cellular Sema3A induced HUVEC repulsion. The BI Antibodies TPP-30788, TPP-30798, TPP-30790 and TPP-30791 exhibited slightly higher EC50 values (2-5-fold). The Samsung Antibody TPP-17755 has a significantly lower potency than the TPP-23298 (50-fold). The Chiome Antibody TPP-11489 did only show inhibitory activity at the highest tested concentrations resulting in a predicted EC50 value >400-fold above antibody according to the present disclosure.

That shows that under conditions, that resembles a native environment without any spiked exogenous, recombinant semaphorin3A, the antibodies according to the present disclosure inhibit Sema3A-induced cell repulsion with the strongest activity, as shown in Table 10 and 10a.

Example 12: In Vivo Assay for Detecting
Protective Renal Effects: Inhibition of
Sema3a-Induced Albuminuria in Mice

Sema3A inhibitors decrease urinary albumin excretion induced via systemic injection of recombinant Sema3A. The beneficial effect of the compounds on albuminuria reduction were investigated in a Sema3A-induced albuminuria model as follows:

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Male C57B1/6 mice (8- to 10-wk-old) purchased from Taconic were injected intravenously with anti-Sema3A antibodies. Thirty minutes after antibody application albuminuria was induced by intravenous injection of human recombinant Sema3A (1.0 mg/kg, R&D Systems). Animals were placed into metabolic cages and urine was collected for 4 h. Urinary creatinine was measured via clinical biochemistry analyzer (Pentra400). For the assessment of urinary albumin, a mouse specific Albumin ELISA (Abcam) was used according to manufacturer's protocol. Both urinary creatinine and albumin were used to calculate urinary albumin to creatine ratio (ACR). Differences between groups were analyzed by one-way ANOVA with Dunnett's corrections for multiple comparisons. Statistical significance is defined as $p < 0.05$. All statistical analyses were done using GraphPad Prism 8.

Table 11-15a show dose-response experiments with TPP-15370, TPP-15374, TPP-11489, TPP-17755, TPP-30788 and TPP-23298 in the Sema3A-induced albuminuria model in mice. Effects on albuminuria reduction with TPP-15370, TPP-23298 in comparison to TPP-11489 and/or TPP-17755 and/or TPP-30788 are shown in FIGS. 1A-2C.

The antibodies according to the present disclosure reduce Sema3A-induced urinary Albumin excretion.

TABLE 11

Dose-response of Sema3A-induced albuminuria reduction after treatment with TPP-15370	
	urinary albumin to creatinine ratio [$\mu\text{g}/\text{mg}$]
control; Mean $\pm$ SD	345.30 $\pm$ 102.15****
15 [mg/kg] isotype control; Mean $\pm$ SD	1392.80 $\pm$ 350.70
1 [mg/kg] TPP-15370; Mean $\pm$ SD	1030.80 $\pm$ 216.27**
5 [mg/kg] TPP-15370; Mean $\pm$ SD	693.84 $\pm$ 203.18****
15 [mg/kg] TPP-15370; Mean $\pm$ SD	273.10 $\pm$ 146.02****
8-10 animal/group, One-way ANOVA with Dunnett's corrections for multiple comparisons, **/****/*****= significant with $p < 0.05/0.01/0.001/0.0001$ vs isotype control	

TABLE 12

Dose-response of Sema3A-induced albuminuria reduction after treatment with TPP-15374	
	urinary albumin to creatinine ratio [$\mu\text{g}/\text{mg}$]
Control; Mean $\pm$ SD	226.40 $\pm$ 65.50****
15 [mg/kg] isotype control; Mean $\pm$ SD	1061.43 $\pm$ 216.47
1 [mg/kg] TPP-15374; Mean $\pm$ SD	782.60 $\pm$ 122.43**
5 [mg/kg] TPP-15374; Mean $\pm$ SD	690.19 $\pm$ 190.27****
15 [mg/kg] TPP-15374; Mean $\pm$ SD	592.87 $\pm$ 123.93****
8-10 animal/group, One-way ANOVA with Dunnett's corrections for multiple comparisons, **/****/*****= significant with $p < 0.05/0.01/0.001/0.0001$ vs isotype control	

TABLE 13

Dose-response of Sema3A-induced albuminuria reduction after treatment with TPP-23298	
	urinary albumin to creatinine ratio [$\mu\text{g}/\text{mg}$]
Control; Mean $\pm$ SD	345.30 $\pm$ 102.15****
15 [mg/kg] isotype control; Mean $\pm$ SD	1281.65 $\pm$ 447.14
1 [mg/kg] TPP-23298; Mean $\pm$ SD	623.37 $\pm$ 240.41****

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TABLE 13-continued

Dose-response of Sema3A-induced albuminuria reduction after treatment with TPP-23298	
	urinary albumin to creatinine ratio [$\mu\text{g}/\text{mg}$]
5 [mg/kg] TPP-23298; Mean $\pm$ SD	471.07 $\pm$ 164.97****
15 [mg/kg] TPP-23298; Mean $\pm$ SD	320.60 $\pm$ 166.36****
8-10 animal/group, One-way ANOVA with Dunnett's corrections for multiple comparisons, **/****/*****= significant with $p < 0.05/0.01/0.001/0.0001$ vs isotype control	

TABLE 14

Dose-response of Sema3A-induced albuminuria reduction after treatment with TPP-11489	
	urinary albumin to creatinine ratio [$\mu\text{g}/\text{mg}$]
Control; Mean $\pm$ SD	237.23 $\pm$ 92.61****
15 [mg/kg] isotype control; Mean $\pm$ SD	1404.81 $\pm$ 411.55
1 [mg/kg] TPP-11489; Mean $\pm$ SD	1204.81 $\pm$ 426.64
5 [mg/kg] TPP-11489; Mean $\pm$ SD	664.02 $\pm$ 228.96****
15 [mg/kg] TPP-11489; Mean $\pm$ SD	572.42 $\pm$ 211.05****
8-10 animal/group, One-way ANOVA with Dunnett's corrections for multiple comparisons, **/****/*****= significant with $p < 0.05/0.01/0.001/0.0001$ vs isotype control	

TABLE 15

Dose-response of Sema3A-induced albuminuria reduction after treatment with TPP-17755	
	urinary albumin to creatinine ratio [$\mu\text{g}/\text{mg}$]
Control; Mean $\pm$ SD	298.02 $\pm$ 91.06****
15 [mg/kg] isotype control; Mean $\pm$ SD	1053.75 $\pm$ 162.28
1 [mg/kg] TPP-17755; Mean $\pm$ SD	932.57 $\pm$ 221.09
5 [mg/kg] TPP-17755; Mean $\pm$ SD	823.11 $\pm$ 196.93*
15 [mg/kg] TPP-17755; Mean $\pm$ SD	711.09 $\pm$ 181.65***
8-10 animal/group, One-way ANOVA with Dunnett's corrections for multiple comparisons, **/****/*****= significant with $p < 0.05/0.01/0.001/0.0001$ vs isotype control	

TABLE 15a

Dose-response of Sema3A-induced albuminuria reduction after treatment with TPP-30788	
	urinary albumin to creatinine ratio [$\mu\text{g}/\text{mg}$]
Control; Mean $\pm$ SD	266.67 $\pm$ 115.66****
15 [mg/kg] isotype control; Mean $\pm$ SD	1546.59 $\pm$ 312.43
1 [mg/kg] TPP-30788; Mean $\pm$ SD	1234.13 $\pm$ 353.48
5 [mg/kg] TPP-30788; Mean $\pm$ SD	958.30 $\pm$ 196.93***
15 [mg/kg] TPP-30788; Mean $\pm$ SD	841.46 $\pm$ 438.51****
8-10 animal/group, One-way ANOVA with Dunnett's corrections for multiple comparisons, **/****/*****= significant with $p < 0.05/0.01/0.001/0.0001$ vs isotype control	

Example 13: In Vivo Assay for Detecting Protective Renal Effects: Acute Ischemia/Reperfusion Injury (I/RI) Model in Mice

Unilaterally nephrectomized mice may benefit from treatment with Sema3A inhibitors after ischemia reperfusion injury. The beneficial effect of Sema3A antibodies on kidney function was investigated in a kidney ischemia-reperfusion injury model in mice as follows:

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Laboratory bred male C57B1/6J mice 6-8 weeks old were obtained from Charles River. Mice were maintained under standard laboratory conditions, 12-hour light-dark cycles with access to normal chow and drinking water at libitum. For the ischemia reperfusion injury model, a total of 8-10 was used in each control and experimental group.

Animals were anesthetized with continuous inhaled isoflurane. Right nephrectomy was performed through a right flank incision 7 days before the ischemic procedures in the contralateral kidneys. One-hour before the initiation of renal ischemia antibodies and adequate isotype control were administered to mice via i.v. injection. Mice were anesthetized and a left flank incision was made. Renal vessels were exposed by dissection of the left renal pedicle. Non-traumatic vascular clamps were used to stop blood flow (artery and vein) during 25 min (mice) of ischemia. Reperfusion was established by removing the clamps. The abdominal wall (muscular layer and skin) was closed with 5.0 polypropylene sutures. Temgesic® (Buprenorphin, 0.025 mg/kg s.c.) was applied as an analgesic.

Urine of each animal was collected in metabolic cages over night before sacrifice at 24 h post ischemia. Urinary creatinine was measured by a clinical biochemistry analyzer (Pentra400). For the assessment of urinary albumin, a mouse specific Albumin Kit (Hitachi) was used within the Pentra analyzer. Both urinary creatinine and albumin were used to determine Albuminuria (albumin/creatinine ratio). Upon sacrifice, blood samples were obtained under terminal anesthesia. After centrifugation of the blood samples, serum was isolated. Both serum creatinine and serum urea were measured via clinical biochemistry analyzer (Pentra 400). Differences between groups were analyzed by one-way ANOVA with Dunnett's corrections for multiple comparisons. Statistical significance is defined as $p < 0.05$. All statistical analyses were done using GraphPad Prism 8.

Table 16-20 show dose-response experiments with TPP-15370, TPP-15374, TPP-11489, TPP-17755 and TPP-23298 in an acute renal ischemia/reperfusion injury model in mice. FIGS. 3A-3C show the efficacy of TPP-23374, TPP-23298 and TPP-15370 after treatment with 15 mg/kg in the I/R model. Treatment effects with TPP-15370, TPP-23298 and TPP-15374 in comparison to TPP-11489 and/or TPP-17755 are shown in FIGS. 4A-6C.

The antibodies attenuated ischemia/reperfusion induced kidney damage by reducing serum creatinine and serum urea (surrogates for glomerular filtration rate) and excretion of urinary albumin.

TABLE 16

Dose-response of TPP-15370 in mouse I/R injury model			
	serum creatinine [mg/dl]	serum urea [mg/dl]	urinary albumin to creatinine ratio [μg/mg]
SHAM	0.34 ±	102.78 ±	58.50 ±
Mean ± SD	0.05****	9.45****	19.22****
15 [mg/kg]	1.72 ±	385.63 ±	1699.47 ±
isotype control	0.30	41.69	461.60
Mean ± SD			
1 [mg/kg]	1.61 ±	396.51 ±	1165.37 ±
TPP-15370	0.52	86.91	445.50**
Mean ± SD			
5 [mg/kg]	1.22 ±	297.92 ±	705.21 ±
TPP-15370	0.32*	70.02**	192.26**
Mean ± SD			

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TABLE 16-continued

Dose-response of TPP-15370 in mouse I/R injury model			
	serum creatinine [mg/dl]	serum urea [mg/dl]	urinary albumin to creatinine ratio [μg/mg]
15 [mg/kg]	0.89 ±	261.95 ±	554.52 ±
TPP-15370	0.27****	27.76***	133.99****
Mean ± SD			

8-10 animal/group, One-way ANOVA with Dunnett's corrections for multiple comparisons,

*/**/***/****= significant with $p < 0.05/0.01/0.001/0.0001$ vs isotype control

TABLE 17

Dose-response of TPP-15374 in mouse I/R injury model			
	serum creatinine [mg/dl]	serum urea [mg/dl]	urinary albumin to creatinine ratio [μg/mg]
SHAM	0.26 ±	113.90 ±	39.36 ±
Mean ± SD	0.02****	29.95****	10.19****
15 [mg/kg]	2.09 ±	494.52 ±	3942.50 ±
isotype control	0.19	29.75	1790.29
Mean ± SD			
1 [mg/kg]	1.84 ±	478.10 ±	2774.43 ±
TPP-15374	0.39	66.55	946.18
Mean ± SD			
5 [mg/kg]	1.66 ±	416.49 ±	2195.95 ±
TPP-15374	0.32*	98.47*	900.56*
Mean ± SD			
15 [mg/kg]	1.43 ±	389.02 ±	1495.88 ±
TPP-15374	0.34****	5128**	560.06**
Mean ± SD			

8-10 animal/group, One-way ANOVA with Dunnett's corrections for multiple comparisons,

*/**/***/****= significant with $p < 0.05/0.01/0.001/0.0001$ vs isotype control

TABLE 18

Dose-response of TPP-11489 in mouse I/R injury model			
	serum creatinine [mg/dl]	serum urea [mg/dl]	urinary albumin to creatinine ratio [μg/mg]
SHAM	0.22 ±	57.64 ±	27.87 ±
Mean ± SD	0.02****	14.62****	13.55****
15 [mg/kg]	1.99 ±	410.18 ±	1569.47 ±
isotype control	0.29	39.80	277.70
Mean ± SD			
1 [mg/kg]	2.00 ±	453.84 ±	1600.96 ±
TPP-11489	0.12	26.54	338.48
Mean ± SD			
5 [mg/kg]	1.92 ±	416.87 ±	1437.08 ±
TPP-11489	0.16	49.81	323.46
Mean ± SD			
15 [mg/kg]	1.68 ±	367.67 ±	1186.32 ±
TPP-11489	0.42*	39.32	366.49*
Mean ± SD			

8-10 animal/group, One-way ANOVA with Dunnett's corrections for multiple comparisons, **/***/****= significant with $p < 0.05/0.01/0.001/0.0001$ vs isotype control

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TABLE 19

Dose-response of TPP-17755 antibody in mouse I/R injury model			
	serum creatinine [mg/dl]	serum urea [mg/dl]	urinary albumin to creatinine ratio [μg/mg]
SHAM	0.21 ±	91.20 ±	75.45 ±
Mean ± SD	0.06****	34.20****	42.78****
15 [mg/kg]	1.75 ±	444.25 ±	1791.23 ±
isotype control	0.30	64.25	543.46
Mean ± SD			
1 [mg/kg]	1.74 ±	430.30 ±	1659.08 ±
TPP-17755	0.27	75.96	577.99
Mean ± SD			
5 [mg/kg]	1.84 ±	439.83 ±	1661.14 ±
TPP-17755	0.24	73.68	460.41
Mean ± SD			
15 [mg/kg]	1.31 ±	346.62 ±	1351.64 ±
TPP-17755	0.37**	78.14**	795.59
Mean ± SD			

8-10 animal/group, One-way ANOVA with Dunnett's corrections for multiple comparisons, ****/****/**** = significant with $p < 0.05/0.01/0.001/0.0001$ vs isotype control

TABLE 20

Dose-response of TPP-23298 antibody in mouse I/R injury model			
	serum creatinine [mg/dl]	serum urea [mg/dl]	urinary albumin to creatinine ratio [μg/mg]
SHAM	0.26 ±	115.80 ±	71.05 ±
Mean ± SD	0.04****	6.76****	865.39****
15 [mg/kg]	2.53 ±	498.92 ±	3968.71 ±
isotype control	0.15	45.45	453.52
Mean ± SD			
1 [mg/kg]	2.38 ±	482.06 ±	2383.77 ±
TPP-23298	0.22	25.84	1111.94**
Mean ± SD			
5 [mg/kg]	2.20 ±	425.64 ±	1966.11 ±
TPP-23298	0.36*	58.85*	677.69****
Mean ± SD			
15 [mg/kg]	2.02 ±	422.79 ±	1949.56 ±
TPP-23298	0.28***	71.44**	700.58****
Mean ± SD			

8-10 animal/group, One-way ANOVA with Dunnett's corrections for multiple comparisons, ****/****/**** = significant with $p < 0.05/0.01/0.001/0.0001$ vs isotype control

Example 14: In Vivo Assay for Detecting Protective Renal Effects: Alport Syndrome Model (Col4α3 Deficient Mice)

The phenotype of Alport mice is similar to that of Alport patients, including characteristic thickening and splitting of the glomerular basement membrane as well as strong proteinuria. Alport mice may benefit from treatment with Sema3A inhibitors due to increased Sema3A expression in kidneys of those mice. The beneficial effect of Sema3A blocking antibodies on kidney function was investigated in the Alport mouse model as follows: A colony of knockout Col4α3 (129-Col4α3^{tm1Dec}/J) mice (Jackson Laboratory, USA) was established by mating heterozygous animals within the breeding facilities at Bayer A G, Wuppertal, Germany. Male and female homozygous and wild-type Col4α3 mice at an age of 4-5 weeks were obtained from the animal breeding facilities at Bayer A G and used for this study.

The homozygous mice (HOM) were randomized into groups (n=10 each group) according to their age and gender. Mice were dosed once weekly with isotype control and

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TPP-15370 and TPP-23298. TPP-11489 was administered biweekly. Urine of each animal was collected in metabolic cages once weekly starting before initiation of treatment. Urinary creatinine as well as total protein was measured by a clinical biochemistry analyzer (Pentra400). Both urinary creatinine and albumin were used to determine proteinuria (protein/creatinine ratio). Upon sacrifice at day 21 or day 28 post treatment start, blood samples were obtained under terminal anesthesia. After centrifugation of the blood samples, serum was isolated. Both serum creatinine and serum urea were measured via clinical biochemistry analyzer (Pentra 400).

Kidneys were collected and divided in two parts. One part was snap-frozen in liquid nitrogen for mRNA analysis. The other part was stored in Davidson's fixative for the preparation of histological sections. Total RNA was isolated from parts of harvested kidneys. Kidney tissue was homogenized, and RNA was obtained and transcribed to cDNA. Using TaqMan real time PCR renal mRNA expression of pro-fibrotic markers was analyzed in kidney tissues. For the assessment of fibrosis on the protein level paraffin tissue sections were stained with alpha-smooth muscle actin (αSMA) and Sirius Red/Fast Green Collagen staining using standard procedures.

Quantitative measurements of alpha-smooth muscle actin (αSMA)-positive as well as Sirius Red (Collagen) positive areas within the kidneys were obtained by computer image analysis using the Axio Scan Z1 (Zeiss) microscope and the Zen software.

All data are expressed as means±S.D. Differences between groups were analyzed by one-way ANOVA with Dunnett's corrections for multiple comparisons. Statistical significance was defined as $p < 0.05$. All statistical analyses were done using GraphPad Prism 8.

Tables 21A-21C and 22A-22C show effects on proteinuria, kidney function and kidney fibrosis obtained after treatment with TPP-15370 and TPP-23298 in the Alport model. Effects after treatment with TPP-15370 in comparison to TPP-11489 on proteinuria, kidney function and kidney fibrosis are displayed in FIGS. 7 and 8A-8D.

The antibodies according to the present disclosure stopped the progression of kidney disease in a mouse model of Alport syndrome. The antibodies according to the present disclosure reduced the excretion of urinary protein, reduced creatinine and serum urea (surrogates for glomerular filtration rate) as well as fibrosis quantified via myofibroblasts staining and collagen deposition.

TABLE 21A

Effects of TPP-15370 on proteinuria progression in Alport mouse model				
	urinary protein to creatinine ratio [%] increase from baseline			
	baseline	day 7	day 14	day 21
HOM	100.00 ±	118.65 ±	167.49 ±	192.03 ±
15 [mg/kg]	53.71	47.18	55.77	40.23
isotype control				
Mean ± SD				
HOM	100.00 ±	114.61 ±	149.35 ±	164.92 ±
5 [mg/kg]	54.02	50.48	95.41	47.18
TPP-15370				
Mean ± SD				

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TABLE 21A-continued

Effects of TPP-15370 on proteinuria progression in Alport mouse model				
	urinary protein to creatinine ratio [%] increase from baseline			
	baseline	day 7	day 14	day 21
HOM	100.00 ±	114.61 ±	95.41 ±	93.04 ±
15 [mg/kg]	65.59	50.48	52.50**	31.26****
TPP-15370				
Mean ± SD				

10 animal/group, data are expressed as relative means ± SD percentage values calculated vs. baseline (set to 100). Differences between groups were analyzed by one-way ANOVA with Dunnett's corrections for multiple comparisons. Statistical significance was defined as $p \leq 0.05$.

TABLE 21B

Effects of TPP-15370 on functional parameters at day 21 in Alport mouse model		
	serum creatinine [mg/dl]	serum urea [mg/dl]
HOM	0.71 ± 0.26	380.61 ± 120.28
15 [mg/kg]		
isotype control		
Mean ± SD		
HOM	0.39 ± 0.16**	255.25 ± 56.80**
5 [mg/kg]		
TPP-15370		
Mean ± SD		
HOM	0.44 ± 0.21**	256.71 ± 95.03**
15 [mg/kg]		
TPP-15370		
Mean ± SD		

10-15 animal/group, One-way ANOVA with Dunnett's corrections for multiple comparisons. One-way ANOVA with Dunnett's corrections for multiple comparisons, **/****/*****= significant with $p < 0.05/0.01/0.001/0.0001$ vs isotype control

TABLE 21C

Effects of TPP-15370 on fibrosis at day 28 in Alport mouse model		
	Myofibroblasts % αSMA reduction	Collagen % Sirius Red reduction
HOM	100.00 ± 53.53	100.00 ± 47.78
15 [mg/kg]		
isotype control		
Mean ± SD		
HOM	50.18 ± 21.00**	80.08 ± 51.58
5 [mg/kg]		
TPP-15370		
Mean ± SD		
HOM	54.86 ± 17.60**	100.26 ± 50.97
15 [mg/kg]		
TPP-15370		
Mean ± SD		

10-15 animal/group, One-way ANOVA with Dunnett's corrections for multiple comparisons. One-way ANOVA with Dunnett's corrections for multiple comparisons, **/****/*****= significant with $p < 0.05/0.01/0.001/0.0001$ vs isotype control

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TABLE 22A

Effects of TPP-23298 on proteinuria progression in Alport mouse model				
	urinary protein to creatinine ratio [%] increase from baseline			
	baseline	day 14	day 21	day 28
HOM	100.00 ±	185.29 ±	228.62 ±	283.62 ±
15 [mg/kg]	70.94	88.09	160.68	77.37
isotype control				
Mean ± SD				
HOM	100.00 ±	148.01 ±	155.25 ±	151.82 ±
5 [mg/kg]	55.72	77.13	61.60	45.84****
TPP-23298				
Mean ± SD				
HOM	100.00 ±	154.58 ±	120.54 ±	125.71 ±
15 [mg/kg]	56.02	91.21	37.21****	34.25****
TPP-23298				
Mean ± SD				

10 animal/group, data are expressed as relative means ± SD percentage values calculated vs. baseline (set to 100). Differences between groups were analyzed by one-way ANOVA with Dunnett's corrections for multiple comparisons. Statistical significance was defined as $p \leq 0.05$.

TABLE 22B

Effects of TPP-23298 on functional parameters at day 28 in Alport mouse model		
	serum creatinine [mg/dl]	serum urea [mg/dl]
HOM	0.29 ± 0.07	208.89 ± 0.07
15 [mg/kg]		
isotype control		
Mean ± SD		
HOM	0.22 ± 0.09*	175.54 ± 0.03
5 [mg/kg]		
TPP-23298		
Mean ± SD		
HOM	0.19 ± 0.03***	141.84 ± 0.03***
15 [mg/kg]		
TPP-23298		
Mean ± SD		

10-15 animal/group, One-way ANOVA with Dunnett's corrections for multiple comparisons. One-way ANOVA with Dunnett's corrections for multiple comparisons, **/****/*****= significant with $p < 0.05/0.01/0.001/0.0001$ vs isotype control

TABLE 22C

Effects of TPP-23298 on fibrosis at day 28 in Alport mouse model		
	Myofibroblasts % αSMA positive area	Collagen % Sirius Red positive area
HOM	100.00 ± 53.53	100.00 ± 47.78
15 [mg/kg]		
isotype control		
Mean ± SD		
HOM	50.18 ± 21.00**	80.08 ± 51.58
5 [mg/kg]		
TPP-23298		
Mean ± SD		
HOM	54.86 ± 17.60**	100.26 ± 50.97
15 [mg/kg]		
TPP-23298		
Mean ± SD		

10-15 animal/group, One-way ANOVA with Dunnett's corrections for multiple comparisons. One-way ANOVA with Dunnett's corrections for multiple comparisons, **/****/*****= significant with $p < 0.05/0.01/0.001/0.0001$ vs isotype control

Example 15: In Vivo Away for Detecting Protective Renal Effects: Unilateral Kidney IRI Model in Pig

TPP-23298 was tested in a minimal invasive, unilateral kidney artery balloon-catheter occlusion model in adult minipigs with a post-reperfusion follow-up of about 24 hours. Female Göttingen mini pigs of a body weight range 14 to 17 kg (Ellegaard, Denmark) were used for the experiments. Animals were randomly assigned to experimental groups.

TPP-23298 was administered in a blinded, controlled study to 6 animals in comparison to 6 matched IgG-treated controls. Animals which were subjected to all treatment procedures without kidney artery occlusion and received phosphate buffered saline vehicle only served as sham treated reference group.

TPP-23298 was administered at weight adjusted doses in a final volume of 1 ml/kg phosphate buffered saline as a bolus by slow intravenous injection before start of kidney artery occlusion (preventive setting).

For the intervention on day 1 of experimentation pigs were anesthetized by a combination of Propofol and Fentanyl and artificially ventilated over an oro-tracheal tube under muscular relaxation by Pancuronium. Volume was continuously substituted by continuous infusion of Ringer lactate solution. Before starting surgery, antibiotic and thrombosis prophylaxis were provided by administration of Enrofloxacin i.m. and Heparin i.v., respectively. Blood pressure and heart rate were monitored with a non-invasive veterinary device equipped with a foreleg cuff.

All following interventions were performed under strictly aseptic conditions. A catheter was tunneled subcutaneously through the dorsal neck skin to a jugular vein for drug administration. A sheath was placed into the—preferably left—femoral artery and fixed, through which a hockey-stick catheter with a balloon catheter inside was advanced upstream into the abdominal aorta and inserted with its tip into the orifice of the left or right kidney artery. The balloon catheter was then protruded, and the balloon was inflated to interrupt blood flow to the kidney. Correct positioning of the balloon was controlled by Doppler ultrasound using a commercial ultrasound diagnostic apparatus. Plasma samples were collected at baseline and 2 h after start of ischemia.

Kidney ischemia was relieved exactly at pre-defined time points after start of occlusion (ranging from 90 to 120 min) by deflating the balloon and withdrawing the catheter and the sheath. After vascular suture and wound closure animals were re-awakened from anesthesia and after onset of spontaneous breathing extubated.

About 22 to 23 hours after the kidney artery occlusion animals were re-anesthetized by a combination of Ketaset/Dormicum and Pancuronium and artificially ventilated as described. Blood pressure and heart rate were invasively monitored via a carotid artery catheter. Volume substitution was provided at a flow rate of 10 ml/kg/h Ringer Lactate intravenously. Via a small incision in the lower abdomen both ureters were dissected on the urinary bladder wall and catheters were inserted to collect urine side separately for volume determination and urinalysis. Recordings and sample collections were started when all parameters were stable, which was typically the case 24 hours after occlusion. Blood samples were collected at baseline and every hour for three hours (24-27 h interval). In parallel urine was collected for three intervals of 1 h.

After urine volume flow (V_U) and urinary creatinine concentrations ($[Crea]_U$) were determined creatinine clearance (CL_{Crea}) was calculated side separately according to

the standard formula $CL_{Crea} = V_U * [Crea]_U / [Crea]_{Pl}$ in which $[Crea]_{Pl}$ stands for plasma creatinine concentration. Global CL_{Crea} was calculated by adding CL_{Crea} of left and right kidney of each animal.

The results are depicted in FIGS. 9A-9D. TPP-23298 when administered in a preventive manner 30 min before occlusion prevented deterioration of ischemia/reperfusion-induced creatinine clearance significantly in this experimental setting after a unilateral kidney artery occlusion of 105 min.

Example 16: Expression Titer of Anti-Sema3A Antibodies in Mammalian Cell Culture

HEK293-6E cells were transfected with pTT5 plasmids coding for the heavy and light chain of anti-Sema3A antibodies or with the Fab fragment of TPP-30792 (TPP-31357). Two days prior to transfection, HEK293-6E cells were split to a density of 5×10^5 cells/mL in FreeStyle™ F17 Expression Medium (Gibco, A1383501) with 0.1% Pluronic F68 (Gibco, 24040032) and 4 mM GlutaMax (Gibco, 35050061) in a shake flask, making up 90% of the desired expression volume. HEK293-6E cells were cultivated at 37° C., 5% CO₂ shaking at 75 rpm.

For transfection, the DNA and polyethylenimine (Polysciences, 29366) are mixed in FreeStyle™ F17 Expression Medium (Gibco, A1383501) with 4 mM GlutaMax (Gibco, 35050061) making up 10% of the final expression volume. The solution is incubated for 10 minutes and added to the shake flask.

24 hours after transfection, 1% (v/v) ultra-low IgG FBS (Gibco, 16250078) and 0.05% (v/v) 1N valproic acid (Sigma, P4543) are added to the shake flask.

The cell viability and density are monitored every day starting 4 days post transfection, the supernatant is harvested by centrifugation and sterile filtration when the viability is determined to be 70%. To determine the production titer, 100 μ L of the harvested supernatant are loaded to a 0.1 mL Poros A affinity column (Thermo Scientific, 2100100) via HPLC-system (Agilent, 1100 HPLC system) using 50 mM sodium phosphate (Sigma, S0751, S9763), 150 mM NaCl (Sigma, S6546), 5% 2-propanol (sigma, 34863), pH 7.2 as running buffer. Subsequently, the protein is eluted using 12 mM HCl (Sigma, H9892), 150 mM NaCl, 5% 2-propanol pH 2. A calibration curve from 5 μ g/mL to 150 μ g/mL is set up using a protein of known size and is applied to the Poros A column via HPLC-system as well. Taking the size and extinction coefficient of the protein in the supernatant into consideration, the exact titer can be calculated using the standard curve. Expression in CHO is similar to HEK cells except that plasmid pTT22AKT was used for TPP-30792.

TABLE 23

Expression Titer of anti-Sema3A antibodies in mammalian cells in mg/L	
	Titer [mg/L]
TPP-23298	203.6
TPP-17755 (Samsung)	277.0
TPP-11489 (Chiome)	132.0
TPP-30791 (BI clone IV)	333.0
TPP-30790 (BI clone III)	160.9
TPP-30789 (BI clone II)	187.6
TPP-30788 (BI clone I)	240.2
TPP-30792 (3H4 Univ Ramot)	3.0 (HEK), 3.2 (CHO)
TPP-31357 (Fab of TPP-30792)	Not determined

The antibody of the present disclosure as well as all prior art antibodies except TPP-30792 can be produced with high titers in mammalian cells, as shown in Table 23. TPP-30792 could not be expressed in a significant amount in either HEK or CHO cells. In total 125 µg of TPP-30792 could be purified out of 4.5 liters of HEK293 cell culture. Similarly, the Fab fragment of TPP30792 (TPP-31357) yielded only 200 µg purified Fab out of 5 liters HEK293 cell culture.

Example 17: Analysis of CMC Parameter Stability and Solubility of Anti-Sema3A Antibodies

It is known that high concentrated protein solutions of more than 50 mg/ml usually exhibit also higher viscosities compared to lower concentrated protein solutions. Increased viscosity negatively affects the deliverability of the protein solutions especially in low application volumes and it may increase the injection time and pain at the site of injection. In addition to that, high viscosity impacts high-scale protein production in the industry. Thus, reducing viscosity of high concentrated protein solutions while maintaining stability for a long shelf life is i.a. important for the therapeutical in vivo setting.

Proteins in high concentrated solutions are often less stable than in diluted solutions, since the proteins tend to aggregate and may reversibly self-associate at higher concentrations. Aggregation may negatively impact structural integrity and therefore also the amount of functional, bio-available protein in the therapeutical in vivo setting. This further complicates delivery by injection.

Solubility of proteins is another important quality criterion. Increased solubility of the isolated protein allows for the preparation of highly concentrated solutions required for the therapeutical in vivo setting.

Thus, providing a high concentrated protein solution with reduced viscosity and increased stability and solubility is beneficial for therapeutic applicability of therapeutic molecules.

To assess the CMC (Chemistry, Manufacturing, Control) parameters stability, solubility and viscosity of anti-Sema3A antibodies for potential therapeutic use, antibodies TPP-23289 and TPP-30788 (BI clone I) were diluted in PBS to 25 mg/ml and incubated at 700 rpm and 40° C. for two weeks. While antibodies are usually stored at 4°–10° C. for short-term or frozen at ≤–18° C. or ≥–81° C. for long term

an exposure of mammalian antibodies to temperatures higher than ≥40° C. (mammalian average body temperature is 36° C.–39° C.) resembles a thermal stress condition. In this thermal stress condition accelerated protein stability/stress stability is tested. Analysis of stability was assessed by size-exclusion chromatography using a Superdex 200 column (Cytiva) coupled to an Äkta system (Cytiva) in PBS buffer as well as capillary gel electrophoresis using a Caliper system (Perkin Elmer). Changes in profile were calculated as percentage to non-stressed starting material. Solubility was determined by concentrating anti-Sema3A antibodies using an Amicon spin filter (Millipore) with a cut-off of 30 kDa in PBS buffer. The solubility was determined at 90% recovery from the concentrator and protein concentration was measured using Absorption at UV280 nm.

TABLE 24

Overview of CMC parameters for TPP-23298 and TPP-30788; SEC = size-exclusion chromatography, cGE = capillary gel electrophoresis				
CMC Parameter	Method	Analysis	TPP-23298	TPP-30788 (BI clone I)
Stability at 40° C.	SEC*	Δ % monomer	1	–5.5
	cGE**	Δ % LC ± HC	<1	–4.7
Solubility	concentrator	mg/ml at 90% recovery	225	105
Viscosity	SEC*	Δ % monomer	<1	<1
	Viscosizer	cP	5.1 (150 mg/ml)	5.3 (127 mg/ml)

*SEC = Size exclusion chromatography;

**cGE = capillary gel electrophoresis

Stability, solubility and viscosity are critical CMC parameters for therapeutic molecules as described above. The structural integrity after a thermal stress condition, like exposure to 40° C., or concentrating step is analyzed via SEC and/or cGE to see the effect of the applied stress on the structural integrity. Less than 1% change after the applied stress compared to the start points to a stable molecule whereas deviations >1% points to instabilities in the molecule. TPP-23289 shows a much higher solubility in PBS compared to TPP-30788 by a factor >2 which is very beneficial for e.g. enabling low application volume. Furthermore, TPP-23298 is more stable and more resistant to heat stress than TPP-30788 and is less viscous in PBS buffer.

TABLE 1

Amino acid sequences and nucleic acid sequences of preferred antibodies according to the present disclosure and of three prior art antibodies. TPP-11489 corresponds to Chiome antibody Humanized-2 derived of clone No. 4-2 strain (WO 2014/123186); TPP-15051 represents a murine IgG1 variant thereof. TPP-30788-TPP-30791 corresponds to Böhlinger Ingelheim antibody (BI) Clone I-IV (WO 2020/225400). TPP-30792 corresponds to University Ramot antibody clone I (WO 2020/261281).				
TPP ID	Antibody Description	Sequence Region	Sequence Type	SEQ ID
TPP-11489	Chiome Prior Art (hIgG1)	VH	PRT	SEQ ID NO: 1
TPP-11489	Chiome Prior Art (hIgG1)	HCDR1	PRT	SEQ ID NO: 2
TPP-11489	Chiome Prior Art (hIgG1)	HCDR2	PRT	SEQ ID NO: 3
TPP-11489	Chiome Prior Art (hIgG1)	HCDR3	PRT	SEQ ID NO: 4
TPP-11489	Chiome Prior Art (hIgG1)	VL	PRT	SEQ ID NO: 5
TPP-11489	Chiome Prior Art (hIgG1)	LCDR1	PRT	SEQ ID NO: 6
TPP-11489	Chiome Prior Art (hIgG1)	LCDR2	PRT	SEQ ID NO: 7
TPP-11489	Chiome Prior Art (hIgG1)	LCDR3	PRT	SEQ ID NO: 8
TPP-11489	Chiome Prior Art (hIgG1)	VH	DNA	SEQ ID NO: 9
TPP-11489	Chiome Prior Art (hIgG1)	HCDR1	DNA	SEQ ID NO: 10
TPP-11489	Chiome Prior Art (hIgG1)	HCDR2	DNA	SEQ ID NO: 11
TPP-11489	Chiome Prior Art (hIgG1)	HCDR3	DNA	SEQ ID NO: 12
TPP-11489	Chiome Prior Art (hIgG1)	VL	DNA	SEQ ID NO: 13
TPP-11489	Chiome Prior Art (hIgG1)	LCDR1	DNA	SEQ ID NO: 14

TABLE 1-continued

Amino acid sequences and nucleic acid sequences of preferred antibodies according to the present disclosure and of three prior art antibodies. TPP-11489 corresponds to Chiome antibody Humanized-2 derived of clone No. 4-2 strain (WO 2014/123186); TPP-15051 represents a murine IgG1 variant thereof. TPP-30788-TPP-30791 corresponds to Böhlinger Ingelheim antibody (BI) Clone I-IV (WO 2020/225400). TPP-30792 corresponds to University Ramot antibody clone I (WO 2020/261281).				
TPP ID	Antibody Description	Sequence Region	Sequence Type	SEQ ID
TPP-11489	Chiome Prior Art (hIgG1)	LCDR2	DNA	SEQ ID NO: 15
TPP-11489	Chiome Prior Art (hIgG1)	LCDR3	DNA	SEQ ID NO: 16
TPP-11489	Chiome Prior Art (hIgG1)	Heavy Chain	PRT	SEQ ID NO: 17
TPP-11489	Chiome Prior Art (hIgG1)	Light Chain	PRT	SEQ ID NO: 18
TPP-11489	Chiome Prior Art (hIgG1)	Heavy Chain	DNA	SEQ ID NO: 19
TPP-11489	Chiome Prior Art (hIgG1)	Light Chain	DNA	SEQ ID NO: 20
TPP-15051	Chiome Prior Art (mIgG1)	VH	PRT	SEQ ID NO: 21
TPP-15051	Chiome Prior Art (mIgG1)	HCDR1	PRT	SEQ ID NO: 22
TPP-15051	Chiome Prior Art (mIgG1)	HCDR2	PRT	SEQ ID NO: 23
TPP-15051	Chiome Prior Art (mIgG1)	HCDR3	PRT	SEQ ID NO: 24
TPP-15051	Chiome Prior Art (mIgG1)	VL	PRT	SEQ ID NO: 25
TPP-15051	Chiome Prior Art (mIgG1)	LCDR1	PRT	SEQ ID NO: 26
TPP-15051	Chiome Prior Art (mIgG1)	LCDR2	PRT	SEQ ID NO: 27
TPP-15051	Chiome Prior Art (mIgG1)	LCDR3	PRT	SEQ ID NO: 28
TPP-15051	Chiome Prior Art (mIgG1)	VH	DNA	SEQ ID NO: 29
TPP-15051	Chiome Prior Art (mIgG1)	HCDR1	DNA	SEQ ID NO: 30
TPP-15051	Chiome Prior Art (mIgG1)	HCDR2	DNA	SEQ ID NO: 31
TPP-15051	Chiome Prior Art (mIgG1)	HCDR3	DNA	SEQ ID NO: 32
TPP-15051	Chiome Prior Art (mIgG1)	VL	DNA	SEQ ID NO: 33
TPP-15051	Chiome Prior Art (mIgG1)	LCDR1	DNA	SEQ ID NO: 34
TPP-15051	Chiome Prior Art (mIgG1)	LCDR2	DNA	SEQ ID NO: 35
TPP-15051	Chiome Prior Art (mIgG1)	LCDR3	DNA	SEQ ID NO: 36
TPP-15051	Chiome Prior Art (mIgG1)	Heavy Chain	PRT	SEQ ID NO: 37
TPP-15051	Chiome Prior Art (mIgG1)	Light Chain	PRT	SEQ ID NO: 38
TPP-15051	Chiome Prior Art (mIgG1)	Heavy Chain	DNA	SEQ ID NO: 39
TPP-15051	Chiome Prior Art (mIgG1)	Light Chain	DNA	SEQ ID NO: 40
TPP-15370	IgG1, hit from panning	VH	PRT	SEQ ID NO: 41
TPP-15370	IgG1, hit from panning	HCDR1	PRT	SEQ ID NO: 42
TPP-15370	IgG1, hit from panning	HCDR2	PRT	SEQ ID NO: 43
TPP-15370	IgG1, hit from panning	HCDR3	PRT	SEQ ID NO: 44
TPP-15370	IgG1, hit from panning	VL	PRT	SEQ ID NO: 45
TPP-15370	IgG1, hit from panning	LCDR1	PRT	SEQ ID NO: 46
TPP-15370	IgG1, hit from panning	LCDR2	PRT	SEQ ID NO: 47
TPP-15370	IgG1, hit from panning	LCDR3	PRT	SEQ ID NO: 48
TPP-15370	IgG1, hit from panning	VH	DNA	SEQ ID NO: 49
TPP-15370	IgG1, hit from panning	HCDR1	DNA	SEQ ID NO: 50
TPP-15370	IgG1, hit from panning	HCDR2	DNA	SEQ ID NO: 51
TPP-15370	IgG1, hit from panning	HCDR3	DNA	SEQ ID NO: 52
TPP-15370	IgG1, hit from panning	VL	DNA	SEQ ID NO: 53
TPP-15370	IgG1, hit from panning	LCDR1	DNA	SEQ ID NO: 54
TPP-15370	IgG1, hit from panning	LCDR2	DNA	SEQ ID NO: 55
TPP-15370	IgG1, hit from panning	LCDR3	DNA	SEQ ID NO: 56
TPP-15370	IgG1, hit from panning	Heavy Chain	PRT	SEQ ID NO: 57
TPP-15370	IgG1, hit from panning	Light Chain	PRT	SEQ ID NO: 58
TPP-15370	IgG1, hit from panning	Heavy Chain	DNA	SEQ ID NO: 59
TPP-15370	IgG1, hit from panning	Light Chain	DNA	SEQ ID NO: 60
TPP-15374	IgG1, hit from panning	VH	PRT	SEQ ID NO: 61
TPP-15374	IgG1, hit from panning	HCDR1	PRT	SEQ ID NO: 62
TPP-15374	IgG1, hit from panning	HCDR2	PRT	SEQ ID NO: 63
TPP-15374	IgG1, hit from panning	HCDR3	PRT	SEQ ID NO: 64
TPP-15374	IgG1, hit from panning	VL	PRT	SEQ ID NO: 65
TPP-15374	IgG1, hit from panning	LCDR1	PRT	SEQ ID NO: 66
TPP-15374	IgG1, hit from panning	LCDR2	PRT	SEQ ID NO: 67
TPP-15374	IgG1, hit from panning	LCDR3	PRT	SEQ ID NO: 68
TPP-15374	IgG1, hit from panning	VH	DNA	SEQ ID NO: 69
TPP-15374	IgG1, hit from panning	HCDR1	DNA	SEQ ID NO: 70
TPP-15374	IgG1, hit from panning	HCDR2	DNA	SEQ ID NO: 71
TPP-15374	IgG1, hit from panning	HCDR3	DNA	SEQ ID NO: 72
TPP-15374	IgG1, hit from panning	VL	DNA	SEQ ID NO: 73
TPP-15374	IgG1, hit from panning	LCDR1	DNA	SEQ ID NO: 74
TPP-15374	IgG1, hit from panning	LCDR2	DNA	SEQ ID NO: 75
TPP-15374	IgG1, hit from panning	LCDR3	DNA	SEQ ID NO: 76
TPP-15374	IgG1, hit from panning	Heavy Chain	PRT	SEQ ID NO: 77
TPP-15374	IgG1, hit from panning	Light Chain	PRT	SEQ ID NO: 78
TPP-15374	IgG1, hit from panning	Heavy Chain	DNA	SEQ ID NO: 79
TPP-15374	IgG1, hit from panning	Light Chain	DNA	SEQ ID NO: 80
TPP-17755	Samsung Prior Art F11	VH	PRT	SEQ ID NO: 81
TPP-17755	Samsung Prior Art F11	HCDR1	PRT	SEQ ID NO: 82
TPP-17755	Samsung Prior Art F11	HCDR2	PRT	SEQ ID NO: 83
TPP-17755	Samsung Prior Art F11	HCDR3	PRT	SEQ ID NO: 84
TPP-17755	Samsung Prior Art F11	VL	PRT	SEQ ID NO: 85

TABLE 1-continued

Amino acid sequences and nucleic acid sequences of preferred antibodies according to the present disclosure and of three prior art antibodies. TPP-11489 corresponds to Chiome antibody Humanized-2 derived of clone No. 4-2 strain (WO 2014/123186); TPP-15051 represents a murine IgG1 variant thereof. TPP-30788-TPP-30791 corresponds to Böhlinger Ingelheim antibody (BI) Clone I-IV (WO 2020/225400). TPP-30792 corresponds to University Ramot antibody clone I (WO 2020/261281).				
TPP ID	Antibody Description	Sequence Region	Sequence Type	SEQ ID
TPP-17755	Samsung Prior Art F11	LCDR1	PRT	SEQ ID NO: 86
TPP-17755	Samsung Prior Art F11	LCDR2	PRT	SEQ ID NO: 87
TPP-17755	Samsung Prior Art F11	LCDR3	PRT	SEQ ID NO: 88
TPP-17755	Samsung Prior Art F11	VH	DNA	SEQ ID NO: 89
TPP-17755	Samsung Prior Art F11	HCDR1	DNA	SEQ ID NO: 90
TPP-17755	Samsung Prior Art F11	HCDR2	DNA	SEQ ID NO: 91
TPP-17755	Samsung Prior Art F11	HCDR3	DNA	SEQ ID NO: 92
TPP-17755	Samsung Prior Art F11	VL	DNA	SEQ ID NO: 93
TPP-17755	Samsung Prior Art F11	LCDR1	DNA	SEQ ID NO: 94
TPP-17755	Samsung Prior Art F11	LCDR2	DNA	SEQ ID NO: 95
TPP-17755	Samsung Prior Art F11	LCDR3	DNA	SEQ ID NO: 96
TPP-17755	Samsung Prior Art F11	Heavy Chain	PRT	SEQ ID NO: 97
TPP-17755	Samsung Prior Art F11	Light Chain	PRT	SEQ ID NO: 98
TPP-17755	Samsung Prior Art F11	Heavy Chain	DNA	SEQ ID NO: 99
TPP-17755	Samsung Prior Art F11	Light Chain	DNA	SEQ ID NO: 100
TPP-18533	germline IgG1 of TPP-15374	VH	PRT	SEQ ID NO: 101
TPP-18533	germline IgG1 of TPP-15374	HCDR1	PRT	SEQ ID NO: 102
TPP-18533	germline IgG1 of TPP-15374	HCDR2	PRT	SEQ ID NO: 103
TPP-18533	germline IgG1 of TPP-15374	HCDR3	PRT	SEQ ID NO: 104
TPP-18533	germline IgG1 of TPP-15374	VL	PRT	SEQ ID NO: 105
TPP-18533	germline IgG1 of TPP-15374	LCDR1	PRT	SEQ ID NO: 106
TPP-18533	germline IgG1 of TPP-15374	LCDR2	PRT	SEQ ID NO: 107
TPP-18533	germline IgG1 of TPP-15374	LCDR3	PRT	SEQ ID NO: 108
TPP-18533	germline IgG1 of TPP-15374	VH	DNA	SEQ ID NO: 109
TPP-18533	germline IgG1 of TPP-15374	HCDR1	DNA	SEQ ID NO: 110
TPP-18533	germline IgG1 of TPP-15374	HCDR2	DNA	SEQ ID NO: 111
TPP-18533	germline IgG1 of TPP-15374	HCDR3	DNA	SEQ ID NO: 112
TPP-18533	germline IgG1 of TPP-15374	VL	DNA	SEQ ID NO: 113
TPP-18533	germline IgG1 of TPP-15374	LCDR1	DNA	SEQ ID NO: 114
TPP-18533	germline IgG1 of TPP-15374	LCDR2	DNA	SEQ ID NO: 115
TPP-18533	germline IgG1 of TPP-15374	LCDR3	DNA	SEQ ID NO: 116
TPP-18533	germline IgG1 of TPP-15374	Heavy Chain	PRT	SEQ ID NO: 117
TPP-18533	germline IgG1 of TPP-15374	Light Chain	PRT	SEQ ID NO: 118
TPP-18533	germline IgG1 of TPP-15374	Heavy Chain	DNA	SEQ ID NO: 119
TPP-18533	germline IgG1 of TPP-15374	Light Chain	DNA	SEQ ID NO: 120
TPP-21565	germline IgG1 of TPP-15370	VH	PRT	SEQ ID NO: 121
TPP-21565	germline IgG1 of TPP-15370	HCDR1	PRT	SEQ ID NO: 122
TPP-21565	germline IgG1 of TPP-15370	HCDR2	PRT	SEQ ID NO: 123
TPP-21565	germline IgG1 of TPP-15370	HCDR3	PRT	SEQ ID NO: 124
TPP-21565	germline IgG1 of TPP-15370	VL	PRT	SEQ ID NO: 125
TPP-21565	germline IgG1 of TPP-15370	LCDR1	PRT	SEQ ID NO: 126
TPP-21565	germline IgG1 of TPP-15370	LCDR2	PRT	SEQ ID NO: 127
TPP-21565	germline IgG1 of TPP-15370	LCDR3	PRT	SEQ ID NO: 128
TPP-21565	germline IgG1 of TPP-15370	VH	DNA	SEQ ID NO: 129
TPP-21565	germline IgG1 of TPP-15370	HCDR1	DNA	SEQ ID NO: 130
TPP-21565	germline IgG1 of TPP-15370	HCDR2	DNA	SEQ ID NO: 131
TPP-21565	germline IgG1 of TPP-15370	HCDR3	DNA	SEQ ID NO: 132
TPP-21565	germline IgG1 of TPP-15370	VL	DNA	SEQ ID NO: 133
TPP-21565	germline IgG1 of TPP-15370	LCDR1	DNA	SEQ ID NO: 134
TPP-21565	germline IgG1 of TPP-15370	LCDR2	DNA	SEQ ID NO: 135
TPP-21565	germline IgG1 of TPP-15370	LCDR3	DNA	SEQ ID NO: 136
TPP-21565	germline IgG1 of TPP-15370	Heavy Chain	PRT	SEQ ID NO: 137
TPP-21565	germline IgG1 of TPP-15370	Light Chain	PRT	SEQ ID NO: 138
TPP-21565	germline IgG1 of TPP-15370	Heavy Chain	DNA	SEQ ID NO: 139
TPP-21565	germline IgG1 of TPP-15370	Light Chain	DNA	SEQ ID NO: 140
TPP-23298	Recombi Variant of TPP-15370	VH	PRT	SEQ ID NO: 141
TPP-23298	Recombi Variant of TPP-15370	HCDR1	PRT	SEQ ID NO: 142
TPP-23298	Recombi Variant of TPP-15370	HCDR2	PRT	SEQ ID NO: 143
TPP-23298	Recombi Variant of TPP-15370	HCDR3	PRT	SEQ ID NO: 144
TPP-23298	Recombi Variant of TPP-15370	VL	PRT	SEQ ID NO: 145
TPP-23298	Recombi Variant of TPP-15370	LCDR1	PRT	SEQ ID NO: 146
TPP-23298	Recombi Variant of TPP-15370	LCDR2	PRT	SEQ ID NO: 147
TPP-23298	Recombi Variant of TPP-15370	LCDR3	PRT	SEQ ID NO: 148
TPP-23298	Recombi Variant of TPP-15370	VH	DNA	SEQ ID NO: 149
TPP-23298	Recombi Variant of TPP-15370	HCDR1	DNA	SEQ ID NO: 150
TPP-23298	Recombi Variant of TPP-15370	HCDR2	DNA	SEQ ID NO: 151
TPP-23298	Recombi Variant of TPP-15370	HCDR3	DNA	SEQ ID NO: 152
TPP-23298	Recombi Variant of TPP-15370	VL	DNA	SEQ ID NO: 153
TPP-23298	Recombi Variant of TPP-15370	LCDR1	DNA	SEQ ID NO: 154
TPP-23298	Recombi Variant of TPP-15370	LCDR2	DNA	SEQ ID NO: 155
TPP-23298	Recombi Variant of TPP-15370	LCDR3	DNA	SEQ ID NO: 156

TABLE 1-continued

Amino acid sequences and nucleic acid sequences of preferred antibodies according to the present disclosure and of three prior art antibodies. TPP-11489 corresponds to Chiome antibody Humanized-2 derived of clone No. 4-2 strain (WO 2014/123186); TPP-15051 represents a murine IgG1 variant thereof. TPP-30788-TPP-30791 corresponds to Böhlinger Ingelheim antibody (BI) Clone I-IV (WO 2020/225400). TPP-30792 corresponds to University Ramot antibody clone I (WO 2020/261281).					
TPP ID	Antibody Description	Sequence Region	Sequence Type	SEQ ID	
TPP-23298	Recombi Variant of TPP-15370	Heavy Chain	PRT	SEQ ID NO: 157	
TPP-23298	Recombi Variant of TPP-15370	Light Chain	PRT	SEQ ID NO: 158	
TPP-23298	Recombi Variant of TPP-15370	Heavy Chain	DNA	SEQ ID NO: 159	
TPP-23298	Recombi Variant of TPP-15370	Light Chain	DNA	SEQ ID NO: 160	
TPP-23334	Recombi Variant of TPP-15370	VH	PRT	SEQ ID NO: 161	
TPP-23334	Recombi Variant of TPP-15370	HCDR1	PRT	SEQ ID NO: 162	
TPP-23334	Recombi Variant of TPP-15370	HCDR2	PRT	SEQ ID NO: 163	
TPP-23334	Recombi Variant of TPP-15370	HCDR3	PRT	SEQ ID NO: 164	
TPP-23334	Recombi Variant of TPP-15370	VL	PRT	SEQ ID NO: 165	
TPP-23334	Recombi Variant of TPP-15370	LCDR1	PRT	SEQ ID NO: 166	
TPP-23334	Recombi Variant of TPP-15370	LCDR2	PRT	SEQ ID NO: 167	
TPP-23334	Recombi Variant of TPP-15370	LCDR3	PRT	SEQ ID NO: 168	
TPP-23334	Recombi Variant of TPP-15370	VH	DNA	SEQ ID NO: 169	
TPP-23334	Recombi Variant of TPP-15370	HCDR1	DNA	SEQ ID NO: 170	
TPP-23334	Recombi Variant of TPP-15370	HCDR2	DNA	SEQ ID NO: 171	
TPP-23334	Recombi Variant of TPP-15370	HCDR3	DNA	SEQ ID NO: 172	
TPP-23334	Recombi Variant of TPP-15370	VL	DNA	SEQ ID NO: 173	
TPP-23334	Recombi Variant of TPP-15370	LCDR1	DNA	SEQ ID NO: 174	
TPP-23334	Recombi Variant of TPP-15370	LCDR2	DNA	SEQ ID NO: 175	
TPP-23334	Recombi Variant of TPP-15370	LCDR3	DNA	SEQ ID NO: 176	
TPP-23334	Recombi Variant of TPP-15370	Heavy Chain	PRT	SEQ ID NO: 177	
TPP-23334	Recombi Variant of TPP-15370	Light Chain	PRT	SEQ ID NO: 178	
TPP-23334	Recombi Variant of TPP-15370	Heavy Chain	DNA	SEQ ID NO: 179	
TPP-23334	Recombi Variant of TPP-15370	Light Chain	DNA	SEQ ID NO: 180	
TPP-23337	Recombi Variant of TPP-15370	VH	PRT	SEQ ID NO: 181	
TPP-23337	Recombi Variant of TPP-15370	HCDR1	PRT	SEQ ID NO: 182	
TPP-23337	Recombi Variant of TPP-15370	HCDR2	PRT	SEQ ID NO: 183	
TPP-23337	Recombi Variant of TPP-15370	HCDR3	PRT	SEQ ID NO: 184	
TPP-23337	Recombi Variant of TPP-15370	VL	PRT	SEQ ID NO: 185	
TPP-23337	Recombi Variant of TPP-15370	LCDR1	PRT	SEQ ID NO: 186	
TPP-23337	Recombi Variant of TPP-15370	LCDR2	PRT	SEQ ID NO: 187	
TPP-23337	Recombi Variant of TPP-15370	LCDR3	PRT	SEQ ID NO: 188	
TPP-23337	Recombi Variant of TPP-15370	VH	DNA	SEQ ID NO: 189	
TPP-23337	Recombi Variant of TPP-15370	HCDR1	DNA	SEQ ID NO: 190	
TPP-23337	Recombi Variant of TPP-15370	HCDR2	DNA	SEQ ID NO: 191	
TPP-23337	Recombi Variant of TPP-15370	HCDR3	DNA	SEQ ID NO: 192	
TPP-23337	Recombi Variant of TPP-15370	VL	DNA	SEQ ID NO: 193	
TPP-23337	Recombi Variant of TPP-15370	LCDR1	DNA	SEQ ID NO: 194	
TPP-23337	Recombi Variant of TPP-15370	LCDR2	DNA	SEQ ID NO: 195	
TPP-23337	Recombi Variant of TPP-15370	LCDR3	DNA	SEQ ID NO: 196	
TPP-23337	Recombi Variant of TPP-15370	Heavy Chain	PRT	SEQ ID NO: 197	
TPP-23337	Recombi Variant of TPP-15370	Light Chain	PRT	SEQ ID NO: 198	
TPP-23337	Recombi Variant of TPP-15370	Heavy Chain	DNA	SEQ ID NO: 199	
TPP-23337	Recombi Variant of TPP-15370	Light Chain	DNA	SEQ ID NO: 200	
TPP-23338	Recombi Variant of TPP-15370	VH	PRT	SEQ ID NO: 201	
TPP-23338	Recombi Variant of TPP-15370	HCDR1	PRT	SEQ ID NO: 202	
TPP-23338	Recombi Variant of TPP-15370	HCDR2	PRT	SEQ ID NO: 203	
TPP-23338	Recombi Variant of TPP-15370	HCDR3	PRT	SEQ ID NO: 204	
TPP-23338	Recombi Variant of TPP-15370	VL	PRT	SEQ ID NO: 205	
TPP-23338	Recombi Variant of TPP-15370	LCDR1	PRT	SEQ ID NO: 206	
TPP-23338	Recombi Variant of TPP-15370	LCDR2	PRT	SEQ ID NO: 207	
TPP-23338	Recombi Variant of TPP-15370	LCDR3	PRT	SEQ ID NO: 208	
TPP-23338	Recombi Variant of TPP-15370	VH	DNA	SEQ ID NO: 209	
TPP-23338	Recombi Variant of TPP-15370	HCDR1	DNA	SEQ ID NO: 210	
TPP-23338	Recombi Variant of TPP-15370	HCDR2	DNA	SEQ ID NO: 211	
TPP-23338	Recombi Variant of TPP-15370	HCDR3	DNA	SEQ ID NO: 212	
TPP-23338	Recombi Variant of TPP-15370	VL	DNA	SEQ ID NO: 213	
TPP-23338	Recombi Variant of TPP-15370	LCDR1	DNA	SEQ ID NO: 214	
TPP-23338	Recombi Variant of TPP-15370	LCDR2	DNA	SEQ ID NO: 215	
TPP-23338	Recombi Variant of TPP-15370	LCDR3	DNA	SEQ ID NO: 216	
TPP-23338	Recombi Variant of TPP-15370	Heavy Chain	PRT	SEQ ID NO: 217	
TPP-23338	Recombi Variant of TPP-15370	Light Chain	PRT	SEQ ID NO: 218	
TPP-23338	Recombi Variant of TPP-15370	Heavy Chain	DNA	SEQ ID NO: 219	
TPP-23338	Recombi Variant of TPP-15370	Light Chain	DNA	SEQ ID NO: 220	
TPP-23340	Recombi Variant of TPP-15370	VH	PRT	SEQ ID NO: 221	
TPP-23340	Recombi Variant of TPP-15370	HCDR1	PRT	SEQ ID NO: 222	
TPP-23340	Recombi Variant of TPP-15370	HCDR2	PRT	SEQ ID NO: 223	
TPP-23340	Recombi Variant of TPP-15370	HCDR3	PRT	SEQ ID NO: 224	
TPP-23340	Recombi Variant of TPP-15370	VL	PRT	SEQ ID NO: 225	
TPP-23340	Recombi Variant of TPP-15370	LCDR1	PRT	SEQ ID NO: 226	
TPP-23340	Recombi Variant of TPP-15370	LCDR2	PRT	SEQ ID NO: 227	

TABLE 1-continued

Amino acid sequences and nucleic acid sequences of preferred antibodies according to the present disclosure and of three prior art antibodies. TPP-11489 corresponds to Chiome antibody Humanized-2 derived of clone No. 4-2 strain (WO 2014/123186); TPP-15051 represents a murine IgG1 variant thereof. TPP-30788-TPP-30791 corresponds to Böhlinger Ingelheim antibody (BI) Clone I-IV (WO 2020/225400). TPP-30792 corresponds to University Ramot antibody clone I (WO 2020/261281).				
TPP ID	Antibody Description	Sequence Region	Sequence Type	SEQ ID
TPP-23340	Recombi Variant of TPP-15370	LCDR3	PRT	SEQ ID NO: 228
TPP-23340	Recombi Variant of TPP-15370	VH	DNA	SEQ ID NO: 229
TPP-23340	Recombi Variant of TPP-15370	HCDR1	DNA	SEQ ID NO: 230
TPP-23340	Recombi Variant of TPP-15370	HCDR2	DNA	SEQ ID NO: 231
TPP-23340	Recombi Variant of TPP-15370	HCDR3	DNA	SEQ ID NO: 232
TPP-23340	Recombi Variant of TPP-15370	VL	DNA	SEQ ID NO: 233
TPP-23340	Recombi Variant of TPP-15370	LCDR1	DNA	SEQ ID NO: 234
TPP-23340	Recombi Variant of TPP-15370	LCDR2	DNA	SEQ ID NO: 235
TPP-23340	Recombi Variant of TPP-15370	LCDR3	DNA	SEQ ID NO: 236
TPP-23340	Recombi Variant of TPP-15370	Heavy Chain	PRT	SEQ ID NO: 237
TPP-23340	Recombi Variant of TPP-15370	Light Chain	PRT	SEQ ID NO: 238
TPP-23340	Recombi Variant of TPP-15370	Heavy Chain	DNA	SEQ ID NO: 239
TPP-23340	Recombi Variant of TPP-15370	Light Chain	DNA	SEQ ID NO: 240
TPP-23341	Recombi Variant of TPP-15370	VH	PRT	SEQ ID NO: 241
TPP-23341	Recombi Variant of TPP-15370	HCDR1	PRT	SEQ ID NO: 242
TPP-23341	Recombi Variant of TPP-15370	HCDR2	PRT	SEQ ID NO: 243
TPP-23341	Recombi Variant of TPP-15370	HCDR3	PRT	SEQ ID NO: 244
TPP-23341	Recombi Variant of TPP-15370	VL	PRT	SEQ ID NO: 245
TPP-23341	Recombi Variant of TPP-15370	LCDR1	PRT	SEQ ID NO: 246
TPP-23341	Recombi Variant of TPP-15370	LCDR2	PRT	SEQ ID NO: 247
TPP-23341	Recombi Variant of TPP-15370	LCDR3	PRT	SEQ ID NO: 248
TPP-23341	Recombi Variant of TPP-15370	VH	DNA	SEQ ID NO: 249
TPP-23341	Recombi Variant of TPP-15370	HCDR1	DNA	SEQ ID NO: 250
TPP-23341	Recombi Variant of TPP-15370	HCDR2	DNA	SEQ ID NO: 251
TPP-23341	Recombi Variant of TPP-15370	HCDR3	DNA	SEQ ID NO: 252
TPP-23341	Recombi Variant of TPP-15370	VL	DNA	SEQ ID NO: 253
TPP-23341	Recombi Variant of TPP-15370	LCDR1	DNA	SEQ ID NO: 254
TPP-23341	Recombi Variant of TPP-15370	LCDR2	DNA	SEQ ID NO: 255
TPP-23341	Recombi Variant of TPP-15370	LCDR3	DNA	SEQ ID NO: 256
TPP-23341	Recombi Variant of TPP-15370	Heavy Chain	PRT	SEQ ID NO: 257
TPP-23341	Recombi Variant of TPP-15370	Light Chain	PRT	SEQ ID NO: 258
TPP-23341	Recombi Variant of TPP-15370	Heavy Chain	DNA	SEQ ID NO: 259
TPP-23341	Recombi Variant of TPP-15370	Light Chain	DNA	SEQ ID NO: 260
TPP-23345	Recombi Variant of TPP-15370	VH	PRT	SEQ ID NO: 261
TPP-23345	Recombi Variant of TPP-15370	HCDR1	PRT	SEQ ID NO: 262
TPP-23345	Recombi Variant of TPP-15370	HCDR2	PRT	SEQ ID NO: 263
TPP-23345	Recombi Variant of TPP-15370	HCDR3	PRT	SEQ ID NO: 264
TPP-23345	Recombi Variant of TPP-15370	VL	PRT	SEQ ID NO: 265
TPP-23345	Recombi Variant of TPP-15370	LCDR1	PRT	SEQ ID NO: 266
TPP-23345	Recombi Variant of TPP-15370	LCDR2	PRT	SEQ ID NO: 267
TPP-23345	Recombi Variant of TPP-15370	LCDR3	PRT	SEQ ID NO: 268
TPP-23345	Recombi Variant of TPP-15370	VH	DNA	SEQ ID NO: 269
TPP-23345	Recombi Variant of TPP-15370	HCDR1	DNA	SEQ ID NO: 270
TPP-23345	Recombi Variant of TPP-15370	HCDR2	DNA	SEQ ID NO: 271
TPP-23345	Recombi Variant of TPP-15370	HCDR3	DNA	SEQ ID NO: 272
TPP-23345	Recombi Variant of TPP-15370	VL	DNA	SEQ ID NO: 273
TPP-23345	Recombi Variant of TPP-15370	LCDR1	DNA	SEQ ID NO: 274
TPP-23345	Recombi Variant of TPP-15370	LCDR2	DNA	SEQ ID NO: 275
TPP-23345	Recombi Variant of TPP-15370	LCDR3	DNA	SEQ ID NO: 276
TPP-23345	Recombi Variant of TPP-15370	Heavy Chain	PRT	SEQ ID NO: 277
TPP-23345	Recombi Variant of TPP-15370	Light Chain	PRT	SEQ ID NO: 278
TPP-23345	Recombi Variant of TPP-15370	Heavy Chain	DNA	SEQ ID NO: 279
TPP-23345	Recombi Variant of TPP-15370	Light Chain	DNA	SEQ ID NO: 280
TPP-23346	Recombi Variant of TPP-15370	VH	PRT	SEQ ID NO: 281
TPP-23346	Recombi Variant of TPP-15370	HCDR1	PRT	SEQ ID NO: 282
TPP-23346	Recombi Variant of TPP-15370	HCDR2	PRT	SEQ ID NO: 283
TPP-23346	Recombi Variant of TPP-15370	HCDR3	PRT	SEQ ID NO: 284
TPP-23346	Recombi Variant of TPP-15370	VL	PRT	SEQ ID NO: 285
TPP-23346	Recombi Variant of TPP-15370	LCDR1	PRT	SEQ ID NO: 286
TPP-23346	Recombi Variant of TPP-15370	LCDR2	PRT	SEQ ID NO: 287
TPP-23346	Recombi Variant of TPP-15370	LCDR3	PRT	SEQ ID NO: 288
TPP-23346	Recombi Variant of TPP-15370	VH	DNA	SEQ ID NO: 289
TPP-23346	Recombi Variant of TPP-15370	HCDR1	DNA	SEQ ID NO: 290
TPP-23346	Recombi Variant of TPP-15370	HCDR2	DNA	SEQ ID NO: 291
TPP-23346	Recombi Variant of TPP-15370	HCDR3	DNA	SEQ ID NO: 292
TPP-23346	Recombi Variant of TPP-15370	VL	DNA	SEQ ID NO: 293
TPP-23346	Recombi Variant of TPP-15370	LCDR1	DNA	SEQ ID NO: 294
TPP-23346	Recombi Variant of TPP-15370	LCDR2	DNA	SEQ ID NO: 295
TPP-23346	Recombi Variant of TPP-15370	LCDR3	DNA	SEQ ID NO: 296
TPP-23346	Recombi Variant of TPP-15370	Heavy Chain	PRT	SEQ ID NO: 297
TPP-23346	Recombi Variant of TPP-15370	Light Chain	PRT	SEQ ID NO: 298

TABLE 1-continued

Amino acid sequences and nucleic acid sequences of preferred antibodies according to the present disclosure and of three prior art antibodies. TPP-11489 corresponds to Chiome antibody Humanized-2 derived of clone No. 4-2 strain (WO 2014/123186); TPP-15051 represents a murine IgG1 variant thereof. TPP-30788-TPP-30791 corresponds to Böhlinger Ingelheim antibody (BI) Clone I-IV (WO 2020/225400). TPP-30792 corresponds to University Ramot antibody clone I (WO 2020/261281).					
TPP ID	Antibody Description	Sequence Region	Sequence Type	SEQ ID	
TPP-23346	Recombi Variant of TPP-15370	Heavy Chain	DNA	SEQ ID NO: 299	
TPP-23346	Recombi Variant of TPP-15370	Light Chain	DNA	SEQ ID NO: 300	
TPP-23347	Recombi Variant of TPP-15370	VH	PRT	SEQ ID NO: 301	
TPP-23347	Recombi Variant of TPP-15370	HCDR1	PRT	SEQ ID NO: 302	
TPP-23347	Recombi Variant of TPP-15370	HCDR2	PRT	SEQ ID NO: 303	
TPP-23347	Recombi Variant of TPP-15370	HCDR3	PRT	SEQ ID NO: 304	
TPP-23347	Recombi Variant of TPP-15370	VL	PRT	SEQ ID NO: 305	
TPP-23347	Recombi Variant of TPP-15370	LCDR1	PRT	SEQ ID NO: 306	
TPP-23347	Recombi Variant of TPP-15370	LCDR2	PRT	SEQ ID NO: 307	
TPP-23347	Recombi Variant of TPP-15370	LCDR3	PRT	SEQ ID NO: 308	
TPP-23347	Recombi Variant of TPP-15370	VH	DNA	SEQ ID NO: 309	
TPP-23347	Recombi Variant of TPP-15370	HCDR1	DNA	SEQ ID NO: 310	
TPP-23347	Recombi Variant of TPP-15370	HCDR2	DNA	SEQ ID NO: 311	
TPP-23347	Recombi Variant of TPP-15370	HCDR3	DNA	SEQ ID NO: 312	
TPP-23347	Recombi Variant of TPP-15370	VL	DNA	SEQ ID NO: 313	
TPP-23347	Recombi Variant of TPP-15370	LCDR1	DNA	SEQ ID NO: 314	
TPP-23347	Recombi Variant of TPP-15370	LCDR2	DNA	SEQ ID NO: 315	
TPP-23347	Recombi Variant of TPP-15370	LCDR3	DNA	SEQ ID NO: 316	
TPP-23347	Recombi Variant of TPP-15370	Heavy Chain	PRT	SEQ ID NO: 317	
TPP-23347	Recombi Variant of TPP-15370	Light Chain	PRT	SEQ ID NO: 318	
TPP-23347	Recombi Variant of TPP-15370	Heavy Chain	DNA	SEQ ID NO: 319	
TPP-23347	Recombi Variant of TPP-15370	Light Chain	DNA	SEQ ID NO: 320	
TPP-23373	Recombi Variant of TPP-15370	VH	PRT	SEQ ID NO: 321	
TPP-23373	Recombi Variant of TPP-15370	HCDR1	PRT	SEQ ID NO: 322	
TPP-23373	Recombi Variant of TPP-15370	HCDR2	PRT	SEQ ID NO: 323	
TPP-23373	Recombi Variant of TPP-15370	HCDR3	PRT	SEQ ID NO: 324	
TPP-23373	Recombi Variant of TPP-15370	VL	PRT	SEQ ID NO: 325	
TPP-23373	Recombi Variant of TPP-15370	LCDR1	PRT	SEQ ID NO: 326	
TPP-23373	Recombi Variant of TPP-15370	LCDR2	PRT	SEQ ID NO: 327	
TPP-23373	Recombi Variant of TPP-15370	LCDR3	PRT	SEQ ID NO: 328	
TPP-23373	Recombi Variant of TPP-15370	VH	DNA	SEQ ID NO: 329	
TPP-23373	Recombi Variant of TPP-15370	HCDR1	DNA	SEQ ID NO: 330	
TPP-23373	Recombi Variant of TPP-15370	HCDR2	DNA	SEQ ID NO: 331	
TPP-23373	Recombi Variant of TPP-15370	HCDR3	DNA	SEQ ID NO: 332	
TPP-23373	Recombi Variant of TPP-15370	VL	DNA	SEQ ID NO: 333	
TPP-23373	Recombi Variant of TPP-15370	LCDR1	DNA	SEQ ID NO: 334	
TPP-23373	Recombi Variant of TPP-15370	LCDR2	DNA	SEQ ID NO: 335	
TPP-23373	Recombi Variant of TPP-15370	LCDR3	DNA	SEQ ID NO: 336	
TPP-23373	Recombi Variant of TPP-15370	Heavy Chain	PRT	SEQ ID NO: 337	
TPP-23373	Recombi Variant of TPP-15370	Light Chain	PRT	SEQ ID NO: 338	
TPP-23373	Recombi Variant of TPP-15370	Heavy Chain	DNA	SEQ ID NO: 339	
TPP-23373	Recombi Variant of TPP-15370	Light Chain	DNA	SEQ ID NO: 340	
TPP-23374	Recombi Variant of TPP-15370	VH	PRT	SEQ ID NO: 341	
TPP-23374	Recombi Variant of TPP-15370	HCDR1	PRT	SEQ ID NO: 342	
TPP-23374	Recombi Variant of TPP-15370	HCDR2	PRT	SEQ ID NO: 343	
TPP-23374	Recombi Variant of TPP-15370	HCDR3	PRT	SEQ ID NO: 344	
TPP-23374	Recombi Variant of TPP-15370	VL	PRT	SEQ ID NO: 345	
TPP-23374	Recombi Variant of TPP-15370	LCDR1	PRT	SEQ ID NO: 346	
TPP-23374	Recombi Variant of TPP-15370	LCDR2	PRT	SEQ ID NO: 347	
TPP-23374	Recombi Variant of TPP-15370	LCDR3	PRT	SEQ ID NO: 348	
TPP-23374	Recombi Variant of TPP-15370	VH	DNA	SEQ ID NO: 349	
TPP-23374	Recombi Variant of TPP-15370	HCDR1	DNA	SEQ ID NO: 350	
TPP-23374	Recombi Variant of TPP-15370	HCDR2	DNA	SEQ ID NO: 351	
TPP-23374	Recombi Variant of TPP-15370	HCDR3	DNA	SEQ ID NO: 352	
TPP-23374	Recombi Variant of TPP-15370	VL	DNA	SEQ ID NO: 353	
TPP-23374	Recombi Variant of TPP-15370	LCDR1	DNA	SEQ ID NO: 354	
TPP-23374	Recombi Variant of TPP-15370	LCDR2	DNA	SEQ ID NO: 355	
TPP-23374	Recombi Variant of TPP-15370	LCDR3	DNA	SEQ ID NO: 356	
TPP-23374	Recombi Variant of TPP-15370	Heavy Chain	PRT	SEQ ID NO: 357	
TPP-23374	Recombi Variant of TPP-15370	Light Chain	PRT	SEQ ID NO: 358	
TPP-23374	Recombi Variant of TPP-15370	Heavy Chain	DNA	SEQ ID NO: 359	
TPP-23374	Recombi Variant of TPP-15370	Light Chain	DNA	SEQ ID NO: 360	
TPP-23375	Recombi Variant of TPP-15370	VH	PRT	SEQ ID NO: 361	
TPP-23375	Recombi Variant of TPP-15370	HCDR1	PRT	SEQ ID NO: 362	
TPP-23375	Recombi Variant of TPP-15370	HCDR2	PRT	SEQ ID NO: 363	
TPP-23375	Recombi Variant of TPP-15370	HCDR3	PRT	SEQ ID NO: 364	
TPP-23375	Recombi Variant of TPP-15370	VL	PRT	SEQ ID NO: 365	
TPP-23375	Recombi Variant of TPP-15370	LCDR1	PRT	SEQ ID NO: 366	
TPP-23375	Recombi Variant of TPP-15370	LCDR2	PRT	SEQ ID NO: 367	
TPP-23375	Recombi Variant of TPP-15370	LCDR3	PRT	SEQ ID NO: 368	
TPP-23375	Recombi Variant of TPP-15370	VH	DNA	SEQ ID NO: 369	

TABLE 1-continued

Amino acid sequences and nucleic acid sequences of preferred antibodies according to the present disclosure and of three prior art antibodies. TPP-11489 corresponds to Chiome antibody Humanized-2 derived of clone No. 4-2 strain (WO 2014/123186); TPP-15051 represents a murine IgG1 variant thereof. TPP-30788-TPP-30791 corresponds to Böhlinger Ingelheim antibody (BI) Clone I-IV (WO 2020/225400). TPP-30792 corresponds to University Ramot antibody clone I (WO 2020/261281).				
TPP ID	Antibody Description	Sequence Region	Sequence Type	SEQ ID
TPP-23375	Recombi Variant of TPP-15370	HCDR1	DNA	SEQ ID NO: 370
TPP-23375	Recombi Variant of TPP-15370	HCDR2	DNA	SEQ ID NO: 371
TPP-23375	Recombi Variant of TPP-15370	HCDR3	DNA	SEQ ID NO: 372
TPP-23375	Recombi Variant of TPP-15370	VL	DNA	SEQ ID NO: 373
TPP-23375	Recombi Variant of TPP-15370	LCDR1	DNA	SEQ ID NO: 374
TPP-23375	Recombi Variant of TPP-15370	LCDR2	DNA	SEQ ID NO: 375
TPP-23375	Recombi Variant of TPP-15370	LCDR3	DNA	SEQ ID NO: 376
TPP-23375	Recombi Variant of TPP-15370	Heavy Chain	PRT	SEQ ID NO: 377
TPP-23375	Recombi Variant of TPP-15370	Light Chain	PRT	SEQ ID NO: 378
TPP-23375	Recombi Variant of TPP-15370	Heavy Chain	DNA	SEQ ID NO: 379
TPP-23375	Recombi Variant of TPP-15370	Light Chain	DNA	SEQ ID NO: 380
TPP-25064	Recombi Variant of TPP-15374	VH	PRT	SEQ ID NO: 381
TPP-25064	Recombi Variant of TPP-15374	HCDR1	PRT	SEQ ID NO: 382
TPP-25064	Recombi Variant of TPP-15374	HCDR2	PRT	SEQ ID NO: 383
TPP-25064	Recombi Variant of TPP-15374	HCDR3	PRT	SEQ ID NO: 384
TPP-25064	Recombi Variant of TPP-15374	VL	PRT	SEQ ID NO: 385
TPP-25064	Recombi Variant of TPP-15374	LCDR1	PRT	SEQ ID NO: 386
TPP-25064	Recombi Variant of TPP-15374	LCDR2	PRT	SEQ ID NO: 387
TPP-25064	Recombi Variant of TPP-15374	LCDR3	PRT	SEQ ID NO: 388
TPP-25064	Recombi Variant of TPP-15374	VH	DNA	SEQ ID NO: 389
TPP-25064	Recombi Variant of TPP-15374	HCDR1	DNA	SEQ ID NO: 390
TPP-25064	Recombi Variant of TPP-15374	HCDR2	DNA	SEQ ID NO: 391
TPP-25064	Recombi Variant of TPP-15374	HCDR3	DNA	SEQ ID NO: 392
TPP-25064	Recombi Variant of TPP-15374	VL	DNA	SEQ ID NO: 393
TPP-25064	Recombi Variant of TPP-15374	LCDR1	DNA	SEQ ID NO: 394
TPP-25064	Recombi Variant of TPP-15374	LCDR2	DNA	SEQ ID NO: 395
TPP-25064	Recombi Variant of TPP-15374	LCDR3	DNA	SEQ ID NO: 396
TPP-25064	Recombi Variant of TPP-15374	Heavy Chain	PRT	SEQ ID NO: 397
TPP-25064	Recombi Variant of TPP-15374	Light Chain	PRT	SEQ ID NO: 398
TPP-25064	Recombi Variant of TPP-15374	Heavy Chain	DNA	SEQ ID NO: 399
TPP-25064	Recombi Variant of TPP-15374	Light Chain	DNA	SEQ ID NO: 400
TPP-25224	Recombi Variant of TPP-15374	VH	PRT	SEQ ID NO: 401
TPP-25224	Recombi Variant of TPP-15374	HCDR1	PRT	SEQ ID NO: 402
TPP-25224	Recombi Variant of TPP-15374	HCDR2	PRT	SEQ ID NO: 403
TPP-25224	Recombi Variant of TPP-15374	HCDR3	PRT	SEQ ID NO: 404
TPP-25224	Recombi Variant of TPP-15374	VL	PRT	SEQ ID NO: 405
TPP-25224	Recombi Variant of TPP-15374	LCDR1	PRT	SEQ ID NO: 406
TPP-25224	Recombi Variant of TPP-15374	LCDR2	PRT	SEQ ID NO: 407
TPP-25224	Recombi Variant of TPP-15374	LCDR3	PRT	SEQ ID NO: 408
TPP-25224	Recombi Variant of TPP-15374	VH	DNA	SEQ ID NO: 409
TPP-25224	Recombi Variant of TPP-15374	HCDR1	DNA	SEQ ID NO: 410
TPP-25224	Recombi Variant of TPP-15374	HCDR2	DNA	SEQ ID NO: 411
TPP-25224	Recombi Variant of TPP-15374	HCDR3	DNA	SEQ ID NO: 412
TPP-25224	Recombi Variant of TPP-15374	VL	DNA	SEQ ID NO: 413
TPP-25224	Recombi Variant of TPP-15374	LCDR1	DNA	SEQ ID NO: 414
TPP-25224	Recombi Variant of TPP-15374	LCDR2	DNA	SEQ ID NO: 415
TPP-25224	Recombi Variant of TPP-15374	LCDR3	DNA	SEQ ID NO: 416
TPP-25224	Recombi Variant of TPP-15374	Heavy Chain	PRT	SEQ ID NO: 417
TPP-25224	Recombi Variant of TPP-15374	Light Chain	PRT	SEQ ID NO: 418
TPP-25224	Recombi Variant of TPP-15374	Heavy Chain	DNA	SEQ ID NO: 419
TPP-25224	Recombi Variant of TPP-15374	Light Chain	DNA	SEQ ID NO: 420
TPP-25248	Recombi Variant of TPP-15374	VH	PRT	SEQ ID NO: 421
TPP-25248	Recombi Variant of TPP-15374	HCDR1	PRT	SEQ ID NO: 422
TPP-25248	Recombi Variant of TPP-15374	HCDR2	PRT	SEQ ID NO: 423
TPP-25248	Recombi Variant of TPP-15374	HCDR3	PRT	SEQ ID NO: 424
TPP-25248	Recombi Variant of TPP-15374	VL	PRT	SEQ ID NO: 425
TPP-25248	Recombi Variant of TPP-15374	LCDR1	PRT	SEQ ID NO: 426
TPP-25248	Recombi Variant of TPP-15374	LCDR2	PRT	SEQ ID NO: 427
TPP-25248	Recombi Variant of TPP-15374	LCDR3	PRT	SEQ ID NO: 428
TPP-25248	Recombi Variant of TPP-15374	VH	DNA	SEQ ID NO: 429
TPP-25248	Recombi Variant of TPP-15374	HCDR1	DNA	SEQ ID NO: 430
TPP-25248	Recombi Variant of TPP-15374	HCDR2	DNA	SEQ ID NO: 431
TPP-25248	Recombi Variant of TPP-15374	HCDR3	DNA	SEQ ID NO: 432
TPP-25248	Recombi Variant of TPP-15374	VL	DNA	SEQ ID NO: 433
TPP-25248	Recombi Variant of TPP-15374	LCDR1	DNA	SEQ ID NO: 434
TPP-25248	Recombi Variant of TPP-15374	LCDR2	DNA	SEQ ID NO: 435
TPP-25248	Recombi Variant of TPP-15374	LCDR3	DNA	SEQ ID NO: 436
TPP-25248	Recombi Variant of TPP-15374	Heavy Chain	PRT	SEQ ID NO: 437
TPP-25248	Recombi Variant of TPP-15374	Light Chain	PRT	SEQ ID NO: 438
TPP-25248	Recombi Variant of TPP-15374	Heavy Chain	DNA	SEQ ID NO: 439
TPP-25248	Recombi Variant of TPP-15374	Light Chain	DNA	SEQ ID NO: 440

TABLE 1-continued

Amino acid sequences and nucleic acid sequences of preferred antibodies according to the present disclosure and of three prior art antibodies. TPP-11489 corresponds to Chiome antibody Humanized-2 derived of clone No. 4-2 strain (WO 2014/123186); TPP-15051 represents a murine IgG1 variant thereof. TPP-30788-TPP-30791 corresponds to Böhlinger Ingelheim antibody (BI) Clone I-IV (WO 2020/225400). TPP-30792 corresponds to University Ramot antibody clone I (WO 2020/261281).				
TPP ID	Antibody Description	Sequence Region	Sequence Type	SEQ ID
TPP-25255	Recombi Variant of TPP-15374	VH	PRT	SEQ ID NO: 441
TPP-25255	Recombi Variant of TPP-15374	HCDR1	PRT	SEQ ID NO: 442
TPP-25255	Recombi Variant of TPP-15374	HCDR2	PRT	SEQ ID NO: 443
TPP-25255	Recombi Variant of TPP-15374	HCDR3	PRT	SEQ ID NO: 444
TPP-25255	Recombi Variant of TPP-15374	VL	PRT	SEQ ID NO: 445
TPP-25255	Recombi Variant of TPP-15374	LCDR1	PRT	SEQ ID NO: 446
TPP-25255	Recombi Variant of TPP-15374	LCDR2	PRT	SEQ ID NO: 447
TPP-25255	Recombi Variant of TPP-15374	LCDR3	PRT	SEQ ID NO: 448
TPP-25255	Recombi Variant of TPP-15374	VH	DNA	SEQ ID NO: 449
TPP-25255	Recombi Variant of TPP-15374	HCDR1	DNA	SEQ ID NO: 450
TPP-25255	Recombi Variant of TPP-15374	HCDR2	DNA	SEQ ID NO: 451
TPP-25255	Recombi Variant of TPP-15374	HCDR3	DNA	SEQ ID NO: 452
TPP-25255	Recombi Variant of TPP-15374	VL	DNA	SEQ ID NO: 453
TPP-25255	Recombi Variant of TPP-15374	LCDR1	DNA	SEQ ID NO: 454
TPP-25255	Recombi Variant of TPP-15374	LCDR2	DNA	SEQ ID NO: 455
TPP-25255	Recombi Variant of TPP-15374	LCDR3	DNA	SEQ ID NO: 456
TPP-25255	Recombi Variant of TPP-15374	Heavy Chain	PRT	SEQ ID NO: 457
TPP-25255	Recombi Variant of TPP-15374	Light Chain	PRT	SEQ ID NO: 458
TPP-25255	Recombi Variant of TPP-15374	Heavy Chain	DNA	SEQ ID NO: 459
TPP-25255	Recombi Variant of TPP-15374	Light Chain	DNA	SEQ ID NO: 460
TPP-25256	Recombi Variant of TPP-15374	VH	PRT	SEQ ID NO: 461
TPP-25256	Recombi Variant of TPP-15374	HCDR1	PRT	SEQ ID NO: 462
TPP-25256	Recombi Variant of TPP-15374	HCDR2	PRT	SEQ ID NO: 463
TPP-25256	Recombi Variant of TPP-15374	HCDR3	PRT	SEQ ID NO: 464
TPP-25256	Recombi Variant of TPP-15374	VL	PRT	SEQ ID NO: 465
TPP-25256	Recombi Variant of TPP-15374	LCDR1	PRT	SEQ ID NO: 466
TPP-25256	Recombi Variant of TPP-15374	LCDR2	PRT	SEQ ID NO: 467
TPP-25256	Recombi Variant of TPP-15374	LCDR3	PRT	SEQ ID NO: 468
TPP-25256	Recombi Variant of TPP-15374	VH	DNA	SEQ ID NO: 469
TPP-25256	Recombi Variant of TPP-15374	HCDR1	DNA	SEQ ID NO: 470
TPP-25256	Recombi Variant of TPP-15374	HCDR2	DNA	SEQ ID NO: 471
TPP-25256	Recombi Variant of TPP-15374	HCDR3	DNA	SEQ ID NO: 472
TPP-25256	Recombi Variant of TPP-15374	VL	DNA	SEQ ID NO: 473
TPP-25256	Recombi Variant of TPP-15374	LCDR1	DNA	SEQ ID NO: 474
TPP-25256	Recombi Variant of TPP-15374	LCDR2	DNA	SEQ ID NO: 475
TPP-25256	Recombi Variant of TPP-15374	LCDR3	DNA	SEQ ID NO: 476
TPP-25256	Recombi Variant of TPP-15374	Heavy Chain	PRT	SEQ ID NO: 477
TPP-25256	Recombi Variant of TPP-15374	Light Chain	PRT	SEQ ID NO: 478
TPP-25256	Recombi Variant of TPP-15374	Heavy Chain	DNA	SEQ ID NO: 479
TPP-25256	Recombi Variant of TPP-15374	Light Chain	DNA	SEQ ID NO: 480
TPP-25257	Recombi Variant of TPP-15374	VH	PRT	SEQ ID NO: 481
TPP-25257	Recombi Variant of TPP-15374	HCDR1	PRT	SEQ ID NO: 482
TPP-25257	Recombi Variant of TPP-15374	HCDR2	PRT	SEQ ID NO: 483
TPP-25257	Recombi Variant of TPP-15374	HCDR3	PRT	SEQ ID NO: 484
TPP-25257	Recombi Variant of TPP-15374	VL	PRT	SEQ ID NO: 485
TPP-25257	Recombi Variant of TPP-15374	LCDR1	PRT	SEQ ID NO: 486
TPP-25257	Recombi Variant of TPP-15374	LCDR2	PRT	SEQ ID NO: 487
TPP-25257	Recombi Variant of TPP-15374	LCDR3	PRT	SEQ ID NO: 488
TPP-25257	Recombi Variant of TPP-15374	VH	DNA	SEQ ID NO: 489
TPP-25257	Recombi Variant of TPP-15374	HCDR1	DNA	SEQ ID NO: 490
TPP-25257	Recombi Variant of TPP-15374	HCDR2	DNA	SEQ ID NO: 491
TPP-25257	Recombi Variant of TPP-15374	HCDR3	DNA	SEQ ID NO: 492
TPP-25257	Recombi Variant of TPP-15374	VL	DNA	SEQ ID NO: 493
TPP-25257	Recombi Variant of TPP-15374	LCDR1	DNA	SEQ ID NO: 494
TPP-25257	Recombi Variant of TPP-15374	LCDR2	DNA	SEQ ID NO: 495
TPP-25257	Recombi Variant of TPP-15374	LCDR3	DNA	SEQ ID NO: 496
TPP-25257	Recombi Variant of TPP-15374	Heavy Chain	PRT	SEQ ID NO: 497
TPP-25257	Recombi Variant of TPP-15374	Light Chain	PRT	SEQ ID NO: 498
TPP-25257	Recombi Variant of TPP-15374	Heavy Chain	DNA	SEQ ID NO: 499
TPP-25257	Recombi Variant of TPP-15374	Light Chain	DNA	SEQ ID NO: 500
TPP-25448	Recombi Variant of TPP-15374	VH	PRT	SEQ ID NO: 501
TPP-25448	Recombi Variant of TPP-15374	HCDR1	PRT	SEQ ID NO: 502
TPP-25448	Recombi Variant of TPP-15374	HCDR2	PRT	SEQ ID NO: 503
TPP-25448	Recombi Variant of TPP-15374	HCDR3	PRT	SEQ ID NO: 504
TPP-25448	Recombi Variant of TPP-15374	VL	PRT	SEQ ID NO: 505
TPP-25448	Recombi Variant of TPP-15374	LCDR1	PRT	SEQ ID NO: 506
TPP-25448	Recombi Variant of TPP-15374	LCDR2	PRT	SEQ ID NO: 507
TPP-25448	Recombi Variant of TPP-15374	LCDR3	PRT	SEQ ID NO: 508
TPP-25448	Recombi Variant of TPP-15374	VH	DNA	SEQ ID NO: 509
TPP-25448	Recombi Variant of TPP-15374	HCDR1	DNA	SEQ ID NO: 510
TPP-25448	Recombi Variant of TPP-15374	HCDR2	DNA	SEQ ID NO: 511

TABLE 1-continued

Amino acid sequences and nucleic acid sequences of preferred antibodies according to the present disclosure and of three prior art antibodies. TPP-11489 corresponds to Chiome antibody Humanized-2 derived of clone No. 4-2 strain (WO 2014/123186); TPP-15051 represents a murine IgG1 variant thereof. TPP-30788-TPP-30791 corresponds to Böhlinger Ingelheim antibody (BI) Clone I-IV (WO 2020/225400). TPP-30792 corresponds to University Ramot antibody clone I (WO 2020/261281).				
TPP ID	Antibody Description	Sequence Region	Sequence Type	SEQ ID
TPP-25448	Recombi Variant of TPP-15374	HCDR3	DNA	SEQ ID NO: 512
TPP-25448	Recombi Variant of TPP-15374	VL	DNA	SEQ ID NO: 513
TPP-25448	Recombi Variant of TPP-15374	LCDR1	DNA	SEQ ID NO: 514
TPP-25448	Recombi Variant of TPP-15374	LCDR2	DNA	SEQ ID NO: 515
TPP-25448	Recombi Variant of TPP-15374	LCDR3	DNA	SEQ ID NO: 516
TPP-25448	Recombi Variant of TPP-15374	Heavy Chain	PRT	SEQ ID NO: 517
TPP-25448	Recombi Variant of TPP-15374	Light Chain	PRT	SEQ ID NO: 518
TPP-25448	Recombi Variant of TPP-15374	Heavy Chain	DNA	SEQ ID NO: 519
TPP-25448	Recombi Variant of TPP-15374	Light Chain	DNA	SEQ ID NO: 520
TPP-25497	Recombi Variant of TPP-15374	VH	PRT	SEQ ID NO: 521
TPP-25497	Recombi Variant of TPP-15374	HCDR1	PRT	SEQ ID NO: 522
TPP-25497	Recombi Variant of TPP-15374	HCDR2	PRT	SEQ ID NO: 523
TPP-25497	Recombi Variant of TPP-15374	HCDR3	PRT	SEQ ID NO: 524
TPP-25497	Recombi Variant of TPP-15374	VL	PRT	SEQ ID NO: 525
TPP-25497	Recombi Variant of TPP-15374	LCDR1	PRT	SEQ ID NO: 526
TPP-25497	Recombi Variant of TPP-15374	LCDR2	PRT	SEQ ID NO: 527
TPP-25497	Recombi Variant of TPP-15374	LCDR3	PRT	SEQ ID NO: 528
TPP-25497	Recombi Variant of TPP-15374	VH	DNA	SEQ ID NO: 529
TPP-25497	Recombi Variant of TPP-15374	HCDR1	DNA	SEQ ID NO: 530
TPP-25497	Recombi Variant of TPP-15374	HCDR2	DNA	SEQ ID NO: 531
TPP-25497	Recombi Variant of TPP-15374	HCDR3	DNA	SEQ ID NO: 532
TPP-25497	Recombi Variant of TPP-15374	VL	DNA	SEQ ID NO: 533
TPP-25497	Recombi Variant of TPP-15374	LCDR1	DNA	SEQ ID NO: 534
TPP-25497	Recombi Variant of TPP-15374	LCDR2	DNA	SEQ ID NO: 535
TPP-25497	Recombi Variant of TPP-15374	LCDR3	DNA	SEQ ID NO: 536
TPP-25497	Recombi Variant of TPP-15374	Heavy Chain	PRT	SEQ ID NO: 537
TPP-25497	Recombi Variant of TPP-15374	Light Chain	PRT	SEQ ID NO: 538
TPP-25497	Recombi Variant of TPP-15374	Heavy Chain	DNA	SEQ ID NO: 539
TPP-25497	Recombi Variant of TPP-15374	Light Chain	DNA	SEQ ID NO: 540
TPP-25655	Recombi Variant of TPP-15374	VH	PRT	SEQ ID NO: 541
TPP-25655	Recombi Variant of TPP-15374	HCDR1	PRT	SEQ ID NO: 542
TPP-25655	Recombi Variant of TPP-15374	HCDR2	PRT	SEQ ID NO: 543
TPP-25655	Recombi Variant of TPP-15374	HCDR3	PRT	SEQ ID NO: 544
TPP-25655	Recombi Variant of TPP-15374	VL	PRT	SEQ ID NO: 545
TPP-25655	Recombi Variant of TPP-15374	LCDR1	PRT	SEQ ID NO: 546
TPP-25655	Recombi Variant of TPP-15374	LCDR2	PRT	SEQ ID NO: 547
TPP-25655	Recombi Variant of TPP-15374	LCDR3	PRT	SEQ ID NO: 548
TPP-25655	Recombi Variant of TPP-15374	VH	DNA	SEQ ID NO: 549
TPP-25655	Recombi Variant of TPP-15374	HCDR1	DNA	SEQ ID NO: 550
TPP-25655	Recombi Variant of TPP-15374	HCDR2	DNA	SEQ ID NO: 551
TPP-25655	Recombi Variant of TPP-15374	HCDR3	DNA	SEQ ID NO: 552
TPP-25655	Recombi Variant of TPP-15374	VL	DNA	SEQ ID NO: 553
TPP-25655	Recombi Variant of TPP-15374	LCDR1	DNA	SEQ ID NO: 554
TPP-25655	Recombi Variant of TPP-15374	LCDR2	DNA	SEQ ID NO: 555
TPP-25655	Recombi Variant of TPP-15374	LCDR3	DNA	SEQ ID NO: 556
TPP-25655	Recombi Variant of TPP-15374	Heavy Chain	PRT	SEQ ID NO: 557
TPP-25655	Recombi Variant of TPP-15374	Light Chain	PRT	SEQ ID NO: 558
TPP-25655	Recombi Variant of TPP-15374	Heavy Chain	DNA	SEQ ID NO: 559
TPP-25655	Recombi Variant of TPP-15374	Light Chain	DNA	SEQ ID NO: 560
TPP-26111	Recombi Variant of TPP-15374	VH	PRT	SEQ ID NO: 561
TPP-26111	Recombi Variant of TPP-15374	HCDR1	PRT	SEQ ID NO: 562
TPP-26111	Recombi Variant of TPP-15374	HCDR2	PRT	SEQ ID NO: 563
TPP-26111	Recombi Variant of TPP-15374	HCDR3	PRT	SEQ ID NO: 564
TPP-26111	Recombi Variant of TPP-15374	VL	PRT	SEQ ID NO: 565
TPP-26111	Recombi Variant of TPP-15374	LCDR1	PRT	SEQ ID NO: 566
TPP-26111	Recombi Variant of TPP-15374	LCDR2	PRT	SEQ ID NO: 567
TPP-26111	Recombi Variant of TPP-15374	LCDR3	PRT	SEQ ID NO: 568
TPP-26111	Recombi Variant of TPP-15374	VH	DNA	SEQ ID NO: 569
TPP-26111	Recombi Variant of TPP-15374	HCDR1	DNA	SEQ ID NO: 570
TPP-26111	Recombi Variant of TPP-15374	HCDR2	DNA	SEQ ID NO: 571
TPP-26111	Recombi Variant of TPP-15374	HCDR3	DNA	SEQ ID NO: 572
TPP-26111	Recombi Variant of TPP-15374	VL	DNA	SEQ ID NO: 573
TPP-26111	Recombi Variant of TPP-15374	LCDR1	DNA	SEQ ID NO: 574
TPP-26111	Recombi Variant of TPP-15374	LCDR2	DNA	SEQ ID NO: 575
TPP-26111	Recombi Variant of TPP-15374	LCDR3	DNA	SEQ ID NO: 576
TPP-26111	Recombi Variant of TPP-15374	Heavy Chain	PRT	SEQ ID NO: 577
TPP-26111	Recombi Variant of TPP-15374	Light Chain	PRT	SEQ ID NO: 578
TPP-26111	Recombi Variant of TPP-15374	Heavy Chain	DNA	SEQ ID NO: 579
TPP-26111	Recombi Variant of TPP-15374	Light Chain	DNA	SEQ ID NO: 580
TPP-13211	huSema3a_FXaFc	Chain 1	PRT	SEQ ID NO: 581
TPP-19068	human Sema3a_FXaHis6	Chain 1	PRT	SEQ ID NO: 582

TABLE 1-continued

Amino acid sequences and nucleic acid sequences of preferred antibodies according to the present disclosure and of three prior art antibodies. TPP-11489 corresponds to Chiome antibody Humanized-2 derived of clone No. 4-2 strain (WO 2014/123186); TPP-15051 represents a murine IgG1 variant thereof. TPP-30788-TPP-30791 corresponds to Böhlinger Ingelheim antibody (BI) Clone I-IV (WO 2020/225400). TPP-30792 corresponds to University Ramot antibody clone I (WO 2020/261281).				
TPP ID	Antibody Description	Sequence Region	Sequence Type	SEQ ID
TPP-19069	mouse Sema3a_FXaHis6	Chain 1	PRT	SEQ ID NO: 583
TPP-19120	rat-Sema3a_FXaHis6	Chain 1	PRT	SEQ ID NO: 584
TPP-19121	dog-Sema3a_FXaHis6	Chain 1	PRT	SEQ ID NO: 585
TPP-19122	cyno-Sema3a_FXaHis6	Chain 1	PRT	SEQ ID NO: 586
TPP-20176	pigSema3A_FXaHis6	Chain 1	PRT	SEQ ID NO: 587
TPP-30788	Böhlinger (BI) Clone I	VH	PRT	SEQ ID NO: 800
TPP-30788	Böhlinger (BI) Clone I	HCDR1	PRT	SEQ ID NO: 801
TPP-30788	Böhlinger (BI) Clone I	HCDR2	PRT	SEQ ID NO: 802
TPP-30788	Böhlinger (BI) Clone I	HCDR3	PRT	SEQ ID NO: 803
TPP-30788	Böhlinger (BI) Clone I	VL	PRT	SEQ ID NO: 804
TPP-30788	Böhlinger (BI) Clone I	LCDR1	PRT	SEQ ID NO: 805
TPP-30788	Böhlinger (BI) Clone I	LCDR2	PRT	SEQ ID NO: 806
TPP-30788	Böhlinger (BI) Clone I	LCDR3	PRT	SEQ ID NO: 807
TPP-30788	Böhlinger (BI) Clone I	VH	DNA	SEQ ID NO: 808
TPP-30788	Böhlinger (BI) Clone I	VL	DNA	SEQ ID NO: 809
TPP-30788	Böhlinger (BI) Clone I	Heavy Chain	PRT	SEQ ID NO: 810
TPP-30788	Böhlinger (BI) Clone I	Light Chain	PRT	SEQ ID NO: 811
TPP-30788	Böhlinger (BI) Clone I	Heavy Chain	DNA	SEQ ID NO: 812
TPP-30788	Böhlinger (BI) Clone I	Light Chain	DNA	SEQ ID NO: 813
TPP-30789	Böhlinger (BI) Clone II	VH	PRT	SEQ ID NO: 814
TPP-30789	Böhlinger (BI) Clone II	HCDR1	PRT	SEQ ID NO: 815
TPP-30789	Böhlinger (BI) Clone II	HCDR2	PRT	SEQ ID NO: 816
TPP-30789	Böhlinger (BI) Clone II	HCDR3	PRT	SEQ ID NO: 817
TPP-30789	Böhlinger (BI) Clone II	VL	PRT	SEQ ID NO: 818
TPP-30789	Böhlinger (BI) Clone II	LCDR1	PRT	SEQ ID NO: 819
TPP-30789	Böhlinger (BI) Clone II	LCDR2	PRT	SEQ ID NO: 820
TPP-30789	Böhlinger (BI) Clone II	LCDR3	PRT	SEQ ID NO: 821
TPP-30789	Böhlinger (BI) Clone II	VH	DNA	SEQ ID NO: 822
TPP-30789	Böhlinger (BI) Clone II	VL	DNA	SEQ ID NO: 823
TPP-30789	Böhlinger (BI) Clone II	Heavy Chain	PRT	SEQ ID NO: 824
TPP-30789	Böhlinger (BI) Clone II	Light Chain	PRT	SEQ ID NO: 825
TPP-30789	Böhlinger (BI) Clone II	Heavy Chain	DNA	SEQ ID NO: 826
TPP-30789	Böhlinger (BI) Clone II	Light Chain	DNA	SEQ ID NO: 827
TPP-30790	Böhlinger (BI) Clone III	VH	PRT	SEQ ID NO: 828
TPP-30790	Böhlinger (BI) Clone III	HCDR1	PRT	SEQ ID NO: 829
TPP-30790	Böhlinger (BI) Clone III	HCDR2	PRT	SEQ ID NO: 830
TPP-30790	Böhlinger (BI) Clone III	HCDR3	PRT	SEQ ID NO: 831
TPP-30790	Böhlinger (BI) Clone III	VL	PRT	SEQ ID NO: 832
TPP-30790	Böhlinger (BI) Clone III	LCDR1	PRT	SEQ ID NO: 833
TPP-30790	Böhlinger (BI) Clone III	LCDR2	PRT	SEQ ID NO: 834
TPP-30790	Böhlinger (BI) Clone III	LCDR3	PRT	SEQ ID NO: 835
TPP-30790	Böhlinger (BI) Clone III	VH	DNA	SEQ ID NO: 836
TPP-30790	Böhlinger (BI) Clone III	VL	DNA	SEQ ID NO: 837
TPP-30790	Böhlinger (BI) Clone III	Heavy Chain	PRT	SEQ ID NO: 838
TPP-30790	Böhlinger (BI) Clone III	Light Chain	PRT	SEQ ID NO: 839
TPP-30790	Böhlinger (BI) Clone III	Heavy Chain	DNA	SEQ ID NO: 840
TPP-30790	Böhlinger (BI) Clone III	Light Chain	DNA	SEQ ID NO: 841
TPP-30791	Böhlinger (BI) Clone IV	VH	PRT	SEQ ID NO: 842
TPP-30791	Böhlinger (BI) Clone IV	HCDR1	PRT	SEQ ID NO: 843
TPP-30791	Böhlinger (BI) Clone IV	HCDR2	PRT	SEQ ID NO: 844
TPP-30791	Böhlinger (BI) Clone IV	HCDR3	PRT	SEQ ID NO: 845
TPP-30791	Böhlinger (BI) Clone IV	VL	PRT	SEQ ID NO: 846
TPP-30791	Böhlinger (BI) Clone IV	LCDR1	PRT	SEQ ID NO: 847
TPP-30791	Böhlinger (BI) Clone IV	LCDR2	PRT	SEQ ID NO: 848
TPP-30791	Böhlinger (BI) Clone IV	LCDR3	PRT	SEQ ID NO: 849
TPP-30791	Böhlinger (BI) Clone IV	VH	DNA	SEQ ID NO: 850
TPP-30791	Böhlinger (BI) Clone IV	VL	DNA	SEQ ID NO: 851
TPP-30791	Böhlinger (BI) Clone IV	Heavy Chain	PRT	SEQ ID NO: 852
TPP-30791	Böhlinger (BI) Clone IV	Light Chain	PRT	SEQ ID NO: 853
TPP-30791	Böhlinger (BI) Clone IV	Heavy Chain	DNA	SEQ ID NO: 854
TPP-30791	Böhlinger (BI) Clone IV	Light Chain	DNA	SEQ ID NO: 855
TPP-30792	3H4 (Ramot) Clon I	VH	PRT	SEQ ID NO: 856
TPP-30792	3H4 (Ramot) Clon I	HCDR1	PRT	SEQ ID NO: 857
TPP-30792	3H4 (Ramot) Clon I	HCDR2	PRT	SEQ ID NO: 858
TPP-30792	3H4 (Ramot) Clon I	HCDR3	PRT	SEQ ID NO: 859
TPP-30792	3H4 (Ramot) Clon I	VL	PRT	SEQ ID NO: 860
TPP-30792	3H4 (Ramot) Clon I	LCDR1	PRT	SEQ ID NO: 861
TPP-30792	3H4 (Ramot) Clon I	LCDR2	PRT	SEQ ID NO: 862
TPP-30792	3H4 (Ramot) Clon I	LCDR3	PRT	SEQ ID NO: 863
TPP-30792	3H4 (Ramot) Clon I	VH	DNA	SEQ ID NO: 864
TPP-30792	3H4 (Ramot) Clon I	VL	DNA	SEQ ID NO: 865

TABLE 1-continued

Amino acid sequences and nucleic acid sequences of preferred antibodies according to the present disclosure and of three prior art antibodies. TPP-11489 corresponds to Chiome antibody Humanized-2 derived of clone No. 4-2 strain (WO 2014/123186); TPP-15051 represents a murine IgG1 variant thereof. TPP-30788-TPP-30791 corresponds to Böhlinger Ingelheim antibody (BI) Clone I-IV (WO 2020/225400). TPP-30792 corresponds to University Ramot antibody clone I (WO 2020/261281).				
TPP ID	Antibody Description	Sequence Region	Sequence Type	SEQ ID
TPP-30792	3H4 (Ramot) Clon I	Heavy Chain	PRT	SEQ ID NO: 866
TPP-30792	3H4 (Ramot) Clon I	Light Chain	PRT	SEQ ID NO: 867
TPP-30792	3H4 (Ramot) Clon I	Heavy Chain	DNA	SEQ ID NO: 868
TPP-30792	3H4 (Ramot) Clon I	Light Chain	DNA	SEQ ID NO: 869
TPP-31357	Fab of 3H4 Univ Ramot	VH	PRT	SEQ ID NO: 870
TPP-31357	Fab of 3H4 Univ Ramot	HCDR1	PRT	SEQ ID NO: 871
TPP-31357	Fab of 3H4 Univ Ramot	HCDR2	PRT	SEQ ID NO: 872
TPP-31357	Fab of 3H4 Univ Ramot	HCDR3	PRT	SEQ ID NO: 873
TPP-31357	Fab of 3H4 Univ Ramot	VL	PRT	SEQ ID NO: 874
TPP-31357	Fab of 3H4 Univ Ramot	LCDR1	PRT	SEQ ID NO: 875
TPP-31357	Fab of 3H4 Univ Ramot	LCDR2	PRT	SEQ ID NO: 876
TPP-31357	Fab of 3H4 Univ Ramot	LCDR3	PRT	SEQ ID NO: 877
TPP-31357	Fab of 3H4 Univ Ramot	VH	DNA	SEQ ID NO: 878
TPP-31357	Fab of 3H4 Univ Ramot	VL	DNA	SEQ ID NO: 879
TPP-31357	Fab of 3H4 Univ Ramot	Heavy Chain	PRT	SEQ ID NO: 880
TPP-31357	Fab of 3H4 Univ Ramot	Light Chain	PRT	SEQ ID NO: 881
TPP-31357	Fab of 3H4 Univ Ramot	Heavy Chain	DNA	SEQ ID NO: 882
TPP-31357	Fab of 3H4 Univ Ramot	Light Chain	DNA	SEQ ID NO: 883

TABLE 1A

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
1	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFSYPMGWVRQAPGKGLEWV AGIDDDGSDTRYAPAVKGRATISRDNKNTVYLQMNSLRAEDTAVYY CAKHTGIGANSAGSIDAWGQGTLLVTVSS
2	PRT	SYPMG
3	PRT	GIDDDGSDTRYAPAVKG
4	PRT	HTGIGANSAGSIDA
5	PRT	SYELTQPPSVSVSPGQTARITCSGGGSYTGSIYYGWYQQKPGQAPVTVI YYNNKRPSDIPERFSGSLSGTTNTLTISGVQAEDEADYYCGSADNSGDAF GTGTKVTVL
6	PRT	SGGGSYTGSIYYG
7	PRT	YYNNKRPS
8	PRT	GSADNSGDA
9	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCACCTTTAGCAGCT ATCCTATGGGCTGGGTCCGACAGGCCCTTGGCAAAGGACTTGAATGG GTGGCCGGCATCGACGACGATGGCGATAGCGATACAAGATACGCC CTGCCGTGAAGGCGAGAGCCACCATCTCCAGAGACAACAGCAAGAA CACCGTGTAACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCC GTGTACTATTGTGCCAAGCACACAGGCATCGGCGCCAATTCTGCCGG CTCTATTGATGCCTGGGCCAGGGAACACTGGTCACAGTTTCTTCA
10	DNA	AGCTATCCTATGGGC
11	DNA	GGCATCGACGACGATGGCGATAGCGATACAAGATACGCCCTGCCGT GAAGGGC
12	DNA	CACACAGGCATCGGCGCCAATTCTGCCGGCTCTATTGATGCC
13	DNA	AGCTATGAGCTGACACAGCCTCCAAGCGTGTCCGTGTCTCCTGGACA GACCGCCAGAATCACATGTAGCGCGGAGGACGCTACACCGGCAGC

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
		TACTACTATGGCTGGTATCAGCAGAAGCCCGACAGGCCCTGTGAC CGTGATCTACTACAACAACAGCGGCCAGCGACATCCCCGAGAGAT TTTCTGGCTCTCTGAGCGGCACCAACACACTGACAATCTCTGGC GTGCAGGCCGAGGACGAGGCCGATTACTATTGTGGCAGCGCCGATAA TAGCGGCGACGCCTTTGGCACCAGCACCAAGTTACAGTGCTA
14	DNA	AGCGGGCGGAGGCAGCTACACCGGCAGCTACTACTATGGC
15	DNA	TACAACAACAAGCGGCCAGC
16	DNA	GGCAGCGCCGATAATAGCGGCGACGCC
17	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYPMGWVRQAPGKLEWV AGIDDDGDS DTRYAPAVKGRATISRDN SKNTVYLQMNSLRAEDTAVYY CAKHTGIGANSAGSIDAWGGTLVTVSSASTKGPSVFPLAPSSKSTSGGT AALGCLVKDYFPEPVTWNSGALTSGVHTFPAVLQSSGLYSLSSVTV PSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKHTCTPCPAPPELLGGP SVFLFPPKPKDTLMISRTPEVTVVVDVSHEDPEVKFNWYVDGVEVHNA KTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKT I SKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNG QPENNYKTTTPPVLDSDGSFPLYSKLTVDKSRWQQGNVFCFSVMHEALH NHYTQKSLSLSPG
18	PRT	SYELTQPPSVSVSPGQTARITCSGGGSYTGSIYYGWYQQKPGQAPVTVI YYNNKRPSDIPERFSGSLSGTTNTLTISGVQAEDEADYYCGSADNSGDAP GTGKVTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTVA WKADSPVKAGVETTTPSKQSNNKYAASSYLSLTPEQWKSHRYSYSCQV THEGSTVEKTVAPTECS
19	DNA	GAAGTTCAGCTGCTGGAATCTGGCGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCGCAGCGCTTCACCTTAGCAGCT ATCCTATGGGCTGGGTCCGACAGGCCCTGGCAAAGGACTTGAATGG GTGGCCGGCATCGACGACGATGGCGATAGCGATAACAAGATACGCCC CTGCCGTGAAGGGCAGAGCCACCATCTCCAGAGACAACAGCAAGAA CACCGTGTAACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCC GTGTACTATTGTGCCAAGCACACAGGCATCGGCGCCAATTCTGCCGG CTCTATTGATGCCTGGGGCCAGGGAACACTGGTCACAGTTTCTTCAG CCAGCACCAAGGGCCCCAGCGTGTTCCTCTGGCCCTTAGCAGCAAG AGCACATCTGGCGGAACAGCCGCCCTGGGCTGCCTCGTGAAGGACTA CTTCCCGAGCCCGTGACCGTGTCTGGAACTCTGGCGCTCTGACAA GCGGCGTGACACCTTTCCAGCCGTGCTGCAGAGCAGCGGCTGTAC TCTCTGAGCAGCGTCTGTGACAGTGCCAGCAGCTCTCTGGGCACCCA GACCTACATCTGCAACGTGAACACAAAGCCAGCAACACCAAGGTG GACAAGAAGGTGAACCCAAAGAGCTGCGACAAGACCCACACCTGTC CCCCTGTCTGCCCCGAACTGCTGGGAGGCCCTTCCGTGTTCTCTGT TCCCCCAAGCCCAAGGACACCTGTATGATCAGCCGGAACCCCCGAA GTGACCTGCGTGGTGGTGGATGTGTCCACAGGAGCCCTGAAGTGAA GTTCAATTGGTACGTGGACGGCGTGGAAAGTGACAAACGCCAAGACCA AGCCTAGAGAGGAACAGTACAACAGCACCTACCGGGTGGTGTCCGT GCTGACAGTGCTGCACCAAGGACTGGCTGAACGGCAAGAGTACAAG TGCAAGGTGTCCAACAAGGCCCTGCCTGCCCCATCGAGAAAACCAT CAGCAAGGCCAAGGGCCAGCCCCGCGAAACCCAGGTGTACACACTG CCCCAAGCAGGGACGAGCTGACCAAGAACCAGGTGTCCCTGACCTG TCTCGTGAAAGGCTTCTACCCCTCCGATATCGCCGTGGAATGGGAGA GCAACGGCCAGCCCGAGAACTACAAGACCAACCCCCCTGTGCTG GACAGCGACGGCTCATTTCTCTGTACAGCAAGCTGACCGTGGACAA GTCCCGGTGGCAGCAGGGCAACGTGTTACAGCTGCAGCGTATGCACG AGGCCCTGCACAACCACTACACCCAGAAGTCCCTGAGCCTGAGCCCT GGC
20	DNA	AGCTATGAGCTGACACAGCCTCCAAGCGTGTCCGTGTCTCCTGGACA GACCGCCAGAATCACATGTAGCGGCGGAGGCAGCTACACCGGCAGC TACTACTATGGCTGGTATCAGCAGAAGCCCGACAGGCCCTGTGAC CGTGATCTACTACAACAACAAGCGGCCAGCGACATCCCCGAGAGAT TTTCTGGCTCTCTGAGCGGCACCAACACACTGACAATCTCTGGC GTGCAGGCCGAGGACGAGGCCGATTACTATTGTGGCAGCGCCGATAA TAGCGCGACGCCTTTGGCACCGGCACCAAGTTACAGTGCTAGGCC AGCCTAAAGCCGCCCTAGCGTGACCTGTTCCCTCCAAGCAGCGAG GAACTGCAGGCCAACAAGGCCACCCCTCGTGTGCCTGATCAGCGACTT CTATCCTGGCGCCGTGACCGTGGCCTGGAAGGCCGATAGCTCTCCTG TGAAGGCCGCGCTGGAACCAACACCCCTAGCAAGCAGAGCAACAA

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
		CAAATACGCCGCCAGCAGCTACCTGAGCCTGACCCCCGAGCAGTGGA AGTCCACAGATCCTACAGCTGCCAAGTGACCCACGAGGGCAGCACC GTGAAAAGACAGTGGCCCCTACCGAGTGCGAC
21	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYPMGWVRQAPGKGLEWV AGIDDDGSDTRYAPAVKGRATISRDN SKNTVYLQMNLSRAEDTAVYY CAKHTGIGANSAGSIDAWGQGLTVTVSS
22	PRT	SYPMG
23	PRT	GIDDDGSDTRYAPAVKG
24	PRT	HTGIGANSAGSIDA
25	PRT	SYELTQPPSVSVSPGQTARITCSGGGSYTGSIYYGWYQQKPGQAPVTVI YYNNKRPSDIPERFSGSLSGTTNTLTISGVQAEDEADYYCGSADNSGDAF GTGTKVTVL
26	PRT	SGGGSYTGSIYYG
27	PRT	YYNNKRPS
28	PRT	GSADNSGDA
29	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTACCTTTAGCAGCT ATCCTATGGGCTGGGTCCGACAGGCCCTGGCAAAGGACTTGAATGG GTGGCCGGCATCGACGACGATGGCGATAGCGATACAAGATACGCC CTGCCGTGAAGGGCAGAGCCACCATCTCCAGAGACAACAGCAAGAA CACCGTGTACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCC GTGTACTATTGTGCCAAGCACACAGGCATCGGCGCCAATTCTGCCGG CTCTATTGATGCCTGGGGCCAGGGAACACTGGTCACAGTTTCTTCA
30	DNA	AGCTATCCTATGGGC
31	DNA	GGCATCGACGACGATGGCGATAGCGATACAAGATACGCCCTGCCGT GAAGGGC
32	DNA	CACACAGGCATCGGCGCCAATTCTGCCGGCTCTATTGATGCC
33	DNA	AGCTATGAGCTGACACAGCCTCCAAGCGTGTCCGTGTCTCTGGACA GACCGCCAGAATCACATGTAGCGGCGGAGGCAGCTACACCGGCAGC TACTACTATGGCTGGTATCAGCAGAAGCCGGACAGGCCCTGTGAC CGTGATCTACTACAACAACAAGCGGCCAGCGACATCCCGAGAGAT TTTCTGGCTCTCTGAGCGGCACCAACAACACTGACAATCTCTGGC GTGACGCCGAGGACGAGGCGGATTAATTGTGGCAGCGCCGATAA TAGCGGCGACGCCTTTGGCACCGGCACCAAGTTACAGTGCTA
34	DNA	AGCGGCGGAGGCAGCTACACCGGCAGCTACTACTATGGC
35	DNA	TACAACAACAAGCGGCCAGC
36	DNA	GGCAGCGCCGATAATAGCGGCGACGCC
37	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYPMGWVRQAPGKGLEWV AGIDDDGSDTRYAPAVKGRATISRDN SKNTVYLQMNLSRAEDTAVYY CAKHTGIGANSAGSIDAWGQGLTVTVSSAKTTPPSVYPLAPGSAQTNS MVTGLCLVKGYFPEPVTVTWNSGSLSSGVHTFPAVLQSDLYTLSSSVTV PSSTWPSSETVT CNVAHPASSTKVDKKIVPRDCGCKPCICTVPEVSSVFI PKPKDVLITITLTPKVTGVVVDISKDDPEVQFSWFVDDVEVHTAQTQPRE EQFNSTFRSVSELPIMHQDWLNGKEFKCRVNSAAPPAPIEKTISKTKGRP KAPQVYTI PPPKEQMAKDKVSLTCMITDFFPEDITVEWQWNGQPAENYK NTQPIMDTDGSYFVYSKLVNQS NWEAGNTFTCSVLHLEGLHNHHTKS LSHSPGK
38	PRT	SYELTQPPSVSVSPGQTARITCSGGGSYTGSIYYGWYQQKPGQAPVTVI YYNNKRPSDIPERFSGSLSGTTNTLTISGVQAEDEADYYCGSADNSGDAF GTGTKVTVLGQPKSSPSVTLFPPSSSELETNKATLVCTITDFYPGVTVTD WKVDGTPVTQGMETTPQSKQSNKYMAYSLTLTARAWERHSSYSQC VTHEGHTVEKSLSRADCS

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
39	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCACCTTTAGCAGCT ATCCTATGGGCTGGGTCCGACAGGCCCTGGCAAAGGACTTGAATGG GTGGCCGGCATCGACGACGATGGCGATAGCGATACAAGATACGCC CTGCCGTGAAGGGCAGAGCCACCATCTCCAGAGACAACAGCAAGAA CACCCTGTACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCC GTGTACTATTGTGCCAAGCACACAGGCATCGGCGCCAATTCTGCCGG CTCTATTGATGCCCTGGGGCCAGGGAACTGGTCACAGTTTCTTCAG CCAAGACCAACCCCCCAGCGTGTACCCTCTGGCTCCTGGATCTGCC CCCCAGACCAACAGCATGGTCAACCTGGGCTGCCTCGTGAAGGGCTA CTTCCCTGAGCCTGTGACCGTGACCTGGAACAGCGCTCTCTGTCTAG CGGCGTGCACACCTTTCCAGCCGTGCTGCAGAGCGACCTGTACACCC TGAGCAGCAGCGTGACCGTGCCCTAGCAGCACCTGGCCTAGCGAGACA GTGACCTGCAACGTGGCCCCACCTGCCAGCAGCACAAAGGTGGACA AGAAAAATCGTGCCCGGGACTGCGGCTGCAAGCCCTGTATCTGTACC GTGCCCGAGGTGTCCAGCGTGTTCATCTTCCACCCAAGCCCAAGGA CGTGCTGACCATCACCTGACCCCCAAAGTGACCTGTGTGGTGGTGG ACATCAGCAAGGACGACCCCGAGGTGCGATTCAAGTTGGTTCGTGGAC GAGTGGAAAGTGACACAGCCCCAGACCCAGCCAGAGAGGAACAGT TCACAGCACCTTCAGAGCGTGTCCGAGCTGCCCATCATGCACCCAG GACTGGCTGAACGGCAAAGAGTTCAAGTGACAGTGAACAGCGCCG CCTTCCCTGCCCCCATCGAGAAAACCATCTCCAAGACCAAGGGCAGA CCCAAGGCCCTCAGGTGTACAAATCCCCCACCACAAAGAACAGAT GGCCAAGGACAAGGTGTCCCTGACCTGCATGATCACCGATTCTTCTCC CAGAGGACATCACCGTGGAATGGCAGTGGAAACGGCCAGCCCGCCGA GAACTACAAGAACACCCAGCCTATCATGGACACCGACGGCAGCTACT TCGTGTACAGCAAGCTGAACGTGCAGAAAGTCCAACTGGGAGGCCGG CAACACCTTCACCTGTAGCGTGTGTGCACGAGGGCCTGCACAATCACC ACCCGAGAAGTCCCTGTCCACAGCCCTGGCAAG
40	DNA	AGCTATGAGCTGACACAGCCTCCAAGCGTGTCCGTGTCTCTGGACA GACCGCCAGAAATCACATGTAGCGCGGAGGCGAGCTACACCGGCAGC TACTACTATGGCTGGTATCAGCAGAAGCCCGACAGGCCCTGTGAC CGTGATCTACTACAACAACAAGCGGCCAGCGACATCCCCGAGAGAT TTTCTGGCTCTCTGAGCGGCACCAACACACTGACAATCTCTGGC GTGCAGGCCGAGGACGAGGCCGATTACTATTGTGGCAGCGCCGATAA TAGCGGCGACGCCTTTGGCACCGGCACCAAGTTACAGTGTAGGCC AGCCCAAGAGCAGCCCTAGCGTGACCTGTTCCCTCCAAGCAGCGAG GAACTGGAACAACAAGGCCACCCCTCGTGTGCACCATCACCGACTT CTACCCCGGCGTGTGACCGTGGAAGGTGGACGGCACCCAG TGACCCAGGGCATGGAACACCCAGCCAGCAAGCAGAGCAACAA CAAGTACATGGCCAGCAGCTACCTGACCCCTGACCGCCAGAGCCTGGG AGAGACACAGCTCTACAGCTGCCAAGTGACCCAGAGGGCCACAC CGTGGAAAAGAGCCTGAGCAGAGCCGACTGCAGC
41	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYGMHWVRQAPGKLEWV SAIGTGGDTYYADSVMGRTISRDNKNTLYLQMNSLRAEDTAVYYCA RRDDYTSRDAFDVWGQTLTVTVSS
42	PRT	SYGMH
43	PRT	AIGTGGDTYYADSVMG
44	PRT	RRDDYTSRDAFDV
45	PRT	QSVLTQPPSASGTPGQRVITISCGSSSNIGSNTVNWYQQLPGTAPKLLIYY DDLPLSGVPDRFSGSKSGTSASLAISGLRSEADYYCAAWDDSLNGYV VFGGGTKLTVL
46	PRT	SGSSSNIGSNTVN
47	PRT	YDDLPS
48	PRT	AAWDDSLNGYV
49	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCACCTTTAGCAGCT ATGGCATGCACTGGGTCCGACAGGCCCTGGCAAAGGACTTGAATGG GTGTCCGCCATCGGCACAGCGCGCGATACCTACTATGCCGATAGCGT GATGGGCAGATTACCATCAGCCGGGACAACAGCAAGAACACCTTG TACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCGTGACTA

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
		TTGCGCCAGAAAGGACGACTACACCAGCAGGGATGCCTTCGATGTGT GGGGCCAGGAACACTGGTTACCGTTTCTTCA
50	DNA	AGCTATGGCATGCAC
51	DNA	GCCATCGGCACAGGCGGCGATACCTACTATGCCGATAGCGTGATGGG C
52	DNA	AGGGACGACTACACCAGCAGGGATGCCTTCGATGTG
53	DNA	CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTAGCGGCAGCAGCTCCAACATCGGCAGC AACACCGTGAAGTGGTATCAGCAGCTGCCTGGCACAGCCCCCTAACT GCTGATCTACTACGACGACCTGTGCTAGCGCGCTGCCCGATAGAT TTTCTGGCAGCAAGAGCGGCACAAGCGCCAGCCTGGCTATCTCTGGA CTGAGATCTGAGGACGAGGCGGACTACTATTGCGCCGCTGGGACGA TAGCCTGAACGGCTATGTGGTTTTCGGCGGAGGCACCAAGCTGACCG TGCTA
54	DNA	AGCGGCAGCAGCTCCAACATCGGCAGCAACACCGTGAAC
55	DNA	TACGACGACCTGCTGCCTAGC
56	DNA	GCCGCCTGGGACGATAGCCTGAACGGCTATGTGGTT
57	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYGMHWVRQAPGKGLEWV SAIGTGGDTYYADSVMRFTISRDNSENKNTLYLQMNSLRAEDTAVYYCA RRDDYTSRDADFVWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAAL GCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSS LGTQTYICNVNHKPSNTKVDKKVEPKSCDKHTHTCPPCPAPELLGGPSVFL FPPKPKDTLMI SRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTK PREQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPI EKTISKAK KGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPE NNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFPSCSMHEALHNHY TQKLSLSLSPG
58	PRT	QSVLTQPPSASGTPGQRTVISCSGSSSNIGSNTVNWYQLPGTAPKLLIYY DDLPSGVPDRFSGSKSGTSASLAISGLRSEDEADYYCAAWDDSLNGYV VFGGGTKLTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVT VANKADSPVKAGVETTPSKQSNNKYAASSYLSLTPEQWKSHRSYSY QVTHEGSTVEKTVAPTECS
59	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCACCTTTAGCAGCT ATGGCATGCACTGGGTCCGACAGGCCCTGGCAAAGGACTTGAATGG GTGTCGCGCATCGGCACAGGCGGCGATACCTACTATGCCGATAGCGT GATGGGCAGATTACCATCAGCCGGGACAAAGCAAGAACAACCCCTG TACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCGTGTACTA TTGCGCCAGAAAGGACGACTACACCAGCAGGGATGCCTTCGATGTGT GGGGCCAGGGAACTGGTTACCGTTTCTTCAGCCAGCACCAAGGGC CCCAGCGTGTTCCCTCTGGCCCCCTAGCAGCAAGAGCACATCTGGCGG AACAGCCGCCCTGGGCTGCCTCGTGAAGGACTACTTTCCGAGCCCG TGACCGTGTCTGGAACCTGGCGCTCTGACAAGCGCGTGCACACC TTTCCAGCCGTGCTGCAGAGCAGCGGCTGTACTCTCTGAGCAGCGT CTGACAGTGCACAGCAGCTCTCTGGGCACCCAGACCTACATCTGCA ACGTGAACCAAGCCAGCAACACCAAGGTGGACAAGAAGGTGGA ACCCAAGAGCTGCGACAAGACCCACACCTGTCCCCCTGTCTCTGCCCC CCGAAGTGTGGGAGGCCCTTCCGTGTTCTGTTCCTCCCAAGGCCA AGGACACCTGATGATCAGCCGACCCCGAAGTGACCTGCGTGGTG GTGGATGTGTCCCACGAGGACCTGAAGTGAAGTTCAATTGGTACGT GGACGCGTGGAAAGTGACAACGCCAAGACCAAGCCTAGAGAGGAA CAGTACAACAGCACCTACCGGGTGGTGTCCGTGCTGACAGTGTGCA CCAGGACTGGCTGAACGGCAAGAGTACAAGTGAAGGTGTCCAAC AAGGCCCTGCCTGCCCCATCGAGAAAACCATCAGCAAGGCCAAGG GCCAGCCCCGCAACCCAGGTGTACACACTGCCCCCAAGCAGGGAC GAGCTGACCAAGAACCAGGTGTCCCTGACCTGTCTCGTGAAAGGCTT CTACCCCTCCGATATCGCGTGAATGGGAGAGCAACGGCCAGCCCG AGAACAACCTACAAGACACCCCTGTGCTGGACAGCGACGGCTCA TTCTTCTGTACAGCAAGCTGACCGTGGACAAGTCCCGGTGGCAGCA GGGCAACGTGTTAGCTGCAGCGTGATGCACGAGGCCCTGCACAACC ACTACACCCAGAAGTCCCTGAGCCTGAGCCCTGGC

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
60	DNA	CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTAGCGGCAGCAGCTCCAACATCGGCAGC AACACCGTGAAGTGGTATCAGCAGCTGCCTGGCACAGCCCTAAACT GCTGATCTACTACGACGACCTGTGCTAGCGCGCTGCCGATAGAT TTTCTGGCAGCAAGAGCGGCACAAGCGCCAGCCTGGCTATCTCTGGA CTGAGATCTGAGGACGAGGCCGACTACTATTGCGCCGCTGGGACGA TAGCCTGAACGGCTATGTGGTTTTCGGCGGAGGCACCAAGCTGACCG TGCTAGGCCAGCCTAAAGCGCCCTAGCGTGACCTGTTCCCTCCA AGCAGCGAGGAAGTGCAGGCCAACAAGGCCACCTCGTGTGCCTGAT CAGCGACTTCTATCCTGGCGCCGTGACCGTGCCCTGGAAGGCCGATA GCTCTCCTGTGAAGGCCGGCGTGGAAACCACACCCCTAGCAAGCAG AGCAACAACAAATACGCCGCCAGCAGCTACCTGAGCCTGACCCCGA GCAGTGAAGTCCACAGATCCTACAGCTGCCAAGTGACCCACGAGG GCAGCACCGTGGAAAAGACAGTGGCCCTACCGAGTGACGC
61	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYEMNWVRQAPGKLEWV SGISWNSGSIYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCA RSGYSSSWFDPDFDYWGQGLTVTVSS
62	PRT	SYEMN
63	PRT	GISWNSGSIYADSVKG
64	PRT	SGYSSSWFDPDFDY
65	PRT	QSVLTQPPSASGTPGQRTVITSGTSSSNIGAGYDVHWYQQLPGTAPKLLI YGNSNRPSGVPRFSGSKSGTSASLAISGLRSEDEADYYCSSYAGSNPYV VFGGGTKLTVL
66	PRT	TGSSSNIGAGYDVH
67	PRT	GNSNRPS
68	PRT	SSYAGSNPYVV
69	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCACTTTAGCAGCT ACGAGATGAAGTGGGTCCGACAGGCCCTGGCAAAGGCCCTGAATG GGTGTCCGGCATCAGCTGGAATAGCGGCTCTATCGGCTACGCCGACA GCGTGAAGGGCAGATTACCATCAGCCGGGACAACAGCAAGAACAC CCTGTACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCGTGT ACTACTGTGCCAGAAGCGGCTACAGCAGCTCTTGGTTTGACCCCGAC TTCGACTATTGGGGCCAGGGCACACTGGTCACAGTCTCTTCA
70	DNA	AGCTACGAGATGAAC
71	DNA	GGCATCAGCTGGAATAGCGGCTCTATCGGCTACGCCGACAGCGTGAA GGC
72	DNA	AGCGGCTACAGCAGCTCTTGGTTTGACCCCGACTTCGACTAT
73	DNA	CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTACCGGCAGCAGCTCCAATATCGGAGCCG GCTATGACGTGCCTGGTATCAGCAGCTGCCTGGCACAGCCCTAAA CTGCTGATCTACGGCAACAGCAACAGACCCAGCGGCGTGCCGATAG ATTTTCCGGCTCTAAGAGCGGCACAAGCGCCAGCCTGGCTATCTCTG GACTGAGATCTGAGGACGAGGCCGACTACTACTGCAGCAGCTATGCC GGCAGCAACCCCTACGTTGTGTTGGCGGAGGCACCAAGCTGACCGT TCTA
74	DNA	ACCGGCAGCAGCTCCAATATCGGAGCCGGCTATGACGTGCAC
75	DNA	GGCAACAGCAACAGACCCAGC
76	DNA	AGCAGCTATGCCGGCAGCAACCCCTACGTTGTG
77	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYEMNWVRQAPGKLEWV SGISWNSGSIYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCA RSGYSSSWFDPDFDYWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGTA ALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSVTVTP

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
		SSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTI SKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFPLYSKLTVDKSRWQQGNVFSQSVSMHEALH NHYTQKSLSLSPG
78	PRT	QSVLTQPPSASGTPGQRVITISCTGSSSNIGAGYDVHWYQQLPGTAPKLLI YGNSNRPSGVPDRFSGSKSGTSASLAISGLRSEADYYCSSYAGSNPYV VFGGGTKLTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVT VAWKADSSPVKAGVETTPSKQSNKYAASSYLSLTPEQWKSRSYSC QVTHEGSTVEKTVAPTECS
79	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTACCTTTAGCAGCT ACAGATGAACCTGGGTCCGACAGGCCCTGGCAAAGGCTTGAATG GGTGTCCGGCATCAGCTGGAATAGCGGCTCTATCGGCTACGCCGACA GCGTGAAGGGCAGATTACCATCAGCCGGGACAAACAGCAAGAACAC CCTGTACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCGGTGT ACTACTGTGCCAGAAGCGGCTACAGCAGCTCTTGGTTTGACCCGAC TTCGACTATTGGGGCCAGGGCACACTGGTTCACAGTCTCTTCAGCCAG CACCAAGGGCCCCAGCGTGTCCCTCTGGCCCCCTAGCAGCAAGAGCA CATCTGGCGGAACAGCCGCCCTGGGCTGCCTCGTGAAGGACTACTTT CCCGAGCCCGTGACCGTGTCTCTGGAACCTGCGGCGCTCTGACAAGCGG CGTGACACACCTTTCCAGCCGTGCTGCAGAGCAGCGCCGTACTCTCT GAGCAGCGTCTGACAGTGCCAGCAGCTCTCTGGGCACCCAGACCT ACATCTGCAACGTGAACCAAGCCAGCAACACCAAGGTGGACAA GAAGGTGGAACCAAGAGCTGCGACAAAGACCCACACCTGTCCCCCTT GTCTTGCCTCCCAACTGCTGGGAGGCCCTTCCGTGTCTCTGTCCCCC CAAAGCCCAAGGACACCTGATGATCAGCCGGACCCCGAAGTGAC CTGCGTGGTGGTGATGTGTCCACGAGGACCTGAAGTGAAGTTCA ATTGGTACGTGGACGGCGTGAAGTGCAACGCCAAGACCAAGCC TAGAGAGGAACAGTACAACAGCACCTACCGGGTGGTGTCCGTGCTGA CAGTGTGTCACCAAGGACTGGCTGAACGGCAAAGAGTACAAGTGCAA GGTGTCCAACAAGGCCCTGCCTGCCCCATCAGAGAAACCATCAGCA AGGCCAAGGGCCAGCCCCGCAACCCAGGTGTACACACTGCCCCCA AGCAGGGACGAGCTGACCAAGAACCAGGTGTCCCTGACCTGTCTCGT GAAAGGCTTCTACCCCTCCGATATCGCCGTGGAATGGGAGAGCAACG GCCAGCCCGAGAACTACAAGACACCCCCCTGTGCTGGACAGC GACGGCTCATTTCTCTGTACAGCAAGCTGACCGTGGACAAGTCCCG GTGGCAGCAGGGCAACGTGTTCAGCTGCAGCGTGATGCACGAGGCCC TGCACAACCACTACACCCAGAAGTCCCTGAGCCTGAGCCCTGGC
80	DNA	CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTACCGGCAGCAGCTCCAATATCGGAGCCG GCTATGACGTGCACCTGGTATCAGCAGCTGCCTGGCACAGCCCCATAA CTGCTGATCTACGGCAACAGCAACAGACCCAGCGCGGTGCCCGATAG ATTTTCCGGCTCTAAGAGCGGCACAAGCGCCAGCCTGGCTATCTCTG GACTGAGATCTGAGGACGAGGCCGACTACTACTGCAGCAGCTATGCC GGCAGCAACCCCTACGTTGTGTTTGGCGGAGGCACCAAGCTGACCGT TCTAGGCCAGCCTAAAGCCGCCCTAGCGTGACCTGTTCCTCCCAA GCAGCGAGGAAGTGCAGGCCAAAGGCCACCCCTCGTGTGCCGTGATC AGCGACTTCTATCCTGGCGCCGTGACCGTGGCCTGGAAGGCCGATAG CTCTCTGTGAAGGCCGGCGTGGAAACCACACCCCTAGCAAGCAGA GCAACAAACAATAACGCCGCCAGCAGCTACCTGAGCCTGACCCCGAG CAGTGGAAGTCCACAGATCTACAGCTGCCAAGTGACCCACGAGGG CAGCACCGTGGAAAAGACAGTGGCCCTACCGAGTGCAGC
81	PRT	EVQLLESGGGLVQTGGSRLRLSCAASGFTFSDYAMSWVRQAPGKLEWV SWIYYDSGSKYYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYC AKLNGDFDYWGQGLTVTVSS
82	PRT	DYAMS
83	PRT	WIYYDSGSKYYADSVK
84	PRT	LNGDFDY
85	PRT	QSVLTQPPSASGTPGQRVITISCSGSSSNIGNNDVSWYQQLPGTAPKLLIY ADSHRPSGVPDRFSGSKSGTSASLAISGLRSEADYYCGAWDSSLSGY VFGGGTKLTVL

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
86	PRT	SGSSSNIGNNDVS
87	PRT	ADSHRPS
88	PRT	GAWDSSLSGYV
89	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAAACAGGCGGCTCTCTGAGACTGAGCTGTGCCGCCTCTGGCTTCACTTCAGCGATTA CGCCATGAGCTGGGTCCGACAGGCCCTGGAAAAGGCCTGAAATGGG TGTCTGGATCTACTACGACAGCGGCAGCAAGTACTACGCCGACAGC GTGAAGGGCAGATTACCATCAGCCGGGACAAACAGCAAGAACACCC TGTACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCGTGTAC TATTGCCCAAGCTGAACGGCGACTTCGACTATTGGGGCCAGGGCAC ACTGGTCACAGTCTCTTCA
90	DNA	GATTACGCCATGAGC
91	DNA	TGGATCTACTACGACAGCGGCAGCAAGTACTACGCCGACAGCGTGAA GGGC
92	DNA	CTGAACGGCGACTTCGACTAT
93	DNA	CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTAGCGGCAGCAGCTCCAACATCGGCAAC AACGACGTGTCCTGGTATCAGCAGCTGCCTGGCACAGCCCTAAACT GCTGATCTACGCCGACAGCCACAGACCTAGCGGCGTGCCAGATAGAT TCAGCGGCTCTAAGAGCGGCACATCTGCCAGCCTGGCCATCTCTGGA CTGAGATCTGAGGACGAGCCGACTACTATTGCGGCGCCTGGGATTCT TAGCCTGAGCGGCTATGTTTTTGGCGGAGGCACCAAGCTGACCGTGC TA
94	DNA	AGCGGCAGCAGCTCCAACATCGGCAACAACGACGTGTCC
95	DNA	GCCGACAGCCACAGACCTAGC
96	DNA	GGCGCCTGGGATTCTAGCCTGAGCGGCTATGTT
97	PRT	EVQLLESGGGLVQTGGSRLRLSCAASGFTFSDYAMSWVRQAPGKGLEWV SWIYYDSGSKYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYC AKLNGDFDYWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAAALGCLV KDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQ TYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPK PKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREE QYNSTYRVVSVLTVLHQDWLNGKEYCKCKVSNKALPAPIEKTISKAKGQ PREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNY KTPPVLDSGDFLYSKLTVDKSRWQQGNVFSCSVMEALHNHYTQK SLSLSPG
98	PRT	QSVLTQPPSASGTPGQRVTISCSGSSSNIGNNDVSWYQQLPGTAPKLLIY ADSHRPSGVDPDRFSGSKSGTSASLAISGLRSEDEADYICGAWDSSLSGY VFGGGTKLTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVT VAWKADSPVKAGVETTPSKQSNNKYAASSYLSLTPEQWKSHRSYSC QVTHEGSTVEKTVAPTECS
99	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAAACAGGCGGCTCTCTGAGACTGAGCTGTGCCGCCTCTGGCTTCACTTCAGCGATTA CGCCATGAGCTGGGTCCGACAGGCCCTGGAAAAGGCCTGAAATGGG TGTCTGGATCTACTACGACAGCGGCAGCAAGTACTACGCCGACAGC GTGAAGGGCAGATTACCATCAGCCGGGACAAACAGCAAGAACACCC TGTACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCGTGTAC TATTGCCCAAGCTGAACGGCGACTTCGACTATTGGGGCCAGGGCAC ACTGGTCACAGTCTCTTACGCCAGCACCAAGGGCCCCAGCGTGTTC CTCTGGCCCCCTAGCAGCAAGAGCACATCTGGCGGAACAGCCGCCCTG GGCTGCCCTCGTGAAGGACTACTTTCCGAGCCCGTGACCGTGTCTCTG GAACTCTGGCGCTCTGACAAGCGGCGTGACACACCTTTCCAGCCGTGC TGCAGAGCAGCGGCTGTACTCTCTGAGCAGCGTCTGTGACAGTGCC AGCAGCTCTCTGGGCACCCAGACCTACATCTGCAACGTGAACACCAA GCCAGCAACACCAAGGTGGACAAGAAGGTGGAACCAAGAGCTGC GACAAGACCCACACCTGTCCCTTGTCTGCCCCGAACTGCTGGG AGGCCCTTCGTGTTCTGTTCCTTCCCAAGCCCAAGGACACCTGAT

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
		GATCAGCCGGACCCCCGAAGTGACCTGCGTGGTGGTGGATGTGTCCC ACGAGGACCCCTGAAGTGAAGTTCAATTGGTACGTGGACGGCGTGGA AGTGCACAACGCCAAGACCAAGCCTAGAGAGGAACAGTACAACAGC ACCTACCGGGTGGTGTCCGTGCTGACAGTGTGACACAGGACTGGCT GAACGGCAAAGAGTACAAGTGCAAGGTGTCCAACAAGGCCCTGCCT GCCCCATCGAGAAAACCATCAGCAAGGCCAAGGGCCAGCCCCGCG AACCCAGGTGTACACACTGCCCCAAGCAGGGACGAGCTGACCAA GAACCAAGGTGTCCCTGACCTGTCTCGTGAAAGGCTTCTACCCCTCCG ATATCGCCGTGGAATGGGAGAGCAACGGCCAGCCCCGAGAACACTA CAAGACCACCCCCCTGTGCTGGACAGCGACGGCTCATTCTTCCTGT ACAGCAAGCTGACCGTGGACAAGTCCCGGTGGCAGCAGGGCAACGT GTTTCAGCTGCAGCGTGATGCACGAGGCCCTGCACAACCACTACACCC AGAAGTCCCTGAGCCTGAGCCCTGGC
100	DNA	CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTAGCGGCAGCAGCTCCAACATCGGCAAC AACGACGTGTCTGGTATCAGCAGCTGCCTGGCACAGCCCCCTAAACT GCTGATCTACGCCGACAGCCACAGACCTAGCGCGTGCAGATAGAT TCAGCGGCTCTAAGAGCGGCACATCTGCCAGCCTGGCCATCTCTGGA CTGAGATCTGAGGACGAGGCGGACTACTATTGCGGCGCCTGGGATTCT TAGCCTGAGCGGCTATGTTTTTGGCGGAGGCCACCAAGCTGACCGTGC TAGGCCAGCCTAAAGCCGCCCCCTAGCGTGACCTGTTCCCTCCAAGC AGCGAGGAACTGCAGGCCAACAAGGCCACCTCGTGTGCCTGATCAG CGACTTCTATCCTGGCGCCGTGACCGTGGCCTGGAAGGCCGATAGCT CTCCTGTGAAGGCCGCGCTGGAACCACACCCCTAGCAAGCAGAGC AACAAACAATACGCCGCCAGCAGCTACCTGAGCCTGACCCCCGAGCA GTGGAAGTCCCACAGATCCTACAGCTGCCAAGTGACCCACGAGGGCA GCACCGTGGAAAAGACAGTGGCCCTACCGAGTGCAGC
101	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYEMNWVRQAPGKLEWV SGISWNSGSIQYADSVKGRFTISRDNKNTLYLQMNSLRAREDYVYYCA RSGYSSSWFDPDFDYWQQGLTVTVSS
102	PRT	SYEMN
103	PRT	GISWNSGSIQYADSVKG
104	PRT	SGYSSSWFDPDFDY
105	PRT	QSVLTQPPSVSGAPGQRTISCTGSSSNIGAGYDVHWYQQLPGTAPKLLI YGNNSNRPSGVPDRFSGSKSGTSASLAITGLQAEDEADYYCSSYAGSNPY VVFPGGKLTVL
106	PRT	TGSSSNIGAGYDVH
107	PRT	GNSNRPS
108	PRT	SSYAGSNPYVV
109	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTACCTTTAGCAGCT ACGAGATGAACCTGGGTCCGACAGGCCCTGGCAAAGGCCCTGAATG GGTGTCCGGCATCAGCTGGAATAGCGGCTCTATCGGCTACGCCGACA GCGTGAAGGGCAGATTACCATCAGCCGGGACAAACAGCAAGAACAC CCTGTACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCGTGT ACTACTGTGCCAGAAGCGGCTACAGCAGCTCTTGGTTTGACCCCGAC TTCGACTATTGGGGCCAGGGCACACTGGTCACAGTCTCTTCA
110	DNA	AGCTACGAGATGAAC
111	DNA	GGCATCAGCTGGAATAGCGGCTCTATCGGCTACGCCGACAGCGTGAA GGC
112	DNA	AGCGGCTACAGCAGCTCTTGGTTTGACCCCGACTTCGACTAT

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
113	DNA	CAGTCTGTTCTGACACAGCCTCCATCTGTGTCTGGCGCCCTGGACAG AGAGTGACCATCAGCTGTACAGGCAGCAGCTCCAATATCGGAGCCGG CTATGACGTGCACTGGTATCAGCAGCTGCCTGGCACAGCCCTAAAC TGCTGATCTACGGCAACAGCAACAGACCCAGCGGCGTGCCCGATAGA TTTCCGGCTCTAAGAGCGGCACAAGCGCCAGCCTGGCTATTACTGG ACTGCAGGCCGAGGACGAGGCCGACTACTACTGTTCTAGCTACGCCG GCAGCAACCCCTACGTGGTGTGTTGGCGGAGGCACCAAGCTGACAGTT CTA
114	DNA	ACAGGCAGCAGCTCCAATATCGGAGCCGGCTATGACGTGCAC
115	DNA	GGCAACAGCAACAGACCCAGC
116	DNA	TCTAGCTACGCCGGCAGCAACCCCTACGTGGTG
117	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTSSYEMNWVRQAPGKLEWV SGISWNSGSIQYADSVKGRFTISRDNKNTLYLQMNSLRADTAVYYCA RSQYSSSWFDPDFDYWGQGLTVTVSSASTKGPSVFPPLAPSSKSTSGGTA ALGCLVKDYFPEPTVTSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTP SSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPELGGPS VFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNA KTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKT I SKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNG QPENNYKTTTPVLDSDGSFPLYSKLTVDKSRWQQGNVFCFSVMHEALH NHYTQKSLSLSPG
118	PRT	QSVLTQPPSVSGAPGQRVTISCTGSSSNIGAGYDVHWYQQLPGTAPKLLI YGNSNRPSGVPDRFSGSKSGTSASLAITGLQAEDEADYYCSSYAGSNPY VVGGGTKLTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAV TVAWKADSSPVKAGVETTTPSKQSNKNKYAASSYLSLTPEQWKSHRYS CQVTHEGSTVEKTVAPTECS
119	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCACTTTAGCAGCT ACGAGATGAACCTGGGTCGACAGGCCCTGGCAAAGCCCTGAATG GGTGTCCGGCATCAGCTGGAATAGCGGCTCTATCGGCTACGCCGACA CGGTGAAGGCAGATTACCATCAGCCGGGACAAACAGCAAGAACAC CCTGTACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCGTGT ACTACTGTGCCAGAAGCGGCTACAGCAGCTCTTGGTTTGACCCCGAC TTCGACTATTGGGGCCAGGGCACACTGGTCACAGTCTCTTCAGCCAG CACCAGGGGCCAGCGTGTTCCTCTGGGCCCTAGCAGCAAGAGCA CATCTGGCGGAACAGCCGCCCTGGGCTGCCTCGTGAAGGACTACTTT CCCGAGCCCGTGACCGTGTCTGGAACCTTGCGGCTCTGACAAGCGG CGTGACACACCTTTCAGCCGTGTGTCAGAGCAGCGCCTGTACTCTCT GAGCAGCGTGTGACAGTGCCAGCAGCTCTCTGGGCACCCAGACCT ACATCTGCAACGTGAACCAAGCCAGCAACCAAGGTGGACAA GAAGGTGGAACCAAGAGCTGCACAGACCCACACCTGTCCCCCTT GTCCTGCCCCGAACTGCTGGGAGGCCCTTCCGTGTTCTGTTCCTCC CAAAGCCCAAGGACACCTTGATGATCAGCCGGACCCCGAAGTGAC CTGCGTGGTGGTGGATGTGTCCACAGAGACCTGAAGTGAAGTTCA ATTGGTACGTGGACGGCGTGAAGTGCACAACGCCAAGACCAAGCC TAGAGAGGAACAGTACAACAGCACCTACCGGTGGTGTCCGTGCTGA CAGTGCTGCACAGGACTGGCTGAACGGCAAGAGTACAAGTGCAA GGTGTCCAACAAGGCCCTGCCTGCCCTATCGAGAAAACCATCAGCA AGGCCAAGGGCCAGCCCCGCAACCCAGGTGTACACACTGCCCCCA AGCAGGGACGAGCTGACCAAGAACCAGGTGTCCCTGACCTGTCTCGT GAAAGGCTTCTACCCCTCCGATATCGCCGTGGAATGGGAGAGCAACG CCAGCCCGAGAACAACTACAAGACCAACCCCTGTGCTGGACAGC GACGGCTCATTTCTTCTGTACAGCAAGCTGACCGTGGACAAGTCCCG GTGGCAGCAGGGCAACGTGTTACAGCTGCAGCGTGTATGCACGAGGCC TGACAAACCACTACACCCAGAAGTCCCTGAGCCTGAGCCCTGGC
120	DNA	CAGTCTGTTCTGACACAGCCTCCATCTGTGTCTGGCGCCCTGGACAG AGAGTGACCATCAGCTGTACAGGCAGCAGCTCCAATATCGGAGCCGG CTATGACGTGCACTGGTATCAGCAGCTGCCTGGCACAGCCCTAAAC TGCTGATCTACGGCAACAGCAACAGACCCAGCGGCGTGCCCGATAGA TTTCCGGCTCTAAGAGCGGCACAAGCGCCAGCCTGGCTATTACTGG ACTGCAGGCCGAGGACGAGGCCGACTACTACTGTTCTAGCTACGCCG GCAGCAACCCCTACGTGGTGTGTTGGCGGAGGCACCAAGCTGACAGTT CTAGCCAGCCTAAAGCCGCCCTAGCGTGACCTGTTCCTTCCAAG CAGCGAGGAAGTGCAGGCCAACAGGCCACCCCTCGTGTGCTGATCA

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
		GCGACTTCTATCCTGGCGCCGTGACCGTGGCCTGGAAGGCCGATAGC TCTCCTGTGAAGGCCGGCGTGGAAACCACCACCCCTAGCAAGCAGAG CAACAACAAATACGCCGCCAGCAGCTACCTGAGCCTGACCCCCGAGC AGTGGAAAGTCCACAGATCCTACAGCTGCCAAGTGACCCACGAGGGC AGCACCGTGAAAAGACAGTGGCCCTACCGAGTGCAGC
121	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYAMSWVRQAPGKLEWV SAIGTGGDTYYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCA RRDDYTSRDAFDYWGQGLTVTVSS
122	PRT	SYAMS
123	PRT	AIGTGGDTYYADSVKG
124	PRT	RDDYTSRDAFDY
125	PRT	QSVLTQPPSASGTPGQRTVITSCGSSSNIGSNTVNWYQQLPGTAPKLLIYY DDL RPSGV PDRFSGSKSGTSASLAISGLQSEDEADYYCAAWDDSLNGYV VFGGGTKLTVL
126	PRT	SGSSSNIGSNTVN
127	PRT	YDDL RPS
128	PRT	AAWDDSLNGYVV
129	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCACTTTAGCAGCT ACGCCATGAGCTGGGTCCGACAGGCTCCTGGCAAAGCCCTTGAATGG GTGTCCGCCATTGGCACAGGCGGCGATACCTACTACGCCGACTCTGT GAAGGGCAGATTACCATCAGCCGGGACAACAGCAAGAACACCCCTG TACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCGTGTACTA TTGCGCCAGAAGGGACGACTACACCAGCAGGGACGCCCTTCGATTATT GGGGCCAGGGCACACTGGTCACCGTTTCTTCA
130	DNA	AGCTACGCCATGAGC
131	DNA	GCCATTGGCACAGGCGGCGATACCTACTACGCCGACTCTGTGAAGGG C
132	DNA	AGGGACGACTACACCAGCAGGGACGCCTTCGATTAT
133	DNA	CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTAGCGGCAGCAGCTCCAACATCGGCAGC AACACCGTGAACTGGTATCAGCAGCTGCCTGGCACAGCCCTAAACT GCTGATCTACTACGACGACCTGCGGCCTAGCGCGTGCCAGATAGAT TTTCTGGCAGCAAGAGCGGCACCTCTGCCAGCCTGGCTATTTCTGGA CTGCAGAGCAGGACGAGGCGGACTATTATTGTGCCGCGCTGGGACGA CAGCCTGAACGGCTATGTTGTTTTTCGGCGGAGGCACCAAGCTGACCG TTCTA
134	DNA	AGCGGCAGCAGCTCCAACATCGGCAGCAACACCGTGAAC
135	DNA	TACGACGACCTGCGGCCTAGC
136	DNA	GCCGCCTGGGACGACAGCCTGAACGGCTATGTTGTT
137	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYAMSWVRQAPGKLEWV SAIGTGGDTYYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCA RRDDYTSRDAFDYWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAAL GCLVKDYFPEPTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSS LGTQTYICNVNHKPSNTKVDKKVEPKSCDKHTHTCPPCPAPPELLGGPSVFL FPPKPKDTLMI SRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTK PREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKA KGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPE NNYKTTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHY TQKSLSLSPG
138	PRT	QSVLTQPPSASGTPGQRTVITSCGSSSNIGSNTVNWYQQLPGTAPKLLIYY DDL RPSGV PDRFSGSKSGTSASLAISGLQSEDEADYYCAAWDDSLNGYV VFGGGTKLTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVT

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
		VAWKADSSPVKAGVETTPSKQSNKNYAASSYLSLTPQWKSHRSYSYSC QVTHEGSTVEKTVAPTECS
139	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTACCTTTAGCAGCT ACGCCATGAGCTGGGTCCGACAGGCTCCTGGCAAAGGCCCTTGAATGG GTGTCCGCCATTGGCACAGGCGGCGATACCTACTACGCCGACTCTGT GAAGGGCAGATTACCATCAGCCGGGACAAACAGCAAGAACACACCTG TACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCGTGTACTA TTGCGCCAGAAGGGACGACTACACAGCAGGGACGCTTCGATTATT GGGGCCAGGGCACACTGGTCACCGTTCTTTCAGCCAGCACAAGGGC CCCAGCGTGTTCCTCTGGCCCCCTAGCAGCAAGAGCACATCTGGCGG AACAGCCGCCCTGGGCTGCCTCGTGAAGGACTACTTCCCGAGCCCG TGACCGTGTCTTGAACCTCTGGCGCTCTGACAAGCGGCGTGACACCC TTTCCAGCCGTGCTGCAGAGCAGCGGCTGTACTCTCTGAGCAGCGT CGTGACAGTGCCCGAGCAGCTCTCTGGGCACCCAGACCTACATCTGCA ACGTGAACCACAAGCCAGCAACACCAAGGTGGACAAGAGGTGGA ACCCAAGAGCTGCGACAAGACCCACACCTGTCCCCCTTGTCTGCCCC CCGAAGTGTCTGGGAGGCCCTTCCGTGTTCTGTTCCCCCAAAGCCCA AGGACACCCGTGATGATCAGCCGGACCCCGAAGTGACCTGCGTGGTG GTGGATGTGTCCCACGAGGACCTGAAGTGAAGTTCAATTGGTACGT GGACGGCGTGGAAGTGACACACGCCAAGACCAAGCCTAGAGAGGAA CAGTACAACAGCACCTACCGGGTGGTGTCCGTGCTGACAGTGTGTGCA CCAGGACTGGCTGAACGGCAAGAGTACAAGTGAAGGTGTCCAAC AAGGCCCTGCCTGCCCCATCGAGAAAACCATCAGCAAGGCCAAGG GCCAGCCCCGCAACCCAGGTGTACACACTGCCCCCAAGCAGGGAC GAGCTGACCAAGAACCAGGTGTCCCTGACCTGTCTCGTGAAAGGCTT CTACCCCTCCGATATCGCCGTGGAATGGGAGAGCAACGGCCAGCCCG AGAACAACCTACAAGACCACCCCCCTGTGTCTGACAGCGACGGCTCA TTCTTCTGTACAGCAAGCTGACCGTGGACAAGTCCCGGTGGCAGCA GGGCAACGTGTTTACGTGCAGCGTGATGCACGAGGCCCTGCACAACC ACTACACCCAGAAGTCCCTGAGCCTGAGCCCTGGC
140	DNA	CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTAGCGGCAGCAGCTCCAACATCGGCAGC AACACCGTGAACCTGGTATCAGCAGCTGCCTGGCACAGCCCCCTAACT GCTGATCTACTACGACGACCTGCGGCCTAGCGCGTGCCAGATAGAT TTTCTGGCAGCAAGAGCGGCACCTCTGCCAGCCTGGCTATTTCTGGA CTGCAGAGCGAGGACGAGGCCGACTATTATTGTGCCGCTGGGACGA CAGCCTGAACGGCTATGTTGTTTTTCGGCGGAGGCACCAAGCTGACCG TTCTAGGCCAGCCTAAAGCCGCCCTAGCGTGACCTGTTCCTCCCTCAA GCAGCGAGGAACTGCAGGCCAACAGGCCACCCCTCGTGTGCCTGATC AGCGACTTCTATCCTGGCGCGTGACCGTGGCTTGAAGGCCGATAG CTCTCCTGTGAAGGCCGGCGTGGAACACCACCCCTAGCAAGCAGA GCAACAACAATAACGCCCGCAGCAGCTACCTGAGCCTGACCCCGAG CAGTGAAGTCCACAGATCCTACAGCTGCCAAGTGACCCACGAGGG CAGCACCGTGGAAAAGACAGTGGCCCCCTACCGAGTGCAGC
141	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFYSYAMSWVRQAPGKLEWV SAIGTGGDTYYADSVKGRFTISRDNKNTLYLQMNSLRADTAVYYCA RRDDYTSRDAFDYWGQTLVTVSS
142	PRT	SYAMS
143	PRT	AIGTGGDTYYADSVKG
144	PRT	RDDYTSRDAFDY
145	PRT	QSVLTQPPSASGTPGQRVITISCSGSSSNIGSNTVNWYQQLPGTAPKLLIYY DDLRLSPGVPRFSGSKSGTSASLAISGLQSEDEADYYCAAWDDSLNDYV VFGGGTKLTVL
146	PRT	SGSSSNIGSNTVN
147	PRT	YDDLRLPS
148	PRT	AAWDDSLNDYVV
149	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTACCTTTTACAGCTA CGCCATGAGCTGGGTCCGACAGGCCCTGGAAAAGGCCTTGAATGGG

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
		TGTCGCCATCGGCACAGGCGGCGATACCTACTATGCCGACTCTGTG AAGGGCAGATTACCATCAGCCGGGACACAGCAAGAACCCTGT ACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCGTGTACTAT TGCGCCAGAAGGGACGACTACACAGCAGGGACGCCCTTCGATTATTG GGCCAGGGCACACTGGTCACCGTTTCTTCA
150	DNA	AGCTACGCCATGAGC
151	DNA	GCCATCGGCACAGGCGGCGATACCTACTATGCCGACTCTGTGAAGGG C
152	DNA	AGGGACGACTACACAGCAGGGACGCCCTTCGATTAT
153	DNA	CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTAGCGGCAGCAGCTCCAACATCGGCAGC AACACCGTGAAGTGGTATCAGCAGCTGCCTGGCACAGCCCCATAACT GCTGATCTACTACGACGACCTGCGGCCTAGCGCGTGCAGATAGAT TTTCTGGCAGCAAGAGCGGCACCTCTGCCAGCCTGGCTATTCTGGA CTGCAGAGCGAGGACGAGGCCGACTATTATTGCGCGCTGGGACGA CAGCCTGAACGACTACGTTGTGTTGGCGGAGGCCAACGCTGACCG TTCTA
154	DNA	AGCGGCAGCAGCTCCAACATCGGCAGCAACACCGTGAAC
155	DNA	TACGACGACCTGCGGCCTAGC
156	DNA	GCCGCCTGGGACGACAGCCTGAACGACTACGTTGTG
157	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFYYSAMSWVRQAPGKLEWV SAIGTGGDTYYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCA RRDDYTSRDAFDYWGQGLVTVSSASTKGPSVFPLAPCSRSTSESTAAL GCLVKDYFPEPTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSS LGKTKYITCNVDHKPSNTKVDKRVESKYGGPPCPAPAEFLGGPSVFLPPP KPKDTLMISRTPEVTCVVVDVSDPEVQFNWYVDGVEVHNAKTKPRE EQFNSTYRVVSVLTVLHQDNLNGKEYKCKVSNKGLPSSIEKTIKAKGQ PREPQVYTLPPSQEEMTKNQVSLTCLVKGPYPSDIAVEWESNGQPENNY KTTTPVLDSDGSFFLYSRLTVDKSRWQEGNVFSCSMHEALHNHYTQK SLSLSLGK
158	PRT	QSVLTQPPSASGTPGQRTVITSCGSSSNIGSNTVNWYQLPGTAPKLLIYY DDL RPSGVPDRFSGSKSGTSASLAISGLQSEDEADYYCAWDDSLNDYV VFGGGTKLTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVT VAWKADSPVKAGVETTPSKQSNKYAASSYLSTPEQWKSHRSYSY QVTHEGSTVEKTVAPTECS
159	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCACTTTTACAGCTA CGCCATGAGCTGGGTCGACAGGCCCTGGAAAAGGCCTTGAATGGG TGTCGCCCATCGGCACAGGCGGCGATACCTACTATGCCGACTCTGTG AAGGGCAGATTACCATCAGCCGGGACACAGCAAGAACCCTGT ACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCGTGTACTAT TGCGCCAGAAGGGACGACTACACAGCAGGGACGCCCTTCGATTATTG GGCCAGGGCACACTGGTCACCGTTTCTTCAGCCAGCACCAAGGGCC CCAGCGTGTTCCTCTGGCCCCCTTGTAGCAGAAGCACCAGCGAGTCT ACAGCCGCCCTGGGCTGCCTCGTGAAGGACTACTTCCCGAGCCCGT GACCGTGTCTGGAATCTGGCGCTCTGACAAGCGCGGTGCACACCT TTCAGCCCGTGTGTCAGAGCAGCGGCTGTACTCTCTGAGCAGCGTC GTGACAGTGCCAGCAGCAGCCTGGGCACCAAGACCTACACCTGTAA CGTGGACCAACAAGCCAGCAACACCAAGGTGGACAAGCGGGTGGAA TCTAAGTACGGCCCTCCCTGCCCTCCTTGCCAGCCCTGAATTTCTG GCGGACCCCTCCGTGTTCTGTTCCCCCAAGCCCAAGGACACCCCT GATGATCAGCCGACCCCGAAGTGACCTGCGTGGTGGTGGATGTGT CCCAGGAAGATCCCAGGTGCAGTTCAATTGGTACGTGGACGGCGTG GAAGTGACACAACGCCAAGACCAAGCCAGAGAGGAACAGTTCAACA GCACCTACCGGGTGGTGTCCGTGCTGACAGTGTGACACAGGACTGG CTGAACGGCAAGAGGTACAAGTGCAAGGTGTCCAACAAGGGCCTGC CCAGCTCCATCGAGAAAACCATCAGCAAGGCCAAGGGCCAGCCCCG CGAACCCAGGTGTACACACTGCCTCCAAGCCAGGAAGAGATGACC AAGAACCAGGTGTCCCTGACCTGTCTCGTGAAAGGCTTCTACCCCTC CGATATCGCCGTGGAATGGGAGAGCAACGGCCAGCCCGAGAACAAC TACAAGACCACCCCCCTGTGCTGGACAGCGACGGCTATTCTTCCT

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
160	DNA	GTACAGCAGACTGACCGTGGACAAGAGCCGGTGGCAGGAAGGCAAC GTGTTGAGCTGCAGCGTGATGCACGAGGCCCTGCACAACCACTACAC CCAGAAGTCCCTGTCTCTGAGCCTGGGCAAG CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTAGCGGCAGCAGCTCCAACATCGGCAGC AACACCGTGAAGTGGTATCAGCAGCTGCCTGGCACAGCCCCTAAACT GCTGATCTACTACGACGACCTGCGGCCTAGCGCGTGGCCAGATAGAT TTTCTGGCAGCAAGAGCGGCACCTCTGCCAGCCTGGCTATTTCTGGA CTGCAGAGCGAGGACGAGGCCGACTATTATTGTGCCGCTGGGACGA CAGCCTGAACGACTACGTTGTGTTGGCGGAGGCACCAAGCTGACCG TTCTAGGCCAGCCTAAAGCCGCCCTAGCGTGACCTGTTCCCTCCAA GCAGCGAGGAAGTGCAGGCCAACAGGCCACCTCGTGTGCTGATC AGCGACTTCTATCCTGGCGCCGTGACCGTGGCCTGGAAGGCCGATAG CTCTCTGTGAAGGCCGCGTGGAAACCAACCCCTAGCAAGCAGA GCAACAACAAATACGCCGCCAGCAGCTACCTGAGCCTGACCCCGAG CAGTGGGAAGTCCACAGATCCTACAGCTGCCAAGTGACCCACGAGGG CAGCACCGTGGAAAAGACAGTGGCCCTACCGAGTGCAGC
161	PRT	EVQLLESGLVQPGGSLRLSCAASGFTFSSYAMSWVRQAPGKLEWV SAIGYGGDTYYADSVKGRFTISRDNISKNTLYLQMNSLRAEDTAVYYCA RRDYSRDAFDYWGQTLTVSS
162	PRT	SYAMS
163	PRT	AIGYGGDTYYADSVKG
164	PRT	RDDYSRDAFDY
165	PRT	QSVLTQPPSASGTPGQRTVISCSSSSNIGSNNTVNWYQQLPGTAPKLLIYY DDLRLPSGVPRDFSGSKSGTSASLAISGLQSEDEADYYCAWDDSLNDIV VFGGGTKLTVL
166	PRT	SGSSSNIGSNNTVN
167	PRT	YDDLRLPS
168	PRT	AAWDDSLNDIV
169	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCACTTTAGCAGCT ACGCCATGAGCTGGGTCCGACAGGCTCCTGGCAAAGGCCCTGAATGG GTGTCGCCCATCGGCTATGGCGGCGATACCTACTACGCCGACTCTGT GAAGGGCAGATTACCATCAGCCGGGACAAACAGCAAGAACACCCCTG TACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCGTGACTA TTGCGCCAGAAGGGACGACTACACAGCAGGGACGCCTTCGATTATT GGGGCCAGGGCACACTGGTCACCGTTTCTTCA
170	DNA	AGCTACGCCATGAGC
171	DNA	GCCATCGGCTATGGCGGCGATACCTACTACGCCGACTCTGTGAAGGG C
172	DNA	AGGGACGACTACACCAGCAGGGACGCCTTCGATTAT
173	DNA	CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTAGCGGCAGCAGCTCCAACATCGGCAGC AACACCGTGAAGTGGTATCAGCAGCTGCCTGGCACAGCCCCTAAACT GCTGATCTACTACGACGACCTGCGGCCTAGCGCGTGGCCAGATAGAT TTTCTGGCAGCAAGAGCGGCACCTCTGCCAGCCTGGCTATTTCTGGA CTGCAGAGCGAGGACGAGGCCGACTATTATTGTGCCGCTGGGACGA CAGCCTGAACGACATCGTTGTTTTGCGCGGAGGCACCAAGCTGACCG TTCTA
174	DNA	AGCGGCAGCAGCTCCAACATCGGCAGCAACCCGTGAAC
175	DNA	TACGACGACCTGCGGCCTAGC
176	DNA	GCCGCTGGGACGACAGCCTGAACGACATCGTTGTT

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
177	PRT	EVQLLESGGGLVQPGGSLRLSCLASGFTFSSYAMSWVRQAPGKGLEWV SAIGYGGDTYYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCA RRDDYTSRDAFDYWGQGLTVTVSSASTKGPSVFPLAPCSRSTSESTAAL GCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSS LGTKTYTNCNVDHKPSNTKVDKRVESKYGPPCPPCPAPEFLGGPSVFLFPP KPKDTLMISRTPEVTCVVDVVSQEDPEVQFNWYVDGVEVHNAKTKPRE EQFNTYRVVSVLTVLHQDWLNGKEYKCKVSNKGLPSSIEKTIKAKGQ PREPQVYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNY KTTTPPVLDSDGSFFLYSRLTVDKSRWQEGNVFSCSVMEALHNHYTQK SLSLSLGK
178	PRT	QSVLTQPPSASGTPGQRVTISCSGSSSNIGSNTVNWYQQLPGTAPKLLIYY DDLRLPSGVPRDFSGSKSGTSASLAISGLQSEDEADYYCAWDDSLNDIV VFGGGTKLTVLQGPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVT VAWKADSSPVKAGVETTPSKQSNKNYAASSYLSLTPEQWKSRSYSC QVTHEGSTVEKTVAPTECS
179	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCGCAGCGCTTCACTTTAGCAGCT ACGCCATGAGCTGGGTCCGACAGGCTCCTGGCAAAGGCCTTGAATGG GTGTCGCCCATCGGCTATGGCGGCGATACCTACTACGCCGACTCTGT GAAGGGCAGATTACCATCAGCCGGGACAAACAGCAAGAACACCCCTG TACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCGGTGTA TTGCGCCAGAAGGGACGACTACACCAGCAGGACGCGCTTCGATTATT GGGGCCAGGGCACACTGGTCACCGTTTCTTACGCCAGCACCAAGGGC CCAGCGGTGTTCCTCTGGCCCCCTTGTAGCAGAAGCACCAGCGAGTC TACAGCGCCCTGGGCTGCCTCGTGAAGGACTACTTTCCGAGCCCG TGACCGTGTCTTGAACCTCGGCGCTCTGACAAGCGCGGTGCACACC TTTCAGCGCTGTGTCAGAGCAGCGGCTGTACTCTCTGAGCAGCGT CGTGACAGTGCCAGCAGCAGCCTGGGCACCAAGACCTACACCTGTA ACGTGGACCACAAGCCAGCAACACCAAGGTGGACAAGCGGGTGA ATCTAAGTACGGCCCTCCCTGCCTCCTTGCCAGCCCTGAATTTCT GGGCGGACCCTCCGTGTTCTGTTCCCCCAAGCCCAAGGACACCC TGATGATCAGCCGGACCCCGAAGTGACCTGCGTGGTGGTGGATGTG TCCAGGAAGATCCCGAGGTGCAGTTCAATTGGTACGTGGACGGCGT GGAAGTGCAACGCCAAGACCAAGCCAGAGAGGAACAGTTCAAC AGCACCTACCGGGTGGTGTCCGTGCTGACAGTGCTGCACAGGACTG GCTGAACGGCAAAGAGTACAAGTGCAAGGTGTCCAACAAGGGCCTG CCCAGCTCCATCGAGAAAACCATCAGCAAGGCCAAGGGCCAGCCCC CGGAACCCAGGTGTACACACTGCCTCCAAGCCAGGAAGAGATGAC CAAGAACCAGGTGTCCCTGACCTGTCTCGTGAAGGCTTCTACCCCT CCGATATCGCCGTGGAATGGGAGAGCAACGGCCAGCCCGAGAACAA CTACAAGACCACCCCCCTGTGCTGGACAGCGACGCTATTCTTCTT GTACAGCAGACTGACCGTGGACAAGAGCCGCTGGCAGGAAGGCAAC GTGTTCAAGTGCAGCGTGATGCACGAGGCCCTGCACAACCACTACAC CCAGAAGTCCCTGTCTCTGAGCCTGGCAAG
180	DNA	CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTAGCGGCAGCAGCTCCAACATCGGCAGC AACACCGTGAACCTGGTATCAGCAGCTGCCTGGCACAGCCCTAAACT GCTGATCTACTACGACGACCTGCGGCCTAGCGGCGTGCCAGATAGAT TTTCTGGCAGCAAGAGCGGCACCTCTGCCAGCCTGGCTATTTCTGGA CTGCAGAGCGAGGACGAGGCCGACTATTATTGTGCCGCTGGGACGA CAGCCTGAACGACATCGTTGTTTTCGGCGGAGGCACCAAGCTGACCG TTCTAGGCCAGCCTAAAGCCGCCCTAGCGTGACCTGTTCCTCCAA GCAGCGAGGAACCTGACGGCCAAACAGGCCACCCCTCGTGTGCTGATC AGCGACTTCTATCTGGCGCGGTGACCGTGGCCTGGAAGGCCGATAG CTCTCTGTGAAGGCCGGCGTGAAACCAACACCCCTAGCAAGCAGA GCAACAACAATAACGCCCGCAGCAGCTACCTGAGCCTGACCCCGAG CAGTGAAGTCCACAGATCCTACAGCTGCCAAGTGACCCACGAGGG CAGCACCGTGAAAAGACAGTGGCCCTACCGAGTGCAGC
181	PRT	EVQLLESGGGLVQPGGSLRLSCLASGFTFSSYAMSWVRQAPGKGLEWV SAIGYGGDTYYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCA RRDDYTSRDAFDYWGQGLTVTVSS
182	PRT	SYAMS
183	PRT	AIGYGGDTYYADSVKG
184	PRT	RDDYTSRDAFDY

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
185	PRT	QSVLTQPPSASGTPGQRTVITSCSGSSSNIGSNTVNWYQQLPGTAPKLLIYY DDLRP SGVPDRFSGSKSGTSASLAISGLQSEDEADYYCAAWDDSLNVYP VFGGGTKLTVL
186	PRT	SGSSSNIGSNTVN
187	PRT	YDDL RPS
188	PRT	AAWDDSLNVYPV
189	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCGCAGCGCTTCACCTTTAGCAGCT ACGCCATGAGCTGGGTCCGACAGGCTCCTGGCAAAGCCCTTGAATGG GTGTCCGCCATCGGCTATGGCGGCGATACCTACTACGCCGACTCTGT GAAGGGCAGATTACCATCAGCCGGGACAAACAGCAAGAACACCCCTG TACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCGTGTACTA TTGCGCCAGAAGGGACGACTACACCAGCAGGGACGCCCTTCGATTATT GGGCCAGGGCACACTGGTCACCGTTTCTTCA
190	DNA	AGCTACGCCATGAGC
191	DNA	GCCATCGGCTATGGCGGCGATACCTACTACGCCGACTCTGTGAAGGG C
192	DNA	AGGGACGACTACACCAGCAGGGACGCCTTCGATTAT
193	DNA	CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTAGCGGCAGCAGCTCCAACATCGGCAGC AACACCGTGAACCTGGTATCAGCAGCTGCCTGGCACAGCCCCCTAACT CTGATCTACTACGACGACCTGCGGCTAGCGCGTGCAGATAGAT TTTCTGGCAGCAAGAGCGGCACCTCTGCCAGCCTGGCTATTTCTGGA CTGCAGAGCGAGGACGAGGCCGACTATTATTGTGCCGCTGGGACGA CAGCCTGAACGTGTACCTGTTTGGCGGAGGCACCAAGCTGACCG TTCTA
194	DNA	AGCGGCAGCAGCTCCAACATCGGCAGCAACACCGTGAAC
195	DNA	TACGACGACCTGCGGCCTAGC
196	DNA	GCCGCTGGGACGACAGCCTGAACGTGTACCTGTT
197	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYAMSWVRQAPGKLEWV SAIGYGGDTYYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCA RRDDYTSRDAFDYWGQTLVTVSSASTKGPSVFPLAPCSRSTSESTAAL GCLVKDYFPEPTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSS LGTKYTCNVDHKPSNTKVDKRVESKYGPCCPPCPAPEFLGGPSVFLFPP KPKDTLMISRTPEVTCVVVDVSDQEDPEVFQFNWYVDGVEVHNAKTKPRE EQFNSYTRVSVLTVLHQDWLNGKEYKCKVSNKGLPSSIEKTIKAKGQ PREPQVYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNY KTTTPVLDSDGSFFLYSRLTVDKSRWQEGNVFSCVMHEALHNHYTQK SLSLSLGK
198	PRT	QSVLTQPPSASGTPGQRTVITSCSGSSSNIGSNTVNWYQQLPGTAPKLLIYY DDLRP SGVPDRFSGSKSGTSASLAISGLQSEDEADYYCAAWDDSLNVYP VFGGGTKLTVLGQPKAAPSVTLPFPSSSEELQANKATLVCLISDFYPGAVT VANKADSSPVKAGVETTPSKQSNKYAASSYLSLTPEQWKSHRSYSC QVTHEGSTVEKTVAPTECS
199	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCGCAGCGCTTCACCTTTAGCAGCT ACGCCATGAGCTGGGTCCGACAGGCTCCTGGCAAAGCCCTTGAATGG GTGTCCGCCATCGGCTATGGCGGCGATACCTACTACGCCGACTCTGT GAAGGGCAGATTACCATCAGCCGGGACAAACAGCAAGAACACCCCTG TACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCGTGTACTA TTGCGCCAGAAGGGACGACTACACCAGCAGGGACGCCCTTCGATTATT GGGCCAGGGCACACTGGTCACCGTTTCTTACGCCAGCAACCAAGGGC CCAGCGTGTTCCTCTGGCCCCCTGTAGCAGAAGCACCAGCGAGTC TACAGCGCCCTGGGCTGCTCGTGAAGGACTACTTTCCGAGCCCG TGACCGTGTCTGGAACCTGGCGCTCTGACAAGCGCGTGCACACC TTCCAGCGCTGCTGCAGAGCAGCGCCTGTACTCTCTGAGCAGCGT

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
		CGTGACAGTGCCAGCAGCAGCCTGGGCACCAAGACCTACACCTGTA ACGTGGACCACAAGCCCAGCAACCAAGGTGGACAGCGGTGGA ATCTAAGTACGGCCCTCCCTGCCCTCCTTGCCAGCCCCCTGAATTTCT GGCGGACCCCTCCGTGTTCTGTTCCTCCCAAGCCCAAGGACACCC TGATGATCAGCCGACCCCGAAGTGACCTGCGTGGTGGTGGATGTG TCCAGGAAGATCCCGAGGTGCAAGTTCAATTGGTACGTGGACGGCGT GGAAGTGCAACACGCCAAGCCAAGCCAGAGAGGAACAGTTCAAC AGCACCTACCGGTGGTGTCCGTGCTGACAGTGTGCACCGAGACTG GCTGAACGGCAAAGAGTACAAGTGCAAGGTGTCCAACAAGGGCCTG CCAGCTCCATCGAGAAAACCATCAGCAAGGCCAAGGCCAGCCCC GCGAACCCAGGTGTACACACTGCCTCCAAGCCAGGAAGAGATGAC CAAGAACCAGGTGTCCTGACCTGTCTCGTGAAGGCTTCTACCCCT CCGATATCGCCGTGGAATGGGAGAGCAACGCCAGCCCGAGAACA CTACAAGACCACCCCCCTGTGCTGGACAGCGACGGCTCATTCTCCT GTACAGCAGACTGACCGTGACAAGAGCCGGTGGCAGGAAGGCAAC GTGTTGAGCTGCAGCGTGATGCACGAGGCCCTGCACAACCACTACAC CCAGAAGTCCCTGTCTCTGAGCCTGGGCAAG
200	DNA	CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTAGCGGCAGCAGCTCCAACATCGGCAGC AACACCGTGAACCTGGTATCAGCAGCTGCCTGGCACAGCCCCCTAACT GCTGATCTACTACGACGACCTGCGGCCTAGCGCGTGCAGATAGAT TTTCTGGCAGCAAGAGCGGCACCTCTGCCAGCCTGGCTATTTCTGGA CTGCAGAGCGAGGACGAGGCCGACTATTATTGTGCGCCCTGGGACGA CAGCCTGAACGTGTACCTGTTTTTGGCGGAGGCACCAAGCTGACCG TTCTAGGCCAGCCTAAAGCCGCCCTAGCGTGACCTGTTCCTCCAA GCAGCGAGGAACCTGCAGGCCAACAGGCCACCCCTCGTGTGCCTGATC AGCGACTTCTATCCTGGCGCCGTGACCGTGGCCTGGAAGGCCGATAG CTCTCCTGTGAAGGCCGGCGTGGAACCAACACCCCTAGCAAGCAGA GCAACAACAATAACGCCGCCAGCAGCTACCTGAGCCTGACCCCGAG CAGTGAAGTCCACAGATCCTACAGCTGCCAAGTGACCCACGAGGG CAGCACCGTGGAAAAGACAGTGGCCCTACCGAGTGCAGC
201	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYAMSWVRQAPGKLEWV SAIGYGGDTYYADSVKGRFTISRDNKNTLYLQMNSLRABDTAVYYCA RRDDYTSRDAFDYWGQTLTVTVSS
202	PRT	SYAMS
203	PRT	AIGYGGDTYYADSVKG
204	PRT	RDDYTSRDAFDY
205	PRT	QSVLTQPPSASGTPGQRVITISCSGSSSNIGSNTNVNYYQLPGTAPKLLIYY DDLRLSPGVPRFSGSKGSASLAISGLQSEADYYCHAWDDSLNDIV VFGGGTKLTVL
206	PRT	SGSSSNIGSNTVN
207	PRT	YDDLRLPS
208	PRT	HAWDDSLNDIV
209	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGCTTCACCTTTAGCAGCT ACGCCATGAGCTGGGTCCGACAGGCTCCTGGCAAAGGCCTTGAATGG GTGTCCGCCATCGGCTATGGCGGCGATACCTACTACGCCGACTCTGT GAAGGGCAGATTACCATCAGCCGGGACAAACAGCAAGAACACCTG TACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCGTGTACTA TTGCGCCAGAAGGGACGACTACACCAGCAGGGACCGCCTTCGATTATT GGGCGCAGGGCACACTGGTCACCGTTTCTTCA
210	DNA	AGCTACGCCATGAGC
211	DNA	GCCATCGGCTATGGCGGCGATACCTACTACGCCGACTCTGTGAAGGG C
212	DNA	AGGGACGACTACACCAGCAGGGACGCCTTCGATTAT
213	DNA	CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTAGCGGCAGCAGCTCCAACATCGGCAGC

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
		AACACCGTGAACCTGGTATCAGCAGCTGCCTGGCACAGCCCCCTAAACT GCTGATCTACTACGACGACCTGCGGCCTAGCGGCTGCCAGATAGAT TTTCTGGCAGCAAGAGCGGCACCTCTGCCAGCCTGGCTATTTCTGGA CTGCAGAGCGAGGACGAGGCCGACTACTATTGTCAGCCTGGGACGA CAGCCTGAACGACATCGTGGTTTTTGGCGGAGGCACCAAGCTGACCG TTCTA
214	DNA	AGCGGCAGCAGCTCCAACATCGGCAGCAACACCGTGAAC
215	DNA	TACGACGACCTGCGGCCTAGC
216	DNA	CACGCCTGGGACGACAGCCTGAACGACATCGTGGTT
217	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYAMSWVRQAPGKGLEWV SAIGYGGDTYYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCA RRDDYTSRDAFDYWGQGLTVTVSSASTKGPSVFPLAPCSRSTSESTAAL GCLVKDYFPEPTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSS LGKTYTTCNVDHKPSNTKVDKRVESKYGPPCPPCPAPEFLGGPSVFLPPP KPKDTLMISRTPEVTCVVVDVSDPEVQFNWYVDGVEVHNAKTKPRE EQPMSTYRVVSVLTVLHQDWLNGKEYKCKVSNKGLPSSIEKTIKAKGQ PREPQVYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNY KTTTPVLDSDGSFFLYSRLTVDKSRWQEGNVFSCSVMHEALHNHYTQK SLSLSLGK
218	PRT	QSVLTQPPSASGTPGQRTVISCSGSSSNIGSNTVNWYQQLPGTAPKLLIYY DDL RPSGVPDRFSGSKSGTSASLAISGLQSEDEADYYCHAWDDSLNDIV VFGGGTKLTVLGGPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVT VAWKADSPVKAGVETTPSKQSNKYAASSYLSTPEQWKSHRSYSYC QVTHEGSTVEKTVAPTECS
219	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCACTTTAGCAGCT ACGCCATGAGCTGGGTCCGACAGGCTCCTGGCAAAGGCCCTGAATGG GTGTCGCCCATCGGCTATGGCGGCGATACCTACTACGCCGACTCTGT GAAGGGCAGATTCAACATCAGCCGGGACAAACAGCAAGAACACCCCTG TACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCGTGTACTA TTGCGCCAGAAGGGACGACTACACAGCAGGGACGCTTCGATTATT GGGGCCAGGGCACACTGGTCACCGTTTCTTACGCCAGCACCAAGGGC CCCAGCGTGTTCCTCTGGCCCCCTGTAGCAGAAGCACCGCAGCTG TACAGCCGCCCTGGGCTGCCTCGTGAAGGACTACTTCCCGAGCCCG TGACCGTGTCTGGAACCTGGCGCTCTGACAAGCGGCGTGACACCC TTTCCAGCCGTGCTGCAGAGCAGCGGCTGTACTCTCTGAGCAGCGT CGTGACAGTGCCAGCAGCAGCCTGGGCACCAAGACCTACACCTGTA ACGTGGACCACAAGCCAGCAACCAAGGTGACAGAGCGGCTGGA ATCTAAGTACGGCCCTCCCTGCCCTCCTTGCCAGCCCTGAATTTCT GGCGGAGCCCTCCGTGTTCTGTTCCCCCAAGCCCAAGGACACCC TGATGATCAGCCGAGCCCGAAGTGACCTGCGTGGTGGTGGATGTG TCCAGGAAGATCCCGAGGTGCAGTTCAATTGGTACGTGGACGGCGT GGAAGTGCAACGCCAAGACCAAGCCAGAGAGGAACAGTTCAAC AGCACCTACCGGGTGGTGTCCGTGCTGACAGTGTGTCACCGAGGACTG GCTGAACGGCAAAGAGTACAAGTGCAAGGTGTCCAACAAGGGCCTG CCAGCTCCATCGAGAAAACATCAGCAAGGCCAAGGGCCAGCCCC GCGAACCCAGGTGTACACACTGCCTCCAAGCCAGGAAGAGATGAC CAAGAACCAGGTGTCCCTGACCTGTCTCTGTAAGGCTTCTACCCCT CCGATATCGCCGTGGAATGGGAGAGCAACGGCCAGCCCGAGAACA CTACAAGACCACCCCCCTGTGCTGGACAGCGACGCTATTCTTCCT GTACAGCAGACTGACCGTGGACAAGAGCCGGTGGCAGGAAGGCAAC GTGTTAGCTGCAGCGTGATGCACGAGGCCCTGCACAACCACTACAC CCAGAAGTCCCTGTCTCTGAGCCTGGGCAAG
220	DNA	CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTAGCGGCAGCAGCTCCAACATCGGCAGC AACACCGTGAACCTGGTATCAGCAGCTGCCTGGCACAGCCCCCTAAACT GCTGATCTACTACGACGACCTGCGGCCTAGCGGCTGCCAGATAGAT TTTCTGGCAGCAAGAGCGGCACCTCTGCCAGCCTGGCTATTTCTGGA CTGCAGAGCGAGGACGAGGCCGACTACTATTGTCAGCCTGGGACGA CAGCCTGAACGACATCGTGGTTTTTGGCGGAGGCACCAAGCTGACCG TTCTAGGCCAGCCTAAAGCCGCCCTAGCGTGACCTGTTCCTCCAA GCAGCGAGGAACTGCAGGCCAACAAGGCCACCTCTGTGCTGCTGATC AGCGACTTCTATCCTGGCGCGTGACCGTGGCCTGGAAGGCCGATAG CTCTCTGTGAAGGCCGCGTGGAACCAACACCCCTAGCAAGCAGA

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
		GCAACAACAAATACGCCGCCAGCTACCTGAGCCTGACCCCCGAG CAGTGGAGTCCCACAGATCTACAGCTGCCAAGTGACCCACGAGGG CAGCACCGTGGAAAAGACAGTGGCCCTACCGAGTGCAGC
221	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYAMSWVRQAPGKGLEWV SAIGYGGDTYYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCA RRDDYTSRDAFDYWGQGLTVTVSS
222	PRT	SYAMS
223	PRT	AIGYGGDTYYADSVKG
224	PRT	RDDYTSRDAFDY
225	PRT	QSVLTQPPSASGTPGQRVITISGSSSNIGSNTVNWYQLPGTAPKLLIYY DDL RPSGVPDRFSGSKSGTSASLAISGLQSEDEADYYCHAWDDSLNDYP VFGGGTKLTVL
226	PRT	SGSSSNIGSNTVN
227	PRT	YDDL RPS
228	PRT	HAWDDSLNDYPV
229	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGCTTCACCTTTAGCAGCT ACGCCATGAGCTGGGTCCGACAGGCTCCTGGCAAAGGCTTGAATGG GTGTCGCCCATCGGCTATGGCGGCGATACCTACTACGCCGACTCTGT GAAGGGCAGATTACCATCAGCCGGGACAAACAGCAAGAACACCCCTG TACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCGTGACTA TTGCGCCAGAAGGGACGACTACACCAGCAGGGACGCCTTCGATTATT GGGGCCAGGCACACTGGTCACCGTTTCTTCA
230	DNA	AGCTACGCCATGAGC
231	DNA	GCCATCGGCTATGGCGGCGATACCTACTACGCCGACTCTGTGAAGGG C
232	DNA	AGGGACGACTACACCAGCAGGGACGCCTTCGATTAT
233	DNA	CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTAGCGGCAGCAGCTCCAACATCGGCAGC AACACCGTGAAGTGGTATCAGCAGCTGCCTGGCACAGCCCCATAACT GCTGATCTACTACGACGACCTGCGGCCTAGCGGCGTGCCAGATAGAT TTTCTGGCAGCAAGAGCGGCACCTCTGCCAGCCTGGCTATTCTGGA CTGCAGAGCGAGGACGAGGCGGACTACTATTGTACGCGCTGGGACGA CAGCCTGAACGACTACCTGTTTTTGGCGGAGGCACCAAGCTGACCG TTCTA
234	DNA	AGCGGCAGCAGCTCCAACATCGGCAGCAACACCGTGAAC
235	DNA	TACGACGACCTGCGGCCTAGC
236	DNA	CACGCCTGGGACGACAGCCTGAACGACTACCTGTT
237	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYAMSWVRQAPGKGLEWV SAIGYGGDTYYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCA RRDDYTSRDAFDYWGQGLTVTVSSASTKGPSVFPLAPCSRSTSESTAAL GCLVKDYFPEPTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSS LGKTKYTCNV DHPKPSNTKVDKRVESKYGPPCPPCPAPEFLGGPSVFLFPP PKDTLMISRTPEVTCVVVDVSDPEVQFNWYVDGVEVHNAKTKPRE EQFNTSYRVVSVLTVLHQDNLNGKEYKCKVSNKGLPSSIEKTIKAKGQ PREPQVYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNY KTTTPVLDSDGSFFLYSRLLTVDKSRWQEGNVFSCSVMHREALHNHYTQK SLSLSLGK
238	PRT	QSVLTQPPSASGTPGQRVITISGSSSNIGSNTVNWYQLPGTAPKLLIYY DDL RPSGVPDRFSGSKSGTSASLAISGLQSEDEADYYCHAWDDSLNDYP VFGGGTKLTVLGPQAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVT VAWKADSSPVKAGVETTPSKQSN NKYAASSYLSLTPEQWKSHRSYSC QVTHEGSTVEKTVAPTECS

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
239	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCACCTTTAGCAGCT ACGCCATGAGCTGGGTCCGACAGGCTCCTGGCAAAGGCCCTGAATGG GTGTCCGCCATCGGCTATGGCGGCGATACCTACTACGCCGACTCTGT GAAGGGCAGATTACCATCAGCCGGGACAACAGCAAGAACACCCCTG TACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCGTGTACTA TTGCGCCAGAAGGGACGACTACACCAGCAGGGACGCCTTCGATTATT GGGGCCAGGGCACACTGGTCACCGTTTCTTCAGCCAGCACCAAGGGC CCCAGCGTGTTCCTCTGGCCCCCTTGTAGCAGAAGCACCAGCGAGTC TACAGCCGCCCTGGGCTGCCTCGTGAAGGACTACTTCCCGAGCCCG TGACCGTGTCTGGAACCTGGGCGCTCTGACAAGCGGCGTGCACACC TTTCCAGCCGTGTGCAGAGCAGCGCCCTGTACTCTCTGAGCAGCGT CGTGACAGTGCCAGCAGCAGCCTGGGCACCAAGACCTACACCTGTA ACGTGGACCACAAGCCAGCAACACCAAGGTGGACAAGCGGGTGGGA ATCTAAGTACGGCCCTCCCTGCCCTCCTTGCCAGCCCCGTAATTTCT GGGCGGACCCCTCCGTGTTCCTGTTCCTCCCAAAGCCCAAGGACACCC TGATGATCAGCCGGACCCCCGAAGTGACCTGCGTGGTGGTGGATGTG TCCAGGAAGATCCCGAGGTGCAGTTCAATTGGTACGTGGACGGCGT GGAAGTGCAACAACGCCAAGACCAAGCCAGAGAGGAACAGTTCAAC AGCACCTACCGGGTGGTGTCCGTGTGACAGTGTCTGCACCAGGACTG GCTGAACGGCAAAGAGTACAAGTGCAAGGTGTCCAACAAGGGCTG CCAGCTCCATCGAGAAAACCATCAGCAAGGCCAAGGGCCAGCCCC GCGAACCCAGGTGTACACACTGCCTCCAAGCCAGGAAGAGATGAC CAAGAACCAGGTGTCCCTGACCTGTCTCGTGAAAGGCTTCTACCCCT CCGATATCGCCGTGGAATGGGAGAGCAACGGCCAGCCCGAGAACAA CTACAAGACCACCCCCCTGTGCTGGACAGCGACGGCTCATTCTTCCT GTACAGCAGACTGACCGTGGACAAGAGCCGGTGGCAGGAAGGCAAC GTGTTCAAGTGCAGCGTGATGCACGAGGCCCTGCACAACCACTACAC CCAGAAGTCCCTGTCTCTGAGCCTGGGCAAG
240	DNA	CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTAGCGGCAGCAGCTCCAACATCGGCAGC AACACCGTGAAGTGGTATCAGCAGCTGCCCTGGCACAGCCCCATAACT GCTGATCTACTACGACGACCTGCGGCCTAGCGGCGTGCCAGATAGAT TTTCTGGCAGCAAGAGCGGCACCTCTGCCAGCCTGGCTATTCTTGGA CTGCAGAGCGAGGACGAGGCCGACTACTATTGTACGCGCTGGGACGA CAGCCTGAACGACTACCTGTTTTTGGCGGAGGCACCAAGCTGACCG TTCTAGCCAGCCTAAAGCGGCCCTAGCGTGACCCCTGTTCCCTCCAA GCAGCGAGGAAGTGCAGGCCAACAGGCCACCTCGTGTGCTGATC AGCGACTTCTATCTGGCGCGGTGACCGTGGCCTGGAAGGCCGATAG CTCTCTGTGAAGGCCGCGTGGAAACCAACCCCTAGCAAGCAGA GCAACAACAATAACGCGCCAGCAGCTACCTGAGCCTGACCCCCGAG CAGTGGAAAGTCCACAGATCTACAGCTGCCAAGTGACCCACGAGGG CAGCACCGTGGAAAAGACAGTGGCCCTACCGAGTGCAGC
241	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYAMSWVRQAPGKLEWV SAIGYGGDTYYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCA RRDDYTSRDAFDYWGQGLTVTVSS
242	PRT	SYAMS
243	PRT	AIGYGGDTYYADSVKG
244	PRT	RDDYTSRDAFDY
245	PRT	QSVLTQPPSASGTPGQRTVISCSSSSNIGSNVTNWNYYQLPGTAPKLLIYY DDLRLPSGVPDRFSGSKSGTSASLAISGLQSEDEADYYCHAWDDSLNVYP VFGGGTKLTVL
246	PRT	SGSSSNIGSNVTN
247	PRT	YDDLRLPS
248	PRT	HAWDDSLNVYPV
249	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCACCTTTAGCAGCT ACGCCATGAGCTGGGTCCGACAGGCTCCTGGCAAAGGCCCTGAATGG GTGTCCGCCATCGGCTATGGCGGCGATACCTACTACGCCGACTCTGT GAAGGGCAGATTACCATCAGCCGGGACAACAGCAAGAACACCCCTG

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
		TACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCGTGTA TTGCGCCAGAAGGGACGACTACACCAGCAGGACGCCTTCGATTAT GGGCCAGGCACACTGGTCACCGTTTCTTCA
250	DNA	AGCTACGCCATGAGC
251	DNA	GCCATCGGCTATGGCGGCGATACCTACTACGCCGACTCTGTGAAGG C
252	DNA	AGGGACGACTACACCAGCAGGGACGCCTTCGATTAT
253	DNA	CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTAGCGGCAGCAGCTCCAACATCGGCAGC AACACCGTGAACCTGGTATCAGCAGCTGCCTGGCACAGCCCTAACT GCTGATCTACTACGACGACCTGCGGCCTAGCGCGTGGCAGATAGAT TTTCTGGCAGCAAGAGCGGCACCTCTGCCAGCCTGGCTATTTCTGGA CTGCAGAGCGAGGACGAGGCCGACTACTATTGTACGCGCTGGGACGA CAGCCTGAACGTGTACCCTGTTTTTGGCGGAGGCACCAAGCTGACCG TTCTA
254	DNA	AGCGGCAGCAGCTCCAACATCGGCAGCAACACCGTGAAC
255	DNA	TACGACGACCTGCGGCCTAGC
256	DNA	CACGCCTGGGACGACAGCCTGAACGTGTACCCTGTT
257	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYAMSWVRQAPGKLEWV SAIGYGGDTYYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCA RRDDYTSRDAFDYWGQGTLVTVSSASTKGPSVFPLAPCSRSTSESTAAL GCLVKDYFPEPTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSS LGTKTYTNCVNDHKPSNTKVDKRVESKYGPPCPPCPAPEFLGGPSVFLFPP KPKDTLMISRTPEVTVVVDVVSQEDPEVQFNWYVDGVEVHNAKTKPRE EQFNTYTRRVSVLTVLHQDWLNGKEYKCKVSNKGLPSSIEKTIKAKGQ PREPQVYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNY KTTTPVLDSDGSFPLYSRLLTVDKSRWQEGNVFCSVMHEALHNHYTQK SLSLSLGK
258	PRT	QSVLTQPPSASGTPGQRVITISCSGSSSNIGSNTNVNYYQLPGTAPKLLIYY DDLRLSPGVDRFSGSKSGTSASLAISGLQSEDEADYYCHAWDDSLNVYP VFGGKTLTLVGLQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVT VAWKADSSPVKAGVETTPSKQSNKYAASSYLSLTPEQWKSHRYSYC QVTHEGSTVEKTVAPTECS
259	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCACCTTTAGCAGCT ACGCCATGAGCTGGGTCCGACAGGCTCCTGGCAAAGGCCTGAATGG GTGTCCGCATCGGCTATGGCGGCGATACCTACTACGCCGACTCTGT GAAGGGCAGATTACCATCAGCCGGGACAAACAGCAAGAACACCTG TACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCGTGTA TTGCGCCAGAAGGGACGACTACACCAGCAGGGACGCCTTCGATTAT GGGGCCAGGGCACACTGGTCACCGTTTCTTACGCCAGCACCAGGGC CCAGCGTGTTCCTCTGGCCCTTGTAGCAGAAGCACCAGCGAGTC TACAGCCGCCCTGGGCTGCCTCGTGAAGGACTACTTTCCGAGCCCG TGACCGTGTCTGGAACCTGGCGCTCTGACAAGCGGCGTGCACACC TTTCCAGCCGTGCTGCAGAGCAGCGGCTGTACTCTCTGAGCAGCGT CGTGACAGTGCCAGCAGCAGCCTGGGCACCAAGACCTACACCTGTA ACGTGGACCACAAGCCAGCAACACCAAGGTGGACAAGCGGGTGA ATCTAAGTACGGCCCTCCCTGCCCTCCTTGCCAGCCCTGAATTTCT GGGCGGACCTCCGTGTTCTGTTCCTCCCAAGCCCAAGGACACCC TGATGATCAGCCGACCCCAAGTGACCTGCGTGGTGGTGGATGTG TCCCAGGAAGATCCCGAGGTGCAGTTCAATTGGTACGTGGACGGCGT GGAAGTGCAACGCCAAGACCAAGCCAGAGAGGAACAGTTCAAC AGCACCTACCGGTGGTGTCCGTGCTGACAGTGTCTGCACGAGACTG CTGAACGGCAAAGAGTACAAGTGCAAGGTGTCCAACAAGGGCTG CCAGCTCCATCGAGAAAACCATCAGCAAGGCCAAGGGCCAGCCCC

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
		GCGAACCCAGGTGTACACACTGCCTCCAAGCCAGGAAGAGATGAC CAGAACCAGGTGTCCCTGACCTGTCTCGTGAAAGGCTTCTACCCCT CCGATATCGCCGTGGAATGGGAGAGCAACGGCCAGCCCGAGAACA CTACAAGACCACCCCCCTGTGCTGGACAGCGACGGCTCATTCTTCCT GTACAGCAGACTGACCGTGGACAAGAGCCGTTGGCAGGAAGGCAAC GTGTTACAGCTGCAGCGTGATGCACGAGGCCCTGCACAACCACTACAC CCAGAAGTCCCTGTCTCTGAGCCTGGGCAAG
260	DNA	CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTAGCGGCAGCAGCTCCAACATCGGCAGC AACACCGTGAACCTGGTATCAGCAGCTGCCTGGCACAGCCCTAAACT GCTGATCTACTACGACGACCTGCGGCCTAGCGCGCTGCCAGATAGAT TTTCTGGCAGCAAGAGCGGCACCTCTGCCAGCCTGGCTATTTCTGGA CTGCAGAGCGAGGACGAGGCCGACTACTATTGTACGCCCTGGGACGA CAGCCTGAACGTGTACCCTGTTTTTGGCGAGGCCACCAAGCTGACCG TTCTAGGCCAGCCTAAAGCCGCCCTAGCGTGACCCCTGTCCCTCCAA GCAGCGAGGAACCTGCAGGCCAACCAAGGCCACCTCGTGTGCCTGATC AGCGACTTCTATCCTGGCGCCGTGACCGTGGCCTGGAAGGCCGATAG CTCTCCTGTGAAGCCGGCGTGGAAACCACCCCTAGCAAGCAGA GCAACAACAATAACGCCGCCAGCAGCTACCTGAGCCTGACCCCGGAG CAGTGGAGTCCCACAGATCCTACAGCTGCCAAGTGACCCACGAGGG CAGCACCGTGGAAAAGACAGTGGCCCTACCGAGTGCAGC
261	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYAMSWVRQAPGKLEWV SAIGYGGDTYYADSVKGRFTISRDNKNTLYLQMNSLRADTVVYYCA RRDDYTSRDAFDYWGQTLVTVSS
262	PRT	SYAMS
263	PRT	AIGYGGDTYYADSVKG
264	PRT	RDDYTSRDAFDY
265	PRT	QSVLTQPPSASGTPGQRTVITSCGSSSNIGSNTVNWYQQLPGTAPKLLIYY DDLRLPSGVPDRFSGSKGTSASLAISGLQSEDEADYYCHAWDDSLNVIPIV FGGGTKLTVL
266	PRT	SGSSSNIGSNTVN
267	PRT	YDDLRLPS
268	PRT	HAWDDSLNVIPIV
269	DNA	GAAGTTACAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCACCTTTAGCAGCT ACGCCATGAGCTGGGTCCGACAGGCTCCTGGCAAAGGCCCTGAATGG GTGTCCGCCATCGGCTATGGCGGCGATACCTACTACGCCGACTCTGT GAAGGGCAGATTACCATCAGCCGGGACAAACAGCAAGAACACCCCTG TACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCGTGACTA TTGCGCCAGAAGGGACGACTACACCAGCAGGGACGCCCTTCGATTATT GGGGCCAGGGCACACTGGTCACCGTTTCTTCA
270	DNA	AGCTACGCCATGAGC
271	DNA	GCCATCGGCTATGGCGGCGATACCTACTACGCCGACTCTGTGAAGGG C
272	DNA	AGGGACGACTACACCAGCAGGGACGCCTTCGATTAT
273	DNA	CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTAGCGGCAGCAGCTCCAACATCGGCAGC AACACCGTGAACCTGGTATCAGCAGCTGCCTGGCACAGCCCTAAACT GCTGATCTACTACGACGACCTGCGGCCTAGCGCGCTGCCAGATAGAT TTTCTGGCAGCAAGAGCGGCACCTCTGCCAGCCTGGCTATTTCTGGA CTGCAGAGCGAGGACGAGGCCGACTACTATTGTACGCCCTGGGACGA CAGCCTGAACGTGATCCCTGTTTTTGGCGGAGGCCACCAAGCTGACCG TTCTA
274	DNA	AGCGGCAGCAGCTCCAACATCGGCAGCAACACCGTGAAC
275	DNA	TACGACGACCTGCGGCCTAGC

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
276	DNA	CACGCCTGGGACGACAGCCTGAACGTGATCCCTGTT
277	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYAMSWVRQAPGKLEWV SAIGYGGDTYYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCA RRDDYTSRDAFDYWGQGLTVTVSSATKGPSVFPLAPCSRSTSESTAAL GCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSS LGKTKYTCNVDPKPSNTKVDKRVESKYGPCCPAPAEFLGGPSVFLFPP KPKDTLMISRTPEVTCVVVDVSDPEVQFNWYVDGVEVHNAKTKPRE EQFNTYRVVSVLTVLHQDWLNGKEYKCKVSNKGLPSSIEKTIKAKGQ PREPQVYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNY KTTTPPVLDSDGSFFLYSRLTVDKSRWQEGNVPFSCVMHEALHNHYTQK SLSLSLGK
278	PRT	QSVLTQPPSASGTPGQRVTISCSGSSSNIGSNTVNWYQQLPGTAPKLLIYY DDLRRPSGVPRDRFSGSKSGTSASLAISGLQSEDEADYYCHAWDDSLNVI FVGKTKLTVLQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTV AWKADSSPVKAGVETTPPSKQSNKYAASSYLSLTPEQWKSHRYSYSCQ VTHEGSTVEKTVAPTECS
279	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCGCAGCGCTTACCTTTAGCAGCT ACGCCATGAGCTGGGTCCGACAGGCTCCTGGCAAAGGCTTGAATGG GTGTCCGCCATCGGCTATGGCGGCGATACCTACTACGCCGACTCTGT GAAGGGCAGATTACCATCAGCGGGACAAACAGCAAGAACACCTTG TACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCGGTGTA TTGCGCCAGAAGGGACGACTACACAGCAGGGACGCGCTTCGATTATT GGGGCCAGGGCACACTGGTCACCGTTTCTTCAGCCAGCACCAAGGGC CCAGCGGTGTTCCCTCTGGCCCCCTTAGCAGAAAGCAGCAGCAGT TACAGCGCGCCTGGGCTGCTCGTGAAGGACTACTTTCCGAGCCCG TGACCGTGTCTTGAACCTCGGCGCTCTGACAAGCGCGGTGCACACC TTTCCAGCGGTGCTGCAGAGCAGCGGCTGACTCTCTGAGCAGCGT CGTGACAGTGCCAGCAGCAGCCTGGGCACCAAGACCTACACCTGTA ACGTGGACCAAGCCAGCAACACCAAGGTGGACAAGCGGGTGGGA ATCTAAGTACGGCCCTCCCTGCCTCCTTGCCAGCCCTGAATTTCT GGGCGGACCCTCCGTGTTCTGTTCCCCCAAGCCCAAGGACACCC TGATGATCAGCGGACCCCGAAGTGACCTGCGTGGTGGTGGATGTG TCCAGGAAGATCCCGAGGTGCAGTTCAATTGGTACGTGGACGGCGT GGAAGTGCAACGCCAAGACCAAGCCAGAGAGGAACAGTTCAAC AGCACCTACCGGGTGGTGTCCGTGCTGACAGTGTGCACAGGACTG GCTGAACGGCAAAGAGTACAAGTGCAAGGTGTCCAACAGGGCCTG CCCAGCTCCATCGAGAAAACATCAGCAAGGCCAAGGCCAGCCCC CGGAACCCAGGTGTACACACTGCCTCAAGCCAGGAAGAGATGAC CAAGAACCAGGTGTCCCTGACCTGTCTCGTGAAGGCTTCTACCCCT CCGATATCGCCGTGGAATGGGAGAGCAACGGCCAGCCCGAGAACAA CTACAAGACCACCCCCCTGTGCTGGACAGCGACGGCTCATTCTTCCT GTACAGCAGACTGACCGTGGACAGAGCCGCTGGCAGGAAGGCAAC GTGTTCAGCTGCAGCGTGATGCACGAGGCCCTGCACAACCACTACAC CCAGAAGTCCCTGTCTCTGAGCCTGGGCAAG
280	DNA	CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTAGCGGCAGCAGCTCCAACATCGGCAGC AACACCGTGAACCTGGTATCAGCAGCTGCCTGGCACAGCCCCCTAACT GCTGATCTACTACGACGACCTGCGGCCTAGCGCGCTGCCAGATAGAT TTTCTGGCAGCAAGAGCGGCACCTCTGCCAGCCTGGCTATTTCTGGA CTGCAGAGCGAGGACGAGGCGGACTACTATTGTACGCGCTGGGACGA CAGCCTGAACGTGATCCCTGTTTTGGCGGAGGCACCAAGCTGACCG TTCTAGGCCAGCCTAAAGCCGCCCTAGCGTGACCTGTTCCTCCAA GCAGCGAGGAACCTGACGGCCAAACAGGCCACCTCGTGTGCTGATC AGCGACTTCTATCCTGGCGCCGTGACCGTGGCCTGGAAGGCCGATAG CTCTCTGTGAAGGCCGCGGTGGAACCAACACCCCTAGCAAGCAGA GCAACAACAATAACGCCGCCAGCAGCTACCTGAGCCTGACCCCGAG CAGTGGAGTCCACAGATCCTACAGCTGCCAAGTGACCCACGAGGG CAGCACCTGGAAAAGACAGTGGCCCCCTACCGAGTGCAGC
281	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFYAMSLWVRQAPGKLEWV SAIGTGGDTYYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCA RRDDYTSRDAFDYWGQGLTVTVSS
282	PRT	SYAML

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
283	PRT	AIGTGDTYYADSVKG
284	PRT	RDDYTSRDAFDY
285	PRT	QSVLTQPPSASGTPGQRTVITSCSGSSSNIGSNTVNWYQQLPGTAPKLLIYY DDLRP SGVPDRFSGSKSGTSASLAISGLQSEDEADYYCAAWDDSLNDYV VFGGGTKLTVL
286	PRT	SGSSSNIGSNTVN
287	PRT	YDDL RPS
288	PRT	AAWDDSLNDYVV
289	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCACCTTTTACAGCTA CGCCATGCTGTGGGTCCGACAGGCCCTGGAAAAGGCCTTGAATGGG TGTCGCCATCGGCACAGGCGGCGATACCTACTATGCCGACTCTGTG AAGGGCAGATTACCATCAGCCGGGACAACAGCAAGAACACCTGT ACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCGTGTACTAT TGCGCCAGAAGGGACGACTACACAGCAGGACGCCTTCGATTATTG GGCCAGGGCACACTGGTCACCGTTTCTTCA
290	DNA	AGCTACGCCATGCTG
291	DNA	GCCATCGGCACAGGCGGCGATACCTACTATGCCGACTCTGTGAAGGG C
292	DNA	AGGGACGACTACACCAGCAGGGACGCCTTCGATTAT
293	DNA	CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTAGCGGCAGCAGCTCCAACATCGGCAGC AACACCGTGAACCTGGTATCAGCAGCTGCCTGGCACAGCCCCCTAACT GCTGATCTACTACGACGACCTGCGGCCTAGCGCGTGCAGATAGAT TTTCTGGCAGCAAGAGCGGCACCTCTGCCAGCCTGGCTATTTCTGGA CTGCAGAGCGAGGACGAGGCGGACTATTATTGTGCCGCGCTGGGACGA CAGCCTGAACGACTACGTTGTGTTTGGCGGAGGACCAAGCTGACCG TTCTA
294	DNA	AGCGGCAGCAGCTCCAACATCGGCAGCAACACCGTGAAC
295	DNA	TACGACGACCTGCGGCCTAGC
296	DNA	GCCGCTGGGACGACAGCCTGAACGACTACGTTGTG
297	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFYSYAMLWVRQAPGKLEWV SAIGTGDTYYADSVKGRFTISRDNKNTLYLQMNLSRAEDTAVYYCA RRDDYTSRDAFDYWGQTLVTVSSASTKGPSVFPLAPCSRSTSESTAAL GCLVKDYFPEPTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSS LGKTKYTCNVDHKPSNTKVDKRVESKYGPPCPPCPAPEFLGGPSVFLFPP KPKDTLMISRTPEVTVVVDVVSQEDPEVQFNWYVDGVEVHNAKTKPRE EQFNSYTRVVSFLTVLHQDWLNGKEYKCKVSNKGLPSSIEKTIKAKGQ PREPQVYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNY KTTTPVLDSDGSFFLYSRLTVDKSRWQEGNVFSCSVMHREALHNYHTQK SLSLSLGK
298	PRT	QSVLTQPPSASGTPGQRTVITSCSGSSSNIGSNTVNWYQQLPGTAPKLLIYY DDLRP SGVPDRFSGSKSGTSASLAISGLQSEDEADYYCAAWDDSLNDYV VFGGGTKLTVLGQPKAAPSVKTLFPPSSEELQANKATLVCLISDFYPGAVT VANKADSSPVKAGVETTPSKQSNKYAASSYLSTPEQWKSHRSYSC QVTHEGSTVEKTVAPTECS
299	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCACCTTTTACAGCTA CGCCATGCTGTGGGTCCGACAGGCCCTGGAAAAGGCCTTGAATGGG TGTCGCCATCGGCACAGGCGGCGATACCTACTATGCCGACTCTGTG AAGGGCAGATTACCATCAGCCGGGACAACAGCAAGAACACCTGT ACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCGTGTACTAT TGCGCCAGAAGGGACGACTACACAGCAGGACGCCTTCGATTATTG GGCCAGGGCACACTGGTCACCGTTTCTTACGCAGCACCAAGGGCC CCAGCGTGTTCCTCTGGCCCCCTGTAGCAGAAGCACCAGCGAGTCT

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
		ACAGCCGCCCTGGGCTGCCTCGTGAAGGACTACTTTCCCGAGCCCGT GACCGTGCTCTGGAACCTCTGGCGCTCTGACAAGCGCGGTGCACACCT TTCAGGCCGTGCTGCAGAGCAGCGGCTGTACTCTCTGAGCAGCGTC GTGACAGTGCCAGCAGCAGCCTGGGCACCAAGACCTACACCTGTAA CGTGGACCACAAGCCAGCAACACCAAGGTGGACAAGCGGGTGGAA TCTAAGTACGGCCCTCCCTGCCCTCCTTGCCAGCCCTGAATTTCTG GGCGGACCCCTCCGTGTTCTGTTCCTCCCTCCAAAGCCCAAGGACACCT GATGATCAGCCGAGCCCTCGAAGTGACCTGCGTGGTGGTGGATGTGT CCCAGGAAGATCCCGAGGTGCAGTTCAATTGGTACGTGGACGGCGTG AAGTGCACAAACGCCAAGACCAAGCCAGAGAGGAACAGTTCAACA GCACCTACCGGGTGGTGTCCGTGCTGACAGTGCTGACACAGGACTGG CTGAACGGCAAAGAGTACAAGTGCAAGGTGTCCAACAAGGGCTGCG CCAGCTCCATCGAGAAAACCATCAGCAAGGCCAAGGGCCAGCCCG CGAACCCAGGTGTACACACTGCCTCCAAGCCAGGAAGAGATGACC AAGAACCAGGTGTCCCTGACCTGTCTCGTAAAGGCTTCTACCCCTC CGATATCGCCGTGGAATGGGAGAGCAACGGCCAGCCCGAGAAACAAC TACAAGACCACCCCTGTGCTGGACAGCGACGGCTCATTTCTTCT GTACAGCAGACTGACCGTGGACAAGAGCCGTTGGCAGGAAGGCAAC GTGTTGAGCTGCAGCGTGATGCACGAGGCCCTGCACAACCACTACAC CCAGAAGTCCCTGTCTCTGAGCCTGGGCAAG
300	DNA	CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTAGCGGCAGCAGCTCCAACATCGGCAGC AACACCGTGAACCTGGTATCAGCAGCTGCCTGGCACAGCCCTAAACT CTGATCTACTACGACGACCTGCGGCCTAGCGCGTGCAGATAGAT TTTCTGGCAGCAAGAGCGGCACCTCTGCCAGCCTGGCTATTTCTGGA CTGCAGAGCGAGGACGAGGCCGACTATTATTGTGCGCGCTGGGACGA CAGCCTGAACGACTACGTTGTGTTTGGCGGAGGCACCAAGCTGACCG TTCTAGGCCAGCCTAAAGCCGCCCTAGCGTGACCTGTTCCCTCCAA GCAGCGAGGAACCTGCAGGCCAACAGGCCACCTCTGTGTGCTGATC AGCGACTTCTATCCTGGCGCCGTGACCGTGGCTTGAAGGCCGATAG CTCTCCTGTGAAGGCCGCGCTGGAACCAACACCCCTAGCAAGCAGA GCAACAACAAATACGCCGCCAGCAGCTACCTGAGCCTGACCCCGGAG CAGTGAAGTCCACAGATCCTACAGCTGCCAAGTGACCCACGAGGG CAGCACCGTGGAAAAGACAGTGGCCCTACCGAGTGCAGC
301	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFYSYAMLWVRQAPGKLEWV SAIGTGGDTYYADSVKGRFTISRDNKNTLYLQMNSLRADTAVYYCA RRDDYTSRDAFDYWGQTLTVTVSS
302	PRT	SYAML
303	PRT	AIGTGGDTYYADSVKG
304	PRT	RDDYTSRDAFDY
305	PRT	QSVLTQPPSASGTPGQRVITISCGSSSNIGSNTVNWYQQLPGTAPKLLIYY DDLRLPSGVPDRFSGSKGTSASLAISGLQSEADYYCAAWDDSLNVYV VFGGGTKLTVL
306	PRT	SGSSSNIGSNTVN
307	PRT	YDDLRLPS
308	PRT	AAWDDSLNVYVV
309	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTTGAGACTGAGCTGTGCCGCCAGCGGCTTACCTTTACAGCTA CGCCATGCTGTGGGTCCGACAGGCCCTGGAAAAGGCCTGAATGGG TGTCGCCATCGGCACAGGCGGCGATACCTACTATGCCGACTCTGTG AAGGGCAGATTACCATCAGCCGGGACAACAGCAAGAACACCTGT ACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCGTGTACTAT TGCGCCAGAAGGAGGACTACACAGCAGGACGCGCTTCGATTATTG GGCCAGGGCACACTGGTCACCGTTTCTTCA
310	DNA	AGCTACGCCATGCTG
311	DNA	GCCATCGGCACAGGCGGCGATACCTACTATGCCGACTCTGTGAAGGG C
312	DNA	AGGGACGACTACACCAGCAGGGACGCGCTTCGATTAT

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
313	DNA	CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTAGCGGCAGCAGCTCCAACATCGGCAGC AACACCGTGAACCTGGTATCAGCAGCTGCCTGGCACAGCCCCCTAAACT GCTGATCTACTACGACGACCTGCGGCCTAGCGCGTGCACAGATAGAT TTTCTGGCAGCAAGAGCGGCACCTCTGCCAGCCTGGCTATTCTGGA CTGCAGAGCGAGGACGAGGCGCACTATTATTGTGCCGCTGGGACGA CAGCCTGAACGTGTACGTTGTGTTGGCGGAGGCACCAAGCTGACCG TTCTA
314	DNA	AGCGGCAGCAGCTCCAACATCGGCAGCAACACCGTGAAC
315	DNA	TACGACGACCTGCGGCCTAGC
316	DNA	GCCGCCTGGGACGACAGCCTGAACGTGTACGTTGTG
317	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFYSYAMLWVRQAPGKGLEWV SAIGTGGDTYYADSVKGRFTISRDNSENTLYLQMNLSRAEDTAVYYCA RRDYSRDAFDYWGQGLTVTVSSASTKGPSVFPLAPCSRSTSESTAAL GCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSS LGTKYITCNVDHKPSNTKVDKRVESKYGPCCPAPPEFLGGPSVFLFPP KPKDTLMISRTPEVTCVVVDVSDPEVQFNWYVDGVEVHNAKTKPRE EQFNTSTYRVSVLTVLHQDWLNGKEYKCKVSNKGLPSSIEKTIKAKGQ PREPQVYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNY KTTTPVLDSDGSFFLYSRLLTVDKSRWQEGNVFSCSVMHEALHNHYTQK SLSLSLGK
318	PRT	QSVLTQPPSASGTPGQRTTISCSGSSSNIGSNTVNWYQLPGTAPKLLIYY DDLRPSPVDRFSGSGSKSGTSASLAISGLQSEDEADYYCAAWDDSLNVYV VFGGGTKLTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVT VAWKADSPVKAGVETTPSKQSNKYAAASSYLSLTPEQWKSHRSYSC QVTHEGSTVEKTVAPTECS
319	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCACTTTTACAGCTA CGCCATGCTGTGGGTCCGACAGGCCCTGGAAAAGGCCTTGAATGGG TGTCCGCCATCGGCACAGGCGGCGATACCTACTATGCCGACTCTGTG AAGGGCAGATTACCATCAGCCGGGACAACAGCAAGAACACCTGT ACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCGTGTACTAT TGCGCCAGAAGGGACGACTACACCAGCAGGACGCGCTTCGATTATTG GGGCCAGGGCACACTGGTCACCGTTTCTTCAGCCAGCACCAAGGGCC CCAGCGTGTTCCTCTGGCCCCCTTGTAGCAGAAGCACAGCGAGTCT ACAGCCGCCCTGGGCTGCCTCGTGAAGGACTACTTCCCGAGCCCGT GACCGTGTCTGGAATCTGGCGCTCTGACAAGCGCGTGCACACCT TTCACGCCGTGTGCAGAGCAGCGGCTGTACTCTCTGAGCAGCGTC GTGACAGTGCCAGCAGCAGCCTGGGCACCAAGACCTACACCTGTAA CGTGGACCACAAGCCAGCAACACCAAGGTGGACAAGCGGGTGGAA TCTAAGTACGGCCCTCCCTGCCCTCCTTGCCAGCCCTGAATTTCTG GGCGGACCCCTCCGTGTTCTGTTCCCCCAAAGCCCAAGGACACCT GATGATCAGCCGACCCCCGAAGTGACCTGCGTGGTGGTGGATGTGT CCCAGGAAGATCCCGAGGTGCAGTTCAATTGGTACGTGGACGGCGTG GAAGTGCACAACGCCAAGACCAAGCCAGAGAGGAACAGTTCAACA GCACCTACCGGGTGGTGTCCGTGCTGACAGTGTGACACAGGACTGG CTGAACGGCAAAGAGTACAAGTGCAAGGTGTCCAACAAGGGCTTGC CCAGCTCCATCGAGAAAACCATCAGCAAGGCCAAGGGCCAGCCCCG CGAACCCAGGTGTACACACTGCCTCCAAGCCAGGAAGAGATGACC AAGAACCAGGTGTCCCTGACCTGTCTCGTGAAAGGCTTCTACCCCTC CGATATCGCCGTGGAATGGGAGAGCAACGGCCAGCCCGAGAACAA TACAAGACCACCCCCCTGTGCTGGACAGCGACGGCTCATTCTCTCT GTACAGCAGACTGACCGTGACAAAGAGCCGGTGGCAGGAAGGCAAC GTGTTGAGCTGCAGCGTGATGCACGAGGCCCTGCACAACCACTACAC CCAGAAGTCCCTGTCTCTGAGCCTGGGCAAG
320	DNA	CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTAGCGGCAGCAGCTCCAACATCGGCAGC AACACCGTGAACCTGGTATCAGCAGCTGCCTGGCACAGCCCCCTAAACT GCTGATCTACTACGACGACCTGCGGCCTAGCGCGTGCACAGATAGAT TTTCTGGCAGCAAGAGCGGCACCTCTGCCAGCCTGGCTATTCTGGA CTGCAGAGCGAGGACGAGGCGCACTATTATTGTGCCGCTGGGACGA CAGCCTGAACGTGTACGTTGTGTTGGCGGAGGCACCAAGCTGACCG TTCTAGGCCAGCCTAAAGCCGCCCTAGCGTGACCTGTTCCTCCCTCAA

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
		GCAGCGAGGAAGTGCAGGCCAACAAAGGCCACCCCTCGTGTGCCTGATC AGCGACTTCTATCCTGGCGCCGTGACCGTGGCCTGGAAGGCCGATAG CTCTCTGTGAAGGCCGCGTGGAAACCACCAACCCCTAGCAAGCAGA GCAACAACAATAACGCCGCCAGCAGCTACCTGAGCCTGACCCCGAG CAGTGGAAAGTCCACAGATCTACAGCTGCCAAGTGACCCACGAGGG CAGCACCGTGGAAAAGACAGTGGCCCTACCGAGTGCAGC
321	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFYSYAMSWVRQAPGKLEWV SAIGYGGDTYYADSVKGRFTISRDNSENTLYLQMNLSRAEDTAVYYCA RRDDYTSRDAFDYWGQGLTVTVSS
322	PRT	SYAMS
323	PRT	AIGYGGDTYYADSVKG
324	PRT	RDDYTSRDAFDY
325	PRT	QSVLTQPPSASGTPGQRVITISGSSSSNIGSNTVNWYQQLPGTAPKLLIYY DDLRLPSGVPDRFSGSKSGTSASLAISGLQSEDEADYYCHAWDDSLNVYP VFGGGTKLTVL
326	PRT	SGSSSNIGSNTVN
327	PRT	YDDLRLPS
328	PRT	HAWDDSLNVYPV
329	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGCTTCACCTTTTACAGCTA CGCCATGAGCTGGGTCCGACAGGCCCTTGAAAAGGCCTTGAATGGG TGTCGCCCATCGGCTATGGCGGCGATACCTACTACGCCGACTCTGTG AAGGGCAGATTACCATCAGCCGGGACAACAGCAAGAACACCCCTGT ACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCCGCCGTGTACTAT TGCGCCAGAAGGGACGACTACACCAGCAGGGACGCCTTCGATTATTG GGGCCAGGGCACACTGGTCACCGTTTCTTCA
330	DNA	AGCTACGCCATGAGC
331	DNA	GCCATCGGCTATGGCGGCGATACCTACTACGCCGACTCTGTGAAGGG C
332	DNA	AGGGACGACTACACCAGCAGGGACGCCTTCGATTAT
333	DNA	CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTAGCGGCAGCAGCTCCAACATCGGCAGC AACACCGTGAAGTGGTATCAGCAGCTGCCGACAGCCCTAAACT GCTGATCTACTACGACGACCTGCGGCCTAGCGGCGTGCCAGATAGAT TTTCTGGCAGCAAGAGCGGCACCTCTGCCAGCCTGGCTATTCTGGA CTGCAGAGCGAGGACGAGGCGGACTACTATTGTACGCCTGGGACGA CAGCCTGAACGTGTACCTGTTTTTGGCGGAGGCACCAAGCTGACCG TTCTA
334	DNA	AGCGGCAGCAGCTCCAACATCGGCAGCAACACCGTGAAC
335	DNA	TACGACGACCTGCGGCTAGC
336	DNA	CACGCCTGGGACGACAGCCTGAACGTGTACCCGTGT
337	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFYSYAMSWVRQAPGKLEWV SAIGYGGDTYYADSVKGRFTISRDNSENTLYLQMNLSRAEDTAVYYCA RRDDYTSRDAFDYWGQGLTVTVSSASTKGPSVFPLAPCSRSTSESTAAL GCLVKDYFPEPTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSS LGKTKYTCNVDHKPSNTKVDKRVESKYGPPCPPCPAPEFLGGPSVFLFPP KPKDTLMISRTPEVTCVVVDVSDPEVQFNWYVDGVEVHNAKTKPRE EQFNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKGLPSSIEKTIKAKGQ PREPQVYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNY KTTTPVLDSDGSFFLYSRLLTVDKSRWQEGNVFSCSVMHEALHNHYTQK SLSLSLGK
338	PRT	QSVLTQPPSASGTPGQRVITISGSSSSNIGSNTVNWYQQLPGTAPKLLIYY DDLRLPSGVPDRFSGSKSGTSASLAISGLQSEDEADYYCHAWDDSLNVYP

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
		VFGGGKLTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVT VANKADSSPVKAGVETTPSKQSNKNYAASSYLSLTPEQWKSHRSYSC QVTHEGSTVEKTVAPTECS
339	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTACACCTTTTACAGCTA CGCCATGAGCTGGGTCCGACAGGCCCTTGAAAAGGCCTTGAATGGG TGTCGCCATCGGCTATGGCGGCGATACCTACTACGCCGACTCTGTG AAGGGCAGATTACCATCAGCCGGGACAAACAGCAAGAACACCTGT ACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCGTGTACTAT TGCGCCAGAAGGGACGACTACACAGCAGGGACGCCTTCGATTATTG GGGCCAGGGCACACTGGTCACCGTTTCTTCAGCCAGCACCAAGGGCC CCAGCGTGTTCCTCTGGCCCTTGTAGCAGAAGCACCAGCGAGTCT ACAGCCGCCCTGGGCTGCCTCGTGAAGGACTACTTTCCCGAGCCCGT GACCGTGTCTTGGAACTCTGGCGCTCTGACAAGCGCGTGCACACCT TTCCAGCCGTGCTGCAGAGCAGCGCCTGTACTCTCTGAGCAGCGTC GTGACAGTGCCAGCAGCAGCCTGGGCACCAAGACCTACACCTGTAA CGTGGACCAAGCCAGCAACACCAAGGTGGACAAGCGGTGGAA TCTAAGTACGGCCCTCCCTGCCCTCCTTGCCAGCCCTGAATTTCTG GGCGGACCCCTCCGTGTCTCTGTTCCCTCCAAAGCCCAAGGACACCT GATGATCAGCCGACCCCGAAGTGACCTGCGTGGTGGTGGATGTGT CCAGGAAGATCCCGAGGTGCAGTTCAATTGGTACGTGGACGGCGTG GAAGTGACAAACGCCAAGACCAAGCCAGAGAGGAACAGTTCAACA GCACCTACCGGGTGGTGTCCGTGCTGACAGTGTGCACACGAGACTGG CTGAACGGCAAGAGTACAAGTGCAAGGTGTCCAACAAGGGCTGC CCAGCTCCATCGAGAAAACCATCAGCAAGGCCAAGGGCCAGCCCCG CGAACCCAGGTGTACACACTGCCTCCAAGCCAGGAAGAGATGACC AAGAACCAGGTGTCCCTGACCTGTCTCGTGAAAGGCTTCTACCCCTC CGATATCGCCGTGGAATGGGAGAGCAACGGCCAGCCCGAGAACAC TACAAGACCACCCCTGTGCTGGACAGCGACGGCTCATTCTTCCT GTACAGCAGACTGACCGTGGACAAGAGCCGGTGGCAGGAAGGCAAC GTGTTAGCTGCAGCGTGATGCACGAGGCCCTGCACAACCACTACAC CCAGAAGTCCCTGTCTCTGAGCCTGGGCAAG
340	DNA	CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTAGCGGCAGCAGCTCCAACATCGGCAGC AACACCGTGAAGTGGTATCAGCAGCTGCCTGGCACAGCCCTAAACT GCTGATCTACTACGACGACCTGCGGCCTAGCGGCGTGCCAGATAGAT TTTCTGGCAGCAAGAGCGGCACCTCTGCCAGCCTGGCTATTCTTGA CTGCAGAGCGAGGACGAGGCCGACTACTATTGTACGCTGGGACGA CAGCCTGAACGTGTACCTGTTTTTGGCGAGGCACCAAGCTGACCG TTCTAGGCCAGCCTAAAGCGCCCTTAGCGTGACCTGTTCCTCTCAA GCAGCGAGGAAGTGCAGGCCAACAGGCCACCTCGTGTGCTGATC AGCGACTTCTATCTGGCGCCGTGACCGTGGCCTGGAAGGCCGATAG CTCTCTGTGAAGGCCGCGTGGAAACCAACCCCTAGCAAGCAGA GCAACAAACAAATACGCCGCCAGCAGCTACCTGAGCCTGACCCCGAG CAGTGGAAAGTCCACAGATCTACAGCTGCCAAGTGACCCACGAGGG CAGCACCGTGGAAAGACAGTGGCCCTACCGAGTGCAGC
341	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFYSYAMSWVRQAPGKLEWV SAIGYGGDTYYADSVKGRFTISRDNKNTLYLQMNLSRAEDTAVYYCA RRDYYTSRDAFDYWGQGLTVTVSS
342	PRT	SYAMS
343	PRT	AIGYGGDTYYADSVKG
344	PRT	RDDYTSRDAFDY
345	PRT	QSVLTQPPSASGTPGQRTVITSCGSSSNIGSNTVNWYQQLPGTAPKLLIYY DDLRLPSGVPDRFSGSKSGTSASLAISGLQSEDEADYYCAAWDDSLNDIPV FGGGKLTVL
346	PRT	SGSSSNIGSNTVN
347	PRT	YDDLRLPS
348	PRT	AAWDDSLNDIPV
349	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTACACCTTTTACAGCTA

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
		CGCCATGAGCTGGGTCCGACAGGCCCTGGAAAAGGCCTGGAATGGG TGTCGCCATCGGCTATGGCGGCGATACCTACTACGCCGACTCTGTG AAGGGCAGATTACCATCAGCCGGGACAACAGCAAGAACACCCCTGT ACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCGTGTACTAT TGCGCCAGAAGGGACGACTACACCAGCAGGGACGCCTTCGATTATTG GGGCCAGGGCACACTGGTCACCGTTTCTTCA
350	DNA	AGCTACGCCATGAGC
351	DNA	GCCATCGGCTATGGCGGCGATACCTACTACGCCGACTCTGTGAAGGG C
352	DNA	AGGGACGACTACACCAGCAGGGACGCCTTCGATTAT
353	DNA	CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTAGCGGCGAGCAGCTCCAACATCGGCAGC AACACCGTGAACCTGGTATCAGCAGCTGCCTGGCACAGCCCTAAACT GCTGATCTACTACGACGACCTGCGGCCTAGCGCGGTGCCAGATAGAT TTTCTGGCAGCAAGAGCGGCACCTCTGCCAGCCTGGCTATTTCTGGA CTGCAGAGCGAGGACGAGGCCGACTATTATTGTGCCGCTGGGACGA CAGCCTGAACGACATCCCTGTTTTTGGCGGAGGCCAACAGCTGACCG TTCTA
354	DNA	AGCGGCAGCAGCTCCAACATCGGCAGCAACACCGTGAAC
355	DNA	TACGACGACCTGCGGCCTAGC
356	DNA	GCCGCCTGGGACGACAGCCTGAACGACATCCCTGTT
357	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTTFYSYAMSWVRQAPGKLEWV SAIGYGGDTYYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCA RRDDYTSRDAFDYWGQGLTVTVSSASTKGPSVFPLAPCSRSTSESTAAL GCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSS LGTKTYTCNVDHKPSNTKVDKRVESKYGPCCPPCPAPEFLGGPSVFLFPP KPKDTLMISRTPEVTCVVVDVSDQEDPEVFQFNWYVDGVEVHNAKTKPRE EQFNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKGLPSSIEKTIKAKGQ PREPQVYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNY KTPPVLDSGDSFPLYSLRTVDKSRWQEGNVFSCSVMEALHNHYTQK SLSLSLGK
358	PRT	QSVLTQPPSASGTPGQRVITISCSGSSSNIGSNTNVNYYQLPGTAPKLLIYY DDLRP SGV PDRFSGSKSGTSASLAISGLQSEDEADYYCAAWDDSLNDIPV FGGGTKLTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTV AWKADSSPVKAGVETTPSKQSNNKYAASSYLSLTPEQWKS HRSYSCQ VTHEGSTVEKTVAPTECS
359	DNA	GAAGTTACAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTACACCTTTTACAGCTA GCCATGAGCTGGGTCCGACAGGCCCTGGAAAAGGCCTGGAATGGG TGTCGCCATCGGCTATGGCGGCGATACCTACTACGCCGACTCTGTG AAGGGCAGATTACCATCAGCCGGGACAACAGCAAGAACACCCCTGT ACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCGTGTACTAT TGCGCCAGAAGGGACGACTACACCAGCAGGGACGCCTTCGATTATTG GGGCCAGGGCACACTGGTCACCGTTTCTTTCAGCCAGCACCAAGGGCC CCAGCGTGTTCCTCTGGCCCCCTGTAGCAGAAGCACACGCGAGTCT ACAGCCGCCCTGGGCTGCCTCGTGAAGGACTACTTTCCCGAGCCCGT GACCGTGTCTCTGGAACCTCTGGCGCTCTGACAAGCGCGCTGCACACCT TTCCAGCCGTGCTGCAGAGCAGCGGCTGTACTCTCTGAGCAGCGTC GTGACAGTGCCAGCAGCAGCCTGGGCACCAAGACCTACACCTGTAA CGTGGACCACAAGCCCAGCAACACCAAGGTGGACAAGCGGGTGGAA TCTAAGTACGGCCCTCCCTGCCCTCCTTGCCAGCCCTGAATTCTGTG GGCGGACCCCTCCGTGTTCTGTTCCTCCCAAGCCCAAGGACACCCCT GATGATCAGCCGAGCCCGAAGTGACCTGCGTGGTGGTGGATGTGT CCAGGAAGATCCCGAGGTGCAGTTCAATTGGTACGTGGACGCGCTG GAAGTGACACAACGCCAAGACCAAGCCAGAGAGGAACAGTTCAACA GCACCTACCGGGTGGTGTCCGTGCTGACAGTGTGTCACCGAGACTGG CTGAACGGCAAGAGTACAAGTGCAAGGTGTCCAACAAGGGCCTGC CCAGCTCCATCGAGAAAACCATCAGCAAGGCCAAGGGCCAGCCCCG CGAACCCAGGTGTACACACTGCCTCCAAGCCAGGAAGAGATGACC AAGAACCAGGTGTCCCTGACCTGTCTCGTGAAGGCTTCTACCCCTC CGATATCGCCGTGGAATGGGAGAGCAACGGCCAGCCCGAGAACAAAC

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
		TACAAGACCACCCCTGTGCTGGACAGCGACGGCTCATTTCTTCCT GTACAGCAGACTGACCGTGGACAAGAGCCGTGGCAGGAAGGCAAC GTGTTTCTGAGCTGACGCTGATGCACGAGGCCCTGCACAACCACTACAC CCAGAAGTCCCTGTCTCTGAGCCTGGGCAAG
360	DNA	CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTAGCGGCAGCAGCTCCAACATCGGCAGC AACACCGTGAACCTGGTATCAGCAGCTGCCTGGCACAGCCCTAAACT GCTGATCTACTACGACGACCTGCGGCCTAGCGGCGTGCCAGATAGAT TTTCTGGCAGCAAGAGCGGCACCTCTGCCAGCCTGGCTATTTCTGGA CTGCAGAGCGAGGACGAGCCGACTATTATTGTGCCGCTGGGACGA CAGCCTGAACGACATCCCTGTTTTTGGCGGAGGCCAACAGCTGACCG TTCTAGGCCAGCCTAAAGCCGCCCTAGCGTGACCCCTGTTCCCTCCAA GCAGCGAGGAACCTGCAGGCCAACAGGCCACCCCTCGTGTGCCTGATC AGCGACTTCTATCCTGGCGCCGTGACCGTGGCCTGGAAGGCCGATAG CTCTCCTGTGAAGGCCGGCGTGGAAACCAACCCCTAGCAAGCAGA GCAACAAATAACGCCGCCAGCAGCTACCTGAGCCTGACCCCGAG CAGTGGAAAGTCCCACAGATCCTACAGCTGCCAAGTGACCCACGAGGG CAGCACCGTGGAAAAGACAGTGGCCCTACCGAGTGCAGC
361	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFYSYAMSWVRQAPGKLEWV SAIGYGGDTYYADSVKGRFTISRDNKNTLYLQMNSLRADTAVYYCA RRDDYTSRDAFDYWGQTLVTVSS
362	PRT	SYAMS
363	PRT	AIGYGGDTYYADSVKG
364	PRT	RDDYTSRDAFDY
365	PRT	QSVLTQPPSASGTPGQRVITISCGSSSNIGSNTVNWYQQLPGTAPKLLIYY DDLRLPSGVPDRFSGSKGTSASLAISGLQSEADYYCAAWDDSLNVIPIV FGGGTKLTVL
366	PRT	SGSSSNIGSNTVN
367	PRT	YDDLRLPS
368	PRT	AAWDDSLNVIPIV
369	DNA	GAAGTTCTAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCACTTTTACAGCTA CGCCATGAGCTGGGTCCGACAGGCCCTGGAAAAGGCCTGAAATGGG TGTCGCCATCGGCTATGGCGGCGATACCTACTACGCCGACTCTGTG AAGGGCAGATTACCATCAGCCGGGCAACAGCAAGAACACCCCTGT ACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCGTGTACTAT TGCGCCAGAAGGGACGACTACACAGCAGGGACGCCCTTCGATTATTG GGGCCAGGGCACACTGGTCACCGTTTCTTCA
370	DNA	AGCTACGCCATGAGC
371	DNA	GCCATCGGCTATGGCGGCGATACCTACTACGCCGACTCTGTGAAGGG C
372	DNA	AGGGACGACTACACAGCAGGGACGCCTTCGATTAT
373	DNA	CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTAGCGGCAGCAGCTCCAACATCGGCAGC AACACCGTGAACCTGGTATCAGCAGCTGCCTGGCACAGCCCTAAACT GCTGATCTACTACGACGACCTGCGGCCTAGCGGCGTGCCAGATAGAT TTTCTGGCAGCAAGAGCGGCACCTCTGCCAGCCTGGCTATTTCTGGA CTGCAGAGCGAGGACGAGCCGACTATTATTGTGCCGCTGGGACGA CAGCCTGAACGTGATCCCTGTTTTTGGCGGAGGCCAACAGCTGACCG TTCTA
374	DNA	AGCGGCAGCAGCTCCAACATCGGCAGCAACACCGTGAAC
375	DNA	TACGACGACCTGCGGCCTAGC
376	DNA	GCCGCCCTGGGACGACAGCCTGAACGTGATCCCTGTT

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
377	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFYYSAMSWVRQAPGKGLEWV SAIGYGGDTYYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCA RRDDYTSRDAFDYWGQGLTVTVSSASTKGPSVFPLAPCSRSTSESTAAL GCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSS LGTKTYTNCNVDHKPSNTKVDKRVESKYGPPCPPCPAPEFLGGPSVFLFPP KPKDTLMISRTPEVTCVVDVVSQEDPEVQFNWYVDGVEVHNAKTKPRE EQFNSYTRVVSVLTVLHQDWLNGKEYKCKVSNKGLPSSIEKTIKAKGQ PREPQVYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNY KTTTPPVLDSDGSFFLYSRLTVDKSRWQEGNVFSCSVMHLEHNHYTQK SLSLSLGK
378	PRT	QSVLTQPPSASGTPGQRVTISCSGSSSNIGSNTVNWYQQLPGTAPKLLIYY DDLRLPSGVPRDRFSGSKSGTSASLAISGLQSEDEADYYCAWDDSLNVI FVGGGKTLTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTV AWKADSSPVKAGVETTPPSKQSNKYAASSYLSLTPEQWKSRSYSQC VTHEGSTVEKTVAPTECS
379	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCGCAGCGCTTCACTTTTACAGCTA CGCCATGAGCTGGGTCCGACAGGCCCTTGAAAGGCCTTGAATGGG TGTCGCCCATCGGCTATGGCGGCGGATACCTACTACGCCGACTCTGTG AAGGCGAGATTACCATCAGCCGGGACACAGCAAGAACACCTGT ACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCCCGGTGTACTAT TGCGCCAGAAGGACGACTACACAGCAGGACGCGCTTCGATTATTG GGCCAGGGCACACTGGTCACCGTTTCTTACGACAGCACCAGGGCC CCAGCGTGTTCCTCTGGCCCCCTGTAGCAGAAGCACCAGCGAGTCT ACAGCCGCCCTGGGCTGCCTCGTGAAGGACTACTTTCCCGAGCCCGT GACCGTGTCTCGAACTCTGGCGCTCTGACAAGCGCGTGACACCT TTCCAGCCGTGTGTCAGAGCAGCGGCTGTACTCTCTGAGCAGCGTC GTGACAGTGCCAGCAGCAGCCTGGGCACCAAGACCTACACCTGTAA CGTGGACCAACAAGCCAGCAACACCAAGGTGGACAAGCGGGTGGAA TCTAAGTACGGCCCTCCCTGCCCTCTTGCCAGCCCTGAATTTCTG GGCGGACCCCTCCGTGTTCTGTTCCTCCCAAGCCCAAGGACACCT GATGATCAGCCGAGACCCCGAAGTGACCTGCGTGGTGGTGGATGTGT CCAGGAAGATCCCGAGGTGCAGTTCAATTGGTACGTGGACGGCGTG GAAGTGACACAACGCCAAGACCAAGCCAGAGAGGAACAGTTCAACA GCACCTACCGGGTGGTGTCCGTGCTGACAGTGCTGACACAGGACTGG CTGAACGGCAAGAGTACAAGTGCAAGGTGTCCAACAAGGGCCTGC CCAGCTCCATCGAGAAAACCATCAGCAAGGCCAAGGGCCAGCCCCG CGAACCCAGGTGTACACACTGCCTCAAGCCAGGAAGAGATGACC AAGAACCAGGTGTCCCTGACCTGTCTCGTGAAAGGCTTCTACCCCTC CGATATCGCCGTGGAATGGGAGAGCAACGGCCAGCCCGAGAACAC TACAAGACCAACCCCTGTGCTGGACAGCGACGGCTCATTCTTCTCT GTACAGCAGACTGACCGTGGACAAGAGCCGCTGGCAGGAAGGCAAC GTGTTCAAGTGCAGCGTGATGCACGAGGCCCTGCACAACCACTACAC CCAGAAGTCCCTGTCTCTGAGCCTGGCAAG
380	DNA	CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTAGCGGCAGCAGCTCCAACATCGGCAGC AACACCGTGAACCTGGTATCAGCAGCTGCCTGGCACAGCCCCCTAACT GCTGATCTACTACGACGACCTGCGGCCTAGCGCGCTGCCAGATAGAT TTTCTGGCAGCAAGAGCGGCACCTCTGCCAGCCTGGCTATTTCTGGA CTGCAGAGCGAGGACGAGGCCGACTATTATTGTGCCGCTGGGACGA CAGCCTGAACGTGATCCCTGTTTTTGGCGGAGGCACCAAGCTGACCG TTCTAGGCCAGCCTAAAGCCGCCCTAGCGTGACCTGTTCCTCCAA GCAGCGAGGAACCTGACGGCCAAAGGCCACCCCTCGTGTGCTGATC AGCGACTTCTATCTGGCGCCGTGACCGTGGCCTGGAAGGCCGATAG CTCTCTGTGAAGGCCGGCGTGAAACCAACACCCCTAGCAAGCAGA GCAACAACAATAACGCCCGCAGCAGCTACCTGAGCCTGACCCCGAG CAGTGGAGTCCACAGATCCTACAGCTGCCAAGTGACCCACGAGGG CAGCACCGTGGAAAAGACAGTGGCCCTACCGAGTGCAGC
381	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFDSYEMNWVRQAPGKGLEWV SGISWNSGWIDYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYC ARSGYSSSWFDPDFDYWGQGLTVTVSS
382	PRT	SYEMN
383	PRT	GISWNSGWIDYADSVKG
384	PRT	SGYSSSWFDPDFDY

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
385	PRT	QSVLTQPPSVSGAPGQRVITISCTGSSSDIGAGYDVHWYQQLPGTAPKLLI YGNNSNRPSGVPDRFSGSKSGTSASLAITGLQAEDEADYYCSSYAGPNPY VVFGGGTKLTVL
386	PRT	TGSSSDIGAGYDVH
387	PRT	GNSNRPS
388	PRT	SSYAGPNPYV
389	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCACCTTCGATAGCT ACGAGATGAACTGGGTCCGACAGGCCCTTGGCAAAGGCCTTGAATG GGTGTCCGGCATCAGCTGGAATAGCGGCTGGATCGACTACGCCGACA CGGTGAAGGGCAGATTACCATCAGCCGGGACAAACAGCAAGAACAC CCTGTACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCGTGT ACTACTGTGCCAGAAGCGGCTACAGCAGCTCTTGGTTTGACCCCGAC TTCGACTATTGGGGCCAGGGCACACTGGTCACAGTCTCTTCA
390	DNA	AGCTACGAGATGAAC
391	DNA	GGCATCAGCTGGAATAGCGGCTGGATCGACTACGCCGACAGCGTGA AGGGC
392	DNA	AGCGGCTACAGCAGCTCTTGGTTTGACCCCGACTTCGACTAT
393	DNA	CAGTCTGTTCTGACACAGCCTCCATCTGTGTCTGGCGCCCTGGACAG AGAGTGACCATCAGCTGTACAGGCAGCAGCTCCGATATTGGCGCCG ATACGACGTGCACTGGTATCAGCAACTGCCTGGCACAGCCCTAAGC TGCTGATCTACGGCAACAGCAACAGACCTAGCGGCGTGCCTGATAGA TTCAGCGGCTCTAAGTCTGGCAACAGCGCCAGCCTGGCCATTACTGG ACTGCAGGCCGAAGATGAGGCCGACTACTACTGCAGCAGCTACGCTG GCCCAATCCTTACGTGGTGTGGCGGCGGAACAAAGCTGACCGTT CTA
394	DNA	ACAGGCAGCAGCTCCGATATTGGCGCCGGATACGACGTGCAC
395	DNA	GGCAACAGCAACAGACCTAGC
396	DNA	AGCAGCTACGCTGGCCCCAATCCTTACGTGGTG
397	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFDSYEMNWVRQAPGKLEWV SGISWNSGWIDYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYC ARSGYSSSWFDPDFDYWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGT AALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTV PSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKHTCTCPCPAPELLGGP SVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNA KTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKT I SKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNG QPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALH NHYTQKSLSLSPGK
398	PRT	QSVLTQPPSVSGAPGQRVITISCTGSSSDIGAGYDVHWYQQLPGTAPKLLI YGNNSNRPSGVPDRFSGSKSGTSASLAITGLQAEDEADYYCSSYAGPNPY VVFGGGTKLTVLQGPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAV TVAWKADSSPVKAGVETTTPSKQSNKYAASSYLSLTPEQWKSHRYS CQVTHEGSTVEKTVAPTECS
399	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCACCTTCGATAGCT ACGAGATGAACTGGGTCCGACAGGCCCTTGGCAAAGGCCTTGAATG GGTGTCCGGCATCAGCTGGAATAGCGGCTGGATCGACTACGCCGACA CGGTGAAGGGCAGATTACCATCAGCCGGGACAAACAGCAAGAACAC CCTGTACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCGTGT ACTACTGTGCCAGAAGCGGCTACAGCAGCTCTTGGTTTGACCCCGAC TTCGACTATTGGGGCCAGGGCACACTGGTCACAGTCTCTTCAGCCAG CACCAAGGGCCCCAGCGTGTTCCTCTGGCCCTAGCAGCAAGAGCA CATCTGGCGGAACAGCCGCCCTGGGCTGCCTCGTGAAGGACTACTTT CCCGAGCCCGTGACCGTGTCTGGAACCTGGCGCTCTGACAAGCGG CGTGCACACCTTTCCAGCCGTGCTGCAGAGCAGCGGCTGTACTCTCT

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
		GAGCAGCGTCGTGACAGTGCCCGAGCCTCTCTGGGCACCCAGACCT ACATCTGCAACGTGAACCACAAGCCAGCAACACCAAGGTGGACAA GAAGGTGGAACCCAAGAGCTGCGACAAGACCCACACCTGTCCCCCTT GTCCCTGCCCCGAACTGCTGGGAGGCCCTTCCGTGTTCTGTTCCTCC CAAAGCCCAAGGACACCTGATGATCAGCCGACCCCCGAAGTGAC CTGCGTGGTGGTGGATGTGTCCACGAGGACCTGAAGTGAAGTTCA ATTGGTACGTGGACGGCGTGAAGTGACACAACGCCAAGACCAAGCC TAGAGAGGAACAGTACAACAGCACCTACCGGTGGTGTCCGTGCTGA CAGTGTGTCACAGGACTGGCTGAACGGCAAGAGTACAAGTGCAA GGTGTCCAACAAGGCCCTGCCTGCCCCATCGAGAAAACCATCAGCA AGGCCAAGGGCCAGCCCCGCAACCCAGGTGTACACACTGCCCCCA AGCAGGGACGAGCTGACCAAGAACCAGGTGTCCCTGACCTGTCTCGT GAAAGGCTTCTACCCCTCCGATATCGCCGTGGAATGGGAGAGCAACG GCCAGCCCCGAGAACTACAAGACCACCCCTGTGTGGACAGC GACGGCTCATTCTTCTGTACAGCAAGCTGACCGTGGACAAGTCCCG GTGGCAGCAGGGCAACGTGTTTCAGCTGCAGCGTGATGCACGAGGCC TGCACAACCACTACACCCAGAAGTCCCTGAGCCTGAGCCCTGGCAAG
400	DNA	CAGTCTGTTCTGACACAGCCTCCATCTGTGTCTGGCGCCCTGGACAG AGAGTGACCATCAGCTGTACAGGCAGCAGCTCCGATATTGGCGCCGG ATACGACGTGCACTGGTATCAGCAACTGCCTGGCAGACCCCTAAGC TGCTGATCTACGGCAACAGCAACAGACCTAGCGGCGTGCCGATAGA TTCAGCGGCTCTAAGTCTGGCACAAGCGCCAGCCTGGCCATTACTGG ACTGCAAGGCCAAGATGAGGCCGACTACTACTGCAGCAGCTACGCTG GCCCCAATCCTTACGTGGTGTGTTGGCGGCGGAACAAAGCTGACCGTT CTAGGCCAGCCTAAAGCCGCCCCTAGCGTGACCTGTTCCTCCAAG CAGCGAGGAAGTGCAGGCCAACAAAGGCCACCTCGTGTGCCTGATCA GCGACTTCTATCCTGGCGCCGTGACCGTGGCCTGGAAGGCCGATAGC TCTCCTGTGAAGCCGCGCTGGAAACCACACCCCTAGCAAGCAGAG CAACAACAAATACGCGCCAGCAGCTACCTGAGCCTGACCCCGAGC AGTGGAAAGTCCACAGATCCTACAGCTGCCAAGTGACCCACGAGGGC AGCACCGTGAAAAGACAGTGGCCCCTACCGAGTGCAGC
401	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFDSYEMNWVRQAPGKLEWV SGISWNSGWIDYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYC ARSGYSSSWFDPDFDYWGQTLTVTVSS
402	PRT	SYEMN
403	PRT	GISWNSGWIDYADSVKG
404	PRT	SGYSSSWFDPDFDY
405	PRT	QSVLTQPPSVSGAPGQRTVTSCTGSSSNIGAGYDVHWYQQLPGTAPKLLI YGNNSNRPSGVPDFRFGSKSGTSASLAITGLQAEDEADYYCQSYAGINPY VVFGGGKLTVL
406	PRT	TGSSSNIGAGYDVH
407	PRT	GNSNRPS
408	PRT	QSYAGINPYV
409	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCACTTCGATAGCT ACGAGATGAACTGGGTCCGACAGGCCCTTGGCAAAGGCCCTGAATG GGTGTCCGGCATCAGCTGGAATAGCGGCTGGATCGACTACGCCGACA CGGTGAAGGGCAGATTACCATCAGCCGGGACAAACGCAAGAACAC CCTGTACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCGTGT ACTACTGTGCCAGAAGCGGCTACAGCAGCTCTTGGTTTGACCCCGAC TTCGACTATTGGGGCCAGGGCACACTGGTCACAGTCTCTTCA
410	DNA	AGCTACGAGATGAAC
411	DNA	GGCATCAGCTGGAATAGCGGCTGGATCGACTACGCCGACAGCGTGA AGGGC
412	DNA	AGCGGCTACAGCAGCTCTTGGTTTGACCCGACTTCGACTAT
413	DNA	CAGTCTGTTCTGACACAGCCTCCATCTGTGTCTGGCGCCCTGGACAG AGAGTGACCATCAGCTGTACAGGCAGCAGCTCCAATATCGGAGCCGG

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
		CTATGACGTGCACTGGTATCAGCAGCTGCCTGGCACAGCCCCCTAAAC TGCTGATCTACGGCAACAGCAACAGACCCAGCGGCGTGCCCGATAGA TTTTCCGGCTCTAAGAGCGGCACAAGCGCCAGCCTGGCTATTACTGG ACTGCAGGCCGAGGACGAGGCCGACTACTACTGTCTAGAGCTACGCCG GCATCAACCCCTACGTGGTGTGCGGAGGCACCAAGCTGACAGTT CTA
414	DNA	ACAGGCAGCAGCTCCAATATCGGAGCCGGCTATGACGTGCAC
415	DNA	GGCAACAGCAACAGACCCAGC
416	DNA	CAGAGCTACGCCGGCATCAACCCCTACGTGGTG
417	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFDSYEMNWVRQAPGKGLEWV SGISWNSGWIDYADSVKGRFTISRDNKNTLYLQMNSLRADTAVYYC ARSGYSSSWFDPDFDYWGQGLTVTVSSASTKGPSVFLAPSSKSTSGGT AALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTV PSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKHTHTCPPCPAPELGGP SVFLPPPKPDTLMI SRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNA KTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKT SKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNG QPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCSCVMHEALH NHYTQKSLSLSPGK
418	PRT	QSVLTQPPSVSGAPGQRTVISCTGSSSNIGAGYDVHWYQQLPGTAPKLLI YGNNSNRPSGVDPDRFSGSKGTSASLAITGLQAEDEADYQCQSYAGINPY VVFPGGKTGLTVLQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAV TVAWKADSSPVKAGVETTTPSKQSNNKYAASSYLSLTPEQWKSHRYS CQVTHEGSTVEKTVAPTECS
419	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCACTTCGATAGCT ACGAGATGAACTGGGTCCGACAGGCCCTTGGCAAAGGCCCTGAATG GGTGTCCGGCATCAGCTGGAATAGCGGCTGGATCGACTACGCCGACA GCGTGAAGGGCAGATTACCATCAGCCGGACCAACAGCAAGAACAC CCTGTACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCGTGT ACTACTGTGCCAGAAGCGGCTACAGCAGCTCTTGGTTTGACCCCGAC TTCGACTATTGGGGCCAGGGCACACTGGTCACAGTCTCTTCAGCCAG CACCAAGGGCCCCAGCGTGTTCCTCTGGCCCCCTAGCAGCAAGAGCA CATCTGGCGGAACAGCCGCTTGGGCTGCCTCGTGAAGGACTACTTT CCCGAGCCCGTGACCGTGTCTTGAAGTCTGGCGCTCTGACAAGCGG CGTGACACCTTTCCAGCCGTGTGTCAGAGCAGCGGCTGTACTCTCT GAGCAGCGTGTGACAGTGCCAGCAGCTCTCTGGGCACCCAGACCT ACATCTGCAACGTGAACCAAGCCAGCAACACCAAGGTGGACAA GAAGGTGGAACCAAGAGCTGCGACAAGACCCACACCTGTCCCCCTT GTCTGCCCCCGAAGTGTCTGGGAGGCCCTTCGTGTCTCTGTCCCCC CAAAGCCCAAGGACACCTGATGATCAGCCGACCCCGAAGTGAC CTGCGTGGTGGTGGATGTGTCCACGAGGACCTGAAGTGAAGTTCA ATTGGTACGTGGACGGCGTGAAGTGCAACGCCAAGACCAAGCC TAGAGAGGAACAGTACAACAGCACCTACCGGGTGGTGTCTGCTGTA CAGTGTGACACAGGACTGGCTGAACGGCAAAGAGTACAAGTGCAA GGTGTCCAACAAGGCCCTGCCTGCCCCATCGAGAAAACCATCAGCA AGGCCAAGGGCCAGCCCCGCAACCCAGGTGTACACACTGCCCCCA AGCAGGGACGAGCTGACCAAGAACCAGGTGTCCCTGACCTGTCTCGT GAAAGGCTTCTACCCCTCGATATCGCCGTGGAATGGGAGAGCAACG GCCAGCCCGAGAACTACAAGACCACCCCTGTGTGGACACG GACGGCTCATTCTTCTGTACAGCAAGCTGACCGTGGACAAGTCCCG GTGGCAGCAGGGCAACGTGTTAGCTGCAGCGTGATGCACGAGGCC TGACAAACCTACACCCAGAAGTCCCTGAGCTGAGCCCTGGCAAG
420	DNA	CAGTCTGTTCTGACACAGCCTCCATCTGTGTCTGGCGCCCCCTGGACAG AGAGTGACCATCAGCTGTACAGGCAGCAGCTCCAATATCGGAGCCGG CTATGACGTGCACTGGTATCAGCAGCTGCCTGGCACAGCCCCCTAAAC TGCTGATCTACGGCAACAGCAACAGACCCAGCGGCGTGCCCGATAGA TTTTCCGGCTCTAAGAGCGGCACAAGCGCCAGCCTGGCTATTACTGG ACTGCAGGCCGAGGACGAGGCCGACTACTACTGTCTAGAGCTACGCCG GCATCAACCCCTACGTGGTGTGCGGAGGCAACCAAGCTGACAGTT CTAGGCCAGCCTAAAGCCGCCCTAGCGTGACCTGTTCCTCCAAG CAGCGAGGAAGTGCAGGCCAACAAGGCCACCTCGTGTGCCTGATCA GCGACTTCTATCTTGGCGCGGTGACCGTGGCCTGGAAGGCCGATAGC TCTCTGTGAAGGCCGGCGTGGAAACCAACCCCTAGCAAGCAGAG

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
		CAACAACAAATACGCCGCCAGCAGCTACCTGAGCCTGACCCCGAGC AGTGGAGTCCCACAGATCCTACAGCTGCCAAGTGACCCACGAGGGC AGCACCGTGGAAAAGACAGTGGCCCCCTACCGAGTGCGAC
421	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFDSSYEMNWVRQAPGKLEWV SGISWNSGWIGYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYC ARSGYSSSWFDPDFDYWGQGLTVTVSS
422	PRT	SYEMN
423	PRT	GISWNSGWIGYADSVKG
424	PRT	SGYSSSWFDPDFDY
425	PRT	QSVLTQPPSVSGAPGQRTISCTGSSSNIGAGYDVHWYQQLPGTAPKLLI YGNSNRPSGVPDRFSGSKGTSASLAITGLQAEDEADYYCQSYAGPNPY VVFGGGKLTVL
426	PRT	TGSSSNIGAGYDVH
427	PRT	GNSNRPS
428	PRT	QSYAGPNPYV
429	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCGATTTTCAGCAGCT ACGAGATGAACTGGGTCCGACAGGCCCTTGGCAAAGGCTTGAATG GGTGTCCGGCATCAGCTGGAATAGCGGCTGGATCGGCTACGCCGATA GCGTGAAGGGCAGATTACCATCAGCCGGACAAACAGCAAGAACAC CCTGTACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCGTGT ACTACTGTGCCAGAAGCGGCTACAGCAGCTCTTGGTTTGACCCCGAC TTCGACTATTGGGGCCAGGGCACACTGGTCACAGTCTCTTCA
430	DNA	AGCTACGAGATGAAC
431	DNA	GGCATCAGCTGGAATAGCGGCTGGATCGGCTACGCCGATAGCGTGAA GGC
432	DNA	AGCGGCTACAGCAGCTCTTGGTTTGACCCCGACTTCGACTAT
433	DNA	CAGTCTGTTCTGACACAGCCTCCATCTGTGTCTGGCGCCCTGGACAG AGAGTGACCATCAGCTGTACAGGCAGCAGCTCCAATATCGGAGCCGG CTATGACGTGCACTGGTATCAGCAGCTGCCTGGCACAGCCCTAAAC TGCTGATCTACGGCAACAGCAACAGACCAGCGGCGTGCCCGATAGA TTTTCCGGCTCTAAGAGCGGCACAAGCGCCAGCCTGGCTATTACTGG ACTGCAGGCCGAGGACGAGGCCGACTACTACTGTCACTTACGCTG GCCCAATCCTTACGTGGTGTGGCGGCGGAACAAAGCTGACCGTT CTA
434	DNA	ACAGGCAGCAGCTCCAATATCGGAGCCGGCTATGACGTGCAC
435	DNA	GGCAACAGCAACAGACCCAGC
436	DNA	CAGTCTTACGCTGGCCCCAATCCTTACGTGGTG
437	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFDSSYEMNWVRQAPGKLEWV SGISWNSGWIGYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYC ARSGYSSSWFDPDFDYWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGT AALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTV PSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKHTHTCPPCPAPPELLGGP SVFLPPPKPDTLMI SRTPEVTVVVDVSHEDPEVKFNWYVDGVEVHNA KTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKT I SKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNG QPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSQVSMHEALH NHYTQKSLSLSPGK
438	PRT	QSVLTQPPSVSGAPGQRTISCTGSSSNIGAGYDVHWYQQLPGTAPKLLI YGNSNRPSGVPDRFSGSKGTSASLAITGLQAEDEADYYCQSYAGPNPY VVFGGGKLTVLGQPKAAPSVTLPFPPSSEELQANKATLVCLISDFYPGAV TVAWKADSSPVKAGVETTPSKQSNKNYAASSYLSLTPEQWKSHRYS CQVTHEGSTVEKTVAPTECS

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
439	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCGATTTTCAGCAGCT ACGAGATGAACTGGGTCCGACAGGCCCTGGCAAAGGCCTTGAATG GGTGTCGGCATCAGCTGGAATAGCGGCTGGATCGGCTACGCCGATA GCGTGAAGGCGAGATTACCATCAGCCGGGACAAACAGCAAGAACAC CCTGTACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCGTGT ACTACTGTGCCAGAAGCGGTACAGCAGCTCTTGGTTGACCCCGAC TTCGACTATTGGGGCCAGGGCACACTGGTCACAGTCTCTTCAGCCAG CACCAGGGCCCCAGCGTGTCCCTCTGGCCCTAGCAGCAAGAGCA CATCTGGCGGAACAGCCGCCCTGGGCTGCCTCGTGAAGGACTACTTT CCCCAGCCCGTGACCGTGTCTCTGGAACCTGCGCGCTCTGACAAGCGG CGTGACACACCTTTCCAGCCGTGCTGCAGAGCAGCGCCTGTACTCTCT GAGCAGCGTCTGACAGTGCCAGCAGCTCTCTGGGCACCCAGACCT ACATCTGCAACGTGAACCAAGCCAGCAACACCAAGGTGGACAA GAAGGTGGAACCCAAGAGCTGCGACAAGACCCACACCTGTCCCTT GTCTGCCCCGAACCTGCTGGGAGGCCCTTCCTGTCTCTGTTCCTT CAAAGCCCAAGGACACCCCTGATGATCAGCCGAGCCCCGAAGTGAC CTGCGTGGTGGTGATGTGTCCACGAGGACCTGAAGTGAAGTTCA ATTGGTACGTGGACGGCGTGAAGTGACCAACGCCAAGCAAGCC TAGAGAGGAACAGTACAACAGCACCTACCGGGTGGTGTCCGTGCTGA CAGTGTGTCACCCAGGACTGGCTGAACGGCAAGAGTACAAGTGCAA GGTGTCCAACAAGGCCCTGCCTGCCCCATCAGAGAAACCATCAGCA AGGCCAAGGGCCAGCCCCGCAACCCAGGTGTACACACTGCCCCCA AGCAGGGACGAGCTGACCAAGAACCAGGTGTCCCTGACCTGTCTCGT GAAAGGCTTCTACCCCTCCGATATCGCCGTGGAATGGGAGAGCAACG GCCAGCCCCGAGAACTACAAGACACCCCCCTGTGCTGGACAGC GACGGCTCATTCTTCTGTACAGCAAGCTGACCGTGGACAAGTCCCG GTGGCAGCAGGGCAACGTGTTTACGTGACGCTGATGACAGGGCCC TGCAACAACCTACACCCAGAAGTCCCTGAGCCTGAGCCTGGCAAG
440	DNA	CAGTCTGTTCTGACACAGCCTCCATCTGTGTCTGGCGCCCTGGACAG AGAGTGACCATCAGCTGTACAGGCAGCAGCTCCAATATCGGAGCCGG CTATGACGTGCACTGGTATCAGCAGCTGCCTGGCAGCCCTAAAC TGCTGATCTACGGCAACAGCAACAGCCAGCGGCGTGCCCGATAGA TTTTCCGGCTCTAAGAGCGGCACAAGCGCCAGCCTGGCTATTACTGG ACTGAGGCCGAGGACGAGGCCGACTACTACTGTCTAGTCTTACGCTG GCCCAATCCTTACGTGGTGTGTTGGCGGCGGAACAAGCTGACCGTT CTAGGCCAGCCTAAAGCCGCCCCTAGCGTGACCCCTGTTCCCTCCAAG CAGCGAGGAACTGCAGGCCAACAAGGCCACCCCTCGTGTGCTGATCA GCGACTTCTATCTTGGCGCCGTGACCGTGGCTGGAAGGCCGATAGC TCTCCTGTGAAGGCCGCGTGGAAACCACACCCCTAGCAAGCAGAG CAACAACAAATACGCCGCCAGCAGCTACCTGAGCCTGACCCCGAGC AGTGAAGTCCACAGATCCTACAGCTGCCAAGTGACCCACGAGGGC AGCACCGTGGAAGACAGTGGCCCCCTACCGAGTGCAGC
441	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFDFSSYEMNWVRQAPGKLEWV SGISWNSGWIDYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYC ARSGYSSSWFDPDFDYWGQGLTVTVSS
442	PRT	SYEMN
443	PRT	GISWNSGWIDYADSVKG
444	PRT	SGYSSSWFDPDFDY
445	PRT	QSVLTQPPSVSGAPGQRVTISCTGSSSNIGAGYDVHWYQQLPGTAPKLLI YGNSNRPSGVPDRFSGSKGTSASLAITGLQAEDEADYYQSYAGPNPY VVFGGTKLTVL
446	PRT	TGSSSNIGAGYDVH
447	PRT	GNSNRPS
448	PRT	QSYAGPNPYV
449	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCGATTTTCAGCAGCT ACGAGATGAACTGGGTCCGACAGGCCCTGGCAAAGGCCTTGAATG GGTGTCGGCATCAGCTGGAATAGCGGCTGGATCGACTACGCCGACA GCGTGAAGGCGAGATTACCATCAGCCGGGACAAACAGCAAGAACAC

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
		CCTGTACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCGTGT ACTACTGTGCCAGAAGCGGCTACAGCAGCTCTTGGTTTGACCCCGAC TTCGACTATTGGGGCCAGGGCACACTGGTCACAGTCTCTTCA
450	DNA	AGCTACGAGATGAAC
451	DNA	GGCATCAGCTGGAATAGCGGCTGGATCGACTACGCCGACAGCGTGA AGGGC
452	DNA	AGCGGCTACAGCAGCTCTTGGTTTGACCCCGACTTCGACTAT
453	DNA	CAGTCTGTTCTGACACAGCCTCCATCTGTGTCTGGCGCCCTGGACAG AGAGTGACCATCAGCTGTACAGGCAGCAGCTCCAATATCGAGAGCCGG CTATGACGTGCACTGGTATCAGCAGCTGCCTGGCAGAGCCCTAAAC TGCTGATCTACGGCAACAGCAACAGACCCAGCGGCGTGCCCGATAGA TTTTCGGCTCTAAGAGCGGCACAAGCGCCAGCCTGGCTATTACTGG ACTGCAGGCCGAGGACGAGGCCGACTACTACTGTCTAGTCTTACGCTG GCCCAATCCTTACGTGGTGTGGCGGCGGAACAAAGCTGACCGTT CTA
454	DNA	ACAGGCAGCAGCTCCAATATCGGAGCCGGCTATGACGTGCAC
455	DNA	GGCAACAGCAACAGACCCAGC
456	DNA	CAGTCTTACGCTGGCCCCAATCCTTACGTGGTG
457	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFDSSYEMNWVRQAPGKLEWV SGISWNSGWIDYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYC ARSGYSSSWFDPDFDYWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGT AALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTV PSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKHTKCPPELGGP SVFLFPPKPKDTLMI SRTPEVTVVVDVSHEDPEVKFNWYVDGVEVHNA KTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKT ISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNG QPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSQSVMEALH NHYTQKSLSLSPGK
458	PRT	QSVLTQPPSVSGAPGQRVITISCTGSSSNIGAGYDVHWYQQLPGTAPKLLI YGNNSNRPSGVPDRFSGSKSGTSASLAITGLQAEADYYCQSYAGPNPY VVFGGGTLTVLGGPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAV TVAWKADSSPVKAGVETTPSKQSNKYAASSYLSLTPEQWKSHRYS CQVTHEGSTVEKTVAPTECS
459	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGCTTCGATTTAGCAGCT ACGAGATGAACTGGGTCCGACAGGCCCTGGCAAAGGCCCTGAATG GGTGTCCGGCATCAGCTGGAATAGCGGCTGGATCGACTACGCCGACA GCGTGAAGGCGAGATTACCATCAGCCGGGACAAACAGCAAGAACAC CCTGTACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCGTGT ACTACTGTGCCAGAAGCGGCTACAGCAGCTCTTGGTTTGACCCCGAC TTCGACTATTGGGGCCAGGGCACACTGGTCACAGTCTCTTACGCCAG CACCAGGGGCCAGCGTGTCCCTCTGGGCCCTAGCAGCAAGAGCA CATCTGGCGGAACAGCCGCCCTGGGCTGCCTCGTGAAGGACTACTTT CCGAGCCCCGTGACCGTGTCTGGAACCTGGCGCTCTGACAAGCGG CGTGACACACTTCCAGCCGTGCTGCAGAGCAGCGCCTGTACTCTCT GAGCAGCGTCGTGACAGTGCCAGCAGCTCTCTGGGCACCCAGACCT ACATCTGCAACGTGAACCAAGCCAGCAACACCAAGGTGGACAA GAAGGTGGAACCAAGAGCTGCGACAAGACCCACACCTGTCCCCCTT GTCCTGCCCCCCAAGTCTGGGAGGCCCTTCCTGTCTCTGTCCCCC CAAAGCCCCAAGGACACCTGATGATCAGCCGACCCCGAAGTGAC CTGCGTGGTGGTGGATGTGTCCACGAGGACCTGAAGTGAAGTTCA ATTGGTACGTGGACGGCGTGGAGTGCACAACGCCAAGACCAAGCC TAGAGAGGAACAGTACAACAGCACCTACCGGGTGGTGTCCGTGCTGA CAGTGTCTGCACAGGACTGGCTGAACGGCAAAGAGTACAAGTGCAA GGTGTCACAAGGCCCTGCCTGCCCCCATCGAGAAAACCATCAGCA AGGCCAAGGCCAGCCCCGGAACCCAGGTGTACACACTGCCCCCA

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
460	DNA	AGCAGGGACGAGCTGACCAAGAACCAGGTGTCCCTGACCTGTCTCGT GAAAGGCTTCTACCCCTCCGATATCGCCGTGGAATGGGAGAGCAACG GCCAGCCCGAGAACAACTACAAGACCACCCCCCTGTGCTGGACAGC GACGGCTCATTTCTTCTGTACAGCAAGCTGACCGTGGACAAGTCCCG GTGGCAGCAGGGCAACGTGTTCAGCTGCAGCGTGATGCACGAGGCC TGCACAACCACTACACCCAGAAGTCCCTGAGCCTGAGCCTGGCAAG CAGTCTGTTCTGACACAGCCTCCATCTGTGTCTGGCGCCCTGGACAG AGAGTGACCATCAGCTGTACAGGCAGCAGCTCCAATATCGGAGCCGG CTATGACGTGCACTGGTATCAGCAGCTGCCTGGCACAGCCCCATAAC TGCTGATCTACGGCAACAGCAACAGACCCAGCGGCGTGCCCGATAGA TTTTCCGGCTCTAAGAGCGGCACAAGCGCCAGCCTGGCTATTACTGG ACTGCAGGCCGAGGACGAGGCCGACTACTACTGTCTAGTCTTACGCTG GCCCAATCCTTACGTGGTGTGTTGGCGGCGGAACAAGCTGACCGTT CTAGGCCAGCCTAAAGCCGCCCCCTAGCGTGACCCCTGTCCCTCCAAG CAGCGAGGAACTGCAGGCCAACAAAGGCCACCCCTCGTGTGCTGATCA GCGACTTCTATCTTGGCGCCGTGACCGTGGCTGGAAGGCCGATAGC TCTCCTGTGAAGGCCGCGTGGAAACCACACCCCTAGCAAGCAGAG CAACAACAAATACGCCGCCAGCAGCTACTGAGCCTGACCCCGAGC AGTGGAACTCCACAGATCCTACAGCTGCCAAGTGACCCACGAGGGC AGCACCGTGGAAAAGACAGTGGCCCCCTACCGAGTGACGC
461	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFDSYEMNWRQAPGKLEWV SGISWNSGWIDYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYC ARSGYSSSWFDPDFDYWGQGLTVTVSS
462	PRT	SYEMN
463	PRT	GISWNSGWIDYADSVKG
464	PRT	SGYSSSWFDPDFDY
465	PRT	QSVLTQPPSVSGAPGQRVTISCTGSSSNIGAGYDVHWYQQLPGTAPKLLI YGNSNRPSGVPDRFSGSKSGTSASLAITGLQAEDEADYYCQSYAGPNPY VVFGGTKLTVL
466	PRT	TGSSSNIGAGYDVH
467	PRT	GNSNRPS
468	PRT	QSYAGPNPYV
469	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCACTTCGATAGCT ACGAGATGAAGTGGTCCGACAGGCCCTGGCAAAGGCCCTGAATG GGTGTCCGGCATCAGCTGGAATAGCGGCTGGATCGACTACGCCGACA GCGTGAAGGGCAGATTACCATCAGCCGGGACAACAGCAAGAACAC CCTGTACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCGTGT ACTACTGTGCCAGAAGCGGCTACAGCAGCTCTTGGTTTGACCCCGAC TTCGACTATTGGGGCCAGGGCACACTGGTCACAGTCTCTTCA
470	DNA	AGCTACGAGATGAAC
471	DNA	GGCATCAGCTGGAATAGCGGCTGGATCGACTACGCCGACAGCGTGA AGGGC
472	DNA	AGCGGCTACAGCAGCTCTTGGTTTGACCCCGACTTCGACTAT
473	DNA	CAGTCTGTTCTGACACAGCCTCCATCTGTGTCTGGCGCCCTGGACAG AGAGTGACCATCAGCTGTACAGGCAGCAGCTCCAATATCGGAGCCGG CTATGACGTGCACTGGTATCAGCAGCTGCCTGGCACAGCCCCATAAC TGCTGATCTACGGCAACAGCAACAGACCCAGCGGCGTGCCCGATAGA TTTTCCGGCTCTAAGAGCGGCACAAGCGCCAGCCTGGCTATTACTGG ACTGCAGGCCGAGGACGAGGCCGACTACTACTGTCTAGTCTTACGCTG GCCCAATCCTTACGTGGTGTGTTGGCGGCGGAACAAGCTGACCGTT CTA
474	DNA	ACAGGCAGCAGCTCCAATATCGGAGCCGGCTATGACGTGCAC
475	DNA	GGCAACAGCAACAGACCCAGC

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
476	DNA	CAGTCTTACGCTGGCCCCAATCCTTACGTGGTG
477	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFDSYEMNWVRQAPGKGLEWV SGISWNSGWIDYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYC ARSGYSSSWFDPDFDYWGQGLVTVSSASTKGPSVFPLAPSSKSTSGGT AALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTV PSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPELLGGP SVFLPPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNA KTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKT ISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNG QPENNYKTTTPVLDSDGSFPLYSKLTVDKSRWQQGNVFCSCVMHEALH NHYTQKSLSLSPGK
478	PRT	QSVLTQPPSVSGAPGQRVTISCTGSSSNIGAGYDVHWYQQLPGTAPKLLI YGNNSNRPSGVPDRFSGSKSGTSASLAITGLQAEADYYCQSYAGPNPY VVFGGGTKLTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAV TVAWKADSSPVKAGVETTTPSKQSNKYAASSYLSLTPEQWKSHRYSYS CQVTHEGSTVEKTVAPTECS
479	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCACCTTCGATAGCT ACGAGATGAACTGGGTCCGACAGGCCCTGGCAAAGGCTTGAATG GGTGTCGCGCATCAGCTGGAATAGCGGCTGGATCGACTACGCCGACA GCGTGAAGGGCAGATTACCATCAGCCGGGACAAACAGCAAGAACAC CCTGTACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCGTGT ACTACTGTGCCAGAAGCGGCTACAGCAGCTCTTGGTTTGACCCCGAC TTCGACTATTGGGGCCAGGGCACACTGGTCACAGTCTCTTCAGCCAG CACCAGGGCCCCAGCGTGTCCCTCTGGCCCCAGCAGCAAGAGCA CATCTGGCGGAACAGCCGCCCTGGGCTGCCTCGTGAAGGACTACTTT CCCCAGCCCCGTGACCGTGTCTTGAACCTCTGGCGCTCTGACAAGCGG CGTGACACACCTTTCCAGCCGTGCTGCAGAGCAGCGGCTGTACTCTCT GAGCAGCGTCTGACAGTGCCCGAGCAGCTCTCTGGGCACCCAGACCT ACATCTGCAACGTGAACCAAGCCAGCAACACCAAGGTGGACAA GAAGGTGGAACCAAGAGCTGCGACAAGACCCACACCTGTCCCCCTT GTCCTGCCCCCCAACTGCTGGGAGGCCCTTCCTGTCTCTGTTCCCCC CAAAGCCCAAGGACACCCGTGATGATCAGCCGAGCCCCGAAGTGAC CTGCGTGGTGGTGGATGTGTCCACGAGGACCTGAAGTGAAGTTCA ATTGGTACGTGGACGGCGTGAAGTGACAACGCCAAGACCAAGCC TAGAGAGGAACAGTACAACAGCACCTACCGGGTGGTGTCCGTGCTGA CAGTGTGTCACCGGACTGGCTGAACGGCAAGAGTACAAGTGCAA GGTGTCCAACAAGGCCCTGCCTGCCCCATCGAGAAAACCATCAGCA AGGCCAAGGGCCAGCCCCGCAACCCAGGTGTACACACTGCCCCCA AGCAGGGACGAGCTGACCAAGAACCAGGTGTCCCTGACCTGTCTCGT GAAAGGCTTCTACCCCTCCGATATCGCCGTGGAATGGGAGAGCAACG GCCAGCCCCGAGAACTACAAGACACCCCCCTGTGCTGGACAGC GACGGCTCATTCTTCTGTACAGCAAGCTGACCGTGGACAAGTCCCG GTGGCAGCAGGGCAACGTGTTCAGCTGCAGCGTGTGACGAGGCC TGCAACAACCTACACCCAGAAGTCCCTGAGCCTGAGCCCTGGCAAG
480	DNA	CAGTCTGTTCTGACACAGCCTCCATCTGTGTCTGGCGCCCCCTGGACAG AGAGTGACCATCAGCTGTACAGGCAGCAGCTCCAATATCGGAGCCGG CTATGACGTGCACTGGTATCAGCAGCTGCCTGGCACAGCCCCATAAC TGCTGATCTACGGCAACAGCAACAGACCCAGCGGCGTGCCCGATAGA TTTTCCGGCTCTAAGAGCGGCACAAGCGCCAGCCTGGCTATTACTGG ACTGCAGGCCGAGGACGAGGCCGACTACTGTCTAGTCTTACGCTG GCCCAATCCTTACGTGGTGTGTTGGCGGCGGAACAAGCTGACCGTT CTAGGCCAGCCTAAAGCCGCCCTAGCGTGACCTGTTCCTCCAAG CAGCGAGGAACCTGCAGGCCAACAAGGCCACCTCGTGTGCTGTATCA GCGACTTCTATCTTGGCGCGCTGACCGTGGCTGGAAGGCCGATAGC TCTCCTGTGAAGCCGGCGTGGAAACACACCCCTAGCAAGCAGAG CAACAACAAATACGCCGCCAGCAGCTACCTGAGCCTGACCCCGAGC AGTGGAAAGTCCACAGATCTTACAGCTGCCAAGTGACCCACGAGGGC AGCACCGTGGAAAAGACAGTGGCCCCCTACCGAGTGCAGC
481	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFDSYEMNWVRQAPGKGLEW VSGISWNSGWIDYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYY CARSGYSSSWFDPDFDYWGQGLVTVSS
482	PRT	SYEMN
483	PRT	GISWNSGWIDYADSVKG

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
484	PRT	SGYSSSWFDPDFDY
485	PRT	QSVLTQPPSVSGAPGQRVTI SCTGSSSNIGAGYDVHWYQQLPGTAPKLLI YGNNSNRPSGVPDRFSGSKSGTSASLAITGLQAEDEADYYCQSYAGPNPY VVFGGGTKLTVL
486	PRT	TGSSSNIGAGYDVH
487	PRT	GNSNRPS
488	PRT	QSYAGPNPYVV
489	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCGACTTCGATAGCT ACGAGATGAACTGGGTCCGACAGGCCCTTGCCAAAGGCCTTGAATG GGTGTCGGCATCAGCTGGAATAGCGGCTGGATCGACTACGCCGACA CGGTGAAGGGCAGATTACCATCAGCCGGGACAAACAGCAAGAACAC CCTGTACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCGTGT ACTACTGTGCCAGAAGCGGCTACAGCAGCTCTTGGTTTGACCCGAC TTCGACTATTGGGGCCAGGGCACACTGGTCACAGTCTCTTCA
490	DNA	AGCTACGAGATGAAC
491	DNA	GGCATCAGCTGGAATAGCGGCTGGATCGACTACGCCGACAGCGTGA AGGGC
492	DNA	AGCGGCTACAGCAGCTCTTGGTTTGACCCGACTTCGACTAT
493	DNA	CAGTCTGTTCTGACACAGCCTCCATCTGTGTCTGGCGCCCTGGACAG AGAGTGACCATCAGCTGTACAGGCAGCAGCTCCAATATCGAGCGCG CTATGACGTGCACTGGTATCAGCAGCTGCCTGGCACAGCCCTAAAC TGCTGATCTACGGCAACAGCAACAGACCCAGCGGCGTGCCCGATAGA TTTTCCGGCTCTAAGAGCGGCACAAGCGCAGCCTGGCTATTACTGG ACTGCAGGCCGAGGACGAGCGGCACTACTACTGTCTAGTCTTACGCTG GCCCAATCCTTACGTGGTGTGGCGGCGGAACAAAGCTGACCGTT CTA
494	DNA	ACAGGCAGCAGCTCCAATATCGGAGCCGGCTATGACGTGCAC
495	DNA	GGCAACAGCAACAGACCCAGC
496	DNA	CAGTCTTACGCTGGCCCCAATCCTTACGTGGTG
497	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFDSDSYEMNWVRQAPGKGLEW VSGISWNSGWIDYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYY CARSGYSSSWFDPDFDYWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGG TAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVT VPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPELLGG PSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHN AKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEK TISKAKQPREPVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESN GQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMEAL HNHYTQKSLSLSPGK
498	PRT	QSVLTQPPSVSGAPGQRVTI SCTGSSSNIGAGYDVHWYQQLPGTAPKLLI YGNNSNRPSGVPDRFSGSKSGTSASLAITGLQAEDEADYYCQSYAGPNPY VVFGGGTKLTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAV TVAWKADSSPVKAGVETTTPSKQSNNKYAASSYLSLTPEQWKSHRYS CQVTHEGSTVEKTVAPTECS
499	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCGACTTCGATAGCT ACGAGATGAACTGGGTCCGACAGGCCCTTGCCAAAGGCCTTGAATG GGTGTCGGCATCAGCTGGAATAGCGGCTGGATCGACTACGCCGACA CGGTGAAGGGCAGATTACCATCAGCCGGGACAAACAGCAAGAACAC CCTGTACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCGTGT ACTACTGTGCCAGAAGCGGCTACAGCAGCTCTTGGTTTGACCCGAC TTCGACTATTGGGGCCAGGGCACACTGGTCACAGTCTCTTCAGCCAG CACCAGGGCCCCAGCGTGTTCCTCTGGCCCTAGCAGCAAGAGCA CATCTGGCGGAACAGCCGCCCTGGGCTGCCTCGTGAAGGACTACTTT

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
		CCGGAGCCCGTGACCGTGTCTGGAACCTCTGGCGCTCTGACAAGCGG CGTGCACACCTTTCCAGCCGTGTGCAGAGCAGCGCCTGTACTCTCT GAGCAGCGTCGTGACAGTGCCAGCAGCTCTCTGGGCACCCAGACCT ACATCTGCAACGTGAACCAAGCCAGCAACACCAAGGTGGACAA GAAGGTGGAACCCAAGAGCTGCGACAAGACCCACACCTGTCCCCCTT GTCCTGCCCCGAACTGCTGGGAGGCCCTTCCGTGTTCTGTTCCTCC CAAAGCCCAAGGACACCTGTATGATCAGCCGAGCCCCGAAGTGAC CTGCGTGGTGGTGGATGTGCCACGAGGACCTGAAGTGAAGTTCA ATTGGTACGTGGACGGCGTGAAGTGCACAACGCCAAGACCAAGCC TAGAGAGGAACAGTACAACAGCACCTACCGGTGGTGTCCGTGCTGA CAGTGCTGCACAGGACTGGCTGAACGGCAAGAGTACAAGTGCAA GGTGTCCAACAAGGCCCTGCCTGCCCCATCGAGAAAACCATCAGCA AGGCCAAGGGCCAGCCCCGCAACCCAGGTGTACACACTGCCCCCA AGCAGGGACGAGCTGACCAAGAACCAGGTGTCCCTGACCTGTCTCGT GAAAGGCTTCTACCCCTCCGATATCGCCGTGGAATGGGAGAGCAACG GCCAGCCCGAGAACTACAAGACCACCCCTGTGCTGGACAGC GACGGCTCATTTCTCTGTACAGCAAGCTGACCGTGGACAAGTCCCG GTGGCAGCAGGGCAACGTGTTAGCTGCAGCGTGTGACGAGGCC TGCACAACCACTACACCCAGAAGTCCCTGAGCCTGAGCCCTGGCAAG
500	DNA	CAGTCTGTTCTGACACAGCCTCCATCTGTGTCTGGCGCCCTGGACAG AGAGTGACCATCAGCTGTACAGGCAGCAGCTCCAATATCGAGGCCGG CTATGACGTGCACTGGTATCAGCAGCTGCCTGGCAGAGCCCTAAAC TGCTGATCTACGGCAACAGCAACAGACCCAGCGGCGTGCCCGATAGA TTTTCCGGCTCTAAGAGCGGCACAAGCGCCAGCCTGGCTATTACTGG ACTGCAGGCCGAGGACGAGGCCGACTACTACTGTCAGTCTTACGCTG GCCCAATCCTTACGTGGTGTGTTGGCGGCGGAACAAAGCTGACCGTT CTAGGCCAGCCTAAAGCCGCCCTAGCGTGACCCTGTTCCTCCAAG CAGCGAGGAACCTGACGGCCAAACAGGCCACCCTCGTGTGCTGATCA GCGACTTCTATCTGGCGCCGTGACCGTGGCTGGAAGGCCGATAGC TCTCCTGTGAAGCGCGCGTGGAAACCACACCCCTAGCAAGCAGAG CAACAACAAATACGCCGCCAGCAGCTACCTGAGCCTGACCCCGAGC AGTGAAGTCCCACAGATCTACAGCTGCCAAGTGACCCACGAGGGC AGCACCGTGGAAAGACAGTGGCCCTACCGAGTGACG
501	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFDSYEMNWVRQAPGKLEWV SGISWNSGWIDYADSVKGRFTISRDNKNTLYLQMNSLR AEDTAVYYC ARSGYSSSWFDPDFDYWGQTLTVSS
502	PRT	SYEMN
503	PRT	GISWNSGWIDYADSVKG
504	PRT	SGYSSSWFDPDFDY
505	PRT	QSVLTQPPSVSGAPGQRTISCTGSSSDIGAGYDVHWYQQLPGTAPKLLI YGNSNRPSGVDPDRFSGSKGTSASLAITGLQAEDEADYYCQSYAGINPY VVFGGQTKLTVL
506	PRT	TGSSSDIGAGYDVH
507	PRT	GNSNRPS
508	PRT	QSYAGINPYVV
509	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCACCTTCGATAGCT ACGAGATGAACCTGGGTCCGACAGGCCCTGGCAAAGGCTTGAATG GGTGTCCGGCATCAGCTGGAATAGCGGCTGGATCGACTACGCCGACA CGGTGAAGGGCAGATTACCATCAGCCGGGACAACAGCAAGAACAC CCGTACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCGCTGT ACTACTGTGCCAGAAGCGGCTACAGCAGCTCTTGGTTGACCCCGAC TTCGACTATTGGGGCCAGGGCACACTGGTCACAGTCTCTTCA
510	DNA	AGCTACGAGATGAAC
511	DNA	GGCATCAGCTGGAATAGCGGCTGGATCGACTACGCCGACAGCGTGAAGGCG
512	DNA	AGCGGCTACAGCAGCTCTTGGTTTGACCCGACTTCGACTAT

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
513	DNA	CAGTCTGTTCTGACACAGCCTCCATCTGTGTCTGGCGCCCTGGACAG AGAGTGACCATCAGCTGTACAGGCAGCAGCTCCGATATTGGCGCCGG ATACGACGTGCACTGGTATCAGCAACTGCCTGGCACAGCCCTAAGC TGCTGATCTACGGCAACAGCAACAGACCTAGCGGCGTGCCCGATAGA TTCAGCGGCTCTAAGTCTGGCACAAGCGCCAGCCTGGCCATTACTGG ACTGCAGGCCGAAGATGAGGCCGACTACTACTGTCAGAGCTACGCCG GCATCAACCCCTACGTGGTGTGTTGGCGGAGGCCACCAAGCTGACAGTT CTA
514	DNA	ACAGGCAGCAGCTCCGATATTGGCGCCGGATACGACGTGCAC
515	DNA	GGCAACAGCAACAGACCTAGC
516	DNA	CAGAGCTACGCCGGCATCAACCCCTACGTGGTG
517	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFDSYEMNWVRQAPGKLEWV SGISWNSGWIDYADSVKGRFTISRDN SKNTLYLQMNSLR AEDTAVYYC ARSGYSSSWFDPDFDWGQGT LTVVSSASTKGPSVFPLAPSSKSTSGGT AALGCLVKDYFPEPVT VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTV PSSSLGTQTYI CNVNHKPSNTKVDKKVEPKSCDKHTCTPPCPAPPELLGGP SVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNA KTKPREEQYNSTYRVVSVLTVLHQDNLNGKEYKCKVSNKALPAPIEKT I SKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNG QPENNYKTTTPPVLDSDGSFPLYSKLTVDKSRWQQGNV FSCSVMEALH NHYTQKSLSLSPGK
518	PRT	QSVLTQPPSVSGAPGQRVTISCTGSSSDIGAGYDVHWYQQLPGTAPKLLI YGNSNRPSGVPDRFSGSKSGTSASLAITGLQAEDEADYYCQSYAGINPY VVGFGGTKLTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAV TVAWKADSSPVKAGVETTTPSKQSNKNKYAASSYLSLTPEQWKSHRYSYS CQVTHEGSTVEKTVAPTECS
519	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCACCTTCGATAGCT ACGAGATGAACCTGGGTCGACAGGCCCTGGCAAAGGCCCTGAATG GGTGTCCGGCATCAGCTGGAATAGCGGCTGGATCGACTACGCCGACA CGGTGAAGGCGAGATTACCATCAGCCGGGACAAACAGCAAGAACAC CCTGTACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCGTGT ACTACTGTGCCAGAAGCGGCTACAGCAGCTCTTGGTTTGACCCCGAC TTCGACTATTGGGGCCAGGGCACACTGGTCACAGTCTCTTCAGCCAG CACCAAGGGCCCCAGCGTGTTCCTCTGGGCCCTAGCAGCAAGAGCA CATCTGGCGGAACAGCCGCCCTGGGCTGCCTCGTGAAGGACTACTTT CCCGAGCCCGTGACCGTGTCTGGAACCTTGCGGCTCTGACAAGCGG CGTGACACACCTTTCAGCCGTGTGTCAGAGCAGCGGCTGTACTCTCT GAGCAGCGTGTGACAGTGCCAGCAGCTCTCTGGGCACCCAGACCT ACATCTGCAACGTGAACCAAGCCAGCAACACCAAGGTGGACAA GAAGGTGGAACCCAAGAGCTGCACAGACCCACACCTGTCCCCCTT GTCCTGCCCCGAACTGCTGGGAGGCCCTTCCGTGTTCTGTTCCTCCC CAAAGCCCAAGGACACCTTGATGATCAGCCGGACCCCGAAGTGAC CTGCGTGGTGGTGGATGTGTCCACAGAGACCTGAAGTGAAGTTCA ATTGGTACGTGGACGGCGTGAAGTGCACAACGCCAAGACCAAGCC TAGAGAGGAACAGTACAACAGCACCTACCGGTGGTGTCCGTGCTGA CAGTGCTGCACAGGACTGGCTGAACGGCAAGAGTACAAGTGCAA GGTGTCCAACAAGGCCCTGCCTGCCCTATCGAGAAAACCATCAGCA AGGCCAAGGGCCAGCCCCGCAACCCAGGTGTACACACTGCCCCCA AGCAGGGACGAGCTGACCAAGAACCAGGTGTCCCTGACCTGTCTCGT GAAAGGCTTCTACCCCTCCGATATCGCCGTGGAATGGGAGAGCAACG CCAGCCCGAGAACTACAAGACCAACCCCTGTGCTGGACAGC GACGGCTCATTTCTTCTGTACAGCAAGCTGACCGTGGACAAGTCCCG GTGGCAGCAGGGCAACGTGTTAGCTGCAGCGTGTATGCACGAGGCC TGACAAACCACTACACCCAGAAGTCCCTGAGCCTGAGCCCTGGCAAG
520	DNA	CAGTCTGTTCTGACACAGCCTCCATCTGTGTCTGGCGCCCTGGACAG AGAGTGACCATCAGCTGTACAGGCAGCAGCTCCGATATTGGCGCCGG ATACGACGTGCACTGGTATCAGCAACTGCCTGGCACAGCCCTAAGC TGCTGATCTACGGCAACAGCAACAGACCTAGCGGCGTGCCCGATAGA TTCAGCGGCTCTAAGTCTGGCACAAGCGCCAGCCTGGCCATTACTGG ACTGCAGGCCGAAGATGAGGCCGACTACTACTGTCAGAGCTACGCCG GCATCAACCCCTACGTGGTGTGTTGGCGGAGGCCACCAAGCTGACAGTT CTAGGCCAGCCTAAAGCCGCCCTAGCGTGACCTGTTCCTCCAAG CAGCGAGGAAGTGCAGGCCAACAGGCCACCCCTCGTGTGCTGATCA

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
		GCGACTTCTATCCTGGCGCCGTGACCGTGGCCTGGAAGGCCGATAGC TCTCCTGTGAAGCCGCGCTGGAACACCACCCCTAGCAAGCAGAG CAACAACAAATACGCCGCCAGCAGCTACCTGAGCCTGACCCCGAGC AGTGGAAAGTCCACAGATCCTACAGCTGCCAAGTGACCCACGAGGGC AGCACCGTGAAAAGACAGTGGCCCTACCGAGTGCAGC
521	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFDFDSYEMNWVRQAPGKGLEW VSGISWNSGWIDYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYY CARSGYSSSWFDPDFDYWGQGLTVTVSS
522	PRT	SYEMN
523	PRT	GISWNSGWIDYADSVKG
524	PRT	SGYSSSWFDPDFDY
525	PRT	QSVLTQPPSVSGAPGQRVTISCTGSSSDIGAGYDVHWYQQLPGTAPKLLI YGNSNRPSGVPDRFSGSKSGTSASLAITGLQAEDEADYYCSSYEGINPYV VFGGGTKLTVL
526	PRT	TGSSSDIGAGYDVH
527	PRT	GNSNRPS
528	PRT	SSYEGINPYV
529	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCGACTTCGATAGCT ACGAGATGAAGTGGGTCCGACAGGCCCTTGGCAAAGCCCTTGAATG GGTGTCCGGCATCAGCTGGAATAGCGGCTGGATCGACTACGCCGACA GCGTGAAGGGCAGATTCAACCATCAGCCGGGACAAACAGCAAGAACAC CCTGTACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCGTGT ACTACTGTGCCAGAAGCGGCTACAGCAGCTCTTGGTTTGACCCCGAC TTCGACTATTGGGGCCAGGGCACACTGGTCACAGTCTCTTCA
530	DNA	AGCTACGAGATGAAC
531	DNA	GGCATCAGCTGGAATAGCGGCTGGATCGACTACGCCGACAGCGTGA AGGGC
532	DNA	AGCGGCTACAGCAGCTCTTGGTTTGACCCCGACTTCGACTAT
533	DNA	CAGTCTGTTCTGACACAGCCTCCATCTGTGTCTGGCGCCCTTGACAG AGAGTGACCATCAGCTGTACAGGCAGCAGCTCCGATATTGGCGCCG ATACGACGTGCACTGGTATCAGCAACTGCCTGGCAGACCCCTAAGC TGCTGATCTACGGCAACAGCAACAGACCTAGCGGCGTGCCGATAGA TTCAGCGGCTCTAAGTCTGGCAAGCGCCAGCCTGGCCATTACTGG ACTGACGGCCGAAGATGAGGCCGACTACTACTGCAGCAGCTACGAG GGCATCAACCCCTACGTGGTGTCTGGCGGCGGAACAAAGCTGACCGT TCTA
534	DNA	ACAGGCAGCAGCTCCGATATTGGCGCCGATACGACGTGCAC
535	DNA	GGCAACAGCAACAGACCTAGC
536	DNA	AGCAGCTACGAGGCATCAACCCCTACGTGGTG
537	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFDFDSYEMNWVRQAPGKGLEW VSGISWNSGWIDYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYY CARSGYSSSWFDPDFDYWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGG TAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVT VPSSSLGTQTYICNVNHKPSNTKVDKKEPKSCDKHTHTCPPCPAPELLGG PSVFLFPPPKDKTLKISRTPEVTCVVDVSHEDPEVKFNWYVDGVEVHN AKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEK TISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESN GQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEAL HNHYTQKSLSLSPGK
538	PRT	QSVLTQPPSVSGAPGQRVTISCTGSSSDIGAGYDVHWYQQLPGTAPKLLI YGNSNRPSGVPDRFSGSKSGTSASLAITGLQAEDEADYYCSSYEGINPYV VFGGGTKLTVLQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVT

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
		VANKADSSPVKAGVETTPSKQSNKNYAASSYLSLTPQWKSHRYSYC QVTHEGSTVEKTVAPTECS
539	DNA	GAAGTTCAGCTGCTGGAATCTGGCGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCGACTTCGATAGCT ACGAGATGAACTGGGTCCGACAGGCCCTTGGCAAAGGCCTTGAATG GGTGTCCGGCATCAGCTGGAATAGCGGCTGGATCGACTACGCCGACA GCGTGAAGGGCAGATTACCATCAGCCGGGACAAACAGCAAGAACAC CCTGTACTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCGTGT ACTACTGTGCCAGAAGCGGCTACAGCAGCTCTTGGTTTGACCCGAC TTCGACTATTGGGGCCAGGGCACACTGGTCACAGTCTCTTCAGCCAG CACCAAGGGCCCCAGCGTGTTCCTCTGGCCCCCTAGCAGCAAGAGCA CATCTGGCGGAACAGCCGCCCTGGGCTGCCTCGTGAAGGACTACTTT CCCGAGCCCCGTGACCGTGTCTCTGGAACCTCTGGCGCTCTGACAAGCGG CGTGACACACCTTTCAGCCGTGTCTGCAGAGCAGCGCCCTGTACTCTCT GAGCAGCGTCGTGACAGTGCCCGCAGCAGCTCTCTGGGCACCCAGACCT ACATCTGCAACGTGAACCACAAGCCAGCAACACCAAGGTGGACAA GAAGGTGGAACCAAGAGCTGCGACAAGACCCACACCTGTCCCCCTT GTCTTGCCCCCGAAGTGTCTGGAGGGCCCTTCGTGTTCTGTTCCTT CAAAGCCCAAGGACACCCCTGATGATCAGCCGAGCCCCCGAAGTGAC CTGCGTGGTGGTGGATGTGTCCACGAGGACCCCTGAAGTGAAGTTCA ATTGGTACGTGGACGGCGTGGAAAGTGCAACGCCAAGACCAAGCC TAGAGAGGAACAGTACAACAGCACCTACCGGGTGGTGTCTGCTGTA CAGTGTGTCACAGGACTGGCTGAACGGCAAGAGTACAAGTGCAA GGTGTCCAACAAGGCCCTGCCTGCCCCATCGAGAAAACCATCAGCA AGGCCAAGGGCCAGCCCCGCAACCCAGGTGTACACACTGCCCCCA AGCAGGGACGAGCTGACCAAGAACCAGGTGTCCCTGACCTGTCTCGT GAAAGGCTTCTACCCCTCCGATATCGCCGTGGAATGGGAGAGCAACG GCCAGCCCCGAGAACTACAAGACCACCCCTGTGTGGACAGC GACGGCTCATTCTTCTGTACAGCAAGCTGACCGTGGACAAGTCCCG GTGGCAGCAGGGCAACGTGTTACGCTGCAGCGTGATGCACGAGGCC TGCACAACCACTACACCCAGAAGTCCCTGAGCCTGAGCCCTGGCAAG
540	DNA	CAGTCTGTTCTGACACAGCCTCCATCTGTGTCTGGCGCCCCCTGGACAG AGAGTGACCATCAGCTGTACAGGCAGCAGCTCCGATATTGGCGCCCG ATACGACGTGCACTGGTATCAGCAACTGCCTGGCACAGCCCCTAAGC TGCTGATCTACGGCAACAGCAACAGACCTAGCGGCGTGCCTGATAGA TTCAGCGGCTCTAAGTCTGGCACAAGCGCCAGCCTGGCCATTACTGG ACTGCAGGCCGAAGATGAGGCCGACTACTACTGCAGCAGCTACGAG GGCATCAACCCCTACGTGGTGTGTTGGCGCGGAACAAAGCTGACCGT TCTAGGCCAGCCTAAAGCCGCCCTAGCGTGACCCCTGTTCCCTCCAA GCAGCGAGGAACCTGCAGGCCAACAGGCCACCCCTCGTGTGCCTGATC AGCGACTTCTATCCTGGCGCCGTGACCGTGGCCTGGAAGGCCGATAG CTCTCCTGTGAAGGCCGCGCTGGAAACCAACCCCTAGCAAGCAGA GCAACAACAAATACGCCCGCAGCAGCTACCTGAGCCTGACCCCGAG CAGTGAAGTCCACAGATCTACAGCTGCCAAGTGACCCACGAGGG CAGCACCGTGGAAAAGACAGTGGCCCCCTACCGAGTGCAGC
541	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFDSSYEMNWVRQAPGKLEWV SGISWNSGWIDYADSVKGRFTISRDN SKNTLYLQMNSLR AEDTAVYYC ARSGYSSSWFDPDFDYWGQTLTVTSS
542	PRT	SYEMN
543	PRT	GISWNSGWIDYADSVKG
544	PRT	SGYSSSWFDPDFDY
545	PRT	QSVLTQPPSVSGAPGQRTVTSCTGSSSNIGAGYDVHWYQQLPGTAPKLLI YGASNRPSGVPDRFSGSKSGTSASLAITGLQAEDEADYYCSSYEGPNPYV VFGGGTKLTVL
546	PRT	TGSSSNIGAGYDVH
547	PRT	GASNRPS
548	PRT	SSYEGPNPYVV
549	DNA	GAAGTTCAGCTGCTGGAATCTGGCGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCGATTTACAGCAGT ACGAGATGAACTGGGTCCGACAGGCCCTTGGCAAAGGCCTTGAATG

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
		GGTGTCGGCATCAGCTGGAATAGCGGCTGGATCGACTACGCCGACA CGGTGAAGGGCAGATTACCATCAGCCGGGACAACAGCAAGAACAC CCTGTACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCGTGT ACTACTGTGCCAGAAGCGGCTACAGCAGCTCTTGGTTTGACCCCGAC TTCGACTATTGGGGCCAGGGCACACTGGTCACAGTCTCTTCA
550	DNA	AGCTACGAGATGAAC
551	DNA	GGCATCAGCTGGAATAGCGGCTGGATCGACTACGCCGACAGCGTGA AGGGC
552	DNA	AGCGGCTACAGCAGCTCTTGGTTTGACCCCGACTTCGACTAT
553	DNA	CAGTCTGTTCTGACACAGCCTCCATCTGTGTCTGGCGCCCTGGACAG AGAGTGACCATCAGCTGTACAGGCAGCAGCTCCAATATCGGAGCCGG CTATGACGTGCACTGGTATCAGCAGCTGCCTGGCACAGCCCTAAAC TGCTGATCTACGGCGCCAGCAATAGACCTAGCGGCGTGCCCGATAGA TTCAGCGGCTCTAAGTCTGGCACAAGCGCCAGCCTGGCCATTACTGG ACTGCAGGCCGAAGATGAGGCCGACTACTACTGCAGCAGCTACGAG GGCCCCAATCCTTACGTGGTGTGGCGGCGAACAAGCTGACCGT TCTA
554	DNA	ACAGGCAGCAGCTCCAATATCGGAGCCGGCTATGACGTGCAC
555	DNA	GGCGCCAGCAATAGACCTAGC
556	DNA	AGCAGCTACGAGGGCCCCAATCCTTACGTGGTG
557	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFDSSYEMNWVRQAPGKGLEWV SGISWNSGWIDYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYC ARSGYSSSWFDPDFYWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGT AALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTV PSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKHTHTCPPCPAPELLGPP SVFLPPPKPDTLMI SRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNA KTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKT SKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNG QPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCSCVMHEALH NHYTQKSLSLSPGK
558	PRT	QSVLTQPPSVSGAPGQRVTISCTGSSSNIGAGYDVHWYQQLPGTAPKLLI YGASNRPSGVDPDRFSGSKSGTSASLAITGLQAEDEADYYCSSYEGPNPV VFGGGTKLTVLGGPKAASVTLFPPSSEELQANKATLVCLISDFYPGAVT VAWKADSPVKAGVETTPSKQSNKYAASSYLSLTPEQWKSRRSYSC QVTHEGSTVEKTVAPTECS
559	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCGATTTAGCAGCT ACGAGATGAAGTGGGTCCGACAGGCCCTGGCAAAGGCCCTGAATG GGTGTCCGGCATCAGCTGGAATAGCGGCTGGATCGACTACGCCGACA CGGTGAAGGGCAGATTACCATCAGCCGGGACAACAGCAAGAACAC CCTGTACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCGTGT ACTACTGTGCCAGAAGCGGCTACAGCAGCTCTTGGTTTGACCCCGAC TTCGACTATTGGGGCCAGGGCACACTGGTCACAGTCTCTTCAGCCAG CACCAAGGGCCCCAGCGTGTTCCTCTGGCCCCCTAGCAGCAAGAGCA CATCTGGCGGAACAGCCGCTGGGCTGCCTCGTGAAGGACTACTTT CCCGAGCCCGTGACCGTGTCTGGAACCTCTGGCGCTCTGACAAGCGG CGTGACACACCTTTCCAGCCGTGTGTCAGAGCAGCGGCTGACTCTCT GAGCAGCGTCGTGACAGTGCCAGCAGCTCTCTGGGCACCCAGACCT ACATCTGCAACGTGAACCACAAGCCAGCAACACCAAGGTGGACAA GAAGGTGGAACCAAGAGCTGCGACAAGACCCACACCTGTCCCCCTT GTCTGCCCCCGAAGTGTCTGGGAGGCCCTCCGTGTTCTGTTCCTT CAAAGCCCAAGGACACCTGATGATCAGCCGAGCCCCGAAGTGAC CTGCGTGGTGGTGGATGTGTCCACGAGGACCTGAAGTGAAGTTCA ATTGGTACGTGGACGGCGTGAAGTGACACAACGCCAAGCAAGGCC TAGAGAGGAACAGTACAACAGCACCTACCGGGTGGTGTCTGCTGTA CAGTGTGTCACAGGACTGGCTGAACGGCAAAGAGTACAAGTGCAA GGTGTCCAACAAGGCCCTGCCTGCCCCATCGAGAAAACCATCAGCA AGGCCAAGGGCCAGCCCCGCAACCCAGGTGTACACACTGCCCCCA AGCAGGGACGAGCTGACCAAGAACCAGGTGTCCCTGACCTGTCTCGT GAAAGGCTTCTACCCCTCCGATATCGCCGTGGAATGGGAGAGCAACG GCCAGCCCGAGAACAACTACAAGACCACCCCTGTGCTGGACAGC

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
560	DNA	GACGGCTCATTCTTCTGTACAGCAAGCTGACCGTGGACAAGTCCCG GTGGCAGCAGGGCAACGTGTTCAGCTGCAGCGTGATGCACGAGGCC TGCACAACCACTACACCCAGAAGTCCCTGAGCCTGAGCCCTGGCAAG CAGTCTGTTCTGACACAGCCTCCATCTGTGTCTGGCGCCCTGGACAG AGAGTGACCATCAGCTGTACAGGCAGCAGCTCCAATATCGGAGCCGG CTATGACGTGCACTGGTATCAGCAGCTGCCTGGCACAGCCCCATAAC TGCTGATCTACGGCGCCAGCAATAGACCTAGCGGCGTGCCCGATAGA TTCAGCGGCTCTAAGCTGGCACAAGCGCCAGCCTGGCCATTACTGG ACTGCAGGCCGAAGATGAGGCCGACTACTACTGCAGCAGCTACGAG GGCCCCAATCCTTACGTGGTGTGGCGGCGGAACAAAGCTGACCGT TCTAGGCCAGCCTAAAGCCGCCCTAGCGTGACCCCTGTTCCCTCCAA GCAGCGAGGAAGTGCAGGCCAACAGGCCACCCCTCGTGTGCCTGATC AGCGACTTCTATCCTGGCGCCGTGACCGTGGCCTGGAAGGCCGATAG CTCTCTGTGAAGGCCGCGTGGAAACCAACCCCTAGCAAGCAGA GCAACAACAAATACGCCGCCAGCAGCTACCTGAGCCTGACCCCCGAG CAGTGGAAAGTCCACAGATCCTACAGCTGCCAAGTGACCCACGAGGG CAGCACCGTGGAAAAGACAGTGGCCCTACCGAGTGCAGC
561	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFDSYEMNWVRQAPGKGLEWV SGISWNSGWIDYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYC ARSGYSSSWFDPDFDYWGQTLTVSS
562	PRT	SYEMN
563	PRT	GISWNSGWIDYADSVKG
564	PRT	SGYSSSWFDPDFDY
565	PRT	QSVLTQPPSVSGAPGQRTVISCTGSSSNIGAGYDVHWYQQLPGTAPKLLI YGNSNRPSGVPDRFSGSKGTSASLAITGLQAEDEADYYCSSYAGPNPY VVFGGGKTLTVL
566	PRT	TGSSSNIGAGYDVH
567	PRT	GNSNRPS
568	PRT	SSYAGPNPYVV
569	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCACTTCGATAGCT ACGAGATGAAGTGGTCCGACAGGCCCTGGCAAAGGCCCTGAATG GGTGTCCGGCATCAGCTGGAATAGCGGCTGGATCGACTACGCCGACA CGGTGAAGGGCAGATTACCATCAGCCGGGACACAGCAAGAACAC CCTGTACTCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCGTGT ACTACTGTGCCAGAAGCGGCTACAGCAGCTCTTGGTTTGACCCCGAC TTCGACTATTGGGGCCAGGGCACACTGGTCACAGTCTCTTCA
570	DNA	AGCTACGAGATGAAC
571	DNA	GGCATCAGCTGGAATAGCGGCTGGATCGACTACGCCACAGCGTGA AGGGC
572	DNA	AGCGGCTACAGCAGCTCTTGGTTTGACCCGACTTCGACTAT
573	DNA	CAGTCTGTTCTGACACAGCCTCCATCTGTGTCTGGCGCCCTGGACAG AGAGTGACCATCAGCTGTACAGGCAGCAGCTCCAATATCGGAGCCGG CTATGACGTGCACTGGTATCAGCAGCTGCCTGGCACAGCCCCATAAC TGCTGATCTACGGCAACAGCAACAGACCCAGCGGCGTGCCCGATAGA TTTTCCGGCTCTAAGAGCGGCACAAGCGCCAGCCTGGCTATTACTGG ACTGCAGGCCGAGGACGAGGCCGACTACTACTGTAGCTCTTACGCTG GCCCCAATCCTTACGTGGTGTGGCGGCGGAACAAAGCTGACCGTT CTA
574	DNA	ACAGGCAGCAGCTCCAATATCGGAGCCGGCTATGACGTGCAC
575	DNA	GGCAACAGCAACAGACCCAGC

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
576	DNA	AGCTCTTACGCTGGCCCCAATCCTTACGTGGTG
577	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFDSYEMNWVRQAPGKGLEWV SGISWNSGWIDYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYC ARSGYSSSWFDPDFYWGQGLVTVSSASTKGPSVFPLAPSSKSTSGGT AALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTV PSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKHTHTCPPCPAPELLGGP SVFLPPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNA KTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKT ISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNG QPENNYKTTTPVLDSDGSFPLYKLTVDKSRWQQGNVFCSCVMHEALH NHYTQKSLSLSPGK
578	PRT	QSVLTQPPSVSGAPGQRVTISCTGSSSNIGAGYDVHWYQQLPGTAPKLLI YGNNSNRPSGVPDRFSGSKSGTSASLAITGLQAEADYYCSSYAGPNPY VVFGGGTKLTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAV TVAWKADSSPVKAGVETTTPSKQSNKYAASSYLSLTPEQWKSHRYSYS CQVTHEGSTVEKTVAPTECS
579	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCACCTTCGATAGCT ACGAGATGAACTGGGTCCGACAGGCCCTGGCAAAGGCTTGAATG GGTGTCCGGCATCAGCTGGAATAGCGGCTGGATCGACTACGCCGACA GCGTGAAGGGCAGATTACCATCAGCCGGGACAAACAGCAAGAACAC CCTGTACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCGTGT ACTACTGTGCCAGAAGCGGCTACAGCAGCTCTTGGTTTGACCCCGAC TTCGACTATTGGGGCCAGGGCACACTGGTCACAGTCTCTTCAGCCAG CACCAGGGCCCCAGCGTGTTCCTCTGGCCCCAGCAGCAAGAGCA CATCTGGCGGAACAGCCGCCCTGGGCTGCCTCGTGAAGGACTACTTT CCCGAGCCCGTGACCGTGTCTTGAACCTCTGGCGCTCTGACAAGCGG CGTGACACACCTTTCCAGCCGTGCTGCAGAGCAGCGCCTGTACTCTCT GAGCAGCGTCGTGACAGTGCCCGAGCAGCTCTCTGGGCACCCAGACCT ACATCTGCAACGTGAACCACAAGCCCAGCAACACCAAGGTGGACAA GAAGGTGGAACCAAGAGCTGCGACAGACCCACACCTGTCCCCCTT GTCCTGCCCCCCAACTGCTGGGAGGCCCTTCCTGTCTCTGTTCCCCC CAAAGCCCAAGGACACCCGTGATGATCAGCCGAGCCCCGAAGTGAC CTGCGTGGTGGTGGATGTGTCCACGAGGACCCCTGAAGTGAAGTTCA ATTGGTACGTGGACGGCGTGAAGTGACAAACGCCAAGACCAAGCC TAGAGAGGAACAGTACAACAGCACCTACCGGGTGGTGTCCGTGCTGA CAGTGTGTCACCGGACTGGCTGAACGGCAAGAGTACAAGTGCAA GGTGTCCAACAAGGCCCTGCCTGCCCCATCGAGAAAACCATCAGCA AGGCCAAGGGCCAGCCCCGCAACCCAGGTGTACACACTGCCCCCA AGCAGGGACGAGCTGACCAAGAACCAGGTGTCCCTGACCTGTCTCGT GAAAGGCTTCTACCCCTCCGATATCGCCGTGGAATGGGAGAGCAAG GCCAGCCCGGAGAACTACAAGACACCCCCCTGTGCTGGACAGC GACGGCTCATTCTTCTGTACAGCAAGCTGACCGTGGACAAGTCCCG GTGGCAGCAGGGCAACGTGTTCAGCTGCAGCGTGATGCACGAGGCC TGCAACAACCTACACCCAGAAGTCCCTGAGCCTGAGCCCTGGCAAG
580	DNA	CAGTCTGTTCTGACACAGCCTCCATCTGTGTCTGGCGCCCCCTGGACAG AGAGTGACCATCAGCTGTACAGGCAGCAGCTCCAATATCGGAGCCGG CTATGACGTGCACTGGTATCAGCAGCTGCCTGGCACAGCCCCATAAC TGCTGATCTACGGCAACAGCAACAGACCCAGCGGCGTGCCCGATAGA TTTTCCGGCTCTAAGAGCGGCACAAGCGCCAGCCTGGCTATTACTGG ACTGCAGGCCGAGGACGAGGCCGACTACTACTGTAGCTCTTACGCTG GCCCAATCCTTACGTGGTGTGGCGGCGGAACAAGCTGACCCGT CTAGGCCAGCCTAAAGCCGCCCTAGCGTGACCCGTGTCCCTCCAAG CAGCGAGGAACCTGCAGGCCAACAAGGCCACCCCTCGTGTGCTGTATCA GCGACTTCTATCTTGGCGCGCTGACCGTGGCTGGAAGGCCGATAGC TCTCCTGTGAAGCCGGCGTGGAAACCAACCCCTAGCAAGCAGAG CAACAACAAATACGCGCCAGCAGCTACCTGAGCCTGACCCCGAGC AGTGGAAAGTCCACAGATCCTACAGCTGCCAAGTGACCCACGAGGGC AGCACCCGTGGAAAAGACAGTGGCCCCCTACCGAGTGCAGC
581	PRT	HHHHHHKNNVPRKLKSYKEMLESNNVITFNGLANSSSYHTFLLEERSR LYVGAKDHI FSPDLVNIKDFQKIVWPVSYTRRDECKWAGKILKECANF IKVLKAYNQTHLYACGTGAFHP ICTYIEIGHHPEDNIFKLENSHFENGRG KSPYDPKLLTASLLIDGELYSGTAADFMRDFAIFRTLGHHPHPIRTEQHD SRWLNDPKFISAHLISESDNPEDDKVYFFRENAIDGEHSGKATHARIGQI CKNDFGGHRSLVNWKWTFLLKARLICSVPNGIDTHFDELQDVFLMNFK DPKNPVVYGVFTTSSNIFKGSVCMYSMSDVRVRVFLGPYAHRDGPNYQ

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
		WVPYQGRVPYPRPGTCPSKTFGGFDSTKDLRDDVITFARSHPAMYNPVF PMNNRPVIVKTDVNYQFTQIVVDRVDAEDGQYDVMFIGTDVGTVLKVV SIPKETWYDLEEVLLEEMTVFREPTAISAMELSTKQQQLYIGSTAGVAQL PLHRCDIYGKACAECCCLARDPYCAWDGSACSRYFPTAKARTRAQDIRN GDPLTHCSDLHHDNHHGHSPEERI IYGVENSSTFLECSPKSQRALVYWQF QRRNEERKEEIRVDDHIIRTDQGLLLRS LQQKDSGNYLCHAVEHGFQITL LKVTLEVIDTEHLEELLHKDDDDGSGSKTEMSNSMTPSQKVWYRDFMQ LINHPNLNTMDEFCEQVWKRDRKQRRQRP GHTPGNSNKWKHLQENKK GRNRRTHEFERAPRSVDIEGRMDPKSCDKTHTCPPCPAPELLGGPSVFLF PPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKP REEQYNSTYRVVSVLTVLHQDWLNGKEYCKCKVSNKALPAPIEKTISKAK GQPREPQVYITLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPEN NYKTTTPVLDSGDSFFLYSKLTVDKSRWQQGNVFSQVMHEALHNNHYT QKSLSLSPGK
582	PRT	NYQNGKNNVPRKLKLSYKEMLESNNVITFNGLANSSSYHTFLDDEERSRL YVGAKDHIFSFNVLNIDFQKIVWPVSYTRRDECKWAGKDI LKECANFI KVLKAYNQTHLYACGTGAFHPICITYIEIGHHPEDNIFKLENSHFENGRGK SPYDPKLLTASLLIDGELYSGTAADFMGRDFAIFRTLGHHPHPIRTEQHDS RWLNDPKFISAHLISESDNPEDDKVYFFFRENAIDGEHSGKATHARIGQIC KNDFGGHRS LVNKWTTFLKARLICSVPGPNGIDTHFDELQDVFLMNFKD PKNPVYGVFTTSSNIFKGSAVCMYSMSDVRVFLGPYAHRDGPNYQW VPYQGRVPYPRPGTCPSKTFGGFDSTKDLRDDVITFARSHPAMYNPVFP MNNRPVIVKTDVNYQFTQIVVDRVDAEDGQYDVMFIGTDVGTVLKVVS I PKETWYDLEEVLLEEMTVFREPTAISAMELSTKQQQLYIGSTAGVAQLP LHRCDIYGKACAECCCLARDPYCAWDGSACSRYFPTAKARTRAQDIRNG DPLTHCSDGGIEGRMDHHHHHH
583	PRT	NYANGKNNVPRKLKLSYKEMLESNNVITFNGLANSSSYHTFLDDEERSRL YVGAKDHIFSFNVLNIDFQKIVWPVSYTRRDECKWAGKDI LKECANFI KVLKAYNQTHLYACGTGAFHPICITYIEVGHHPEDNIFKLQDSHFENGRG KSPYDPKLLTASLLIDGELYSGTAADFMGRDFAIFRTLGHHPHPIRTEQHD SRWLNDPRFISAHLIPESDNPEDDKVYFFFRENAIDGEHSGKATHARIGQI CKNDFGGHRS LVNKWTTFLKARLICSVPGPNGIDTHFDELQDVFLMNSK DPKNPIVYGVFTTSSNIFKGSAVCMYSMSDVRVFLGPYAHRDGPNYQ WVPYQGRVPYPRPGTCPSKTFGGFDSTKDLRDDVITFARSHPAMYNPVF PINNRPIIMIKTDVNYQFTQIVVDRVDAEDGQYDVMFIGTDVGTVLKVVS VPKETWHDLEEVLLEEMTVFREPTAISAMELSTKQQQLYIGSTAGVAQLP LHRCDIYGKACAECCCLARDPYCAWDGSSCSRYFPTAKARTRAQDIRNG DPLTHCSDGGIEGRMDHHHHHH
584	PRT	NYANGKNNVPRKLKLSYKEMLESNNVITFNGLANSSSYHTFLDDEERSRL YVGAKDHIFSFNVLNIDFQKIVWPVSYTRRDECKWAGKDI LKECANFI KVLKAYNQTHLYACGTGAFHPICITYIEVGHHPEDNIFKLQDSHFENGRG KSPYDPKLLTASLLIDGELYSGTAADFMGRDFAIFRTLGHHPHPIRTEQHD SRWLNDPRFISAHLIPESDNPEDDKVYFFFRENAIDGEHSGKATHARIGQI CKNDFGGHRS LVNKWTTFLKARLICSVPGPNGIDTHFDELQDVFLMNSK DPKNPIVYGVFTTSSNIFKGSAVCMYSMSDVRVFLGPYAHRDGPNYQ WVPYQGRVPYPRPGTCPSKTFGGFDSTKDLRDDVITFARSHPAMYNPVF PINNRPIIMIKTDVNYQFTQIVVDRVDAEDGQYDVMFIGTDVGTVLKVVS VPKETWHDLEEVLLEEMTVFREPTAISAMELSTKQQQLYIGSTAGVAQL PLHRCDIYGKACAECCCLARDPYCAWDGSSCSRYFPTAKARTRAQDIRNG DPLTHCSDGGIEGRMDHHHHHH
585	PRT	NYQNGKNNVPRKLKLSYKEMLESNSVITFNGLANSSSYHTFLDDEERSRL YVGAKDHIFSFNVLNIDFQKIVWPVSYTRRDECKWAGKDI LKECANFI KVLKAYNQTHLYACGTGAFHPICITYIEIGHHPEDNIFKLEDSHFENGRGK SPYDPKLLTASLLIDGELYSGTAADFMGRDFAIFRTLGHHPHPIRTEQHDS RWLNDPRFISAHLIPESDNPEDDKVYFFFRENAIDGEHTGKATHARIGQIC KNDFGGHRS LVNKWTTFLKARLICSVPGPNGIDTHFDELQDVFLMNSK PKNPVYGVFTTSSNIFKGSAVCMYSMSDVRVFLGPYAHRDGPNYQW VPYQGRVPYPRPGTCPSKTFGGFDSTKDLRDDVITFARSHPAMYNPVFP PINNRPIIMIKTDVNYQFTQIVVDRVDAEDGQYDVMFIGTDVGTVLKVVSIP KETWHDLEEVLLEEMTVFREPTAISAMELSTKQHLYAGSPAGLAQLPL QRCAAYGRACAECCCLARDPYCAWDGACACSRYFPAAKARTRAQDIRNG DPLTHCSDGGIEGRMDHHHHHH
586	PRT	NYQNGKNNVPRKLKLSYKEMLESNNVITFNGLANSSSYHTFLDDEERSRL YVGAKDHIFSFNVLNIDFQKIVWPVSYTRRDECKWAGKDI LKECANFI KVLKAYNQTHLYACGTGAFHPICITYIEIGHHPEDNIFKLENSHFENGRGK SPYDPKLLTASLLIDGELYSGTAADFMGRDFAIFRTLGHHPHPIRTEQHDS

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
		RWLNDPRFISAHLIPESDNPEDDKVYFFFRENAIDGEHSGKATHARIGQIC KNDFGGHRSLVNKWTTFCLKARLICSVPGPNGIDTHFDELQDVFLMNFKD PKNPVYGVFTTSSNIFKGSAVCMYSMSDVRVFLGPYAHRDGPNYQW VPYQGRVPYPRPGTCSKTFGGFDSTKDLRDDVITFARSHPAMYNPVFPI NNRPIMIKTDVNYQFTQIVVDRVDAEDGQYDVMFIGTDVGTVLKVVSI KETWHDLEEVLEEMTVFREPTTISAMELSTKQQQLYIGSTAGIAQLPLH RCDIYGKACAECCCLARDPYCAWDGSSCSRYFPTAKARTRAQDIRNGDPL THCSDGGIEGRMDHHHHHH
587	PRT	NYQNGKNNVPRKLKLSYKEMLESNNVITFNGLANSSSYHTFLDDEERSRL YVGAKDHIFSFNLVNIKDFQKIVWPVSYTRRDECKWAGKDILKECANFI KVLKAYNQTHLYACGTGAPHPICITYIEIGHHPEDNIFKLEDSEHFNENGRK SPYDPKLLTASLLIDGELYSGTAADFMGRDFAIFRTLGHHPHPIRTEQHDS RWLNDPRFISAHLIPESDNPEDDKVYFFFRENAIDGEHTGKATHARIGQIC KNDFGGHRSLVNKWTTFCLKARLICSVPGPNGIDTHFDELQDVFLMNSKD PKNPVYGVFTTSSNIFKGSAVCMYSMSDVRVFLGPYAHRDGPNYQW VPYQGRVPYPRPGTCSKTFGGFDSTKDLRDDVITFARSHPAMYNPVFPI NNRPIMIKTDVNYQFTQIVVDRVDAEDGQYDVMFIGTDVGTVLKVVSI KETWHDLEEVLEEMTVFREPTTISAMELSTKQQQLYVGSAGVAQLPL HRCDIYGKACAECCCLARDPYCAWDGSSCSRYFPTAKARTRAQDIRNGD PLTHCSDGGIEGRMDHHHHHH
800	PRT	EVQLVESGGGLVQPQGLSLRLSCAASGFTFSSYYMSWVRQAPGKGLEWV STIIKSGGYAYYPDSVKDRFTISRDNKNTLYLQMSSLRAEDTAVYYCVR GGQGAMDYWGQGTITVTVSS
801	PRT	SYYS
802	PRT	TIKSGGYAYYPDSVKD
803	PRT	GGQGAMDY
804	PRT	EIVLTQSPATLSLSPGERATLSCRASQSIGDYLHWYQKPGQAPRLIKY ASQSIGIPARFSGSGSGTDFTLTITSLPEDEFAVYYCQGYSPYTFGGG TKLEIK
805	PRT	RASQSIGDYLH
806	PRT	YASQSIG
807	PRT	QQGYSPYT
808	DNA	GAAGTGCAGCTGGTGAATCTGGCGGAGGACTGGTTCAACCTGGCGG CTCTCTGAGACTGTCTTGTGCCGCCAGCGGCTTACCTTCAGCAGCTA CTACATGAGCTGGGTCGACAGGCCCTGGCAAAGGACTTGAATGGG GTCCACCATCATCAAGAGCGGCGGCTACGCTACTATCCGACAGC GTGAAGGACCGGTTACCATCTCCAGAGACAACAGCAAGAACCCCT GTACCTGCAGATGAGCAGCTGAGAGCCGAGGATACCGCGTGTACT ACTGTGTAGAGGCGGACAGGGGCCATGGATTATTGGGGCCAGGG AACCACAGTGACCGTGTATCA
809	DNA	GAGATTGTGCTGACACAGTCTCCCGCCACACTGTCTCTTAGCCCTGGC GAAAGAGCCCACTGAGCTGTAGAGCCAGCCAGAGCATCGGCGATT ACCTGCACTGGTATCAGCAGAAGCCTGGACAGGCCCTCGGCTGCTG ATTAAGTACGCCAGCCAGTCCATCAGCGGCATCCCTGCCAGATTTCT GGCAGCGGCTCTGGCACCGATTTACCCCTGACCATCACCAGCTGGA ACCTGAGGACTTCGCCGTGTACTACTGCCAGCAGGGCTACAGCTTCC CCTACACATTTGGCGGAGGCACCAAGCTGGAAATCAAA
810	PRT	EVQLVESGGGLVQPQGLSLRLSCAASGFTFSSYYMSWVRQAPGKGLEWV STIIKSGGYAYYPDSVKDRFTISRDNKNTLYLQMSSLRAEDTAVYYCVR GGQGAMDYWGQGTITVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLV KDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQ TYICNVNHKPSNTKVDKVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPK PKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREE QYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTKAKGQ PREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNY KTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNHYTQK SLSLSPGK

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
811	PRT	EIVLTQSPATLSLSPGERATLSCRASQSIGDYLHWYQQKPGQAPRLLIKY ASQSIGIPARFSGSGSGTDFTLTITSLPEPDAVYYCQQGYSFPYTFGGG TKLEIKRTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVD NALQSGNSQESVTEQDSKDSSTYLSSTLTLSKADYKHKVYACEVTHQG LSSPVTKSFNRGEC
812	DNA	GAAGTGCAGCTGGTGAATCTGGCGGAGGACTGGTTCAACCTGGCGG CTCTCTGAGACTGTCTTGTGCCGCCAGCGGCTTACCTTCAGCAGCTA CTACATGAGCTGGGTCGACAGGCCCCCTGGCAAAGGACTTGAATGGG TGTCACCATCATCAAGAGCGGGCGGTACGCCTACTATCCCGACAGC GTGAAGGACCGGTTACCATCTCCAGAGACAACAGCAAGAACCCCT GTACCTGCAGATGAGCAGCCTGAGAGCCGAGGATACCGCCGTGTACT ACTGTGTTAGAGGCGGACAGGGCGCATGGATTATTGGGGCCAGGG AACCACAGTGACCGTGTATCAGCCAGCACCAGGGCCCCAGCGTGT TCCCTCTGGCCCCCTAGCAGCAAGAGCACATCTGGCGGAACAGCCGCC CTGGGCTGCCTCGTGAAGGACTACTTTCAGAGCCCGTGACCGTGTCT CTGGAACCTCTGGCGCTCTGACAAGCGGCGTGACACCTTTCCAGCCG TGCTGCAGAGCAGCGGCTGTACTCTCTGAGCAGCGTCTGTACAGTG CCAGCAGCTCTCTGGGCACCCAGACTACATCTGCAACGTGAACCA CAAGCCCAGCAACACCAAGGTGGACAAGAGGTGGAACCCAGAGC TGCAGACAAGACCCACACCTGTCCCCCTTGTCTGCCCCGAACCTGCTG GGAGGCCCTTCCGTGTTCTGTTCCTTCCCCCAAAGCCCAAGGACACCT GATGATCAGCCGACCCCCGAAGTGACCTGCGTGGTGGTGGATGTGT CCCACGAGGACCTGAAGTGAAGTTCAATTGGTACGTGGACGGCGTG GAAGTGCACAACGCCAAGACCAAGCCTAGAGAGGAACAGTACAACA GCACCTACCGGGTGGTGTCCGTGCTGACAGTGTGACACAGGACTGG CTGAACGGCAAAGAGTACAAGTGCAAGGTGTCCAACAAGGCCCTGCT TGCCCCCATCGAGAAAACCATCAGCAAGGCCAAGGGCCAGCCCCG CGAACCCAGGTGTACACACTGCCCCAAGCAGGGACGAGCTGACC AAGAACCAGGTGTCCCTGACCTGTCTCGTGAAAGGCTTCTACCCCTC CGATATCGCCGTGGAATGGGAGAGCAACGGCCAGCCCGAGAACAA TACAAGACCACCCCCCTGTGCTGGACAGCGACGGCTCATTCTTCCT GTACAGCAAGCTGACCGTGGACAAGTCCCGGTGGCAGCAGGGCAAC GTGTTCAGCTGCAGCGTGATGCACGAGGCCCTGCACAACCACTACAC CCAGAAGTCCCTGAGCCTGAGCCCTGGCAAG
813	DNA	GAGATTGTGCTGACACAGTCTCCCGCCACACTGTCTCTTAGCCCTGGC GAAAGAGCCACACTGAGCTGTAGAGCCAGCCAGAGCATCGGCGATT ACCTGCACTGGTATCAGCAGAAGCCTGGACAGGCCCTCGGCTGCTG ATTAAGTACGCCAGCCAGTCCATCAGCGGCATCCCTGCCAGATTTCT GGCAGCGGCTCTGGCACCGATTTACCCCTGACCATCACCAGCCTGGA ACCTGAGGACTTCGCCGTGTACTACTGCCAGCAGGGCTACAGCTTCC CTACACATTTGGCGGAGGCACCAAGCTGGAAATCAAACGAACCGTG GCCGCTCCCAGCGTGTTCATCTTCCACCTAGCGACGAGCAGCTGAA GTCCGCGCACAGCCTCTGTCGTGTGCCGTGTAACAACCTTCTACCCCG CGAGGCCAAGGTGCAGTGGAAAGGTGGACAATGCCCTGCAGAGCGGC AACAGCCAGGAAAGCGTGACCGAGCAGGACAGCAAGGACTCCACCT ACAGCCTGAGCAGCACCCCTGACCTGAGCAAGGCCGACTACGAGAA GCACAAGGTGTACGCTGCGAAGTGACCCACCAGGGCCTGTCTAGCC CCGTGACCAAGAGCTTCAACCGGGCGAGTGT
814	PRT	EVQLVESGGGLVQPGGSLRLSCAASGFPFSSYYMSWVRQAPGKLEWV STIIKSGGYAYYPDSVKDRFTISRDNKNTLYLQMSLR AEDTAVYYCVR GGQGAMDYWGQTTVTVSS
815	PRT	SYYS
816	PRT	TIKSGGYAYYPDSVKD
817	PRT	GGQGAMDY
818	PRT	EIVLTQSPATLSLSPGERATLSCRASQSIGDYLHWYQQKPGQAPRLLIKY ASQSIGIPARFSGSGSGTDFTLTITSLPEPDAVYYCQQGYSFPYTFGGG TKLEIK
819	PRT	RASQSIGDYLH
820	PRT	YASQSIG
821	PRT	QQGYSFPYT

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
822	DNA	GAAGTGCAGCTGGTGAATCTGGCGGAGGACTGGTTCAACCTGGCGG CTCTCTGAGACTGTCTTGTGCCGCTCTGGCTTCCCATTGAGCAGCTA CTACATGAGCTGGGTCCGACAGGCCCTGGCAAAGGACTTGAATGGG TGTCACCATCATCAAGAGCGGCGGCTACGCCTACTATCCCGACAGC GTGAAGGACCGGTTACCATCAGCCGGGACAACAGCAAGAACACCC TGTACCTGCAGATGAGCAGCCTGAGAGCCGAGGATACCGCCGTGTAC TACTGTGTTAGAGGCGGACAGGGCGCCATGGATTATTGGGGCCAGGG AACCACAGTGACCGTGTCATCA
823	DNA	GAGATTGTGCTGACACAGTCTCCCGCCACACTGTCTCTTAGCCCTGGC GAAAGAGCCACACTGAGCTGTAGAGCCAGCCAGAGCATCGGCGATT ACCTGCACTGGTATCAGCAGAAGCCTGGACAGGCCCTCGGCTGCTG ATTAAGTACGCCAGCCAGTCCATCAGCGGCATCCCTGCCAGATTTCT GGCAGCGGCTCTGGCACCGATTTCACCTGACCATCACCAGCCTGGA ACCTGAGGACTTCGCCGTGTACTACTGCCAGCAGGGCTACAGCTTCC CCTACACATTTGGCGGAGGCACCAAGCTGGAATCAAA
824	PRT	EVQLVESGGGLVQPGGSLRLSCAASGFPFSSYYMSWVRQAPGKLEWV STIIKSGGYAYYPDVKDRFTIIRDNSKNTLYLQMSSLRAEDTAVYYCVR GGQGAMDYWGQGTIVTVSSASTKGPSVFPLAPSSKSTSGGTAAALGLV KDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQ TYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPK PKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREE QYNSTYRVVSVLTVLHQDNLNGKEYKCKVSNKALPAPIEKTISKAKGQ PREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNY KTTTPVLDSDGSFPLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQK SLSLSPGK
825	PRT	EIVLTQSPATLSLSPGERATLSCRASQSIGDYLHWYQQKPGQAPRLIKY ASQSIGIPARFSGSGSGTDFTLTITISLEPEDFAVYYCQGYSPYTFGGG TKLEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNMFYPREAKVQWKVD NALQSGNSQESVTEQDSKDSSTYLSSTLTLSKADYEKHKVYACEVTHQG LSSPVTKSFNRGEC
826	DNA	GAAGTGCAGCTGGTGAATCTGGCGGAGGACTGGTTCAACCTGGCGG CTCTCTGAGACTGTCTTGTGCCGCTCTGGCTTCCCATTGAGCAGCTA CTACATGAGCTGGGTCCGACAGGCCCTGGCAAAGGACTTGAATGGG TGTCACCATCATCAAGAGCGGCGGCTACGCCTACTATCCCGACAGC GTGAAGGACCGGTTACCATCAGCCGGGACAACAGCAAGAACACCC TGTACCTGCAGATGAGCAGCCTGAGAGCCGAGGATACCGCCGTGTAC TACTGTGTTAGAGGCGGACAGGGCGCCATGGATTATTGGGGCCAGGG AACCACAGTGACCGTGTCATCAGCCAGCACCAAGGGCCCCAGCGTGT TCCCTCTGGCCCCTAGCAGCAAGAGCACATCTGGCGGAACAGCCGCG CTGGGCTGCCTCGTGAAGGACTACTTCCCGAGCCCGTGACCGTGTC CTGGAACCTCTGGCGCTCTGACAAGCGCGCTGCACACCTTTCCAGCCG TGCTGCAGAGCAGCGGCTGTACTCTCTGAGCAGCGCTCGTGACAGTG CCCAGCAGCTCTCTGGGCACCCAGACCTACATCTGCAACGTGAACCA CAAGCCCAGCAACACCAAGTGGACAAGAAGGTGGAACCAAGAGC TGCACAAAGACCCACACCTGTCCCCCTTGTCTGCCCCGAACTGCTG GGAGGCCCTTCCGTGTTCTGTTCCTCCCAAGGCCAAGGACACCCCT GATGATCAGCCGACCCCGAAGTGACCTGCGTGGTGGTGGATGTGT CCACAGGAGCCCTGAAGTGAAGTTCAATTGGTACGTGGACGCGCTG GAAGTGACAACGCCAAGACCAAGCCTAGAGAGGAACAGTACAACA GCACCTACCGGGTGGTGTCCGTGCTGACAGTGTGACACAGGACTGG CTGAACGGCAAGAGTACAAGTGCAAGGTGTCCAACAAGGCCCTGC CTGCCCATCATCGAGAAAACCATCAGCAAGGCCAAGGGCCAGCCCCG CGAACCCAGGTGTACACACTGCCCCCAAGCAGGGACGAGCTGACC AAGAACCAGGTGTCCCTGACCTGTCTCGTGAAGGCTTCTACCCCTC CGATATCGCCGTGGAATGGGAGAGCAACGGCCAGCCCGAGAACAAAC TACAAGACCACCCCTCTGTCTGGACAGCAGCGGCTATTCTTCCT GTACAGCAAGCTGACCGTGGACAAGTCCCGGTGGCAGCAGGGCAAC GTGTTAGCTGCAGCGTGATGCACGAGGCCCTGCACAACCACTACAC CCAGAAGTCCCTGAGCCTGAGCCCTGGCAAG
827	DNA	GAGATTGTGCTGACACAGTCTCCCGCCACACTGTCTCTTAGCCCTGGC GAAAGAGCCACACTGAGCTGTAGAGCCAGCCAGAGCATCGGCGATT ACCTGCACTGGTATCAGCAGAAGCCTGGACAGGCCCTCGGCTGCTG ATTAAGTACGCCAGCCAGTCCATCAGCGGCATCCCTGCCAGATTTCT GGCAGCGGCTCTGGCACCGATTTCACCTGACCATCACCAGCCTGGA ACCTGAGGACTTCGCCGTGTACTACTGCCAGCAGGGCTACAGCTTCC CCTACACATTTGGCGGAGGCACCAAGCTGGAATCAAACGAACCGTG

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
		GCCGCTCCAGCGTGTTCATCTTCCACCTAGCGACGAGCAGCTGAA GTCGGGCACAGCCTCTGTCTGTGCTGCTGAACAACTTCTACCCCG CGAGGCCAAGGTGCAGTGAAGGTGGACAATGCCCTGCAGAGCGGC AACAGCCAGGAAAGCGTGACCGAGCAGGACAGCAAGGACTCCACCT ACAGCCTGAGCAGCACCTGACCTGAGCAAGGCCGACTACGAGAA GCACAAGGTGTACGCTGCGAAGTGACCCACCAGGGCCTGTCTAGCC CCGTGACCAAGAGCTTCAACCGGGCGAGTGT
828	PRT	EVQLVESGGGLVQLGGSRLRLSCAASGFTFSSYYMSWVRQAPGKGLEWV STIIKSGGYAYYPDSVKDRFTISRDNKNTLYLQMNSLRAEDTAVYYCV KGGQGAMDYWGQGTITVTVSS
829	PRT	SYMS
830	PRT	TIKSGGYAYYPDSVKD
831	PRT	GGQGAMDY
832	PRT	EIVLTQSPATLSLSPGERATLSCRASQSIGDYLHWYQQKPGQAPRLIIYY ASQSIGIPARFSGSGSGTDFTLTISLEPEDFAVYYCQQGYSPYTFGGGT KLEIK
833	PRT	RASQSIGDYLH
834	PRT	YASQSIG
835	PRT	QQGYSPYTF
836	DNA	GAAGTGCAGCTGGTGAATCTGGCGGAGGACTGGTTCAGCTCGGCGG ATCTCTGAGACTGTCTTGTGCCGCCAGCGGCTTCACTTCAGCAGCTA CTACATGAGCTGGGTCCGACAGGCCCTGGCAAAGGACTGAATGGG TGTCACCATCATCAAGAGCGGCGGTACGCCTACTATCCCGACAGC GTGAAGGACCGGTTACCATCTCCAGAGACAACAGCAAGAACACCT GTACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCGTGTACT ACTGTGTGAAAGGTGGACAGGGCGCCATGGAATATTGGGGCCAGG AACACAGTGACCGTGTCCTCA
837	DNA	GAGATTGTGCTGACACAGTCTCCCGCCACACTGTCTCTTAGCCCTGGC GAAAGAGCCACACTGAGCTGTAGAGCCAGCCAGAGCATCGGCGATT ACCTGCACTGGTATCAGCAGAAGCCTGGACAGGCCCTCGGCTGCTG ATCTACTATGCCAGCAGTCCATCAGCGGCATCCCGCCAGATTTTCT GGCAGCGGCTCTGGCACCGATTTCACCTGACCATAAGCAGCCTGGA ACCTGAGGACTTCGCGGTGTACTACTGCCAGCAGGGCTACAGCTTCC CCTACACATTTGGCGGAGGCACCAAGCTGGAATCAAA
838	PRT	EVQLVESGGGLVQLGGSRLRLSCAASGFTFSSYYMSWVRQAPGKGLEWV STIIKSGGYAYYPDSVKDRFTISRDNKNTLYLQMNSLRAEDTAVYYCV KGGQGAMDYWGQGTITVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCL VKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGT QTYICNVNHKPSNTKVDKKVEPKSCDKHTCTPPCPAPELGGPSVFLFPP KPKDTLMISRTPEVTVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPRE EQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKG QPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPENN YKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNYHQ KSLSLSPGK
839	PRT	EIVLTQSPATLSLSPGERATLSCRASQSIGDYLHWYQQKPGQAPRLIIYY ASQSIGIPARFSGSGSGTDFTLTISLEPEDFAVYYCQQGYSPYTFGGGT KLEIKRTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVD NALQSGNSQESVTEQDSKSTYSLSSTLTLSKADYEKHKVYACEVTHQG LSSPVTKSFNRGEC
840	DNA	GAAGTGCAGCTGGTGAATCTGGCGGAGGACTGGTTCAGCTCGGCGG ATCTCTGAGACTGTCTTGTGCCGCCAGCGGCTTCACTTCAGCAGCTA CTACATGAGCTGGGTCCGACAGGCCCTGGCAAAGGACTGAATGGG TGTCACCATCATCAAGAGCGGCGGTACGCCTACTATCCCGACAGC GTGAAGGACCGGTTACCATCTCCAGAGACAACAGCAAGAACACCT GTACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCGTGTACT ACTGTGTGAAAGGTGGACAGGGCGCCATGGAATATTGGGGCCAGG AACACAGTGACCGTGTCCTCAGCCAGCACCAAGGGCCCGAGCGTGT TCCCTCTGGCCCTTAGCAGCAAGAGCACATCTGGCGGAACAGCCGCC

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
		CTGGGCTGCCTCGTGAAGGACTACTTTCCTCCGAGCCCGTGACCGTGTCTGGAACTCTGGCGCTCTGACAAGCGCGTGACACCTTTCCAGCCGTGCTGCAGAGCAGCGGCTGTACTCTCTGAGCAGCGTGTGACAGTGCCAGCAGCTCTCTGGGCACCCAGACCTACATCTGCAACGTGAACCA CAAGCCCAGCAACCAAGTGGACAAGAAGGTGGAACCAAGAGCTGCGACAAGACCCACACCTGTCCCCCTTGTCTGCCCCGAACTGCTGGAGGCCCTTCCGTGTCTGTTCCTGTTCCTCCCAAGCCCAAGGACACCCCTGATGATCAGCCGAGCCCCGAAGTGACCTGCGTGGTGGTGGATGTGTCCACGAGGACCTGAAGTGAAGTTCAATTGGTACGTGGACGGCGTGAAGTGACAACGCCAAGACCAAGCCTAGAGAGGAACAGTACAACAGCCTACCGGGTGGTGTCCGTGCTGACAGTGTGACCAAGGACTGGCTGAACGGCAAAGAGTACAAGTGCAAGGTGTCCAACAAGGCCCTGCTGCCCCCATCGAGAAAACCATCAGCAAGGCCAAGGCCAGCCCCGCGAACCCAGGTGTACACACTGCCCCAAGCAGGGACGAGCTGACC AAGAACCAGGTGTCCCTGACCTGTCTCGTGAAAGGCTTCTACCCCTCGATATCGCCGTGGAATGGGAGAGCAACGGCCAGCCCGAGAACAAC TACAAGACCACCCCTGTGTGTCGACAGCGACGGCTATTCTTCTGTACAGCAAGCTGACCGTGGACAAGTCCCGGTGGCAGCAGGGCAAC GTGTTGAGCTGCAGCGTGATGCACGAGGCCCTGCACAACCACTACAC CAGAAGTCCCTGAGCCTGAGCCTGGCAAG
841	DNA	GAGATTGTGCTGACACAGTCTCCCGCCACACTGTCTCTTAGCCCTGGC GAAAGAGCCACACTGAGCTGTAGAGCCAGCCAGAGCATCGGCGATT ACCTGCACTGGTATCAGCAGAAGCCTGGACAGGCCCTCGGCTGCTG ATCTACTATGCCAGCCAGTCCATCAGCGGCATCCCCGCCAGATTTTCT GGCAGCGGCTCTGGCACCGATTTCACCTGACCATAAGCAGCCTGGA ACCTGAGGACTTCGCGGTGTACTACTGCCAGCAGGGCTACAGCTTCC CTTACACATTTGGCGGAGGCACCAAGCTGAAATCAAACGAACCGTG GCCGCTCCAGCGTGTTCATCTTCCACCTAGCGACGAGCAGCTGAA GTCCGGCACAGCCTCTGTCTGTGCTGTGTAACAACCTTCTACCCCG CGAGGCCAAGGTGCAGTGGAAAGTGGACAATGCCCTGCAGAGCGGC AACAGCCAGGAAAGCGTGACCGAGCAGGACAGCAAGGACTCCACCT ACAGCCTGAGCAGCACCTGACCTGAGCAAGGCCGACTACGAGAA GCACAAGGTGTACGCTGCGAAGTGACCCACCAGGGCCTGTCTAGCC CCGTGACCAAGAGCTTCAACCGGGCGAGTGT
842	PRT	EVQLVESGGGLLQLGGSRLRLSCAASGFTFSSYYMSWVRQAPGKLEWV STIIKSGGYAYYPDSVKDRFTISRDNKNTLNLQMNSLRAEDTAVYYCV KGGQGAMDYWGQGTTVTVSS
843	PRT	SYYS
844	PRT	TIIKSGGYAYYPDSVKD
845	PRT	GGQGAMDY
846	PRT	EIVLTQSPATLSLSPGERATLSCRASQSIGDYLHWYQQKPGQAPRLIKY ASQSIGIPARFSGSGSGTDFTLTISLEPEDFAVYYCQQGYSPPYTFGGGT KLEIK
847	PRT	RASQSIGDYLH
848	PRT	YASQSIG
849	PRT	QQGYSPPYT
850	DNA	GAGTGCAGCTGGTGAATCTGGCGGAGGACTGCTGCAGCTTGCGCG ATCTCTGAGACTGTCTTGTGCCCGCAGCGCTTCACTTTCAGCAGCTA CTACATGAGCTGGGTCCGACAGGCCCTGGCAAGGACTTGAATGGG GTCCACCATCATCAAGAGCGGCGCTACGCTTACTATCCCGACAGC GTGAAGGACCGGTTCACCATCTCCAGAGACAACAGCAAGAACACCCCT GAACCTGCAGATGAACGCCTGAGAGCCGAGGACACCGCGTGTAC TACTGTGTGAAAGGTGGACAGGGCGCCATGGACTATTGGGGCCAGG GAACAACAGTGACCGTGTCTCTCA

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
851	DNA	GAGATTGTGCTGACACAGTCTCCCGCCACACTGTCTCTTAGCCCTGGC GAAAGAGCCACACTGAGCTGTAGAGCCAGCCAGAGCATCGGCGATT ACCTGCACTGGTATCAGCAGAAGCCTGGACAGGCCCTCGGCTGTCTG ATTAAGTACGCCAGCCAGTCCATCAGCGGCATCCCTGCCAGATTTTCT GGCAGCGGCTCTGGCACCATTTCACCTGACCATAAGCAGCCTGGA ACCTGAGGACTTCGCGGTGTACTACTGCCAGCAGGGCTACAGCTTCC CCTACACATTGGCGGAGGCACCAAGCTGGAATCAAA
852	PRT	EVQLVESGGGLLQLGGSRLRLSCAASGFTFSSYYMSWVRQAPGKGLEWV STIIKSGGYAYYPDSVKDRFTISRDNKNTLNLMNSLRAEDTAVYYCV KGGQGAMDYWGQGTITVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCL VKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGT QTYICNVNHKPSNTKVDKKVEPKSCDKHTCTPPCPAPELLGGPSVFLFPP KPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPRE EQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKG QPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPENN YKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVPSCSVMHEALHNHYTQ KSLSLSPGK
853	PRT	EIVLTQSPATLSLSPGERATLSCRASQSIGDYLHWYQQKPGQAPRLLIKY ASQSIGIPARFSGSGSGTDFTLTISLEPEDFAVYYCQGYSPPYTFGGGT KLEIKRTVAAPSVFIFPPSDEQLKSGTASVCLLNNFYPREAKVQWKVD NALQSGNSQESVTEQDSKDSSTYSLSSTLTLSKADYEKKHYACEVTHQG LSSPVTKSFNRGEC
854	DNA	GAAGTGCAGCTGGTGAATCTGGCGGAGGACTGCTGCAGCTTGGCGG ATCTCTGAGACTGTCTTGTGCCGCCAGCGGCTTCACCTTCAGCAGCTA CTACATGAGCTGGGTCCGACAGGCCCTTGCCAAAGGACTGAATGGG TGTCACCATCATCAAGAGCGGCGGTACGCCTACTATCCCGACAGC GTGAAGACCGGTTACCATCTCCAGAGACAACAGCAAGAACACCCCT GAACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCGCTGTAC TACTGTGTGAAAGGTGGACAGGGCGCCATGGACTATTGGGGCCAGG GAACAACAGTGACCGTGTCTCAGCCAGCACCAGGGCCCCAGCGTG TTCCCTCTGGCCCCCTAGCAGCAAGAGCACATCTGGCGGAACAGCCGC CCTGGGCTGCCTCGTGAAGGACTACTTCCCGAGCCCGTGACCGTGT CCTGGAACCTCTGGCGCTCTGACAAGCGCGGTGCACACCTTTCAGCC GTGCTGCAGAGCAGCGGCTGTACTCTCTGAGCAGCGTGTGACAGT GCCAGCAGCTCTCTGGGCAACCAGACCTACATCTGCAACGTGAACC ACAAGCCCAGCAACACCAAGGTGGACAAGAAGGTGAACCCAAGAG CTCGCAACAAGACCCACACCTGTCCCCCTTGTCTGCCCCGAACTGT GGGAGGCCCTTCCGTGTCTCTGTTCCTCCCAAGCCCAAGGACACCC TGATGATCAGCCGACCCCCGAAGTGACCTGCGTGGTGGTGGATGTG TCCACAGGAGACCTGAAGTGAAGTTCAATTGGTACGTGGACGGCGT GGAAGTGCAACAAGCCCAAGACCAAGCCTAGAGAGGAACAGTACAAC AGCACCTACCGGCTGGTGTCCGTGTGACAGTGTCTGCACAGGACTG GCTGAACGGCAAAGAGTACAAGTGCAAGGTGTCCAACAAGGCCCTG CCTGCCCCATCGAGAAAACCATCAGCAAGGCCAAGGGCCAGCCCC GCGAACCCAGGTGTACACACTGCCCCCAAGCAGGGACGAGCTGAC CAAGAACCAGGTGTCCCTGACCTGTCTCGTGAAAGGCTTCTACCCCT CCGATATCGCCGTGGAATGGGAGAGCAACGGCCAGCCCGAGAACAA CTACAAGACACCCCCCTGTGCTGGACAGCAGCGCTCATTCTTCCT GTACAGCAAGCTGACCGTGGACAAGTCCCGTGGCAGCAGGGCAAC GTGTTAGCTGCAGCGTGATGCACAGGCCCTGCACAACCACTACAC CCAGAAGTCCCTGAGCTGAGCCCTGGCAAG
855	DNA	GAGATTGTGCTGACACAGTCTCCCGCCACACTGTCTCTTAGCCCTGGC GAAAGAGCCACACTGAGCTGTAGAGCCAGCCAGAGCATCGGCGATT ACCTGCACTGGTATCAGCAGAAGCCTGGACAGGCCCTCGGCTGTCTG ATTAAGTACGCCAGCCAGTCCATCAGCGGCATCCCTGCCAGATTTTCT GGCAGCGGCTCTGGCACCATTTCACCTGACCATAAGCAGCCTGGA ACCTGAGGACTTCGCGGTGTACTACTGCCAGCAGGGCTACAGCTTCC CCTACACATTGGCGGAGGCACCAAGCTGGAATCAAACGAACCGTG GCCGCTCCCAGCGTGTTCATCTTCCACCTAGCGACGAGCAGCTGAA GTCCGGCACAGCCTCTGTCTGTGCTGTGTAACAACCTTCTACCCCCG CGAGGCCAAGGTGCAGTGGAAAGGTGGACAATGCCCTGCAGAGCGGC AACAGCCAGGAAAGCGTGACCGAGCAGGACAGCAAGGACTCCACCT ACAGCCTGAGCAGCACCTTACCTGAGCAAGGCCGACTACGAGAA GCACAAGGTGTACGCTGCGAAGTGACCCACAGGGCCTGTCTAGCC CCGTGACCAAGAGCTTCAACCGGGCGAGTGT

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
856	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFRSYAVHWVRQAPGKGLEWV SSTEGSGVGTSYTDVSKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCA RMLGGGNPLDYLDYWGQGTLLVTVSS
857	PRT	SYAVH
858	PRT	STEGSGVGTSYTDVSKG
859	PRT	MLGGGNPLDYLDY
860	PRT	QSVLTQPPSASGTPGQRVITISCSGSSSNLGEYDVHWYQQLPGKAPKLLI YYSDFRPSGVSDRFSGSKSGTSASLAISGLQSEDEADYYCAWDDSLSSQ VFGGGTQVTVL
861	PRT	SGSSSNLGEYDVH
862	PRT	YSDFRPS
863	PRT	AAWDDSLSSQV
864	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCACCTTTAGAAGCT ATGCCGTGCACTGGGTCCGACAGGCCCTTGAAAAGGACTGGAATG GGTGTCCAGCACCGAAGGCTCTGGCGTGGGCACAAGCTACACCGATT CTGTGAAGGGCAGATTACCATCAGCCGGGACAAACAGCAAGAACAC CCTGTACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCGTGT ACTACTGTGCCAGAATGCTCGGCGGAGGCAACCTCTGGACTACCTG GATTATTGGGGCCAGGGCACCTGGTCACAGTCTCTTCA
865	DNA	CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTAGCGGCAGCAGCTCCAATCTCGGCGAGG GCTATGACGTGCACTGGTATCAGCAGCTGCCTGGCAAGGCCCTTAA CTGCTGATCTACTACAGCGACTTCAGACCCAGCGGCGTGTCCGATAG ATTGAGCGGCTCTAAGAGCGGCACATCTGCCAGCCTGGCCATCTCTG GACTGCAGAGCGAAGATGAGGCCGACTACTATTGCCCGCCTGGGAT GATAGCCTGAGCAGCCAAGTTTTTGGCGGCGGAACCAAGTGACCGT GCTA
866	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFRSYAVHWVRQAPGKGLEWV SSTEGSGVGTSYTDVSKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCA RMLGGGNPLDYLDYWGQGTLLVTVSSASTKGPSVFPLAPSSKSTSGGTA LGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSS SLGTQTYICNVNHKPSNTKVDKKEPKSCDKHTCTPPCPAPELLGGPSVF LFPPKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWYVDGVEVHNAKT KPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISK AKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQP ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFPSCSMHEALHNH YTQKSLSLSPGK
867	PRT	QSVLTQPPSASGTPGQRVITISCSGSSSNLGEYDVHWYQQLPGKAPKLLI YYSDFRPSGVSDRFSGSKSGTSASLAISGLQSEDEADYYCAWDDSLSSQ VFGGGTQVTVLGQPKAAPSVTLPSPSSEELQANKATLVCLISDFYPGAVT VAWKADSPVKAGVETTPSKQSNKYYAASSYLSLTPEQWKSHRSYSC QVTHEGSTVEKTVAPTECS
868	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCACCTTTAGAAGCT ATGCCGTGCACTGGGTCCGACAGGCCCTTGAAAAGGACTGGAATG GGTGTCCAGCACCGAAGGCTCTGGCGTGGGCACAAGCTACACCGATT CTGTGAAGGGCAGATTACCATCAGCCGGGACAAACAGCAAGAACAC CCTGTACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCGTGT ACTACTGTGCCAGAATGCTCGGCGGAGGCAACCTCTGGACTACCTG GATTATTGGGGCCAGGGCACCTGGTCACAGTCTCTTACGCCAGCAC CAAGGGCCCCAGCGTGTTCCTCTGGCCCTAGCAGCAAGAGCACAT CTGGCGGAACAGCCGCCCTGGGCTGCCTCGTGAAGGACTACTTTCCC GAGCCCGTGACCGTGTCCTGGAACCTCTGGCGCTCTGACAAGCGGCGT GCACACCTTTCCAGCGGTGCTGCAGAGCAGCGGCTGTACTCTCTGA GCAGCGTCGTGACAGTGCACAGCAGCTCTCTGGGCACCCAGACCTAC ATCTGCAACGTGAACCAAGCCAGCAACCAAGTGGAACAAGA AGGTGGAACCAAGAGCTGCGACAAGACCAACACCTGTCCCCCTGT CCTGCCCCGAACCTGCTGGGAGGCCCTTCGTGTCTCTGTTCCCCCA

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
		AAGCCCAAGGACACCCCTGATGATCAGCCGGACCCCCGAAGTGACCTG CGTGGTGGTGGATGTGTCCACGAGGACCCCTGAAGTGAAGTTCAATT GGTACGTGGACGGCGTGAAGTGCACAACGCCAAGACCAAGCCTAG AGAGGAACAGTACAACAGCACCTACCGGGTGGTGTCCGTGCTGACA GTGCTGCACCAAGGACTGGCTGAACGGCAAAGAGTACAAGTGCAAGG TGTCACAACAGGCCCTGCCTGCCCCCATCGAGAAAACCATCAGCAAG GCCAAGGGCCAGCCCCGCGAACCCTCAGGTGTACACACTGCCCCAAG CAGGGACGAGCTGACCAAGAACCAGGTGTCCCTGACCTGTCTCGTGA AAGGCTTCTACCCCTCCGATATCGCCGTGGAATGGGAGAGCAACGGC CAGCCCGAGAACAACTACAAGACCACCCCTGTGCTGGACAGCGA CGGCTCATTCTTCTGTACAGCAAGCTGACCGTGGACAGTCCCGGT GGCAGCAGGGCAACGTGTTTCTGCTGACGCTGATGCACGAGGCCCTG CACACCCTACACCCAGAAGTCCCTGAGCCTGAGCCCTGGCAAG
869	DNA	CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTAGCGGCAGCAGCTCCAATCTCGGCGAGG GCTATGACGTGCACTGGTATCAGCAGCTGCCTGGCAAGGCCCTAAA CTGCTGATCTACTACAGCGACTTCAGACCAGCGGCGTGTCCGATAG ATTACGCGGCTCTAAGAGCGGCACATCTGCCAGCCTGGCCATCTCTG GACTGCAGAGCGAAGATGAGGCCGACTACTATTGCGCCGCTGGGAT GATAGCCTGAGCAGCCAAGTTTTTGGCGGCGGAACCAAGTGACCGT GCTAGGCCAGCCTAAAGCCGCCCCCTAGCGTGACCTGTTCCCTCCAA GCAGCGAGGAACTGCAGGCCAACAGGCCACCCCTCGTGTGCCTGATC AGCGACTTCTATCCTGGCGCCGTGACCGTGGCCTGGAAGGCCGATAG CTCTCCTGTGAAGGCCGCGGTGGAACCAACCCCTAGCAAGCAGA GCAACAACAATAACGCCGCCAGCAGCTACCTGAGCCTGACCCCGAG CAGTGGAAAGTCCCACAGATCCTACAGCTGCCAAGTGACCCACGAGGG CAGCACCGTGAAAAGACAGTGGCCCTACCGAGTGCAGC
870	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFRSYAVHWVRQAPGKLEWV SSTEGSGVGTSTYDTSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCA RMLGGGNPLDYLDYWGQGLTVTVSS
871	PRT	SYAVH
872	PRT	STEGSGVGTSTYDTSVK
873	PRT	MLGGGNPLDYLDY
874	PRT	QSVLTQPPSASGTPGQRTTISCSGSSSNLGEYDVHWYQQLPGKAPKLLI YYSDFRPSGVSDRFSKSGTSASLAISGLQSEADYCAAWDDSLSSQ VFGGGTQVTVL
875	PRT	SGSSSNLGEYDVH
876	PRT	YSDFRPS
877	PRT	AAWDDSLSSQV
878	DNA	GAAGTTTACAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCACTTTAGAAAGCT ATGCCGTGCACTGGGTCCGACAGGCCCTTGAAAAGGACTGGAATG GGTGTCCAGCACCGAAGGCTCTGGCGTGGGCACAAGCTACACCGATT CTGTGAAGGGCAGATTACCATCAGCCGGGACAACAGCAAGAACAC CCTGTACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCGTGT ACTACTGTGCCAGAATGCTCGGCGGAGGCAACCCTCTGGACTACCTG GATTATTGGGGCCAGGGCACCCCTGGTCACAGTCTCTTCA
879	DNA	CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTAGCGGCAGCAGCTCCAATCTCGGCGAGG GCTATGACGTGCACTGGTATCAGCAGCTGCCTGGCAAGGCCCTAAA CTGCTGATCTACTACAGCGACTTCAGACCAGCGGCGTGTCCGATAG ATTACGCGGCTCTAAGAGCGGCACATCTGCCAGCCTGGCCATCTCTG GACTGCAGAGCGAAGATGAGGCCGACTACTATTGCGCCGCTGGGAT GATAGCCTGAGCAGCCAAGTTTTTGGCGGCGGAACCAAGTGACCGT GCTA
880	PRT	EVQLLESGGGLVQPGGSLRLSCAASGFTFRSYAVHWVRQAPGKLEWV SSTEGSGVGTSTYDTSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCA RMLGGGNPLDYLDYWGQGLTVTVSSASTKGPSVFPPLAPSKSTSGGTAA LGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSS

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
		SLGTQTYICNVNHNKPSNTKVDKKVEPKSCAAGSEQKLISEEDLSGSAAA HHHHHH
881	PRT	QSVLTQPPSASGTPGQRTVISCSGSSSNLGEYDVHWYQQLPGKAPKLLI YYSDFRPSGVSDRFSKSGTSASLAISGLQSEADYVCAWDDSLSSQ VFGGGTQVTVLGGPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVT VAWKADSSPVKAGVETTPSKQSNKYAASSYLSTPEQWKSHRSYSYSC QVTHEGSTVEKTVAPTECS
882	DNA	GAAGTTCAGCTGCTGGAATCTGGCGGCGGACTGGTTCAACCTGGCGG ATCTCTGAGACTGAGCTGTGCCGCCAGCGGCTTCACCTTTAGAAAGCT ATGCCGTGCACTGGGTCCGACAGGCCCTTGGAAAAGGACTGGAATG GGTGTCCAGCACCGAAGGCTCTGGCGTGGGCACAAGCTACACCGATT CTGTGAAGGGCAGATTACCATCAGCCGGGACAACAGCAGAACAAC CCTGTACCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCGTGT ACTACTGTGCCAGAATGCTCGGCGGAGGCAACCCTCTGGACTACCTG GATTATTGGGGCCAGGGCACCCCTGGTCACAGTCTCTTCAGCCTCCAC CAAGGGCCCATCGGTGTTCCCTTGGCACCTCTCTCAAGAGCACCT CTGGGGGCACAGCGGCCCTGGGCTGCTTGGTCAAGGACTACTTCCCC GAACCGGTGACGGTGTCTGTGAAGTCTAGGCGCCCTGACAGCGGCGT GCACACCTTCCCGGCTGTCTACAGTCTCTCAGGACTCTACTCCCTCAG CAGCGTGGTGACCGTGCCCTCCAGCAGCTTGGGCACCCAGACCTACA TCTGCAACGTGAATCACAAGCCAGCAACACCAAGGTGGACAAGAA AGTTGAGCCCCAAATCTTGTGCAGCGGGTTCTGAACAAAACTCATCT CAGAAGAGGATCTGTCTGGATCAGCGGCCCGCATCATCATCATCAT CAT
883	DNA	CAGTCTGTTCTGACACAGCCTCCTAGCGCCTCTGGCACACCTGGACA GAGAGTGACCATCAGCTGTAGCGGCAGCAGCTCCAATCTCGCGGAGG GCTATGACGTGCACTGGTATCAGCAGCTGCCTGGCAAGGCCCTTAAA CTGCTGATCTACTACAGCGACTTCAGACCCAGCGGCGTGTCCGATAG ATTGAGCGGCTCTAAGAGCGGCACATCTGCCAGCCTGGCCATCTCTG GACTGCAGAGCGAAGATGAGGCCGACTACTATTGCGCCGCTGGGAT GATAGCCTGAGCAGCCAAAGTTTTGGCGGCGGAACCAAGTGACCGT GCTAGGCCAGCCTAAAGCGGCCCTAGCGTGACCTGTTCCCTCCAA GCAGCGAGGAACCTGCAGGCCAACAGGCCACCTCTGTGCTGATC AGCGACTTCTATCTGCGCGCGTGACCGTGGCCTGGAAGGCCGATAG CTCTCTGTGAAGGCCGCGCTGGAAACCAACCCCTAGCAAGCAGA GCAACAACAAATACGCCGCCAGCAGCTACCTGAGCCTGACCCCGGAG CAGTGAAGTCCACAGATCTTACAGCTGCCAAGTGACCCACGAGGG CAGCACCGTGGAAAAGACAGTGGCCCTACCGAGTGCAGC
884	PRT	NYQNGKNNVPRKLKSYKEMLESNNVITFNGLANSSSYHTFLDEERSRL YVGAKDHIIFSNLVNIKDFQKIAPVSYTRRDECKWAGKDI LRECANFI KVLKVYNQTHLYACGTGAFHPICTYVIGIHHPEDNIFKLEDSHFENGRG KSPYDPKLLTASLLIDGELYSGTAADFMRGRFAIFRTLQGHHPIRTEQHD SRWLNDPRFISAHLPESDNPEDDKVYFFRENAIDGEHSGKATHARIGQI CKNDFGHRSLVNKWTFLKARLICSVPNGNIDTHFDELQDVFLMNSK DPKNPIVYGVFTTSSNIFRGSAVCMYSMSDVRVFLGPYAHRDGPNYQ WVPPQGRVPYPRPGTSPSKTFGGFESTKDLDPDDVITFARSHPAMYNPVFP INNRPIMVKTDVNYQFTQIVVDRVDAEDGQYDVMFIGTDVGTVLKVVS I PKETWHDLEEVLLEEMTVFREPTTISAMELSTKQQQLYVGSAGVAQLP LHRCDIYGKACAECC LARDPYCAWDGSSCSRYFPTAKRRTRRQDIRNG DPLTHCSDGGIEGRMDHHHHHH
885	DNA	AACATATCAGAACGGCAAGAACAACGTGCCCCGGCTGAAGCTGAGCT ACAAAGAGATGTGGAAGCAACAACGTGATCACCTTCAACGGCCT GGCCAACAGCAGCAGCTACACACCTTTCTGCTGGACGAGGAACGGT CCAGACTGTACGTGGGAGCCAAAGACACATCTTCAGCTTCAACCTG GTCAACATCAAGGACTTCCAGAAAATCGCCTGGCCTGTGTCTACAC CAGACGGGATGAGTGTAATGGGCGGCAAGGACATCCTGAGAGAG TGCGCCAACTTATCAAGGTGCTGAAGGTGTACAATCAGACCCACCT GTACGCTGTGGCACCGGCGCTTTTACCTATCTGTACCTATGTCTCGG CATCGGCCACCATCTTGAGGACAATATCTTCAAGCTCGAGGACAGCC ACTTCGAGAACGGCAGAGGCAAGAGCCCCCTACGATCCCAAACCTGCTG ACAGCCTCTCTGCTGATCGACGGCGAGCTGTATTCTGGCACAGCCGC CGATTTTCATGGGCAGAGACTTCGCCATCTTCAAGACCTGGGCCAGC ATCACCCATCAGAACCAGCAGCAGCAGCAGCAGATGGCTGAACGA CCCCAGATTATCAGCGCCCATCTGATCCCCGAGAGCGACAACCCCG AGGACGACAAGGTGTACTTCTTCTCCGGGAAAACGCCATCGACGGG GAGCACTCTGGAAAAGCCACACAGCCAGAACTCGGCCAGATCTGCA

TABLE 1A-continued

Corresponding amino acid sequences and nucleic acid sequences of antibodies according to the present disclosure mentioned in table 1 under the respective SEQ IDs. SEQ ID 581 to 587 being the corresponding Sema3A protein sequences from <i>Homo sapiens</i> (SEQ ID 581, 582), <i>Mus Musculus</i> (SEQ ID 583), <i>Rattus norvegicus</i> (SEQ ID 584), <i>Canis lupus familiaris</i> (SEQ ID 585), <i>Macaca fascicularis</i> (SEQ ID 586), <i>Sus scrofa</i> (SEQ ID 587).		
SEQ ID No	SEQ Type	SEQUENCE
		AGAACGACTTCGGCGGCCACAGATCCCTCGTGAACAAGTGGACCACC TTCCTGAAGCCCCGGCTGATCTGTTCTGTGCCCGGACCTAATGGCATC GATACCCACTTCGACGAGCTGCAGGACGTGTTCTCTGATGAACAGCAA GGACCCCAAGAATCCCATCGTGTACGGCGTGTTCAACCACCAGCAGCA ACATCTTTAGAGGCAGCGCCGTGTGCATGTACAGCATGTCCGATGTG CGGAGAGTGTCTTCTGGGCCCTACGCTCACAGAGATGGCCCCAATTA TCAGTGGGTGCCATTCCAGGGCAGAGTGCCCTATCCTAGACCTGGCA CCTGTCCCTAGCAAGACCTTTGGCGGCTTCGAGAGCACCAAGGACCTG CCTGACGATGTGATTACCTTCGCCAGATCTCACCCCGCCATGTACAAC CCTGTGTTCCTCCATCAACAACAGGCCCATCATGGTCAAGACCGACGT GAACTACCAAGTTCACCCAGATCGTGGTGGACAGAGTGGATGCCGAGG ACGCCAGTACGACGTGATGTTTCATCGGCACCGATGTGGGCACCGTG CTGAAAGTGGTGTCTATCCCCAAGAGACATGGCAGCACCTGGAAGA GGTGCTGCTGGAAGAGATGACCGTGTTCAGAGAGCCCCACCACCATCT CCGCCATGGAAGTGAACACAAACAGCAACAGCTGTATGTGGGCTCC GCCGCTGGTGTGCTCAACTGCCTCTGCACAGATGCGACATCTACGG CAAGCCTGCGCCGAGTGTTCCTGGCCAGAGATCCTTACTGTGCCT GGGATGGCAGCAGCTGCAGCAGATACTTTCCACCGCCAAGCGGAG AACCAGACGGCAGGATATCAGAAACGGCGACCTCTGACACACTGC AGCGACGGTGGCATCGAGGGCCGCATGGATCATCATCATCACCATCA T

SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 885

<210> SEQ ID NO 1

<211> LENGTH: 124

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 1

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
 20 25 30

Pro Met Gly Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ala Gly Ile Asp Asp Asp Gly Asp Ser Asp Thr Arg Tyr Ala Pro Ala
 50 55 60

Val Lys Gly Arg Ala Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Val
 65 70 75 80

Tyr Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr
 85 90 95

Cys Ala Lys His Thr Gly Ile Gly Ala Asn Ser Ala Gly Ser Ile Asp
 100 105 110

Ala Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 2

<211> LENGTH: 5

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 2

Ser Tyr Pro Met Gly
1 5

<210> SEQ ID NO 3

<211> LENGTH: 18

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 3

Gly Ile Asp Asp Asp Gly Asp Ser Asp Thr Arg Tyr Ala Pro Ala Val
1 5 10 15

Lys Gly

<210> SEQ ID NO 4

<211> LENGTH: 14

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 4

His Thr Gly Ile Gly Ala Asn Ser Ala Gly Ser Ile Asp Ala
1 5 10

<210> SEQ ID NO 5

<211> LENGTH: 108

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 5

Ser Tyr Glu Leu Thr Gln Pro Pro Ser Val Ser Val Ser Pro Gly Gln
1 5 10 15

Thr Ala Arg Ile Thr Cys Ser Gly Gly Gly Ser Tyr Thr Gly Ser Tyr
20 25 30

Tyr Tyr Gly Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Val Thr Val
35 40 45

Ile Tyr Tyr Asn Asn Lys Arg Pro Ser Asp Ile Pro Glu Arg Phe Ser
50 55 60

Gly Ser Leu Ser Gly Thr Thr Asn Thr Leu Thr Ile Ser Gly Val Gln
65 70 75 80

Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Gly Ser Ala Asp Asn Ser Gly
85 90 95

Asp Ala Phe Gly Thr Gly Thr Lys Val Thr Val Leu
100 105

<210> SEQ ID NO 6

<211> LENGTH: 13

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 6

Ser Gly Gly Gly Ser Tyr Thr Gly Ser Tyr Tyr Tyr Gly
1 5 10

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<210> SEQ ID NO 7
 <211> LENGTH: 7
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 7

Tyr Asn Asn Lys Arg Pro Ser
 1 5

<210> SEQ ID NO 8
 <211> LENGTH: 9
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 8

Gly Ser Ala Asp Asn Ser Gly Asp Ala
 1 5

<210> SEQ ID NO 9
 <211> LENGTH: 372
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 9

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg	60
agctgtgccc ccagcggtt cacttttagc agctatccta tgggctgggt ccgacaggcc	120
cctggcaaa gacttgaatg ggtggccggc atcgacgacg atggcgatag cgatacaaga	180
tacgccccctg ccgtgaaggg cagagccacc atctccagag acaacagcaa gaacaccgtg	240
tacctgcaga tgaacagcct gagagccgag gacaccgccc tgtactattg tgccaagcac	300
acaggcatcg gcgccaattc tgccggtctt attgatgcct ggggccaggg aacactggtc	360
acagtttctt ca	372

<210> SEQ ID NO 10
 <211> LENGTH: 15
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 10

agctatccta tgggc	15
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<210> SEQ ID NO 11
 <211> LENGTH: 54
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 11

ggcatcgacg acgatggcga tagcgataca agatacgccc ctgccgtgaa gggc	54
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<210> SEQ ID NO 12
 <211> LENGTH: 42
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:

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<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 12

cacacaggca tcggcgccaa ttctgccggc tctattgatg cc 42

<210> SEQ ID NO 13

<211> LENGTH: 324

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 13

agctatgagc tgacacagcc tccaagcgtg tccgtgtctc ctggacagac cgccagaatc 60

acatgtagcg gcggaggcag ctacaccggc agctactact atggctggta tcagcagaag 120

cccgacagc cccctgtgac cgtgatctac tacaacaaca agcggcccag cgacatcccc 180

gagagatttt ctggtctctc gagcggcacc accaacacac tgacaatctc tggcgtgcag 240

gccgaggacg aggcgagatta ctattgtggc agcgcgata atagcggcga cgcctttggc 300

accggcacca aagttacagt gcta 324

<210> SEQ ID NO 14

<211> LENGTH: 39

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 14

agcggcggag gcagctacac cggcagctac tactatggc 39

<210> SEQ ID NO 15

<211> LENGTH: 21

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 15

tacaacaaca agcggcccag c 21

<210> SEQ ID NO 16

<211> LENGTH: 27

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 16

ggcagcgccg ataatagcgg cgacgcc 27

<210> SEQ ID NO 17

<211> LENGTH: 453

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 17

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20 25 30

Pro	Met	Gly	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Val	
35																
Ala	Gly	Ile	Asp	Asp	Asp	Gly	Asp	Ser	Asp	Thr	Arg	Tyr	Ala	Pro	Ala	
50																
Val	Lys	Gly	Arg	Ala	Thr	Ile	Ser	Arg	Asp	Asn	Ser	Lys	Asn	Thr	Val	
65					70					75					80	
Tyr	Leu	Gln	Met	Asn	Ser	Leu	Arg	Ala	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	
85																
Cys	Ala	Lys	His	Thr	Gly	Ile	Gly	Ala	Asn	Ser	Ala	Gly	Ser	Ile	Asp	
100				105												
Ala	Trp	Gly	Gln	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	Lys	
115		120														
Gly	Pro	Ser	Val	Phe	Pro	Leu	Ala	Pro	Ser	Ser	Lys	Ser	Thr	Ser	Gly	
130		135														
Gly	Thr	Ala	Ala	Leu	Gly	Cys	Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu	Pro	
145					150					155					160	
Val	Thr	Val	Ser	Trp	Asn	Ser	Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr	
165				170												
Phe	Pro	Ala	Val	Leu	Gln	Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	
180				185												
Val	Thr	Val	Pro	Ser	Ser	Ser	Leu	Gly	Thr	Gln	Thr	Tyr	Ile	Cys	Asn	
195		200														
Val	Asn	His	Lys	Pro	Ser	Asn	Thr	Lys	Val	Asp	Lys	Lys	Val	Glu	Pro	
210		215														
Lys	Ser	Cys	Asp	Lys	Thr	His	Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	
225					230					235					240	
Leu	Leu	Gly	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	
245				250												
Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	
260					265											
Val	Ser	His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	
275				280												
Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	
290		295														
Ser	Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	
305					310					315					320	
Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	
325				330												
Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	
340				345												
Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Asp	Glu	Leu	Thr	Lys	Asn	
355		360														
Gln	Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	
370		375														
Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	
385					390					395					400	
Thr	Pro	Pro	Val	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	
405				410												
Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	
420				425												
Ser	Val	Met	His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	
435		440														
445																

-continued

Ser Leu Ser Pro Gly
450

<210> SEQ ID NO 18
<211> LENGTH: 214
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 18

Ser Tyr Glu Leu Thr Gln Pro Pro Ser Val Ser Val Ser Pro Gly Gln
1 5 10 15
Thr Ala Arg Ile Thr Cys Ser Gly Gly Gly Ser Tyr Thr Gly Ser Tyr
20 25 30
Tyr Tyr Gly Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Val Thr Val
35 40 45
Ile Tyr Tyr Asn Asn Lys Arg Pro Ser Asp Ile Pro Glu Arg Phe Ser
50 55 60
Gly Ser Leu Ser Gly Thr Thr Asn Thr Leu Thr Ile Ser Gly Val Gln
65 70 75 80
Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Gly Ser Ala Asp Asn Ser Gly
85 90 95
Asp Ala Phe Gly Thr Gly Thr Lys Val Thr Val Leu Gly Gln Pro Lys
100 105 110
Ala Ala Pro Ser Val Thr Leu Phe Pro Pro Ser Ser Glu Glu Leu Gln
115 120 125
Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp Phe Tyr Pro Gly
130 135 140
Ala Val Thr Val Ala Trp Lys Ala Asp Ser Ser Pro Val Lys Ala Gly
145 150 155 160
Val Glu Thr Thr Thr Pro Ser Lys Gln Ser Asn Asn Lys Tyr Ala Ala
165 170 175
Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys Ser His Arg Ser
180 185 190
Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val Glu Lys Thr Val
195 200 205
Ala Pro Thr Glu Cys Ser
210

<210> SEQ ID NO 19
<211> LENGTH: 1359
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 19

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg 60
agctgtgccg ccagcggett cacctttagc agctatccta tgggctgggt ccgacaggcc 120
cctggcaaag gacttgaatg ggtggccggc atcgacgacg atggcgatag cgatacaaga 180
tacgccccctg ccgtgaaggg cagagccacc atctccagag acaacagcaa gaacaccgtg 240
tacctgcaga tgaacagcct gagagccgag gacaccgccg tgtactattg tgccaagcac 300
acaggcatcg gcgccaattc tgccggtctc attgatgcct ggggccaggg aacactggtc 360
acagtttctt cagccagcac caagggcccc agcgtgttcc ctctggcccc tagcagcaag 420

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agcacatctg gcggaacagc cgccctgggc tgectcgtga aggactactt tcccagagccc 480
gtgaccgtgt cctggaactc tggcgctctg acaagcggcg tgcacacctt tccagccgtg 540
ctgcagagca gcggcctgta ctctctgagc agcgtcgtga cagtgccag cagctctctg 600
ggcaccacaga cctacatctg caacgtgaac cacaagccca gcaacaccaa ggtggacaag 660
aaggtggaac ccaagagctg cgacaagacc cacacctgtc ccccttgtcc tgcccccgaa 720
ctgctgggag gcccttcogt gttcctgttc ccccaaaagc ccaaggacac cctgatgatc 780
agccggagccc ccgaagtgac ctgcgtggtg gtggatgtgt cccacgagga ccctgaagtg 840
aagttcaatt ggtacgtgga cggcgtggaa gtgcacaacg ccaagaccaa gcctagagag 900
gaacagtaca acagcaccta ccgggtggtg tccgtgctga cagtgtgca ccaggactgg 960
ctgaacggca aagagtacaa gtgcaagggtg tccaacaagg ccctgcctgc ccccatcgag 1020
aaaaccatca gcaaggccaa gggccagccc cgcaacccc aggtgtacac actgccccca 1080
agcagggagc agctgaccaa gaaccagggtg tccctgacct gtctcgtgaa aggcttctac 1140
ccctccgata tcgcogtgga atgggagagc aacggccagc ccgagaacaa ctacaagacc 1200
acccccctg tgctggacag cgacggctca ttcttctgt acagcaagct gaccgtggac 1260
aagtcctggt ggcagcaggg caacgtgttc agctgcagcg tgatgcacga ggccctgcac 1320
aaccactaca cccagaagtc cctgagcctg agccctggc 1359

```

```

<210> SEQ ID NO 20
<211> LENGTH: 642
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 20

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agctatgagc tgacacagcc tccaagcgtg tccgtgtctc ctggacagac cgccagaatc 60
acatgtagcg gcggaggcag ctacacccgc agctactact atggctggta tcagcagaag 120
cccgagacag cccctgtgac cgtgatctac tacaacaaca agcggcccag cgacatcccc 180
gagagatttt ctggtctctc gagcggcacc accaacacac tgacaatctc tggcgtgcag 240
gccgaggagc aggcggatta ctattgtggc agcggcgata atagcggcga cgcctttggc 300
accggcacca aagttacagt gctaggccag cctaagccg cccctagcgt gaccctgttc 360
cctccaagca gcgaggaact gcaggccaac aaggccaccc tcgtgtgcct gatcagcgac 420
ttctatcctg gcgcogtgac cgtggcctgg aaggccgata gctctcctgt gaaggccggc 480
gtggaaacca ccaccctag caagcagagc aacaacaaat acgcccagcag cagctacctg 540
agcctgaccc ccgagcagtg gaagtccac agatcctaca gctgccaaat gaccacagag 600
ggcagcaccc tggaaaagac agtggcccct accgagtgca gc 642

```

```

<210> SEQ ID NO 21
<211> LENGTH: 124
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 21

```

```

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1           5           10           15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20           25           30

```

Tyr Tyr Gly Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Val Thr Val
35 40 45

-continued

```
Ile Tyr Tyr Asn Asn Lys Arg Pro Ser Asp Ile Pro Glu Arg Phe Ser
    50                      55                      60
```

```
Gly Ser Leu Ser Gly Thr Thr Asn Thr Leu Thr Ile Ser Gly Val Gln
    65                      70                      75                      80
```

```
Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Gly Ser Ala Asp Asn Ser Gly
    85                      90                      95
```

```
Asp Ala Phe Gly Thr Gly Thr Lys Val Thr Val Leu
    100                      105
```

```
<210> SEQ ID NO 26
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence
```

```
<400> SEQUENCE: 26
```

```
Ser Gly Gly Gly Ser Tyr Thr Gly Ser Tyr Tyr Tyr Gly
    1                      5                      10
```

```
<210> SEQ ID NO 27
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence
```

```
<400> SEQUENCE: 27
```

```
Tyr Asn Asn Lys Arg Pro Ser
    1                      5
```

```
<210> SEQ ID NO 28
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence
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```
<400> SEQUENCE: 28
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```
Gly Ser Ala Asp Asn Ser Gly Asp Ala
    1                      5
```

```
<210> SEQ ID NO 29
<211> LENGTH: 372
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence
```

```
<400> SEQUENCE: 29
```

```
gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg      60
agctgtgccg ccagcggcctt cacctttagc agctatccta tgggctgggt ccgacaggcc      120
cctggcaaag gacttgaatg ggtggccggc atcgacgacg atggcgatag cgatacaaga      180
tacgcccctg ccgtgaaggg cagagccacc atctccagag acaacagcaa gaacaccgtg      240
tacctgcaga tgaacagcct gagagccgag gacaccgccc tgtactattg tgccaagcac      300
acaggcatcg gcgccaattc tgccggctct attgatgcct ggggccaggg aacactggtc      360
acagtttctt ca                                     372
```

```
<210> SEQ ID NO 30
<211> LENGTH: 15
<212> TYPE: DNA
```

-continued

```

<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 30

agctatccta tgggc                                     15

<210> SEQ ID NO 31
<211> LENGTH: 54
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 31

ggcatcgacg acgatggcga tagcgataca agatacgccc ctgccgtgaa gggc       54

<210> SEQ ID NO 32
<211> LENGTH: 42
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 32

cacacaggca tcggcgccaa ttctgccggc tctattgatg cc                     42

<210> SEQ ID NO 33
<211> LENGTH: 324
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 33

agctatgagc tgacacagcc tccaagcgtg tccgtgtctc ctggacagac cgccagaatc   60
acatgtagcg gcggaggcag ctacacccgc agctactact atggctggta tcagcagaag   120
cccgacagc cccctgtgac cgtgatctac tacaacaaca agcggcccag cgacatcccc   180
gagagatttt ctggtctctc gagcggcacc accaacacac tgacaatctc tggcgtgcag   240
gccgaggacg aggcggatta ctattgtggc agcggcgata atagcggcga cgcctttggc   300
accggcacca aagttacagt gcta                                           324

<210> SEQ ID NO 34
<211> LENGTH: 39
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 34

agcggcggag gcagctacac cggcagctac tactatggc                         39

<210> SEQ ID NO 35
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 35

tacaacaaca agcggcccag c                                             21

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-continued

<210> SEQ ID NO 36
 <211> LENGTH: 27
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 36

ggcagcgccg ataatagcgg cgacgcc

27

<210> SEQ ID NO 37
 <211> LENGTH: 448
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 37

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
 20 25 30

Pro Met Gly Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ala Gly Ile Asp Asp Asp Gly Asp Ser Asp Thr Arg Tyr Ala Pro Ala
 50 55 60

Val Lys Gly Arg Ala Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Val
 65 70 75 80

Tyr Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr
 85 90 95

Cys Ala Lys His Thr Gly Ile Gly Ala Asn Ser Ala Gly Ser Ile Asp
 100 105 110

Ala Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Lys Thr Thr
 115 120 125

Pro Pro Ser Val Tyr Pro Leu Ala Pro Gly Ser Ala Ala Gln Thr Asn
 130 135 140

Ser Met Val Thr Leu Gly Cys Leu Val Lys Gly Tyr Phe Pro Glu Pro
 145 150 155 160

Val Thr Val Thr Trp Asn Ser Gly Ser Leu Ser Ser Gly Val His Thr
 165 170 175

Phe Pro Ala Val Leu Gln Ser Asp Leu Tyr Thr Leu Ser Ser Ser Val
 180 185 190

Thr Val Pro Ser Ser Thr Trp Pro Ser Glu Thr Val Thr Cys Asn Val
 195 200 205

Ala His Pro Ala Ser Ser Thr Lys Val Asp Lys Lys Ile Val Pro Arg
 210 215 220

Asp Cys Gly Cys Lys Pro Cys Ile Cys Thr Val Pro Glu Val Ser Ser
 225 230 235 240

Val Phe Ile Phe Pro Pro Lys Pro Lys Asp Val Leu Thr Ile Thr Leu
 245 250 255

Thr Pro Lys Val Thr Cys Val Val Val Asp Ile Ser Lys Asp Asp Pro
 260 265 270

Glu Val Gln Phe Ser Trp Phe Val Asp Asp Val Glu Val His Thr Ala
 275 280 285

Gln Thr Gln Pro Arg Glu Glu Gln Phe Asn Ser Thr Phe Arg Ser Val
 290 295 300

Ser Glu Leu Pro Ile Met His Gln Asp Trp Leu Asn Gly Lys Glu Phe

-continued

305	310	315	320
Lys Cys Arg Val Asn Ser Ala Ala Phe Pro Ala Pro Ile Glu Lys Thr	325	330	335
Ile Ser Lys Thr Lys Gly Arg Pro Lys Ala Pro Gln Val Tyr Thr Ile	340	345	350
Pro Pro Pro Lys Glu Gln Met Ala Lys Asp Lys Val Ser Leu Thr Cys	355	360	365
Met Ile Thr Asp Phe Phe Pro Glu Asp Ile Thr Val Glu Trp Gln Trp	370	375	380
Asn Gly Gln Pro Ala Glu Asn Tyr Lys Asn Thr Gln Pro Ile Met Asp	385	390	395
Thr Asp Gly Ser Tyr Phe Val Tyr Ser Lys Leu Asn Val Gln Lys Ser	405	410	415
Asn Trp Glu Ala Gly Asn Thr Phe Thr Cys Ser Val Leu His Glu Gly	420	425	430
Leu His Asn His His Thr Glu Lys Ser Leu Ser His Ser Pro Gly Lys	435	440	445

<210> SEQ ID NO 38
 <211> LENGTH: 214
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 38

Ser Tyr Glu Leu Thr Gln Pro Pro Ser Val Ser Val Ser Pro Gly Gln	1	5	10	15
Thr Ala Arg Ile Thr Cys Ser Gly Gly Gly Ser Tyr Thr Gly Ser Tyr	20	25	30	
Tyr Tyr Gly Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Val Thr Val	35	40	45	
Ile Tyr Tyr Asn Asn Lys Arg Pro Ser Asp Ile Pro Glu Arg Phe Ser	50	55	60	
Gly Ser Leu Ser Gly Thr Thr Asn Thr Leu Thr Ile Ser Gly Val Gln	65	70	75	80
Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Gly Ser Ala Asp Asn Ser Gly	85	90	95	
Asp Ala Phe Gly Thr Gly Thr Lys Val Thr Val Leu Gly Gln Pro Lys	100	105	110	
Ser Ser Pro Ser Val Thr Leu Phe Pro Pro Ser Ser Glu Glu Leu Glu	115	120	125	
Thr Asn Lys Ala Thr Leu Val Cys Thr Ile Thr Asp Phe Tyr Pro Gly	130	135	140	
Val Val Thr Val Asp Trp Lys Val Asp Gly Thr Pro Val Thr Gln Gly	145	150	155	160
Met Glu Thr Thr Gln Pro Ser Lys Gln Ser Asn Asn Lys Tyr Met Ala	165	170	175	
Ser Ser Tyr Leu Thr Leu Thr Ala Arg Ala Trp Glu Arg His Ser Ser	180	185	190	
Tyr Ser Cys Gln Val Thr His Glu Gly His Thr Val Glu Lys Ser Leu	195	200	205	
Ser Arg Ala Asp Cys Ser	210			

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<210> SEQ ID NO 39
 <211> LENGTH: 1344
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 39

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg	60
agctgtgccg ccagcggett cacccttagc agctatccta tgggctgggt ccgacaggcc	120
cctggcaaag gacttgaatg ggtggccggc atcgacgacg atggcgatag cgatacaaga	180
tacgccccctg ccgtgaaggg cagagccacc atctccagag acaacagcaa gaacaccgtg	240
tacctgcaga tgaacagcct gagagccgag gacaccgccg tgtactattg tgccaagcac	300
acaggcatcg gcgccaattc tgccggtctt attgatgcct ggggccaggg aacactggtc	360
acagtttctt cagccaagac ccccccccc agcgtgtacc ctctggctcc tggatctgcc	420
gcccagacca acagcatggt caccctgggc tgcctcgtga agggctactt ccctgagcct	480
gtgaccgtga cctggaacag cggtctctctg tctagcggcg tgcacacctt tccagccgtg	540
ctgcagagcg acctgtacac cctgagcagc agcgtgaccg tgcctagcag caccctggcct	600
agcgagacag tgacctgcaa cgtggcccac cctgccagca gcacaaaggt ggacaagaaa	660
atcgtgcccc gggactgagg ctgcaagccc tgtatctgta ccgtgccga ggtgtccagc	720
gtgttcactt tcccacccaa gccaaggac gtgctgacca tcacctgac ccccaaagtg	780
acctgtgtgg tggtgacat cagcaaggac gaccccgagg tgcagttcag ttggttcgtg	840
gacgacgtgg aagtgcacac agcccagacc cagcccagag aggaacagtt caacagcacc	900
ttcagaagcg tgtccgagct gcccacatg caccaggact ggctgaacgg caaagagttc	960
aagtgcagag tgaacagcgc cgcttccct gcccaccatg agaaaacat ctccaagacc	1020
aagggcagac ccaaggcccc tcaggtgtac acaatcccc caccacaaga acagatggcc	1080
aaggacaagg tgtccctgac ctgcctgac accgatttct tcccagagga catcacctg	1140
gaatggcagt ggaacggcca gcccgccgag aactacaaga acaccagcc tatcatggac	1200
accgacggca gctacttcgt gtacagcaag ctgaacgtgc agaagtccaa ctgggaggcc	1260
ggcaacacct tcacctgtag cgtgctgcac gagggcctgc acaatcacca caccgagaag	1320
tccctgtccc acagccctgg caag	1344

<210> SEQ ID NO 40
 <211> LENGTH: 642
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 40

agctatgagc tgacacagcc tccaagcgtg tccgtgtctc ctggacagac cgccagaatc	60
acatgtagcg gcggaggcag ctacaccggc agctactact atggctggta tcagcagaag	120
cccgacaggg cccctgtgac cgtgatctac tacaacaaca agcggcccag cgacatcccc	180
gagagatttt ctggctctct gagcggcacc accaacacac tgacaatctc tggcgtgcag	240
gccgaggacg aggcgatta ctattgtggc agcgcgata atagcggcga cgcccttggc	300
accggcacca aagttacagt gctaggccag cccaagagca gccctagcgt gaccctgttc	360
cctccaagca gcgaggaact ggaaacaaac aaggccaccc tcgtgtgcac catcacccgac	420

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ttctaccccg gcgtcgtgac cgtggactgg aaggtggacg gcacccagc gaccagggc 480
atggaaacca cccagccag caagcagagc aacaacaagt acatggccag cagctacctg 540
acctgacgg ccagagcctg ggagagacac agctcctaca gctgccaagt gaccacgag 600
ggccacaccg tggaaaagag cctgagcaga gccgactgca gc 642

```

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<210> SEQ ID NO 41
<211> LENGTH: 120
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 41

```

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Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1           5           10          15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
          20          25          30
Gly Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
          35          40          45
Ser Ala Ile Gly Thr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Met
          50          55          60
Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
65          70          75          80
Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
          85          90          95
Arg Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Val Trp Gly Gln
          100         105         110
Gly Thr Leu Val Thr Val Ser Ser
          115         120

```

```

<210> SEQ ID NO 42
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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```

<400> SEQUENCE: 42

```

```

Ser Tyr Gly Met His
1           5

```

```

<210> SEQ ID NO 43
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 43

```

```

Ala Ile Gly Thr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Met Gly
1           5           10          15

```

```

<210> SEQ ID NO 44
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 44

```

```

Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Val

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-continued

1	5	10
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<210> SEQ ID NO 45
<211> LENGTH: 111
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 45

Gln	Ser	Val	Leu	Thr	Gln	Pro	Pro	Ser	Ala	Ser	Gly	Thr	Pro	Gly	Gln
1				5					10					15	
Arg	Val	Thr	Ile	Ser	Cys	Ser	Gly	Ser	Ser	Ser	Asn	Ile	Gly	Ser	Asn
			20					25					30		
Thr	Val	Asn	Trp	Tyr	Gln	Gln	Leu	Pro	Gly	Thr	Ala	Pro	Lys	Leu	Leu
			35				40					45			
Ile	Tyr	Tyr	Asp	Asp	Leu	Leu	Pro	Ser	Gly	Val	Pro	Asp	Arg	Phe	Ser
	50					55					60				
Gly	Ser	Lys	Ser	Gly	Thr	Ser	Ala	Ser	Leu	Ala	Ile	Ser	Gly	Leu	Arg
65					70					75				80	
Ser	Glu	Asp	Glu	Ala	Asp	Tyr	Tyr	Cys	Ala	Ala	Trp	Asp	Asp	Ser	Leu
			85						90					95	
Asn	Gly	Tyr	Val	Val	Phe	Gly	Gly	Gly	Thr	Lys	Leu	Thr	Val	Leu	
			100					105					110		

<210> SEQ ID NO 46
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 46

Ser	Gly	Ser	Ser	Ser	Asn	Ile	Gly	Ser	Asn	Thr	Val	Asn
1					5					10		

<210> SEQ ID NO 47
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 47

Tyr	Asp	Asp	Leu	Leu	Pro	Ser
1				5		

<210> SEQ ID NO 48
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 48

Ala	Ala	Trp	Asp	Asp	Ser	Leu	Asn	Gly	Tyr	Val	Val
1					5				10		

<210> SEQ ID NO 49
<211> LENGTH: 360
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

-continued

<400> SEQUENCE: 49

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg	60
agctgtgccg ccagcggctt caccttttagc agctatggca tgcactgggt ccgacaggcc	120
cctggcaaa gacttgaatg ggtgtccgcc atcggcacag gcggcgatac ctactatgcc	180
gatagcgtga tgggcagatt caccatcagc cgggacaaca gcaagaacac cctgtacctg	240
cagatgaaca gcctgagagc cgaggacacc gccgtgtact attgcgccag aagggacgac	300
tacaccagca gggatgcctt cgatgtgtgg ggccaggga cactggttac cgtttcttca	360

<210> SEQ ID NO 50

<211> LENGTH: 15

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 50

agctatggca tgcac	15
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<210> SEQ ID NO 51

<211> LENGTH: 48

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 51

gccatcgga caggcggcga tacctactat gccgatagcg tgatgggc	48
---	----

<210> SEQ ID NO 52

<211> LENGTH: 36

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 52

agggacgact acaccagcag ggatgccttc gatgtg	36
---	----

<210> SEQ ID NO 53

<211> LENGTH: 333

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 53

cagtctgttc tgacacagcc tcctagcgcc tctggcacac ctggacagag agtgaccatc	60
agctgtagcg gcagcagctc caacatcgcc agcaacaccg tgaactggta tcagcagctg	120
cctggcacag cccctaaact gctgatctac tacgacgacc tgctgcctag cggcgtgccc	180
gatagatttt ctggcagcaa gagcggcaca agcggcagcc tggctatctc tggactgaga	240
tctgaggacg aggcgacta ctattgcgcc gcctgggacg atagcctgaa cggctatgtg	300
gttttcggcg gaggcaccaa gctgaccgtg cta	333

<210> SEQ ID NO 54

<211> LENGTH: 39

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

-continued

```

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 54

agcggcagca gctccaacat cggcagcaac accgtgaac

39

<210> SEQ ID NO 55
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 55

tacgacgacc tgctgcctag c

21

<210> SEQ ID NO 56
<211> LENGTH: 36
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 56

gccgcctggg acgatagcct gaacggctat gtggtt

36

<210> SEQ ID NO 57
<211> LENGTH: 449
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 57

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1          5          10          15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20          25          30

Gly Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35          40          45

Ser Ala Ile Gly Thr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Met
50          55          60

Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
65          70          75          80

Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
85          90          95

Arg Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Val Trp Gly Gln
100         105         110

Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val
115         120         125

Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala Ala
130         135         140

Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser
145         150         155         160

Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val
165         170         175

Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro
180         185         190

Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His Lys
195         200         205

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Pro Ser Asn Thr Lys Val Asp Lys Lys Val Glu Pro Lys Ser Cys Asp
 210                215                220

Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly
 225                230                235                240

Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile
                245                250                255

Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu
                260                265                270

Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His
 275                280                285

Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg
 290                295                300

Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys
 305                310                315                320

Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu
                325                330                335

Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr
                340                345                350

Thr Leu Pro Pro Ser Arg Asp Glu Leu Thr Lys Asn Gln Val Ser Leu
 355                360                365

Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp
 370                375                380

Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val
 385                390                395                400

Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp
                405                410                415

Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His
                420                425                430

Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro
 435                440                445

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Gly

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<210> SEQ ID NO 58
<211> LENGTH: 217
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 58

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Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
 1                5                10                15

Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn
 20                25                30

Thr Val Asn Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu
 35                40                45

Ile Tyr Tyr Asp Asp Leu Leu Pro Ser Gly Val Pro Asp Arg Phe Ser
 50                55                60

Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Arg
 65                70                75                80

Ser Glu Asp Glu Ala Asp Tyr Tyr Cys Ala Ala Trp Asp Asp Ser Leu
 85                90                95

Asn Gly Tyr Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu Gly
 100               105               110

Gln Pro Lys Ala Ala Pro Ser Val Thr Leu Phe Pro Pro Ser Ser Glu

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115	120	125	
Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys	Leu Ile Ser Asp Phe		
130	135	140	
Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Ser Ser Pro Val			
145	150	155	160
Lys Ala Gly Val Glu Thr Thr Thr Pro Ser Lys Gln Ser Asn Asn Lys			
165	170	175	
Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys Ser			
180	185	190	
His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val Glu			
195	200	205	
Lys Thr Val Ala Pro Thr Glu Cys Ser			
210	215		

<210> SEQ ID NO 59
 <211> LENGTH: 1347
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

 <400> SEQUENCE: 59

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg	60
agctgtgccg ccagcggcgt cacccttagc agctatggca tgcactgggt ccgacaggcc	120
cctggcaaag gacttgaatg ggtgtccgcc atcggcacag gcggcgatac ctactatgcc	180
gatagcgtga tgggcagatt caccatcagc cgggacaaca gcaagaacac cctgtacctg	240
cagatgaaca gcctgagagc cgaggacacc gccgtgtact attgcgccag aagggacgac	300
tacaccagca gggatgcctt cgatgtgtgg ggccaggga cactggttac cgtttcttca	360
gccagcacca agggccccag cgtgttcct ctggccccta gcagcaagag cacatctggc	420
ggaacagccg ccctgggctg cctcgtgaag gactacttcc ccgagcccgt gaccgtgtcc	480
tggaaactctg gcgctctgac aagcggcgtg cacaccttcc cagccgtgct gcagagcagc	540
ggcctgtact ctctgagcag cgtcgtgaca gtgcccagca gctctctggg caccagacc	600
tacatctgca acgtgaacca caagcccagc aacaccaagg tggacaagaa ggtggaaccc	660
aagagctgcg acaagaccca cacctgtccc ccttgtctctg cccccaact gctgggaggc	720
ccttcctgtg tctgttccc cccaaagccc aaggacaccc tgatgatcag ccggaccccc	780
gaagtgaact gcgtgggtgt ggtatgtgtc cagcaggacc ctgaagtga gttcaattgg	840
tacgtggacg gcgtggaagt gcacaacgcc aagaccaagc ctagagagga acagtacaac	900
agcacctacc ggggtgtgtc cgtgctgaca gtgctgcacc aggactggct gaacggcaaa	960
gagtacaagt gcaagggtgc caacaaggcc ctgcctgccc ccatcgagaa aaccatcagc	1020
aaggccaagg gccagccccg cgaaccccag gtgtacacac tgccccaag cagggacgag	1080
ctgaccaaga accaggtgtc cctgacctgt ctcgtgaaag gcttctaccc ctccgatatc	1140
gccgtggaat gggagagcaa cgccagccc gagaacaact acaagaccac cccccctgtg	1200
ctggacagcg acggctcatt ctctctgtac agcaagctga ccgtggacaa gtcccgtgtg	1260
cagcagggca acgtgttcag ctgcagcgtg atgcacgagg ccctgcacaa ccactacacc	1320
cagaagtccc tgagcctgag cctgtggc	1347

<210> SEQ ID NO 60
 <211> LENGTH: 651

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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 60
cagtctgttc tgacacagcc tcttagcgcc tctggcacac ctggacagag agtgaccatc   60
agctgttagcg gcagcagctc caacatcggc agcaacaccg tgaactggta tcagcagctg   120
cctggcacag cccctaaact gctgatctac tacgacgacc tgctgcctag cggcgtgccc   180
gatagatttt ctggcagcaa gagcggcaca agcgccagcc tggtatcttc tggaactgaga   240
tctgaggacg aggcgacta ctattgcgcc gcctgggacg atagcctgaa cggetatgtg   300
gttttcggcg gaggcaccaa gctgaccgtg ctaggccagc ctaaagccgc ccctagcgtg   360
acctgtttcc ctccaagcag cgaggaactg caggccaaca aggccaccct cgtgtgcctg   420
atcagcgact tctatcctgg cgccgtgacc gtggcctgga aggcgcatag ctctcctgtg   480
aaggccggcg tggaaccac caccctagc aagcagagca acaacaaata cgccgccagc   540
agctacctga gcctgacccc cgagcagtgg aagtcaccaca gatcctacag ctgccaagtg   600
acccacgagg gcagcaccgt ggaaaagaca gtggccccta ccgagtgcag c           651

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<210> SEQ ID NO 61
<211> LENGTH: 123
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 61
Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1           5           10          15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20          25          30
Glu Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35          40          45
Ser Gly Ile Ser Trp Asn Ser Gly Ser Ile Gly Tyr Ala Asp Ser Val
50          55          60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65          70          75          80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85          90          95
Ala Arg Ser Gly Tyr Ser Ser Ser Trp Phe Asp Pro Asp Phe Asp Tyr
100         105         110
Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
115         120

```

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<210> SEQ ID NO 62
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 62
Ser Tyr Glu Met Asn
1           5

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<210> SEQ ID NO 63
<211> LENGTH: 17

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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 63

Gly Ile Ser Trp Asn Ser Gly Ser Ile Gly Tyr Ala Asp Ser Val Lys
1 5 10 15

Gly

<210> SEQ ID NO 64
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 64

Ser Gly Tyr Ser Ser Ser Trp Phe Asp Pro Asp Phe Asp Tyr
1 5 10

<210> SEQ ID NO 65
<211> LENGTH: 111
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 65

Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
1 5 10 15

Arg Val Thr Ile Ser Cys Thr Gly Ser Ser Ser Asn Ile Gly Ala Gly
20 25 30

Tyr Asp Val His Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu
35 40 45

Leu Ile Tyr Gly Asn Ser Asn Arg Pro Ser Gly Val Pro Asp Arg Phe
50 55 60

Ser Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu
65 70 75 80

Arg Ser Glu Asp Glu Ala Asp Tyr Tyr Cys Ser Ser Tyr Ala Gly Ser
85 90 95

Asn Pro Tyr Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
100 105 110

<210> SEQ ID NO 66
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 66

Thr Gly Ser Ser Ser Asn Ile Gly Ala Gly Tyr Asp Val His
1 5 10

<210> SEQ ID NO 67
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 67

-continued

Gly Asn Ser Asn Arg Pro Ser
1 5

<210> SEQ ID NO 68
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 68

Ser Ser Tyr Ala Gly Ser Asn Pro Tyr Val Val
1 5 10

<210> SEQ ID NO 69
<211> LENGTH: 369
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 69

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg 60
agctgtgccg ccagcggett cacctttagc agctacgaga tgaactgggt ccgacaggcc 120
cctggcaaag gccttgaatg ggtgtccggc atcagctgga atagcggctc tateggctac 180
gccgacagcg tgaagggcag attcaccatc agccgggaca acagcaagaa caccctgtac 240
ctgcagatga acagcctgag agccgaggac accgccgtgt actactgtgc cagaagcggc 300
tacagcagct cttggtttga ccccgacttc gactattggg gccagggcac actggtcaca 360
gtctcttca 369

<210> SEQ ID NO 70
<211> LENGTH: 15
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 70

agctacgaga tgaac 15

<210> SEQ ID NO 71
<211> LENGTH: 51
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 71

ggcatcagct ggaatagcgg ctctatcggc tacgccgaca gcgtgaaggg c 51

<210> SEQ ID NO 72
<211> LENGTH: 42
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 72

agcggctaca gcagctcttg gtttgacccc gacttogact at 42

<210> SEQ ID NO 73
<211> LENGTH: 333

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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 73

cagtctgttc tgacacagcc tctagcgcc tctggcacac ctggacagag agtgaccatc      60
agctgtaccg gcagcagctc caatatcgga gccggctatg acgtgcactg gtatcagcag      120
ctgcctggca cagcccctaa actgctgatac tacggcaaca gcaacagacc cagcggcgctg      180
cccgatagat tttcgggtc taagagcggc acaagcgcca gcctggctat ctctggactg      240
agatctgagg acgaggccga ctactactgc agcagctatg ccggcagcaa cccctacgtt      300
gtgtttggcg gaggcaccaa gctgaccgtt cta                                     333

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<210> SEQ ID NO 74
<211> LENGTH: 42
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 74

accggcagca gctccaatat cggagccggc tatgacgtgc ac                               42

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<210> SEQ ID NO 75
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 75

ggcaacagca acagaccag c                                                         21

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<210> SEQ ID NO 76
<211> LENGTH: 33
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 76

agcagctatg ccggcagcaa cccctacgtt gtg                                       33

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<210> SEQ ID NO 77
<211> LENGTH: 452
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 77

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1          5          10          15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20         25         30
Glu Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35         40         45
Ser Gly Ile Ser Trp Asn Ser Gly Ser Ile Gly Tyr Ala Asp Ser Val
50         55         60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65         70         75         80

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[illegible]

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<210> SEQ ID NO 78
<211> LENGTH: 217
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 78

Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
 1 5 10 15
 Arg Val Thr Ile Ser Cys Thr Gly Ser Ser Ser Asn Ile Gly Ala Gly
 20 25 30
 Tyr Asp Val His Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu
 35 40 45
 Leu Ile Tyr Gly Asn Ser Asn Arg Pro Ser Gly Val Pro Asp Arg Phe
 50 55 60
 Ser Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu
 65 70 75 80
 Arg Ser Glu Asp Glu Ala Asp Tyr Tyr Cys Ser Ser Tyr Ala Gly Ser
 85 90 95
 Asn Pro Tyr Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu Gly
 100 105 110
 Gln Pro Lys Ala Ala Pro Ser Val Thr Leu Phe Pro Pro Ser Ser Glu
 115 120 125
 Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp Phe
 130 135 140
 Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Ser Ser Pro Val
 145 150 155 160
 Lys Ala Gly Val Glu Thr Thr Thr Pro Ser Lys Gln Ser Asn Asn Lys
 165 170 175
 Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys Ser
 180 185 190
 His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val Glu
 195 200 205
 Lys Thr Val Ala Pro Thr Glu Cys Ser
 210 215

<210> SEQ ID NO 79

<211> LENGTH: 1356

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 79

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg 60
 agctgtgccg ccagcgggett cacctttagc agctacgaga tgaactgggt ccgacaggcc 120
 cctggcaaag gccttgaatg ggtgtccggc atcagctgga atagcggctc tatcggtac 180
 gccgacagcg tgaagggcag attcaccatc agccgggaca acagcaagaa caccctgtac 240
 ctgcagatga acagcctgag agccgaggac accgccgtgt actactgtgc cagaagcggc 300
 tacagcagct cttggtttga ccccgacttc gactattggg gccagggcac actggtcaca 360
 gtctcttcag ccagcaccaa gggccccagc gtgttccttc tggcccctag cagcaagagc 420
 acatctggcg gaacagccgc cctgggctgc ctctgaagg actactttcc cgagcccgctg 480
 accgtgtcct ggaactctgg cgctctgaca agcggcgtgc acacctttcc agcctgtgctg 540
 cagagcagcg gcctgtactc tctgagcagc gtcgtgacag tgcccagcag ctctctgggc 600
 acccagacct acatctgcaa cgtgaaccac aagcccagca acaccaaggt ggacaagaag 660
 gtggaaccca agagctgcga caagaccac acctgtcccc cttgtcctgc ccccgaaactg 720

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ctgggaggcc cttcogtggt cctgttcccc ccaaagccca aggacacct gatgatcagc 780
cggacccccg aagtgacctg cgtggtggtg gatgtgtccc acgaggacct tgaagtgaag 840
ttcaattggt acgtggacgg cgtggaagtg cacaacgcca agaccaagcc tagagaggaa 900
cagtacaaca gcacctaccg ggtggtgtcc gtgtgtgacag tgtgtcacca ggactggctg 960
aacggcaaaag agtacaagtg caaggtgtcc aacaaggccc tgctgtcccc catcgagaaa 1020
accatcagca aggccaaaggg ccagccccgc gaaccccagg tgtacacact gcccccaagc 1080
agggacgagc tgaccaagaa ccaggtgtcc ctgacctgtc tcgtgaaagg cttctacccc 1140
tccgatatcg ccgtggaatg ggagagcaac ggccagcccc agaacaacta caagaccacc 1200
ccccctgtgc tggacagcga cggtctattc ttcctgtaca gcaagctgac cgtggacaag 1260
tcccgggtggc agcagggcaa cgtgttcagc tgcagcgtga tgcacgaggg cctgcacaac 1320
cactacaccc agaagtcctt gagcctgagc cctggc 1356

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<210> SEQ ID NO 80
<211> LENGTH: 651
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 80
cagtctgttc tgacacagcc tcctagcgcc tctggcacac ctggacagag agtgaccatc 60
agctgtaccg gcagcagctc caatatcgga gccggctatg acgtgcactg gtatcagcag 120
ctgctggcca cagcccctaa actgtgtatc tacggcaaca gcaacagacc cagcggcgtg 180
cccgatagat ttccggctc taagagcggc acaagcgcca gcctggctat ctctggactg 240
agatctgagg acgaggccga ctactactgc agcagctatg ccggcagcaa ccctacgtt 300
gtgtttggcg gaggcaccaa gctgaccgtt ctaggccagc ctaaagccgc ccctagcgtg 360
accctgttcc ctccaagcag cgaggaaactg caggccaaca agggccacct cgtgtgcttg 420
atcagcgact tctatcctgg cgccgtgacc gtggcctgga agggcgatag ctctcctgtg 480
aaggccggcg tggaaaccac caccctagc aagcagagca acaacaaata cgccgccagc 540
agctacctga gcctgacccc cgagcagtgg aagtcacaca gatcctacag ctgccaagtg 600
accacagagg gcagcacctt ggaaaagaca gtggccccta ccgagtgcag c 651

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<210> SEQ ID NO 81
<211> LENGTH: 116
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 81
Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Thr Gly Gly
1      5      10      15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asp Tyr
20     25     30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35     40     45
Ser Trp Ile Tyr Tyr Asp Ser Gly Ser Lys Tyr Tyr Ala Asp Ser Val
50     55     60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65     70     75     80

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Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95

Ala Lys Leu Asn Gly Asp Phe Asp Tyr Trp Gly Gln Gly Thr Leu Val
100 105 110

Thr Val Ser Ser
115

<210> SEQ ID NO 82
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 82

Asp Tyr Ala Met Ser
1 5

<210> SEQ ID NO 83
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 83

Trp Ile Tyr Tyr Asp Ser Gly Ser Lys Tyr Tyr Ala Asp Ser Val Lys
1 5 10 15

Gly

<210> SEQ ID NO 84
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 84

Leu Asn Gly Asp Phe Asp Tyr
1 5

<210> SEQ ID NO 85
<211> LENGTH: 110
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 85

Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
1 5 10 15

Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Asn Asn
20 25 30

Asp Val Ser Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu
35 40 45

Ile Tyr Ala Asp Ser His Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
50 55 60

Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Arg
65 70 75 80

Ser Glu Asp Glu Ala Asp Tyr Tyr Cys Gly Ala Trp Asp Ser Ser Leu
85 90 95

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Ser Gly Tyr Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
 100 105 110

<210> SEQ ID NO 86
 <211> LENGTH: 13
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 86

Ser Gly Ser Ser Ser Asn Ile Gly Asn Asn Asp Val Ser
 1 5 10

<210> SEQ ID NO 87
 <211> LENGTH: 7
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 87

Ala Asp Ser His Arg Pro Ser
 1 5

<210> SEQ ID NO 88
 <211> LENGTH: 11
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 88

Gly Ala Trp Asp Ser Ser Leu Ser Gly Tyr Val
 1 5 10

<210> SEQ ID NO 89
 <211> LENGTH: 348
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 89

gaagttcagc tgctggaatc tggcggcgga ctggttcaaa caggcggctc tctgagactg	60
agctgtgccg cctctggctt caccttcagc gattacgcca tgagctgggt ccgacaggcc	120
cctggaaaag gccttgaatg ggtgtcctgg atctactacg acagcggcag caagtactac	180
gccgacagcg tgaagggcag attcaccatc agccgggaca acagcaagaa caccctgtac	240
ctgcagatga acagcctgag agccgaggac accgccgtgt actattgcgc caagctgaac	300
ggcgacttcg actattgggg ccagggcaca ctggtcacag tctcttca	348

<210> SEQ ID NO 90
 <211> LENGTH: 15
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 90

gattacgcca tgagc	15
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<210> SEQ ID NO 91
 <211> LENGTH: 51

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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 91

tggatctact acgacagcgg cagcaagtac tacgccgaca gcgtgaaggg c      51

<210> SEQ ID NO 92
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 92

ctgaacggcg acttcgacta t      21

<210> SEQ ID NO 93
<211> LENGTH: 330
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 93

cagtctgttc tgacacagcc tcttagcgcc tctggcacac ctggacagag agtgaccatc      60
agctgttagcg gcagcagctc caacatcggc aacaacgacg tgtcctggta tcagcagctg      120
cctggcacag cccctaaact gctgatctac gccgacagcc acagacctag cggcgtgcc      180
gatagattca gcggctctaa gagcggcaca tctgccagcc tggccatctc tggactgaga      240
tctgaggacg aggcgacta ctattcgggc gcctgggatt ctacgctgag cggetatgtt      300
tttggcggag gcaccaagct gaccgtgcta      330

<210> SEQ ID NO 94
<211> LENGTH: 39
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 94

agcggcagca gctccaacat cggcaacaac gacgtgtcc      39

<210> SEQ ID NO 95
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 95

gccgacagcc acagacctag c      21

<210> SEQ ID NO 96
<211> LENGTH: 33
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 96

ggcgccctggg attctagcct gagcggctat gtt      33

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<210> SEQ ID NO 97
<211> LENGTH: 445
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 97

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Thr Gly Gly
1          5          10          15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asp Tyr
20          25          30

Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35          40          45

Ser Trp Ile Tyr Tyr Asp Ser Gly Ser Lys Tyr Tyr Ala Asp Ser Val
50          55          60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65          70          75          80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85          90          95

Ala Lys Leu Asn Gly Asp Phe Asp Tyr Trp Gly Gln Gly Thr Leu Val
100         105         110

Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala
115         120         125

Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly Cys Leu
130         135         140

Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly
145         150         155         160

Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser
165         170         175

Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser Leu
180         185         190

Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His Lys Pro Ser Asn Thr
195         200         205

Lys Val Asp Lys Lys Val Glu Pro Lys Ser Cys Asp Lys Thr His Thr
210         215         220

Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly Pro Ser Val Phe
225         230         235         240

Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro
245         250         255

Glu Val Thr Cys Val Val Val Asp Val Ser His Glu Asp Pro Glu Val
260         265         270

Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys Thr
275         280         285

Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg Val Val Ser Val
290         295         300

Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys
305         310         315         320

Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser
325         330         335

Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro
340         345         350

Ser Arg Asp Glu Leu Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val
355         360         365

```

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Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly
 370 375 380

Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp
 385 390 395 400

Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg Trp
 405 410 415

Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu His
 420 425 430

Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly
 435 440 445

<210> SEQ ID NO 98
 <211> LENGTH: 216
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 98

Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
 1 5 10 15

Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Asn Asn
 20 25 30

Asp Val Ser Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu
 35 40 45

Ile Tyr Ala Asp Ser His Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
 50 55 60

Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Arg
 65 70 75 80

Ser Glu Asp Glu Ala Asp Tyr Tyr Cys Gly Ala Trp Asp Ser Ser Leu
 85 90 95

Ser Gly Tyr Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu Gly Gln
 100 105 110

Pro Lys Ala Ala Pro Ser Val Thr Leu Phe Pro Pro Ser Ser Glu Glu
 115 120 125

Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp Phe Tyr
 130 135 140

Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Ser Ser Pro Val Lys
 145 150 155 160

Ala Gly Val Glu Thr Thr Thr Pro Ser Lys Gln Ser Asn Asn Lys Tyr
 165 170 175

Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys Ser His
 180 185 190

Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val Glu Lys
 195 200 205

Thr Val Ala Pro Thr Glu Cys Ser
 210 215

<210> SEQ ID NO 99
 <211> LENGTH: 1335
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 99

gaagttcagc tgctggaatc tggcggcgga ctggttcaaa caggcggctc tctgagactg

60

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agctgtgccg cctctggctt caccttcagc gattacgcca tgagctgggt ccgacaggcc 120
cctggaaaag gccttgaatg ggtgtcctgg atctactacg acagcggcag caagtactac 180
gccgacagcg tgaagggcag attcaccatc agccgggaca acagcaagaa caccctgtac 240
ctgcagatga acagcctgag agccgaggac accgccgtgt actattgcgc caagctgaac 300
ggcgacttcg actattgggg ccagggcaca ctggtcacag tctcttcagc cagcaccaag 360
ggccccagcg tgttcctctt ggccccctagc agcaagagca catctggcgg aacagccgcc 420
ctgggctgcc tcgtgaagga ctactttccc gagcccgta ccggtgtcctg gaactctggc 480
gtcttgacaa gcggcgtgca cacctttcca gccgtgtgc agagcagcgg cctgtactct 540
ctgagcagcg tcgtgacagt gccagcagc tctctgggca ccagaccta catctgcaac 600
gtgaaccaca agcccagcaa caccaagggt gacaagaagg tggaacccaa gagctgcgac 660
aagaccacaa cctgtccccc ttgtcctgcc ccgaaactgc tgggaggccc ttcgtgttc 720
ctgttccccc caaagcccaa ggacacctg atgacagcc ggacccccga agtgacctgc 780
gtggtggtgg atgtgtccca cgaggacctt gaagtgaagt tcaattggta cgtggacggc 840
gtggaagtgc acaacgcaa gaccaagcct agagaggaa agtacaacag cacctaccgg 900
gtggtgtccg tctgacagt gctgcaccag gactggctga acggcaaaga gtacaagtgc 960
aaggtgtcca acaaggccct gcctgcccc atcgagaaaa ccatcagcaa ggccaagggc 1020
cagcccccg aaccccaggt gtacacactg cccccaagca gggacgagct gaccaagaac 1080
caggtgtccc tgacctgtct cgtgaaaggc ttctaccctt ccgatatgc cgtggaatgg 1140
gagagcaacg gccagccga gaacaactac aagaccacc cccctgtgct ggacagcgac 1200
ggctcattct tcctgtacag caagctgacc gtggacaagt cccggtggca gcagggcaac 1260
gtgttcagct gcagcgtgat gcacgaggcc ctgcacaacc actacacca gaagtcctg 1320
agcctgagcc ctggc 1335

```

<210> SEQ ID NO 100

<211> LENGTH: 648

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 100

```

cagtctgttc tgacacagcc tcctagcgcc tctggcacac ctggacagag agtgaccatc 60
agctgtagcg gcagcagctc caacatcggc aacaacgacg tgtcctggta tcagcagctg 120
cctggcacag cccctaaact gctgatctac gccgacagcc acagacctag cggcgtgcca 180
gatagattca gcggtctaa gagcggcaca tctgccagcc tggccatctc tggactgaga 240
tctgaggacg aggcgacta ctattcggc gcctgggatt ctagcctgag cggctatgtt 300
tttggcggag gcaccaagct gaccgtgcta ggccagccta aagccgcccc tagcgtgacc 360
ctgttcctc caagcagcga ggaactgcag gccacaagg ccacctcgt gtgectgate 420
agcgacttct atcctggcgc cgtgaccgtg gcctggaagg ccgatagctc tcctgtgaag 480
gcggcgtgag aaaccaccac ccctagcaag cagagcaaca acaatacgc gccagcagc 540
tacctgagcc tgacccccga gcagtggaag tcccacagat cctacagctg ccaagtgacc 600
cacgagggca gcacggtgga aaagacagtg gccctaccg agtgcagc 648

```

<210> SEQ ID NO 101

<211> LENGTH: 123

-continued

```

<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 101

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1           5           10           15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20          25          30
Glu Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35          40          45
Ser Gly Ile Ser Trp Asn Ser Gly Ser Ile Gly Tyr Ala Asp Ser Val
50          55          60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65          70          75          80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85          90          95
Ala Arg Ser Gly Tyr Ser Ser Ser Trp Phe Asp Pro Asp Phe Asp Tyr
100         105         110
Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
115         120

```

```

<210> SEQ ID NO 102
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 102

```

```

Ser Tyr Glu Met Asn
1           5

```

```

<210> SEQ ID NO 103
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 103

```

```

Gly Ile Ser Trp Asn Ser Gly Ser Ile Gly Tyr Ala Asp Ser Val Lys
1           5           10           15
Gly

```

```

<210> SEQ ID NO 104
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 104

```

```

Ser Gly Tyr Ser Ser Ser Trp Phe Asp Pro Asp Phe Asp Tyr
1           5           10

```

```

<210> SEQ ID NO 105
<211> LENGTH: 111
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

-continued

<400> SEQUENCE: 105

```

Gln Ser Val Leu Thr Gln Pro Pro Ser Val Ser Gly Ala Pro Gly Gln
1           5           10           15

Arg Val Thr Ile Ser Cys Thr Gly Ser Ser Ser Asn Ile Gly Ala Gly
          20           25           30

Tyr Asp Val His Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu
          35           40           45

Leu Ile Tyr Gly Asn Ser Asn Arg Pro Ser Gly Val Pro Asp Arg Phe
          50           55           60

Ser Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Thr Gly Leu
          65           70           75           80

Gln Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Ser Ser Tyr Ala Gly Ser
          85           90           95

Asn Pro Tyr Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
          100          105          110

```

<210> SEQ ID NO 106

<211> LENGTH: 14

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 106

```

Thr Gly Ser Ser Ser Asn Ile Gly Ala Gly Tyr Asp Val His
1           5           10

```

<210> SEQ ID NO 107

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 107

```

Gly Asn Ser Asn Arg Pro Ser
1           5

```

<210> SEQ ID NO 108

<211> LENGTH: 11

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 108

```

Ser Ser Tyr Ala Gly Ser Asn Pro Tyr Val Val
1           5           10

```

<210> SEQ ID NO 109

<211> LENGTH: 369

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 109

```

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg      60
agctgtgccg ccagcggett cacctttagc agctacgaga tgaactgggt ccgacaggcc      120
cctggcaaag gccttgaatg ggtgtccggc atcagctgga atagcggctc tatcggtac      180

```

-continued

```

gccgacagcg tgaagggcag attcaccatc agccggggaca acagcaagaa caccctgtac 240
ctgcagatga acagcctgag agccgaggac accgccgtgt actactgtgc cagaagcggc 300
tacagcagct cttggtttga ccccgacttc gactattggg gccagggcac actggtcaca 360
gtctcttca 369

```

```

<210> SEQ ID NO 110
<211> LENGTH: 15
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 110

```

```

agctacgaga tgaac 15

```

```

<210> SEQ ID NO 111
<211> LENGTH: 51
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 111

```

```

ggcatcagct ggaatagcgg ctctatcggc tacgccgaca gcgtgaaggg c 51

```

```

<210> SEQ ID NO 112
<211> LENGTH: 42
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 112

```

```

agcggctaca gcagctcttg gtttgacccc gacttcgact at 42

```

```

<210> SEQ ID NO 113
<211> LENGTH: 333
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 113

```

```

cagtctgttc tgacacagcc tccatctgtg tctggcgccc ctggacagag agtgaccatc 60
agctgtacag gcagcagctc caatatcgga gccggctatg acgtgcactg gtatcagcag 120
ctgcctggca cagcccctaa actgctgac tacggcaaca gcaacagacc cagcggcgtg 180
cccgatagat ttccggctc taagagcggc acaagcgcca gcctggctat tactggactg 240
caggccgagg acgaggccga ctactactgt tctagctacg ccggcagcaa cccctacgtg 300
gtgtttggcg gaggcaccaa gctgacagtt cta 333

```

```

<210> SEQ ID NO 114
<211> LENGTH: 42
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 114

```

```

acaggcagca gctccaatat cggagccggc tatgacgtgc ac 42

```

-continued

<210> SEQ ID NO 115
 <211> LENGTH: 21
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 115

ggcaacagca acagaccag c 21

<210> SEQ ID NO 116
 <211> LENGTH: 33
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 116

tctagctacg ccggcagcaa cccctacgtg gtg 33

<210> SEQ ID NO 117
 <211> LENGTH: 452
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 117

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
 20 25 30

Glu Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ser Gly Ile Ser Trp Asn Ser Gly Ser Ile Gly Tyr Ala Asp Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
 65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Ala Arg Ser Gly Tyr Ser Ser Ser Trp Phe Asp Pro Asp Phe Asp Tyr
 100 105 110

Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly
 115 120 125

Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly
 130 135 140

Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val
 145 150 155 160

Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe
 165 170 175

Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val
 180 185 190

Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val
 195 200 205

Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys Lys Val Glu Pro Lys
 210 215 220

Ser Cys Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu
 225 230 235 240

Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr

-continued

245					250					255					
Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val
			260						265				270		
Ser	His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val
		275						280					285		
Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser
		290						295				300			
Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu
305						310					315				320
Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala
				325					330					335	
Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro
			340					345					350		
Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Asp	Glu	Leu	Thr	Lys	Asn	Gln
		355						360					365		
Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala
		370					375				380				
Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr
385					390					395					400
Pro	Pro	Val	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu
			405						410					415	
Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser
		420						425					430		
Val	Met	His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser
		435					440					445			
Leu	Ser	Pro	Gly												
		450													

<210> SEQ ID NO 118

<211> LENGTH: 217

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 118

Gln	Ser	Val	Leu	Thr	Gln	Pro	Pro	Ser	Val	Ser	Gly	Ala	Pro	Gly	Gln
1				5					10					15	
Arg	Val	Thr	Ile	Ser	Cys	Thr	Gly	Ser	Ser	Ser	Asn	Ile	Gly	Ala	Gly
			20					25					30		
Tyr	Asp	Val	His	Trp	Tyr	Gln	Gln	Leu	Pro	Gly	Thr	Ala	Pro	Lys	Leu
		35					40					45			
Leu	Ile	Tyr	Gly	Asn	Ser	Asn	Arg	Pro	Ser	Gly	Val	Pro	Asp	Arg	Phe
	50					55					60				
Ser	Gly	Ser	Lys	Ser	Gly	Thr	Ser	Ala	Ser	Leu	Ala	Ile	Thr	Gly	Leu
65					70					75				80	
Gln	Ala	Glu	Asp	Glu	Ala	Asp	Tyr	Tyr	Cys	Ser	Ser	Tyr	Ala	Gly	Ser
			85					90						95	
Asn	Pro	Tyr	Val	Val	Phe	Gly	Gly	Gly	Thr	Lys	Leu	Thr	Val	Leu	Gly
		100						105					110		
Gln	Pro	Lys	Ala	Ala	Pro	Ser	Val	Thr	Leu	Phe	Pro	Pro	Ser	Ser	Glu
		115					120					125			
Glu	Leu	Gln	Ala	Asn	Lys	Ala	Thr	Leu	Val	Cys	Leu	Ile	Ser	Asp	Phe
	130					135					140				
Tyr	Pro	Gly	Ala	Val	Thr	Val	Ala	Trp	Lys	Ala	Asp	Ser	Ser	Pro	Val

-continued

145	150	155	160
Lys Ala Gly Val Glu Thr Thr Thr Pro Ser Lys Gln Ser Asn Asn Lys			
	165	170	175
Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys Ser			
	180	185	190
His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val Glu			
	195	200	205
Lys Thr Val Ala Pro Thr Glu Cys Ser			
	210	215	

<210> SEQ ID NO 119
 <211> LENGTH: 1356
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 119

gaagttcagc tgctggaatc tggcgccgga ctggttcaac ctggcggatc tctgagactg	60
agctgtgccg ccagcggctt cacctttagc agctacgaga tgaactgggt ccgacaggcc	120
cctggcaaag gccttgaatg ggtgtccggc atcagctgga atagcggctc tatcggtac	180
gccgacagcg tgaagggcag attcaccatc agccgggaca acagcaagaa caccctgtac	240
ctgcagatga acagcctgag agccgaggac accgccgtgt actactgtgc cagaagcggc	300
tacagcagct cttggtttga ccccgacttc gactattggg gccagggcac actggtcaca	360
gtctcttcag ccagcaccaa gggccccagc gtgttcctc tggcccctag cagcaagagc	420
acatctggcg gaacagccgc cctgggctgc ctctgaagg actactttcc cgagcccgtg	480
accgtgtcct ggaactctgg cgctctgaca agcggcgtgc acacctttcc agcgtgtctg	540
cagagcagcg gcctgtactc tctgagcagc gtcgtgacag tgcccagcag ctctctgggc	600
accagacct acatctgcaa cgtgaaccac aagcccagca acaccaaggt ggacaagaag	660
gtggaaccca agagctgcga caagaccac acctgtcccc cttgtcctgc ccccgaactg	720
ctgggaggcc cttccgtgtt cctgttcccc ccaaagccca aggacacct gatgatcagc	780
cggacccccg aagtgaacctg cgtggtggtg gatgtgtccc acgaggaccc tgaagtgaag	840
ttcaattggt acgtggacgg cgtggaagtg cacaacgcca agaccaagcc tagagaggaa	900
cagtacaaca gcacctaccg ggtggtgtcc gtgctgacag tgctgcacca ggactggctg	960
aacggcaaaag agtacaagtg caagggtgtcc aacaaggccc tgctgtcccc catcgagaaa	1020
accatcagca agggcaaggg ccagccccgc gaaccccagg tgtacacact gcccccaagc	1080
agggacgagc tgaccaagaa ccagggtgtcc ctgacctgtc tcgtgaaagg cttctacccc	1140
tccgatatcg ccgtggaatg ggagagcaac ggccagcccc agaacaacta caagaccacc	1200
ccccctgtgc tggacagcga cggtctattc ttctgtgata gcaagctgac cgtggacaag	1260
tcccgggtggc agcagggcaa cgtgttcagc tgcagcgtga tgcacgaggc cctgcacaa	1320
cactacaccc agaagtcctt gagcctgagc cctggc	1356

<210> SEQ ID NO 120
 <211> LENGTH: 651
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 120

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```

cagtctgttc tgacacagcc tccatctgtg tctggcgccc ctggacagag agtgaccatc      60
agctgtacag gcagcagctc caatatcgga gccggctatg acgtgcactg gtatcagcag      120
ctgcctggca cagcccctaa actgctgata tacggcaaca gcaacagacc cagcggcgtg      180
cccgatagat ttccggctc taagagcggc acaagcgcca gcctggctat tactggactg      240
caggccgagg acgaggccga ctactactgt tctagctacg ccggcagcaa cccctacgtg      300
gtgtttggcg gaggcaccaa gctgacagtt ctaggccagc ctaaagccgc ccctagcgtg      360
acctgtttcc ctccaagcag cgaggaactg caggccaaca agggccacct cgtgtgcctg      420
atcagcgact tctatcctgg cgccgtgacc gtggcctgga aggcgcatag ctctcctgtg      480
aaggccggcg tggaaccac caccctagc aagcagagca acaacaaata cgccgccagc      540
agctacctga gcctgacccc cgagcagtggt aagtcacaca gatcctacag ctgccaagtg      600
acccacgagg gcagcaccgt ggaaaagaca gtggccccta ccgagtgcag c              651

```

```

<210> SEQ ID NO 121
<211> LENGTH: 120
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 121

```

```

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1             5             10            15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20            25            30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35            40            45
Ser Ala Ile Gly Thr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys
50            55            60
Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
65            70            75            80
Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
85            90            95
Arg Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr Trp Gly Gln
100           105           110
Gly Thr Leu Val Thr Val Ser Ser
115           120

```

```

<210> SEQ ID NO 122
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 122

```

```

Ser Tyr Ala Met Ser
1             5

```

```

<210> SEQ ID NO 123
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 123

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Ala Ile Gly Thr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly
1 5 10 15

<210> SEQ ID NO 124
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 124

Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr
1 5 10

<210> SEQ ID NO 125
<211> LENGTH: 111
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 125

Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
1 5 10 15

Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn
20 25 30

Thr Val Asn Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu
35 40 45

Ile Tyr Tyr Asp Asp Leu Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
50 55 60

Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Gln
65 70 75 80

Ser Glu Asp Glu Ala Asp Tyr Tyr Cys Ala Ala Trp Asp Asp Ser Leu
85 90 95

Asn Gly Tyr Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
100 105 110

<210> SEQ ID NO 126
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 126

Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn Thr Val Asn
1 5 10

<210> SEQ ID NO 127
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 127

Tyr Asp Asp Leu Arg Pro Ser
1 5

<210> SEQ ID NO 128
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 128

Ala Ala Trp Asp Asp Ser Leu Asn Gly Tyr Val Val
 1 5 10

<210> SEQ ID NO 129
 <211> LENGTH: 360
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 129

gaagttcagc tgctggaatc tggcgccgga ctggttcaac ctggcggatc tctgagactg 60
 agctgtgccg ccagcggtt cacttttagc agctacgcca tgagctgggt ccgacaggct 120
 cctggcaaag gccttgaatg ggtgtccgcc attggcacag gcggcgatac ctactacgcc 180
 gactctgtga agggcagatt caccatcagc cgggacaaca gcaagaacac cctgtacctg 240
 cagatgaaca gcctgagagc cgaggacacc gccgtgtact attgcgccag aaggacgac 300
 tacaccagca gggacgcott cgattattgg ggccagggca cactggtcac cgttttctca 360

<210> SEQ ID NO 130
 <211> LENGTH: 15
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 130

agctacgcca tgagc 15

<210> SEQ ID NO 131
 <211> LENGTH: 48
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 131

gccattggca caggcggcga tacctactac gccgactctg tgaagggc 48

<210> SEQ ID NO 132
 <211> LENGTH: 36
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 132

agggacgact acaccagcag ggacgccttc gattat 36

<210> SEQ ID NO 133
 <211> LENGTH: 333
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 133

cagtctgttc tgacacagcc tcctagcgcc tctggcacac ctggacagag agtgaccatc 60
 agctgtagcg gcagcagctc caacatcgcc agcaacaccg tgaactggta tcagcagctg 120

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cctggcacag cccctaaact gctgatctac tacgacgacc tgcggcctag cggcgtgccca 180
gatagatttt ctggcagcaa gageggcacc tctgccagcc tggctatttc tggactgcag 240
agcgaggacg aggcgacta ttattgtgcc gcctgggacg acagcctgaa cggctatgtt 300
gttttcggcg gaggcaccaa gctgaccgtt cta 333

```

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<210> SEQ ID NO 134
<211> LENGTH: 39
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 134

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agcggcagca gctccaacat cggcagcaac accgtgaac 39

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<210> SEQ ID NO 135
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 135

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tacgacgacc tgcggcctag c 21

```

```

<210> SEQ ID NO 136
<211> LENGTH: 36
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 136

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gccgcctggg acgacagcct gaacggctat gttgtt 36

```

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<210> SEQ ID NO 137
<211> LENGTH: 449
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 137

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Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1          5          10          15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20          25          30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35          40          45
Ser Ala Ile Gly Thr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys
50          55          60
Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
65          70          75          80
Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
85          90          95
Arg Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr Trp Gly Gln
100         105         110
Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val
115         120         125

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Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala Ala
 130          135          140

Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser
 145          150          155          160

Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val
          165          170          175

Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro
          180          185          190

Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His Lys
 195          200          205

Pro Ser Asn Thr Lys Val Asp Lys Lys Val Glu Pro Lys Ser Cys Asp
 210          215          220

Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly
 225          230          235          240

Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile
          245          250          255

Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu
          260          265          270

Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His
 275          280          285

Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg
 290          295          300

Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys
 305          310          315          320

Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu
          325          330          335

Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr
          340          345          350

Thr Leu Pro Pro Ser Arg Asp Glu Leu Thr Lys Asn Gln Val Ser Leu
          355          360          365

Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp
 370          375          380

Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val
 385          390          395          400

Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp
          405          410          415

Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His
          420          425          430

Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro
 435          440          445

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Gly

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<210> SEQ ID NO 138
<211> LENGTH: 217
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 138

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```

Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
 1          5          10          15

Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn
 20          25          30

Thr Val Asn Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu

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35	40	45
Ile Tyr Tyr Asp Asp Leu Arg Pro Ser Gly Val Pro Asp Arg Phe Ser		
50	55	60
Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Gln		
65	70	75
Ser Glu Asp Glu Ala Asp Tyr Tyr Cys Ala Ala Trp Asp Asp Ser Leu		
	85	90
Asn Gly Tyr Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu Gly		
	100	105
Gln Pro Lys Ala Ala Pro Ser Val Thr Leu Phe Pro Pro Ser Ser Glu		
	115	120
Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp Phe		
	130	135
Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Ser Ser Pro Val		
145	150	155
Lys Ala Gly Val Glu Thr Thr Thr Pro Ser Lys Gln Ser Asn Asn Lys		
	165	170
Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys Ser		
	180	185
His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val Glu		
	195	200
Lys Thr Val Ala Pro Thr Glu Cys Ser		
	210	215

<210> SEQ ID NO 139

<211> LENGTH: 1347

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 139

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg	60
agctgtgccg ccagcggtt cactttagc agctaccca tgagctgggt ccgacaggct	120
cctggcaaag gccttgaatg ggtgtccgcc attggcacag gcggcgatac ctactacgcc	180
gactctgtga agggcagatt caccatcagc cgggacaaca gcaagaacac cctgtacctg	240
cagatgaaca gcctgagagc cgaggacacc gccgtgtact attgcgccag aagggacgac	300
tacaccagca gggacgcctt cgattattgg gggcagggca cactggtcac cgttttctca	360
gccagcacca agggccccag cgtgttcct ctggccccta gcagcaagag cacatctggc	420
ggaacagccg ccctgggtcg cctcgtgaag gactactttc ccgagcccg gaccgtgtcc	480
tgggaactctg gcgctctgac aagcggcgtg cacaccttc cagccgtgct gcagagcagc	540
ggcctgtact ctctgagcag cgtcgtgaca gtgcccagca gctctctggg caccagacc	600
tacatctgca acgtgaacca caagcccagc aacaccaagg tggacaagaa ggtggaacct	660
aagagctcgc acaagaccca cacctgtccc ccttgtctg ccccggaact gctgggaggc	720
ccttcctgtg tcctgttccc cccaaagccc aaggaccccc tgatgatcag ccggaccccc	780
gaagtgcact gcgtggtggt ggatgtgtcc caccaggacc ctgaagtga gttcaattgg	840
tacgtggacg gcgtggaagt gcacaacgcc aagaccaagc ctagagagga acagtacaac	900
agcacctacc ggggtggtgc cgtgctgaca gtgctgcacc aggactggct gaacggcaaa	960
gagtacaagt gcaagggtgc caacaaggcc ctgcctgccc ccatcgagaa aaccatcagc	1020

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aaggccaagg gccagcccg cgaaccccg gtgtacacac tgccccaag cagggacgag 1080
ctgaccaaga accaggtgtc cctgacctgt ctctgaaag gcttctaccc ctccgatatc 1140
gccgtggaat gggagagcaa cggccagccc gagaacaact acaagaccac cccccctgtg 1200
ctggacagcg acggtcatt ctctctgtac agcaagctga ccgtggacaa gtcccgggtg 1260
cagcagggca acgtgttcag ctgcagcgtg atgcacgagg ccctgcacaa ccactacacc 1320
cagaagtccc tgagcctgag ccctggc 1347

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<210> SEQ ID NO 140
<211> LENGTH: 651
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 140

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cagtctgttc tgacacagcc tctagcgcc tctggcacac ctggacagag agtgaccatc 60
agctgtagcg gcagcagctc caacatcggc agcaacaccg tgaactggta tcagcagctg 120
cctggcacag cccctaaact gctgatctac tacgacgacc tgccggcctag cggcgtgcc 180
gatagatttt ctggcagcaa gagcggcacc tctgccagcc tggtatttct tggactgcag 240
agcgaggacg aggcgacta ttattgtgcc gcctgggacg acagcctgaa cggetatgtt 300
gttttcggcg gaggcaccaa gctgaccgtt ctaggccagc ctaaagccgc ccctagcgtg 360
acctgtttcc ctccaagcag cgaggaactg caggccaaca aggccaccct cgtgtgctg 420
atcagcgact tctatcctgg cgccgtgacc gtggcctgga aggcgatag ctctcctgtg 480
aaggccggcg tggaaccac caccctagc aagcagagca acaacaaata cgccgccagc 540
agctacctga gcctgacccc cgagcagtggt aagtcacaca gatcctacag ctgccaagtg 600
acccacgagg gcagcaccgt ggaaaagaca gtggccccta ccgagtgcag c 651

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<210> SEQ ID NO 141
<211> LENGTH: 120
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 141

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Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1           5           10          15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Tyr Ser Tyr
20          25          30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35          40          45
Ser Ala Ile Gly Thr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys
50          55          60
Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
65          70          75          80
Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
85          90          95
Arg Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr Trp Gly Gln
100         105         110
Gly Thr Leu Val Thr Val Ser Ser
115         120

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<210> SEQ ID NO 142
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 142

Ser Tyr Ala Met Ser
 1 5

<210> SEQ ID NO 143
 <211> LENGTH: 16
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 143

Ala Ile Gly Thr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly
 1 5 10 15

<210> SEQ ID NO 144
 <211> LENGTH: 12
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 144

Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr
 1 5 10

<210> SEQ ID NO 145
 <211> LENGTH: 111
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 145

Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
 1 5 10 15

Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn
 20 25 30

Thr Val Asn Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu
 35 40 45

Ile Tyr Tyr Asp Asp Leu Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
 50 55 60

Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Gln
 65 70 75 80

Ser Glu Asp Glu Ala Asp Tyr Tyr Cys Ala Ala Trp Asp Asp Ser Leu
 85 90 95

Asn Asp Tyr Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
 100 105 110

<210> SEQ ID NO 146
 <211> LENGTH: 13
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 146

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Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn Thr Val Asn
1 5 10

<210> SEQ ID NO 147
 <211> LENGTH: 7
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence
 <400> SEQUENCE: 147

Tyr Asp Asp Leu Arg Pro Ser
1 5

<210> SEQ ID NO 148
 <211> LENGTH: 12
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence
 <400> SEQUENCE: 148

Ala Ala Trp Asp Asp Ser Leu Asn Asp Tyr Val Val
1 5 10

<210> SEQ ID NO 149
 <211> LENGTH: 360
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence
 <400> SEQUENCE: 149

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg	60
agctgtgccg ccagcggcgt caccctttac agctacgcca tgagctgggt ccgacaggcc	120
cctggaaaag gccttgaatg ggtgtccgcc atcggcacag gcggcgatac ctactatgcc	180
gactctgtga agggcagatt caccatcagc cgggacaaca gcaagaacac cctgtacctg	240
cagatgaaca gcctgagagc cgaggacacc gccgtgtact attgcgccag aagggacgac	300
tacaccagca gggacgcctt cgattattgg ggccagggca cactggtcac cgttttctca	360

<210> SEQ ID NO 150
 <211> LENGTH: 15
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence
 <400> SEQUENCE: 150

agctacgcca tgagc	15
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<210> SEQ ID NO 151
 <211> LENGTH: 48
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence
 <400> SEQUENCE: 151

gccatcgga caggcggcga tacctactat gccgactctg tgaagggc	48
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<210> SEQ ID NO 152
 <211> LENGTH: 36
 <212> TYPE: DNA

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 152

agggacgact acaccagcag ggacgccttc gattat          36

<210> SEQ ID NO 153
<211> LENGTH: 333
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 153

cagtctgttc tgacacagcc tcctagcgcc tctggcacac ctggacagag agtgaccatc          60
agctgttagcg gcagcagctc caacatcggc agcaacaccg tgaactggta tcagcagctg          120
cctggcacag cccctaaact gctgatctac tacgacgacc tgcggcctag cggcgtgcca          180
gatagatttt ctggcagcaa gagcggcacc tctgccagcc tggctatttc tggactgcag          240
agcgaggacg aggcgcgacta ttattgtgcc gcctgggacg acagcctgaa cgactacgtt          300
gtgtttggcg gaggcaccaa gctgacgctt cta          333

<210> SEQ ID NO 154
<211> LENGTH: 39
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 154

agcggcagca gctccaacat cggcagcaac accgtgaac          39

<210> SEQ ID NO 155
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 155

tacgacgacc tgcggcctag c          21

<210> SEQ ID NO 156
<211> LENGTH: 36
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 156

gccgcctggg acgacagcct gaacgactac gttgtg          36

<210> SEQ ID NO 157
<211> LENGTH: 447
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 157

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1          5          10          15

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Ser	Leu	Arg	Leu	Ser	Cys	Ala	Ala	Ser	Gly	Phe	Thr	Phe	Tyr	Ser	Tyr	
			20													
Ala	Met	Ser	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Val	
			35													
Ser	Ala	Ile	Gly	Thr	Gly	Gly	Asp	Thr	Tyr	Tyr	Ala	Asp	Ser	Val	Lys	
			50	55			60									
Gly	Arg	Phe	Thr	Ile	Ser	Arg	Asp	Asn	Ser	Lys	Asn	Thr	Leu	Tyr	Leu	
65					70	75										80
Gln	Met	Asn	Ser	Leu	Arg	Ala	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys	Ala	
				85	90											95
Arg	Arg	Asp	Asp	Tyr	Thr	Ser	Arg	Asp	Ala	Phe	Asp	Tyr	Trp	Gly	Gln	
			100	105												110
Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	Lys	Gly	Pro	Ser	Val	
			115	120							125					
Phe	Pro	Leu	Ala	Pro	Cys	Ser	Arg	Ser	Thr	Ser	Glu	Ser	Thr	Ala	Ala	
			130	135							140					
Leu	Gly	Cys	Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu	Pro	Val	Thr	Val	Ser	
145					150	155										160
Trp	Asn	Ser	Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr	Phe	Pro	Ala	Val	
			165	170										175		
Leu	Gln	Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val	Thr	Val	Pro	
			180	185							190					
Ser	Ser	Ser	Leu	Gly	Thr	Lys	Thr	Tyr	Thr	Cys	Asn	Val	Asp	His	Lys	
			195	200							205					
Pro	Ser	Asn	Thr	Lys	Val	Asp	Lys	Arg	Val	Glu	Ser	Lys	Tyr	Gly	Pro	
			210	215							220					
Pro	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Phe	Leu	Gly	Gly	Pro	Ser	Val	
225					230	235										240
Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	
			245	250												255
Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	Gln	Glu	Asp	Pro	Glu	
			260	265							270					
Val	Gln	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	
			275	280							285					
Thr	Lys	Pro	Arg	Glu	Glu	Gln	Phe	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser	
			290	295							300					
Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	
305					310	315										320
Cys	Lys	Val	Ser	Asn	Lys	Gly	Leu	Pro	Ser	Ser	Ile	Glu	Lys	Thr	Ile	
			325	330												335
Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	
			340	345							350					
Pro	Ser	Gln	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	
			355	360							365					
Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	
			370	375							380					
Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp	Ser	
385					390	395										400
Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Arg	Leu	Thr	Val	Asp	Lys	Ser	Arg	
			405	410												415
Trp	Gln	Glu	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	
			420	425							430					
His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Leu	Gly	Lys		

435
440
445

<210> SEQ ID NO 158
 <211> LENGTH: 217
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 158

Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
 1 5 10 15
 Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn
 20 25 30
 Thr Val Asn Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu
 35 40 45
 Ile Tyr Tyr Asp Asp Leu Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
 50 55 60
 Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Gln
 65 70 75 80
 Ser Glu Asp Glu Ala Asp Tyr Tyr Cys Ala Ala Trp Asp Asp Ser Leu
 85 90 95
 Asn Asp Tyr Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu Gly
 100 105 110
 Gln Pro Lys Ala Ala Pro Ser Val Thr Leu Phe Pro Pro Ser Ser Glu
 115 120 125
 Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp Phe
 130 135 140
 Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Ser Ser Pro Val
 145 150 155 160
 Lys Ala Gly Val Glu Thr Thr Thr Pro Ser Lys Gln Ser Asn Asn Lys
 165 170 175
 Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys Ser
 180 185 190
 His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val Glu
 195 200 205
 Lys Thr Val Ala Pro Thr Glu Cys Ser
 210 215

<210> SEQ ID NO 159
 <211> LENGTH: 1341
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 159

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg 60
 agctgtgccc ccagcggcct cacccttttac agctacgccca tgagctgggt ccgacaggcc 120
 cctggaaaag gccttgaatg ggtgtccgcc atcggcacag gcggcgatac ctactatgcc 180
 gactctgtga agggcagatt caccatcagc cgggacaaca gcaagaacac cctgtacctg 240
 cagatgaaca gcctgagagc cgaggacacc gccgtgtact attgcgccag aagggaacgac 300
 tacaccagca gggacgcctt cgattattgg ggccagggca cactggtcac cgttttcttca 360
 gccagcacca agggcccccag cgtgttcctt ctggcccctt gtagcagaag caccagcgag 420
 tctacagccg ccttgggctg cctcgtgaag gactactttc ccgagcccgt gaccgtgtcc 480

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tggaactctg gcgctctgac aagcggcgtg cacacctttc cagccgtgct gcagagcagc 540
ggcctgtact ctctgagcag cgtcgtgaca gtgcccagca gcagcctggg caccaagacc 600
tacacctgta acgtggacca caagcccagc aacaccaagg tggacaagcg ggtggaatct 660
aagtacggcc ctcctgccc tcttgccca gcccctgaat ttctggggcg accctccgtg 720
ttcctgttcc ccccaaagcc caaggacacc ctgatgatca gccggacccc cgaagtgacc 780
tgctgtgtgg tggatgtgtc ccaggaagat cccgaggtgc agttcaattg gtacgtggac 840
ggcgtggaag tgcacaacgc caagaccaag cccagagagg aacagttcaa cagcacctac 900
cgggtggtgt ccgtgctgac agtgtgtcac caggactggc tgaacggcaa agagtacaag 960
tgcaagggtg ccaacaaggg cctgcccagc tccatcgaga aaaccatcag caaggccaag 1020
ggccagcccc gcgaacccca ggtgtacaca ctgcctccaa gccaggaaga gatgaccaag 1080
aaccaggtgt ccctgacctg tctcgtgaaa ggcttctacc cctccgatat cgccgtggaa 1140
tgggagagca acggccagcc cgagaacaac tacaagacca cccccctgt gctggacagc 1200
gacggctcat tcttctgta cagcagactg accgtggaca agagccgggtg gcaggaaggc 1260
aacgtgttca gctgcagcgt gatgcacgag gccctgcaca accactacac ccagaagtcc 1320
ctgtctctga gcctgggcaa g 1341

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<210> SEQ ID NO 160
<211> LENGTH: 651
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 160

```

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cagttctgtt tgacacagcc tcttagcgcc tctggcacac ctggacagag agtgaccatc 60
agctgtagcg gcagcagctc caacatcggc agcaacacgc tgaactggta tcagcagctg 120
cctggcacag cccctaaact gctgatctac tacgacgacc tgcggcctag cggcgtgcca 180
gatagatttt ctggcagcaa gagcggcacc tctgccagcc tggctatttc tggactgcag 240
agcgaggacg agggcgacta ttattgtgcc gcctgggacg acagcctgaa cgactacgtt 300
gtgtttggcg gaggcaccaa gctgaccgtt ctaggccagc ctaaagccgc ccctagcgtg 360
accctgttcc ctccaagcag cgaggaaactg caggccaaca agggccacct cgtgtgcctg 420
atcagcgact tctatcctgg cgccgtgacc gtggcctgga agggcgatag ctctcctgtg 480
aaggccggcg tggaaaccac caccctagc aagcagagca acaacaata cgccgccagc 540
agctacctga gcctgacccc cgagcagtggt aagtcacaca gatcctacag ctgccaagtg 600
accacagagg gcagcaccgt ggaaaagaca gtggccccta ccgagtgcag c 651

```

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<210> SEQ ID NO 161
<211> LENGTH: 120
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 161

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Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1           5           10          15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20          25          30

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-continued

Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ser Ala Ile Gly Tyr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys
 50 55 60

Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
 65 70 75 80

Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
 85 90 95

Arg Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr Trp Gly Gln
 100 105 110

Gly Thr Leu Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 162
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 162

Ser Tyr Ala Met Ser
 1 5

<210> SEQ ID NO 163
 <211> LENGTH: 16
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 163

Ala Ile Gly Tyr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly
 1 5 10 15

<210> SEQ ID NO 164
 <211> LENGTH: 12
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 164

Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr
 1 5 10

<210> SEQ ID NO 165
 <211> LENGTH: 111
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 165

Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
 1 5 10 15

Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn
 20 25 30

Thr Val Asn Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu
 35 40 45

Ile Tyr Tyr Asp Asp Leu Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
 50 55 60

-continued

Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Gln
65 70 75 80

Ser Glu Asp Glu Ala Asp Tyr Tyr Cys Ala Ala Trp Asp Asp Ser Leu
85 90 95

Asn Asp Ile Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
100 105 110

<210> SEQ ID NO 166
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 166

Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn Thr Val Asn
1 5 10

<210> SEQ ID NO 167
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 167

Tyr Asp Asp Leu Arg Pro Ser
1 5

<210> SEQ ID NO 168
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 168

Ala Ala Trp Asp Asp Ser Leu Asn Asp Ile Val Val
1 5 10

<210> SEQ ID NO 169
<211> LENGTH: 360
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 169

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg	60
agctgtgccg ccagcggcctt cacctttagc agctacgcca tgagctgggt ccgacaggct	120
cctggcaaag gccttgaatg ggtgtccgcc atcggctatg gcggcgatac ctactacgcc	180
gactctgtga agggcagatt caccatcagc cgggacaaca gcaagaacac cctgtacctg	240
cagatgaaca gcctgagagc cgaggacacc gccgtgtact attgcgccag aagggaacgac	300
tacaccagca gggacgcctt cgattattgg ggccagggca cactggtcac cgttttctca	360

<210> SEQ ID NO 170
<211> LENGTH: 15
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 170

-continued

agctacgccca tgagc 15

<210> SEQ ID NO 171
<211> LENGTH: 48
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 171

gccatcggt atggcggcga tacctactac gccgactctg tgaagggc 48

<210> SEQ ID NO 172
<211> LENGTH: 36
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 172

agggacgact acaccagcag ggacgccttc gattat 36

<210> SEQ ID NO 173
<211> LENGTH: 333
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 173

cagtctgttc tgacacagcc tcctagcgcc tctggcacac ctggacagag agtgaccatc 60

agctgtagcg gcagcagctc caacatcggc agcaacaccg tgaactggta tcagcagctg 120

cctggcacag cccctaaact gctgatctac tacgacgacc tgccggcctag cggcgtgcc 180

gatatatttt ctggcagcaa gagcggcacc tctgccagcc tggtatttc tggactgcag 240

agcaggagcag aggcgacta ttattgtgcc gcctgggacg acagcctgaa cgacatcggt 300

gttttcggcg gaggcaccaa gctgaccgtt cta 333

<210> SEQ ID NO 174
<211> LENGTH: 39
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 174

agcggcagca gctccaacat cggcagcaac accgtgaac 39

<210> SEQ ID NO 175
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 175

tacgacgacc tgcggcctag c 21

<210> SEQ ID NO 176
<211> LENGTH: 36
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:

-continued

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 176

gcgcctggg acgacagcct gaacgacatc gttgtt

36

<210> SEQ ID NO 177

<211> LENGTH: 447

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 177

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20 25 30

Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45

Ser Ala Ile Gly Tyr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys
50 55 60

Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
65 70 75 80

Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
85 90 95

Arg Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr Trp Gly Gln
100 105 110

Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val
115 120 125

Phe Pro Leu Ala Pro Cys Ser Arg Ser Thr Ser Glu Ser Thr Ala Ala
130 135 140

Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser
145 150 155 160

Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val
165 170 175

Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro
180 185 190

Ser Ser Ser Leu Gly Thr Lys Thr Tyr Thr Cys Asn Val Asp His Lys
195 200 205

Pro Ser Asn Thr Lys Val Asp Lys Arg Val Glu Ser Lys Tyr Gly Pro
210 215 220

Pro Cys Pro Pro Cys Pro Ala Pro Glu Phe Leu Gly Gly Pro Ser Val
225 230 235 240

Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr
245 250 255

Pro Glu Val Thr Cys Val Val Val Asp Val Ser Gln Glu Asp Pro Glu
260 265 270

Val Gln Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys
275 280 285

Thr Lys Pro Arg Glu Glu Gln Phe Asn Ser Thr Tyr Arg Val Val Ser
290 295 300

Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys
305 310 315 320

Cys Lys Val Ser Asn Lys Gly Leu Pro Ser Ser Ile Glu Lys Thr Ile
325 330 335

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Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro
 340 345 350

Pro Ser Gln Glu Glu Met Thr Lys Asn Gln Val Ser Leu Thr Cys Leu
 355 360 365

Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn
 370 375 380

Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser
 385 390 395 400

Asp Gly Ser Phe Phe Leu Tyr Ser Arg Leu Thr Val Asp Lys Ser Arg
 405 410 415

Trp Gln Glu Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu
 420 425 430

His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Leu Gly Lys
 435 440 445

<210> SEQ ID NO 178
 <211> LENGTH: 217
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 178

Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
 1 5 10 15

Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn
 20 25 30

Thr Val Asn Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu
 35 40 45

Ile Tyr Tyr Asp Asp Leu Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
 50 55 60

Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Gln
 65 70 75 80

Ser Glu Asp Glu Ala Asp Tyr Tyr Cys Ala Ala Trp Asp Asp Ser Leu
 85 90 95

Asn Asp Ile Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu Gly
 100 105 110

Gln Pro Lys Ala Ala Pro Ser Val Thr Leu Phe Pro Pro Ser Ser Glu
 115 120 125

Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp Phe
 130 135 140

Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Ser Ser Pro Val
 145 150 155 160

Lys Ala Gly Val Glu Thr Thr Thr Pro Ser Lys Gln Ser Asn Asn Lys
 165 170 175

Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys Ser
 180 185 190

His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val Glu
 195 200 205

Lys Thr Val Ala Pro Thr Glu Cys Ser
 210 215

<210> SEQ ID NO 179
 <211> LENGTH: 1341
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:

-continued

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 179

gaagttcagc	tgctggaatc	tggcggcggg	ctgggtcaac	ctggcggatc	tctgagactg	60
agctgtgccg	ccagcggctt	cacctttagc	agctacgcc	tgagctgggt	ccgacaggct	120
cctggcaaag	gccttgaatg	ggtgtccgcc	atcggtatg	gcggcgatac	ctactacgcc	180
gactctgtga	agggcagatt	caccatcagc	cgggacaaca	gcaagaacac	cctgtacctg	240
cagatgaaca	gcctgagagc	cgaggacacc	gccgtgtact	attgcgccag	aagggaagac	300
tacaccagca	gggacgcctt	cgattattgg	ggccagggca	cactggtcac	cgttttctca	360
gccagcacca	agggccccag	cgtgttcctt	ctggccccct	gtagcagaag	caccagcgag	420
tctacagccg	ccctgggctg	cctcgtgaag	gactactttc	ccgagcccgt	gaccgtgtcc	480
tggaaactctg	gcgctctgac	aagcggcgtg	cacacctttc	cagccgtgct	gcagagcagc	540
ggcctgtact	ctctgagcag	cgtcgtgaca	gtgcccagca	gcagcctggg	caccaagacc	600
tacacctgta	acgtggacca	caagcccagc	aacaccaagg	tggacaagcg	ggtggaatct	660
aagtacggcc	ctccctgccc	tccttgccca	gccctgaaat	ttctgggcgg	accctccgtg	720
ttcctgttcc	ccccaaagcc	caaggacacc	ctgatgatca	gccggacccc	cgaagtgacc	780
tgcgtggtgg	tggatgtgtc	ccaggaagat	cccagaggtc	agttcaattg	gtacgtggac	840
ggcgtggaag	tgcacaacgc	caagaccaag	cccagagagg	aacagttcaa	cagcacctac	900
cgggtggtgt	ccgtgtgac	agtgtgcac	caggactggc	tgaacggcaa	agagtacaag	960
tgcaaggtgt	ccaacaaggg	cctgcccagc	tccatcgaga	aaaccatcag	caaggccaag	1020
ggccagcccc	gcgaacccca	ggtgtacaca	ctgcctccaa	gccaggaaga	gatgaccaag	1080
aaccaggtgt	ccctgacctg	tctcgtgaaa	ggcttctacc	cctccgatat	cgcctgggaa	1140
tgggagagca	acggccagcc	cgagaacaac	tacaagacca	ccccccctgt	gctggacagc	1200
gacggctcat	tcttctgtga	cagcagactg	accgtggaca	agagccgggtg	gcaggaaggc	1260
aacgtgttca	gctgcagcgt	gatgcacgag	gccctgcaca	accactacac	ccagaagtcc	1320
ctgtctctga	gcctgggcaa	g				1341

<210> SEQ ID NO 180

<211> LENGTH: 651

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 180

cagtctgttc	tgacacagcc	tcctagcgcc	tctggcacac	ctggacagag	agtgaccatc	60
agctgtagcg	gcagcagctc	caacatcggc	agcaacaccg	tgaactggta	tcagcagctg	120
cctggcacag	cccctaaact	gctgatctac	tacgacgacc	tgcggcctag	cggcgtgcca	180
gatagatttt	ctggcagcaa	gagcggcacc	tctgccagcc	tggctatttc	tggactgcag	240
agcgaggacg	aggccgacta	ttattgtgcc	gcctgggacg	acagcctgaa	cgacatcggt	300
gttttcggcg	gaggcaccaa	gctgaccggt	ctaggccagc	ctaaagccgc	ccctagcgtg	360
accctgttcc	ctccaagcag	cgaggaaactg	caggccaaca	aggccaccct	cgtgtgcctg	420
atcagcgact	tctatcctgg	cgccgtgacc	gtggcctgga	aggccgatag	ctctcctgtg	480
aaggccggcg	tggaaaccac	cacccttagc	aagcagagca	acaacaaata	cgccgccagc	540
agctacctga	gcctgacccc	cgagcagtg	aagtcacaca	gacccctacg	ctgccaagtg	600

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accacgagg gcagcacgt ggaaaagaca gtggccccta ccgagtgcag c

651

<210> SEQ ID NO 181
<211> LENGTH: 120
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 181

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20 25 30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45
Ser Ala Ile Gly Tyr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys
50 55 60
Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
65 70 75 80
Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
85 90 95
Arg Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr Trp Gly Gln
100 105 110
Gly Thr Leu Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 182
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 182

Ser Tyr Ala Met Ser
1 5

<210> SEQ ID NO 183
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 183

Ala Ile Gly Tyr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly
1 5 10 15

<210> SEQ ID NO 184
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 184

Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr
1 5 10

<210> SEQ ID NO 185
<211> LENGTH: 111

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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 185

Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
1           5           10           15

Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn
20          25          30

Thr Val Asn Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu
35          40          45

Ile Tyr Tyr Asp Asp Leu Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
50          55          60

Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Gln
65          70          75          80

Ser Glu Asp Glu Ala Asp Tyr Tyr Cys Ala Ala Trp Asp Asp Ser Leu
85          90          95

Asn Val Tyr Pro Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
100         105         110

```

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<210> SEQ ID NO 186
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 186

Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn Thr Val Asn
1           5           10

```

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<210> SEQ ID NO 187
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 187

Tyr Asp Asp Leu Arg Pro Ser
1           5

```

```

<210> SEQ ID NO 188
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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```

<400> SEQUENCE: 188

Ala Ala Trp Asp Asp Ser Leu Asn Val Tyr Pro Val
1           5           10

```

```

<210> SEQ ID NO 189
<211> LENGTH: 360
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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```

<400> SEQUENCE: 189

```

```

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg 60

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agctgtgccg ccagcggcctt caccttttagc agctacgccca tgagctgggt ccgacaggct	120
cctggcaaag gccttgaatg ggtgtccgcc atcggtatg gcggcgatac ctactacgcc	180
gactctgtga agggcagatt caccatcagc cgggacaaca gcaagaacac cctgtacctg	240
cagatgaaca gcctgagagc cgaggacacc gccgtgtact attgcgccag aagggacgac	300
tacaccagca gggacgcctt cgattattgg ggccagggca cactggtcac cgtttcttca	360

<210> SEQ ID NO 190
 <211> LENGTH: 15
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 190

agctacgccca tgagc	15
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<210> SEQ ID NO 191
 <211> LENGTH: 48
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 191

gccatcggt atggcgcgca tacctactac gccgactctg tgaagggc	48
---	----

<210> SEQ ID NO 192
 <211> LENGTH: 36
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 192

agggacgact acaccagcag ggacgccttc gattat	36
---	----

<210> SEQ ID NO 193
 <211> LENGTH: 333
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 193

cagtctgttc tgacacagcc tcctagcgcc tctggcacac ctggacagag agtgaccatc	60
agctgtagcg gcagcagctc caacatcggc agcaacaccg tgaactggta tcagcagctg	120
cctggcacag cccctaaact gctgatctac tacgacgacc tgcggcctag cggcgtgcc	180
gatagatttt ctggcagcaa gagcggcacc tctgccagcc tggctatttc tggactgcag	240
agcggaggacg agggcgacta ttattgtgcc gcctgggacg acagcctgaa cgtgtaccct	300
gtttttggcg gaggcaccaa gctgaccgtt cta	333

<210> SEQ ID NO 194
 <211> LENGTH: 39
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 194

agcggcagca gctccaacat cggcagcaac accgtgaac	39
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<210> SEQ ID NO 195
 <211> LENGTH: 21
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 195

tacgacgacc tgcggcctag c

21

<210> SEQ ID NO 196
 <211> LENGTH: 36
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 196

gccgcctggg acgacagcct gaacgtgtac cctgtt

36

<210> SEQ ID NO 197
 <211> LENGTH: 447
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 197

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
 20 25 30

Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ser Ala Ile Gly Tyr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys
 50 55 60

Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
 65 70 75 80

Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
 85 90 95

Arg Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr Trp Gly Gln
 100 105 110

Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val
 115 120 125

Phe Pro Leu Ala Pro Cys Ser Arg Ser Thr Ser Glu Ser Thr Ala Ala
 130 135 140

Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser
 145 150 155 160

Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val
 165 170 175

Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro
 180 185 190

Ser Ser Ser Leu Gly Thr Lys Thr Tyr Thr Cys Asn Val Asp His Lys
 195 200 205

Pro Ser Asn Thr Lys Val Asp Lys Arg Val Glu Ser Lys Tyr Gly Pro
 210 215 220

Pro Cys Pro Pro Cys Pro Ala Pro Glu Phe Leu Gly Gly Pro Ser Val
 225 230 235 240

Phe	Leu	Phe	Pro	Pro 245	Lys	Pro	Lys	Asp	Thr 250	Leu	Met	Ile	Ser	Arg	Thr 255
Pro	Glu	Val	Thr 260	Cys	Val	Val	Val	Asp 265	Val	Ser	Gln	Glu	Asp 270	Pro	Glu
Val	Gln	Phe 275	Asn	Trp	Tyr	Val	Asp 280	Gly	Val	Glu	Val	His 285	Asn	Ala	Lys
Thr	Lys 290	Pro	Arg	Glu	Glu	Gln	Phe 295	Asn	Ser	Thr	Tyr 300	Arg	Val	Val	Ser
Val 305	Leu	Thr	Val	Leu	His 310	Gln	Asp	Trp	Leu	Asn 315	Gly	Lys	Glu	Tyr	Lys 320
Cys	Lys	Val	Ser	Asn 325	Lys	Gly	Leu	Pro	Ser 330	Ser	Ile	Glu	Lys	Thr 335	Ile
Ser	Lys	Ala	Lys 340	Gly	Gln	Pro	Arg	Glu 345	Pro	Gln	Val	Tyr	Thr 350	Leu	Pro
Pro	Ser	Gln	Glu	Glu	Met	Thr	Lys 360	Asn	Gln	Val	Ser	Leu 365	Thr	Cys	Leu
Val	Lys 370	Gly	Phe	Tyr	Pro	Ser	Asp 375	Ile	Ala	Val	Glu 380	Trp	Glu	Ser	Asn
Gly 385	Gln	Pro	Glu	Asn 390	Asn	Tyr	Lys	Thr	Thr 395	Pro	Pro	Val	Leu	Asp 400	Ser
Asp	Gly	Ser	Phe 405	Phe	Leu	Tyr	Ser	Arg	Leu 410	Thr	Val	Asp	Lys	Ser 415	Arg
Trp	Gln	Glu	Gly 420	Asn	Val	Phe	Ser	Cys 425	Ser	Val	Met	His 430	Glu	Ala	Leu
His	Asn 435	His	Tyr	Thr	Gln	Lys	Ser 440	Leu	Ser	Leu	Ser	Leu 445	Gly	Lys	

Gln 1	Ser	Val	Leu	Thr 5	Gln	Pro	Pro	Ser	Ala 10	Ser	Gly	Thr	Pro	Gly 15	Gln
Arg	Val	Thr	Ile 20	Ser	Cys	Ser	Gly	Ser 25	Ser	Ser	Asn	Ile	Gly 30	Ser	Asn
Thr	Val	Asn	Trp 35	Tyr	Gln	Gln	Leu 40	Pro	Gly	Thr	Ala	Pro 45	Lys	Leu	Leu
Ile	Tyr 50	Tyr	Asp	Asp	Leu	Arg 55	Pro	Ser	Gly	Val	Pro 60	Asp	Arg	Phe	Ser
Gly 65	Ser	Lys	Ser	Gly	Thr 70	Ser	Ala	Ser	Leu	Ala 75	Ile	Ser	Gly	Leu 80	Gln
Ser	Glu	Asp	Glu	Ala 85	Asp	Tyr	Tyr	Cys	Ala 90	Ala	Trp	Asp	Asp	Ser 95	Leu
Asn	Val	Tyr	Pro 100	Val	Phe	Gly	Gly	Gly 105	Thr	Lys	Leu	Thr	Val 110	Leu	Gly
Gln	Pro	Lys 115	Ala	Ala	Pro	Ser	Val	Thr 120	Leu	Phe	Pro	Pro 125	Ser	Ser	Glu
Glu 130	Leu	Gln	Ala	Asn	Lys	Ala 135	Thr	Leu	Val	Cys	Leu 140	Ile	Ser	Asp	Phe
Tyr 145	Pro	Gly	Ala	Val 150	Thr	Val	Ala	Trp	Lys	Ala 155	Asp	Ser	Ser	Pro	Val 160

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Lys Ala Gly Val Glu Thr Thr Thr Pro Ser Lys Gln Ser Asn Asn Lys
 165 170 175

Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys Ser
 180 185 190

His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val Glu
 195 200 205

Lys Thr Val Ala Pro Thr Glu Cys Ser
 210 215

<210> SEQ ID NO 199

<211> LENGTH: 1341

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 199

```

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg      60
agctgtgccg ccagcggett cacctttagc agctacgcc tgaagctgggt ccgacaggct      120
cctggcaaa gacctgaatg ggtgtccgcc atcggtatg gcggcgatac ctactacgcc      180
gactctgtga agggcagatt caccatcagc cgggacaaca gcaagaacac cctgtacctg      240
cagatgaaca gcctgagagc cgaggacacc gccgtgtact attgcgccag aagggaagac      300
tacaccagca gggacgcctt cgattattgg ggccagggca cactggtcac cgtttcttca      360
gccagcacca agggccccag cgtgttcctt ctggccccct gtagcagaag caccagcgag      420
tctacagccg ccttgggctg cctcgtgaag gactactttc ccgagcccg gaccgtgtcc      480
tggaactctg gcgctctgac aagcggcgtg cacaccttc cagccgtgct gcagagcagc      540
ggcctgtact ctctgagcag cgtcgtgaca gtgccagca gcagcctggg caccaagacc      600
tacacctgta acgtggacca caagcccagc aacaccaagg tggacaagcg ggtggaatct      660
aagtaaggcc ctccctgccc tccttgccca gccctgaat ttctgggcgg acctccgtg      720
ttcctgttcc cccaaaagcc caaggacacc ctgatgatca gccggacccc cgaagtgacc      780
tgcggtgttg tggatgtgtc ccaggaagat cccaggtgct agttcaattg gtacgtggac      840
ggcgtggaag tgcacaacgc caagaccaag cccagagagg aacagttcaa cagcacctac      900
cgggtgtgtg ccgtgctgac agtgctgcac caggactggc tgaacggcaa agagtacaag      960
tgcaaggtgt ccaacaaggg cctgcccagc tccatcgaga aaaccatcag caaggccaag      1020
ggccagcccc gcgaacccca ggtgtacaca ctgcctccaa gccaggaaga gatgaccaag      1080
aaccaggtgt ccctgacctg tctcgtgaaa ggcttctacc cctccgatat cgcctgggaa      1140
tggaagagca acggccagcc cgagaacaac tacaagacca cccccctgt gctggacagc      1200
gacggctcat tcttctgta cagcagactg accgtggaca agagccggtg gcaggaaggc      1260
aacgtgttca gctgcagcgt gatgcacgag gccctgcaca accactacac ccagaagtcc      1320
ctgtctctga gcctgggcaa g                                     1341

```

<210> SEQ ID NO 200

<211> LENGTH: 651

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 200

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```

cagtctgttc tgacacagcc tcctagcgcc tctggcacac ctggacagag agtgaccatc      60
agctgtagcg gcagcagctc caacatcggc agcaacaccg tgaactggta tcagcagctg      120
cctggcacag cccctaaact gctgatctac tacgacgacc tgcggcctag cggcgtgcca      180
gatagatttt ctggcagcaa gagcggcacc tctgccagcc tggctatttc tggactgcag      240
agcgaggacg aggcgcgacta ttattgtgcc gcctgggacg acagcctgaa cgtgtaccct      300
gtttttggcg gaggcaccaa gctgaccgtt ctaggccagc ctaaagccgc ccctagcgtg      360
accctgttcc ctccaagcag cgaggaactg caggccaaca aggccaccct cgtgtgctg      420
atcagcgact tctatcctgg cgccgtgacc gtggcctgga aggcgatag ctctcctgtg      480
aaggccggcg tggaaaccac caccctagc aagcagagca acaacaata cgccgccagc      540
agctacctga gcctgacccc cgagcagtgg aagtcacaca gatcctacag ctgccaagtg      600
accacagagg gcagcaccgt ggaaaagaca gtggccccta ccgagtgcag c              651

```

```

<210> SEQ ID NO 201
<211> LENGTH: 120
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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```

<400> SEQUENCE: 201

```

```

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1          5          10          15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20        25        30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35        40        45
Ser Ala Ile Gly Tyr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys
50        55        60
Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
65        70        75        80
Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
85        90        95
Arg Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr Trp Gly Gln
100       105       110
Gly Thr Leu Val Thr Val Ser Ser
115       120

```

```

<210> SEQ ID NO 202
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 202

```

```

Ser Tyr Ala Met Ser
1          5

```

```

<210> SEQ ID NO 203
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 203

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-continued

Ala Ile Gly Tyr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly
1 5 10 15

<210> SEQ ID NO 204
 <211> LENGTH: 12
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 204

Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr
1 5 10

<210> SEQ ID NO 205
 <211> LENGTH: 111
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 205

Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
1 5 10 15

Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn
20 25 30

Thr Val Asn Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu
35 40 45

Ile Tyr Tyr Asp Asp Leu Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
50 55 60

Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Gln
65 70 75 80

Ser Glu Asp Glu Ala Asp Tyr Tyr Cys His Ala Trp Asp Asp Ser Leu
85 90 95

Asn Asp Ile Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
100 105 110

<210> SEQ ID NO 206
 <211> LENGTH: 13
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 206

Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn Thr Val Asn
1 5 10

<210> SEQ ID NO 207
 <211> LENGTH: 7
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 207

Tyr Asp Asp Leu Arg Pro Ser
1 5

<210> SEQ ID NO 208
 <211> LENGTH: 12
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:

-continued

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 208

His Ala Trp Asp Asp Ser Leu Asn Asp Ile Val Val
 1 5 10

<210> SEQ ID NO 209

<211> LENGTH: 360

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 209

```

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg      60
agctgtgccc ccagcggett cacccttagc agctacgcca tgagctgggt ccgacaggct      120
cctggcaaag gccttgaatg ggtgtccgcc atcggtatg gcggcgatac ctactacgcc      180
gactctgtga agggcagatt caccatcagc cgggacaaca gcaagaacac cctgtacctg      240
cagatgaaca gcctgagagc cgaggacacc gccgtgtact attgcgccag aagggaagcag      300
tacaccagca gggacgcctt cgattattgg ggccagggca cactggtcac cgtttcttca      360

```

<210> SEQ ID NO 210

<211> LENGTH: 15

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 210

```

agctacgcca tgagc      15

```

<210> SEQ ID NO 211

<211> LENGTH: 48

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 211

```

gccatcggt atggcggcga tacctactac gccgactctg tgaagggc      48

```

<210> SEQ ID NO 212

<211> LENGTH: 36

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 212

```

agggacgact acaccagcag ggacgccttc gattat      36

```

<210> SEQ ID NO 213

<211> LENGTH: 333

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 213

```

cagtcgtgtc tgacacagcc tcctagcgcc tctggcacac ctggacagag agtgaccatc      60
agctgtagcg gcagcagctc caacatcggc agcaacaccg tgaactggta tcagcagctg      120

```

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```

cctggcacag cccctaaact gctgatctac tacgacgacc tgcggcctag cggcgtgcca 180
gatagatttt ctggcagcaa gagcggcacc tctgccagcc tggctatttc tggactgcag 240
agcgaggacg aggcgacta ctattgtcac gcctgggacg acagcctgaa cgacatcgtg 300
gtttttggcg gaggcaccaa gctgaccgtt cta 333

```

```

<210> SEQ ID NO 214
<211> LENGTH: 39
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 214

```

```

agcggcagca gctccaacat cggcagcaac accgtgaac 39

```

```

<210> SEQ ID NO 215
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 215

```

```

tacgacgacc tgcggcctag c 21

```

```

<210> SEQ ID NO 216
<211> LENGTH: 36
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 216

```

```

cacgcctggg acgacagcct gaacgacatc gtggtt 36

```

```

<210> SEQ ID NO 217
<211> LENGTH: 447
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 217

```

```

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1           5           10          15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20          25          30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35          40          45
Ser Ala Ile Gly Tyr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys
50          55          60
Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
65          70          75          80
Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
85          90          95
Arg Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr Trp Gly Gln
100         105         110
Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val
115         120         125
Phe Pro Leu Ala Pro Cys Ser Arg Ser Thr Ser Glu Ser Thr Ala Ala

```

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130	135	140
Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser		
145	150	155 160
Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val		
	165	170 175
Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro		
	180	185 190
Ser Ser Ser Leu Gly Thr Lys Thr Tyr Thr Cys Asn Val Asp His Lys		
	195	200 205
Pro Ser Asn Thr Lys Val Asp Lys Arg Val Glu Ser Lys Tyr Gly Pro		
	210	215 220
Pro Cys Pro Pro Cys Pro Ala Pro Glu Phe Leu Gly Gly Pro Ser Val		
	225	230 235 240
Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr		
	245	250 255
Pro Glu Val Thr Cys Val Val Val Asp Val Ser Gln Glu Asp Pro Glu		
	260	265 270
Val Gln Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys		
	275	280 285
Thr Lys Pro Arg Glu Glu Gln Phe Asn Ser Thr Tyr Arg Val Val Ser		
	290	295 300
Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys		
	305	310 315 320
Cys Lys Val Ser Asn Lys Gly Leu Pro Ser Ser Ile Glu Lys Thr Ile		
	325	330 335
Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro		
	340	345 350
Pro Ser Gln Glu Glu Met Thr Lys Asn Gln Val Ser Leu Thr Cys Leu		
	355	360 365
Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn		
	370	375 380
Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser		
	385	390 395 400
Asp Gly Ser Phe Phe Leu Tyr Ser Arg Leu Thr Val Asp Lys Ser Arg		
	405	410 415
Trp Gln Glu Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu		
	420	425 430
His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Leu Gly Lys		
	435	440 445

<210> SEQ ID NO 218

<211> LENGTH: 217

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 218

Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln		
1	5	10 15
Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn		
	20	25 30
Thr Val Asn Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu		
	35	40 45
Ile Tyr Tyr Asp Asp Leu Arg Pro Ser Gly Val Pro Asp Arg Phe Ser		

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50	55	60
Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Gln		
65	70	75 80
Ser Glu Asp Glu Ala Asp Tyr Tyr Cys His Ala Trp Asp Asp Ser Leu		
	85	90 95
Asn Asp Ile Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu Gly		
	100	105 110
Gln Pro Lys Ala Ala Pro Ser Val Thr Leu Phe Pro Pro Ser Ser Glu		
	115	120 125
Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp Phe		
	130	135 140
Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Ser Ser Pro Val		
	145	150 155 160
Lys Ala Gly Val Glu Thr Thr Thr Pro Ser Lys Gln Ser Asn Asn Lys		
	165	170 175
Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys Ser		
	180	185 190
His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val Glu		
	195	200 205
Lys Thr Val Ala Pro Thr Glu Cys Ser		
	210	215

<210> SEQ ID NO 219

<211> LENGTH: 1341

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 219

gaagttcagc	tgctggaatc	tggcggcgga	ctggttcaac	ctggcggatc	tctgagactg	60
agctgtgccg	ccagcggett	cacctttagc	agctacgcca	tgagctgggt	ccgacaggct	120
cctggcaaag	gccttgaatg	ggtgtccgcc	atcggtatg	gcggcgatac	ctactacgcc	180
gactctgtga	agggcagatt	caccatcagc	cgggacaaca	gcaagaacac	cctgtacctg	240
cagatgaaca	gcctgagagc	cgaggacacc	gccgtgtact	attgcgccag	aaggggacgac	300
tacaccagca	gggacgcctt	cgattattgg	ggccagggca	cactggctac	cgttttcttca	360
gccagcacca	agggccccag	cgtgttcctt	ctggcccctt	gtagcagaag	caccagcgag	420
tctacagccg	ccctgggctg	cctcgtgaag	gactactttc	ccgagcccg	gacctgtgcc	480
tggaaactctg	gcgctctgac	aagcggcggtg	cacacctttc	cagccgtgct	gcagagcagc	540
ggcctgtact	ctctgagcag	cgtcgtgaca	gtgcccagca	gcagcctggg	caccaagacc	600
tacacctgta	acgtggacca	caagcccagc	aacaccaagg	tggacaagcg	ggtggaatct	660
aagtacggcc	ctccctgccc	tccttgcccc	gcccctgaat	ttctgggcgg	acctccgtg	720
ttcctgttcc	ccccaaagcc	caaggacacc	ctgatgatca	gccggacccc	cgaagtgacc	780
tgctggtgg	tggatgtgtc	ccaggaagat	cccaggtg	agttcaattg	gtacgtggac	840
ggcgtggaag	tgacacaacg	caagaccaag	cccagagagg	aacagttcaa	cagcacctac	900
cgggtggtgt	ccgtgctgac	agtgtgtcac	caggactggc	tgaacggcaa	agagtacaag	960
tgcaagggtg	ccaacaagg	cctgcccagc	tccatcgaga	aaaccatcag	caaggccaag	1020
ggccagcccc	gcgaaccccc	ggtgtacaca	ctgcctccaa	gccaggaaga	gatgaccaag	1080
aaccaggtgt	ccctgacctg	tctcgtgaaa	ggcttctacc	cctccgatat	cgccgtggaa	1140

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tgggagagca acggccagcc cgagaacaac tacaagacca cccccctgt gctggacagc 1200
gacggctcat tcttctgtga cagcagactg accgtggaca agagccggtg gcaggaaggc 1260
aacgtgttca gctgcagcgt gatgcacgag gccctgcaca accactacac ccagaagtcc 1320
ctgtctctga gcctgggcaa g                                     1341

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<210> SEQ ID NO 220
<211> LENGTH: 651
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 220

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cagtctgttc tgacacagcc tcctagcgcc tctggcacac ctggacagag agtgaccatc 60
agctgtagcg gcagcagctc caacatcggc agcaacaccg tgaactggta tcagcagctg 120
cctggcacag cccctaaact gctgatctac tacgacgacc tgcggcctag cggcgtgcc 180
gatagatttt ctggcagcaa gagcggcacc tctgccagcc tggctatttc tggactgcag 240
agcgaggacg aggcgacta ctattgtcac gcctgggacg acagcctgaa cgacatcgtg 300
gtttttggcg gaggcaccaa gctgaccgtt ctaggccagc ctaaagccgc ccctagcgtg 360
acctgtttcc ctccaagcag cgaggaactg caggccaaca aggccaccct cgtgtgctg 420
atcagcgact tctatcctgg cgccgtgacc gtggcctgga aggcgatag ctctcctgtg 480
aaggccggcg tggaaccac caccctagc aagcagagca acaacaaata cgccgccagc 540
agctacctga gcctgacccc cgagcagtggt aagtcccaca gatcctacag ctgccaaagt 600
acccacgagg gcagcacctg ggaaaagaca gtggccccta ccgagtgcag c 651

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<210> SEQ ID NO 221
<211> LENGTH: 120
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 221

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Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1           5           10          15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20          25          30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35          40          45
Ser Ala Ile Gly Tyr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys
50          55          60
Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
65          70          75          80
Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
85          90          95
Arg Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr Trp Gly Gln
100         105         110
Gly Thr Leu Val Thr Val Ser Ser
115         120

```

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<210> SEQ ID NO 222
<211> LENGTH: 5
<212> TYPE: PRT

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-continued

<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 222

Ser Tyr Ala Met Ser
1 5

<210> SEQ ID NO 223
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 223

Ala Ile Gly Tyr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly
1 5 10 15

<210> SEQ ID NO 224
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 224

Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr
1 5 10

<210> SEQ ID NO 225
<211> LENGTH: 111
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 225

Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
1 5 10 15

Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn
20 25 30

Thr Val Asn Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu
35 40 45

Ile Tyr Tyr Asp Asp Leu Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
50 55 60

Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Gln
65 70 75 80

Ser Glu Asp Glu Ala Asp Tyr Tyr Cys His Ala Trp Asp Asp Ser Leu
85 90 95

Asn Asp Tyr Pro Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
100 105 110

<210> SEQ ID NO 226
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 226

Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn Thr Val Asn
1 5 10

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<210> SEQ ID NO 227
 <211> LENGTH: 7
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 227

Tyr Asp Asp Leu Arg Pro Ser
 1 5

<210> SEQ ID NO 228
 <211> LENGTH: 12
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 228

His Ala Trp Asp Asp Ser Leu Asn Asp Tyr Pro Val
 1 5 10

<210> SEQ ID NO 229
 <211> LENGTH: 360
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 229

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg	60
agctgtgccg ccagcggcgt cacccttagc agctacgcca tgagctgggt ccgacaggct	120
cctggcaaag gccttgaatg ggtgtccgcc atcggtatg gcggcgatac ctactacgcc	180
gactctgtga agggcagatt caccatcagc cgggacaaca gcaagaacac cctgtacctg	240
cagatgaaca gcctgagagc cgaggacacc gccgtgtact attgcgccag aagggacgac	300
tacaccagca gggacgcctt cgattattgg ggccagggca cactgggtcac cgttttctca	360

<210> SEQ ID NO 230
 <211> LENGTH: 15
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 230

agctacgcca tgagc	15
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<210> SEQ ID NO 231
 <211> LENGTH: 48
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 231

gccatcggt atggcgcgga tacctactac gccgactctg tgaagggc	48
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<210> SEQ ID NO 232
 <211> LENGTH: 36
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 232

agggacgact acaccagcag ggacgccttc gattat 36

<210> SEQ ID NO 233

<211> LENGTH: 333

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 233

cagtcgttc tgacacagcc tcctagcgcc tctggcacac ctggacagag agtgaccatc 60

agctgtagcg gcagcagctc caacatcggc agcaacaccg tgaactggta tcagcagctg 120

cctggcacag cccctaaact gctgatctac tacgacgacc tgcggcctag cggcgtgcc 180

gatagatttt ctggcagcaa gagcggcacc tctgccagcc tggctatttc tggactgcag 240

agcgaggacg aggcgcgacta ctattgtcac gcctgggacg acagcctgaa cgactaccct 300

gtttttggcg gaggcaccaa gctgacggtt cta 333

<210> SEQ ID NO 234

<211> LENGTH: 39

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 234

agcggcagca gctccaacat cggcagcaac accgtgaac 39

<210> SEQ ID NO 235

<211> LENGTH: 21

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 235

tacgacgacc tgcggcctag c 21

<210> SEQ ID NO 236

<211> LENGTH: 36

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 236

cacgcctggg acgacagcct gaacgactac cctggt 36

<210> SEQ ID NO 237

<211> LENGTH: 447

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 237

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
 20 25 30

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Ala	Met	Ser	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Val
	35						40					45			
Ser	Ala	Ile	Gly	Tyr	Gly	Gly	Asp	Thr	Tyr	Tyr	Ala	Asp	Ser	Val	Lys
	50					55					60				
Gly	Arg	Phe	Thr	Ile	Ser	Arg	Asp	Asn	Ser	Lys	Asn	Thr	Leu	Tyr	Leu
	65				70					75					80
Gln	Met	Asn	Ser	Leu	Arg	Ala	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys	Ala
			85						90					95	
Arg	Arg	Asp	Asp	Tyr	Thr	Ser	Arg	Asp	Ala	Phe	Asp	Tyr	Trp	Gly	Gln
		100						105					110		
Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	Lys	Gly	Pro	Ser	Val
	115						120					125			
Phe	Pro	Leu	Ala	Pro	Cys	Ser	Arg	Ser	Thr	Ser	Glu	Ser	Thr	Ala	Ala
	130					135					140				
Leu	Gly	Cys	Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu	Pro	Val	Thr	Val	Ser
	145				150					155					160
Trp	Asn	Ser	Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr	Phe	Pro	Ala	Val
			165					170						175	
Leu	Gln	Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val	Thr	Val	Pro
		180						185					190		
Ser	Ser	Ser	Leu	Gly	Thr	Lys	Thr	Tyr	Thr	Cys	Asn	Val	Asp	His	Lys
	195					200						205			
Pro	Ser	Asn	Thr	Lys	Val	Asp	Lys	Arg	Val	Glu	Ser	Lys	Tyr	Gly	Pro
	210					215					220				
Pro	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Phe	Leu	Gly	Gly	Pro	Ser	Val
	225				230					235					240
Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr
			245						250					255	
Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	Gln	Glu	Asp	Pro	Glu
		260						265					270		
Val	Gln	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys
		275					280					285			
Thr	Lys	Pro	Arg	Glu	Glu	Gln	Phe	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser
	290					295					300				
Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys
	305				310					315					320
Cys	Lys	Val	Ser	Asn	Lys	Gly	Leu	Pro	Ser	Ser	Ile	Glu	Lys	Thr	Ile
			325						330					335	
Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro
		340						345					350		
Pro	Ser	Gln	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu
		355					360					365			
Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn
	370					375					380				
Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp	Ser
	385				390					395					400
Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Arg	Leu	Thr	Val	Asp	Lys	Ser	Arg
			405						410					415	
Trp	Gln	Glu	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu
		420						425					430		
His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Leu	Gly	Lys	
	435						440					445			

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<210> SEQ ID NO 238
<211> LENGTH: 217
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 238

Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
1          5          10          15

Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn
20          25          30

Thr Val Asn Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu
35          40          45

Ile Tyr Tyr Asp Asp Leu Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
50          55          60

Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Gln
65          70          75          80

Ser Glu Asp Glu Ala Asp Tyr Tyr Cys His Ala Trp Asp Asp Ser Leu
85          90          95

Asn Asp Tyr Pro Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu Gly
100         105         110

Gln Pro Lys Ala Ala Pro Ser Val Thr Leu Phe Pro Pro Ser Ser Glu
115         120         125

Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp Phe
130         135         140

Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Ser Ser Pro Val
145         150         155         160

Lys Ala Gly Val Glu Thr Thr Thr Pro Ser Lys Gln Ser Asn Asn Lys
165         170         175

Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys Ser
180         185         190

His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val Glu
195         200         205

Lys Thr Val Ala Pro Thr Glu Cys Ser
210         215

<210> SEQ ID NO 239
<211> LENGTH: 1341
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 239

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg      60
agctgtgccg ccagcggcctt cacctttagc agctacgccca tgagctgggt ccgacaggct      120
cctggcaaag gccttgaatg ggtgtccgcc atcggtatg gcggcgatac ctactacgcc      180
gactctgtga agggcagatt caccatcagc cgggacaaca gcaagaacac cctgtacctg      240
cagatgaaca gcctgagagc cgaggacacc gccgtgtact attgcgccag aagggacgac      300
tacaccagca gggacgcctt cgattattgg ggccagggca cactggtcac cgtttcttca      360
gccagcacca agggccccag cgtgttcct ctggcccctt gtagcagaag caccagcgag      420
tctacagccg ccttggtgtg cctcgtgaag gactactttc ccgagcccg gacctgtcc      480
tggaactctg gcctctgac aagcggcgtg cacaccttcc cagccgtgct gcagagcagc      540

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ggcctgtact ctctgagcag cgtcgtgaca gtgcccagca gcagcctggg caccaagacc 600
tacacctgta acgtggacca caagcccagc aacaccaagg tggacaagcg ggtggaatct 660
aagtacggcc ctccctgccc tcttgccca gcccctgaat ttctgggagg accctccgtg 720
ttcctgttcc ccccaaagcc caaggacacc ctgatgatca gccggacccc cgaagtgacc 780
tgcgtggtgg tggatgtgtc ccaggaagat cccgaggtgc agttcaattg gtacgtggac 840
ggcgtggaag tgcacaacgc caagaccaag cccagagagg aacagttcaa cagcacctac 900
cgggtggtgt ccgtgctgac agtgtgtcac caggactggc tgaacggcaa agagtacaag 960
tgcaaggtgt ccaacaaggg cctgcccagc tccatcgaga aaaccatcag caaggccaag 1020
ggccagcccc gcgaacccca ggtgtacaca ctgcctcaa gccaggaaga gatgaccaag 1080
aaccaggtgt ccctgacctg tctcgtgaaa ggcttctacc cctccgatat cgccgtggaa 1140
tgggagagca acgcccagcc cgagaacaac tacaagacca cccccctgt gctggacagc 1200
gacggctcat tcttctgta cagcagactg accgtggaca agagccgggtg gcaggaaggc 1260
aacgtgttca gctgcagcgt gatgcacgag gcctgcaca accactacac ccagaagtcc 1320
ctgtctctga gcctgggcaa g 1341

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<210> SEQ ID NO 240
<211> LENGTH: 651
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 240

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cagctctgtc tgacacagcc tcctagcgcc tctggcacac ctggacagag agtgaccatc 60
agctgtagcg gcagcagctc caacatcggc agcaacacgg tgaactggta tcagcagctg 120
cctggcacag cccctaaact gctgatctac tacgacgacc tgcggcctag cggcgtgcca 180
gatagatttt ctggcagcaa gagcggcacc tctgccagcc tggctatttc tggactgcag 240
agcgaggacg aggcgcagta ctattgtcac gcctgggacg acagcctgaa cgactaccct 300
gtttttggcg gaggcaccaa gctgaccgtt ctaggccagc ctaaagccgc ccctagcgtg 360
accctgttcc ctccaagcag cgaggaaactg caggccaaca agggccacct cgtgtgcctg 420
atcagcgact tctatcctgg cgccgtgacc gtggcctgga aggcgatag ctctcctgtg 480
aaggccggcg tggaaaccac caccctagc aagcagagca acaacaata cgccgccagc 540
agctacctga gcctgacccc cgagcagtggt aagtcacaca gatcctacag ctgccaagtg 600
accacgaggg gcagcaccgt ggaaaagaca gtggccccta ccgagtgcag c 651

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<210> SEQ ID NO 241
<211> LENGTH: 120
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 241

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Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1           5           10           15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20           25           30

Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35           40           45

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Ser Ala Ile Gly Tyr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys
50 55 60

Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
65 70 75 80

Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
85 90 95

Arg Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr Trp Gly Gln
100 105 110

Gly Thr Leu Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 242
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 242

Ser Tyr Ala Met Ser
1 5

<210> SEQ ID NO 243
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 243

Ala Ile Gly Tyr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly
1 5 10 15

<210> SEQ ID NO 244
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 244

Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr
1 5 10

<210> SEQ ID NO 245
<211> LENGTH: 111
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 245

Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
1 5 10 15

Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn
20 25 30

Thr Val Asn Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu
35 40 45

Ile Tyr Tyr Asp Asp Leu Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
50 55 60

Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Gln
65 70 75 80

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Ser Glu Asp Glu Ala Asp Tyr Tyr Cys His Ala Trp Asp Asp Ser Leu
85 90 95

Asn Val Tyr Pro Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
100 105 110

<210> SEQ ID NO 246
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 246

Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn Thr Val Asn
1 5 10

<210> SEQ ID NO 247
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 247

Tyr Asp Asp Leu Arg Pro Ser
1 5

<210> SEQ ID NO 248
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 248

His Ala Trp Asp Asp Ser Leu Asn Val Tyr Pro Val
1 5 10

<210> SEQ ID NO 249
<211> LENGTH: 360
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 249

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg	60
agctgtgccg ccagcggcgt cacccttagc agctacgcc aagctgggt ccgacaggct	120
cctggcaaag gccttgaatg ggtgtccgcc atcggtatg gcggcgatac ctactacgcc	180
gactctgtga agggcagatt caccatcagc cgggacaaca gcaagaacac cctgtacctg	240
cagatgaaca gcctgagagc cgaggacacc gccgtgtact attgcgccag aagggaacgac	300
tacaccagca gggacgcctt cgattattgg ggccagggca cactggtcac cgttttctca	360

<210> SEQ ID NO 250
<211> LENGTH: 15
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 250

agctacgcc aagc

15

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<210> SEQ ID NO 251
<211> LENGTH: 48
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 251

gccatcggt atggcggcga tacctactac gccgactctg tgaagggc 48

<210> SEQ ID NO 252
<211> LENGTH: 36
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 252

agggacgact acaccagcag ggacgccttc gattat 36

<210> SEQ ID NO 253
<211> LENGTH: 333
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 253

cagtctgttc tgacacagcc tctagcgcc tctggcacac ctggacagag agtgaccatc 60
agctgtagcg gcagcagctc caacatcggc agcaacaccg tgaactggta tcagcagctg 120
cctggcacag cccctaaact gctgatctac tacgacgacc tgcggcctag cggcgtgcc 180
gatagatttt ctggcagcaa gagcggcacc tctgccagcc tggtatttc tggactgcag 240
agcaggagcag aggcgacta ctattgtcac gcctgggacg acagcctgaa cgtgtaccct 300
gtttttggcg gaggcacaa gctgacggtt cta 333

<210> SEQ ID NO 254
<211> LENGTH: 39
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 254

agcggcagca gctccaacat cggcagcaac accgtgaac 39

<210> SEQ ID NO 255
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 255

tacgacgacc tgcggcctag c 21

<210> SEQ ID NO 256
<211> LENGTH: 36
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 256

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cacgcctggg acgacagcct gaacgtgtac cctggt

36

<210> SEQ ID NO 257

<211> LENGTH: 447

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 257

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
 20 25 30
 Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45
 Ser Ala Ile Gly Tyr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys
 50 55 60
 Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
 65 70 75 80
 Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
 85 90 95
 Arg Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr Trp Gly Gln
 100 105 110
 Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val
 115 120 125
 Phe Pro Leu Ala Pro Cys Ser Arg Ser Thr Ser Glu Ser Thr Ala Ala
 130 135 140
 Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser
 145 150 155 160
 Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val
 165 170 175
 Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro
 180 185 190
 Ser Ser Ser Leu Gly Thr Lys Thr Tyr Thr Cys Asn Val Asp His Lys
 195 200 205
 Pro Ser Asn Thr Lys Val Asp Lys Arg Val Glu Ser Lys Tyr Gly Pro
 210 215 220
 Pro Cys Pro Pro Cys Pro Ala Pro Glu Phe Leu Gly Gly Pro Ser Val
 225 230 235 240
 Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr
 245 250 255
 Pro Glu Val Thr Cys Val Val Val Asp Val Ser Gln Glu Asp Pro Glu
 260 265 270
 Val Gln Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys
 275 280 285
 Thr Lys Pro Arg Glu Glu Gln Phe Asn Ser Thr Tyr Arg Val Val Ser
 290 295 300
 Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys
 305 310 315 320
 Cys Lys Val Ser Asn Lys Gly Leu Pro Ser Ser Ile Glu Lys Thr Ile
 325 330 335
 Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro
 340 345 350

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Pro Ser Gln Glu Glu Met Thr Lys Asn Gln Val Ser Leu Thr Cys Leu
   355                               360               365

Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn
   370                               375               380

Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser
  385                               390               395               400

Asp Gly Ser Phe Phe Leu Tyr Ser Arg Leu Thr Val Asp Lys Ser Arg
           405                               410               415

Trp Gln Glu Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu
           420                               425               430

His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Leu Gly Lys
   435                               440               445

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<210> SEQ ID NO 258
<211> LENGTH: 217
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 258

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Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
 1                               5               10               15

Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn
           20                               25               30

Thr Val Asn Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu
           35                               40               45

Ile Tyr Tyr Asp Asp Leu Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
           50                               55               60

Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Gln
           65                               70               75               80

Ser Glu Asp Glu Ala Asp Tyr Tyr Cys His Ala Trp Asp Asp Ser Leu
           85                               90               95

Asn Val Tyr Pro Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu Gly
          100                               105               110

Gln Pro Lys Ala Ala Pro Ser Val Thr Leu Phe Pro Pro Ser Ser Glu
          115                               120               125

Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp Phe
          130                               135               140

Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Ser Ser Pro Val
          145                               150               155               160

Lys Ala Gly Val Glu Thr Thr Thr Pro Ser Lys Gln Ser Asn Asn Lys
          165                               170               175

Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys Ser
          180                               185               190

His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val Glu
          195                               200               205

Lys Thr Val Ala Pro Thr Glu Cys Ser
          210                               215

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<210> SEQ ID NO 259
<211> LENGTH: 1341
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 259

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gaagttcagc	tgctggaatc	tggcggcgga	ctggttcaac	ctggcggatc	tctgagactg	60
agctgtgccg	ccagcggctt	cacctttagc	agctacgcca	tgagctgggt	ccgacaggct	120
cctggcaaa	gccttgaatg	ggtgtccgcc	atcggtatg	gcggcgatac	ctactacgcc	180
gactctgtga	agggcagatt	caccatcagc	cgggacaaca	gcaagaacac	cctgtacctg	240
cagatgaaca	gcctgagagc	cgaggacacc	gccgtgtact	attgcgccag	aagggacgac	300
tacaccagca	gggacgcctt	cgattattgg	ggccagggca	cactgggtcac	cgttttctca	360
gccagcacca	agggccccag	cgtgttcctt	ctggccctt	gtacagaag	caccagcgag	420
tctacagccg	ccctgggctg	cctcgtgaag	gactactttc	ccgagcccg	gaccgtgtcc	480
tggaaactctg	gcgctctgac	aagcggcgtg	cacacctttc	cagccgtgct	gcagagcagc	540
ggcctgtact	ctctgagcag	cgtcgtgaca	gtgccacga	gcagcctggg	caccaagacc	600
tacacctgta	acgtggacca	caagcccagc	aacaccaagg	tggacaagcg	ggtggaatct	660
aagtacggcc	ctccctgccc	tccttgccca	gccctggaat	ttctgggcgg	accctccgtg	720
ttcctgttcc	ccccaaagcc	caaggacacc	ctgatgatca	gccggacccc	cgaagtgacc	780
tgcgtggtgg	tggatgtgtc	ccaggaagat	cccagaggtc	agttcaattg	gtacgtggac	840
ggcgtggaag	tgcacaacgc	caagaccaag	cccagagagg	aacagttcaa	cagcacctac	900
cgggtggtgt	ccgtgctgac	agtgtgtcac	caggactggc	tgaacggcaa	agagtacaag	960
tgcaagggtg	ccaacaaggg	cctgcccagc	tccatcgaga	aaaccatcag	caaggccaag	1020
ggccagcccc	gcgaacccca	ggtgtacaca	ctgcctccaa	gccaggaaga	gatgaccaag	1080
aaccaggtgt	ccctgacctg	tctcgtgaaa	ggcttctacc	cctccgatat	cggcgtggaa	1140
tgggagagca	acggccagcc	cgagaacaac	tacaagacca	ccccccctgt	gctggacagc	1200
gacggctcat	tcttctgtga	cagcagactg	accgtggaca	agagccgggtg	gcaggaaggc	1260
aacgtgttca	gctgcagcgt	gatgcacgag	gccctgcaca	accactacac	ccagaagtcc	1320
ctgtctctga	gcctgggcaa	g				1341

<210> SEQ ID NO 260

<211> LENGTH: 651

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 260

cagtctgttc	tgacacagcc	tcctagcgcc	tctggcacac	ctggacagag	agtgaccatc	60
agctgtagcg	gcagcagctc	caacatcggc	agcaacacgg	tgaactggta	tcagcagctg	120
cctggcacag	cccctaaact	gctgatctac	tacgacgacc	tgcggcctag	cggcgtgcca	180
gatagatttt	ctggcagcaa	gagcggcacc	tctgccagcc	tggctatttc	tggactgcag	240
agcgaggacg	aggccgacta	ctattgtcac	gcctgggacg	acagccctgaa	cgtgtaccct	300
gtttttggcg	gaggcaccaa	gctgacggtt	ctaggccagc	ctaaagccgc	ccctagcgtg	360
accctgttcc	ctccaagcag	cgaggaactg	caggccaaca	aggccaccct	cgtgtgctctg	420
atcagcgact	tctatcctgg	cgccgtgacc	gtggcctgga	aggccgatag	ctctcctgtg	480
aaggccggcg	tggaaaccac	cacccttagc	aagcagagca	acaacaaata	cgccgcccagc	540
agctacctga	gcctgacccc	cgagcagtg	aagtcacaca	gatcctacag	ctgccaaagt	600
accacgagg	gcagcacctg	ggaaaagaca	gtggccccta	ccgagtgcag	c	651

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<210> SEQ ID NO 261
<211> LENGTH: 120
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 261

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20 25 30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45
Ser Ala Ile Gly Tyr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys
50 55 60
Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
65 70 75 80
Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
85 90 95
Arg Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr Trp Gly Gln
100 105 110
Gly Thr Leu Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 262
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 262

Ser Tyr Ala Met Ser
1 5

<210> SEQ ID NO 263
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 263

Ala Ile Gly Tyr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly
1 5 10 15

<210> SEQ ID NO 264
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 264

Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr
1 5 10

<210> SEQ ID NO 265
<211> LENGTH: 111
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:

-continued

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 265

Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
 1 5 10 15

Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn
 20 25 30

Thr Val Asn Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu
 35 40 45

Ile Tyr Tyr Asp Asp Leu Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
 50 55 60

Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Gln
 65 70 75 80

Ser Glu Asp Glu Ala Asp Tyr Tyr Cys His Ala Trp Asp Asp Ser Leu
 85 90 95

Asn Val Ile Pro Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
 100 105 110

<210> SEQ ID NO 266

<211> LENGTH: 13

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 266

Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn Thr Val Asn
 1 5 10

<210> SEQ ID NO 267

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 267

Tyr Asp Asp Leu Arg Pro Ser
 1 5

<210> SEQ ID NO 268

<211> LENGTH: 12

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 268

His Ala Trp Asp Asp Ser Leu Asn Val Ile Pro Val
 1 5 10

<210> SEQ ID NO 269

<211> LENGTH: 360

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 269

gaagttcagc tgctggaate tggcggcgga ctggttcaac ctggcggate tctgagactg 60

agctgtgccg ccagcggcct caccttttagc agctacgcca tgagctgggt ccgacaggct 120

cctggcaaaag gccttgaatg ggtgtccgcc atcggtatg gcggcgatac ctactacgcc 180

-continued

gactctgtga agggcagatt caccatcagc cgggacaaca gcaagaacac cctgtacctg	240
cagatgaaca gcctgagagc cgaggacacc gccgtgtact attgcgccag aagggaacgac	300
tacaccagca gggacgcctt cgattattgg ggccagggca cactgggtcac cgtttcttca	360

<210> SEQ ID NO 270
 <211> LENGTH: 15
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 270

agctacgccca tgagc	15
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<210> SEQ ID NO 271
 <211> LENGTH: 48
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 271

gccatcggt atggcggcga tacctactac gccgactctg tgaagggc	48
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<210> SEQ ID NO 272
 <211> LENGTH: 36
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 272

agggacgact acaccagcag ggacgccttc gattat	36
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<210> SEQ ID NO 273
 <211> LENGTH: 333
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 273

cagtctgttc tgacacagcc tcctagcgcc tctggcacac ctggacagag agtgaccatc	60
agctgtagcg gcagcagctc caacatcggc agcaacaccg tgaactggta tcagcagctg	120
cctggcacag cccctaaact gctgatctac tacgacgacc tgcggcctag cggcgtgcc	180
gatagatttt ctggcagcaa gagcggcacc tctgccagcc tggctatttc tggactgcag	240
agcagaggacg agggcgacta ctattgtcac gcctgggacg acagcctgaa cgtgatecct	300
gtttttggcg gaggcaccaa gctgaccgtt cta	333

<210> SEQ ID NO 274
 <211> LENGTH: 39
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 274

agcggcagca gctccaacat cggcagcaac accgtgaac	39
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<210> SEQ ID NO 275

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<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 275

tacgacgacc tgcggcctag c                21

<210> SEQ ID NO 276
<211> LENGTH: 36
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 276

cacgcctggg acgacagcct gaacgtgatc cctgtt        36

<210> SEQ ID NO 277
<211> LENGTH: 447
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 277

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1      5      10      15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20     25     30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35     40     45
Ser Ala Ile Gly Tyr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys
50     55     60
Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
65     70     75     80
Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
85     90     95
Arg Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr Trp Gly Gln
100    105    110
Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val
115    120    125
Phe Pro Leu Ala Pro Cys Ser Arg Ser Thr Ser Glu Ser Thr Ala Ala
130    135    140
Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser
145    150    155    160
Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val
165    170    175
Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro
180    185    190
Ser Ser Ser Leu Gly Thr Lys Thr Tyr Thr Cys Asn Val Asp His Lys
195    200    205
Pro Ser Asn Thr Lys Val Asp Lys Arg Val Glu Ser Lys Tyr Gly Pro
210    215    220
Pro Cys Pro Pro Cys Pro Ala Pro Glu Phe Leu Gly Gly Pro Ser Val
225    230    235    240
Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr
245    250    255

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Pro Glu Val Thr Cys Val Val Val Asp Val Ser Gln Glu Asp Pro Glu
    260                      265                      270

Val Gln Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys
    275                      280                      285

Thr Lys Pro Arg Glu Glu Gln Phe Asn Ser Thr Tyr Arg Val Val Ser
    290                      295                      300

Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys
    305                      310                      315                      320

Cys Lys Val Ser Asn Lys Gly Leu Pro Ser Ser Ile Glu Lys Thr Ile
    325                      330                      335

Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro
    340                      345                      350

Pro Ser Gln Glu Glu Met Thr Lys Asn Gln Val Ser Leu Thr Cys Leu
    355                      360                      365

Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn
    370                      375                      380

Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser
    385                      390                      395                      400

Asp Gly Ser Phe Phe Leu Tyr Ser Arg Leu Thr Val Asp Lys Ser Arg
    405                      410                      415

Trp Gln Glu Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu
    420                      425                      430

His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Leu Gly Lys
    435                      440                      445

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<210> SEQ ID NO 278
<211> LENGTH: 217
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 278

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Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
1      5      10      15

Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn
20     25     30

Thr Val Asn Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu
35     40     45

Ile Tyr Tyr Asp Asp Leu Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
50     55     60

Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Gln
65     70     75     80

Ser Glu Asp Glu Ala Asp Tyr Tyr Cys His Ala Trp Asp Asp Ser Leu
85     90     95

Asn Val Ile Pro Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu Gly
100    105    110

Gln Pro Lys Ala Ala Pro Ser Val Thr Leu Phe Pro Pro Ser Ser Glu
115    120    125

Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp Phe
130    135    140

Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Ser Ser Pro Val
145    150    155    160

Lys Ala Gly Val Glu Thr Thr Thr Pro Ser Lys Gln Ser Asn Asn Lys
165    170    175

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Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys Ser
 180 185 190

His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val Glu
 195 200 205

Lys Thr Val Ala Pro Thr Glu Cys Ser
 210 215

<210> SEQ ID NO 279

<211> LENGTH: 1341

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 279

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gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg      60
agctgtgccg ccagcggett cacctttagc agctacgcc tgaagctgggt ccgacaggct      120
cctggcaaa gacctgaatg ggtgtccgcc atcggetatg gcggcgatac ctactacgcc      180
gactctgtga agggcagatt caccatcagc cgggacaaca gcaagaacac cctgtacctg      240
cagatgaaca gcctgagagc cgaggacacc gccgtgtact attgcgccag aagggacgac      300
tacaccagca gggacgcctt cgattattgg ggccagggca cactggtcac cgttttctca      360
gccagcacca agggccccag cgtgttcctt ctggccccct gtagcagaag caccagcgag      420
tctacagccg ccttggtgtg cctcgtgaag gactactttc ccgagcccg gacctgttc      480
tggaactctg gcgctctgac aagcggcgtg cacacctttc cagccgtgct gcagagcagc      540
ggcctgtact ctctgagcag cgtcgtgaca gtgccagca gcagcctggg caccaagacc      600
tacacctgta acgtggacca caagcccagc aacaccaagg tggacaagcg ggtggaatct      660
aagtaaggcc ctccctgcc tcttgccca gccctgaat ttctgggcgg accctccgtg      720
tctcgtttcc cccaaaagcc caaggacacc ctgatgatca gccggacccc cgaagtgacc      780
tgcgtggtgg tggatgtgtc ccaggaagat cccaggtgtc agttcaattg gtacgtggac      840
ggcgtggaag tgcacaacgc caagaccaag cccagagagg aacagttcaa cagcacctac      900
cgggtggtgt ccgtgctgac agtgctgcac caggactggc tgaacggcaa agagtacaag      960
tgcaaggtgt ccaacaaggg cctgccagc tccatcgaga aaaccatcag caaggccaag      1020
ggccagcccc gcgaaccca ggtgtacaca ctgcctcaa gccaggaaga gatgaccaag      1080
aaccaggtgt ccctgacctg tctcgtgaaa ggcttctacc cctccgatat cgcctgggaa      1140
tggaagagca acggccagcc cgagaacaac tacaagacca cccccctgt gctggacagc      1200
gacggtcat tcttctgta cagcagactg accgtggaca agagccggtg gcaggaaggc      1260
aacgtgttca gctgcagcgt gatgcacgag gccctgcaca accactacac ccagaagtcc      1320
ctgtctctga gctggggcaa g                                     1341

```

<210> SEQ ID NO 280

<211> LENGTH: 651

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 280

```

cagtctgttc tgacacagcc tctagcgcc tctggcacac ctggacagag agtgaccatc      60
agctgtagcg gcagcagctc caacatcgcc agcaacaccg tgaactggta tcagcagctg      120

```


-continued

```

cctggcacag cccctaaact gctgatctac tacgacgacc tgcggcctag cggcgtgccca 180
gatagatttt ctggcagcaa gagcggcacc tctgccagcc tggctatttc tggactgcag 240
agcgaggacg aggcgacta ctattgtcac gcctgggacg acagcctgaa cgtgatccct 300
gtttttggcg gaggcaccaa gctgaccgtt ctaggccagc ctaaagccgc cctagcgtg 360
accctgttcc ctccaagcag cgaggaactg caggccaaca aggccaccct cgtgtgctg 420
atcagcgact tctatcctgg cgccgtgacc gtggcctgga aggcgatag ctctcctgtg 480
aaggccggcg tggaaaccac caccctagc aagcagagca acaacaata cgccgccagc 540
agctacctga gcctgacccc cgagcagtgg aagtcccaca gatcctacag ctgccaagtg 600
accacgagg gcagcaccgt ggaaaagaca gtggccccta ccgagtgcag c 651

```

```

<210> SEQ ID NO 281
<211> LENGTH: 120
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 281

```

```

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1           5           10           15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Tyr Ser Tyr
20          25          30
Ala Met Leu Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35          40          45
Ser Ala Ile Gly Thr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys
50          55          60
Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
65          70          75          80
Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
85          90          95
Arg Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr Trp Gly Gln
100         105         110
Gly Thr Leu Val Thr Val Ser Ser
115         120

```

```

<210> SEQ ID NO 282
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 282

```

```

Ser Tyr Ala Met Leu
1           5

```

```

<210> SEQ ID NO 283
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 283

```

```

Ala Ile Gly Thr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly
1           5           10           15

```

-continued

<210> SEQ ID NO 284
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 284

Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr
1 5 10

<210> SEQ ID NO 285
<211> LENGTH: 111
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 285

Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
1 5 10 15
Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn
20 25 30
Thr Val Asn Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu
35 40 45
Ile Tyr Tyr Asp Asp Leu Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
50 55 60
Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Gln
65 70 75 80
Ser Glu Asp Glu Ala Asp Tyr Tyr Cys Ala Ala Trp Asp Asp Ser Leu
85 90 95
Asn Asp Tyr Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
100 105 110

<210> SEQ ID NO 286
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 286

Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn Thr Val Asn
1 5 10

<210> SEQ ID NO 287
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 287

Tyr Asp Asp Leu Arg Pro Ser
1 5

<210> SEQ ID NO 288
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 288

-continued

Ala Ala Trp Asp Asp Ser Leu Asn Asp Tyr Val Val
 1 5 10

<210> SEQ ID NO 289

<211> LENGTH: 360

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 289

```
gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg    60
agctgtgccg ccagcggett cacccttttac agctacgcca tgctgtgggt ccgacaggcc    120
cctggaaaag gccttgaatg ggtgtccgcc atcggcacag gcggcgatac ctactatgcc    180
gactctgtga agggcagatt caccatcagc cgggacaaca gcaagaacac cctgtacctg    240
cagatgaaca gcctgagagc cgaggacacc gccgtgtact attgcgccag aagggcagac    300
tacaccagca gggacgcctt cgattattgg ggccagggca cactggtcac cgttttctca    360
```

<210> SEQ ID NO 290

<211> LENGTH: 15

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 290

```
agctacgcca tgctg    15
```

<210> SEQ ID NO 291

<211> LENGTH: 48

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 291

```
gccatcgcca caggcggcga tacctactat gccgactctg tgaagggc    48
```

<210> SEQ ID NO 292

<211> LENGTH: 36

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 292

```
agggacgact acaccagcag ggacgccttc gattat    36
```

<210> SEQ ID NO 293

<211> LENGTH: 333

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 293

```
cagtctgttc tgacacagcc tcctagcgcc tctggcacac ctggacagag agtgaccatc    60
agctgtagcg gcagcagetc caacatcggc agcaacacgg tgaactggta tcagcagctg    120
cctggcacag ccctaaact gctgatctac tacgacgacc tgcggcctag cggcgtgcca    180
gatagatttt ctggcagcaa gagcggcacc tctgccagcc tggctatttc tggactgcag    240
```

-continued

agcgaggacg aggcgacta ttattgtgcc gcctgggacg acagcctgaa cgactacgtt 300

gtgtttggcg gaggcaccaa gctgaccgtt cta 333

<210> SEQ ID NO 294

<211> LENGTH: 39

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 294

agcggcagca gctccaacat cggcagcaac accgtgaac 39

<210> SEQ ID NO 295

<211> LENGTH: 21

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 295

tacgacgacc tgcggcctag c 21

<210> SEQ ID NO 296

<211> LENGTH: 36

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 296

gccgcctggg acgacagcct gaacgactac gttgtg 36

<210> SEQ ID NO 297

<211> LENGTH: 447

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 297

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Tyr Ser Tyr
20 25 30

Ala Met Leu Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45

Ser Ala Ile Gly Thr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys
50 55 60

Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
65 70 75 80

Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
85 90 95

Arg Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr Trp Gly Gln
100 105 110

Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val
115 120 125

Phe Pro Leu Ala Pro Cys Ser Arg Ser Thr Ser Glu Ser Thr Ala Ala
130 135 140

Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser

-continued

145	150	155	160
Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val			
	165	170	175
Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro			
	180	185	190
Ser Ser Ser Leu Gly Thr Lys Thr Tyr Thr Cys Asn Val Asp His Lys			
	195	200	205
Pro Ser Asn Thr Lys Val Asp Lys Arg Val Glu Ser Lys Tyr Gly Pro			
	210	215	220
Pro Cys Pro Pro Cys Pro Ala Pro Glu Phe Leu Gly Gly Pro Ser Val			
	225	230	235
Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr			
	245	250	255
Pro Glu Val Thr Cys Val Val Val Asp Val Ser Gln Glu Asp Pro Glu			
	260	265	270
Val Gln Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys			
	275	280	285
Thr Lys Pro Arg Glu Glu Gln Phe Asn Ser Thr Tyr Arg Val Val Ser			
	290	295	300
Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys			
	305	310	315
Cys Lys Val Ser Asn Lys Gly Leu Pro Ser Ser Ile Glu Lys Thr Ile			
	325	330	335
Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro			
	340	345	350
Pro Ser Gln Glu Glu Met Thr Lys Asn Gln Val Ser Leu Thr Cys Leu			
	355	360	365
Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn			
	370	375	380
Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser			
	385	390	395
Asp Gly Ser Phe Phe Leu Tyr Ser Arg Leu Thr Val Asp Lys Ser Arg			
	405	410	415
Trp Gln Glu Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu			
	420	425	430
His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Leu Gly Lys			
	435	440	445

<210> SEQ ID NO 298

<211> LENGTH: 217

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 298

Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln			
1	5	10	15
Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn			
	20	25	30
Thr Val Asn Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu			
	35	40	45
Ile Tyr Tyr Asp Asp Leu Arg Pro Ser Gly Val Pro Asp Arg Phe Ser			
	50	55	60
Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Gln			

-continued

65	70	75	80
Ser Glu Asp	Glu Ala Asp Tyr Tyr Cys	Ala Ala Trp Asp Asp	Ser Leu
	85	90	95
Asn Asp Tyr	Val Val Phe Gly Gly	Gly Thr Lys Leu Thr Val	Leu Gly
	100	105	110
Gln Pro Lys	Ala Ala Pro Ser Val	Thr Leu Phe Pro Pro	Ser Ser Glu
	115	120	125
Glu Leu Gln	Ala Asn Lys Ala Thr	Leu Val Cys Leu Ile	Ser Asp Phe
	130	135	140
Tyr Pro Gly	Ala Val Thr Val	Ala Trp Lys Ala Asp	Ser Ser Pro Val
	145	150	155
Lys Ala Gly	Val Glu Thr Thr Thr	Pro Ser Lys Gln Ser	Asn Asn Lys
	165	170	175
Tyr Ala Ala	Ser Ser Tyr Leu Ser	Leu Thr Pro Glu Gln	Trp Lys Ser
	180	185	190
His Arg Ser	Tyr Ser Cys Gln Val	Thr His Glu Gly Ser	Thr Val Glu
	195	200	205
Lys Thr Val	Ala Pro Thr Glu	Cys Ser	
	210	215	

<210> SEQ ID NO 299

<211> LENGTH: 1341

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 299

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg	60
agctgtgccc ccagcggtt cactttttac agctacgcca tgctgtgggt ccgacaggcc	120
cctggaaaag gccttgaatg ggtgtccgcc atcggcacag gcggcgatac ctactatgcc	180
gactctgtga agggcagatt caccatcagc cgggacaaca gcaagaacac cctgtacctg	240
cagatgaaca gcctgagagc cgaggacacc gccgtgtact attgcgccag aagggacgac	300
tacaccagca gggacgcctt cgattattgg ggccagggca cactggctac cgttttctca	360
gccagcacca agggccccag cgtgttcct ctggcccctt gtagcagaag caccagcgag	420
tctacagccg ccctgggctg cctcgtgaag gactacttcc ccgagcccgt gaccgtgtcc	480
tggaaactctg gcgctctgac aagcggcgtg cacaccttcc cagccgtgct gcagagcagc	540
ggcctgtact ctctgagcag cgtcgtgaca gtgcccagca gcagcctggg caccaagacc	600
tacacctgta acgtggacca caagcccagc aacaccaagg tggacaagcg ggtggaatct	660
aagtacggcc ctccctgccc tccttgccca gccctgaat ttctgggcgg acctccgtg	720
ttcctgttcc ccccaaagcc caaggacacc ctgatgatca gccggacccc cgaagtgacc	780
tgctgtgtgg tggatgtgtc ccaggaagat cccgaggtgc agttcaattg gtacgtggac	840
ggcgtggaag tgcacaacgc caagaccaag ccagagaggg aacagttcaa cagcacctac	900
cgggtggtgt ccgtgctgac agtgtgcac caggactggc tgaacggcaa agagtacaag	960
tgcgaagtgt ccaacaaggg cctgcccagc tccatcgaga aaacctcag caaggccaag	1020
ggccagcccc gcgaacccca ggtgtacaca ctgcctccaa gccaggaaga gatgaccaag	1080
aaccaggtgt ccctgacctg tctcgtgaaa ggcttctacc cctccgatat cgccgtggaa	1140
tgggagagca acggccagcc cgagaacaac tacaagacca cccccctgt gctggacagc	1200

-continued

```

gacggctcat tcttctgtga cagcagactg accgtggaca agagccggtg gcaggaaggc 1260
aacgtgttca gctgcagcgt gatgcacgag gccctgcaca accactacac ccagaagtcc 1320
ctgtctctga gcctgggcaa g                                     1341

```

```

<210> SEQ ID NO 300
<211> LENGTH: 651
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 300

```

```

cagtctgttc tgacacagcc tcttagcgcc tctggcacac ctggacagag agtgaccatc      60
agctgtagcg gcagcagctc caacatcggc agcaacaccg tgaactggta tcagcagctg      120
cctggcacag cccctaaact gctgatctac tacgacgacc tgcggcctag cggcgtgccca      180
gatagatttt ctggcagcaa gagcggcacc tctgccagcc tggtctatttc tggactgcag      240
agcgaggacg aggcgacta ttattgtgcc gcctgggacg acagcctgaa cgactacgtt      300
gtgtttggcg gaggcaccaa gctgaccgtt ctaggccagc ctaaagccgc ccctagcgtg      360
acctgtttcc ctccaagcag cgaggaactg caggccaaca agggccacct cgtgtgctg      420
atcagcgact tctatcctgg cgccgtgacc gtggcctgga aggcgatag ctctcctgtg      480
aaggccggcg tggaaccac caccctagc aagcagagca acaacaaata cgccgccagc      540
agctacctga gcctgacccc cgagcagtgg aagtcaccaca gatcctacag ctgccaaagt      600
acccacgagg gcagcaccgt ggaaaagaca gtggccccta ccgagtgcag c                651

```

```

<210> SEQ ID NO 301
<211> LENGTH: 120
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 301

```

```

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1           5           10          15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Tyr Ser Tyr
20          25          30
Ala Met Leu Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35          40          45
Ser Ala Ile Gly Thr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys
50          55          60
Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
65          70          75          80
Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
85          90          95
Arg Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr Trp Gly Gln
100         105         110
Gly Thr Leu Val Thr Val Ser Ser
115         120

```

```

<210> SEQ ID NO 302
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

-continued

<400> SEQUENCE: 302

Ser Tyr Ala Met Leu
1 5

<210> SEQ ID NO 303

<211> LENGTH: 16

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 303

Ala Ile Gly Thr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly
1 5 10 15

<210> SEQ ID NO 304

<211> LENGTH: 12

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 304

Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr
1 5 10

<210> SEQ ID NO 305

<211> LENGTH: 111

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 305

Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
1 5 10 15Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn
20 25 30Thr Val Asn Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu
35 40 45Ile Tyr Tyr Asp Asp Leu Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
50 55 60Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Gln
65 70 75 80Ser Glu Asp Glu Ala Asp Tyr Tyr Cys Ala Ala Trp Asp Asp Ser Leu
85 90 95Asn Val Tyr Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
100 105 110

<210> SEQ ID NO 306

<211> LENGTH: 13

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 306

Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn Thr Val Asn
1 5 10

<210> SEQ ID NO 307

<211> LENGTH: 7

-continued

<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 307

Tyr Asp Asp Leu Arg Pro Ser
1 5

<210> SEQ ID NO 308
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 308

Ala Ala Trp Asp Asp Ser Leu Asn Val Tyr Val Val
1 5 10

<210> SEQ ID NO 309
<211> LENGTH: 360
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 309

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg	60
agctgtgccc ccagcggctt caccttttac agctacgcca tgctgtgggt ccgacaggcc	120
cctggaaaag gccttgaatg ggtgtccgcc atcggcacag gcggcgatac ctactatgcc	180
gactctgtga agggcagatt caccatcagc cgggacaaca gcaagaacac cctgtacctg	240
cagatgaaca gcctgagagc cgaggacacc gccgtgtact attgcgccag aagggacgac	300
tacaccagca gggacgcctt cgattattgg ggccagggca cactggtcac cgttttctca	360

<210> SEQ ID NO 310
<211> LENGTH: 15
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 310

agctacgcca tgctg	15
------------------	----

<210> SEQ ID NO 311
<211> LENGTH: 48
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 311

gccatcgga caggcggcga tacctactat gccgactctg tgaagggc	48
---	----

<210> SEQ ID NO 312
<211> LENGTH: 36
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 312

-continued

agggacgact acaccagcag ggacgccttc gattat 36

<210> SEQ ID NO 313
 <211> LENGTH: 333
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 313

cagtctgttc tgacacagcc tcctagcgcc tctggcacac ctggacagag agtgaccatc 60
 agctgtagcg gcagcagctc caacatcggc agcaacaccg tgaactggta tcagcagctg 120
 cctggcacag cccctaaact gctgatctac tacgacgacc tgcggcctag cggcgtgcca 180
 gatagatttt ctggcagcaa gagcggcacc tctgccagcc tggctatttc tggactgcag 240
 agcgaggacg aggcgacta ttattgtgcc gcctgggacg acagcctgaa cgtgtacgtt 300
 gtgtttggcg gaggcaccaa gctgaccgtt cta 333

<210> SEQ ID NO 314
 <211> LENGTH: 39
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 314

agcggcagca gctccaacat cggcagcaac accgtgaac 39

<210> SEQ ID NO 315
 <211> LENGTH: 21
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 315

tacgacgacc tgcggcctag c 21

<210> SEQ ID NO 316
 <211> LENGTH: 36
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 316

gccgcctggg acgacagcct gaacgtgtac gttgtg 36

<210> SEQ ID NO 317
 <211> LENGTH: 447
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 317

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Tyr Ser Tyr
 20 25 30
 Ala Met Leu Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ser	Ala	Ile	Gly	Thr	Gly	Gly	Asp	Thr	Tyr	Tyr	Ala	Asp	Ser	Val	Lys
505560															
Gly	Arg	Phe	Thr	Ile	Ser	Arg	Asp	Asn	Ser	Lys	Asn	Thr	Leu	Tyr	Leu
65707580															
Gln	Met	Asn	Ser	Leu	Arg	Ala	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys	Ala
859095															
Arg	Arg	Asp	Asp	Tyr	Thr	Ser	Arg	Asp	Ala	Phe	Asp	Tyr	Trp	Gly	Gln
100105110															
Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	Lys	Gly	Pro	Ser	Val
115120125															
Phe	Pro	Leu	Ala	Pro	Cys	Ser	Arg	Ser	Thr	Ser	Glu	Ser	Thr	Ala	Ala
130135140															
Leu	Gly	Cys	Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu	Pro	Val	Thr	Val	Ser
145150155160															
Trp	Asn	Ser	Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr	Phe	Pro	Ala	Val
165170175															
Leu	Gln	Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val	Thr	Val	Pro
180185190															
Ser	Ser	Ser	Leu	Gly	Thr	Lys	Thr	Tyr	Thr	Cys	Asn	Val	Asp	His	Lys
195200205															
Pro	Ser	Asn	Thr	Lys	Val	Asp	Lys	Arg	Val	Glu	Ser	Lys	Tyr	Gly	Pro
210215220															
Pro	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Phe	Leu	Gly	Gly	Pro	Ser	Val
225230235240															
Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr
245250255															
Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	Gln	Glu	Asp	Pro	Glu
260265270															
Val	Gln	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys
275280285															
Thr	Lys	Pro	Arg	Glu	Glu	Gln	Phe	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser
290295300															
Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys
305310315320															
Cys	Lys	Val	Ser	Asn	Lys	Gly	Leu	Pro	Ser	Ser	Ile	Glu	Lys	Thr	Ile
325330335															
Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro
340345350															
Pro	Ser	Gln	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu
355360365															
Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn
370375380															
Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp	Ser
385390395400															
Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Arg	Leu	Thr	Val	Asp	Lys	Ser	Arg
405410415															
Trp	Gln	Glu	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu
420425430															
His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Leu	Gly	Lys	
435440445															

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<210> SEQ ID NO 318
<211> LENGTH: 217
<212> TYPE: PRT
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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 318
Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
1          5          10          15
Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn
20          25          30
Thr Val Asn Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu
35          40          45
Ile Tyr Tyr Asp Asp Leu Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
50          55          60
Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Gln
65          70          75          80
Ser Glu Asp Glu Ala Asp Tyr Tyr Cys Ala Ala Trp Asp Asp Ser Leu
85          90          95
Asn Val Tyr Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu Gly
100         105         110
Gln Pro Lys Ala Ala Pro Ser Val Thr Leu Phe Pro Pro Ser Ser Glu
115         120         125
Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp Phe
130         135         140
Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Ser Ser Pro Val
145         150         155         160
Lys Ala Gly Val Glu Thr Thr Thr Pro Ser Lys Gln Ser Asn Asn Lys
165         170         175
Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys Ser
180         185         190
His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val Glu
195         200         205
Lys Thr Val Ala Pro Thr Glu Cys Ser
210         215

<210> SEQ ID NO 319
<211> LENGTH: 1341
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 319
gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg      60
agctgtgccg ccagcggcct caccctttac agctacgcca tgctgtgggt ccgacaggcc      120
cctggaaaag gccttgaatg ggtgtccgcc atcggcacag gcggcgatac ctactatgcc      180
gactctgtga agggcagatt caccatcagc cgggacaaca gcaagaacac cctgtacctg      240
cagatgaaca gcctgagagc cgaggacacc gccgtgtact attgcgccag aagggacgac      300
tacaccagca gggagcctt cgattattgg ggccagggca cactggtcac cgtttcttca      360
gccagcacca agggccccag cgtgttcctt ctggcccctt gtagcagaag caccagcgag      420
tctacagccg ccctgggctg cctcgtgaag gactactttc ccgagcccg gacctgttcc      480
tggaactctg gcgctctgac aagcggcgtg cacacctttc cagccgtgct gcagagcagc      540
ggcctgtact ctctgagcag cgtcgtgaca gtgcccagca gcagcctggg caccaagacc      600
tacacctgta acgtggacca caagcccagc aacaccaagg tggacaagcg ggtggaatct      660

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aagtaaggcc ctccctgccc tcttgccca gcccctgaat ttctgggagg accctccgtg 720
ttctgtgtcc ccccaaagcc caaggacacc ctgatgatca gccggacccc cgaagtgacc 780
tgcgtggtgg tggatgtgtc ccaggaagat cccgagggtc agttcaattg gtacgtggac 840
ggcgtggaag tgcacaacgc caagaccaag cccagagagg aacagttcaa cagcacctac 900
cgggtggtgt ccgtgctgac agtgtgtcac caggactggc tgaacggcaa agagtacaag 960
tgcaagggtg ccaacaaggg cctgcccagc tccatcgaga aaaccatcag caaggccaag 1020
ggccagcccc gcgaacccca ggtgtacaca ctgcctccaa gccaggaaga gatgaccaag 1080
aaccagggtg cctgacctg tctctgaaa ggcttctacc cctccgatat cgccgtggaa 1140
tggaagagca acgcccagcc cgagaacaac tacaagacca cccccctgt gctggacagc 1200
gacggctcat tcttctgta cagcagactg accgtggaca agagccgggtg gcaggaaggc 1260
aacgtgttca gctgcagcgt gatgcacgag gccctgcaca accactacac ccagaagtcc 1320
ctgtctctga gcctgggcaa g 1341

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<210> SEQ ID NO 320
<211> LENGTH: 651
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 320

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cagctctgtc tgacacagcc tcctagcgcc tctggcacac ctggacagag agtgaccatc 60
agctgtagcg gcagcagctc caacatcggc agcaacacgg tgaactggta tcagcagctg 120
cctggcacag cccctaaact gctgatctac tacgacgacc tgcggcctag cggcgtgcca 180
gatagatttt ctggcagcaa gagcggcacc tctgccagcc tggctatttc tggactgcag 240
agcgaggacg aggcgacta ttattgtgcc gcctgggacg acagcctgaa cgtgtacgtt 300
gtgtttggcg gaggcaccaa gctgaccgtt ctaggccagc ctaaagccgc ccctagcgtg 360
accctgttcc ctccaagcag cgaggaaactg caggccaaca agggccacct cgtgtgcctg 420
atcagcgact tctatcctgg cgccgtgacc gtggcctgga agggcgatag ctctcctgtg 480
aaggccggcg tggaaaccac caccctagc aagcagagca acaacaata cgccgccagc 540
agctacctga gcctgacccc cgagcagtggt aagtcaccaca gatcctacag ctgccaaagt 600
accacagagg gcagcaccgt ggaaaagaca gtggccccta ccgagtgcag c 651

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<210> SEQ ID NO 321
<211> LENGTH: 120
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 321

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Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1           5           10          15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Tyr Ser Tyr
20          25          30

Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35          40          45

Ser Ala Ile Gly Tyr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys
50          55          60

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Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
65 70 75 80

Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
85 90 95

Arg Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr Trp Gly Gln
100 105 110

Gly Thr Leu Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 322
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 322

Ser Tyr Ala Met Ser
1 5

<210> SEQ ID NO 323
 <211> LENGTH: 16
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 323

Ala Ile Gly Tyr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly
1 5 10 15

<210> SEQ ID NO 324
 <211> LENGTH: 12
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 324

Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr
1 5 10

<210> SEQ ID NO 325
 <211> LENGTH: 111
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 325

Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
1 5 10 15

Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn
20 25 30

Thr Val Asn Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu
35 40 45

Ile Tyr Tyr Asp Asp Leu Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
50 55 60

Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Gln
65 70 75 80

Ser Glu Asp Glu Ala Asp Tyr Tyr Cys His Ala Trp Asp Asp Ser Leu
85 90 95

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Asn Val Tyr Pro Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
 100 105 110

<210> SEQ ID NO 326
 <211> LENGTH: 13
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence
 <400> SEQUENCE: 326

Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn Thr Val Asn
 1 5 10

<210> SEQ ID NO 327
 <211> LENGTH: 7
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence
 <400> SEQUENCE: 327

Tyr Asp Asp Leu Arg Pro Ser
 1 5

<210> SEQ ID NO 328
 <211> LENGTH: 12
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence
 <400> SEQUENCE: 328

His Ala Trp Asp Asp Ser Leu Asn Val Tyr Pro Val
 1 5 10

<210> SEQ ID NO 329
 <211> LENGTH: 360
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence
 <400> SEQUENCE: 329

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg	60
agctgtgccg ccagcggcct caccctttac agctacgcca tgagctgggt ccgacaggcc	120
cctggaaaag gccttgaatg ggtgtccgcc atcggtatg gcggcgatac ctactacgcc	180
gactctgtga agggcagatt caccatcagc cgggacaaca gcaagaacac cctgtacctg	240
cagatgaaca gcctgagagc cgaggacacc gccgtgtact attgcgccag aagggacgac	300
tacaccagca gggacgcctt cgattattgg ggccagggca cactggtcac cgtttcttca	360

<210> SEQ ID NO 330
 <211> LENGTH: 15
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence
 <400> SEQUENCE: 330

agctacgcca tgagc	15
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<210> SEQ ID NO 331
 <211> LENGTH: 48

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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 331

gccatcggt atgggggga tacctactac gccgactctg tgaagggc      48

<210> SEQ ID NO 332
<211> LENGTH: 36
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 332

agggacgact acaccagcag ggacgccttc gattat      36

<210> SEQ ID NO 333
<211> LENGTH: 333
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 333

cagtctgttc tgacacagcc tctagcgcc tctggcacac ctggacagag agtgaccatc      60
agctgtagcg gcagcagctc caacatcggc agcaacaccg tgaactggta tcagcagctg      120
cctggcacag cccctaaact gctgatctac tacgacgacc tgcggcctag cggcgtgccca      180
gatagatttt ctggcagcaa gagcggcacc tctgccagcc tggctatttc tggactgcag      240
agcagaggacg aggcgacta ctattgtcac gcctgggacg acagcctgaa cgtgtaccct      300
gtttttggcg gaggcaccaa gctgaccgtt cta      333

<210> SEQ ID NO 334
<211> LENGTH: 39
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 334

agcggcagca gctccaacat cggcagcaac accgtgaac      39

<210> SEQ ID NO 335
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 335

tacgacgacc tgcggcctag c      21

<210> SEQ ID NO 336
<211> LENGTH: 36
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 336

cacgcctggg acgacagcct gaacgtgtac cctggt      36

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<210> SEQ ID NO 337
<211> LENGTH: 447
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 337

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1          5          10          15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Tyr Ser Tyr
20          25          30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35          40          45
Ser Ala Ile Gly Tyr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys
50          55          60
Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
65          70          75          80
Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
85          90          95
Arg Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr Trp Gly Gln
100         105         110
Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val
115         120         125
Phe Pro Leu Ala Pro Cys Ser Arg Ser Thr Ser Glu Ser Thr Ala Ala
130         135         140
Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser
145         150         155         160
Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val
165         170         175
Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro
180         185         190
Ser Ser Ser Leu Gly Thr Lys Thr Tyr Thr Cys Asn Val Asp His Lys
195         200         205
Pro Ser Asn Thr Lys Val Asp Lys Arg Val Glu Ser Lys Tyr Gly Pro
210         215         220
Pro Cys Pro Pro Cys Pro Ala Pro Glu Phe Leu Gly Gly Pro Ser Val
225         230         235         240
Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr
245         250         255
Pro Glu Val Thr Cys Val Val Val Asp Val Ser Gln Glu Asp Pro Glu
260         265         270
Val Gln Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys
275         280         285
Thr Lys Pro Arg Glu Glu Gln Phe Asn Ser Thr Tyr Arg Val Val Ser
290         295         300
Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys
305         310         315         320
Cys Lys Val Ser Asn Lys Gly Leu Pro Ser Ser Ile Glu Lys Thr Ile
325         330         335
Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro
340         345         350
Pro Ser Gln Glu Glu Met Thr Lys Asn Gln Val Ser Leu Thr Cys Leu
355         360         365

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Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn
370						375					380				
Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp	Ser
385					390					395					400
Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Arg	Leu	Thr	Val	Asp	Lys	Ser	Arg
			405						410					415	
Trp	Gln	Glu	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu
			420					425					430		
His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Leu	Gly	Lys	
	435						440						445		

<210> SEQ ID NO 338
 <211> LENGTH: 217
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 338

Gln	Ser	Val	Leu	Thr	Gln	Pro	Pro	Ser	Ala	Ser	Gly	Thr	Pro	Gly	Gln
1				5					10					15	
Arg	Val	Thr	Ile	Ser	Cys	Ser	Gly	Ser	Ser	Ser	Asn	Ile	Gly	Ser	Asn
			20				25						30		
Thr	Val	Asn	Trp	Tyr	Gln	Gln	Leu	Pro	Gly	Thr	Ala	Pro	Lys	Leu	Leu
		35					40				45				
Ile	Tyr	Tyr	Asp	Asp	Leu	Arg	Pro	Ser	Gly	Val	Pro	Asp	Arg	Phe	Ser
	50					55				60					
Gly	Ser	Lys	Ser	Gly	Thr	Ser	Ala	Ser	Leu	Ala	Ile	Ser	Gly	Leu	Gln
65					70					75				80	
Ser	Glu	Asp	Glu	Ala	Asp	Tyr	Tyr	Cys	His	Ala	Trp	Asp	Asp	Ser	Leu
			85					90						95	
Asn	Val	Tyr	Pro	Val	Phe	Gly	Gly	Gly	Thr	Lys	Leu	Thr	Val	Leu	Gly
			100					105					110		
Gln	Pro	Lys	Ala	Ala	Pro	Ser	Val	Thr	Leu	Phe	Pro	Pro	Ser	Ser	Glu
		115				120						125			
Glu	Leu	Gln	Ala	Asn	Lys	Ala	Thr	Leu	Val	Cys	Leu	Ile	Ser	Asp	Phe
	130				135					140					
Tyr	Pro	Gly	Ala	Val	Thr	Val	Ala	Trp	Lys	Ala	Asp	Ser	Ser	Pro	Val
145					150				155					160	
Lys	Ala	Gly	Val	Glu	Thr	Thr	Thr	Pro	Ser	Lys	Gln	Ser	Asn	Asn	Lys
			165					170					175		
Tyr	Ala	Ala	Ser	Ser	Tyr	Leu	Ser	Leu	Thr	Pro	Glu	Gln	Trp	Lys	Ser
			180				185						190		
His	Arg	Ser	Tyr	Ser	Cys	Gln	Val	Thr	His	Glu	Gly	Ser	Thr	Val	Glu
	195					200						205			
Lys	Thr	Val	Ala	Pro	Thr	Glu	Cys	Ser							
	210					215									

<210> SEQ ID NO 339
 <211> LENGTH: 1341
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 339

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg

60

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agctgtgccg ccagcggcctt caccttttac agctacgcca tgagctgggt ccgacaggcc	120
cctggaaaag gccttgaatg ggtgtccgcc atcgctatg gcggcgatac ctactacgcc	180
gactctgtga agggcagatt caccatcagc cgggacaaca gcaagaacac cctgtacctg	240
cagatgaaca gcctgagagc cgaggacacc gccgtgtact attgcgccag aagggaagac	300
tacaccagca gggacgcctt cgattattgg ggccagggca cactggctac cgtttcttca	360
gccagcacca agggccccag cgtgttcctt ctggccccct ttagcagaag caccagcgag	420
tctacagccg ccctgggctg cctcgtgaag gactactttc ccgagcccggt gaccgtgtcc	480
tggaaactctg gcgctctgac aagcggcgtg cacacctttc cagccgtgct gcagagcagc	540
ggcctgtact ctctgagcag cgtcgtgaca gtgccacgca gcagcctggg caccaagacc	600
tacacctgta acgtggacca caagcccagc aacaccaagg tggacaagcg ggtggaatct	660
aagtacggcc ctccctgccc tccttgccca gccctgaat ttctgggcgg accctccgtg	720
ttcctgttcc ccccaaagcc caaggacacc ctgatgatca gccggacccc cgaagtgacc	780
tgcgtggtgg tggatgtgtc ccaggaagat cccgaggtgc agttcaattg gtacgtggac	840
ggcgtggaag tgcacaacgc caagaccaag ccagagaggg aacagttcaa cagcacctac	900
cgggtggtgt ccgtgctgac agtgcctcac caggactggc tgaacggcaa agagtacaag	960
tgcaaggtgt ccaacaaggg cctgcccagc tccatcgaga aaacctcag caaggccaag	1020
ggccagcccc gcgaacccca ggtgtacaca ctgcctccaa gccaggaaga gatgaccaag	1080
aaccaggtgt ccctgacctg tctcgtgaaa ggcttctacc cctccgatat cgcctggaa	1140
tgggagagca acggccagcc cgagaacaac tacaagacca cccccctgt gctggacagc	1200
gacggctcat tcttctgta cagcagactg accgtggaca agagccgggtg gcaggaaggc	1260
aacgtgttca gctgcagcgt gatgcacgag gccctgcaca accactacac ccagaagtcc	1320
ctgtctctga gcctgggcaa g	1341

<210> SEQ ID NO 340

<211> LENGTH: 651

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 340

cagtctgttc tgacacagcc tcctagcgcc tctggcacac ctggacagag agtgaccatc	60
agctgtagcg gcagcagctc caacatcggc agcaacacgg tgaactggta tcagcagctg	120
cctggcacag ccctaaaact gctgatctac tacgacgacc tgccggcctag cggcgtgcca	180
gatagatttt ctggcagcaa gagcggcacc tctgccagcc tggctatttc tggactgcag	240
agcgaggacg agggcgacta ctattgtcac gccctgggacg acagccctgaa cgtgtaccct	300
gtttttggcg gaggcaccaa gctgaccgtt ctaggccagc ctaaagccgc ccctagcgtg	360
accctgttcc ctccaagcag cgaggaaactg caggccaaca agggccacct cgtgtgctg	420
atcagcgact tctatcctgg cgcctgacc gtggcctgga agggcgatag ctctcctgtg	480
aaggccggcg tggaaaccac caccctagc aagcagagca acaacaaata cgccgccagc	540
agctacctga gcctgacccc cgagcagtggt aagtcccaca gatcctacag ctgccaaagt	600
accacagagg gcagcacggt ggaaaagaca gtggccoccta ccgagtgcag c	651

<210> SEQ ID NO 341

<211> LENGTH: 120

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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 341

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Tyr Ser Tyr
20 25 30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45
Ser Ala Ile Gly Tyr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys
50 55 60
Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
65 70 75 80
Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
85 90 95
Arg Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr Trp Gly Gln
100 105 110
Gly Thr Leu Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 342
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 342

Ser Tyr Ala Met Ser
1 5

<210> SEQ ID NO 343
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 343

Ala Ile Gly Tyr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly
1 5 10 15

<210> SEQ ID NO 344
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 344

Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr
1 5 10

<210> SEQ ID NO 345
<211> LENGTH: 111
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 345

-continued

Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
 1 5 10 15
 Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn
 20 25 30
 Thr Val Asn Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu
 35 40 45
 Ile Tyr Tyr Asp Asp Leu Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
 50 55 60
 Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Gln
 65 70 75 80
 Ser Glu Asp Glu Ala Asp Tyr Tyr Cys Ala Ala Trp Asp Asp Ser Leu
 85 90 95
 Asn Asp Ile Pro Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
 100 105 110

<210> SEQ ID NO 346
 <211> LENGTH: 13
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 346

Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn Thr Val Asn
 1 5 10

<210> SEQ ID NO 347
 <211> LENGTH: 7
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 347

Tyr Asp Asp Leu Arg Pro Ser
 1 5

<210> SEQ ID NO 348
 <211> LENGTH: 12
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 348

Ala Ala Trp Asp Asp Ser Leu Asn Asp Ile Pro Val
 1 5 10

<210> SEQ ID NO 349
 <211> LENGTH: 360
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 349

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg 60
 agctgtgccg ccagcggctt caccctttac agctacgcca tgagctgggt ccgacaggcc 120
 cctggaaaag gccttgaatg ggtgtccgcc atcggtatg gcggcgatac ctactacgcc 180
 gactctgtga agggcagatt caccatcagc cgggacaaca gcaagaacac cctgtacctg 240

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cagatgaaca gcctgagagc cgaggacacc gccgtgtact attgcgccag aagggacgac	300
tacaccagca gggacgcctt cgattattgg ggccagggca cactggtcac cgtttcttca	360

<210> SEQ ID NO 350
 <211> LENGTH: 15
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 350

agctacgcca tgagc	15
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<210> SEQ ID NO 351
 <211> LENGTH: 48
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 351

gccatcggt atggcggcga tacctactac gccgactctg tgaagggc	48
---	----

<210> SEQ ID NO 352
 <211> LENGTH: 36
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 352

agggacgact acaccagcag ggacgccttc gattat	36
---	----

<210> SEQ ID NO 353
 <211> LENGTH: 333
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 353

cagtcgttc tgacacagcc tcctagcgcc tctggcacac ctggacagag agtgaccatc	60
agctgtagcg gcagcagctc caacatcggc agcaacaccg tgaactggta tcagcagctg	120
cctggcacag cccctaaact gctgatctac tacgacgacc tgcggcctag cggcgtgcca	180
gatagatttt ctggcagcaa gagcggcacc tctgccagcc tggctatttc tggactgcag	240
agcaggagcg aggcgcgacta ttattgtgcc gcttgggacg acagcctgaa cgacatccct	300
gtttttggcg gaggcaccaa gctgaccgtt cta	333

<210> SEQ ID NO 354
 <211> LENGTH: 39
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 354

agcggcagca gctccaacat cggcagcaac accgtgaac	39
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<210> SEQ ID NO 355
 <211> LENGTH: 21
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 355

tacgacgacc tgcggcctag c

21

<210> SEQ ID NO 356

<211> LENGTH: 36

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 356

gccgcctggg acgacagcct gaacgacatc cctgtt

36

<210> SEQ ID NO 357

<211> LENGTH: 447

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 357

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Tyr Ser Tyr
 20 25 30

Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ser Ala Ile Gly Tyr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys
 50 55 60

Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
 65 70 75 80

Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
 85 90 95

Arg Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr Trp Gly Gln
 100 105 110

Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val
 115 120 125

Phe Pro Leu Ala Pro Cys Ser Arg Ser Thr Ser Glu Ser Thr Ala Ala
 130 135 140

Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser
 145 150 155 160

Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val
 165 170 175

Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro
 180 185 190

Ser Ser Ser Leu Gly Thr Lys Thr Tyr Thr Cys Asn Val Asp His Lys
 195 200 205

Pro Ser Asn Thr Lys Val Asp Lys Arg Val Glu Ser Lys Tyr Gly Pro
 210 215 220

Pro Cys Pro Pro Cys Pro Ala Pro Glu Phe Leu Gly Gly Pro Ser Val
 225 230 235 240

Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr
 245 250 255

Pro Glu Val Thr Cys Val Val Val Asp Val Ser Gln Glu Asp Pro Glu
 260 265 270

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Val Gln Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys
275 280 285

Thr Lys Pro Arg Glu Glu Gln Phe Asn Ser Thr Tyr Arg Val Val Ser
290 295 300

Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys
305 310 315 320

Cys Lys Val Ser Asn Lys Gly Leu Pro Ser Ser Ile Glu Lys Thr Ile
325 330 335

Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro
340 345 350

Pro Ser Gln Glu Glu Met Thr Lys Asn Gln Val Ser Leu Thr Cys Leu
355 360 365

Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn
370 375 380

Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser
385 390 395 400

Asp Gly Ser Phe Phe Leu Tyr Ser Arg Leu Thr Val Asp Lys Ser Arg
405 410 415

Trp Gln Glu Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu
420 425 430

His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Leu Gly Lys
435 440 445

<210> SEQ ID NO 358

<211> LENGTH: 217

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 358

Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
1 5 10 15

Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Asn Ile Gly Ser Asn
20 25 30

Thr Val Asn Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu
35 40 45

Ile Tyr Tyr Asp Asp Leu Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
50 55 60

Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Gln
65 70 75 80

Ser Glu Asp Glu Ala Asp Tyr Tyr Cys Ala Ala Trp Asp Asp Ser Leu
85 90 95

Asn Asp Ile Pro Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu Gly
100 105 110

Gln Pro Lys Ala Ala Pro Ser Val Thr Leu Phe Pro Pro Ser Ser Glu
115 120 125

Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp Phe
130 135 140

Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Ser Ser Pro Val
145 150 155 160

Lys Ala Gly Val Glu Thr Thr Thr Pro Ser Lys Gln Ser Asn Asn Lys
165 170 175

Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys Ser
180 185 190

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His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val Glu
 195 200 205

Lys Thr Val Ala Pro Thr Glu Cys Ser
 210 215

<210> SEQ ID NO 359
 <211> LENGTH: 1341
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 359

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gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg    60
agctgtgccc ccagcggett caccttttac agctacgcca tgagctgggt ccgacaggcc    120
cctggaaaag gccttgaatg ggtgtccgcc atcggtatg gcggcgatac ctactacgcc    180
gactctgtga agggcagatt caccatcagc cgggacaaca gcaagaacac cctgtacctg    240
cagatgaaca gcctgagagc cgaggacacc gccgtgtact attgcgccag aagggacgac    300
tacaccagca gggacgcctt cgattattgg ggccagggca cactggtcac cgttttctca    360
gccagcacca agggccccag cgtgttcctt ctggccccct gtagcagaag caccagcgag    420
tctacagccg ccttggggtg cctcgtgaag gactactttc ccgagcccggt gaccgtgtcc    480
tggaactctg gcgctctgac aagcggcgtg cacacctttc cagccgtgct gcagagcagc    540
ggcctgtact ctctgagcag cgtcgtgaca gtgcccagca gcagcctggg caccaagacc    600
tacacctgta acgtggacca caagcccagc aacaccaagg tggacaagcg ggtggaatct    660
aagtaaggcc ctccctgcc tcttgccca gccctgaat ttctgggcgg accctccgtg    720
tctctgttcc cccaaaagcc caaggacacc ctgatgatca gccggacccc cgaagtgacc    780
tgcggtggtg tggatgtgtc ccaggaagat cccagagtgct agttcaattg gtacgtggac    840
ggcgtggaag tgcacaacgc caagaccaag cccagagagg aacagttcaa cagcacctac    900
cgggtggtgt ccgtgctgac agtgctgcac caggactggc tgaacggcaa agagtacaag    960
tgcaagggtg ccaacaaggg cctgcccagc tccatcgaga aaaccatcag caaggccaag    1020
ggccagcccc gcgaacccca ggtgtacaca ctgcctccaa gccaggaaga gatgaccaag    1080
aaccaggtgt cctgacctg tctcgtgaaa ggcttctacc cctccgatat cgcctgggaa    1140
tgggagagca acggccagcc cgagaacaac tacaagacca cccccctgt gctggacagc    1200
gacggctcat tcttctgta cagcagactg accgtggaca agagccgggt gcaggaaggc    1260
aacgtgttca gctgcagcgt gatgcacgag gccctgcaca accactacac ccagaagtcc    1320
ctgtctctga gcctgggcaa g                                     1341

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<210> SEQ ID NO 360
 <211> LENGTH: 651
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 360

```

cagtctgttc tgacacagcc tcctagcgcc tctggcacac ctggacagag agtgaccatc    60
agctgtagcg gcagcagetc caacatcggc agcaacaccg tgaactggta tcagcagctg    120
cctggcacag cccctaaact gctgatctac tacgacgacc tgccgcctag cggcgtgcca    180

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-continued

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gatagatttt ctggcagcaa gageggcacc tctgccagcc tggctatttc tggactgcag 240
agcgaggacg aggcgacta ttattgtgcc gctggggacg acagcctgaa cgacatccct 300
gtttttggcg gaggcaccaa gctgaccgtt ctaggccagc ctaaagccgc ccctagcgtg 360
acctgtttcc ctccaagcag cgaggaactg caggccaaca agggccacct cgtgtgctg 420
atcagcgact tctatcctgg cgccgtgacc gtggcctgga aggcgatag ctctcctgtg 480
aaggccggcg tggaaaccac caccctagc aagcagagca acaacaata cgccgccagc 540
agctacctga gctgacccc cgagcagtgg aagtcacaca gatcctacag ctgccaagtg 600
accacgagg gcagcaccgt ggaaaagaca gtggccccta ccgagtgcag c 651

```

```

<210> SEQ ID NO 361
<211> LENGTH: 120
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 361

```

```

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1          5          10          15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Tyr Ser Tyr
20        25        30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35        40        45
Ser Ala Ile Gly Tyr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys
50        55        60
Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
65        70        75        80
Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
85        90        95
Arg Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr Trp Gly Gln
100       105       110
Gly Thr Leu Val Thr Val Ser Ser
115       120

```

```

<210> SEQ ID NO 362
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 362

```

```

Ser Tyr Ala Met Ser
1          5

```

```

<210> SEQ ID NO 363
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 363

```

```

Ala Ile Gly Tyr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly
1          5          10          15

```

```

<210> SEQ ID NO 364
<211> LENGTH: 12

```

-continued

<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 364

Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr
1 5 10

<210> SEQ ID NO 365
<211> LENGTH: 111
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 365

Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
1 5 10 15

Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn
20 25 30

Thr Val Asn Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu
35 40 45

Ile Tyr Tyr Asp Asp Leu Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
50 55 60

Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Gln
65 70 75 80

Ser Glu Asp Glu Ala Asp Tyr Tyr Cys Ala Ala Trp Asp Asp Ser Leu
85 90 95

Asn Val Ile Pro Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
100 105 110

<210> SEQ ID NO 366
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 366

Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn Thr Val Asn
1 5 10

<210> SEQ ID NO 367
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 367

Tyr Asp Asp Leu Arg Pro Ser
1 5

<210> SEQ ID NO 368
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 368

Ala Ala Trp Asp Asp Ser Leu Asn Val Ile Pro Val
1 5 10

-continued

<210> SEQ ID NO 369
 <211> LENGTH: 360
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

 <400> SEQUENCE: 369

 gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg 60
 agctgtgccg ccagcgggett caccttttac agctacgcca tgagctgggt ccgacaggcc 120
 cctggaaaag gccttgaatg ggtgtccgcc atcggtatg gcggcgatac ctactacgcc 180
 gactctgtga agggcagatt caccatcagc cgggacaaca gcaagaacac cctgtacctg 240
 cagatgaaca gcctgagagc cgaggacacc gccgtgtact attgcgccag aagggaagac 300
 tacaccagca gggacgcctt cgattattgg ggcaggggca cactgggtcac cgtttcttca 360

 <210> SEQ ID NO 370
 <211> LENGTH: 15
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

 <400> SEQUENCE: 370

 agctacgcca tgagc 15

 <210> SEQ ID NO 371
 <211> LENGTH: 48
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

 <400> SEQUENCE: 371

 gccatcggt atggcggcga tacctactac gccgactctg tgaagggc 48

 <210> SEQ ID NO 372
 <211> LENGTH: 36
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

 <400> SEQUENCE: 372

 agggacgact acaccagcag ggacgccttc gattat 36

 <210> SEQ ID NO 373
 <211> LENGTH: 333
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

 <400> SEQUENCE: 373

 cagtctgttc tgacacagcc tcctagcgcc tctggcacac ctggacagag agtgaccatc 60
 agctgtagcg gcagcagctc caacatcggc agcaacaccg tgaactggta tcagcagctg 120
 cctggcacag ccctaaact gctgatctac tacgacgacc tgcggcctag cgcggtgcc 180
 gatagatttt ctggcagcaa gagcggcacc tctgccagcc tggctatttc tggactgcag 240
 agcgaggacg aggcgcacta ttattgtgcc gcctgggacg acagcctgaa cgtgatccct 300

-continued

gtttttggcg gaggcaccaa gctgaccgtt cta

333

<210> SEQ ID NO 374

<211> LENGTH: 39

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 374

agcggcagca gctccaacat cggcagcaac accgtgaac

39

<210> SEQ ID NO 375

<211> LENGTH: 21

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 375

tacgacgacc tgcggcctag c

21

<210> SEQ ID NO 376

<211> LENGTH: 36

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 376

gccgcctggg acgacagcct gaacgtgatc cctgtt

36

<210> SEQ ID NO 377

<211> LENGTH: 447

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 377

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Tyr Ser Tyr
20 25 30Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45Ser Ala Ile Gly Tyr Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys
50 55 60Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
65 70 75 80Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
85 90 95Arg Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Tyr Trp Gly Gln
100 105 110Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val
115 120 125Phe Pro Leu Ala Pro Cys Ser Arg Ser Thr Ser Glu Ser Thr Ala Ala
130 135 140Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser
145 150 155 160

Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val

165										170					175				
Leu	Gln	Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val	Thr	Val	Pro				
			180				185						190						
Ser	Ser	Ser	Leu	Gly	Thr	Lys	Thr	Tyr	Thr	Cys	Asn	Val	Asp	His	Lys				
			195				200						205						
Pro	Ser	Asn	Thr	Lys	Val	Asp	Lys	Arg	Val	Glu	Ser	Lys	Tyr	Gly	Pro				
			210				215						220						
Pro	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Phe	Leu	Gly	Gly	Pro	Ser	Val				
			225				230						235						
Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr				
			245							250			255						
Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	Gln	Glu	Asp	Pro	Glu				
			260							265			270						
Val	Gln	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys				
			275				280						285						
Thr	Lys	Pro	Arg	Glu	Glu	Gln	Phe	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser				
			290				295						300						
Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys				
			305				310						315						
Cys	Lys	Val	Ser	Asn	Lys	Gly	Leu	Pro	Ser	Ser	Ile	Glu	Lys	Thr	Ile				
			325							330			335						
Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro				
			340				345						350						
Pro	Ser	Gln	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu				
			355				360						365						
Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn				
			370				375						380						
Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp	Ser				
			385				390						395						
Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Arg	Leu	Thr	Val	Asp	Lys	Ser	Arg				
			405							410			415						
Trp	Gln	Glu	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu				
			420				425						430						
His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Leu	Gly	Lys					
			435				440						445						

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<210> SEQ ID NO 378
<211> LENGTH: 217
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence
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<400> SEQUENCE: 378

Gln	Ser	Val	Leu	Thr	Gln	Pro	Pro	Ser	Ala	Ser	Gly	Thr	Pro	Gly	Gln
1				5					10					15	
Arg	Val	Thr	Ile	Ser	Cys	Ser	Gly	Ser	Ser	Ser	Asn	Ile	Gly	Ser	Asn
			20					25					30		
Thr	Val	Asn	Trp	Tyr	Gln	Gln	Leu	Pro	Gly	Thr	Ala	Pro	Lys	Leu	Leu
			35				40					45			
Ile	Tyr	Tyr	Asp	Asp	Leu	Arg	Pro	Ser	Gly	Val	Pro	Asp	Arg	Phe	Ser
	50					55					60				
Gly	Ser	Lys	Ser	Gly	Thr	Ser	Ala	Ser	Leu	Ala	Ile	Ser	Gly	Leu	Gln
65					70					75				80	
Ser	Glu	Asp	Glu	Ala	Asp	Tyr	Tyr	Cys	Ala	Ala	Trp	Asp	Asp	Ser	Leu

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	85		90		95
Asn Val Ile Pro Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu Gly					
	100		105		110
Gln Pro Lys Ala Ala Pro Ser Val Thr Leu Phe Pro Pro Ser Ser Glu					
	115		120		125
Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp Phe					
	130		135		140
Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Ser Ser Pro Val					
	145		150		155
Lys Ala Gly Val Glu Thr Thr Thr Pro Ser Lys Gln Ser Asn Asn Lys					
	165		170		175
Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys Ser					
	180		185		190
His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val Glu					
	195		200		205
Lys Thr Val Ala Pro Thr Glu Cys Ser					
	210		215		

<210> SEQ ID NO 379

<211> LENGTH: 1341

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 379

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg	60
agctgtgccc ccagcggtt cactctttac agctacgcca tgagctgggt ccgacaggcc	120
cctggaaaag gccttgaatg ggtgtccgcc atcggtatg gcggcgatac ctactacgcc	180
gactctgtga agggcagatt caccatcagc cgggacaaca gcaagaacac cctgtacctg	240
cagatgaaca gcctgagagc cgaggacacc gccgtgtact attgcgccag aagggacgac	300
tacaccagca gggagccctt cgattattgg ggcagggca cactggtcac cgtttcttca	360
gccagacca agggccccag cgtgttcct ctggcccctt gtagcagaag caccagcgag	420
tctacagccg ccctgggctg cctcgtgaag gactactttc ccgagcccgt gaccgtgtcc	480
tggaaactctg gcgctctgac aagcggcgtg cacacctttc cagccgtgct gcagagcagc	540
ggcctgtact ctctgagcag cgtcgtgaca gtgcccagca gcagcctggg caccaagacc	600
tacacctgta acgtggacca caagcccagc aacaccaagg tggacaagcg ggtggaatct	660
aagtacggcc ctccctgcc tcttgccca gccctgaat ttctgggcgg acctccgtg	720
ttcctgttcc ccccaaagcc caaggacacc ctgatgatca gccggacccc cgaagtgacc	780
tgctgtgtgg tggatgtgtc ccaggaagat cccgaggtgc agttcaattg gtacgtggac	840
ggcgtggaag tgcacaacgc caagaccaag cccagagagg aacagttcaa cagcacctac	900
cgggtggtgt ccgtgctgac agtgtgcac caggactggc tgaacggcaa agagtacaag	960
tgcaaggtgt ccaacaaggg cctgcccagc tccatcgaga aaaccatcag caaggccaag	1020
ggccagcccc gcgaacccca ggtgtacaca ctgcctccaa gccaggaaga gatgaccaag	1080
aaccaggtgt ccctgacctg tctcgtgaaa ggcttctacc cctccgatat cgcctgggaa	1140
tggaagagca acggccagcc cgagaacaac tacaagacca cccccctgt gctggacagc	1200
gacggctcat tcttctgta cagcagactg accgtggaca agagccgggtg gcaggaaggc	1260
aacgtgttca gctgcagcgt gatgcacgag gccctgcaca accactacac ccagaagtcc	1320

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ctgtctctga gcctgggcaa g 1341

<210> SEQ ID NO 380
 <211> LENGTH: 651
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 380

```

cagtctgttc tgacacagcc tcctagcgcc tctggcacac ctggacagag agtgaccatc   60
agctgttagcg gcagcagctc caacatcggc agcaacaccg tgaactggta tcagcagctg   120
cctggcacag cccctaaact gctgatctac tacgacgacc tgcggcctag cggcgtgccca   180
gatagatttt ctggcagcaa gagcggcacc tctgccagcc tggctatttc tggactgcag   240
agcgaggacg aggcgacta ttattgtgcc gcctgggacg acagcctgaa cgtgatccct   300
gtttttggcg gaggcaccaa gctgaccgtt ctaggccagc ctaaagccgc ccctagcgtg   360
acctgtttcc ctccaagcag cgaggaactg caggccaaca aggccaccct cgtgtgctg   420
atcagcgact tctatcctgg cgccgtgacc gtggcctgga aggcgatag ctctcctgtg   480
aaggccggcg tggaaccac caccctagc aagcagagca acaacaaata cgccgccagc   540
agctacctga gcctgacccc cgagcagtgg aagtcccaca gatcctacag ctgccaagtg   600
acccacgagg gcagcaccgt ggaaaagaca gtggccccta ccgagtgcag c           651

```

<210> SEQ ID NO 381
 <211> LENGTH: 123
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 381

```

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1             5             10             15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asp Ser Tyr
20            25            30
Glu Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35            40            45
Ser Gly Ile Ser Trp Asn Ser Gly Trp Ile Asp Tyr Ala Asp Ser Val
50            55            60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65            70            75            80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85            90            95
Ala Arg Ser Gly Tyr Ser Ser Ser Trp Phe Asp Pro Asp Phe Asp Tyr
100           105           110
Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
115           120

```

<210> SEQ ID NO 382
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 382

-continued

Ser Tyr Glu Met Asn
1 5

<210> SEQ ID NO 383
 <211> LENGTH: 17
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 383

Gly Ile Ser Trp Asn Ser Gly Trp Ile Asp Tyr Ala Asp Ser Val Lys
 1 5 10 15

Gly

<210> SEQ ID NO 384
 <211> LENGTH: 14
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 384

Ser Gly Tyr Ser Ser Ser Trp Phe Asp Pro Asp Phe Asp Tyr
 1 5 10

<210> SEQ ID NO 385
 <211> LENGTH: 111
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 385

Gln Ser Val Leu Thr Gln Pro Pro Ser Val Ser Gly Ala Pro Gly Gln
 1 5 10 15

Arg Val Thr Ile Ser Cys Thr Gly Ser Ser Ser Asp Ile Gly Ala Gly
 20 25 30

Tyr Asp Val His Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu
 35 40 45

Leu Ile Tyr Gly Asn Ser Asn Arg Pro Ser Gly Val Pro Asp Arg Phe
 50 55 60

Ser Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Thr Gly Leu
 65 70 75 80

Gln Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Ser Ser Tyr Ala Gly Pro
 85 90 95

Asn Pro Tyr Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
 100 105 110

<210> SEQ ID NO 386
 <211> LENGTH: 14
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 386

Thr Gly Ser Ser Ser Asp Ile Gly Ala Gly Tyr Asp Val His
 1 5 10

<210> SEQ ID NO 387
 <211> LENGTH: 7
 <212> TYPE: PRT

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 387

Gly Asn Ser Asn Arg Pro Ser
1 5

<210> SEQ ID NO 388
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 388

Ser Ser Tyr Ala Gly Pro Asn Pro Tyr Val Val
1 5 10

<210> SEQ ID NO 389
<211> LENGTH: 369
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 389

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg 60
agctgtgccc ccagcggett caccttcgat agctacgaga tgaactgggt ccgacaggcc 120
cctggcaaag gccttgaatg ggtgtccggc atcagctgga atagcggctg gatcgactac 180
gccgacagcg tgaagggcag attcaccatc agccgggaca acagcaagaa caccctgtac 240
ctgcagatga acagcctgag agccgaggac accgccgtgt actactgtgc cagaagcggc 300
tacagcagct cttggtttga ccccgacttc gactattggg gccagggcac actggtcaca 360
gtctcttca 369

<210> SEQ ID NO 390
<211> LENGTH: 15
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 390

agctacgaga tgaac 15

<210> SEQ ID NO 391
<211> LENGTH: 51
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 391

ggcatcagct ggaatagcgg ctggatcgac tacgccgaca gcgtgaaggg c 51

<210> SEQ ID NO 392
<211> LENGTH: 42
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 392

-continued

agcgggtaca gcagctcttg gtttgacccc gacttogact at 42

<210> SEQ ID NO 393
 <211> LENGTH: 333
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 393

cagtctgttc tgacacagcc tccatctgtg tctggcgccc ctggacagag agtgaccatc 60
 agctgtacag gcagcagctc cgatattggc gccggatacg acgtgcactg gtatcagcaa 120
 ctgcctggca cagcccctaa gctgctgac tacggcaaca gcaacagacc tagcggcggtg 180
 cccgatagat tcagcggctc taagtctggc acaagcgcca gcctggccat tactggactg 240
 caggccgaag atgaggccga ctactactgc agcagctacg ctggccccaa tccttacgtg 300
 gtgtttggcg gcggaacaaa gctgaccgtt cta 333

<210> SEQ ID NO 394
 <211> LENGTH: 42
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 394

acaggcagca gctccgatat tggcgccgga tacgacgtgc ac 42

<210> SEQ ID NO 395
 <211> LENGTH: 21
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 395

ggcaacagca acagacctag c 21

<210> SEQ ID NO 396
 <211> LENGTH: 33
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 396

agcagctacg ctggccccaa tccttacgtg gtg 33

<210> SEQ ID NO 397
 <211> LENGTH: 453
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 397

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asp Ser Tyr
 20 25 30

Glu Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

[illegible]

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<210> SEQ ID NO 398
 <211> LENGTH: 217
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 398

Gln Ser Val Leu Thr Gln Pro Pro Ser Val Ser Gly Ala Pro Gly Gln
 1 5 10 15
 Arg Val Thr Ile Ser Cys Thr Gly Ser Ser Ser Asp Ile Gly Ala Gly
 20 25 30
 Tyr Asp Val His Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu
 35 40 45
 Leu Ile Tyr Gly Asn Ser Asn Arg Pro Ser Gly Val Pro Asp Arg Phe
 50 55 60
 Ser Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Thr Gly Leu
 65 70 75 80
 Gln Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Ser Ser Tyr Ala Gly Pro
 85 90 95
 Asn Pro Tyr Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu Gly
 100 105 110
 Gln Pro Lys Ala Ala Pro Ser Val Thr Leu Phe Pro Pro Ser Ser Glu
 115 120 125
 Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp Phe
 130 135 140
 Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Ser Ser Pro Val
 145 150 155 160
 Lys Ala Gly Val Glu Thr Thr Thr Pro Ser Lys Gln Ser Asn Asn Lys
 165 170 175
 Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys Ser
 180 185 190
 His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val Glu
 195 200 205
 Lys Thr Val Ala Pro Thr Glu Cys Ser
 210 215

<210> SEQ ID NO 399
 <211> LENGTH: 1359
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 399

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg 60
 agctgtgccg ccagcggcgtt caccttcgat agctacgaga tgaactgggt ccgacaggcc 120
 cctggc aaag gccttgaatg ggtgtccggc atcagctgga atagcggctg gategactac 180
 gccgacagcg tgaagggcag attcaccatc agccgggaca acagcaagaa caccctgtac 240
 ctgcagatga acagcctgag agccgaggac accgccgtgt actactgtgc cagaagcggc 300
 tacagcagct cttggtttga ccccgacttc gactattggg gccagggcac actgggtcaca 360
 gtctcttcag ccagcaccaa gggccccagc gtgttccttc tggccctag cagcaagagc 420
 acatctggcg gaacagccgc cctgggctgc ctcgtgaagg actactttcc cgagcccgctg 480
 accgtgtcct ggaactctgg cgctctgaca agcggcgtgc acacctttcc agccgtgctg 540

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cagagcagcg gccgtgactc tctgagcagc gtcgtgacag tgcccagcag ctctctgggc 600
acccagacct acatctgcaa cgtgaaccac aagcccagca acaccaaggt ggacaagaag 660
gtggaaccca agagctgcga caagaccac acctgtcccc cttgtcctgc ccccgaactg 720
ctgggaggcc cttccgtgtt cctgttcccc ccaaagccca aggacacct gatgatcagc 780
cggacccccg aagtgcactg cgtggtggtg gatgtgtccc acgaggaccc tgaagtgaag 840
ttcaattggt acgtggacgg cgtggaagtg cacaacgcca agaccaagcc tagagaggaa 900
cagtacaaca gcacctaccg ggtggtgtcc gtgctgacag tgctgcacca ggactggctg 960
aacggcгааг agtacaagtг caagggtgtcc aacaaggccc tgctgtcccc catcgagaaa 1020
accatcagca aggcгаaggг ccagccccgc gaaccccagg tgtacacact gccccгагс 1080
agggacgagc tgaccaagaa ccagggtgtcc ctgacctgtc tcgtgaaagg cttctacccc 1140
tccgatatcg ccgtggaatg ggagagcaac ggcagccccg agaacaacta caagaccacc 1200
ccccctgtgc tggacagcga cggtctattc ttcctgtaca gcaagctgac cgtggacaag 1260
tcccgggtggc agcaggгсaa cgtgttcagc tgcagcgtga tgcacgaggc cctgcacaac 1320
cactacaccс agaagtcctt gagcctgagc cctggcгаг 1359

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<210> SEQ ID NO 400
<211> LENGTH: 651
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 400

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cagtcgttс tgacacagcc tccatctgtg tctggcgccc ctggacagag agtgaccatс 60
agctgtacag gcagcagctc cgatattggc gccggatacg acgtgcactg gtatcagcaa 120
ctgcctggca cagcccctaa gctgctgac tacggcaaca gcaacagacc tagcggcgtg 180
cccgatagat tcagcggtc taagtctggc acaagcgcca gcctggccat tactggactg 240
caggccгаг atgaggccga ctactactgc agcagctacg ctggccccaa tccttacgtg 300
gtgtttggcg gcggaacaaa gctgaccgtt ctaggccagc ctaaagccgc ccctagcgtg 360
accctgttcc ctccaagcag cgaggaaactg caggccaaca aggccacct cgtgtgcctg 420
atcagcgact tctatcctgg cgccgtgacc gtggcctgga aggcgatag ctctcctgtg 480
aaggccggcg tggaaaccac caccctagc aagcagagca acaacaaata cgccgccagc 540
agctacctga gcctgacccc cgagcagtgг aagtcссaс gatcctacag ctgccaagtг 600
accacgagg gcagcacctt ggaaaagaca gtggccссta ccgagtgcag c 651

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<210> SEQ ID NO 401
<211> LENGTH: 123
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 401

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Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1           5           10          15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asp Ser Tyr
20          25          30

Glu Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35          40          45

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Ser Gly Ile Ser Trp Asn Ser Gly Trp Ile Asp Tyr Ala Asp Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
 65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Ala Arg Ser Gly Tyr Ser Ser Ser Trp Phe Asp Pro Asp Phe Asp Tyr
 100 105 110

Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 402
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 402

Ser Tyr Glu Met Asn
 1 5

<210> SEQ ID NO 403
 <211> LENGTH: 17
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 403

Gly Ile Ser Trp Asn Ser Gly Trp Ile Asp Tyr Ala Asp Ser Val Lys
 1 5 10 15

Gly

<210> SEQ ID NO 404
 <211> LENGTH: 14
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 404

Ser Gly Tyr Ser Ser Ser Trp Phe Asp Pro Asp Phe Asp Tyr
 1 5 10

<210> SEQ ID NO 405
 <211> LENGTH: 111
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 405

Gln Ser Val Leu Thr Gln Pro Pro Ser Val Ser Gly Ala Pro Gly Gln
 1 5 10 15

Arg Val Thr Ile Ser Cys Thr Gly Ser Ser Ser Asn Ile Gly Ala Gly
 20 25 30

Tyr Asp Val His Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu
 35 40 45

Leu Ile Tyr Gly Asn Ser Asn Arg Pro Ser Gly Val Pro Asp Arg Phe
 50 55 60

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Ser Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Thr Gly Leu
65 70 75 80

Gln Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Gln Ser Tyr Ala Gly Ile
85 90 95

Asn Pro Tyr Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
100 105 110

<210> SEQ ID NO 406
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 406

Thr Gly Ser Ser Ser Asn Ile Gly Ala Gly Tyr Asp Val His
1 5 10

<210> SEQ ID NO 407
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 407

Gly Asn Ser Asn Arg Pro Ser
1 5

<210> SEQ ID NO 408
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 408

Gln Ser Tyr Ala Gly Ile Asn Pro Tyr Val Val
1 5 10

<210> SEQ ID NO 409
<211> LENGTH: 369
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 409

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg	60
agctgtgccg ccagcggctt caccttcgat agctacgaga tgaactgggt ccgacaggcc	120
cctggcaaaag gccttgaatg ggtgtccggc atcagctgga atagcggctg gatcgactac	180
gccgacagcg tgaagggcag attcaccatc agccgggaca acagcaagaa caccctgtac	240
ctgcagatga acagcctgag agccgaggac accgccgtgt actactgtgc cagaagcggc	300
tacagcagct cttggtttga ccccgacttc gactattggg gccagggcac actggtcaca	360
gtctcttca	369

<210> SEQ ID NO 410
<211> LENGTH: 15
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 410

agctacgaga tgaac

15

<210> SEQ ID NO 411

<211> LENGTH: 51

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 411

ggcatcagct ggaatagcgg ctggatcgac tacgccgaca gcgtgaaggg c

51

<210> SEQ ID NO 412

<211> LENGTH: 42

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 412

agcggctaca gcagctcttg gtttgacccc gacttcgact at

42

<210> SEQ ID NO 413

<211> LENGTH: 333

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 413

cagctgttgc tgacacagcc tccatctgtg tctggcgccc ctggacagag agtgaccatc

60

agctgtacag gcagcagctc caatatcgga gccggctatg acgtgcactg gtatcagcag

120

ctgcctggca cagcccctaa actgctgac tacggcaaca gcaacagacc cagcggcgtg

180

cccgatagat tttccggctc taagagcggc acaagcgcca gcctggctat tactggactg

240

caggccgagg acgaggccga ctactactgt cagagctacg ccggcatcaa cccctacgtg

300

gtgtttggcg gaggcaccaa gctgacagtt cta

333

<210> SEQ ID NO 414

<211> LENGTH: 42

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 414

acaggcagca gctccaatat cggagccggc tatgacgtgc ac

42

<210> SEQ ID NO 415

<211> LENGTH: 21

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 415

ggcaacagca acagacccag c

21

<210> SEQ ID NO 416

<211> LENGTH: 33

<212> TYPE: DNA

-continued

<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 416

cagagctacg ccggcatcaa cccctacgtg gtg

33

<210> SEQ ID NO 417
<211> LENGTH: 453
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 417

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asp Ser Tyr
20 25 30

Glu Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45

Ser Gly Ile Ser Trp Asn Ser Gly Trp Ile Asp Tyr Ala Asp Ser Val
50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95

Ala Arg Ser Gly Tyr Ser Ser Ser Trp Phe Asp Pro Asp Phe Asp Tyr
100 105 110

Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly
115 120 125

Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly
130 135 140

Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val
145 150 155 160

Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe
165 170 175

Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val
180 185 190

Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val
195 200 205

Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys Lys Val Glu Pro Lys
210 215 220

Ser Cys Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu
225 230 235 240

Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr
245 250 255

Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val
260 265 270

Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val
275 280 285

Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser
290 295 300

Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu
305 310 315 320

Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala

-continued

325	330	335
Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro		
340	345	350
Gln Val Tyr Thr Leu Pro Pro Ser Arg Asp Glu Leu Thr Lys Asn Gln		
355	360	365
Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala		
370	375	380
Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr		
385	390	395
Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu		
405	410	415
Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser		
420	425	430
Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser		
435	440	445
Leu Ser Pro Gly Lys		
450		

<210> SEQ ID NO 418
 <211> LENGTH: 217
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 418

Gln Ser Val Leu Thr Gln Pro Pro Ser Val Ser Gly Ala Pro Gly Gln	
1	15
Arg Val Thr Ile Ser Cys Thr Gly Ser Ser Ser Asn Ile Gly Ala Gly	
20	30
Tyr Asp Val His Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu	
35	45
Leu Ile Tyr Gly Asn Ser Asn Arg Pro Ser Gly Val Pro Asp Arg Phe	
50	60
Ser Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Thr Gly Leu	
65	80
Gln Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Gln Ser Tyr Ala Gly Ile	
85	95
Asn Pro Tyr Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu Gly	
100	110
Gln Pro Lys Ala Ala Pro Ser Val Thr Leu Phe Pro Pro Ser Ser Glu	
115	125
Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp Phe	
130	140
Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Ser Ser Pro Val	
145	160
Lys Ala Gly Val Glu Thr Thr Thr Pro Ser Lys Gln Ser Asn Asn Lys	
165	175
Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys Ser	
180	190
His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val Glu	
195	205
Lys Thr Val Ala Pro Thr Glu Cys Ser	
210	215

-continued

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<210> SEQ ID NO 419
<211> LENGTH: 1359
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 419

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg      60
agctgtgccg ccagcggcgt cactctcgat agctacgaga tgaactgggt ccgacaggcc      120
cctggcaaag gccttgaatg ggtgtccggc atcagctgga atagcggctg gatcgactac      180
gccgacagcg tgaagggcag attcaccatc agccgggaca acagcaagaa caccctgtac      240
ctgcagatga acagcctgag agccgaggac accgccgtgt actactgtgc cagaagcggc      300
tacagcagct cttggtttga ccccgacttc gactattggg gccagggcac actggtcaca      360
gtctcttcag ccagcaccaa gggccccagc gtgttcctc tggccctag cagcaagagc      420
acatctggcg gaacagccgc cctgggctgc ctctgaagg actactttcc cgagcccgctg      480
accgtgtcct ggaactctgg cgctctgaca agcggcgtgc acacctttcc agcgtgctg      540
cagagcagcg gcctgtactc tctgagcagc gtcgtgacag tgcccagcag ctctctgggc      600
accagacct acatctgcaa cgtgaaccac aagcccagca acaccaaggt ggacaagaag      660
gtggaaccca agagctgaga caagaccac accgtgtccc cttgtcctgc cccggaactg      720
ctgggaggcc cttcgtgtt cctgttccc ccaaagccca aggacacct gatgatcagc      780
cggacccccg aagtgacctg cgtggtggtg gatgtgtccc acgaggaccc tgaagtgaag      840
ttcaattggt acgtggacgg cgtggaagtg cacaacgcc aagccaagcc tagagaggaa      900
cagtacaaca gcacctaccg ggtggtgtcc gtgctgacag tgctgcacca ggactggctg      960
aacggcaaaag agtacaagtg caagggtgtcc aacaaggccc tgctgtcccc catcgagaaa     1020
accatcagca aggccaaggg ccagccccgc gaaccccagg tgtacacact gcccccaagc     1080
agggacgagc tgaccaagaa ccagggtgtcc ctgacctgtc tcgtgaaagg cttctacccc     1140
tccgatatcg ccgtggaatg ggagagcaac ggccagcccc agaacaacta caagaccacc     1200
ccccctgtgc tggacagcga cggctcattc ttcctgtaca gcaagctgac cgtggacaag     1260
tcccgggtggc agcagggcaa cgtgttcagc tgcagcgtga tgcacgaggc cctgcacaac     1320
cactacaccc agaagtccct gagcctgagc cctggcaag                               1359

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```

<210> SEQ ID NO 420
<211> LENGTH: 651
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 420

cagttctgtt tgacacagcc tccatctgtg tctggcgccc ctggacagag agtgaccatc      60
agctgtacag gcagcagctc caatatcgga gccggctatg acgtgactg gtatcagcag      120
ctgcctggca cagcccctaa actgtgtatc tacggcaaca gcaacagacc cagcggcgtg      180
cccgatagat tttccggctc taagagcggc acaagcgcca gcctggctat tactggactg      240
caggccgagg acgaggccga ctactactgt cagagctacg ccgcatcaa cccctacgtg      300
gtgtttggcg gaggcaccaa gctgacagtt ctaggccagc ctaaagccgc ccctagcgtg      360
accctgttcc ctccaagcag cgaggaaactg caggccaaca aggccacctt cgtgtgcctg      420

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atcagcgact tctatcctgg cgccgtgacc gtggcctgga aggcgatag ctctcctgtg 480
aaggccggcg tggaaccac caccctagc aagcagagca acaacaata cgccgccagc 540
agctacctga gctgacccc cgagcagtg aagtcacaca gatcctacag ctgccaagtg 600
acccacgagg gcagcacctg ggaaaagaca gtggccccta ccgagtgcag c 651

```

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<210> SEQ ID NO 421
<211> LENGTH: 123
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 421

```

```

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1           5           10          15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Asp Phe Ser Ser Tyr
20          25          30
Glu Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35          40          45
Ser Gly Ile Ser Trp Asn Ser Gly Trp Ile Gly Tyr Ala Asp Ser Val
50          55          60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65          70          75          80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85          90          95
Ala Arg Ser Gly Tyr Ser Ser Ser Trp Phe Asp Pro Asp Phe Asp Tyr
100         105         110
Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
115         120

```

```

<210> SEQ ID NO 422
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 422

```

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Ser Tyr Glu Met Asn
1           5

```

```

<210> SEQ ID NO 423
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

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<400> SEQUENCE: 423

```

```

Gly Ile Ser Trp Asn Ser Gly Trp Ile Gly Tyr Ala Asp Ser Val Lys
1           5           10          15
Gly

```

```

<210> SEQ ID NO 424
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 424

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-continued

Ser Gly Tyr Ser Ser Ser Trp Phe Asp Pro Asp Phe Asp Tyr
1 5 10

<210> SEQ ID NO 425
<211> LENGTH: 111
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 425

Gln Ser Val Leu Thr Gln Pro Pro Ser Val Ser Gly Ala Pro Gly Gln
1 5 10 15
Arg Val Thr Ile Ser Cys Thr Gly Ser Ser Ser Asn Ile Gly Ala Gly
20 25 30
Tyr Asp Val His Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu
35 40 45
Leu Ile Tyr Gly Asn Ser Asn Arg Pro Ser Gly Val Pro Asp Arg Phe
50 55 60
Ser Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Thr Gly Leu
65 70 75 80
Gln Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Gln Ser Tyr Ala Gly Pro
85 90 95
Asn Pro Tyr Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
100 105 110

<210> SEQ ID NO 426
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 426

Thr Gly Ser Ser Ser Asn Ile Gly Ala Gly Tyr Asp Val His
1 5 10

<210> SEQ ID NO 427
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 427

Gly Asn Ser Asn Arg Pro Ser
1 5

<210> SEQ ID NO 428
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 428

Gln Ser Tyr Ala Gly Pro Asn Pro Tyr Val Val
1 5 10

<210> SEQ ID NO 429
<211> LENGTH: 369
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

-continued

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<211> LENGTH: 42
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 434
acaggcagca gctccaatat cggagccggc tatgacgtgc ac                42

<210> SEQ ID NO 435
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 435
ggcaacagca acagaccag c                21

<210> SEQ ID NO 436
<211> LENGTH: 33
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 436
cagtcttacg ctggcccaaa tccttacgtg gtg                33

<210> SEQ ID NO 437
<211> LENGTH: 453
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 437
Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1      5      10      15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Asp Phe Ser Ser Tyr
20     25     30
Glu Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35     40     45
Ser Gly Ile Ser Trp Asn Ser Gly Trp Ile Gly Tyr Ala Asp Ser Val
50     55     60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65     70     75     80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85     90     95
Ala Arg Ser Gly Tyr Ser Ser Ser Trp Phe Asp Pro Asp Phe Asp Tyr
100    105    110
Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly
115    120    125
Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly
130    135    140
Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val
145    150    155    160
Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe
165    170    175
Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val
180    185    190

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Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val
 195 200 205
 Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys Lys Val Glu Pro Lys
 210 215 220
 Ser Cys Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu
 225 230 235 240
 Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr
 245 250 255
 Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val
 260 265 270
 Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val
 275 280 285
 Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser
 290 295 300
 Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu
 305 310 315 320
 Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala
 325 330 335
 Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro
 340 345 350
 Gln Val Tyr Thr Leu Pro Pro Ser Arg Asp Glu Leu Thr Lys Asn Gln
 355 360 365
 Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala
 370 375 380
 Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr
 385 390 395 400
 Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu
 405 410 415
 Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser
 420 425 430
 Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser
 435 440 445
 Leu Ser Pro Gly Lys
 450

<210> SEQ ID NO 438

<211> LENGTH: 217

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 438

Gln Ser Val Leu Thr Gln Pro Pro Ser Val Ser Gly Ala Pro Gly Gln
 1 5 10 15
 Arg Val Thr Ile Ser Cys Thr Gly Ser Ser Ser Asn Ile Gly Ala Gly
 20 25 30
 Tyr Asp Val His Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu
 35 40 45
 Leu Ile Tyr Gly Asn Ser Asn Arg Pro Ser Gly Val Pro Asp Arg Phe
 50 55 60
 Ser Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Thr Gly Leu
 65 70 75 80
 Gln Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Gln Ser Tyr Ala Gly Pro
 85 90 95

-continued

Asn Pro Tyr Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu Gly
 100 105 110
 Gln Pro Lys Ala Ala Pro Ser Val Thr Leu Phe Pro Pro Ser Ser Glu
 115 120 125
 Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp Phe
 130 135 140
 Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Ser Ser Pro Val
 145 150 155 160
 Lys Ala Gly Val Glu Thr Thr Thr Pro Ser Lys Gln Ser Asn Asn Lys
 165 170 175
 Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys Ser
 180 185 190
 His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val Glu
 195 200 205
 Lys Thr Val Ala Pro Thr Glu Cys Ser
 210 215

<210> SEQ ID NO 439

<211> LENGTH: 1359

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 439

```

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg      60
agctgtgccc ccagcggtt cgatttcagc agctacgaga tgaactgggt ccgacaggcc      120
cctggcaaa gacctgaatg ggtgtccggc atcagctgga atagcggtg gatcggtac      180
gccgatagcg tgaagggcag attcaccatc agccgggaca acagcaagaa caccctgtac      240
ctgcagatga acagcctgag agccgaggac accgccgtgt actactgtgc cagaagcggc      300
tacagcagct cttggtttga ccccgacttc gactattggg gccagggcac actggtcaca      360
gtctcttcag ccagcaccaa gggccccagc gtgttccttc tggccctag cagcaagagc      420
acatctggcg gaacagccgc cctgggctgc ctctgtaagg actactttcc cgagccctg      480
accgtgtcct ggaactctgg cgctctgaca agcggcgtgc acacctttcc agcgtgctg      540
cagagcagcg gcctgtactc tctgagcagc gtcgtgacag tgcccagcag ctctctgggc      600
accagacct acatctgcaa cgtgaaccac aagcccagca acaccaaggt ggacaagaag      660
gtggaacca agagctgcca caagaccac acctgtcccc cttgtcctgc cccgaactg      720
ctgggaggcc cttccgtgtt cctgttcccc ccaaagccca aggacacct gatgatcagc      780
cggacccccg aagtgcctg cgtggtggtg gatgtgtccc acgaggacct tgaagtgaag      840
ttcaattggt acgtggacgg cgtggaagtg cacaacgcca agaccaagcc tagagaggaa      900
cagtacaaca gcacctaccg ggtggtgtcc gtgctgacag tgctgcacca ggactggctg      960
aacggcaaa agtacaagtg caaggtgtcc aacaaggccc tgctgcccc catcgagaaa     1020
accatcagca aggccaaggg ccagccccgc gaaccccagg tgtacacact gccccaaagc     1080
agggacgagc tgaccaagaa ccaggtgtcc ctgacctgtc tcgtgaaagg cttctacccc     1140
tccgatatcg ccgtggaatg ggagagcaac ggccagcccc agaacaacta caagaccacc     1200
ccccctgtgc tggacagcga cggtcattc ttctgtaca gcaagctgac cgtggacaag     1260
tcccgggtgc agcagggcaa cgtgttcagc tgcagcgtga tgcacgaggc cctgcacaac     1320

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-continued

cactacaccc agaagtcct gagcctgagc cctggcaag 1359

<210> SEQ ID NO 440
 <211> LENGTH: 651
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 440

```

cagtctgttc tgacacagcc tccatctgtg tctggcgccc ctggacagag agtgaccatc    60
agctgtacag gcagcagctc caatatcgga gccggctatg acgtgcactg gtatcagcag    120
ctgcctggca cagcccctaa actgctgata tacggcaaca gcaacagacc cagcggcgtg    180
cccgatagat ttccggctc taagagcggc acaagcgcca gcctggctat tactggactg    240
caggccgagg acgaggccga ctactactgt cagtcttacg ctggcccaaa tccttacgtg    300
gtgtttggcg gcggaacaaa gctgaccgtt ctaggccagc ctaaagcgcg ccctagcgtg    360
accctgttcc ctccaagcag cgaggaaactg caggccaaca aggccaccct cgtgtgctg    420
atcagcgact tctatcctgg cgccgtgacc gtggcctgga aggcgatag ctctcctgtg    480
aaggccggcg tggaaaccac caccctagc aagcagagca acaacaaata cgccgccagc    540
agctacctga gcctgacccc cgagcagtgg aagtcacaca gatcctacag ctgccaaagt    600
accacagagg gcagcaccgt ggaaaagaca gtggccccta ccgagtgcag c          651

```

<210> SEQ ID NO 441
 <211> LENGTH: 123
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 441

```

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1      5      10      15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Asp Phe Ser Ser Tyr
20     25     30
Glu Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35     40     45
Ser Gly Ile Ser Trp Asn Ser Gly Trp Ile Asp Tyr Ala Asp Ser Val
50     55     60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65     70     75     80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85     90     95
Ala Arg Ser Gly Tyr Ser Ser Ser Trp Phe Asp Pro Asp Phe Asp Tyr
100    105    110
Trp Gly Gly Gly Thr Leu Val Thr Val Ser Ser
115    120

```

<210> SEQ ID NO 442
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 442

Ser Tyr Glu Met Asn

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1           5

<210> SEQ ID NO 443
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

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<400> SEQUENCE: 443

```

```

Gly Ile Ser Trp Asn Ser Gly Trp Ile Asp Tyr Ala Asp Ser Val Lys
1           5           10           15

```

```

Gly

```

```

<210> SEQ ID NO 444
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 444

```

```

Ser Gly Tyr Ser Ser Ser Trp Phe Asp Pro Asp Phe Asp Tyr
1           5           10

```

```

<210> SEQ ID NO 445
<211> LENGTH: 111
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 445

```

```

Gln Ser Val Leu Thr Gln Pro Pro Ser Val Ser Gly Ala Pro Gly Gln
1           5           10           15

```

```

Arg Val Thr Ile Ser Cys Thr Gly Ser Ser Ser Asn Ile Gly Ala Gly
20           25           30

```

```

Tyr Asp Val His Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu
35           40           45

```

```

Leu Ile Tyr Gly Asn Ser Asn Arg Pro Ser Gly Val Pro Asp Arg Phe
50           55           60

```

```

Ser Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Thr Gly Leu
65           70           75           80

```

```

Gln Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Gln Ser Tyr Ala Gly Pro
85           90           95

```

```

Asn Pro Tyr Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
100           105           110

```

```

<210> SEQ ID NO 446
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 446

```

```

Thr Gly Ser Ser Ser Asn Ile Gly Ala Gly Tyr Asp Val His
1           5           10

```

```

<210> SEQ ID NO 447
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence

```

-continued

<220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 447

Gly Asn Ser Asn Arg Pro Ser
 1 5

<210> SEQ ID NO 448
 <211> LENGTH: 11
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 448

Gln Ser Tyr Ala Gly Pro Asn Pro Tyr Val Val
 1 5 10

<210> SEQ ID NO 449
 <211> LENGTH: 369
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 449

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg	60
agctgtgccg ccagcggcctt cgatttcagc agctacgaga tgaactgggt ccgacaggcc	120
cctggcaaag gccttgaatg ggtgtccggc atcagctgga atagcggctg gatcgactac	180
gcccagacgc tgaaggcgag attcaccatc agccgggaca acagcaagaa caccctgtac	240
ctgcagatga acagcctgag agccgaggac accgccgtgt actactgtgc cagaagcggc	300
tacagcagct cttggtttga ccccgacttc gactattggg gccagggcac actggtcaca	360
gtctcttca	369

<210> SEQ ID NO 450
 <211> LENGTH: 15
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 450

agctacgaga tgaac	15
------------------	----

<210> SEQ ID NO 451
 <211> LENGTH: 51
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 451

ggcatcagct ggaatagcgg ctggatcgac tacgccgaca gcgtgaaggg c	51
--	----

<210> SEQ ID NO 452
 <211> LENGTH: 42
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 452

-continued

agcggctaca gcagctcttg gtttgacccc gacttcgact at 42

<210> SEQ ID NO 453
 <211> LENGTH: 333
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 453

cagtctgttc tgacacagcc tccatctgtg tctggcgccc ctggacagag agtgaccatc 60
 agctgtacag gcagcagctc caatatcgga gccggctatg acgtgcactg gtatcagcag 120
 ctgcttgga cagcccctaa actgctgac tacggcaaca gcaacagacc cagcggcgtg 180
 cccgatagat ttccggctc taagagcggc acaagcgcca gcctggctat tactggactg 240
 caggccgagg acgaggccga ctactactgt cagtcttacg ctggccccaa tccttacgtg 300
 gtgtttggcg gcggaacaaa gctgaccgtt cta 333

<210> SEQ ID NO 454
 <211> LENGTH: 42
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 454

acaggcagca gctccaatat cggagccggc tatgacgtgc ac 42

<210> SEQ ID NO 455
 <211> LENGTH: 21
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 455

ggcaacagca acagacccag c 21

<210> SEQ ID NO 456
 <211> LENGTH: 33
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 456

cagtcttacg ctggccccaa tccttacgtg gtg 33

<210> SEQ ID NO 457
 <211> LENGTH: 453
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 457

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Asp Phe Ser Ser Tyr
 20 25 30
 Glu Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ser	Gly	Ile	Ser	Trp	Asn	Ser	Gly	Trp	Ile	Asp	Tyr	Ala	Asp	Ser	Val
50 55 60															
Lys	Gly	Arg	Phe	Thr	Ile	Ser	Arg	Asp	Asn	Ser	Lys	Asn	Thr	Leu	Tyr
65 70 75 80															
Leu	Gln	Met	Asn	Ser	Leu	Arg	Ala	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys
85 90 95															
Ala	Arg	Ser	Gly	Tyr	Ser	Ser	Ser	Trp	Phe	Asp	Pro	Asp	Phe	Asp	Tyr
100 105 110															
Trp	Gly	Gln	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	Lys	Gly
115 120 125															
Pro	Ser	Val	Phe	Pro	Leu	Ala	Pro	Ser	Ser	Lys	Ser	Thr	Ser	Gly	Gly
130 135 140															
Thr	Ala	Ala	Leu	Gly	Cys	Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu	Pro	Val
145 150 155 160															
Thr	Val	Ser	Trp	Asn	Ser	Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr	Phe
165 170 175															
Pro	Ala	Val	Leu	Gln	Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val
180 185 190															
Thr	Val	Pro	Ser	Ser	Ser	Leu	Gly	Thr	Gln	Thr	Tyr	Ile	Cys	Asn	Val
195 200 205															
Asn	His	Lys	Pro	Ser	Asn	Thr	Lys	Val	Asp	Lys	Lys	Val	Glu	Pro	Lys
210 215 220															
Ser	Cys	Asp	Lys	Thr	His	Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Leu
225 230 235 240															
Leu	Gly	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr
245 250 255															
Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val
260 265 270															
Ser	His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val
275 280 285															
Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser
290 295 300															
Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu
305 310 315 320															
Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala
325 330 335															
Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro
340 345 350															
Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Asp	Glu	Leu	Thr	Lys	Asn	Gln
355 360 365															
Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala
370 375 380															
Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr
385 390 395 400															
Pro	Pro	Val	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu
405 410 415															
Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser
420 425 430															
Val	Met	His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser
435 440 445															
Leu	Ser	Pro	Gly	Lys											
450															

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<210> SEQ ID NO 458
 <211> LENGTH: 217
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 458

Gln Ser Val Leu Thr Gln Pro Pro Ser Val Ser Gly Ala Pro Gly Gln
 1 5 10 15
 Arg Val Thr Ile Ser Cys Thr Gly Ser Ser Ser Asn Ile Gly Ala Gly
 20 25 30
 Tyr Asp Val His Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu
 35 40 45
 Leu Ile Tyr Gly Asn Ser Asn Arg Pro Ser Gly Val Pro Asp Arg Phe
 50 55 60
 Ser Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Thr Gly Leu
 65 70 75 80
 Gln Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Gln Ser Tyr Ala Gly Pro
 85 90 95
 Asn Pro Tyr Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu Gly
 100 105 110
 Gln Pro Lys Ala Ala Pro Ser Val Thr Leu Phe Pro Pro Ser Ser Glu
 115 120 125
 Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp Phe
 130 135 140
 Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Ser Ser Pro Val
 145 150 155 160
 Lys Ala Gly Val Glu Thr Thr Thr Pro Ser Lys Gln Ser Asn Asn Lys
 165 170 175
 Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys Ser
 180 185 190
 His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val Glu
 195 200 205
 Lys Thr Val Ala Pro Thr Glu Cys Ser
 210 215

<210> SEQ ID NO 459
 <211> LENGTH: 1359
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 459

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg 60
 agctgtgccg ccagcggcgt cgatttcagc agctacgaga tgaactgggt ccgacaggcc 120
 cctggcaaag gccttgaatg ggtgtccggc atcagctgga atagcggctg gatcgactac 180
 gccgacagcg tgaagggcag attcaccatc agccgggaca acagcaagaa caccctgtac 240
 ctgcagatga acagcctgag agccgaggac accgccgtgt actactgtgc cagaagcggc 300
 tacagcagct cttggtttga ccccgaactt cactattggg gccagggcac actggtcaca 360
 gtctcttcag ccagcaccaa gggccccagc gtgttccttc tggcccctag cagcaagagc 420
 acatctggcg gaacagccgc cctgggctgc ctctgaagg actactttcc cgagcccgctg 480
 accgtgtcct ggaactctgg cgctctgaca agcggcgtgc acacotttcc agccgtgctg 540

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cagagcagcg gcctgtactc tctgagcagc gtctgtgacag tgcccagcag ctctctgggc 600
accagacct acatctgcaa cgtgaaccac aagcccagca acaccaaggt ggacaagaag 660
gtggaaccca agagctgcga caagaccac acctgtcccc cttgtcctgc cccggaactg 720
ctgggaggcc cttccgtgtt cctgttcccc ccaaagccca aggacacct gatgatcagc 780
cggacccccg aagtgaacctg cgtgggtgtg gatgtgtccc acgaggaccc tgaagtgaag 840
ttcaatttgt acgtggacgg cgtggaagtg cacaacgcca agaccaagcc tagagaggaa 900
cagtacaaca gcacctaccg ggtggtgtcc gtgctgacag tgctgcacca ggactggctg 960
aacggcaaa agtacaagtg caaggtgtcc aacaaggccc tgctgtcccc catcgagaaa 1020
accatcagca aggccaaagg ccagcccccgc gaaccccagg tgtacacact gcccccaagc 1080
agggacgagc tgaccaagaa ccaggtgtcc ctgacctgtc tcgtgaaagg cttctacccc 1140
tccgatatcg ccgtggaatg ggagagcaac ggccagcccc agaacaacta caagaccacc 1200
ccccctgtgc tggacagcga cggtcattc ttctgtaca gcaagctgac cgtggacaag 1260
tcccgggtgc agcagggcaa cgtgttcagc tgcagcgtga tgcacgaggc cctgcacaa 1320
cactacaccc agaagtcct gagcctgagc cctggcaag 1359

```

```

<210> SEQ ID NO 460
<211> LENGTH: 651
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 460

```

```

cagtcgttgc tgacacagcc tccatctgtg tctggcgccc ctggacagag agtgaccatc 60
agctgtacag gcagcagctc caatatcgga gccggctatg acgtgcactg gtatcagcag 120
ctgctggcca cagcccctaa actgctgata tacggcaaca gcaacagacc cagcggcgtg 180
cccgatagat ttcccggtc taagagcggc acaagcgcca gcctggctat tactggactg 240
caggccgagg acgaggccga ctactactgt cagtcttacg ctggcccca tccttacgtg 300
gtgtttggcg gcggaacaaa gctgaccgtt ctaggccagc ctaaagccgc ccctagcgtg 360
accctgttcc ctccaagcag cgaggaaactg caggccaaca aggccaccct cgtgtgcctg 420
atcagcgact tctatcctgg cgccgtgacc gtggcctgga aggccgatag ctctcctgtg 480
aaggccggcg tggaaaccac caccctagc aagcagagca acaacaata cgccgccagc 540
agctacctga gcctgacccc cgagcagtggt aagtcaccaca gatcctacag ctgccaagtg 600
accacagagg gcagcaccgt ggaaaagaca gtggccccta ccgagtgcag c 651

```

```

<210> SEQ ID NO 461
<211> LENGTH: 123
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 461

```

```

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1           5           10           15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asp Ser Tyr
20           25           30

Glu Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35           40           45

```

-continued

```

Ser Gly Ile Ser Trp Asn Ser Gly Trp Ile Asp Tyr Ala Asp Ser Val
 50                      55                      60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65                      70                      75                      80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
                      85                      90                      95

Ala Arg Ser Gly Tyr Ser Ser Ser Trp Phe Asp Pro Asp Phe Asp Tyr
100                      105                      110

Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
115                      120

```

```

<210> SEQ ID NO 462
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 462

```

```

Ser Tyr Glu Met Asn
1           5

```

```

<210> SEQ ID NO 463
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 463

```

```

Gly Ile Ser Trp Asn Ser Gly Trp Ile Asp Tyr Ala Asp Ser Val Lys
1           5           10           15

```

```

Gly

```

```

<210> SEQ ID NO 464
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 464

```

```

Ser Gly Tyr Ser Ser Ser Trp Phe Asp Pro Asp Phe Asp Tyr
1           5           10

```

```

<210> SEQ ID NO 465
<211> LENGTH: 111
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 465

```

```

Gln Ser Val Leu Thr Gln Pro Pro Ser Val Ser Gly Ala Pro Gly Gln
1           5           10           15

```

```

Arg Val Thr Ile Ser Cys Thr Gly Ser Ser Ser Asn Ile Gly Ala Gly
20           25           30

```

```

Tyr Asp Val His Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu
35           40           45

```

```

Leu Ile Tyr Gly Asn Ser Asn Arg Pro Ser Gly Val Pro Asp Arg Phe
50           55           60

```

```

Ser Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Thr Gly Leu

```


-continued

<400> SEQUENCE: 470

agctacgaga tgaac

15

<210> SEQ ID NO 471

<211> LENGTH: 51

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 471

ggcatcagct ggaatagcgg ctggatcgac tacgccgaca gcgtgaaggg c

51

<210> SEQ ID NO 472

<211> LENGTH: 42

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 472

agcggctaca gcagctcttg gtttgacccc gacttcgact at

42

<210> SEQ ID NO 473

<211> LENGTH: 333

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 473

cagtcctgttc tgacacagcc tccatctgtg tctggcgccc ctggacagag agtgaccatc

60

agctgtacag gcagcagctc caatatcgga gccggctatg acgtgcactg gtatcagcag

120

ctgcctggca cagcccctaa actgctgac tacggcaaca gcaacagacc cagcggcgtg

180

ccgatagat tttccggctc taagagcggc acaagcgcca gcctggctat tactggactg

240

caggccgagg acgaggccga ctactactgt cagtcttacg ctggccccaa tccttacgtg

300

gtgtttggcg gcggaacaaa gctgaccgtt cta

333

<210> SEQ ID NO 474

<211> LENGTH: 42

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 474

acaggcagca gctccaatat cgagaccggc tatgacgtgc ac

42

<210> SEQ ID NO 475

<211> LENGTH: 21

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 475

ggcaacagca acagaccag c

21

<210> SEQ ID NO 476

<211> LENGTH: 33

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 476

cagtccttacg ctggccccaac tccttacgtg gtg

33

<210> SEQ ID NO 477

<211> LENGTH: 453

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 477

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asp Ser Tyr
 20 25 30

Glu Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ser Gly Ile Ser Trp Asn Ser Gly Trp Ile Asp Tyr Ala Asp Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
 65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Ala Arg Ser Gly Tyr Ser Ser Ser Trp Phe Asp Pro Asp Phe Asp Tyr
 100 105 110

Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly
 115 120 125

Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly
 130 135 140

Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val
 145 150 155 160

Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe
 165 170 175

Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val
 180 185 190

Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val
 195 200 205

Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys Lys Val Glu Pro Lys
 210 215 220

Ser Cys Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu
 225 230 235 240

Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr
 245 250 255

Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val
 260 265 270

Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val
 275 280 285

Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser
 290 295 300

Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu
 305 310 315 320

Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala
 325 330 335

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Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro
 340 345 350

Gln Val Tyr Thr Leu Pro Pro Ser Arg Asp Glu Leu Thr Lys Asn Gln
 355 360 365

Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala
 370 375 380

Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr
 385 390 395 400

Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu
 405 410 415

Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser
 420 425 430

Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser
 435 440 445

Leu Ser Pro Gly Lys
 450

<210> SEQ ID NO 478

<211> LENGTH: 217

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 478

Gln Ser Val Leu Thr Gln Pro Pro Ser Val Ser Gly Ala Pro Gly Gln
 1 5 10 15

Arg Val Thr Ile Ser Cys Thr Gly Ser Ser Ser Asn Ile Gly Ala Gly
 20 25 30

Tyr Asp Val His Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu
 35 40 45

Leu Ile Tyr Gly Asn Ser Asn Arg Pro Ser Gly Val Pro Asp Arg Phe
 50 55 60

Ser Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Thr Gly Leu
 65 70 75 80

Gln Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Gln Ser Tyr Ala Gly Pro
 85 90 95

Asn Pro Tyr Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu Gly
 100 105 110

Gln Pro Lys Ala Ala Pro Ser Val Thr Leu Phe Pro Pro Ser Ser Glu
 115 120 125

Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp Phe
 130 135 140

Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Ser Ser Pro Val
 145 150 155 160

Lys Ala Gly Val Glu Thr Thr Thr Pro Ser Lys Gln Ser Asn Asn Lys
 165 170 175

Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys Ser
 180 185 190

His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val Glu
 195 200 205

Lys Thr Val Ala Pro Thr Glu Cys Ser
 210 215

<210> SEQ ID NO 479

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<211> LENGTH: 1359
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 479
gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg      60
agctgtgccg ccagcgggett caccttcgat agctacgaga tgaactgggt ccgacaggcc      120
cctggcaaag gccttgaatg ggtgtccggc atcagctgga atagcggctg gatcgactac      180
gccgacagcg tgaagggcag attcaccatc agccgggaca acagcaagaa caccctgtac      240
ctgcagatga acagcctgag agccgaggac accgccgtgt actactgtgc cagaagcggc      300
tacagcagct cttggtttga ccccgacttc gactattggg gccagggcac actggtcaca      360
gtctcttcag ccagcaccaa gggccccagc gtgttccctc tggccctag cagcaagagc      420
acatctggcg gaacagccgc cctgggctgc ctgctgaagg actactttcc cgagcccgctg      480
accgtgtcct ggaactctgg cgctctgaca agcggcgtgc acacctttcc agcctgtctg      540
cagagcagcg gcctgtactc tctgagcagc gtcgtgacag tgcccagcag ctctctgggc      600
accagacct acatctgcaa cgtgaaccac aagcccagca acaccaaggt ggacaagaag      660
gtggaacca agagctgcga caagaccac acctgtcccc cttgtcctgc cccgaactg      720
ctgggaggcc ctcccggtt cctgttcccc ccaaagccca aggacacct gatgatcagc      780
cggacccccg aagtgcctg cgtggtggtg gatgtgtccc acgaggaccc tgaagtgaag      840
ttcaattggt acgtggacgg cgtggaagtg cacaacgcca agaccaagcc tagagaggaa      900
cagtacaaca gcacctaccg ggtggtgtcc gtgctgacag tgctgcacca ggactggctg      960
aacggcaaag agtacaagtg caaggtgtcc aacaaggccc tgctgtcccc catcgagaaa     1020
accatcagca agggcaaggg ccagccccgc gaacccagcgtgtacacct gcccccaagc     1080
agggacgagc tgaccaagaa ccagggtgtcc ctgacctgtc tcgtgaaagg cttctacccc     1140
tccgatatcg ccgtggaatg ggagagcaac ggcagcccg agaacaacta caagaccacc     1200
ccccctgtgc tggacagcga cggtcattc ttctgtaca gcaagctgac cgtggacaag     1260
tcccgggtggc agcagggcaa cgtgttcagc tgcagcgtga tgcacgaggc cctgcacaac     1320
cactacaccc agaagtcctt gagcctgagc cctggcaag                               1359

```

```

<210> SEQ ID NO 480
<211> LENGTH: 651
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 480
cagtctgttc tgacacagcc tccatctgtg tctggcggcc ctggacagag agtgaccatc      60
agctgtacag gcagcagctc caatatcgga gccggctatg acgtgcactg gtatcagcag      120
ctgcctggca cagcccctaa actgctgata tacggcaaca gcaacagacc cagcggcgtg      180
cccgatagat tttccggctc taagagcggc acaagcgcca gcctggctat tactggactg      240
caggccgagg acgaggccga ctactactgt cagtcttacg ctggcccca tctttacgtg      300
gtgtttggcg gcggaacaaa gctgacggtt ctaggccagc ctaaagccgc ccctagcgtg      360
accctgttcc ctccaagcag cgaggaaactg caggccaaca aggccaccct cgtgtgcctg      420
atcagcgact tctatcctgg cgccgtgacc gtggcctgga aggccgatag ctctcctgtg      480

```

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```
aaggccggcg tggaaccac caccctagc aagcagagca acaacaaata cgccgccagc 540
agctacctga gcctgacccc cgagcagtgg aagtcccaca gatcctacag ctgccaaagtg 600
accacagagg gcagcaccgt ggaaaagaca gtggccccta ccgagtgcag c 651
```

```
<210> SEQ ID NO 481
<211> LENGTH: 123
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence
```

```
<400> SEQUENCE: 481
```

```
Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1      5      10      15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Asp Phe Asp Ser Tyr
20     25     30
Glu Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35     40     45
Ser Gly Ile Ser Trp Asn Ser Gly Trp Ile Asp Tyr Ala Asp Ser Val
50     55     60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65     70     75     80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85     90     95
Ala Arg Ser Gly Tyr Ser Ser Ser Trp Phe Asp Pro Asp Phe Asp Tyr
100    105    110
Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
115    120
```

```
<210> SEQ ID NO 482
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence
```

```
<400> SEQUENCE: 482
```

```
Ser Tyr Glu Met Asn
1      5
```

```
<210> SEQ ID NO 483
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence
```

```
<400> SEQUENCE: 483
```

```
Gly Ile Ser Trp Asn Ser Gly Trp Ile Asp Tyr Ala Asp Ser Val Lys
1      5      10      15
Gly
```

```
<210> SEQ ID NO 484
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence
```

```
<400> SEQUENCE: 484
```


-continued

Ser Gly Tyr Ser Ser Ser Trp Phe Asp Pro Asp Phe Asp Tyr
1 5 10

<210> SEQ ID NO 485
 <211> LENGTH: 111
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 485

Gln Ser Val Leu Thr Gln Pro Pro Ser Val Ser Gly Ala Pro Gly Gln
1 5 10 15

Arg Val Thr Ile Ser Cys Thr Gly Ser Ser Ser Asn Ile Gly Ala Gly
20 25 30

Tyr Asp Val His Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu
35 40 45

Leu Ile Tyr Gly Asn Ser Asn Arg Pro Ser Gly Val Pro Asp Arg Phe
50 55 60

Ser Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Thr Gly Leu
65 70 75 80

Gln Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Gln Ser Tyr Ala Gly Pro
85 90 95

Asn Pro Tyr Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
100 105 110

<210> SEQ ID NO 486
 <211> LENGTH: 14
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 486

Thr Gly Ser Ser Ser Asn Ile Gly Ala Gly Tyr Asp Val His
1 5 10

<210> SEQ ID NO 487
 <211> LENGTH: 7
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 487

Gly Asn Ser Asn Arg Pro Ser
1 5

<210> SEQ ID NO 488
 <211> LENGTH: 11
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 488

Gln Ser Tyr Ala Gly Pro Asn Pro Tyr Val Val
1 5 10

<210> SEQ ID NO 489
 <211> LENGTH: 369
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:

-continued

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 489

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg	60
agctgtgccg ccagcggcctt cgacttcgat agctacgaga tgaactgggt ccgacaggcc	120
cctggcaaag gccttgaatg ggtgtccggc atcagctgga atagcggctg gatcgactac	180
gccgacagcg tgaaggcgag attcaccatc agccgggaca acagcaagaa caccctgtac	240
ctgcagatga acagcctgag agccgaggac accgccgtgt actactgtgc cagaagcggc	300
tacagcagct cttggtttga ccccgacttc gactattggg gccagggcac actggtcaca	360
gtctcttca	369

<210> SEQ ID NO 490

<211> LENGTH: 15

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 490

agctacgaga tgaac	15
------------------	----

<210> SEQ ID NO 491

<211> LENGTH: 51

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 491

ggcatcagct ggaatagcgg ctggatcgac tacgccgaca gcgtgaaggg c	51
--	----

<210> SEQ ID NO 492

<211> LENGTH: 42

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 492

agcggctaca gcagctcttg gtttgacccc gacttcgact at	42
--	----

<210> SEQ ID NO 493

<211> LENGTH: 333

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 493

cagtctgttc tgacacagcc tccatctgtg tctggcgccc ctggacagag agtgaccatc	60
agctgtacag gcagcagctc caatatcgga gccggctatg acgtgcactg gtatcagcag	120
ctgcctggca cagcccctaa actgctgac tacggcaaca gcaacagacc cagcggcgtg	180
cccgatagat tttccggctc taagagcggc acaagcgcca gcctggctat tactggactg	240
caggccgagg acgaggccga ctactactgt cagtcttacg ctggccccaa tccttacgtg	300
gtgtttggcg gcggaacaaa gctgacggtt cta	333

<210> SEQ ID NO 494

<211> LENGTH: 42

-continued

```

<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 494

acaggcagca gctccaatat cggagccggc tatgacgtgc ac                42

<210> SEQ ID NO 495
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 495

ggcaacagca acagaccag c                21

<210> SEQ ID NO 496
<211> LENGTH: 33
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 496

cagtcttacg ctggcccaaa tccttacgtg gtg                33

<210> SEQ ID NO 497
<211> LENGTH: 453
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 497

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1          5          10          15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Asp Phe Asp Ser Tyr
20          25          30

Glu Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35          40          45

Ser Gly Ile Ser Trp Asn Ser Gly Trp Ile Asp Tyr Ala Asp Ser Val
50          55          60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65          70          75          80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85          90          95

Ala Arg Ser Gly Tyr Ser Ser Ser Trp Phe Asp Pro Asp Phe Asp Tyr
100         105         110

Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly
115         120         125

Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly
130         135         140

Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val
145         150         155         160

Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe
165         170         175

Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val
180         185         190

```

Thr	Val	Pro	Ser	Ser	Ser	Leu	Gly	Thr	Gln	Thr	Tyr	Ile	Cys	Asn	Val
195						200						205			
Asn	His	Lys	Pro	Ser	Asn	Thr	Lys	Val	Asp	Lys	Lys	Val	Glu	Pro	Lys
210						215				220					
Ser	Cys	Asp	Lys	Thr	His	Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Leu
225				230						235				240	
Leu	Gly	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr
				245				250						255	
Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val
		260						265				270			
Ser	His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val
275						280						285			
Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser
290						295				300					
Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu
305				310						315				320	
Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala
				325				330						335	
Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro
		340						345				350			
Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Asp	Glu	Leu	Thr	Lys	Asn	Gln
355						360						365			
Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala
370						375				380					
Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr
385				390						395				400	
Pro	Pro	Val	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu
				405				410						415	
Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser
		420						425				430			
Val	Met	His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser
435						440						445			
Leu	Ser	Pro	Gly	Lys											
450															

```
<210> SEQ ID NO 498
<211> LENGTH: 217
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence
```

<400> SEQUENCE: 498

Gln	Ser	Val	Leu	Thr	Gln	Pro	Pro	Ser	Val	Ser	Gly	Ala	Pro	Gly	Gln
1				5					10					15	
Arg	Val	Thr	Ile	Ser	Cys	Thr	Gly	Ser	Ser	Ser	Asn	Ile	Gly	Ala	Gly
			20					25					30		
Tyr	Asp	Val	His	Trp	Tyr	Gln	Gln	Leu	Pro	Gly	Thr	Ala	Pro	Lys	Leu
		35					40					45			
Leu	Ile	Tyr	Gly	Asn	Ser	Asn	Arg	Pro	Ser	Gly	Val	Pro	Asp	Arg	Phe
	50					55					60				
Ser	Gly	Ser	Lys	Ser	Gly	Thr	Ser	Ala	Ser	Leu	Ala	Ile	Thr	Gly	Leu
65					70					75					80
Gln	Ala	Glu	Asp	Glu	Ala	Asp	Tyr	Tyr	Cys	Gln	Ser	Tyr	Ala	Gly	Pro
				85					90					95	

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Asn Pro Tyr Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu Gly
 100 105 110
 Gln Pro Lys Ala Ala Pro Ser Val Thr Leu Phe Pro Pro Ser Ser Glu
 115 120 125
 Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp Phe
 130 135 140
 Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Ser Ser Pro Val
 145 150 155 160
 Lys Ala Gly Val Glu Thr Thr Thr Pro Ser Lys Gln Ser Asn Asn Lys
 165 170 175
 Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys Ser
 180 185 190
 His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val Glu
 195 200 205
 Lys Thr Val Ala Pro Thr Glu Cys Ser
 210 215

<210> SEQ ID NO 499

<211> LENGTH: 1359

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 499

```

gaagttcagc tgetggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg      60
agctgtgccg ccagcggcctt cgacttcgat agctacgaga tgaactgggt ccgacaggcc      120
cctggcaaa gacctgaatg ggtgtccggc atcagctgga atagcggctg gatcgactac      180
gccgacagcg tgaaggcgag attcaccatc agccgggaca acagcaagaa caccctgtac      240
ctgcagatga acagcctgag agccgaggac accgccgtgt actactgtgc cagaagcggc      300
tacagcagct cttggtttga ccccgacttc gactattggg gccagggcac actggtcaca      360
gtctcttcag ccagcaccaa gggccccagc gtgttccttc tggccctag cagcaagagc      420
acatctggcg gaacagccgc cctgggctgc ctgctgaagg actactttcc cgagcccgctg      480
accgtgtcct ggaactctgg cgctctgaca agcggcgtgc acacctttcc agcctgtctg      540
cagagcagcg gcctgtactc tctgagcagc gtcgtgacag tgcccagcag ctctctgggc      600
acccagacct acatctgcaa cgtgaaccac aagcccagca acaccaaggt ggacaagaag      660
gtggaaccca agagctcgca caagaccac acctgtcccc cttgtectgc ccccgaactg      720
ctgggaggcc cttcgtgttt cctgttcccc ccaaagccca aggacacct gatgatcagc      780
cggacccccg aagtgcctg cgtggtggtg gatgtgtccc acgaggaccc tgaagtgaag      840
ttcaattggt acgtggacgg cgtggaagtg cacaacgcca agaccaagcc tagagaggaa      900
cagtacaaca gcacctaccg ggtggtgtcc gtgctgacag tgctgcacca ggactggctg      960
aacggcaaa agtacaagtg caaggtgtcc aacaaggccc tgctgtcccc catcgagaaa     1020
accatcagca aggccaaagg ccagccccgc gaaccccagg tgtacacact gcccccaagc     1080
agggacgagc tgaccaagaa ccagggtgtcc ctgacctgtc tcgtgaaagg cttctacccc     1140
tccgatatcg ccgtggaatg ggagagcaac ggcagcccg agaacaacta caagaccacc     1200
ccccctgtgc tggacagcga cggtcattc ttctgtaca gcaagctgac cgtggacaag     1260
tcccggtggc agcagggcaa cgtgttcagc tgcagcgtga tgcacgaggc cctgcacaac     1320
cactacacc agaagtcct gagcctgagc cctggcaag                               1359

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<210> SEQ ID NO 500
<211> LENGTH: 651
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 500
cagctctgttc tgacacagcc tccatctgtg tctggcgccc ctggacagag agtgaccatc      60
agctgtacag gcagcagctc caatatcgga gccggctatg acgtgcactg gtatcagcag      120
ctgctgggca cagcccctaa actgctgata tacggcaaca gcaacagacc cagcggcgtg      180
cccgatagat tttccggctc taagagcggc acaagcgcca gcctggctat tactggactg      240
caggccgagg acgaggccga ctactactgt cagtcttacg ctggcccca tctttacgtg      300
gtgtttggcg gcggaacaaa gctgaccgtt ctaggccagc ctaaagccgc ccctagcgtg      360
accctgttcc ctccaagcag cgaggaactg caggccaaca aggccaccct cgtgtgcctg      420
atcagcgact tctatcctgg cgccgtgacc gtggcctgga aggccgatag ctctcctgtg      480
aaggccggcg tggaaaccac caccctagc aagcagagca acaacaaata cgccgccagc      540
agctacctga gcctgacccc cgagcagtggt aagtcaccaca gatcctacag ctgccaaagt      600
accacagagg gcagcacctg ggaaaagaca gtggccccta ccgagtgcag c              651

```

```

<210> SEQ ID NO 501
<211> LENGTH: 123
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 501
Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1          5          10          15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asp Ser Tyr
20          25          30
Glu Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35          40          45
Ser Gly Ile Ser Trp Asn Ser Gly Trp Ile Asp Tyr Ala Asp Ser Val
50          55          60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65          70          75          80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85          90          95
Ala Arg Ser Gly Tyr Ser Ser Ser Trp Phe Asp Pro Asp Phe Asp Tyr
100         105         110
Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
115         120

```

```

<210> SEQ ID NO 502
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 502
Ser Tyr Glu Met Asn
1          5

```

-continued

<210> SEQ ID NO 503
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 503

Gly Ile Ser Trp Asn Ser Gly Trp Ile Asp Tyr Ala Asp Ser Val Lys
1 5 10 15

Gly

<210> SEQ ID NO 504
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 504

Ser Gly Tyr Ser Ser Ser Trp Phe Asp Pro Asp Phe Asp Tyr
1 5 10

<210> SEQ ID NO 505
<211> LENGTH: 111
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 505

Gln Ser Val Leu Thr Gln Pro Pro Ser Val Ser Gly Ala Pro Gly Gln
1 5 10 15

Arg Val Thr Ile Ser Cys Thr Gly Ser Ser Ser Asp Ile Gly Ala Gly
20 25 30

Tyr Asp Val His Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu
35 40 45

Leu Ile Tyr Gly Asn Ser Asn Arg Pro Ser Gly Val Pro Asp Arg Phe
50 55 60

Ser Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Thr Gly Leu
65 70 75 80

Gln Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Gln Ser Tyr Ala Gly Ile
85 90 95

Asn Pro Tyr Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
100 105 110

<210> SEQ ID NO 506
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 506

Thr Gly Ser Ser Ser Asp Ile Gly Ala Gly Tyr Asp Val His
1 5 10

<210> SEQ ID NO 507
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:

-continued

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 507

Gly Asn Ser Asn Arg Pro Ser
1 5

<210> SEQ ID NO 508

<211> LENGTH: 11

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 508

Gln Ser Tyr Ala Gly Ile Asn Pro Tyr Val Val
1 5 10

<210> SEQ ID NO 509

<211> LENGTH: 369

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 509

```

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg      60
agctgtgccc ccagcggcgtt caccttcgat agctacgaga tgaactgggt ccgacaggcc      120
cctggcaaag gccttgaatg ggtgtccggc atcagctgga atagcggctg gatcgactac      180
gccgacagcg tgaagggcag attcaccatc agccgggaca acagcaagaa caccctgtac      240
ctgcagatga acagcctgag agccgaggac accgccgtgt actactgtgc cagaagcggc      300
tacagcagct cttggtttga ccccgacttc gactattggg gccagggcac actggtcaca      360
gtctcttca                                     369

```

<210> SEQ ID NO 510

<211> LENGTH: 15

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 510

```

agctacgaga tgaac                                     15

```

<210> SEQ ID NO 511

<211> LENGTH: 51

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 511

```

ggcatcagct ggaatagcgg ctggatcgac tacgccgaca gcgtgaaggg c                51

```

<210> SEQ ID NO 512

<211> LENGTH: 42

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 512

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agcggctaca gcagctcttg gtttgacccc gacttcgact at                            42

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<210> SEQ ID NO 513
<211> LENGTH: 333
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 513

cagtctgttc tgacacagcc tccatctgtg tctggcgccc ctggacagag agtgaccatc      60
agctgtacag gcagcagctc cgatattggc gccggatacg acgtgcactg gtatcagcaa      120
ctgctgggca cagcccctaa gctgctgata tacggcaaca gcaacagacc tagcggcgtg      180
cccgatagat tcagcggctc taagtctggc acaagcgcca gcctggccat tactggactg      240
caggccgaag atgaggccga ctactactgt cagagctacg ccggcatcaa cccctacgtg      300
gtgtttggcg gaggcaccaa gctgacagtt cta                                     333

```

```

<210> SEQ ID NO 514
<211> LENGTH: 42
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 514

acaggcagca gctccgatat tggcgccgga tacgacgtgc ac                          42

```

```

<210> SEQ ID NO 515
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 515

ggcaacagca acagacctag c                                                  21

```

```

<210> SEQ ID NO 516
<211> LENGTH: 33
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 516

cagagctacg ccggcatcaa cccctacgtg gtg                                    33

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<210> SEQ ID NO 517
<211> LENGTH: 453
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 517

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1          5          10          15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asp Ser Tyr
20        25        30

Glu Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35        40        45

Ser Gly Ile Ser Trp Asn Ser Gly Trp Ile Asp Tyr Ala Asp Ser Val

```

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50	55	60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr		
65	70	75 80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys		
	85	90 95
Ala Arg Ser Gly Tyr Ser Ser Ser Trp Phe Asp Pro Asp Phe Asp Tyr		
	100	105 110
Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly		
	115	120 125
Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly		
	130	135 140
Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val		
	145	150 155 160
Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe		
	165	170 175
Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val		
	180	185 190
Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val		
	195	200 205
Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys Lys Val Glu Pro Lys		
	210	215 220
Ser Cys Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu		
	225	230 235 240
Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr		
	245	250 255
Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val		
	260	265 270
Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val		
	275	280 285
Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser		
	290	295 300
Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu		
	305	310 315 320
Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala		
	325	330 335
Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro		
	340	345 350
Gln Val Tyr Thr Leu Pro Pro Ser Arg Asp Glu Leu Thr Lys Asn Gln		
	355	360 365
Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala		
	370	375 380
Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr		
	385	390 395 400
Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu		
	405	410 415
Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser		
	420	425 430
Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser		
	435	440 445
Leu Ser Pro Gly Lys		
450		

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<211> LENGTH: 217
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 518

Gln Ser Val Leu Thr Gln Pro Pro Ser Val Ser Gly Ala Pro Gly Gln
1      5      10      15
Arg Val Thr Ile Ser Cys Thr Gly Ser Ser Ser Asp Ile Gly Ala Gly
20     25     30
Tyr Asp Val His Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu
35     40     45
Leu Ile Tyr Gly Asn Ser Asn Arg Pro Ser Gly Val Pro Asp Arg Phe
50     55     60
Ser Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Thr Gly Leu
65     70     75     80
Gln Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Gln Ser Tyr Ala Gly Ile
85     90     95
Asn Pro Tyr Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu Gly
100    105    110
Gln Pro Lys Ala Ala Pro Ser Val Thr Leu Phe Pro Pro Ser Ser Glu
115    120    125
Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp Phe
130    135    140
Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Ser Ser Pro Val
145    150    155    160
Lys Ala Gly Val Glu Thr Thr Thr Pro Ser Lys Gln Ser Asn Asn Lys
165    170    175
Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys Ser
180    185    190
His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val Glu
195    200    205

Lys Thr Val Ala Pro Thr Glu Cys Ser
210      215

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<210> SEQ ID NO 519
<211> LENGTH: 1359
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 519

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg      60
agctgtgccg ccagcggctt caccttcgat agctacgaga tgaactgggt ccgacaggcc      120
cctggcaaag gccttgaatg ggtgtccggc atcagctgga atagcggctg gatcgactac      180
gccgacagcg tgaagggcag attcaccatc agccgggaca acagcaagaa caccctgtac      240
ctgcagatga acagcctgag agccgaggac accgccgtgt actactgtgc cagaagcggc      300
tacagcagct cttggtttga ccccgacttc gactattggg gccagggcac actggtcaca      360
gtctcttcag ccagcaccaa gggccccagc gtgttccctc tggcccctag cagcaagagc      420
acatctggcg gaacagccgc cctgggctgc ctctgaagg actactttcc cgagcccggtg      480
accgtgtcct ggaactctgg cgctctgaca agcggcgtgc acacctttcc agccgtgctg      540
cagagcagcg gcctgtactc tctgagcagc gtcgtgacag tgcccagcag ctctctgggc      600

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accagacct acatctgcaa cgtgaaccac aagcccagca acaccaaggt ggacaagaag   660
gtggaacca agagctgcga caagaccac acctgtcccc cttgtcctgc ccccgaaactg   720
ctgggaggcc cttcctgtgt cctgttcccc ccaaagccca aggacaccct gatgatcagc   780
cggacccccg aagtgccttg cgtggtggtg gatgtgtccc acgaggaccc tgaagtgaag   840
ttcaattggt acgtggacgg cgtggaagtg cacaacgccca agaccaagcc tagagaggaa   900
cagtacaaca gcacctaccg ggtggtgtcc gtgctgacag tgcctgcacca ggactggctg   960
aacggcaaaag agtacaagtg caaggtgtcc aacaaggccc tgctgtcccc catcgagaaa  1020
accatcagca agggcaaggg ccagccccgc gaaccccagg tgtacacact gcccccaagc  1080
agggacgagc tgaccaagaa ccaggtgtcc ctgacctgtc tcgtgaaagg cttctacccc  1140
tccgatatcg ccgtggaatg ggagagcaac ggcagccccg agaacaacta caagaccacc  1200
ccccctgtgc tggacagcga cggtcattc ttctgtaca gcaagctgac cgtggacaag  1260
tcccgggtggc agcagggcaa cgtgttcagc tgcagcgtga tgcacgaggc cctgcacaac  1320
cactacaccc agaagtcctt gagcctgagc cctggcaag   1359

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<210> SEQ ID NO 520
<211> LENGTH: 651
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 520

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cagtctgttc tgacacagcc tccatctgtg tctggcgccc ctggacagag agtgaccatc   60
agctgtacag gcagcagctc cgatattggc gccggatacg acgtgcactg gtatcagcaa   120
ctgcctggca cagcccctaa gctgctgata tacggcaaca gcaacagacc tagcggcgtg   180
cccgatagat tcagcggctc taagtctggc acaagcgcca gcctggccat tactggactg   240
caggccgaag atgaggccga ctactactgt cagagctacg ccggcatcaa cccctacgtg   300
gtgtttggcg gaggcaccaa gctgacagtt ctaggccagc ctaaagccgc ccctagcgtg   360
acctgtttcc ctccaagcag cgaggaaactg caggccaaca agggccacct cgtgtgcttg   420
atcagcgact tctatcctgg cgccgtgacc gtggcctgga aggcgatag ctctcctgtg   480
aaggccggcg tggaaaccac caccctagc aagcagagca acaacaaata cgccgccagc   540
agctacctga gctgacccc cgagcagtggt aagtcccaca gatcctacag ctgccaaagt   600
acccacgagg gcagcacctt ggaaaagaca gtggccccta ccgagtgcag c   651

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<210> SEQ ID NO 521
<211> LENGTH: 123
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 521

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Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1           5           10          15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Asp Phe Asp Ser Tyr
20          25          30

Glu Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35          40          45

Ser Gly Ile Ser Trp Asn Ser Gly Trp Ile Asp Tyr Ala Asp Ser Val

```

-continued

50	55	60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr		
65	70	75 80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys		
	85	90 95
Ala Arg Ser Gly Tyr Ser Ser Ser Trp Phe Asp Pro Asp Phe Asp Tyr		
	100	105 110
Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser		
	115	120

<210> SEQ ID NO 522
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 522

Ser Tyr Glu Met Asn
 1 5

<210> SEQ ID NO 523
 <211> LENGTH: 17
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 523

Gly Ile Ser Trp Asn Ser Gly Trp Ile Asp Tyr Ala Asp Ser Val Lys
 1 5 10 15

Gly

<210> SEQ ID NO 524
 <211> LENGTH: 14
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 524

Ser Gly Tyr Ser Ser Ser Trp Phe Asp Pro Asp Phe Asp Tyr
 1 5 10

<210> SEQ ID NO 525
 <211> LENGTH: 111
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 525

Gln Ser Val Leu Thr Gln Pro Pro Ser Val Ser Gly Ala Pro Gly Gln
 1 5 10 15

Arg Val Thr Ile Ser Cys Thr Gly Ser Ser Ser Asp Ile Gly Ala Gly
 20 25 30

Tyr Asp Val His Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu
 35 40 45

Leu Ile Tyr Gly Asn Ser Asn Arg Pro Ser Gly Val Pro Asp Arg Phe
 50 55 60

Ser Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Thr Gly Leu
 65 70 75 80

-continued

Gln Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Ser Ser Tyr Glu Gly Ile
 85 90 95

Asn Pro Tyr Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
 100 105 110

<210> SEQ ID NO 526
 <211> LENGTH: 14
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 526

Thr Gly Ser Ser Ser Asp Ile Gly Ala Gly Tyr Asp Val His
 1 5 10

<210> SEQ ID NO 527
 <211> LENGTH: 7
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 527

Gly Asn Ser Asn Arg Pro Ser
 1 5

<210> SEQ ID NO 528
 <211> LENGTH: 11
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 528

Ser Ser Tyr Glu Gly Ile Asn Pro Tyr Val Val
 1 5 10

<210> SEQ ID NO 529
 <211> LENGTH: 369
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 529

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg 60
 agctgtgccg ccagcgggctt cgacttcgat agctacgaga tgaactgggt ccgacaggcc 120
 cctggcaaaag gccttgaatg ggtgtccggc atcagctgga atagcggctg gatcgactac 180
 gccgacagcg tgaagggcag attcaccatc agccgggaca acagcaagaa caccctgtac 240
 ctgcagatga acagcctgag agccgaggac accgccgtgt actactgtgc cagaagcggc 300
 tacagcagct cttggtttga ccccgacttc gactattggg gccagggcac actggtcaca 360
 gtctcttca 369

<210> SEQ ID NO 530
 <211> LENGTH: 15
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 530

-continued

agctacgaga tgaac 15

<210> SEQ ID NO 531
 <211> LENGTH: 51
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 531

ggcatcagct ggaatagcgg ctggatcgac tacgccgaca gcgtgaaggg c 51

<210> SEQ ID NO 532
 <211> LENGTH: 42
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 532

agcggctaca gcagctcttg gtttgacccc gacttegact at 42

<210> SEQ ID NO 533
 <211> LENGTH: 333
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 533

cagtctgttc tgacacagcc tccatctgtg tctggcgccc ctggacagag agtgaccatc 60

agctgtacag gcagcagctc cgatattggc gccggatacg acgtgcactg gtatcagcaa 120

ctgcctggca cagcccctaa gctgctgac tacggcaaca gcaacagacc tagcggcgtg 180

cccgatagat tcagcggctc taagtctggc acaagcgcca gcctggccat tactggactg 240

caggccgaag atgaggccga ctactactgc agcagctacg agggcatcaa cccctacgtg 300

gtgtttggcg gcggaacaaa gctgaccgtt cta 333

<210> SEQ ID NO 534
 <211> LENGTH: 42
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 534

acaggcagca gctccgatat tggcgccgga tacgacgtgc ac 42

<210> SEQ ID NO 535
 <211> LENGTH: 21
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 535

ggcaacagca acagacctag c 21

<210> SEQ ID NO 536
 <211> LENGTH: 33
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:

-continued

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 536

agcagctacg agggcatcaa cccctacgtg gtg

33

<210> SEQ ID NO 537

<211> LENGTH: 453

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 537

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Asp Phe Asp Ser Tyr
20 25 30

Glu Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45

Ser Gly Ile Ser Trp Asn Ser Gly Trp Ile Asp Tyr Ala Asp Ser Val
50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95

Ala Arg Ser Gly Tyr Ser Ser Ser Trp Phe Asp Pro Asp Phe Asp Tyr
100 105 110

Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly
115 120 125

Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly
130 135 140

Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val
145 150 155 160

Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe
165 170 175

Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val
180 185 190

Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val
195 200 205

Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys Lys Val Glu Pro Lys
210 215 220

Ser Cys Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu
225 230 235 240

Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr
245 250 255

Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val
260 265 270

Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val
275 280 285

Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser
290 295 300

Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu
305 310 315 320

Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala
325 330 335

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Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro
      340              345              350

Gln Val Tyr Thr Leu Pro Pro Ser Arg Asp Glu Leu Thr Lys Asn Gln
      355              360              365

Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala
      370              375              380

Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr
      385              390              395              400

Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu
      405              410              415

Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser
      420              425              430

Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser
      435              440              445

Leu Ser Pro Gly Lys
      450

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<210> SEQ ID NO 538
<211> LENGTH: 217
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 538

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Gln Ser Val Leu Thr Gln Pro Pro Ser Val Ser Gly Ala Pro Gly Gln
1          5          10          15

Arg Val Thr Ile Ser Cys Thr Gly Ser Ser Ser Asp Ile Gly Ala Gly
      20          25          30

Tyr Asp Val His Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu
      35          40          45

Leu Ile Tyr Gly Asn Ser Asn Arg Pro Ser Gly Val Pro Asp Arg Phe
      50          55          60

Ser Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Thr Gly Leu
      65          70          75          80

Gln Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Ser Ser Tyr Glu Gly Ile
      85          90          95

Asn Pro Tyr Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu Gly
      100         105         110

Gln Pro Lys Ala Ala Pro Ser Val Thr Leu Phe Pro Pro Ser Ser Glu
      115         120         125

Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp Phe
      130         135         140

Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Ser Ser Pro Val
      145         150         155         160

Lys Ala Gly Val Glu Thr Thr Thr Pro Ser Lys Gln Ser Asn Asn Lys
      165         170         175

Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys Ser
      180         185         190

His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val Glu
      195         200         205

Lys Thr Val Ala Pro Thr Glu Cys Ser
      210         215

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<210> SEQ ID NO 539
<211> LENGTH: 1359

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-continued

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 539

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gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg      60
agctgtgccg ccagcggcctt cgacttcgat agctacgaga tgaactgggt ccgacaggcc      120
cctggcaaag gccttgaatg ggtgtccggc atcagctgga atagcggctg gatcgactac      180
gccgacagcg tgaaggcgag attcaccatc agccgggaca acagcaagaa caccctgtac      240
ctgcagatga acagcctgag agccgaggac accgccgtgt actactgtgc cagaagcggc      300
tacagcagct cttggtttga ccccgacttc gactattggg gccagggcac actggtcaca      360
gtctcttcag ccagcaccaa gggccccagc gtgttccttc tggccctag cagcaagagc      420
acatctggcg gaacagccgc cctgggctgc ctctgaagg actactttcc cgagcccgctg      480
accgtgtcct ggaactctgg cgctctgaca agcggcgtgc acacctttcc agcgtgctg      540
cagagcagcg gcctgtactc tctgagcagc gtctgtgacag tgcccagcag ctctctgggc      600
acccagacct acatctgcaa cgtgaaccac aagcccagca acaccaaggt ggacaagaag      660
gtggaaccca agagctcgca caagaccac acctgtcccc cttgtcctgc ccccgaaactg      720
ctgggaggcc cttcctgtgt cctgttcccc ccaaagccca aggacacct gatgatcagc      780
cggacccccg aagtgcctg cgtgggtgtg gatgtgtccc acgaggacct tgaagtgaag      840
ttcaattggt acgtggacgg cgtggaagtg cacaacgcca agaccaagcc tagagaggaa      900
cagtacaaca gcacctaccg ggtggtgtcc gtgtgacag tgctgcacca ggactggctg      960
aacggcaaa agtacaagtg caaggtgtcc aacaaggccc tgctgtcccc catcgagaaa     1020
accatcagca aggccaaagg ccagccccgc gaaccccagg tgtacacact gcccccaagc     1080
agggacgagc tgaccaagaa ccaggtgtcc ctgacctgtc tctgaaagg cttctacccc     1140
tccgatatcg ccgtggaatg ggagagcaac ggccagcccc agaacaacta caagaccacc     1200
ccccctgtgc tggacagcga cggctcattc ttcctgtaca gcaagctgac cgtggacaag     1260
tcccgggtggc agcagggcaa cgtgttcagc tgcagcgtga tgcacgaggc cctgcacaac     1320
cactacaccc agaagtcctt gagcctgagc cctggcaag                               1359

```

<210> SEQ ID NO 540

<211> LENGTH: 651

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 540

```

cagtctgttc tgacacagcc tccatctgtg tctggcgccc ctggacagag agtgaccatc      60
agctgtacag gcagcagctc cgatatggc gccgatacag acgtgcactg gtatcagcaa      120
ctgcctggca cagcccctaa gctgtgtatc tacggcaaca gcaacagacc tagcggcgtg      180
cccgatagat tcagcggctc taagtctggc acaagcgcca gcctggccat tactggactg      240
caggccgaag atgaggccga ctactactgc agcagctacg agggcatcaa cccctacgtg      300
gtgtttggcg gcggaacaaa gctgaccgtt ctaggccagc ctaaagccgc ccctagcgtg      360
accctgttcc ctccaagcag cgaggaactg caggccaaca aggccacctt cgtgtgcctg      420
atcagcgact totatcctgg cgccgtgacc gtggcctgga aggcgatag ctctcctgtg      480

```

-continued

```

aaggccggcg tggaaccac caccctagc aagcagagca acaacaata cgccgccagc 540
agctacctga gcctgacccc cgagcagtggt aagtcacaca gatcctacag ctgccaagtg 600
acccacgagg gcagcacctt ggaaaagaca gtggccccta ccgagtgacg c 651

```

```

<210> SEQ ID NO 541
<211> LENGTH: 123
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 541

```

```

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1           5           10           15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Asp Phe Ser Ser Tyr
20          25          30
Glu Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35          40          45
Ser Gly Ile Ser Trp Asn Ser Gly Trp Ile Asp Tyr Ala Asp Ser Val
50          55          60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65          70          75          80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85          90          95
Ala Arg Ser Gly Tyr Ser Ser Ser Trp Phe Asp Pro Asp Phe Asp Tyr
100         105         110
Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
115         120

```

```

<210> SEQ ID NO 542
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 542

```

```

Ser Tyr Glu Met Asn
1           5

```

```

<210> SEQ ID NO 543
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 543

```

```

Gly Ile Ser Trp Asn Ser Gly Trp Ile Asp Tyr Ala Asp Ser Val Lys
1           5           10           15
Gly

```

```

<210> SEQ ID NO 544
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 544

```

```

Ser Gly Tyr Ser Ser Ser Trp Phe Asp Pro Asp Phe Asp Tyr

```

-continued

```

1           5           10

<210> SEQ ID NO 545
<211> LENGTH: 111
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 545

Gln Ser Val Leu Thr Gln Pro Pro Ser Val Ser Gly Ala Pro Gly Gln
1           5           10           15

Arg Val Thr Ile Ser Cys Thr Gly Ser Ser Ser Asn Ile Gly Ala Gly
          20           25           30

Tyr Asp Val His Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu
          35           40           45

Leu Ile Tyr Gly Ala Ser Asn Arg Pro Ser Gly Val Pro Asp Arg Phe
          50           55           60

Ser Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Thr Gly Leu
          65           70           75           80

Gln Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Ser Ser Tyr Glu Gly Pro
          85           90           95

Asn Pro Tyr Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
          100          105          110

```

```

<210> SEQ ID NO 546
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 546

Thr Gly Ser Ser Ser Asn Ile Gly Ala Gly Tyr Asp Val His
1           5           10

```

```

<210> SEQ ID NO 547
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 547

```

```

Gly Ala Ser Asn Arg Pro Ser
1           5

```

```

<210> SEQ ID NO 548
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 548

```

```

Ser Ser Tyr Glu Gly Pro Asn Pro Tyr Val Val
1           5           10

```

```

<210> SEQ ID NO 549
<211> LENGTH: 369
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

-continued

<400> SEQUENCE: 549

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg	60
agctgtgccg ccagcggctt cgatttcagc agctacgaga tgaactgggt ccgacaggcc	120
cctggcaaag gccttgaatg ggtgtccggc atcagctgga atagcggctg gatcgactac	180
gccgacagcg tgaagggcag attcaccatc agccgggaca acagcaagaa caccctgtac	240
ctgcagatga acagcctgag agccgaggac accgccgtgt actactgtgc cagaagcggc	300
tacagcagct cttggtttga ccccgacttc gactattggg gccagggcac actggtcaca	360
gtctcttca	369

<210> SEQ ID NO 550

<211> LENGTH: 15

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 550

agctacgaga tgaac	15
------------------	----

<210> SEQ ID NO 551

<211> LENGTH: 51

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 551

ggcatcagct ggaatagcgg ctggatcgac tacgccgaca gcgtgaaggg c	51
--	----

<210> SEQ ID NO 552

<211> LENGTH: 42

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 552

agcggctaca gcagctcttg gtttgacccc gacttcgact at	42
--	----

<210> SEQ ID NO 553

<211> LENGTH: 333

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 553

cagtctgttc tgacacagcc tccatctgtg tctggcgccc ctggacagag agtgaccatc	60
agctgtacag gcagcagctc caatatcgga gccggctatg acgtgcactg gtatcagcag	120
ctgcctggca cagcccctaa actgctgac tacggcgcca gcaatagacc tagcggcgtg	180
cccgatagat tcagcggctc taagtctggc acaagcgcca gcctggccat tactggactg	240
caggccgaag atgaggccga ctactactgc agcagctacg agggccccaa tccttacgtg	300
gtgtttggcg gcggaacaaa gctgaccgtt cta	333

<210> SEQ ID NO 554

<211> LENGTH: 42

<212> TYPE: DNA

-continued

```

<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 554
acaggcagca gctccaatat cggagccggc tatgacgtgc ac                42

<210> SEQ ID NO 555
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 555
ggcgccagca atagacctag c                21

<210> SEQ ID NO 556
<211> LENGTH: 33
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 556
agcagctacg agggccccaa tccttacgtg gtg                33

<210> SEQ ID NO 557
<211> LENGTH: 453
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 557
Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1          5          10          15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Asp Phe Ser Ser Tyr
20          25          30
Glu Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35          40          45
Ser Gly Ile Ser Trp Asn Ser Gly Trp Ile Asp Tyr Ala Asp Ser Val
50          55          60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65          70          75          80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85          90          95
Ala Arg Ser Gly Tyr Ser Ser Ser Trp Phe Asp Pro Asp Phe Asp Tyr
100         105         110
Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly
115         120         125
Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly
130         135         140
Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val
145         150         155         160
Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe
165         170         175
Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val
180         185         190
Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val

```

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195				200				205							
Asn	His	Lys	Pro	Ser	Asn	Thr	Lys	Val	Asp	Lys	Lys	Val	Glu	Pro	Lys
210						215					220				
Ser	Cys	Asp	Lys	Thr	His	Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Leu
225					230					235					240
Leu	Gly	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr
					245					250				255	
Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val
			260						265					270	
Ser	His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val
		275							280				285		
Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser
	290					295					300				
Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu
305					310						315				320
Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala
					325					330				335	
Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro
			340						345					350	
Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Asp	Glu	Leu	Thr	Lys	Asn	Gln
		355					360						365		
Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala
	370					375					380				
Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr
385					390					395					400
Pro	Pro	Val	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu
				405					410					415	
Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser
		420							425				430		
Val	Met	His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser
		435					440					445			
Leu	Ser	Pro	Gly	Lys											
	450														

<210> SEQ ID NO 558

<211> LENGTH: 217

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 558

Gln	Ser	Val	Leu	Thr	Gln	Pro	Pro	Ser	Val	Ser	Gly	Ala	Pro	Gly	Gln
1				5					10					15	
Arg	Val	Thr	Ile	Ser	Cys	Thr	Gly	Ser	Ser	Ser	Asn	Ile	Gly	Ala	Gly
		20						25					30		
Tyr	Asp	Val	His	Trp	Tyr	Gln	Gln	Leu	Pro	Gly	Thr	Ala	Pro	Lys	Leu
	35						40					45			
Leu	Ile	Tyr	Gly	Ala	Ser	Asn	Arg	Pro	Ser	Gly	Val	Pro	Asp	Arg	Phe
	50					55					60				
Ser	Gly	Ser	Lys	Ser	Gly	Thr	Ser	Ala	Ser	Leu	Ala	Ile	Thr	Gly	Leu
65					70					75				80	
Gln	Ala	Glu	Asp	Glu	Ala	Asp	Tyr	Tyr	Cys	Ser	Ser	Tyr	Glu	Gly	Pro
				85					90					95	
Asn	Pro	Tyr	Val	Val	Phe	Gly	Gly	Gly	Thr	Lys	Leu	Thr	Val	Leu	Gly

-continued

100	105	110
Gln Pro Lys Ala Ala Pro Ser Val Thr Leu Phe Pro Pro Ser Ser Glu		
115	120	125
Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp Phe		
130	135	140
Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Ser Ser Pro Val		
145	150	155
Lys Ala Gly Val Glu Thr Thr Thr Pro Ser Lys Gln Ser Asn Asn Lys		
165	170	175
Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys Ser		
180	185	190
His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val Glu		
195	200	205
Lys Thr Val Ala Pro Thr Glu Cys Ser		
210	215	
<210> SEQ ID NO 559		
<211> LENGTH: 1359		
<212> TYPE: DNA		
<213> ORGANISM: Artificial Sequence		
<220> FEATURE:		
<223> OTHER INFORMATION: antibody sequence		
<400> SEQUENCE: 559		
gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg	60	
agctgtgccg ccagcggcgt cgatttcagc agctacgaga tgaactgggt ccgacaggcc	120	
cctggcaaa gcttgaatg ggtgtccggc atcagctgga atagcggctg gatcgactac	180	
gccgacagcg tgaagggcag attcaccatc agccgggaca acagcaagaa caccctgtac	240	
ctgcagatga acagcctgag agccgaggac accgccgtgt actactgtgc cagaagcggc	300	
tacagcagct cttggtttga ccccgacttc gactattggg gccagggcac actggtcaca	360	
gtctcttcag ccagcaccaa gggccccagc gtgttcctc tggcccctag cagcaagagc	420	
acatctggcg gaacagccgc cctgggctgc ctctgaagg actactttcc cgagcccgtg	480	
accgtgtcct ggaactctgg cgctctgaca agcggcgtgc acacctttcc agccgtgctg	540	
cagagcagcg gcctgtactc tctgagcagc gtcgtgacag tgcccagcag ctctctgggc	600	
accagacct acatctgcaa cgtgaaccac aagcccagca acaccaaggt ggacaagaag	660	
gtggaacca agagctgcga caagaccac acctgtcccc cttgtcctgc ccccgaactg	720	
ctgggaggcc cttccgtgtt cctgttcccc ccaaagccca aggacacct gatgatcagc	780	
cggacccccg aagtgaacctg cgtgggtgtg gatgtgtccc acgaggaccc tgaagtgaag	840	
ttcaattggt acgtggacgg cgtggaagtg cacaacgcca agaccaagcc tagagaggaa	900	
cagtacaaca gcacctaccg ggtggtgtcc gtgctgacag tgctgcacca ggactggctg	960	
aacggcaaa agtacaagtg caaggtgtcc aacaaggccc tgctgtcccc catcgagaaa	1020	
accatcagca agggcaaggg ccagccccgc gaaccccagg tgtacacact gcccccaagc	1080	
agggacgagc tgaccaagaa ccaggtgtcc ctgacctgtc tcgtgaaagg cttctacccc	1140	
tccgatatcg ccgtggaatg ggagagcaac ggcagcccg agaacaacta caagaccacc	1200	
ccccctgtgc tggacagcga cggtcattc ttctgtaca gcaagctgac cgtggacaag	1260	
tcccgggtggc agcagggcaa cgtgttcagc tgcagcgtga tgcacgaggc cctgcacaac	1320	
cactacacc agaagtccct gagcctgagc cctggcaag	1359	

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<210> SEQ ID NO 560
<211> LENGTH: 651
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 560

cagtctgttc tgacacagcc tccatctgtg tctggcgccc ctggacagag agtgaccatc      60
agctgtacag gcagcagctc caatatcgga gccggctatg acgtgcactg gtatcagcag      120
ctgcctggca cagcccctaa actgctgatac tacggcgcca gcaatagacc tagcggcgctg      180
cccgatagat tcagcggctc taagtctggc acaagcgcca gcctggccat tactggactg      240
caggccgaag atgaggccga ctactactgc agcagctacg agggccccaa tccttacgtg      300
gtgtttggcg gcggaacaaa gctgaccgtt ctaggccagc ctaaagccgc ccctagcgtg      360
acctgtttcc ctccaagcag cgaggaactg caggccaaca agggccacct cgtgtgctg      420
atcagcgact tctatcctgg cgccgtgacc gtggcctgga aggcgatag ctctcctgtg      480
aaggccggcg tggaaccac caccctagc aagcagagca acaacaaata cgccgccagc      540
agctacctga gcctgacccc cgagcagtggt aagtcccaca gatcctacag ctgccaagtg      600
acccacgagg gcagcaccgt ggaaaagaca gtggccccta ccgagtgcag c              651

```

```

<210> SEQ ID NO 561
<211> LENGTH: 123
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 561

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1           5           10          15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asp Ser Tyr
20          25          30
Glu Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35          40          45
Ser Gly Ile Ser Trp Asn Ser Gly Trp Ile Asp Tyr Ala Asp Ser Val
50          55          60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65          70          75          80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85          90          95
Ala Arg Ser Gly Tyr Ser Ser Ser Trp Phe Asp Pro Asp Phe Asp Tyr
100         105         110
Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
115         120

```

```

<210> SEQ ID NO 562
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 562

Ser Tyr Glu Met Asn
1           5

```

-continued

<210> SEQ ID NO 563
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 563

Gly Ile Ser Trp Asn Ser Gly Trp Ile Asp Tyr Ala Asp Ser Val Lys
1 5 10 15

Gly

<210> SEQ ID NO 564
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 564

Ser Gly Tyr Ser Ser Ser Trp Phe Asp Pro Asp Phe Asp Tyr
1 5 10

<210> SEQ ID NO 565
<211> LENGTH: 111
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 565

Gln Ser Val Leu Thr Gln Pro Pro Ser Val Ser Gly Ala Pro Gly Gln
1 5 10 15

Arg Val Thr Ile Ser Cys Thr Gly Ser Ser Ser Asn Ile Gly Ala Gly
20 25 30

Tyr Asp Val His Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu
35 40 45

Leu Ile Tyr Gly Asn Ser Asn Arg Pro Ser Gly Val Pro Asp Arg Phe
50 55 60

Ser Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Thr Gly Leu
65 70 75 80

Gln Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Ser Ser Tyr Ala Gly Pro
85 90 95

Asn Pro Tyr Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
100 105 110

<210> SEQ ID NO 566
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 566

Thr Gly Ser Ser Ser Asn Ile Gly Ala Gly Tyr Asp Val His
1 5 10

<210> SEQ ID NO 567
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

-continued

<400> SEQUENCE: 567

Gly Asn Ser Asn Arg Pro Ser
1 5

<210> SEQ ID NO 568

<211> LENGTH: 11

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 568

Ser Ser Tyr Ala Gly Pro Asn Pro Tyr Val Val
1 5 10

<210> SEQ ID NO 569

<211> LENGTH: 369

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 569

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg 60
agctgtgccg ccagcgggett caccttcgat agctacgaga tgaactgggt ccgacaggcc 120
cctggcaaag gccttgaatg ggtgtccggc atcagctgga atagcggctg gatcgactac 180
gccgacagcg tgaagggcag attcaccatc agccgggaca acagcaagaa caccctgtac 240
ctgcagatga acagcctgag agccgaggac accgccgtgt actactgtgc cagaagcggc 300
tacagcagct cttggtttga ccccgacttc gactattggg gccagggcac actggtcaca 360
gtctcttca 369

<210> SEQ ID NO 570

<211> LENGTH: 15

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 570

agctacgaga tgaac 15

<210> SEQ ID NO 571

<211> LENGTH: 51

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 571

ggcatcagct ggaatagcgg ctggatcgac tacgccgaca gcgtgaaggg c 51

<210> SEQ ID NO 572

<211> LENGTH: 42

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 572

agcggctaca gcagctcttg gtttgacccc gacttcgact at 42

-continued

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<210> SEQ ID NO 573
<211> LENGTH: 333
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 573

cagtctgttc tgacacagcc tccatctgtg tctggcgccc ctggacagag agtgaccatc   60
agctgtacag gcagcagctc caatatcgga gccggctatg acgtgcactg gtatcagcag   120
ctgcctggca cagcccctaa actgctgatac tacggcaaca gcaacagacc cagcggcgtg   180
cccgatagat tttccggctc taagagcggc acaagcgcca gcctggctat tactggactg   240
caggccgagg acgagggcca ctactactgt agctcttacg ctggcccaaa tccttacgtg   300
gtgtttggcg gcggaacaaa gctgaccgtt cta                               333

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<210> SEQ ID NO 574
<211> LENGTH: 42
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 574

acaggcagca gctccaatat cggagccggc tatgacgtgc ac                               42

```

```

<210> SEQ ID NO 575
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 575

ggcaacagca acagaccag c                               21

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<210> SEQ ID NO 576
<211> LENGTH: 33
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 576

agctcttacg ctggcccaaa tccttacgtg gtg                               33

```

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<210> SEQ ID NO 577
<211> LENGTH: 453
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 577

Glu Val Gln Leu Leu Glu Ser Gly Gly Leu Val Gln Pro Gly Gly
1           5           10          15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asp Ser Tyr
20          25          30

Glu Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35          40          45

Ser Gly Ile Ser Trp Asn Ser Gly Trp Ile Asp Tyr Ala Asp Ser Val
50          55          60

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Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
 65 70 75 80
 Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Ala Arg Ser Gly Tyr Ser Ser Ser Trp Phe Asp Pro Asp Phe Asp Tyr
 100 105 110
 Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly
 115 120 125
 Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly
 130 135 140
 Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val
 145 150 155 160
 Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe
 165 170 175
 Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val
 180 185 190
 Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val
 195 200 205
 Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys Lys Val Glu Pro Lys
 210 215 220
 Ser Cys Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu
 225 230 235 240
 Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr
 245 250 255
 Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val
 260 265 270
 Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val
 275 280 285
 Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser
 290 295 300
 Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu
 305 310 315 320
 Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala
 325 330 335
 Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro
 340 345 350
 Gln Val Tyr Thr Leu Pro Pro Ser Arg Asp Glu Leu Thr Lys Asn Gln
 355 360 365
 Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala
 370 375 380
 Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr
 385 390 395 400
 Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu
 405 410 415
 Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser
 420 425 430
 Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser
 435 440 445
 Leu Ser Pro Gly Lys
 450

<210> SEQ ID NO 578

<211> LENGTH: 217

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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 578

Gln Ser Val Leu Thr Gln Pro Pro Ser Val Ser Gly Ala Pro Gly Gln
 1             5             10             15
Arg Val Thr Ile Ser Cys Thr Gly Ser Ser Ser Asn Ile Gly Ala Gly
 20            25            30
Tyr Asp Val His Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu
 35            40            45
Leu Ile Tyr Gly Asn Ser Asn Arg Pro Ser Gly Val Pro Asp Arg Phe
 50            55            60
Ser Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Thr Gly Leu
 65            70            75            80
Gln Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Ser Ser Tyr Ala Gly Pro
 85            90            95
Asn Pro Tyr Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu Gly
100           105           110
Gln Pro Lys Ala Ala Pro Ser Val Thr Leu Phe Pro Pro Ser Ser Glu
115           120           125
Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp Phe
130           135           140
Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Ser Ser Pro Val
145           150           155           160
Lys Ala Gly Val Glu Thr Thr Thr Pro Ser Lys Gln Ser Asn Asn Lys
165           170           175
Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys Ser
180           185           190
His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val Glu
195           200           205

Lys Thr Val Ala Pro Thr Glu Cys Ser
210           215

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<210> SEQ ID NO 579
<211> LENGTH: 1359
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 579

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg      60
agctgtgccg ccagcggcgtt caccttcgat agctacgaga tgaactgggt ccgacaggcc      120
cctggcaaa gacctgaatg ggtgtccggc atcagctgga atagcggctg gatcgactac      180
gccgacagcg tgaagggcag attcaccatc agccgggaca acagcaagaa caccctgtac      240
ctgcagatga acagcctgag agccgaggac accgccgtgt actactgtgc cagaagcggc      300
tacagcagct cttggtttga ccccgacttc gactattggg gccagggcac actggtcaca      360
gtctcttcag ccagcaccaa gggccccagc gtgttccttc tggcccctag cagcaagagc      420
acatctggcg gaacagccgc cctgggctgc ctgctgaagg actactttcc cgagcccgctg      480
accgtgtcct ggaactctgg cgctctgaca agcggcgtgc acacctttcc agcgtgctg      540
cagagcagcg gcctgtactc tctgagcagc gtcgtgacag tgcccagcag ctctctgggc      600

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accagacct acatctgcaa cgtgaaccac aagcccagca acaccaaggt ggacaagaag 660
gtggaaccca agagctgcga caagaccacac acctgtcccc cttgtcctgc ccccgaaactg 720
ctgggaggcc cttcogtgtt cctgttcccc ccaaagccca aggacacccct gatgatcagc 780
cggacccccg aagtgcactg cgtgggtgtg gatgtgtccc acgaggaccc tgaagtgaag 840
ttcaattggt acgtggacgg cgtggaagtg cacaacgccca agaccaagcc tagagaggaa 900
cagtacaaca gcacotaccg ggtggtgtcc gtgtgcacag tgctgcacca ggactggctg 960
aacggcaaa agtacaagtg caaggtgtcc aacaaggccc tgctgtcccc catcgagaaa 1020
accatcagca agggcaaggg ccagcccccgc gaaccccagg tgtacacact gcccccaagc 1080
agggacgagc tgaccaagaa ccaggtgtcc ctgacctgtc tcgtgaaagg cttctacccc 1140
tccgatatcg ccgtggaatg ggagagcaac ggcagcccg agaacaacta caagaccacc 1200
ccccctgtgc tggacagcga cggctcattc ttcctgtaca gcaagctgac cgtggacaag 1260
tcccgggtggc agcagggcaa cgtgttcagc tgcagcgtga tgcacgaggc cctgcacaac 1320
cactacaccc agaagtcctt gagcctgagc cctggcaag 1359

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<210> SEQ ID NO 580
<211> LENGTH: 651
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 580
cagtcgttgc tgacacagcc tccatctgtg tctggcgccc ctggacagag agtgaccatc 60
agctgtacag gcagcagctc caatatcgga gccggctatg acgtgcactg gtatcagcag 120
ctgcttgcca cagcccctaa actgctgac tacggcaaca gcaacagacc cagcggcgtg 180
cccgatagat tttccggctc taagagcggc acaagcgcca gcctggctat tactggactg 240
caggccgagg acgaggccga ctactactgt agctcttacg ctggccccc aa tccttacgtg 300
gtgtttggcg gcggaacaaa gctgaccgtt ctaggccagc ctaaagccgc ccctagcgtg 360
accctgttcc ctccaagcag cgaggaaactg caggccaaca agggccacct cgtgtgctg 420
atcagcgact tctatcctgg cgccgtgacc gtggcctgga agggcgatag ctctcctgtg 480
aaggccggcg tggaaaccac caccctagc aagcagagca acaacaaata cgccgccagc 540
agctacctga gcctgacccc cgagcagtggt aagtcaccaca gatcctacag ctgccaagtg 600
accacagagg gcagcacctt ggaaaagaca gtggccccta ccgagtgcag c 651

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<210> SEQ ID NO 581
<211> LENGTH: 990
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

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<400> SEQUENCE: 581
His His His His His His Lys Asn Asn Val Pro Arg Leu Lys Leu Ser
1          5          10          15
Tyr Lys Glu Met Leu Glu Ser Asn Asn Val Ile Thr Phe Asn Gly Leu
20        25        30
Ala Asn Ser Ser Ser Tyr His Thr Phe Leu Leu Asp Glu Glu Arg Ser
35        40        45
Arg Leu Tyr Val Gly Ala Lys Asp His Ile Phe Ser Phe Asp Leu Val
50        55        60
Asn Ile Lys Asp Phe Gln Lys Ile Val Trp Pro Val Ser Tyr Thr Arg

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65					70						75					80
Arg	Asp	Glu	Cys	Lys	Trp	Ala	Gly	Lys	Asp	Ile	Leu	Lys	Glu	Cys	Ala	
				85					90					95		
Asn	Phe	Ile	Lys	Val	Leu	Lys	Ala	Tyr	Asn	Gln	Thr	His	Leu	Tyr	Ala	
			100					105					110			
Cys	Gly	Thr	Gly	Ala	Phe	His	Pro	Ile	Cys	Thr	Tyr	Ile	Glu	Ile	Gly	
		115					120					125				
His	His	Pro	Glu	Asp	Asn	Ile	Phe	Lys	Leu	Glu	Asn	Ser	His	Phe	Glu	
	130					135					140					
Asn	Gly	Arg	Gly	Lys	Ser	Pro	Tyr	Asp	Pro	Lys	Leu	Leu	Thr	Ala	Ser	
145					150					155					160	
Leu	Leu	Ile	Asp	Gly	Glu	Leu	Tyr	Ser	Gly	Thr	Ala	Ala	Asp	Phe	Met	
			165						170					175		
Gly	Arg	Asp	Phe	Ala	Ile	Phe	Arg	Thr	Leu	Gly	His	His	His	Pro	Ile	
			180					185						190		
Arg	Thr	Glu	Gln	His	Asp	Ser	Arg	Trp	Leu	Asn	Asp	Pro	Lys	Phe	Ile	
	195						200					205				
Ser	Ala	His	Leu	Ile	Ser	Glu	Ser	Asp	Asn	Pro	Glu	Asp	Asp	Lys	Val	
	210					215					220					
Tyr	Phe	Phe	Phe	Arg	Glu	Asn	Ala	Ile	Asp	Gly	Glu	His	Ser	Gly	Lys	
225					230					235					240	
Ala	Thr	His	Ala	Arg	Ile	Gly	Gln	Ile	Cys	Lys	Asn	Asp	Phe	Gly	Gly	
			245						250					255		
His	Arg	Ser	Leu	Val	Asn	Lys	Trp	Thr	Thr	Phe	Leu	Lys	Ala	Arg	Leu	
			260					265					270			
Ile	Cys	Ser	Val	Pro	Gly	Pro	Asn	Gly	Ile	Asp	Thr	His	Phe	Asp	Glu	
	275						280					285				
Leu	Gln	Asp	Val	Phe	Leu	Met	Asn	Phe	Lys	Asp	Pro	Lys	Asn	Pro	Val	
	290					295					300					
Val	Tyr	Gly	Val	Phe	Thr	Thr	Ser	Ser	Asn	Ile	Phe	Lys	Gly	Ser	Ala	
305					310					315					320	
Val	Cys	Met	Tyr	Ser	Met	Ser	Asp	Val	Arg	Arg	Val	Phe	Leu	Gly	Pro	
			325						330					335		
Tyr	Ala	His	Arg	Asp	Gly	Pro	Asn	Tyr	Gln	Trp	Val	Pro	Tyr	Gln	Gly	
			340					345					350			
Arg	Val	Pro	Tyr	Pro	Arg	Pro	Gly	Thr	Cys	Pro	Ser	Lys	Thr	Phe	Gly	
	355						360					365				
Gly	Phe	Asp	Ser	Thr	Lys	Asp	Leu	Pro	Asp	Asp	Val	Ile	Thr	Phe	Ala	
	370					375					380					
Arg	Ser	His	Pro	Ala	Met	Tyr	Asn	Pro	Val	Phe	Pro	Met	Asn	Asn	Arg	
385					390					395					400	
Pro	Ile	Val	Ile	Lys	Thr	Asp	Val	Asn	Tyr	Gln	Phe	Thr	Gln	Ile	Val	
			405						410					415		
Val	Asp	Arg	Val	Asp	Ala	Glu	Asp	Gly	Gln	Tyr	Asp	Val	Met	Phe	Ile	
		420						425					430			
Gly	Thr	Asp	Val	Gly	Thr	Val	Leu	Lys	Val	Val	Ser	Ile	Pro	Lys	Glu	
	435						440					445				
Thr	Trp	Tyr	Asp	Leu	Glu	Glu	Val	Leu	Leu	Glu	Glu	Met	Thr	Val	Phe	
	450					455					460					
Arg	Glu	Pro	Thr	Ala	Ile	Ser	Ala	Met	Glu	Leu	Ser	Thr	Lys	Gln	Gln	
465					470					475					480	
Gln	Leu	Tyr	Ile	Gly	Ser	Thr	Ala	Gly	Val	Ala	Gln	Leu	Pro	Leu	His	
			485						490					495		

Arg	Cys	Asp	Ile	Tyr	Gly	Lys	Ala	Cys	Ala	Glu	Cys	Cys	Leu	Ala	Arg
			500				505						510		
Asp	Pro	Tyr	Cys	Ala	Trp	Asp	Gly	Ser	Ala	Cys	Ser	Arg	Tyr	Phe	Pro
			515				520				525				
Thr	Ala	Lys	Arg	Ala	Thr	Arg	Ala	Gln	Asp	Ile	Arg	Asn	Gly	Asp	Pro
			530				535				540				
Leu	Thr	His	Cys	Ser	Asp	Leu	His	His	Asp	Asn	His	His	Gly	His	Ser
			545				550				555			560	
Pro	Glu	Glu	Arg	Ile	Ile	Tyr	Gly	Val	Glu	Asn	Ser	Ser	Thr	Phe	Leu
			565				570						575		
Glu	Cys	Ser	Pro	Lys	Ser	Gln	Arg	Ala	Leu	Val	Tyr	Trp	Gln	Phe	Gln
			580				585						590		
Arg	Arg	Asn	Glu	Glu	Arg	Lys	Glu	Glu	Ile	Arg	Val	Asp	Asp	His	Ile
			595				600				605				
Ile	Arg	Thr	Asp	Gln	Gly	Leu	Leu	Leu	Arg	Ser	Leu	Gln	Gln	Lys	Asp
			610				615				620				
Ser	Gly	Asn	Tyr	Leu	Cys	His	Ala	Val	Glu	His	Gly	Phe	Ile	Gln	Thr
			625				630				635			640	
Leu	Leu	Lys	Val	Thr	Leu	Glu	Val	Ile	Asp	Thr	Glu	His	Leu	Glu	Glu
			645				650						655		
Leu	Leu	His	Lys	Asp	Asp	Asp	Gly	Asp	Gly	Ser	Lys	Thr	Lys	Glu	Met
			660				665						670		
Ser	Asn	Ser	Met	Thr	Pro	Ser	Gln	Lys	Val	Trp	Tyr	Arg	Asp	Phe	Met
			675				680				685				
Gln	Leu	Ile	Asn	His	Pro	Asn	Leu	Asn	Thr	Met	Asp	Glu	Phe	Cys	Glu
			690				695				700				
Gln	Val	Trp	Lys	Arg	Asp	Arg	Lys	Gln	Arg	Arg	Gln	Arg	Pro	Gly	His
			705				710				715			720	
Thr	Pro	Gly	Asn	Ser	Asn	Lys	Trp	Lys	His	Leu	Gln	Glu	Asn	Lys	Lys
			725				730						735		
Gly	Arg	Asn	Arg	Arg	Thr	His	Glu	Phe	Glu	Arg	Ala	Pro	Arg	Ser	Val
			740				745						750		
Asp	Ile	Glu	Gly	Arg	Met	Asp	Pro	Lys	Ser	Cys	Asp	Lys	Thr	His	Thr
			755				760				765				
Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Leu	Leu	Gly	Gly	Pro	Ser	Val	Phe
			770				775				780				
Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro
			785				790				795			800	
Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val
			805				810						815		
Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr
			820				825						830		
Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser	Val
			835				840						845		
Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys
			850				855				860				
Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser
			865				870				875			880	
Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro
			885				890						895		
Ser	Arg	Asp	Glu	Leu	Thr	Lys									

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Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly
	915						920					925			
Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp	Ser	Asp
930						935					940				
Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp
945					950					955					960
Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	His
			965						970					975	
Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys		
		980						985					990		

<210> SEQ ID NO 582

<211> LENGTH: 563

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 582

Asn	Tyr	Gln	Asn	Gly	Lys	Asn	Asn	Val	Pro	Arg	Leu	Lys	Leu	Ser	Tyr
1			5					10					15		
Lys	Glu	Met	Leu	Glu	Ser	Asn	Asn	Val	Ile	Thr	Phe	Asn	Gly	Leu	Ala
	20							25					30		
Asn	Ser	Ser	Ser	Tyr	His	Thr	Phe	Leu	Leu	Asp	Glu	Glu	Arg	Ser	Arg
	35					40					45				
Leu	Tyr	Val	Gly	Ala	Lys	Asp	His	Ile	Phe	Ser	Phe	Asp	Leu	Val	Asn
50					55						60				
Ile	Lys	Asp	Phe	Gln	Lys	Ile	Val	Trp	Pro	Val	Ser	Tyr	Thr	Arg	Arg
65				70					75					80	
Asp	Glu	Cys	Lys	Trp	Ala	Gly	Lys	Asp	Ile	Leu	Lys	Glu	Cys	Ala	Asn
		85						90					95		
Phe	Ile	Lys	Val	Leu	Lys	Ala	Tyr	Asn	Gln	Thr	His	Leu	Tyr	Ala	Cys
	100							105					110		
Gly	Thr	Gly	Ala	Phe	His	Pro	Ile	Cys	Thr	Tyr	Ile	Glu	Ile	Gly	His
	115					120						125			
His	Pro	Glu	Asp	Asn	Ile	Phe	Lys	Leu	Glu	Asn	Ser	His	Phe	Glu	Asn
	130				135						140				
Gly	Arg	Gly	Lys	Ser	Pro	Tyr	Asp	Pro	Lys	Leu	Leu	Thr	Ala	Ser	Leu
145			150					155							160
Leu	Ile	Asp	Gly	Glu	Leu	Tyr	Ser	Gly	Thr	Ala	Ala	Asp	Phe	Met	Gly
		165						170						175	
Arg	Asp	Phe	Ala	Ile	Phe	Arg	Thr	Leu	Gly	His	His	His	Pro	Ile	Arg
	180						185						190		
Thr	Glu	Gln	His	Asp	Ser	Arg	Trp	Leu	Asn	Asp	Pro	Lys	Phe	Ile	Ser
	195					200						205			
Ala	His	Leu	Ile	Ser	Glu	Ser	Asp	Asn	Pro	Glu	Asp	Asp	Lys	Val	Tyr
	210					215					220				
Phe	Phe	Phe	Arg	Glu	Asn	Ala	Ile	Asp	Gly	Glu	His	Ser	Gly	Lys	Ala
225					230				235					240	
Thr	His	Ala	Arg	Ile	Gly	Gln	Ile	Cys	Lys	Asn	Asp	Phe	Gly	Gly	His
		245						250					255		
Arg	Ser	Leu	Val	Asn	Lys	Trp	Thr	Thr	Phe	Leu	Lys	Ala	Arg	Leu	Ile
		260					265						270		
Cys	Ser	Val	Pro	Gly	Pro	Asn	Gly	Ile	Asp	Thr	His	Phe	Asp	Glu	Leu
	275					280						285			
Gln	Asp	Val	Phe	Leu	Met	Asn	Phe	Lys	Asp	Pro	Lys	Asn	Pro	Val	Val
	290					295					300				

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Tyr Gly Val Phe Thr Thr Ser Ser Asn Ile Phe Lys Gly Ser Ala Val
 305 310 315 320
 Cys Met Tyr Ser Met Ser Asp Val Arg Arg Val Phe Leu Gly Pro Tyr
 325 330 335
 Ala His Arg Asp Gly Pro Asn Tyr Gln Trp Val Pro Tyr Gln Gly Arg
 340 345 350
 Val Pro Tyr Pro Arg Pro Gly Thr Cys Pro Ser Lys Thr Phe Gly Gly
 355 360 365
 Phe Asp Ser Thr Lys Asp Leu Pro Asp Asp Val Ile Thr Phe Ala Arg
 370 375 380
 Ser His Pro Ala Met Tyr Asn Pro Val Phe Pro Met Asn Asn Arg Pro
 385 390 395 400
 Ile Val Ile Lys Thr Asp Val Asn Tyr Gln Phe Thr Gln Ile Val Val
 405 410 415
 Asp Arg Val Asp Ala Glu Asp Gly Gln Tyr Asp Val Met Phe Ile Gly
 420 425 430
 Thr Asp Val Gly Thr Val Leu Lys Val Val Ser Ile Pro Lys Glu Thr
 435 440 445
 Trp Tyr Asp Leu Glu Glu Val Leu Leu Glu Glu Met Thr Val Phe Arg
 450 455 460
 Glu Pro Thr Ala Ile Ser Ala Met Glu Leu Ser Thr Lys Gln Gln Gln
 465 470 475 480
 Leu Tyr Ile Gly Ser Thr Ala Gly Val Ala Gln Leu Pro Leu His Arg
 485 490 495
 Cys Asp Ile Tyr Gly Lys Ala Cys Ala Glu Cys Cys Leu Ala Arg Asp
 500 505 510
 Pro Tyr Cys Ala Trp Asp Gly Ser Ala Cys Ser Arg Tyr Phe Pro Thr
 515 520 525
 Ala Lys Ala Arg Thr Arg Ala Gln Asp Ile Arg Asn Gly Asp Pro Leu
 530 535 540
 Thr His Cys Ser Asp Gly Gly Ile Glu Gly Arg Met Asp His His His
 545 550 555 560
 His His His

<210> SEQ ID NO 583

<211> LENGTH: 563

<212> TYPE: PRT

<213> ORGANISM: Mus musculus (Mouse)

<400> SEQUENCE: 583

Asn Tyr Ala Asn Gly Lys Asn Asn Val Pro Arg Leu Lys Leu Ser Tyr
 1 5 10 15
 Lys Glu Met Leu Glu Ser Asn Asn Val Ile Thr Phe Asn Gly Leu Ala
 20 25 30
 Asn Ser Ser Ser Tyr His Thr Phe Leu Leu Asp Glu Glu Arg Ser Arg
 35 40 45
 Leu Tyr Val Gly Ala Lys Asp His Ile Phe Ser Phe Asn Leu Val Asn
 50 55 60
 Ile Lys Asp Phe Gln Lys Ile Val Trp Pro Val Ser Tyr Thr Arg Arg
 65 70 75 80
 Asp Glu Cys Lys Trp Ala Gly Lys Asp Ile Leu Lys Glu Cys Ala Asn
 85 90 95
 Phe Ile Lys Val Leu Glu Ala Tyr Asn Gln Thr His Leu Tyr Ala Cys
 100 105 110

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Gly Thr Gly Ala Phe His Pro Ile Cys Thr Tyr Ile Glu Val Gly His	115	120	125
His Pro Glu Asp Asn Ile Phe Lys Leu Gln Asp Ser His Phe Glu Asn	130	135	140
Gly Arg Gly Lys Ser Pro Tyr Asp Pro Lys Leu Leu Thr Ala Ser Leu	145	150	155
Leu Ile Asp Gly Glu Leu Tyr Ser Gly Thr Ala Ala Asp Phe Met Gly	165	170	175
Arg Asp Phe Ala Ile Phe Arg Thr Leu Gly His His His Pro Ile Arg	180	185	190
Thr Glu Gln His Asp Ser Arg Trp Leu Asn Asp Pro Arg Phe Ile Ser	195	200	205
Ala His Leu Ile Pro Glu Ser Asp Asn Pro Glu Asp Asp Lys Val Tyr	210	215	220
Phe Phe Phe Arg Glu Asn Ala Ile Asp Gly Glu His Ser Gly Lys Ala	225	230	235
Thr His Ala Arg Ile Gly Gln Ile Cys Lys Asn Asp Phe Gly Gly His	245	250	255
Arg Ser Leu Val Asn Lys Trp Thr Thr Phe Leu Lys Ala Arg Leu Ile	260	265	270
Cys Ser Val Pro Gly Pro Asn Gly Ile Asp Thr His Phe Asp Glu Leu	275	280	285
Gln Asp Val Phe Leu Met Asn Ser Lys Asp Pro Lys Asn Pro Ile Val	290	295	300
Tyr Gly Val Phe Thr Thr Ser Ser Asn Ile Phe Lys Gly Ser Ala Val	305	310	315
Cys Met Tyr Ser Met Ser Asp Val Arg Arg Val Phe Leu Gly Pro Tyr	325	330	335
Ala His Arg Asp Gly Pro Asn Tyr Gln Trp Val Pro Tyr Gln Gly Arg	340	345	350
Val Pro Tyr Pro Arg Pro Gly Thr Cys Pro Ser Lys Thr Phe Gly Gly	355	360	365
Phe Asp Ser Thr Lys Asp Leu Pro Asp Asp Val Ile Thr Phe Ala Arg	370	375	380
Ser His Pro Ala Met Tyr Asn Pro Val Phe Pro Ile Asn Asn Arg Pro	385	390	395
Ile Met Ile Lys Thr Asp Val Asn Tyr Gln Phe Thr Gln Ile Val Val	405	410	415
Asp Arg Val Asp Ala Glu Asp Gly Gln Tyr Asp Val Met Phe Ile Gly	420	425	430
Thr Asp Val Gly Thr Val Leu Lys Val Val Ser Val Pro Lys Glu Thr	435	440	445
Trp His Asp Leu Glu Glu Ile Leu Leu Glu Glu Met Thr Val Phe Arg	450	455	460
Glu Pro Thr Thr Ile Ser Ala Met Glu Leu Ser Thr Lys Gln Gln Gln	465	470	475
Leu Tyr Ile Gly Ser Thr Ala Gly Val Ala Gln Leu Pro Leu His Arg	485	490	495
Cys Asp Ile Tyr Gly Lys Ala Cys Ala Glu Cys Cys Leu Ala Arg Asp	500	505	510
Pro Tyr Cys Ala Trp Asp Gly Ser Ser Cys Ser Arg Tyr Phe Pro Thr	515	520	525

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Ala Lys Ala Arg Thr Arg Ala Gln Asp Ile Arg Asn Gly Asp Pro Leu
530 535 540

Thr His Cys Ser Asp Gly Gly Ile Glu Gly Arg Met Asp His His His
545 550 555 560

His His His

<210> SEQ ID NO 584
<211> LENGTH: 563
<212> TYPE: PRT
<213> ORGANISM: Rattus norvegicus (Rat)

<400> SEQUENCE: 584

Asn Tyr Ala Asn Gly Lys Asn Asn Val Pro Arg Leu Lys Leu Ser Tyr
1 5 10 15

Lys Glu Met Leu Glu Ser Asn Asn Val Ile Thr Phe Asn Gly Leu Ala
20 25 30

Asn Ser Ser Ser Tyr His Thr Phe Leu Leu Asp Glu Glu Arg Ser Arg
35 40 45

Leu Tyr Val Gly Ala Lys Asp His Ile Phe Ser Phe Asn Leu Val Asn
50 55 60

Ile Lys Asp Phe Gln Lys Ile Val Trp Pro Val Ser Tyr Thr Arg Arg
65 70 75 80

Asp Glu Cys Lys Trp Ala Gly Lys Asp Ile Leu Lys Glu Cys Ala Asn
85 90 95

Phe Ile Lys Val Leu Lys Ala Tyr Asn Gln Thr His Leu Tyr Ala Cys
100 105 110

Gly Thr Gly Ala Phe His Pro Ile Cys Thr Tyr Ile Glu Val Gly His
115 120 125

His Pro Glu Asp Asn Ile Phe Lys Leu Gln Asp Ser His Phe Glu Asn
130 135 140

Gly Arg Gly Lys Ser Pro Tyr Asp Pro Lys Leu Leu Thr Ala Ser Leu
145 150 155 160

Leu Ile Asp Gly Glu Leu Tyr Ser Gly Thr Ala Ala Asp Phe Met Gly
165 170 175

Arg Asp Phe Ala Ile Phe Arg Thr Leu Gly His His His Pro Ile Arg
180 185 190

Thr Glu Gln His Asp Ser Arg Trp Leu Asn Asp Pro Arg Phe Ile Ser
195 200 205

Ala His Leu Ile Pro Glu Ser Asp Asn Pro Glu Asp Asp Lys Val Tyr
210 215 220

Phe Phe Phe Arg Glu Asn Ala Ile Asp Gly Glu His Ser Gly Lys Ala
225 230 235 240

Thr His Ala Arg Ile Gly Gln Ile Cys Lys Asn Asp Phe Gly Gly His
245 250 255

Arg Ser Leu Val Asn Lys Trp Thr Thr Phe Leu Lys Ala Arg Leu Ile
260 265 270

Cys Ser Val Pro Gly Pro Asn Gly Ile Asp Thr His Phe Asp Glu Leu
275 280 285

Gln Asp Val Phe Leu Met Asn Ser Lys Asp Pro Lys Asn Pro Ile Val
290 295 300

Tyr Gly Val Phe Thr Thr Ser Ser Asn Ile Phe Lys Gly Ser Ala Val
305 310 315 320

Cys Met Tyr Ser Met Ser Asp Val Arg Arg Val Phe Leu Gly Pro Tyr
325 330 335

Ala	His	Arg	Asp	Gly	Pro	Asn	Tyr	Gln	Trp	Val	Pro	Tyr	Gln	Gly	Arg	
			340					345					350			
Val	Pro	Tyr	Pro	Arg	Pro	Gly	Thr	Cys	Pro	Ser	Lys	Thr	Phe	Gly	Gly	
			355					360					365			
Phe	Asp	Ser	Thr	Lys	Asp	Leu	Pro	Asp	Asp	Val	Ile	Thr	Phe	Ala	Arg	
			370					375					380			
Ser	His	Pro	Ala	Met	Tyr	Asn	Pro	Val	Phe	Pro	Ile	Asn	Asn	Arg	Pro	
385						390					395			400		
Ile	Met	Ile	Lys	Thr	Asp	Val	Asn	Tyr	Gln	Phe	Thr	Gln	Ile	Val	Val	
			405								410		415			
Asp	Arg	Val	Asp	Ala	Glu	Asp	Gly	Gln	Tyr	Asp	Val	Met	Phe	Ile	Gly	
			420					425					430			
Thr	Asp	Val	Gly	Thr	Val	Leu	Lys	Val	Val	Ser	Val	Pro	Lys	Glu	Thr	
			435					440					445			
Trp	His	Asp	Leu	Glu	Glu	Val	Leu	Leu	Glu	Glu	Met	Thr	Val	Phe	Arg	
450						455					460					
Glu	Pro	Thr	Thr	Ile	Ser	Ala	Met	Glu	Leu	Ser	Thr	Lys	Gln	Gln	Gln	
465						470					475			480		
Leu	Tyr	Ile	Gly	Ser	Thr	Ala	Gly	Val	Ala	Gln	Leu	Pro	Leu	His	Arg	
			485								490		495			
Cys	Asp	Ile	Tyr	Gly	Lys	Ala	Cys	Ala	Glu	Cys	Cys	Leu	Ala	Arg	Asp	
			500					505					510			
Pro	Tyr	Cys	Ala	Trp	Asp	Gly	Ser	Ser	Cys	Ser	Arg	Tyr	Phe	Pro	Thr	
			515					520					525			
Ala	Lys	Ala	Arg	Thr	Arg	Ala	Gln	Asp	Ile	Arg	Asn	Gly	Asp	Pro	Leu	
530						535					540					
Thr	His	Cys	Ser	Asp	Gly	Gly	Ile	Glu	Gly	Arg	Met	Asp	His	His	His	
545						550					555			560		
His	His	His														

<400> SEQUENCE: 585

Asn	Tyr	Gln	Asn	Gly	Lys	Asn	Asn	Val	Pro	Arg	Leu	Lys	Leu	Ser	Tyr
1				5					10					15	
Lys	Glu	Met	Leu	Glu	Ser	Asn	Ser	Val	Ile	Thr	Phe	Asn	Gly	Leu	Ala
			20					25					30		
Asn	Ser	Ser	Ser	Tyr	His	Thr	Phe	Leu	Leu	Asp	Glu	Glu	Arg	Ser	Arg
		35					40					45			
Leu	Tyr	Val	Gly	Ala	Lys	Asp	His	Ile	Phe	Ser	Phe	Asn	Leu	Val	Asn
	50					55					60				
Ile	Lys	Asp	Phe	Gln	Lys	Ile	Val	Trp	Pro	Val	Ser	Tyr	Thr	Arg	Arg
65					70					75					80
Asp	Glu	Cys	Lys	Trp	Ala	Gly	Lys	Asp	Ile	Gln	Lys	Glu	Cys	Ala	Asn
			85						90					95	
Phe	Ile	Lys	Val	Leu	Lys	Ala	Tyr	Asn	Gln	Thr	His	Leu	Tyr	Ala	Cys
			100					105					110		
Gly	Thr	Gly	Ala	Phe	His	Pro	Ile	Cys	Thr	Tyr	Ile	Glu	Ile	Gly	His
		115					120					125			
His	Pro	Glu	Asp	Asn	Ile	Phe	Lys	Leu	Glu	Asp	Ser	His	Phe	Glu	Asn
	130					135					140				

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Gly	Arg	Gly	Lys	Ser	Pro	Tyr	Asp	Pro	Lys	Leu	Leu	Thr	Ala	Ser	Leu	145	150	155	160
Leu	Ile	Asp	Gly	Glu	Leu	Tyr	Ser	Gly	Thr	Ala	Ala	Asp	Phe	Met	Gly	165	170	175	
Arg	Asp	Phe	Ala	Ile	Phe	Arg	Thr	Leu	Gly	His	His	His	Pro	Ile	Arg	180	185	190	
Thr	Glu	Gln	His	Asp	Ser	Arg	Trp	Leu	Asn	Asp	Pro	Arg	Phe	Ile	Ser	195	200	205	
Ala	His	Leu	Ile	Pro	Glu	Ser	Asp	Asn	Pro	Glu	Asp	Asp	Lys	Val	Tyr	210	215	220	
Phe	Phe	Phe	Arg	Glu	Asn	Ala	Ile	Asp	Gly	Glu	His	Thr	Gly	Lys	Ala	225	230	235	240
Thr	His	Ala	Arg	Ile	Gly	Gln	Ile	Cys	Lys	Asn	Asp	Phe	Gly	Gly	His	245	250	255	
Arg	Ser	Leu	Val	Asn	Lys	Trp	Thr	Thr	Phe	Leu	Lys	Ala	Arg	Leu	Ile	260	265	270	
Cys	Ser	Val	Pro	Gly	Pro	Asn	Gly	Ile	Asp	Thr	His	Phe	Asp	Glu	Leu	275	280	285	
Gln	Asp	Val	Phe	Leu	Met	Asn	Ser	Lys	Asp	Pro	Lys	Asn	Pro	Ile	Val	290	295	300	
Tyr	Gly	Val	Phe	Thr	Thr	Ser	Ser	Asn	Ile	Phe	Lys	Gly	Ser	Ala	Val	305	310	315	320
Cys	Met	Tyr	Ser	Met	Ser	Asp	Val	Arg	Arg	Val	Phe	Leu	Gly	Pro	Tyr	325	330	335	
Ala	His	Arg	Asp	Gly	Pro	Asn	Tyr	Gln	Trp	Val	Pro	Tyr	Gln	Gly	Arg	340	345	350	
Val	Pro	Tyr	Pro	Arg	Pro	Gly	Thr	Cys	Pro	Ser	Lys	Thr	Phe	Gly	Gly	355	360	365	
Phe	Asp	Ser	Thr	Lys	Asp	Leu	Pro	Asp	Asp	Val	Ile	Thr	Phe	Ala	Arg	370	375	380	
Ser	His	Pro	Ala	Met	Tyr	Asn	Pro	Val	Phe	Pro	Ile	Asn	Asn	Arg	Pro	385	390	395	400
Ile	Met	Ile	Lys	Thr	Asp	Val	Asn	Tyr	Gln	Phe	Thr	Gln	Ile	Val	Val	405	410	415	
Asp	Arg	Val	Asp	Ala	Glu	Asp	Gly	Gln	Tyr	Asp	Val	Met	Phe	Ile	Gly	420	425	430	
Thr	Asp	Val	Gly	Thr	Val	Leu	Lys	Val	Val	Ser	Ile	Pro	Lys	Glu	Thr	435	440	445	
Trp	His	Asp	Leu	Glu	Glu	Val	Leu	Leu	Glu	Glu	Met	Thr	Val	Phe	Arg	450	455	460	
Glu	Pro	Thr	Pro	Ile	Ser	Ala	Met	Glu	Leu	Ser	Thr	Lys	Gln	His	Gln	465	470	475	480
Leu	Tyr	Ala	Gly	Ser	Pro	Ala	Gly	Leu	Ala	Gln	Leu	Pro	Leu	Gln	Arg	485	490	495	
Cys	Ala	Ala	Tyr	Gly	Arg	Ala	Cys	Ala	Glu	Cys	Cys	Leu	Ala	Arg	Asp	500	505	510	
Pro	Tyr	Cys	Ala	Trp	Asp	Gly	Ala	Ala	Cys	Ser	Arg	Tyr	Phe	Pro	Ala	515	520	525	
Ala	Lys	Ala	Arg	Thr	Arg	Ala	Gln	Asp	Ile	Arg	Asn	Gly	Asp	Pro	Leu	530	535	540	
Thr	His	Cys	Ser	Asp	Gly	Gly	Ile	Glu	Gly	Arg	Met	Asp	His	His	His	545	550	555	560
His	His	His																	

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<210> SEQ ID NO 586
<211> LENGTH: 563
<212> TYPE: PRT
<213> ORGANISM: Macaca fascicularis (Cynomolgus monkey)

<400> SEQUENCE: 586

Asn Tyr Gln Asn Gly Lys Asn Asn Val Pro Arg Leu Lys Leu Ser Tyr
1          5          10          15

Lys Glu Met Leu Glu Ser Asn Asn Val Ile Thr Phe Asn Gly Leu Ala
20          25          30

Asn Ser Ser Ser Tyr His Thr Phe Leu Leu Asp Glu Glu Arg Ser Arg
35          40          45

Leu Tyr Val Gly Ala Lys Asp His Ile Phe Ser Phe Asn Leu Val Asn
50          55          60

Ile Lys Asp Phe Gln Lys Ile Val Trp Pro Val Ser Tyr Thr Arg Arg
65          70          75          80

Asp Glu Cys Lys Trp Ala Gly Lys Asp Ile Leu Lys Glu Cys Ala Asn
85          90          95

Phe Ile Lys Val Leu Lys Ala Tyr Asn Gln Thr His Leu Tyr Ala Cys
100         105         110

Gly Thr Gly Ala Phe His Pro Ile Cys Thr Tyr Ile Glu Ile Gly His
115         120         125

His Pro Glu Asp Asn Ile Phe Lys Leu Glu Asn Ser His Phe Glu Asn
130         135         140

Gly Arg Gly Lys Ser Pro Tyr Asp Pro Lys Leu Leu Thr Ala Ser Leu
145         150         155         160

Leu Ile Asp Gly Glu Leu Tyr Ser Gly Thr Ala Ala Asp Phe Met Gly
165         170         175

Arg Asp Phe Ala Ile Phe Arg Thr Leu Gly His His His Pro Ile Arg
180         185         190

Thr Glu Gln His Asp Ser Arg Trp Leu Asn Asp Pro Arg Phe Ile Ser
195         200         205

Ala His Leu Ile Pro Glu Ser Asp Asn Pro Glu Asp Asp Lys Val Tyr
210         215         220

Phe Phe Phe Arg Glu Asn Ala Ile Asp Gly Glu His Ser Gly Lys Ala
225         230         235         240

Thr His Ala Arg Ile Gly Gln Ile Cys Lys Asn Asp Phe Gly Gly His
245         250         255

Arg Ser Leu Val Asn Lys Trp Thr Thr Phe Leu Lys Ala Arg Leu Ile
260         265         270

Cys Ser Val Pro Gly Pro Asn Gly Ile Asp Thr His Phe Asp Glu Leu
275         280         285

Gln Asp Val Phe Leu Met Asn Phe Lys Asp Pro Lys Asn Pro Ile Val
290         295         300

Tyr Gly Val Phe Thr Thr Ser Ser Asn Ile Phe Lys Gly Ser Ala Val
305         310         315         320

Cys Met Tyr Ser Met Ser Asp Val Arg Arg Val Phe Leu Gly Pro Tyr
325         330         335

Ala His Arg Asp Gly Pro Asn Tyr Gln Trp Val Pro Tyr Gln Gly Arg
340         345         350

Val Pro Tyr Pro Arg Pro Gly Thr Cys Pro Ser Lys Thr Phe Gly Gly
355         360         365

Phe Asp Ser Thr Lys Asp Leu Pro Asp Asp Val Ile Thr Phe Ala Arg

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370					375					380					
Ser	His	Pro	Ala	Met	Tyr	Asn	Pro	Val	Phe	Pro	Ile	Asn	Asn	Arg	Pro
385					390					395					400
Ile	Met	Ile	Lys	Thr	Asp	Val	Asn	Tyr	Gln	Phe	Thr	Gln	Ile	Val	Val
				405					410					415	
Asp	Arg	Val	Asp	Ala	Glu	Asp	Gly	Gln	Tyr	Asp	Val	Met	Phe	Ile	Gly
			420					425					430		
Thr	Asp	Val	Gly	Thr	Val	Leu	Lys	Val	Val	Ser	Ile	Pro	Lys	Glu	Thr
			435				440					445			
Trp	His	Asp	Leu	Glu	Glu	Val	Leu	Leu	Glu	Glu	Met	Thr	Val	Phe	Arg
			450				455					460			
Glu	Pro	Thr	Thr	Ile	Ser	Ala	Met	Glu	Leu	Ser	Thr	Lys	Gln	Gln	Gln
			465				470					475			480
Leu	Tyr	Ile	Gly	Ser	Thr	Ala	Gly	Ile	Ala	Gln	Leu	Pro	Leu	His	Arg
			485					490						495	
Cys	Asp	Ile	Tyr	Gly	Lys	Ala	Cys	Ala	Glu	Cys	Cys	Leu	Ala	Arg	Asp
			500					505					510		
Pro	Tyr	Cys	Ala	Trp	Asp	Gly	Ser	Ser	Cys	Ser	Arg	Tyr	Phe	Pro	Thr
			515				520					525			
Ala	Lys	Ala	Arg	Thr	Arg	Ala	Gln	Asp	Ile	Arg	Asn	Gly	Asp	Pro	Leu
			530				535					540			
Thr	His	Cys	Ser	Asp	Gly	Gly	Ile	Glu	Gly	Arg	Met	Asp	His	His	His
			545				550					555			560
His	His	His													
<210> SEQ ID NO 587															
<211> LENGTH: 563															
<212> TYPE: PRT															
<213> ORGANISM: Sus scrofa (Pig)															
<400> SEQUENCE: 587															
Asn	Tyr	Gln	Asn	Gly	Lys	Asn	Asn	Val	Pro	Arg	Leu	Lys	Leu	Ser	Tyr
1				5					10					15	
Lys	Glu	Met	Leu	Glu	Ser	Asn	Asn	Val	Ile	Thr	Phe	Asn	Gly	Leu	Ala
			20					25					30		
Asn	Ser	Ser	Ser	Tyr	His	Thr	Phe	Leu	Leu	Asp	Glu	Glu	Arg	Ser	Arg
			35				40					45			
Leu	Tyr	Val	Gly	Ala	Lys	Asp	His	Ile	Phe	Ser	Phe	Asn	Leu	Val	Asn
			50				55					60			
Ile	Lys	Asp	Phe	Gln	Lys	Ile	Val	Trp	Pro	Val	Ser	Tyr	Thr	Arg	Arg
			65				70					75			80
Asp	Glu	Cys	Lys	Trp	Ala	Gly	Lys	Asp	Ile	Leu	Lys	Glu	Cys	Ala	Asn
			85					90						95	
Phe	Ile	Lys	Val	Leu	Lys	Ala	Tyr	Asn	Gln	Thr	His	Leu	Tyr	Ala	Cys
			100					105					110		
Gly	Thr	Gly	Ala	Phe	His	Pro	Ile	Cys	Thr	Tyr	Ile	Glu	Ile	Gly	His
			115					120					125		
His	Pro	Glu	Asp	Asn	Ile	Phe	Lys	Leu	Glu	Asp	Ser	His	Phe	Glu	Asn
			130				135					140			
Gly	Arg	Gly	Lys	Ser	Pro	Tyr	Asp	Pro	Lys	Leu	Leu	Thr	Ala	Ser	Leu
			145				150					155			160
Leu	Ile	Asp	Gly	Glu	Leu	Tyr	Ser	Gly	Thr	Ala	Ala	Asp	Phe	Met	Gly
			165					170						175	
Arg	Asp	Phe	Ala	Ile	Phe	Arg	Thr	Leu	Gly	His	His	His	Pro	Ile	Arg

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180					185					190					
Thr	Glu	Gln	His	Asp	Ser	Arg	Trp	Leu	Asn	Asp	Pro	Arg	Phe	Ile	Ser
	195						200					205			
Ala	His	Leu	Ile	Pro	Glu	Ser	Asp	Asn	Pro	Glu	Asp	Asp	Lys	Val	Tyr
	210					215					220				
Phe	Phe	Phe	Arg	Glu	Asn	Ala	Ile	Asp	Gly	Glu	His	Thr	Gly	Lys	Ala
	225					230					235				240
Thr	His	Ala	Arg	Ile	Gly	Gln	Ile	Cys	Lys	Asn	Asp	Phe	Gly	Gly	His
			245						250					255	
Arg	Ser	Leu	Val	Asn	Lys	Trp	Thr	Thr	Phe	Leu	Lys	Ala	Arg	Leu	Ile
			260						265				270		
Cys	Ser	Val	Pro	Gly	Pro	Asn	Gly	Ile	Asp	Thr	His	Phe	Asp	Glu	Leu
		275					280					285			
Gln	Asp	Val	Phe	Leu	Met	Asn	Ser	Lys	Asp	Pro	Lys	Asn	Pro	Val	Val
	290					295					300				
Tyr	Gly	Val	Phe	Thr	Thr	Ser	Ser	Asn	Ile	Phe	Lys	Gly	Ser	Ala	Val
	305					310					315				320
Cys	Met	Tyr	Ser	Met	Ser	Asp	Val	Arg	Arg	Val	Phe	Leu	Gly	Pro	Tyr
			325					330						335	
Ala	His	Arg	Asp	Gly	Pro	Asn	Tyr	Gln	Trp	Val	Pro	Tyr	Gln	Gly	Arg
			340					345					350		
Val	Pro	Tyr	Pro	Arg	Pro	Gly	Thr	Cys	Pro	Ser	Lys	Thr	Phe	Gly	Gly
		355					360					365			
Phe	Asp	Ser	Thr	Lys	Asp	Leu	Pro	Asp	Asp	Val	Ile	Thr	Phe	Ala	Arg
	370					375					380				
Ser	His	Pro	Ala	Met	Tyr	Asn	Pro	Val	Phe	Pro	Ile	Asn	Asn	Arg	Pro
	385					390					395				400
Ile	Met	Ile	Lys	Thr	Asp	Val	Asn	Tyr	Gln	Phe	Thr	Gln	Ile	Val	Val
			405						410					415	
Asp	Arg	Val	Asp	Ala	Glu	Asp	Gly	Gln	Tyr	Asp	Val	Met	Phe	Ile	Gly
			420					425					430		
Thr	Asp	Val	Gly	Thr	Val	Leu	Lys	Val	Val	Ser	Ile	Pro	Lys	Glu	Thr
	435						440					445			
Trp	His	Asp	Leu	Glu	Glu	Val	Leu	Leu	Glu	Glu	Met	Thr	Val	Phe	Arg
	450					455					460				
Glu	Pro	Thr	Thr	Ile	Ser	Ala	Met	Glu	Leu	Ser	Thr	Lys	Gln	Gln	Gln
	465					470					475				480
Leu	Tyr	Val	Gly	Ser	Ala	Ala	Gly	Val	Ala	Gln	Leu	Pro	Leu	His	Arg
			485					490						495	
Cys	Asp	Ile	Tyr	Gly	Lys	Ala	Cys	Ala	Glu	Cys	Cys	Leu	Ala	Arg	Asp
		500					505						510		
Pro	Tyr	Cys	Ala	Trp	Asp	Gly	Ser	Ser	Cys	Ser	Arg	Tyr	Phe	Pro	Thr
		515					520					525			
Ala	Lys	Ala	Arg	Thr	Arg	Ala	Gln	Asp	Ile	Arg	Asn	Gly	Asp	Pro	Leu
	530					535					540				
Thr	His	Cys	Ser	Asp	Gly	Gly	Ile	Glu	Gly	Arg	Met	Asp	His	His	His
	545					550					555				560
His	His	His													

<210> SEQ ID NO 588

<211> LENGTH: 5

<212> TYPE: PRT

<213> ORGANISM: artificial

<220> FEATURE:

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<223> OTHER INFORMATION: antibody sequence
<220> FEATURE:
<221> NAME/KEY: site
<222> LOCATION: (3)..(3)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: site
<222> LOCATION: (5)..(5)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

<400> SEQUENCE: 588

Ser Tyr Xaa Met Xaa
1 5

<210> SEQ ID NO 589
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: artificial
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence
<220> FEATURE:
<221> NAME/KEY: site
<222> LOCATION: (5)..(5)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

<400> SEQUENCE: 589

Ser Tyr Ala Met Xaa
1 5

<210> SEQ ID NO 590
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: artificial
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence
<220> FEATURE:
<221> NAME/KEY: site
<222> LOCATION: (4)..(4)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: site
<222> LOCATION: (15)..(15)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

<400> SEQUENCE: 590

Ala Ile Gly Xaa Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Xaa Gly
1 5 10 15

<210> SEQ ID NO 591
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: artificial
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence
<220> FEATURE:
<221> NAME/KEY: site
<222> LOCATION: (4)..(4)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

<400> SEQUENCE: 591

Ala Ile Gly Xaa Gly Gly Asp Thr Tyr Tyr Ala Asp Ser Val Lys Gly
1 5 10 15

<210> SEQ ID NO 592
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: artificial
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence
<220> FEATURE:
<221> NAME/KEY: site

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<222> LOCATION: (8)..(8)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: site
<222> LOCATION: (10)..(10)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

```

```

<400> SEQUENCE: 592

```

```

Gly Ile Ser Trp Asn Ser Gly Xaa Ile Xaa Tyr Ala Asp Ser Val Lys
1           5           10           15

```

```

Gly

```

```

<210> SEQ ID NO 593
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: artificial
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence
<220> FEATURE:
<221> NAME/KEY: site
<222> LOCATION: (10)..(10)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

```

```

<400> SEQUENCE: 593

```

```

Gly Ile Ser Trp Asn Ser Gly Trp Ile Xaa Tyr Ala Asp Ser Val Lys
1           5           10           15

```

```

Gly

```

```

<210> SEQ ID NO 594
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: artificial
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence
<220> FEATURE:
<221> NAME/KEY: site
<222> LOCATION: (12)..(12)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

```

```

<400> SEQUENCE: 594

```

```

Arg Asp Asp Tyr Thr Ser Arg Asp Ala Phe Asp Xaa
1           5           10

```

```

<210> SEQ ID NO 595
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: artificial
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence
<220> FEATURE:
<221> NAME/KEY: site
<222> LOCATION: (6)..(6)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

```

```

<400> SEQUENCE: 595

```

```

Thr Gly Ser Ser Ser Xaa Ile Gly Ala Gly Tyr Asp Val His
1           5           10

```

```

<210> SEQ ID NO 596
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence
<220> FEATURE:
<221> NAME/KEY: site
<222> LOCATION: (5)..(5)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

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<400> SEQUENCE: 596

Tyr Asp Asp Leu Xaa Pro Ser
1 5

<210> SEQ ID NO 597

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: artificial

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<220> FEATURE:

<221> NAME/KEY: site

<222> LOCATION: (2)..(2)

<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

<400> SEQUENCE: 597

Gly Xaa Ser Asn Arg Pro Ser
1 5

<210> SEQ ID NO 598

<211> LENGTH: 12

<212> TYPE: PRT

<213> ORGANISM: artificial

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<220> FEATURE:

<221> NAME/KEY: site

<222> LOCATION: (1)..(1)

<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

<220> FEATURE:

<221> NAME/KEY: site

<222> LOCATION: (9)..(11)

<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

<400> SEQUENCE: 598

Xaa Ala Trp Asp Asp Ser Leu Asn Xaa Xaa Xaa Val
1 5 10

<210> SEQ ID NO 599

<211> LENGTH: 11

<212> TYPE: PRT

<213> ORGANISM: artificial

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<220> FEATURE:

<221> NAME/KEY: site

<222> LOCATION: (1)..(1)

<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

<220> FEATURE:

<221> NAME/KEY: site

<222> LOCATION: (4)..(4)

<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

<220> FEATURE:

<221> NAME/KEY: site

<222> LOCATION: (6)..(6)

<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

<400> SEQUENCE: 599

Xaa Ser Tyr Xaa Gly Xaa Asn Pro Tyr Val Val
1 5 10

<210> SEQ ID NO 600

<211> LENGTH: 750

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 600

Tyr Gln Asn Gly Lys Asn Asn Val Pro Arg Leu Lys Leu Ser Tyr Lys
1 5 10 15

Glu	Met	Leu	Glu	Ser	Asn	Asn	Val	Ile	Thr	Phe	Asn	Gly	Leu	Ala	Asn
			20				25						30		
Ser	Ser	Ser	Tyr	His	Thr	Phe	Leu	Leu	Asp	Glu	Glu	Arg	Ser	Arg	Leu
			35				40				45				
Tyr	Val	Gly	Ala	Lys	Asp	His	Ile	Phe	Ser	Phe	Asp	Leu	Val	Asn	Ile
			50				55				60				
Lys	Asp	Phe	Gln	Lys	Ile	Val	Trp	Pro	Val	Ser	Tyr	Thr	Arg	Arg	Asp
65				70			75						80		
Glu	Cys	Lys	Trp	Ala	Gly	Lys	Asp	Ile	Leu	Lys	Glu	Cys	Ala	Asn	Phe
			85			90						95			
Ile	Lys	Val	Leu	Lys	Ala	Tyr	Asn	Gln	Thr	His	Leu	Tyr	Ala	Cys	Gly
			100			105						110			
Thr	Gly	Ala	Phe	His	Pro	Ile	Cys	Thr	Tyr	Ile	Glu	Ile	Gly	His	His
			115			120						125			
Pro	Glu	Asp	Asn	Ile	Phe	Lys	Leu	Glu	Asn	Ser	His	Phe	Glu	Asn	Gly
			130			135						140			
Arg	Gly	Lys	Ser	Pro	Tyr	Asp	Pro	Lys	Leu	Leu	Thr	Ala	Ser	Leu	Leu
145				150			155						160		
Ile	Asp	Gly	Glu	Leu	Tyr	Ser	Gly	Thr	Ala	Ala	Asp	Phe	Met	Gly	Arg
			165			170						175			
Asp	Phe	Ala	Ile	Phe	Arg	Thr	Leu	Gly	His	His	His	Pro	Ile	Arg	Thr
			180			185						190			
Glu	Gln	His	Asp	Ser	Arg	Trp	Leu	Asn	Asp	Pro	Lys	Phe	Ile	Ser	Ala
			195			200						205			
His	Leu	Ile	Ser	Glu	Ser	Asp	Asn	Pro	Glu	Asp	Asp	Lys	Val	Tyr	Phe
			210			215						220			
Phe	Phe	Arg	Glu	Asn	Ala	Ile	Asp	Gly	Glu	His	Ser	Gly	Lys	Ala	Thr
225				230			235						240		
His	Ala	Arg	Ile	Gly	Gln	Ile	Cys	Lys	Asn	Asp	Phe	Gly	Gly	His	Arg
			245			250						255			
Ser	Leu	Val	Asn	Lys	Trp	Thr	Thr	Phe	Leu	Lys	Ala	Arg	Leu	Ile	Cys
			260			265						270			
Ser	Val	Pro	Gly	Pro	Asn	Gly	Ile	Asp	Thr	His	Phe	Asp	Glu	Leu	Gln
			275			280						285			
Asp	Val	Phe	Leu	Met	Asn	Phe	Lys	Asp	Pro	Lys	Asn	Pro	Val	Val	Tyr
			290			295						300			
Gly	Val	Phe	Thr	Thr	Ser	Ser	Asn	Ile	Phe	Lys	Gly	Ser	Ala	Val	Cys
305				310			315						320		
Met	Tyr	Ser	Met	Ser	Asp	Val	Arg	Arg	Val	Phe	Leu	Gly	Pro	Tyr	Ala
			325			330						335			
His	Arg	Asp	Gly	Pro	Asn	Tyr	Gln	Trp	Val	Pro	Tyr	Gln	Gly	Arg	Val
			340			345						350			
Pro	Tyr	Pro	Arg	Pro	Gly	Thr	Cys	Pro	Ser	Lys	Thr	Phe	Gly	Gly	Phe
			355			360						365			
Asp	Ser	Thr	Lys	Asp	Leu	Pro	Asp	Asp	Val	Ile	Thr	Phe	Ala	Arg	Ser
			370			375						380			
His	Pro	Ala	Met	Tyr	Asn	Pro	Val	Phe	Pro	Met	Asn	Asn	Arg	Pro	Ile
385				390			395						400		
Val	Ile	Lys	Thr	Asp	Val	Asn	Tyr	Gln	Phe	Thr	Gln	Ile	Val	Val	Asp
			405			410						415			
Arg	Val	Asp	Ala	Glu	Asp	Gly	Gln	Tyr	Asp	Val	Met	Phe	Ile	Gly	Thr
			420			425						430			
Asp	Val	Gly													

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435					440					445					
Tyr	Asp	Leu	Glu	Glu	Val	Leu	Leu	Glu	Glu	Met	Thr	Val	Phe	Arg	Glu
450						455					460				
Pro	Thr	Ala	Ile	Ser	Ala	Met	Glu	Leu	Ser	Thr	Lys	Gln	Gln	Gln	Leu
465					470					475					480
Tyr	Ile	Gly	Ser	Thr	Ala	Gly	Val	Ala	Gln	Leu	Pro	Leu	His	Arg	Cys
				485						490				495	
Asp	Ile	Tyr	Gly	Lys	Ala	Cys	Ala	Glu	Cys	Cys	Leu	Ala	Arg	Asp	Pro
			500					505					510		
Tyr	Cys	Ala	Trp	Asp	Gly	Ser	Ala	Cys	Ser	Arg	Tyr	Phe	Pro	Thr	Ala
		515					520					525			
Lys	Arg	Arg	Thr	Arg	Arg	Gln	Asp	Ile	Arg	Asn	Gly	Asp	Pro	Leu	Thr
	530					535					540				
His	Cys	Ser	Asp	Leu	His	His	Asp	Asn	His	His	Gly	His	Ser	Pro	Glu
545					550					555					560
Glu	Arg	Ile	Ile	Tyr	Gly	Val	Glu	Asn	Ser	Ser	Thr	Phe	Leu	Glu	Cys
				565					570					575	
Ser	Pro	Lys	Ser	Gln	Arg	Ala	Leu	Val	Tyr	Trp	Gln	Phe	Gln	Arg	Arg
			580					585					590		
Asn	Glu	Glu	Arg	Lys	Glu	Glu	Ile	Arg	Val	Asp	Asp	His	Ile	Ile	Arg
	595						600					605			
Thr	Asp	Gln	Gly	Leu	Leu	Leu	Arg	Ser	Leu	Gln	Gln	Lys	Asp	Ser	Gly
	610					615				620					
Asn	Tyr	Leu	Cys	His	Ala	Val	Glu	His	Gly	Phe	Ile	Gln	Thr	Leu	Leu
625					630					635					640
Lys	Val	Thr	Leu	Glu	Val	Ile	Asp	Thr	Glu	His	Leu	Glu	Glu	Leu	Leu
			645					650					655		
His	Lys	Asp	Asp	Asp	Gly	Asp	Gly	Ser	Lys	Thr	Lys	Glu	Met	Ser	Asn
		660						665					670		
Ser	Met	Thr	Pro	Ser	Gln	Lys	Val	Trp	Tyr	Arg	Asp	Phe	Met	Gln	Leu
	675						680					685			
Ile	Asn	His	Pro	Asn	Leu	Asn	Thr	Met	Asp	Glu	Phe	Cys	Glu	Gln	Val
	690					695					700				
Trp	Lys	Arg	Asp	Arg	Lys	Gln	Arg	Arg	Gln	Arg	Pro	Gly	His	Thr	Pro
705					710					715					720
Gly	Asn	Ser	Asn	Lys	Trp	Lys	His	Leu	Gln	Glu	Asn	Lys	Lys	Gly	Arg
			725					730					735		
Asn	Arg	Arg	Thr	His	Glu	Phe	Glu	Arg	Ala	Pro	Arg	Ser	Val		
			740					745				750			

<210> SEQ ID NO 601

<211> LENGTH: 752

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 601

Asn	Tyr	Ala	Asn	Gly	Lys	Asn	Asn	Val	Pro	Arg	Leu	Lys	Leu	Ser	Tyr
1				5					10					15	
Lys	Glu	Met	Leu	Glu	Ser	Asn	Asn	Val	Ile	Thr	Phe	Asn	Gly	Leu	Ala
		20						25					30		
Asn	Ser	Ser	Ser	Tyr	His	Thr	Phe	Leu	Leu	Asp	Glu	Glu	Arg	Ser	Arg
	35					40					45				
Leu	Tyr	Val	Gly	Ala	Lys	Asp	His	Ile	Phe	Ser	Phe	Asn	Leu	Val	Asn
	50					55					60				

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Ile	Lys	Asp	Phe	Gln	Lys	Ile	Val	Trp	Pro	Val	Ser	Tyr	Thr	Arg	Arg	65	70	75	80
Asp	Glu	Cys	Lys	Trp	Ala	Gly	Lys	Asp	Ile	Leu	Lys	Glu	Cys	Ala	Asn	85	90	95	
Phe	Ile	Lys	Val	Leu	Glu	Ala	Tyr	Asn	Gln	Thr	His	Leu	Tyr	Ala	Cys	100	105	110	
Gly	Thr	Gly	Ala	Phe	His	Pro	Ile	Cys	Thr	Tyr	Ile	Glu	Val	Gly	His	115	120	125	
His	Pro	Glu	Asp	Asn	Ile	Phe	Lys	Leu	Gln	Asp	Ser	His	Phe	Glu	Asn	130	135	140	
Gly	Arg	Gly	Lys	Ser	Pro	Tyr	Asp	Pro	Lys	Leu	Leu	Thr	Ala	Ser	Leu	145	150	155	160
Leu	Ile	Asp	Gly	Glu	Leu	Tyr	Ser	Gly	Thr	Ala	Ala	Asp	Phe	Met	Gly	165	170	175	
Arg	Asp	Phe	Ala	Ile	Phe	Arg	Thr	Leu	Gly	His	His	His	Pro	Ile	Arg	180	185	190	
Thr	Glu	Gln	His	Asp	Ser	Arg	Trp	Leu	Asn	Asp	Pro	Arg	Phe	Ile	Ser	195	200	205	
Ala	His	Leu	Ile	Pro	Glu	Ser	Asp	Asn	Pro	Glu	Asp	Asp	Lys	Val	Tyr	210	215	220	
Phe	Phe	Phe	Arg	Glu	Asn	Ala	Ile	Asp	Gly	Glu	His	Ser	Gly	Lys	Ala	225	230	235	240
Thr	His	Ala	Arg	Ile	Gly	Gln	Ile	Cys	Lys	Asn	Asp	Phe	Gly	Gly	His	245	250	255	
Arg	Ser	Leu	Val	Asn	Lys	Trp	Thr	Thr	Phe	Leu	Lys	Ala	Arg	Leu	Ile	260	265	270	
Cys	Ser	Val	Pro	Gly	Pro	Asn	Gly	Ile	Asp	Thr	His	Phe	Asp	Glu	Leu	275	280	285	
Gln	Asp	Val	Phe	Leu	Met	Asn	Ser	Lys	Asp	Pro	Lys	Asn	Pro	Ile	Val	290	295	300	
Tyr	Gly	Val	Phe	Thr	Thr	Ser	Ser	Asn	Ile	Phe	Lys	Gly	Ser	Ala	Val	305	310	315	320
Cys	Met	Tyr	Ser	Met	Ser	Asp	Val	Arg	Arg	Val	Phe	Leu	Gly	Pro	Tyr	325	330	335	
Ala	His	Arg	Asp	Gly	Pro	Asn	Tyr	Gln	Trp	Val	Pro	Tyr	Gln	Gly	Arg	340	345	350	
Val	Pro	Tyr	Pro	Arg	Pro	Gly	Thr	Cys	Pro	Ser	Lys	Thr	Phe	Gly	Gly	355	360	365	
Phe	Asp	Ser	Thr	Lys	Asp	Leu	Pro	Asp	Asp	Val	Ile	Thr	Phe	Ala	Arg	370	375	380	
Ser	His	Pro	Ala	Met	Tyr	Asn	Pro	Val	Phe	Pro	Ile	Asn	Asn	Arg	Pro	385	390	395	400
Ile	Met	Ile	Lys	Thr	Asp	Val	Asn	Tyr	Gln	Phe	Thr	Gln	Ile	Val	Val	405	410	415	
Asp	Arg	Val	Asp	Ala	Glu	Asp	Gly	Gln	Tyr	Asp	Val	Met	Phe	Ile	Gly	420	425	430	
Thr	Asp	Val	Gly	Thr	Val	Leu	Lys	Val	Val	Ser	Val	Pro	Lys	Glu	Thr	435	440	445	
Trp	His	Asp	Leu	Glu	Glu	Val	Leu	Leu	Glu	Glu	Met	Thr	Val	Phe	Arg	450	455	460	
Glu	Pro	Thr	Thr	Ile	Ser	Ala	Met	Glu	Leu	Ser	Thr	Lys	Gln	Gln	Gln	465	470	475	480
Leu	Tyr	Ile	Gly	Ser	Thr	Ala	Gly	Val	Ala	Gln	Leu	Pro	Leu	His	Arg				

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485					490					495					
Cys	Asp	Ile	Tyr	Gly	Lys	Ala	Cys	Ala	Glu	Cys	Cys	Leu	Ala	Arg	Asp
			500					505					510		
Pro	Tyr	Cys	Ala	Trp	Asp	Gly	Ser	Ser	Cys	Ser	Arg	Tyr	Phe	Pro	Thr
		515					520					525			
Ala	Lys	Arg	Arg	Thr	Arg	Arg	Gln	Asp	Ile	Arg	Asn	Gly	Asp	Pro	Leu
		530					535					540			
Thr	His	Cys	Ser	Asp	Leu	Gln	His	His	Asp	Asn	His	His	Gly	Pro	Ser
		545					550					555			560
Leu	Glu	Glu	Arg	Ile	Ile	Tyr	Gly	Val	Glu	Asn	Ser	Ser	Thr	Phe	Leu
			565						570						575
Glu	Cys	Ser	Pro	Lys	Ser	Gln	Arg	Ala	Leu	Val	Tyr	Trp	Gln	Phe	Gln
			580					585					590		
Arg	Arg	Asn	Glu	Asp	Arg	Lys	Glu	Glu	Ile	Arg	Met	Gly	Asp	His	Ile
		595					600					605			
Ile	Arg	Thr	Glu	Gln	Gly	Leu	Leu	Leu	Arg	Ser	Leu	Gln	Lys	Lys	Asp
		610					615					620			
Ser	Gly	Asn	Tyr	Leu	Cys	His	Ala	Val	Glu	His	Gly	Phe	Met	Gln	Thr
		625					630					635			640
Leu	Leu	Lys	Val	Thr	Leu	Glu	Val	Ile	Asp	Thr	Glu	His	Leu	Glu	Glu
			645						650						655
Leu	Leu	His	Lys	Asp	Asp	Asp	Gly	Asp	Gly	Ser	Lys	Ile	Lys	Glu	Met
			660					665					670		
Ser	Ser	Ser	Met	Thr	Pro	Ser	Gln	Lys	Val	Trp	Tyr	Arg	Asp	Phe	Met
		675					680					685			
Gln	Leu	Ile	Asn	His	Pro	Asn	Leu	Asn	Thr	Met	Asp	Glu	Phe	Cys	Glu
		690					695					700			
Gln	Val	Trp	Lys	Arg	Asp	Arg	Lys	Gln	Arg	Arg	Gln	Arg	Pro	Gly	His
		705					710					715			720
Ser	Gln	Gly	Ser	Ser	Asn	Lys	Trp	Lys	His	Met	Gln	Glu	Ser	Lys	Lys
			725						730						735
Gly	Arg	Asn	Arg	Arg	Thr	His	Glu	Phe	Glu	Arg	Ala	Pro	Arg	Ser	Val
			740					745					750		
<210> SEQ ID NO 602															
<211> LENGTH: 772															
<212> TYPE: PRT															
<213> ORGANISM: Macaca fascicularis															
<400> SEQUENCE: 602															
Met	Gly	Trp	Leu	Thr	Arg	Ile	Val	Cys	Leu	Phe	Trp	Gly	Val	Leu	Leu
1				5					10					15	
Thr	Ala	Arg	Ala	Asn	Tyr	Gln	Asn	Gly	Lys	Asn	Asn	Val	Pro	Arg	Leu
			20					25					30		
Lys	Leu	Ser	Tyr	Lys	Glu	Met	Leu	Glu	Ser	Asn	Asn	Val	Ile	Thr	Phe
			35				40					45			
Asn	Gly	Leu	Ala	Asn	Ser	Ser	Ser	Tyr	His	Thr	Phe	Leu	Leu	Asp	Glu
		50					55					60			
Glu	Arg	Ser	Arg	Leu	Tyr	Val	Gly	Ala	Lys	Asp	His	Ile	Phe	Ser	Phe
		65					70					75			80
Asn	Leu	Val	Asn	Ile	Lys	Asp	Phe	Gln	Lys	Ile	Val	Trp	Pro	Val	Ser
			85					90						95	
Tyr	Thr	Arg	Arg	Asp	Glu	Cys	Lys	Trp	Ala	Gly	Lys	Asp	Ile	Leu	Lys
			100					105						110	

Glu	Cys	Ala	Asn	Phe	Ile	Lys	Val	Leu	Lys	Ala	Tyr	Asn	Gln	Thr	His
115						120						125			
Leu	Tyr	Ala	Cys	Gly	Thr	Gly	Ala	Phe	His	Pro	Ile	Cys	Thr	Tyr	Ile
130						135				140					
Glu	Ile	Gly	His	His	Pro	Glu	Asp	Asn	Ile	Phe	Lys	Leu	Glu	Asn	Ser
145				150						155				160	
His	Phe	Glu	Asn	Gly	Arg	Gly	Lys	Ser	Pro	Tyr	Asp	Pro	Lys	Leu	Leu
				165				170						175	
Thr	Ala	Ser	Leu	Leu	Ile	Asp	Gly	Glu	Leu	Tyr	Ser	Gly	Thr	Ala	Ala
		180						185				190			
Asp	Phe	Met	Gly	Arg	Asp	Phe	Ala	Ile	Phe	Arg	Thr	Leu	Gly	His	His
195						200						205			
His	Pro	Ile	Arg	Thr	Glu	Gln	His	Asp	Ser	Arg	Trp	Leu	Asn	Asp	Pro
210						215				220					
Arg	Phe	Ile	Ser	Ala	His	Leu	Ile	Pro	Glu	Ser	Asp	Asn	Pro	Glu	Asp
225				230						235				240	
Asp	Lys	Val	Tyr	Phe	Phe	Phe	Arg	Glu	Asn	Ala	Ile	Asp	Gly	Glu	His
				245				250						255	
Ser	Gly	Lys	Ala	Thr	His	Ala	Arg	Ile	Gly	Gln	Ile	Cys	Lys	Asn	Asp
		260				265						270			
Phe	Gly	Gly	His	Arg	Ser	Leu	Val	Asn	Lys	Trp	Thr	Thr	Phe	Leu	Lys
275						280						285			
Ala	Arg	Leu	Ile	Cys	Ser	Val	Pro	Gly	Pro	Asn	Gly	Ile	Asp	Thr	His
290						295				300					
Phe	Asp	Glu	Leu	Gln	Asp	Val	Phe	Leu	Met	Asn	Phe	Lys	Asp	Pro	Lys
305				310						315				320	
Asn	Pro	Ile	Val	Tyr	Gly	Val	Phe	Thr	Thr	Ser	Ser	Asn	Ile	Phe	Lys
				325				330						335	
Gly	Ser	Ala	Val	Cys	Met	Tyr	Ser	Met	Ser	Asp	Val	Arg	Arg	Val	Phe
		340						345				350			
Leu	Gly	Pro	Tyr	Ala	His	Arg	Asp	Gly	Pro	Asn	Tyr	Gln	Trp	Val	Pro
355						360						365			
Tyr	Gln	Gly	Arg	Val	Pro	Tyr	Pro	Arg	Pro	Gly	Thr	Cys	Pro	Ser	Lys
370						375				380					
Thr	Phe	Gly	Gly	Phe	Asp	Ser	Thr	Lys	Asp	Leu	Pro	Asp	Asp	Val	Ile
385				390						395				400	
Thr	Phe	Ala	Arg	Ser	His	Pro	Ala	Met	Tyr	Asn	Pro	Val	Phe	Pro	Ile
				405				410						415	
Asn	Asn	Arg	Pro	Ile	Met	Ile	Lys	Thr	Asp	Val	Asn	Tyr	Gln	Phe	Thr
		420						425				430			
Gln	Ile	Val	Val	Asp	Arg	Val	Asp	Ala	Glu	Asp	Gly	Gln	Tyr	Asp	Val
435						440						445			
Met	Phe	Ile	Gly	Thr	Asp	Val	Gly	Thr	Val	Leu	Lys	Val	Val	Ser	Ile
450						455				460					
Pro	Lys	Glu	Thr	Trp	His	Asp	Leu	Glu	Glu	Val	Leu	Leu	Glu	Glu	Met
465				470						475				480	
Thr	Val	Phe	Arg												

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530	535	540
Tyr Phe Pro Thr Ala Lys Arg Arg Thr Arg Arg Gln Asp Ile Arg Asn		
545	550	555 560
Gly Asp Pro Leu Thr His Cys Ser Asp Leu Gln His His Asp Asn His		
	565	570 575
His Gly His Ser Pro Glu Glu Arg Ile Ile Tyr Gly Val Glu Asn Ser		
	580	585 590
Ser Thr Phe Leu Glu Cys Ser Pro Lys Ser Gln Arg Ala Leu Val Tyr		
	595	600 605
Trp Gln Phe Gln Arg Arg Asn Glu Glu Arg Lys Glu Glu Ile Arg Val		
	610	615 620
Asp Asp His Ile Ile Arg Thr Asp Gln Gly Leu Leu Leu Arg Ser Leu		
625	630	635 640
Gln Arg Lys Asp Ser Gly Ser Tyr Leu Cys His Ala Val Glu His Gly		
	645	650 655
Phe Ile Gln Thr Leu Leu Lys Val Thr Leu Glu Val Ile Asp Thr Glu		
	660	665 670
His Leu Glu Glu Leu Leu His Lys Asp Asp Asp Gly Asp Gly Ser Lys		
	675	680 685
Thr Lys Glu Met Ser Asn Ser Met Thr Pro Ser Gln Lys Val Trp Tyr		
	690	695 700
Arg Asp Phe Met Gln Leu Ile Asn His Pro Asn Leu Asn Thr Met Asp		
705	710	715 720
Glu Phe Cys Glu Gln Val Trp Lys Arg Asp Arg Lys Gln Arg Arg Gln		
	725	730 735
Arg Pro Gly His Thr Gln Gly Asn Ser Asn Lys Trp Lys His Leu Gln		
	740	745 750
Glu Asn Lys Lys Gly Arg Asn Arg Arg Thr His Glu Phe Glu Arg Ala		
	755	760 765
Pro Arg Ser Val		
770		

<210> SEQ ID NO 603
 <211> LENGTH: 752
 <212> TYPE: PRT
 <213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 603

Asn Tyr Ala Asn Gly Lys Asn Asn Val Pro Arg Leu Lys Leu Ser Tyr
1 5 10 15
Lys Glu Met Leu Glu Ser Asn Asn Val Ile Thr Phe Asn Gly Leu Ala
20 25 30
Asn Ser Ser Ser Tyr His Thr Phe Leu Leu Asp Glu Glu Arg Ser Arg
35 40 45
Leu Tyr Val Gly Ala Lys Asp His Ile Phe Ser Phe Asn Leu Val Asn
50 55 60
Ile Lys Asp Phe Gln Lys Ile Val Trp Pro Val Ser Tyr Thr Arg Arg
65 70 75 80
Asp Glu Cys Lys Trp Ala Gly Lys Asp Ile Leu Lys Glu Cys Ala Asn
85 90 95
Phe Ile Lys Val Leu Lys Ala Tyr Asn Gln Thr His Leu Tyr Ala Cys
100 105 110
Gly Thr Gly Ala Phe His Pro Ile Cys Thr Tyr Ile Glu Val Gly His
115 120 125

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His	Pro	Glu	Asp	Asn	Ile	Phe	Lys	Leu	Gln	Asp	Ser	His	Phe	Glu	Asn
130						135					140				
Gly	Arg	Gly	Lys	Ser	Pro	Tyr	Asp	Pro	Lys	Leu	Leu	Thr	Ala	Ser	Leu
145					150					155					160
Leu	Ile	Asp	Gly	Glu	Leu	Tyr	Ser	Gly	Thr	Ala	Ala	Asp	Phe	Met	Gly
			165						170					175	
Arg	Asp	Phe	Ala	Ile	Phe	Arg	Thr	Leu	Gly	His	His	His	Pro	Ile	Arg
		180						185					190		
Thr	Glu	Gln	His	Asp	Ser	Arg	Trp	Leu	Asn	Asp	Pro	Arg	Phe	Ile	Ser
	195						200					205			
Ala	His	Leu	Ile	Pro	Glu	Ser	Asp	Asn	Pro	Glu	Asp	Asp	Lys	Val	Tyr
210						215					220				
Phe	Phe	Phe	Arg	Glu	Asn	Ala	Ile	Asp	Gly	Glu	His	Ser	Gly	Lys	Ala
225					230					235					240
Thr	His	Ala	Arg	Ile	Gly	Gln	Ile	Cys	Lys	Asn	Asp	Phe	Gly	Gly	His
			245					250						255	
Arg	Ser	Leu	Val	Asn	Lys	Trp	Thr	Thr	Phe	Leu	Lys	Ala	Arg	Leu	Ile
		260						265					270		
Cys	Ser	Val	Pro	Gly	Pro	Asn	Gly	Ile	Asp	Thr	His	Phe	Asp	Glu	Leu
	275						280					285			
Gln	Asp	Val	Phe	Leu	Met	Asn	Ser	Lys	Asp	Pro	Lys	Asn	Pro	Ile	Val
290						295					300				
Tyr	Gly	Val	Phe	Thr	Thr	Ser	Ser	Asn	Ile	Phe	Lys	Gly	Ser	Ala	Val
305					310					315					320
Cys	Met	Tyr	Ser	Met	Ser	Asp	Val	Arg	Arg	Val	Phe	Leu	Gly	Pro	Tyr
			325					330						335	
Ala	His	Arg	Asp	Gly	Pro	Asn	Tyr	Gln	Trp	Val	Pro	Tyr	Gln	Gly	Arg
		340						345					350		
Val	Pro	Tyr	Pro	Arg	Pro	Gly	Thr	Cys	Pro	Ser	Lys	Thr	Phe	Gly	Gly
		355					360					365			
Phe	Asp	Ser	Thr	Lys	Asp	Leu	Pro	Asp	Asp	Val	Ile	Thr	Phe	Ala	Arg
370						375					380				
Ser	His	Pro	Ala	Met	Tyr	Asn	Pro	Val	Phe	Pro	Ile	Asn	Asn	Arg	Pro
385					390					395					400
Ile	Met	Ile	Lys	Thr	Asp	Val	Asn	Tyr	Gln	Phe	Thr	Gln	Ile	Val	Val
			405						410					415	
Asp	Arg	Val	Asp	Ala	Glu	Asp	Gly	Gln	Tyr	Asp	Val	Met	Phe	Ile	Gly
			420					425					430		
Thr	Asp	Val	Gly	Thr	Val	Leu	Lys	Val	Val	Ser	Val	Pro	Lys	Glu	Thr
	435						440					445			
Trp	His	Asp	Leu	Glu	Glu	Val	Leu	Leu	Glu	Glu	Met	Thr	Val	Phe	Arg
	450					455					460				
Glu	Pro	Thr	Thr	Ile	Ser	Ala	Met	Glu	Leu	Ser	Thr	Lys	Gln	Gln	Gln
465					470					475					480
Leu	Tyr	Ile	Gly	Ser	Thr	Ala	Gly	Val	Ala	Gln	Leu	Pro	Leu	His	Arg
			485						490					495	
Cys	Asp	Ile	Tyr	Gly	Lys	Ala	Cys	Ala	Glu	Cys	Cys	Leu	Ala	Arg	Asp
		500						505					510		
Pro	Tyr	Cys	Ala	Trp	Asp	Gly	Ser	Ser	Cys	Ser	Arg	Tyr	Phe	Pro	Thr
	515						520					525			
Ala	Lys	Arg	Arg	Thr	Arg	Arg	Gln	Asp	Ile	Arg	Asn	Gly	Asp	Pro	Leu
530						535					540				
Thr	His	Cys	Ser	Asp	Leu	Gln	His	His	Asp	Asn	His	His	Gly	His	Ser

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545	550	555	560
Leu Glu Glu Arg Ile Ile Tyr Gly Val Glu Asn Ser Ser Thr Phe Leu			
	565	570	575
Glu Cys Ser Pro Lys Ser Gln Arg Ala Leu Val Tyr Trp Gln Phe Gln			
	580	585	590
Arg Arg Asn Glu Asp Arg Lys Glu Glu Ile Arg Val Gly Asp His Ile			
	595	600	605
Ile Arg Thr Glu Gln Gly Leu Leu Leu Arg Ser Leu Gln Lys Lys Asp			
	610	615	620
Ser Gly Asn Tyr Leu Cys His Ala Val Glu His Gly Phe Met Gln Thr			
	625	630	635
Leu Leu Lys Val Thr Leu Glu Val Ile Asp Thr Glu His Leu Glu Glu			
	645	650	655
Leu Leu His Lys Asp Asp Asp Gly Asp Gly Ser Lys Thr Lys Glu Met			
	660	665	670
Ser Ser Ser Met Thr Pro Ser Gln Lys Val Trp Tyr Arg Asp Phe Met			
	675	680	685
Gln Leu Ile Asn His Pro Asn Leu Asn Thr Met Asp Glu Phe Cys Glu			
	690	695	700
Gln Val Trp Lys Arg Asp Arg Lys Gln Arg Arg Gln Arg Pro Gly His			
	705	710	715
Ser Gln Gly Ser Ser Asn Lys Trp Lys His Met Gln Glu Ser Lys Lys			
	725	730	735
Gly Arg Asn Arg Arg Thr His Glu Phe Glu Arg Ala Pro Arg Ser Val			
	740	745	750

<210> SEQ ID NO 604

<211> LENGTH: 772

<212> TYPE: PRT

<213> ORGANISM: Sus scrofa

<400> SEQUENCE: 604

Met Gly Trp Phe Ser Arg Ile Val Cys Leu Phe Trp Gly Val Leu Leu			
1	5	10	15
Thr Ala Arg Ala Asn Tyr Gln Asn Gly Lys Asn Asn Val Pro Arg Leu			
	20	25	30
Lys Leu Ser Tyr Lys Glu Met Leu Glu Ser Asn Asn Val Ile Thr Phe			
	35	40	45
Asn Gly Leu Ala Asn Ser Ser Ser Tyr His Thr Phe Leu Leu Asp Glu			
	50	55	60
Glu Arg Ser Arg Leu Tyr Val Gly Ala Lys Asp His Ile Phe Ser Phe			
	65	70	75
Asn Leu Val Asn Ile Lys Asp Phe Gln Lys Ile Val Trp Pro Val Ser			
	85	90	95
Tyr Thr Arg Arg Asp Glu Cys Lys Trp Ala Gly Lys Asp Ile Leu Lys			
	100	105	110
Glu Cys Ala Asn Phe Ile Lys Val Leu Lys Ala Tyr Asn Gln Thr His			
	115	120	125
Leu Tyr Ala Cys Gly Thr Gly Ala Phe His Pro Ile Cys Thr Tyr Ile			
	130	135	140
Glu Ile Gly His His Pro Glu Asp Asn Ile Phe Lys Leu Glu Asp Ser			
	145	150	155
His Phe Glu Asn Gly Arg Gly Lys Ser Pro Tyr Asp Pro Lys Leu Leu			
	165	170	175

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Thr	Ala	Ser	Leu	Leu	Ile	Asp	Gly	Glu	Leu	Tyr	Ser	Gly	Thr	Ala	Ala
			180					185					190		
Asp	Phe	Met	Gly	Arg	Asp	Phe	Ala	Ile	Phe	Arg	Thr	Leu	Gly	His	His
		195					200					205			
His	Pro	Ile	Arg	Thr	Glu	Gln	His	Asp	Ser	Arg	Trp	Leu	Asn	Asp	Pro
	210					215					220				
Arg	Phe	Ile	Ser	Ala	His	Leu	Ile	Pro	Glu	Ser	Asp	Asn	Pro	Glu	Asp
225					230					235				240	
Asp	Lys	Val	Tyr	Phe	Phe	Phe	Arg	Glu	Asn	Ala	Ile	Asp	Gly	Glu	His
				245					250					255	
Thr	Gly	Lys	Ala	Thr	His	Ala	Arg	Ile	Gly	Gln	Ile	Cys	Lys	Asn	Asp
			260					265					270		
Phe	Gly	Gly	His	Arg	Ser	Leu	Val	Asn	Lys	Trp	Thr	Thr	Phe	Leu	Lys
		275					280					285			
Ala	Arg	Leu	Ile	Cys	Ser	Val	Pro	Gly	Pro	Asn	Gly	Ile	Asp	Thr	His
		290				295					300				
Phe	Asp	Glu	Leu	Gln	Asp	Val	Phe	Leu	Met	Asn	Ser	Lys	Asp	Pro	Lys
305					310					315					320
Asn	Pro	Val	Val	Tyr	Gly	Val	Phe	Thr	Thr	Ser	Ser	Asn	Ile	Phe	Lys
				325					330					335	
Gly	Ser	Ala	Val	Cys	Met	Tyr	Ser	Met	Ser	Asp	Val	Arg	Arg	Val	Phe
			340					345					350		
Leu	Gly	Pro	Tyr	Ala	His	Arg	Asp	Gly	Pro	Asn	Tyr	Gln	Trp	Val	Pro
		355					360					365			
Tyr	Gln	Gly	Arg	Val	Pro	Tyr	Pro	Arg	Pro	Gly	Thr	Cys	Pro	Ser	Lys
	370					375					380				
Thr	Phe	Gly	Gly	Phe	Asp	Ser	Thr	Lys	Asp	Leu	Pro	Asp	Asp	Val	Ile
385					390					395					400
Thr	Phe	Ala	Arg	Ser	His	Pro	Ala	Met	Tyr	Asn	Pro	Val	Phe	Pro	Ile
				405					410					415	
Asn	Asn	Arg	Pro	Ile	Met	Ile	Lys	Thr	Asp	Val	Asn	Tyr	Gln	Phe	Thr
			420					425					430		
Gln	Ile	Val	Val	Asp	Arg	Val	Asp	Ala	Glu	Asp	Gly	Gln	Tyr	Asp	Val
		435					440					445			
Met	Phe	Ile	Gly	Thr	Asp	Val	Gly	Thr	Val	Leu	Lys	Val	Val	Ser	Ile
	450				455						460				
Pro	Lys	Glu	Thr	Trp	His	Asp	Leu	Glu	Glu	Val	Leu	Leu	Glu	Glu	Met
465					470					475					480
Thr	Val	Phe	Arg	Glu	Pro	Thr	Thr	Ile	Ser	Ala	Met	Glu	Leu	Ser	Thr
				485					490					495	
Lys	Gln	Gln	Gln	Leu	Tyr	Val	Gly	Ser	Ala	Ala	Gly	Val	Ala	Gln	Leu
			500					505					510		
Pro	Leu	His	Arg	Cys	Asp	Ile	Tyr	Gly	Lys	Ala	Cys	Ala	Glu	Cys	Cys
		515					520					525			
Leu	Ala	Arg	Asp	Pro	Tyr	Cys	Ala	Trp	Asp	Gly	Ser	Ser	Cys	Ser	Arg
	530					535					540				
Tyr	Phe	Pro	Thr	Ala	Lys	Arg	Arg	Thr	Arg	Arg	Gln	Asp	Ile	Arg	Asn
545					550					555					560
Gly	Asp	Pro	Leu	Thr	His	Cys	Ser	Asp	Leu	Gln	His	His	Asp	Asn	His
			565						570					575	
Arg	Gly	His	Asn	Phe	Glu	Glu	Arg	Ile	Ile	Tyr	Gly	Val	Glu	Asn	Ser
			580					585					590		
Ser	Thr	Phe	Leu	Glu	Cys	Ser	Pro	Lys	Ser	Gln	Arg	Ala	Leu	Val	Tyr

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595					600					605					
Trp	Gln	Phe	Gln	Arg	Arg	Asn	Glu	Glu	Arg	Lys	Glu	Glu	Ile	Arg	Val
610						615					620				
Asp	Asp	His	Ile	Ile	Arg	Thr	Glu	Gln	Gly	Leu	Leu	Leu	Arg	Ser	Leu
625					630					635					640
Gln	Arg	Lys	Asp	Ser	Gly	Ser	Tyr	Leu	Cys	His	Ala	Val	Glu	His	Gly
				645					650					655	
Phe	Met	Gln	Thr	Leu	Leu	Lys	Val	Thr	Leu	Glu	Val	Ile	Asp	Thr	Glu
				660					665					670	
His	Leu	Glu	Glu	Leu	Leu	His	Lys	Asp	Asp	Asp	Gly	Asp	Ser	Ser	Lys
		675					680					685			
Thr	Lys	Glu	Met	Ser	Asn	Ser	Met	Thr	Pro	Ser	Gln	Lys	Ile	Trp	Tyr
	690					695					700				
Arg	Asp	Phe	Met	Gln	Leu	Ile	Asn	His	Pro	Asn	Leu	Asn	Thr	Met	Asp
705				710						715					720
Glu	Phe	Cys	Glu	Gln	Val	Trp	Lys	Arg	Asp	Arg	Lys	Gln	Arg	Arg	Gln
				725					730					735	
Arg	Pro	Gly	His	Thr	Gln	Gly	Asn	Ser	Asn	Lys	Trp	Lys	His	Leu	Gln
			740					745					750		
Glu	Asn	Lys	Lys	Cys	Arg	Asn	Arg	Arg	Thr	His	Glu	Phe	Glu	Arg	Ala
		755					760					765			
Pro	Arg	Ser	Val												
			770												

<210> SEQ ID NO 605

<211> LENGTH: 772

<212> TYPE: PRT

<213> ORGANISM: Canis lupus familiaris

<400> SEQUENCE: 605

Met	Gly	Trp	Leu	Ala	Arg	Ile	Ala	Cys	Leu	Phe	Trp	Gly	Val	Leu	Leu
1				5					10					15	
Thr	Ala	Thr	Ala	Asn	Tyr	Gln	Asn	Gly	Lys	Asn	Asn	Val	Pro	Arg	Leu
			20					25					30		
Lys	Leu	Ser	Tyr	Lys	Glu	Met	Leu	Glu	Ser	Asn	Ser	Val	Ile	Thr	Phe
		35					40					45			
Asn	Gly	Leu	Ala	Asn	Ser	Ser	Ser	Tyr	His	Thr	Phe	Leu	Leu	Asp	Glu
	50				55						60				
Glu	Arg	Ser	Arg	Leu	Tyr	Val	Gly	Ala	Lys	Asp	His	Ile	Phe	Ser	Phe
65				70						75					80
Asn	Leu	Val	Asn	Ile	Lys	Asp	Phe	Gln	Lys	Ile	Val	Trp	Pro	Val	Ser
			85					90						95	
Tyr	Thr	Arg	Arg	Asp	Glu	Cys	Lys	Trp	Ala	Gly	Lys	Asp	Ile	Gln	Lys
		100						105					110		
Glu	Cys	Ala	Asn	Phe	Ile	Lys	Val	Leu	Lys	Ala	Tyr	Asn	Gln	Thr	His
		115					120					125			
Leu	Tyr	Ala	Cys	Gly	Thr	Gly	Ala	Phe	His	Pro	Ile	Cys	Thr	Tyr	Ile
		130				135					140				
Glu	Ile	Gly	His	His	Pro	Glu	Asp	Asn	Ile	Phe	Lys	Leu	Glu	Asp	Ser
145				150						155					160
His	Phe	Glu	Asn	Gly	Arg	Gly	Lys	Ser	Pro	Tyr	Asp	Pro	Lys	Leu	Leu
			165					170						175	
Thr	Ala	Ser	Leu	Leu	Ile	Asp	Gly	Glu	Leu	Tyr	Ser	Gly	Thr	Ala	Ala
			180					185						190	

Asp	Phe	Met	Gly	Arg	Asp	Phe	Ala	Ile	Phe	Arg	Thr	Leu	Gly	His	His
	195						200					205			
His	Pro	Ile	Arg	Thr	Glu	Gln	His	Asp	Ser	Arg	Trp	Leu	Asn	Asp	Pro
	210					215					220				
Arg	Phe	Ile	Ser	Ala	His	Leu	Ile	Pro	Glu	Ser	Asp	Asn	Pro	Glu	Asp
225					230					235					240
Asp	Lys	Val	Tyr	Phe	Phe	Phe	Arg	Glu	Asn	Ala	Ile	Asp	Gly	Glu	His
				245					250					255	
Thr	Gly	Lys	Ala	Thr	His	Ala	Arg	Ile	Gly	Gln	Ile	Cys	Lys	Asn	Asp
			260					265					270		
Phe	Gly	Gly	His	Arg	Ser	Leu	Val	Asn	Lys	Trp	Thr	Thr	Phe	Leu	Lys
		275					280					285			
Ala	Arg	Leu	Ile	Cys	Ser	Val	Pro	Gly	Pro	Asn	Gly	Ile	Asp	Thr	His
						295					300				
Phe	Asp	Glu	Leu	Gln	Asp	Val	Phe	Leu	Met	Asn	Ser	Lys	Asp	Pro	Lys
305					310					315					320
Asn	Pro	Ile	Val	Tyr	Gly	Val	Phe	Thr	Thr	Ser	Ser	Asn	Ile	Phe	Lys
				325					330					335	
Gly	Ser	Ala	Val	Cys	Met	Tyr	Ser	Met	Ser	Asp	Val	Arg	Arg	Val	Phe
			340					345					350		
Leu	Gly	Pro	Tyr	Ala	His	Arg	Asp	Gly	Pro	Asn	Tyr	Gln	Trp	Val	Pro
		355					360					365			
Tyr	Gln	Gly	Arg	Val	Pro	Tyr	Pro	Arg	Pro	Gly	Thr	Cys	Pro	Ser	Lys
	370					375					380				
Thr	Phe	Gly	Gly	Phe	Asp	Ser	Thr	Lys	Asp	Leu	Pro	Asp	Asp	Val	Ile
385					390					395					400
Thr	Phe	Ala	Arg	Ser	His	Pro	Ala	Met	Tyr	Asn	Pro	Val	Phe	Pro	Ile
				405					410					415	
Asn	Asn	Arg	Pro	Ile	Met	Ile	Lys	Thr	Asp	Val	Asn	Tyr	Gln	Phe	Thr
			420						425				430		
Gln	Ile	Val	Val	Asp	Arg	Val	Asp	Ala	Glu	Asp	Gly	Gln	Tyr	Asp	Val
		435					440					445			
Met	Phe	Ile	Gly	Thr	Asp	Val	Gly	Thr	Val	Leu	Lys	Val	Val	Ser	Ile
	450					455					460				
Pro	Lys	Glu	Thr	Trp	His	Asp	Leu	Glu	Glu	Val	Leu	Leu	Glu	Glu	Met
465					470					475					480
Thr	Val	Phe	Arg	Glu	Pro	Thr	Pro	Ile	Ser	Ala	Met	Glu	Leu	Ser	Thr
				485					490					495	
Lys	Gln	His	Gln	Leu	Tyr	Ala	Gly	Ser	Pro	Ala	Gly	Leu	Ala	Gln	Leu
			500					505					510		
Pro	Leu	Gln	Arg	Cys	Ala	Ala	Tyr	Gly	Arg	Ala	Cys	Ala	Glu	Cys	Cys
		515					520					525			
Leu	Ala	Arg	Asp	Pro	Tyr	Cys	Ala	Trp	Asp	Gly	Ala	Ala	Cys	Ser	Arg
	530					535					540				
Tyr	Phe	Pro	Ala	Ala</											

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610					615					620					
Asp	Asp	His	Ile	Ile	Arg	Thr	Glu	Gln	Gly	Leu	Leu	Leu	Arg	Ser	Leu
625					630					635					640
Gln	Arg	Lys	Asp	Ser	Gly	Asn	Tyr	Leu	Cys	His	Ala	Val	Glu	His	Gly
			645						650					655	
Phe	Met	Gln	Thr	Leu	Leu	Lys	Val	Thr	Leu	Glu	Val	Ile	Asp	Thr	Glu
			660					665					670		
His	Leu	Glu	Glu	Leu	Leu	His	Lys	Asp	Asp	Asp	Gly	Asp	Gly	Ser	Lys
		675					680					685			
Thr	Lys	Glu	Ile	Ser	Asn	Ser	Met	Thr	Pro	Ser	Gln	Lys	Val	Trp	Tyr
	690					695					700				
Arg	Asp	Phe	Met	Gln	Leu	Ile	Asn	His	Pro	Asn	Leu	Asn	Thr	Met	Asp
705				710						715					720
Glu	Phe	Cys	Glu	Gln	Val	Trp	Lys	Arg	Asp	Arg	Lys	Gln	Arg	Arg	Gln
			725						730					735	
Arg	Pro	Gly	His	Thr	Gln	Gly	Asn	Ser	Asn	Lys	Trp	Lys	His	Leu	Gln
			740				745						750		
Glu	Asn	Lys	Lys	Gly	Arg	Asn	Arg	Arg	Thr	His	Glu	Phe	Glu	Arg	Ala
		755				760						765			
Pro	Arg	Ser	Val												
			770												

<210> SEQ ID NO 606

<400> SEQUENCE: 606

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<210> SEQ ID NO 800

<211> LENGTH: 117

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 800

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
 20 25 30

Tyr Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ser Thr Ile Ile Lys Ser Gly Gly Tyr Ala Tyr Tyr Pro Asp Ser Val
 50 55 60

Lys Asp Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
 65 70 75 80

Leu Gln Met Ser Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Val Arg Gly Gly Gln Gly Ala Met Asp Tyr Trp Gly Gln Gly Thr Thr
 100 105 110

Val Thr Val Ser Ser

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115

<210> SEQ ID NO 801
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 801

Ser Tyr Tyr Met Ser
1 5

<210> SEQ ID NO 802
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 802

Thr Ile Ile Lys Ser Gly Gly Tyr Ala Tyr Tyr Pro Asp Ser Val Lys
1 5 10 15

Asp

<210> SEQ ID NO 803
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 803

Gly Gly Gln Gly Ala Met Asp Tyr
1 5

<210> SEQ ID NO 804
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 804

Glu Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Leu Ser Pro Gly
1 5 10 15

Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Ile Gly Asp Tyr
20 25 30

Leu His Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
35 40 45

Lys Tyr Ala Ser Gln Ser Ile Ser Gly Ile Pro Ala Arg Phe Ser Gly
50 55 60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Thr Ser Leu Glu Pro
65 70 75 80

Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Gly Tyr Ser Phe Pro Tyr
85 90 95

Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys
100 105

<210> SEQ ID NO 805
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence

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<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 805

Arg Ala Ser Gln Ser Ile Gly Asp Tyr Leu His
 1 5 10

<210> SEQ ID NO 806

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 806

Tyr Ala Ser Gln Ser Ile Ser
 1 5

<210> SEQ ID NO 807

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 807

Gln Gln Gly Tyr Ser Phe Pro Tyr Thr
 1 5

<210> SEQ ID NO 808

<211> LENGTH: 351

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 808

gaagtgcagc tggtggaatc tggcggagga ctggttcaac ctggcggctc tctgagactg	60
tcttgtgccg ccagcgggett caccttcagc agctactaca tgagctgggt ccgacaggcc	120
cctggcaaag gacttgaatg ggtgtccacc atcatcaaga gcggcggcta cgctactat	180
cccgacagcg tgaaggaccg gtccaccatc tccagagaca acagcaagaa caccctgtac	240
ctgcagatga gcagcctgag agccgaggat accgccgtgt actactgtgt tagaggcgga	300
cagggcgccca tggattattg gggccaggga accacagtga ccgtgtcatc a	351

<210> SEQ ID NO 809

<211> LENGTH: 321

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 809

gagattgtgc tgacacagtc tcccgccaca ctgtctctta gccctggcga aagagccaca	60
ctgagctgta gagccagcca gagcatcggc gattacctgc actgggtatca gcagaagcct	120
ggacaggccc ctcggtgtgt gattaagtac gccagccagt ccatcagcgg catccctgcc	180
agattttctg gcagcggctc tggcaccgat ttcacctga ccatcaccag cctggaacct	240
gaggacttcg ccgtgtacta ctgccagcag ggctacagct tcccctacac atttgcgga	300
ggcaccaagc tggaaatcaa a	321

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<210> SEQ ID NO 810
<211> LENGTH: 447
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 810

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1          5          10          15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20          25          30
Tyr Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35          40          45
Ser Thr Ile Ile Lys Ser Gly Gly Tyr Ala Tyr Tyr Pro Asp Ser Val
50          55          60
Lys Asp Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65          70          75          80
Leu Gln Met Ser Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85          90          95
Val Arg Gly Gly Gln Gly Ala Met Asp Tyr Trp Gly Gln Gly Thr Thr
100         105         110
Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu
115         120         125
Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly Cys
130         135         140
Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser
145         150         155         160
Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val Leu Gln Ser
165         170         175
Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser
180         185         190
Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His Lys Pro Ser Asn
195         200         205
Thr Lys Val Asp Lys Lys Val Glu Pro Lys Ser Cys Asp Lys Thr His
210         215         220
Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly Pro Ser Val
225         230         235         240
Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr
245         250         255
Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu Asp Pro Glu
260         265         270
Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys
275         280         285
Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg Val Val Ser
290         295         300
Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys
305         310         315         320
Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu Lys Thr Ile
325         330         335
Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro
340         345         350
Pro Ser Arg Asp Glu Leu Thr Lys Asn Gln Val Ser Leu Thr Cys Leu
355         360         365
Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn

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370	375	380
Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser		
385	390	395 400
Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg		
	405	410 415
Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu		
	420	425 430
His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys		
	435	440 445

<210> SEQ ID NO 811
 <211> LENGTH: 214
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 811

Glu Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Leu Ser Pro Gly	
1	5 10 15
Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Ile Gly Asp Tyr	
	20 25 30
Leu His Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile	
	35 40 45
Lys Tyr Ala Ser Gln Ser Ile Ser Gly Ile Pro Ala Arg Phe Ser Gly	
	50 55 60
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Thr Ser Leu Glu Pro	
	65 70 75 80
Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Gly Tyr Ser Phe Pro Tyr	
	85 90 95
Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys Arg Thr Val Ala Ala	
	100 105 110
Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly	
	115 120 125
Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala	
	130 135 140
Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln	
	145 150 155 160
Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser	
	165 170 175
Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr	
	180 185 190
Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser	
	195 200 205
Phe Asn Arg Gly Glu Cys	
	210

<210> SEQ ID NO 812
 <211> LENGTH: 1341
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 812

gaagtgcagc tgggtggaatc tggcggagga ctggttcaac ctggcggctc tctgagactg	60
tcttgtgccg ccagcggcgtt caccttcagc agctactaca tgagctgggt ccgacaggcc	120

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cctggcaaag gacttgaatg ggtgtccacc atcatcaaga gcggcggcta cgctactat 180
cccacacagc tgaaggaccg gttcaccatc tccagagaca acagcaagaa caccctgtac 240
ctgcagatga gcagcctgag agccgaggat accgccgtgt actactgtgt tagaggcgga 300
cagggcgcca tggattattg gggccaggga accacagtga ccgtgtcatc agccagcacc 360
aaggggccca gcgtgttccc tctggcccct agcagcaaga gcacatctgg cggaacagcc 420
gccctgggct gcctcgtgaa ggactacttt cccagagccc tgaccgtgtc ctggaactct 480
ggcgctctga caagcggcgt gcacaccttt ccagccgtgc tgcagagcag cggcctgtac 540
tctctgagca gcgtcgtgac agtgcccagc agctctctgg gcaccagac ctacatctgc 600
aacgtgaacc acaagcccag caacaccaag gtggacaaga aggtggaacc caagagctgc 660
gacaagaccc acacotgtcc ccttgtcct gccccgaac tgetgggagg ccttccgtg 720
ttcctgttcc ccccaaagcc caaggacacc ctgatgatca gccggacccc cgaagtgacc 780
tgctgtgtgg tggatgtgtc ccacgaggac cctgaagtga agttcaattg gtacgtggac 840
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cgggtggtgt ccgtgctgac agtgctgcac caggactggc tgaacggcaa agagtacaag 960
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<210> SEQ ID NO 813
<211> LENGTH: 642
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 813
gagattgtgc tgacacagtc tcccgccaca ctgtctctta gccctggcga aagagccaca 60
ctgagctgta gagccagcca gagcatcggc gattacctgc actggtatca gcagaagcct 120
ggacaggccc ctcggtgctg gattaagtac gccagccagt ccatcagcgg catccctgcc 180
agattttctg gcagcggctc tggcaccgat ttcaccctga ccatcaccag cctggaaacct 240
gaggacttcg ccgtgtacta ctgccagcag ggctacagct tccccacac atttggcgga 300
ggaccaaacg tggaaatcaa acgaaccgtg gccgctccca gcgtgttcat ctccccacct 360
agcgacgagc agctgaagtc cggcacagcc tctgtcgtgt gcctgctgaa caacttctac 420
ccccgcgagg ccaagggtga gtggaagggt gacaatgccc tgcagagcgg caacagccag 480
gaaagcgtga ccgagcagga cagcaaggac tccacctaca gcctgagcag caccctgacc 540
ctgagcaagg ccgactacga gaagcacaag gtgtacgcct gcgaagtgac ccaccagggc 600
ctgtctagcc ccgtgaccaa gagcttcaac cggggcgagt gt 642

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<210> SEQ ID NO 814
<211> LENGTH: 117
<212> TYPE: PRT

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 814

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1           5           10          15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Pro Phe Ser Ser Tyr
          20          25          30

Tyr Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
          35          40          45

Ser Thr Ile Ile Lys Ser Gly Gly Tyr Ala Tyr Tyr Pro Asp Ser Val
          50          55          60

Lys Asp Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65          70          75          80

Leu Gln Met Ser Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
          85          90          95

Val Arg Gly Gly Gln Gly Ala Met Asp Tyr Trp Gly Gln Gly Thr Thr
          100         105         110

Val Thr Val Ser Ser
          115

```

```

<210> SEQ ID NO 815
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 815

```

```

Ser Tyr Tyr Met Ser
1           5

```

```

<210> SEQ ID NO 816
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 816

```

```

Thr Ile Ile Lys Ser Gly Gly Tyr Ala Tyr Tyr Pro Asp Ser Val Lys
1           5           10          15

```

```

Asp

```

```

<210> SEQ ID NO 817
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 817

```

```

Gly Gly Gln Gly Ala Met Asp Tyr
1           5

```

```

<210> SEQ ID NO 818
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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-continued

<400> SEQUENCE: 818

Glu Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Leu Ser Pro Gly
 1 5 10 15
 Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Ile Gly Asp Tyr
 20 25 30
 Leu His Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
 35 40 45
 Lys Tyr Ala Ser Gln Ser Ile Ser Gly Ile Pro Ala Arg Phe Ser Gly
 50 55 60
 Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Thr Ser Leu Glu Pro
 65 70 75 80
 Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Gly Tyr Ser Phe Pro Tyr
 85 90 95
 Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys
 100 105

<210> SEQ ID NO 819

<211> LENGTH: 11

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 819

Arg Ala Ser Gln Ser Ile Gly Asp Tyr Leu His
 1 5 10

<210> SEQ ID NO 820

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 820

Tyr Ala Ser Gln Ser Ile Ser
 1 5

<210> SEQ ID NO 821

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 821

Gln Gln Gly Tyr Ser Phe Pro Tyr Thr
 1 5

<210> SEQ ID NO 822

<211> LENGTH: 351

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 822

gaagtgcagc tgggtggaatc tggcggagga ctggttcaac ctggcggctc tctgagactg 60
 tcttgtgccg cctctggett cccattcagc agctactaca tgagctgggt ccgacaggcc 120
 cctggcaaa gacttgaatg ggtgtccacc atcatcaaga gcgccggcta cgcctactat 180
 cccgacagcg tgaaggaccg gtccaccatc agccgggaca acagcaagaa caccctgtac 240

-continued

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ctgcagatga gcagcctgag agccgaggat accgccgtgt actactgtgt tagaggcgga    300
cagggcgcca tggattattg gggccaggga accacagtga ccgtgtcatc a              351

```

```

<210> SEQ ID NO 823
<211> LENGTH: 321
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 823

```

```

gagattgtgc tgacacagtc tccccccaca ctgtctctta gccctggcga aagagccaca    60
ctgagctgta gagccagcca gagcatcggc gattacctgc actgggtatca gcagaagcct   120
ggacaggccc ctcggtgtct gattaagtac gccagccagt ccatcagcgg catccctgcc   180
agattttctg gcagcggctc tggcaccgat ttcacctga ccatcaccag cctggaacct   240
gaggacttcg ccgtgtacta ctgccagcag ggctacagct tcccctacac atttggcgga   300
ggcaccaagc tggaaatcaa a                                              321

```

```

<210> SEQ ID NO 824
<211> LENGTH: 447
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 824

```

```

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1          5          10          15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Pro Phe Ser Ser Tyr
20         25         30
Tyr Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35         40         45
Ser Thr Ile Ile Lys Ser Gly Gly Tyr Ala Tyr Tyr Pro Asp Ser Val
50         55         60
Lys Asp Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65         70         75         80
Leu Gln Met Ser Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85         90         95
Val Arg Gly Gly Gln Gly Ala Met Asp Tyr Trp Gly Gln Gly Thr Thr
100        105        110
Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu
115        120        125
Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly Cys
130        135        140
Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser
145        150        155        160
Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val Leu Gln Ser
165        170        175
Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser
180        185        190
Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His Lys Pro Ser Asn
195        200        205
Thr Lys Val Asp Lys Lys Val Glu Pro Lys Ser Cys Asp Lys Thr His
210        215        220

```

-continued

Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly Pro Ser Val
 225 230 235 240
 Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr
 245 250 255
 Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu Asp Pro Glu
 260 265 270
 Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys
 275 280 285
 Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg Val Val Ser
 290 295 300
 Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys
 305 310 315 320
 Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu Lys Thr Ile
 325 330 335
 Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro
 340 345 350
 Pro Ser Arg Asp Glu Leu Thr Lys Asn Gln Val Ser Leu Thr Cys Leu
 355 360 365
 Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn
 370 375 380
 Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser
 385 390 395 400
 Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg
 405 410 415
 Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu
 420 425 430
 His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys
 435 440 445

 <210> SEQ ID NO 825
 <211> LENGTH: 214
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

 <400> SEQUENCE: 825

 Glu Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Leu Ser Pro Gly
 1 5 10 15
 Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Ile Gly Asp Tyr
 20 25 30
 Leu His Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
 35 40 45
 Lys Tyr Ala Ser Gln Ser Ile Ser Gly Ile Pro Ala Arg Phe Ser Gly
 50 55 60
 Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Thr Ser Leu Glu Pro
 65 70 75 80
 Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Gly Tyr Ser Phe Pro Tyr
 85 90 95
 Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys Arg Thr Val Ala Ala
 100 105 110
 Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly
 115 120 125
 Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala
 130 135 140

-continued

Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln
145 150 155 160

Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser
165 170 175

Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr
180 185 190

Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser
195 200 205

Phe Asn Arg Gly Glu Cys
210

<210> SEQ ID NO 826

<211> LENGTH: 1341

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 826

```

gaagtgcagc tgggtgaatc tggcggagga ctggttcaac ctggcggctc tctgagactg      60
tcttgtgccg cctctggctt ccattcagc agctactaca tgagctgggt ccgacaggcc      120
cctggcaaag gacttgaatg ggtgtccacc atcatcaaga gcggcggcta cgcctactat      180
cccgacagcg tgaaggaccg gttcaccatc agccgggaca acagcaagaa caccctgtac      240
ctgcagatga gcagctgag agccgaggat accgccgtgt actactgtgt tagaggcgga      300
cagggcgcca tggattattg gggccaggga accacagtga ccgtgtcatc agccagcacc      360
aaggggccca gcgtgttccc tctggcccct agcagcaaga gcacatctgg cggaacagcc      420
gccctgggct gcctcgtgaa ggactacttt cccgagcccg tgaccgtgtc ctggaactct      480
ggcgtcttga caagcggcgt gcacaccttt ccagccgtgc tgcagagcag cggcctgtac      540
tctctgagca gcgtcgtgac agtgcaccag agctctctgg gcacccagac ctacatctgc      600
aacgtgaacc acaagcccag caacaccaag gtggacaaga aggtggaacc caagagctgc      660
gacaagacc acacctgtcc ccttctgtcc gccccgaac tgcctggagg ccttccgtg      720
ttcctgttcc ccccaaagcc caaggacacc ctgatgatca gccggacccc cgaagtgacc      780
tgcgtggtgg tggatgtgtc ccacgaggac cctgaagtga agttcaattg gtacgtggac      840
ggcgtggaag tgcacaacgc caagaccaag cctagagagg aacagtacaa cagcacctac      900
cgggtggtgt ccgtgctgac agtgcgtcac caggactggc tgaacggcaa agagtacaag      960
tgcaaggtgt ccaacaaggc cctgcctgcc cccatcgaga aaaccatcag caaggccaag     1020
ggccagcccc gcgaacccca ggtgtacaca ctgcccccaa gcagggacga gctgaccaag     1080
aaccaggtgt cctgacctg tctcgtgaaa ggcttctacc cctccgatat cgcctgggaa     1140
tggaagagca acggccagcc cgagaacaac tacaagacca cccccctgt gctggacagc     1200
gacggctcat tcttctgtga cagcaagctg accgtggaca agtcccggtg gcagcagggc     1260
aacgtgttca gctgcagcgt gatgcacgag gccctgcaca accactacac ccagaagtcc     1320
ctgagcctga gccctggcaa g                                     1341

```

<210> SEQ ID NO 827

<211> LENGTH: 642

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

-continued

<400> SEQUENCE: 827

```

gagattgtgc tgacacagtc tcccgccaca ctgtctctta gccctggcga aagagccaca      60
ctgagctgta gagccagcca gagcatcggc gattacctgc actggtatca gcagaagcct      120
ggacaggccc ctcggtgctt gattaagtac gccagccagt ccatcagcgg catccctgcc      180
agattttctg gcagcggttc tggcaccgat ttcaccctga ccatcaccag cctggaacct      240
gaggacttcg ccgtgtacta ctgccagcag ggctacagct tcccctacac atttgcgga      300
ggcaccaagc tggaaatcaa acgaaccgtg gccgctccca gcgtgttcat ctteccacct      360
agcgacgagc agctgaagtc cggcacagcc tctgtcgtgt gcctgctgaa caacttctac      420
ccccgcgagg ccaaggtgca gtggaagggt gacaatgccc tgcagagcgg caacagccag      480
gaaagcgtga ccgagcagga cagcaaggac tccacctaca gcctgagcag caccctgacc      540
ctgagcaagg ccgactacga gaagcacaag gtgtacgcct gcgaagtgac ccaccagggc      600
ctgtctagcc ccgtgaccaa gagcttcaac cggggcgagt gt                          642

```

<210> SEQ ID NO 828

<211> LENGTH: 117

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 828

```

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Leu Gly Gly
1           5           10          15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20          25          30
Tyr Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35          40          45
Ser Thr Ile Ile Lys Ser Gly Gly Tyr Ala Tyr Tyr Pro Asp Ser Val
50          55          60
Lys Asp Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65          70          75          80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85          90          95
Val Lys Gly Gly Gln Gly Ala Met Asp Tyr Trp Gly Gln Gly Thr Thr
100         105         110
Val Thr Val Ser Ser
115

```

<210> SEQ ID NO 829

<211> LENGTH: 5

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 829

```

Ser Tyr Tyr Met Ser
1           5

```

<210> SEQ ID NO 830

<211> LENGTH: 17

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

-continued

<400> SEQUENCE: 830

Thr Ile Ile Lys Ser Gly Gly Tyr Ala Tyr Tyr Pro Asp Ser Val Lys
1 5 10 15

Asp

<210> SEQ ID NO 831

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 831

Gly Gly Gln Gly Ala Met Asp Tyr
1 5

<210> SEQ ID NO 832

<211> LENGTH: 107

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 832

Glu Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Leu Ser Pro Gly
1 5 10 15

Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Ile Gly Asp Tyr
20 25 30

Leu His Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
35 40 45

Tyr Tyr Ala Ser Gln Ser Ile Ser Gly Ile Pro Ala Arg Phe Ser Gly
50 55 60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Glu Pro
65 70 75 80

Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Gly Tyr Ser Phe Pro Tyr
85 90 95

Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys
100 105

<210> SEQ ID NO 833

<211> LENGTH: 11

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 833

Arg Ala Ser Gln Ser Ile Gly Asp Tyr Leu His
1 5 10

<210> SEQ ID NO 834

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 834

Tyr Ala Ser Gln Ser Ile Ser
1 5

<210> SEQ ID NO 835

-continued

<211> LENGTH: 9
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 835

Gln Gln Gly Tyr Ser Phe Pro Tyr Thr
 1 5

<210> SEQ ID NO 836
 <211> LENGTH: 351
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 836

gaagtcgagc tgggtggaatc tggcggagga ctggttcagc tcggcggatc tctgagactg 60
 tcttgtgccg ccagcgggtt caccttcagc agctactaca tgagctgggt ccgacaggcc 120
 cctggcaaag gacttgaatg ggtgtccacc atcatcaaga gcggcggcta cgcctactat 180
 cccgacagcg tgaaggaccg gttcaccatc tccagagaca acagcaagaa caccctgtac 240
 ctgcagatga acagcctgag agccgaggac accgccgtgt actactgtgt gaaaggtgga 300
 cagggcgcca tggactattg gggccaggga acaacagtga ccgtgtcctc a 351

<210> SEQ ID NO 837
 <211> LENGTH: 321
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 837

gagattgtgc tgacacagtc tcccgccaca ctgtctctta gccctggcga aagagccaca 60
 ctgagctgta gagccagcca gagcatcggc gattacctgc actggtatca gcagaagcct 120
 ggacaggccc ctcgggtgct gatctactat gccagccagt ccatcagcgg catccccgcc 180
 agattttctg gcagcggctc tggcaccgat ttcacctga ccataagcag cctggaacct 240
 gaggacttcg ccgtgtacta ctgccagcag ggctacagct tcccctacac atttgcgga 300
 ggcaccaagc tggaaatcaa a 321

<210> SEQ ID NO 838
 <211> LENGTH: 447
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 838

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Leu Gly Gly
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
 20 25 30
 Tyr Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45
 Ser Thr Ile Ile Lys Ser Gly Gly Tyr Ala Tyr Tyr Pro Asp Ser Val
 50 55 60
 Lys Asp Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
 65 70 75 80

Leu	Gln	Met	Asn	Ser	Leu	Arg	Ala	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys
85								90				95			
Val	Lys	Gly	Gly	Gln	Gly	Ala	Met	Asp	Tyr	Trp	Gly	Gln	Gly	Thr	Thr
100								105				110			
Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	Lys	Gly	Pro	Ser	Val	Phe	Pro	Leu
115								120				125			
Ala	Pro	Ser	Ser	Lys	Ser	Thr	Ser	Gly	Gly	Thr	Ala	Ala	Leu	Gly	Cys
130								135				140			
Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu	Pro	Val	Thr	Val	Ser	Trp	Asn	Ser
145								150				155			
Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr	Phe	Pro	Ala	Val	Leu	Gln	Ser
165								170				175			
Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val	Thr	Val	Pro	Ser	Ser	Ser
180								185				190			
Leu	Gly	Thr	Gln	Thr	Tyr	Ile	Cys	Asn	Val	Asn	His	Lys	Pro	Ser	Asn
195								200				205			
Thr	Lys	Val	Asp	Lys	Lys	Val	Glu	Pro	Lys	Ser	Cys	Asp	Lys	Thr	His
210								215				220			
Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Leu	Leu	Gly	Gly	Pro	Ser	Val
225								230				235			
Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr
245								250				255			
Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu
260								265				270			
Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys
275								280				285			
Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser
290								295				300			
Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys
305								310				315			
Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile
325								330				335			
Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro
340								345				350			
Pro	Ser	Arg	Asp	Glu	Leu	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu
355								360				365			
Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn
370								375				380			
Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp	Ser
385								390				395			
Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg
405								410				415			
Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu
420								425				430			
His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys	
435								440				445			

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<210> SEQ ID NO 839
<211> LENGTH: 214
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 839
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Glu Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Leu Ser Pro Gly
 1 5 10 15
 Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Ile Gly Asp Tyr
 20 25 30
 Leu His Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
 35 40 45
 Tyr Tyr Ala Ser Gln Ser Ile Ser Gly Ile Pro Ala Arg Phe Ser Gly
 50 55 60
 Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Glu Pro
 65 70 75 80
 Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Gly Tyr Ser Phe Pro Tyr
 85 90 95
 Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys Arg Thr Val Ala Ala
 100 105 110
 Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly
 115 120 125
 Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala
 130 135 140
 Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln
 145 150 155 160
 Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser
 165 170 175
 Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr
 180 185 190
 Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser
 195 200 205
 Phe Asn Arg Gly Glu Cys
 210

<210> SEQ ID NO 840

<211> LENGTH: 1341

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 840

gaagtgcagc tggtggaatc tggcggagga ctggttcagc tcggcggatc tctgagactg 60
 tcttctgtccg ccagcggcgtt caccttcagc agctactaca tgagctgggt ccgacaggcc 120
 cctggcaaaag gacttgaatg ggtgtccacc atcatcaaga gcggcggtta cgcctactat 180
 cccgacagcg tgaaggaccg gttcaccatc tccagagaca acagcaagaa caccctgtac 240
 ctgcagatga acagcctgag agccgaggac accgccgtgt actactgtgt gaaaggtgga 300
 cagggcgcca tggactattg gggccaggga acaacagtga ccgtgtcctc agccagcacc 360
 aagggcccca gcgtgttccc tctggcccct agcagcaaga gcacatctgg cggaacagcc 420
 gccctgggct gcctcgtgaa ggactacttt cccgagcccg tgaccgtgtc ctggaactct 480
 ggcgctctga caagcggcgt gcacaccttt ccagccgtgc tgcagagcag cggcctgtac 540
 tctctgagca gcgtcgtgac agtgcccagc agctctctgg gcacccagac ctacatctgc 600
 aacgtgaacc acaagcccag caacaccaag gtggacaaga aggtggaacc caagagctgc 660
 gacaagaccc acacctgtcc cccttgcctt gccccgaac tgctgggagg cccttccgtg 720
 ttctctgtcc ccccaagacc caaggacacc ctgatgatca gccggacccc cgaagtgacc 780

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tcgctggttg	tgatgtgtc	ccacgaggac	cctgaagtga	agttcaattg	gtacgtggac	840
ggcgtggaag	tgacaaacgc	caagaccaag	cctagagagg	aacagtacaa	cagcacctac	900
cgggtggtgt	ccgtgctgac	agtgtgcac	caggactggc	tgaacggcaa	agagtacaag	960
tgcaagggtg	ccaacaaggc	cctgctgccc	cccacgcaga	aaaccatcag	caaggccaag	1020
ggccagcccc	gcgaacccca	ggtgtacaca	ctgcccccaa	gcagggacga	gctgaccaag	1080
aaaccaggtg	ccctgacctg	tctcgtgaaa	ggcttctacc	cctccgatat	cgccgtggaa	1140
tggaagagca	acggccagcc	cgagaacaac	tacaagacca	ccccccctgt	gctggacagc	1200
gacggctcat	tcttctgta	cagcaagctg	accgtggaca	agtcccggtg	gcagcagggc	1260
aacgtgttca	gctgcagcgt	gatgcacgag	gccctgcaca	accactacac	ccagaagtcc	1320
ctgagcctga	gccctggcaa	g				1341

<210> SEQ ID NO 841
 <211> LENGTH: 642
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 841

gagattgtgc	tgacacagtc	tcccgccaca	ctgtctctta	gccctggcga	aagagccaca	60
ctgagctgta	gagccagcca	gagcatcggc	gattacctgc	actggatatca	gcagaagcct	120
ggacaggccc	ctcggtgct	gatctactat	gccagccagt	ccatcagcgg	catccccgcc	180
agattttctg	gcagcggctc	tggcaccgat	ttcacctga	ccataagcag	cctggaacct	240
gaggacttcg	ccgtgtacta	ctgccagcag	ggctacagct	tcccctacac	atttggcgga	300
ggcaccaagc	tggaaatcaa	acgaaccgtg	gccgtctcca	gcgtgttcat	cttcccacct	360
agcgcagagc	agctgaagtc	cggcacagcc	tctgtcgtgt	gcctgctgaa	caacttctac	420
ccccgcgagg	ccaagggtga	gtggaagggtg	gacaatgccc	tgacagcggg	caacagccag	480
gaaagcgtga	ccgagcagga	cagcaaggac	tccacctaca	gcctgagcag	caccctgacc	540
ctgagcaagg	ccgactacga	gaagcacaag	gtgtacgcct	gcgaagtgc	ccaccagggc	600
ctgtctagcc	ccgtgaccaa	gagcttcaac	cggggcgagt	gt		642

<210> SEQ ID NO 842
 <211> LENGTH: 117
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 842

Glu	Val	Gln	Leu	Val	Glu	Ser	Gly	Gly	Gly	Leu	Leu	Gln	Leu	Gly	Gly
1			5					10					15		
Ser	Leu	Arg	Leu	Ser	Cys	Ala	Ala	Ser	Gly	Phe	Thr	Phe	Ser	Ser	Tyr
		20					25						30		
Tyr	Met	Ser	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Val
		35					40					45			
Ser	Thr	Ile	Ile	Lys	Ser	Gly	Gly	Tyr	Ala	Tyr	Tyr	Pro	Asp	Ser	Val
		50				55					60				
Lys	Asp	Arg	Phe	Thr	Ile	Ser	Arg	Asp	Asn	Ser	Lys	Asn	Thr	Leu	Asn
		65			70				75					80	
Leu	Gln	Met	Asn	Ser	Leu	Arg	Ala	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys
			85					90						95	

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Val Lys Gly Gly Gln Gly Ala Met Asp Tyr Trp Gly Gln Gly Thr Thr
 100 105 110

Val Thr Val Ser Ser
 115

<210> SEQ ID NO 843
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 843

Ser Tyr Tyr Met Ser
 1 5

<210> SEQ ID NO 844
 <211> LENGTH: 17
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 844

Thr Ile Ile Lys Ser Gly Gly Tyr Ala Tyr Tyr Pro Asp Ser Val Lys
 1 5 10 15

Asp

<210> SEQ ID NO 845
 <211> LENGTH: 8
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 845

Gly Gly Gln Gly Ala Met Asp Tyr
 1 5

<210> SEQ ID NO 846
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 846

Glu Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Leu Ser Pro Gly
 1 5 10 15

Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Ile Gly Asp Tyr
 20 25 30

Leu His Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
 35 40 45

Lys Tyr Ala Ser Gln Ser Ile Ser Gly Ile Pro Ala Arg Phe Ser Gly
 50 55 60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Glu Pro
 65 70 75 80

Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Gly Tyr Ser Phe Pro Tyr
 85 90 95

Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys
 100 105

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<210> SEQ ID NO 847
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 847

Arg Ala Ser Gln Ser Ile Gly Asp Tyr Leu His
1 5 10

<210> SEQ ID NO 848
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 848

Tyr Ala Ser Gln Ser Ile Ser
1 5

<210> SEQ ID NO 849
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 849

Gln Gln Gly Tyr Ser Phe Pro Tyr Thr
1 5

<210> SEQ ID NO 850
<211> LENGTH: 351
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 850

gaagtgcagc tgggtggaatc tggcggagga ctgctgcagc ttggcggatc tctgagactg 60
tcttgtgccg ccagcggcctt cacccttcagc agctactaca tgagctgggt ccgacaggcc 120
cctggcaaag gacttgaatg ggtgtccacc atcatcaaga gcggcggcta cgcctactat 180
cccgcagcgc tgaaggaccg gttcaccatc tccagagaca acagcaagaa caccctgaac 240
ctgcagatga acagcctgag agccgaggac accgccgtgt actactgtgt gaaaggtgga 300
cagggcgcca tggactattg gggccaggga acaacagtga ccgtgtcctc a 351

<210> SEQ ID NO 851
<211> LENGTH: 321
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 851

gagattgtgc tgacacagtc tcccgccaca ctgtctctta gccctggcga aagagccaca 60
ctgagctgta gagccagcca gagcatcggc gattacctgc actggatatca gcagaagcct 120
ggacaggccc ctcggctgct gattaagtac gccagccagt ccatcagcgg catccctgcc 180
agattttctg gcagcggctc tggcacccgat ttcaccctga ccataagcag cctggaacct 240

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gaggacttcg ccgtgtacta ctgccagcag ggctacagct tcccctacac atttgccgga 300
 ggcaccaagc tggaatcaa a 321

<210> SEQ ID NO 852
 <211> LENGTH: 447
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 852

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Leu Gln Leu Gly Gly
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
 20 25 30
 Tyr Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45
 Ser Thr Ile Ile Lys Ser Gly Gly Tyr Ala Tyr Tyr Pro Asp Ser Val
 50 55 60
 Lys Asp Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Asn
 65 70 75 80
 Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Val Lys Gly Gly Gln Gly Ala Met Asp Tyr Trp Gly Gln Gly Thr Thr
 100 105 110
 Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu
 115 120 125
 Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly Cys
 130 135 140
 Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser
 145 150 155 160
 Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val Leu Gln Ser
 165 170 175
 Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser
 180 185 190
 Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His Lys Pro Ser Asn
 195 200 205
 Thr Lys Val Asp Lys Lys Val Glu Pro Lys Ser Cys Asp Lys Thr His
 210 215 220
 Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly Pro Ser Val
 225 230 235 240
 Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr
 245 250 255
 Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu Asp Pro Glu
 260 265 270
 Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys
 275 280 285
 Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg Val Val Ser
 290 295 300
 Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys
 305 310 315 320
 Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu Lys Thr Ile
 325 330 335
 Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro
 340 345 350

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Pro Ser Arg Asp Glu Leu Thr Lys Asn Gln Val Ser Leu Thr Cys Leu
   355                               360                   365

Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn
   370                               375                   380

Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser
  385                               390                   395                   400

Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg
   405                               410                   415

Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu
   420                               425                   430

His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys
   435                               440                   445

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<210> SEQ ID NO 853
<211> LENGTH: 214
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 853

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Glu Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Leu Ser Pro Gly
 1           5           10           15

Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Ile Gly Asp Tyr
 20           25           30

Leu His Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
 35           40           45

Lys Tyr Ala Ser Gln Ser Ile Ser Gly Ile Pro Ala Arg Phe Ser Gly
 50           55           60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Glu Pro
 65           70           75           80

Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Gly Tyr Ser Phe Pro Tyr
 85           90           95

Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys Arg Thr Val Ala Ala
100          105          110

Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly
115          120          125

Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala
130          135          140

Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln
145          150          155          160

Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser
165          170          175

Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr
180          185          190

Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser
195          200          205

Phe Asn Arg Gly Glu Cys
210

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<210> SEQ ID NO 854
<211> LENGTH: 1341
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 854

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gaagtgcagc tgggtggaatc tggcggagga ctgctgcagc ttggcggatc tctgagactg      60
tcttgtgccg ccagcggett caccttcagc agctactaca tgagctgggt ccgacaggcc      120
cctggcaaa gacttgaatg ggtgtccacc atcatcaaga gcggcggcta cgcctactat      180
cccgacagcg tgaaggaccg gtccaccatc tccagagaca acagcaagaa caccctgaac      240
ctgcagatga acagcctgag agccgaggac accgccgtgt actactgtgt gaaaggtgga      300
cagggcgccca tggactattg gggccaggga acaacagtga ccgtgtcctc agccagcacc      360
aaggggcccca gcgtgttccc tctggcccct agcagcaaga gcacatctgg cggaacagcc      420
gcccctgggt gcctcgtgaa ggactacttt cccgagcccg tgaccgtgtc ctggaactct      480
ggcgctctga caagcggcgt gcacaccttt ccagccgtgc tgcagagcag cggcctgtac      540
tctctgagca gcgtcgtgac agtgcccagc agctctctgg gcaccagac ctacatctgc      600
aacgtgaacc acaagcccag caacaccaag gtggacaaga aggtggaacc caagagctgc      660
gacaagaccc acacctgtcc cccttgtcct gcccgcgaac tgctgggagg cccttcctgtg      720
ttcctgttcc ccccaagacc caaggacacc ctgatgatca gccggacccc cgaagtgacc      780
tgcggtgttg tggatgtgtc ccacgaggac cctgaagtga agttcaattg gtacgtggac      840
ggcgtggaag tgcacaacgc caagaccaag cctagagagg aacagtacaa cagcacctac      900
cgggtgtgtg ccgtgctgac agtgctgcac caggactggc tgaacggcaa agagtacaag      960
tgcaaggtgt ccaacaaggc cctgcctgcc cccatcgaga aaaccatcag caaggccaag      1020
ggccagcccc gcgaacccca ggtgtacaca ctgcccccaa gcaggagcga gctgaccaag      1080
aaccaggtgt ccctgacctg tctcgtgaaa ggcttctacc cctccgatat cgcctgggaa      1140
tgaggagagca acggccagcc cgagaacaac tacaagacca cccccctgt gctggacagc      1200
gacggctcat tcttctgtga cagcaagctg accgtggaca agtcccgggtg gcagcagggc      1260
aacgtgttca gctgcagcgt gatgcacgag gccctgcaca accactacac ccagaagtc      1320
ctgagcctga gccctggcaa g

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<210> SEQ ID NO 855

<211> LENGTH: 642

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 855

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gagattgtgc tgacacagtc tcccgcacac ctgtctctta gccctggcga aagagccaca      60
ctgagctgta gagccagcca gagcatcggc gattacctgc actggtatca gcagaagcct      120
ggacaggccc ctcggctgct gattaagtac gccagccagt ccatcagcgg catccctgcc      180
agattttctg gcagcggctc tggcaccgat ttcacctga ccataagcag cctggaacct      240
gaggacttcg ccgtgtacta ctgccagcag ggctacagct tcccctacac atttgcgga      300
ggcaccaagc tggaaatcaa acgaaccgtg gccgctccca gcgtgttcat cttcccacct      360
agcgacgagc agctgaagtc cggcacagcc tctgtcgtgt gcctgctgaa caacttctac      420
ccccgcgagg ccaagggtga gtggaagggtg gacaatgccc tgcagagcgg caacagccag      480
gaaagcgtga ccgagcagga cagcaaggac tccacctaca gcctgagcag caccctgacc      540
ctgagcaagg ccgactacga gaagcacaag gtgtacgcct gcgaagtgc ccaccagggc      600
ctgtctagcc ccgtgaccaa gagcttcaac cggggcgagt gt

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<210> SEQ ID NO 856
<211> LENGTH: 122
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 856

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Arg Ser Tyr
20 25 30
Ala Val His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45
Ser Ser Thr Glu Gly Ser Gly Val Gly Thr Ser Tyr Thr Asp Ser Val
50 55 60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65 70 75 80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95
Ala Arg Met Leu Gly Gly Gly Asn Pro Leu Asp Tyr Leu Asp Tyr Trp
100 105 110
Gly Gln Gly Thr Leu Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 857
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 857

Ser Tyr Ala Val His
1 5

<210> SEQ ID NO 858
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 858

Ser Thr Glu Gly Ser Gly Val Gly Thr Ser Tyr Thr Asp Ser Val Lys
1 5 10 15
Gly

<210> SEQ ID NO 859
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 859

Met Leu Gly Gly Gly Asn Pro Leu Asp Tyr Leu Asp Tyr
1 5 10

<210> SEQ ID NO 860
<211> LENGTH: 111

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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 860

Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
1          5          10          15

Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Leu Gly Glu Gly
20         25         30

Tyr Asp Val His Trp Tyr Gln Gln Leu Pro Gly Lys Ala Pro Lys Leu
35         40         45

Leu Ile Tyr Tyr Ser Asp Phe Arg Pro Ser Gly Val Ser Asp Arg Phe
50         55         60

Ser Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu
65         70         75         80

Gln Ser Glu Asp Glu Ala Asp Tyr Tyr Cys Ala Ala Trp Asp Asp Ser
85         90         95

Leu Ser Ser Gln Val Phe Gly Gly Gly Thr Gln Val Thr Val Leu
100        105        110

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<210> SEQ ID NO 861
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 861

Ser Gly Ser Ser Ser Asn Leu Gly Glu Gly Tyr Asp Val His
1          5          10

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<210> SEQ ID NO 862
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 862

Tyr Ser Asp Phe Arg Pro Ser
1          5

```

```

<210> SEQ ID NO 863
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 863

Ala Ala Trp Asp Asp Ser Leu Ser Ser Gln Val
1          5          10

```

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<210> SEQ ID NO 864
<211> LENGTH: 366
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 864

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gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg 60

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agctgtgccg ccagcggcctt cacctttaga agctatgccg tgcactgggt ccgacaggcc 120
cctggaaaag gactggaatg ggtgtccagc accgaaggct ctggcggtggg cacaagctac 180
accgattctg tgaagggcag attcaccatc agccgggaca acagcaagaa caccctgtac 240
ctgcagatga acagcctgag agccgaggac accgccgtgt actactgtgc cagaatgctc 300
ggcggaggca accctctgga ctacctggat tattggggcc agggcaccct ggtcacagtc 360
tcttca 366

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<210> SEQ ID NO 865
<211> LENGTH: 333
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 865

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cagctctgttc tgacacagcc tcttagcgcc tctggcacac ctggacagag agtgaccatc 60
agctgtagcg gcagcagctc caatctcggc gagggctatg acgtgcactg gtatcagcag 120
ctgcctggca agggccctaa actgctgac tactacagcg acttcagacc cagcggcgtg 180
tccgatagat tcagcggctc taagagcggc acatctgccg gcctggccat ctctggactg 240
cagagcgaag atgagggcga ctactattgc gccgcctggg atgatagcct gagcagccaa 300
gtttttggcg gcggaaccca agtgaccgtg cta 333

```

```

<210> SEQ ID NO 866
<211> LENGTH: 452
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

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<400> SEQUENCE: 866

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Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1      5      10      15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Arg Ser Tyr
20     25     30
Ala Val His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35     40     45
Ser Ser Thr Glu Gly Ser Gly Val Gly Thr Ser Tyr Thr Asp Ser Val
50     55     60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65     70     75     80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85     90     95
Ala Arg Met Leu Gly Gly Gly Asn Pro Leu Asp Tyr Leu Asp Tyr Trp
100    105    110
Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro
115    120    125
Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr
130    135    140
Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr
145    150    155    160
Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro
165    170    175
Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr
180    185    190

```

[illegible]

Gln	Ser	Val	Leu	Thr	Gln	Pro	Pro	Ser	Ala	Ser	Gly	Thr	Pro	Gly	Gln
1				5					10					15	
Arg	Val	Thr	Ile	Ser	Cys	Ser	Gly	Ser	Ser	Ser	Asn	Leu	Gly	Glu	Gly
			20					25					30		
Tyr	Asp	Val	His	Trp	Tyr	Gln	Gln	Leu	Pro	Gly	Lys	Ala	Pro	Lys	Leu
		35					40					45			
Leu	Ile	Tyr	Tyr	Ser	Asp	Phe	Arg	Pro	Ser	Gly	Val	Ser	Asp	Arg	Phe
	50					55					60				
Ser	Gly	Ser	Lys	Ser	Gly	Thr	Ser	Ala	Ser	Leu	Ala	Ile	Ser	Gly	Leu
65					70					75					80
Gln	Ser	Glu	Asp	Glu	Ala	Asp	Tyr	Tyr	Cys	Ala	Ala	Trp	Asp	Asp	Ser
				85					90					95	

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Leu Ser Ser Gln Val Phe Gly Gly Gly Thr Gln Val Thr Val Leu Gly
 100 105 110
 Gln Pro Lys Ala Ala Pro Ser Val Thr Leu Phe Pro Pro Ser Ser Glu
 115 120 125
 Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp Phe
 130 135 140
 Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Ser Ser Pro Val
 145 150 155 160
 Lys Ala Gly Val Glu Thr Thr Thr Pro Ser Lys Gln Ser Asn Asn Lys
 165 170 175
 Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys Ser
 180 185 190
 His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val Glu
 195 200 205
 Lys Thr Val Ala Pro Thr Glu Cys Ser
 210 215

<210> SEQ ID NO 868

<211> LENGTH: 1356

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 868

```

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg      60
agctgtgccg ccagcggtt cacccttaga agctatgccg tgcactgggt ccgacaggcc      120
cctggaaaag gactggaatg ggtgtccagc accgaaggct ctggcgtggg cacaagctac      180
accgattctg tgaagggcag attcaccatc agccgggaca acagcaagaa caccctgtac      240
ctgcagatga acagcctgag agccgaggac accgccgtgt actactgtgc cagaatgtc      300
ggcggaggca accctctgga ctacctggat tattggggcc agggcaccct ggtcacagtc      360
tcttcagcca gcaccaaggg cccagcgtg ttcctctctg cccctagcag caagagcaca      420
tctggcggaa cagccgcctt gggctgcttc gtgaaggact actttcccga gcccgtagcc      480
gtgtcctgga actctggcgc tctgacaagc ggcgtagcaca ctttccagc cgtgctgcag      540
agcagcggcc tgtactctct gagcagcgtc gtgacagtgc ccagcagctc tctgggcacc      600
cagacctaca tctgcaacgt gaaccacaag cccagcaaca ccaaggtgga caagaaggtg      660
gaaccaaga gctgcgacaa gaccacaccc tgccccctt gtccctgccc cgaactgctg      720
ggaggccctt ccgtgttctt gttcccccca aagcccaagg acaccctgat gatcagccgg      780
acccccgaag tgacctgcgt ggtggtggat gtgtcccacg aggacctga agtgaagttc      840
aattggtacg tggacggcgt ggaagtgcac aacccaaga ccaagcctag agaggaacag      900
tacaacagca cctaccgggt ggtgtccgtg ctgacagtgc tgcaccagga ctggctgaac      960
ggcaaagagt acaagtgcac ggtgtccaac aaggccctgc ctgcccccat cgagaaaacc     1020
atcagcaagg ccaagggccca gccccgcgaa cccaggtgt acacactgcc ccaagcagg     1080
gacgagctga ccaagaacca ggtgtccctg acctgtctcg tgaaaggctt ctaccctcc     1140
gatatgcccg tggaatggga gagcaacggc cagcccgaga acaactacaa gaccacccc     1200
cctgtgctgg acagcgacgg ctctattctt ctgtacagca agctgacctg ggacaagtcc     1260
cgggtggcagc agggcaacgt gttcagctgc agcgtgatgc acgaggccct gcacaaccac     1320

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-continued

tacacccaga agtccctgag cctgagccct ggcaag

1356

<210> SEQ ID NO 869

<211> LENGTH: 651

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 869

```

cagtctgttc tgacacagcc tcctagcgcc tctggcacac ctggacagag agtgaccatc      60
agctgtagcg gcagcagctc caatctcggc gagggctatg acgtgcaactg gtatcagcag      120
ctgcctggca agggccctaa actgctgata tactacagcg acttcagacc cagcggcgtg      180
tccgatagat tcagcggctc taagagcggc acatctgccg gcctggccat ctctggactg      240
cagagcgcaag atgaggccga ctactattgc gccgcctggg atgatagcct gagcagccaa      300
gtttttggcg gcggaaccca agtgaccgtg ctaggccagc ctaaagccgc ccctagcgtg      360
accctgttcc ctccaagcag cgaggaaactg caggccaaca aggccaccct cgtgtgctg      420
atcagcgact tctatcctgg cgccgtgacc gtggcctgga aggcgatag ctctcctgtg      480
aaggccggcg tggaaaccac caccctagc aagcagagca acaacaaata cgccgccagc      540
agctacctga gcctgacccc cgagcagtgg aagtcacaca gatcctacag ctgccaaagt      600
acccacgagg gcagcaccgt ggaaaagaca gtggccccta ccgagtgcag c              651

```

<210> SEQ ID NO 870

<211> LENGTH: 122

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 870

```

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1      5      10      15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Arg Ser Tyr
20     25     30
Ala Val His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35     40     45
Ser Ser Thr Glu Gly Ser Gly Val Gly Thr Ser Tyr Thr Asp Ser Val
50     55     60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65     70     75     80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85     90     95
Ala Arg Met Leu Gly Gly Gly Asn Pro Leu Asp Tyr Leu Asp Tyr Trp
100    105    110
Gly Gln Gly Thr Leu Val Thr Val Ser Ser
115    120

```

<210> SEQ ID NO 871

<211> LENGTH: 5

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 871

Ser Tyr Ala Val His

-continued

```

1           5

<210> SEQ ID NO 872
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 872

```

```

Ser Thr Glu Gly Ser Gly Val Gly Thr Ser Tyr Thr Asp Ser Val Lys
1           5           10           15

```

```

Gly

```

```

<210> SEQ ID NO 873
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 873

```

```

Met Leu Gly Gly Gly Asn Pro Leu Asp Tyr Leu Asp Tyr
1           5           10

```

```

<210> SEQ ID NO 874
<211> LENGTH: 111
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 874

```

```

Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
1           5           10           15

```

```

Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Leu Gly Glu Gly
20           25           30

```

```

Tyr Asp Val His Trp Tyr Gln Gln Leu Pro Gly Lys Ala Pro Lys Leu
35           40           45

```

```

Leu Ile Tyr Tyr Ser Asp Phe Arg Pro Ser Gly Val Ser Asp Arg Phe
50           55           60

```

```

Ser Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu
65           70           75           80

```

```

Gln Ser Glu Asp Glu Ala Asp Tyr Tyr Cys Ala Ala Trp Asp Asp Ser
85           90           95

```

```

Leu Ser Ser Gln Val Phe Gly Gly Gly Thr Gln Val Thr Val Leu
100           105           110

```

```

<210> SEQ ID NO 875
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: antibody sequence

```

```

<400> SEQUENCE: 875

```

```

Ser Gly Ser Ser Ser Asn Leu Gly Glu Gly Tyr Asp Val His
1           5           10

```

```

<210> SEQ ID NO 876
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence

```

-continued

<220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 876

Tyr Ser Asp Phe Arg Pro Ser
 1 5

<210> SEQ ID NO 877
 <211> LENGTH: 11
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 877

Ala Ala Trp Asp Asp Ser Leu Ser Ser Gln Val
 1 5 10

<210> SEQ ID NO 878
 <211> LENGTH: 366
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 878

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg	60
agctgtgccg ccagcggcctt cacctttaga agctatgccg tgcaactgggt ccgacaggcc	120
cctggaaaag gactggaatg ggtgtccagc accgaaggct ctggcgtggg cacaagctac	180
accgattctg tgaaggcgag attcaccatc agccgggaca acagcaagaa caccctgtac	240
ctgcagatga acagcctgag agccgaggac accgccgtgt actactgtgc cagaatgctc	300
ggcggaggca accctctgga ctacctggat tattggggcc agggcaccct ggtcacagtc	360
tcttca	366

<210> SEQ ID NO 879
 <211> LENGTH: 333
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 879

cagctctgttc tgacacagcc tcctagcgcc tctggcacac ctggacagag agtgaccatc	60
agctgtagcg gcagcagctc caatctcggc gagggctatg acgtgcactg gtatcagcag	120
ctgcctggca agggccctaa actgctgata tactacagcg acttcagacc cagcggcgtg	180
tccgatagat tcagcggctc taagagcggc acatctgcca gcctggccat ctctggactg	240
cagagcgaag atgaggccga ctactattgc gccgcctggg atgatagcct gagcagccaa	300
gtttttggcg gcggaacca agtgaccgtg cta	333

<210> SEQ ID NO 880
 <211> LENGTH: 251
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 880

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15

-continued

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Arg Ser Tyr
 20 25 30
 Ala Val His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45
 Ser Ser Thr Glu Gly Ser Gly Val Gly Thr Ser Tyr Thr Asp Ser Val
 50 55 60
 Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
 65 70 75 80
 Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Ala Arg Met Leu Gly Gly Gly Asn Pro Leu Asp Tyr Leu Asp Tyr Trp
 100 105 110
 Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro
 115 120 125
 Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr
 130 135 140
 Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr
 145 150 155 160
 Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro
 165 170 175
 Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr
 180 185 190
 Val Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn
 195 200 205
 His Lys Pro Ser Asn Thr Lys Val Asp Lys Lys Val Glu Pro Lys Ser
 210 215 220
 Cys Ala Ala Gly Ser Glu Gln Lys Leu Ile Ser Glu Glu Asp Leu Ser
 225 230 235 240
 Gly Ser Ala Ala Ala His His His His His His
 245 250

<210> SEQ ID NO 881

<211> LENGTH: 217

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 881

Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
 1 5 10 15
 Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Leu Gly Glu Gly
 20 25 30
 Tyr Asp Val His Trp Tyr Gln Gln Leu Pro Gly Lys Ala Pro Lys Leu
 35 40 45
 Leu Ile Tyr Tyr Ser Asp Phe Arg Pro Ser Gly Val Ser Asp Arg Phe
 50 55 60
 Ser Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu
 65 70 75 80
 Gln Ser Glu Asp Glu Ala Asp Tyr Tyr Cys Ala Ala Trp Asp Asp Ser
 85 90 95
 Leu Ser Ser Gln Val Phe Gly Gly Gly Thr Gln Val Thr Val Leu Gly
 100 105 110
 Gln Pro Lys Ala Ala Pro Ser Val Thr Leu Phe Pro Pro Ser Ser Glu
 115 120 125

-continued

Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp Phe
130 135 140

Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Ser Ser Pro Val
145 150 155 160

Lys Ala Gly Val Glu Thr Thr Thr Pro Ser Lys Gln Ser Asn Asn Lys
165 170 175

Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys Ser
180 185 190

His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val Glu
195 200 205

Lys Thr Val Ala Pro Thr Glu Cys Ser
210 215

<210> SEQ ID NO 882

<211> LENGTH: 753

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 882

```

gaagttcagc tgctggaatc tggcggcgga ctggttcaac ctggcggatc tctgagactg      60
agctgtgccg ccagcgggett cacctttaga agctatgccg tgcactgggt ccgacaggcc      120
cctggaaaag gactggaatg ggtgtccagc accgaaggct ctggcgtggg cacaagctac      180
accgattctg tgaagggcag attcaccatc agccgggaca acagcaagaa caccctgtac      240
ctgcagatga acagcctgag agccgaggac accgccgtgt actactgtgc cagaatgtc      300
ggcggaggca accctctgga ctacctggat tattggggcc agggcaccct ggtcacagtc      360
tcttcagcct ccaccaaggg cccatcggtg tccccctgg caccctcctc caagagcacc      420
tctgggggca cagcggcctt gggctgctg gtcaaggact acttcccga accggtgacg      480
gtgtcgtgga actcaggcgc cctgaccagc ggcggtgcaca ccttcccggc tgtcctacag      540
tcctcaggac tctactcctc cagcagcgtg gtgaccgtgc cctccagcag cttgggcacc      600
cagacctaca tctgcaacgt gaatcacaag cccagcaaca ccaaggtgga caagaaagt      660
gagcccaaat cttgtgcagc gggttctgaa caaaaactca tctcagaaga ggatctgtct      720
ggatcagcgg ccgcccatca tcatcatcat cat                                     753

```

<210> SEQ ID NO 883

<211> LENGTH: 651

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: antibody sequence

<400> SEQUENCE: 883

```

cagtcgtgtc tgacacagcc tcctagcgcc tctggcacac ctggacagag agtgaccatc      60
agctgtagcg gcagcagctc caatctcggc gagggctatg acgtgcactg gtatcagcag      120
ctgcctggca agggccctaa actgctgata tactacagcg acttcagacc cagcggcgtg      180
tccgatagat tcagcggctc taagagcggc acatctgccg gcctggccat ctctggactg      240
cagagcgaag atgaggccga ctactattgc gccgcctggg atgatagcct gagcagccaa      300
gtttttggcg gcggaacca agtgaccgtg ctaggccagc ctaaagccgc ccctagcgtg      360
accctgttcc ctccaagcag cgaggaaactg caggccaaca aggccaccct cgtgtgcctg      420

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atcagcgact tctatcctgg cgccgtgacc gtggcctgga aggccgatag ctctcctgtg 480
aaggccggcg tggaaccac caccctagc aagcagagca acaacaata cgccgccagc 540
agctacctga gctgacccc cgagcagtg aagtcacaca gatcctacag ctgccaagtg 600
acccacgagg gcagaccgt ggaaaagaca gtggccccta ccgagtgcag c 651

```

<210> SEQ ID NO 884

<211> LENGTH: 563

<212> TYPE: PRT

<213> ORGANISM: *Oryctolagus cuniculus* (Rabbit)

<400> SEQUENCE: 884

```

Asn Tyr Gln Asn Gly Lys Asn Asn Val Pro Arg Leu Lys Leu Ser Tyr
1           5           10           15
Lys Glu Met Leu Glu Ser Asn Asn Val Ile Thr Phe Asn Gly Leu Ala
20          25          30
Asn Ser Ser Ser Tyr His Thr Phe Leu Leu Asp Glu Glu Arg Ser Arg
35          40          45
Leu Tyr Val Gly Ala Lys Asp His Ile Phe Ser Phe Asn Leu Val Asn
50          55          60
Ile Lys Asp Phe Gln Lys Ile Ala Trp Pro Val Ser Tyr Thr Arg Arg
65          70          75          80
Asp Glu Cys Lys Trp Ala Gly Lys Asp Ile Leu Arg Glu Cys Ala Asn
85          90          95
Phe Ile Lys Val Leu Lys Val Tyr Asn Gln Thr His Leu Tyr Ala Cys
100         105        110
Gly Thr Gly Ala Phe His Pro Ile Cys Thr Tyr Val Gly Ile Gly His
115        120        125
His Pro Glu Asp Asn Ile Phe Lys Leu Glu Asp Ser His Phe Glu Asn
130        135        140
Gly Arg Gly Lys Ser Pro Tyr Asp Pro Lys Leu Leu Thr Ala Ser Leu
145        150        155        160
Leu Ile Asp Gly Glu Leu Tyr Ser Gly Thr Ala Ala Asp Phe Met Gly
165        170        175
Arg Asp Phe Ala Ile Phe Arg Thr Leu Gly Gln His His Pro Ile Arg
180        185        190
Thr Glu Gln His Asp Ser Arg Trp Leu Asn Asp Pro Arg Phe Ile Ser
195        200        205
Ala His Leu Ile Pro Glu Ser Asp Asn Pro Glu Asp Asp Lys Val Tyr
210        215        220
Phe Phe Phe Arg Glu Asn Ala Ile Asp Gly Glu His Ser Gly Lys Ala
225        230        235        240
Thr His Ala Arg Ile Gly Gln Ile Cys Lys Asn Asp Phe Gly Gly His
245        250        255
Arg Ser Leu Val Asn Lys Trp Thr Thr Phe Leu Lys Ala Arg Leu Ile
260        265        270
Cys Ser Val Pro Gly Pro Asn Gly Ile Asp Thr His Phe Asp Glu Leu
275        280        285
Gln Asp Val Phe Leu Met Asn Ser Lys Asp Pro Lys Asn Pro Ile Val
290        295        300
Tyr Gly Val Phe Thr Thr Ser Ser Asn Ile Phe Arg Gly Ser Ala Val
305        310        315        320
Cys Met Tyr Ser Met Ser Asp Val Arg Arg Val Phe Leu Gly Pro Tyr
325        330        335

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Ala His Arg Asp Gly Pro Asn Tyr Gln Trp Val Pro Phe Gln Gly Arg
340 345 350

Val Pro Tyr Pro Arg Pro Gly Thr Cys Pro Ser Lys Thr Phe Gly Gly
355 360 365

Phe Glu Ser Thr Lys Asp Leu Pro Asp Asp Val Ile Thr Phe Ala Arg
370 375 380

Ser His Pro Ala Met Tyr Asn Pro Val Phe Pro Ile Asn Asn Arg Pro
385 390 395 400

Ile Met Val Lys Thr Asp Val Asn Tyr Gln Phe Thr Gln Ile Val Val
405 410 415

Asp Arg Val Asp Ala Glu Asp Gly Gln Tyr Asp Val Met Phe Ile Gly
420 425 430

Thr Asp Val Gly Thr Val Leu Lys Val Val Ser Ile Pro Lys Glu Thr
435 440 445

Trp His Asp Leu Glu Glu Val Leu Leu Glu Glu Met Thr Val Phe Arg
450 455 460

Glu Pro Thr Thr Ile Ser Ala Met Glu Leu Ser Thr Lys Gln Gln Gln
465 470 475 480

Leu Tyr Val Gly Ser Ala Ala Gly Val Ala Gln Leu Pro Leu His Arg
485 490 495

Cys Asp Ile Tyr Gly Lys Ala Cys Ala Glu Cys Cys Leu Ala Arg Asp
500 505 510

Pro Tyr Cys Ala Trp Asp Gly Ser Ser Cys Ser Arg Tyr Phe Pro Thr
515 520 525

Ala Lys Arg Arg Thr Arg Arg Gln Asp Ile Arg Asn Gly Asp Pro Leu
530 535 540

Thr His Cys Ser Asp Gly Gly Ile Glu Gly Arg Met Asp His His His
545 550 555 560

His His His

<210> SEQ ID NO 885

<211> LENGTH: 1689

<212> TYPE: DNA

<213> ORGANISM: *Oryctolagus cuniculus* (Rabbit)

<400> SEQUENCE: 885

aactatcaga acggcaagaa caacgtgccc cggctgaagc tgagctacaa agagatgctg	60
gaaagcaaca acgtgatcac cttcaacggc ctggccaaca gcagcagcta ccacaccttt	120
ctgctggagc aggaacggtc cagactgtac gtgggagcca aggaccacat cttcagcttc	180
aacctggtca acatcaagga cttccagaaa atcgccctggc ctgtgtccta caccagacgg	240
gatgagtgtg aatgggccgg caaggacatc ctgagagagt gcgccaactt catcaagggtg	300
ctgaagggtg acaatcagac ccacctgtac gcctgtggca ccggcgcttt tcaccctatc	360
tgtacctatg tcggcatcgg ccaccatcct gaggacaata tcttcaagct cgaggacagc	420
cacttcgaga acggcagagg caagagcccc tacgatccca aactgctgac agcctctctg	480
ctgatcgacg gcgagctgta ttctggcaca gccgcccatt tcatgggcag agacttcgcc	540
atcttcagaa ccttgggcca gcatcacccc atcagaaccg agcagcacga cagcagatgg	600
ctgaacgacc ccagattcat cagcgcccat ctgatcccg agagcgacaa ccccgaggac	660
gacaagggtg actttcttct cgggaaaac gccatcgacg gggagcactc tggaaaagcc	720
acacacgcca gaatcggcc gatctgcaag aacgacttcg gcggccacag atccctcgtg	780
aacaagtgga ccaccttct gaaggcccg ctgatctgtt ctgtgccgg acctaattgc	840

-continued

atcgataccc acttogaaga gctgcaggac gtgttctga tgaacagcaa ggacccaag	900
aatcccatcg tgtaaggcgt gttcaccacc agcagcaaca tcttagagg cagcgccgtg	960
tgcatgtaca gcatgtccga tgtgcggaga gtgtttctgg gccctacgc tcacagagat	1020
ggccccaatt atcagtgggt gccattccag ggcagagtgc cctatcctag acctggcacc	1080
tgtcctagca agacctttgg cggtctcag agcaccaagg acctgcctga cgatgtgatt	1140
accttcgcca gatctcacc cgcatgtac aacctgtgt tcccatcaa caacaggccc	1200
atcatggtca agacggcgt gaactaccag ttcaccaga tcgtgggga cagagtggat	1260
gccgaggacg gccagtaaga cgtgatgttc atcggcaccg atgtgggcac cgtgctgaaa	1320
gtggtgtcta tcccaaaaga gacatggcac gacctggaag aggtgtgtct ggaagagatg	1380
accgtgttca gagagccac caccatctcc gccatggaac tgagcacaaa acagcaacag	1440
ctgtatgtgg gtcgcgcgc tgggtgtgct caactgctc tgcacagatg cgacatctac	1500
ggcaaagcct gcgcgagtg ttgcctggcc agagatcctt actgtgcctg ggatggcagc	1560
agctgcagca gatactttcc caccgccaag cggagaacca gacggcagga tatcagaaac	1620
ggcgaccctc tgacacactg cagcgacggt ggcacgagg gccgcagga tcatcatcat	1680
caccatcat	1689

What is claimed is:

1. An isolated antibody or antigen-binding fragment thereof which binds to human Semaphorin 3A (Sema3A), wherein said isolated antibody or antigen-binding fragment thereof comprises:

- a heavy chain antigen-binding region that comprises an H-CDR1 comprising SEQ ID NO: 42, an H-CDR2 comprising SEQ ID NO: 43, and an H-CDR3 comprising SEQ ID NO: 44 and a light chain antigen-binding region that comprises an L-CDR1 comprising SEQ ID NO: 46, an L-CDR2 comprising SEQ ID NO: 47, and an L-CDR3 comprising SEQ ID NO: 48; or
- a heavy chain antigen-binding region that comprises an H-CDR1 comprising SEQ ID NO: 62, an H-CDR2 comprising SEQ ID NO: 63, and an H-CDR3 comprising SEQ ID NO: 64 and a light chain antigen-binding region that comprises an L-CDR1 comprising SEQ ID NO: 66, an L-CDR2 comprising SEQ ID NO: 67, and an L-CDR3 comprising SEQ ID NO: 68; or
- a heavy chain antigen-binding region that comprises an H-CDR1 comprising SEQ ID NO: 102, an H-CDR2 comprising SEQ ID NO: 103, and an H-CDR3 comprising SEQ ID NO: 104 and a light chain antigen-binding region that comprises an L-CDR1 comprising SEQ ID NO: 106, an L-CDR2 comprising SEQ ID NO: 107, and an L-CDR3 comprising SEQ ID NO: 108; or
- a heavy chain antigen-binding region that comprises an H-CDR1 comprising SEQ ID NO: 122, an H-CDR2 comprising SEQ ID NO: 123, and an H-CDR3 comprising SEQ ID NO: 124 and a light chain antigen-binding region that comprises an L-CDR1 comprising SEQ ID NO: 126, an L-CDR2 comprising SEQ ID NO: 127, and an L-CDR3 comprising SEQ ID NO: 128; or
- a heavy chain antigen-binding region that comprises an H-CDR1 comprising SEQ ID NO: 162, an H-CDR2 comprising SEQ ID NO: 163, and an H-CDR3 comprising SEQ ID NO: 164 and a light chain antigen-binding region that comprises an L-CDR1 comprising

- SEQ ID NO: 166, an L-CDR2 comprising SEQ ID NO: 167, and an L-CDR3 comprising SEQ ID NO: 168; or
- a heavy chain antigen-binding region that comprises an H-CDR1 comprising SEQ ID NO: 182, an H-CDR2 comprising SEQ ID NO: 183, and an H-CDR3 comprising SEQ ID NO: 184 and a light chain antigen-binding region that comprises an L-CDR1 comprising SEQ ID NO: 186, an L-CDR2 comprising SEQ ID NO: 187, and an L-CDR3 comprising SEQ ID NO: 188; or
- a heavy chain antigen-binding region that comprises an H-CDR1 comprising SEQ ID NO: 342, an H-CDR2 comprising SEQ ID NO: 343, and an H-CDR3 comprising SEQ ID NO: 344 and a light chain antigen-binding region that comprises an L-CDR1 comprising SEQ ID NO: 346, an L-CDR2 comprising SEQ ID NO: 347, and an L-CDR3 comprising SEQ ID NO: 348.

2. The isolated antibody or antigen-binding fragment according to claim 1, comprising:

- a variable heavy chain domain that is at least 98% identical to SEQ ID NO: 41 and a variable light chain domain that is at least 98%, identical to SEQ ID NO: 45; or
- a variable heavy chain domain that is at least 98% identical to SEQ ID NO: 61 and a variable light chain domain that is at least 98%, identical to SEQ ID NO: 65; or
- a variable heavy chain domain that is at least 98% identical to SEQ ID NO: 101 and a variable light chain domain that is at least 98%, identical to SEQ ID NO: 105; or
- a variable heavy chain domain that is at least 98% identical to SEQ ID NO: 121 and a variable light chain domain that is at least 98%, identical to SEQ ID NO: 125; or
- a variable heavy chain domain that is at least 98% identical to SEQ ID NO: 161 and a variable light chain domain that is at least 98%, identical to SEQ ID NO: 165;

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- a variable heavy chain domain that is at least 98% identical to SEQ ID NO: 181 and a variable light chain domain that is at least 98%, identical to SEQ ID NO: 185; or
- a variable heavy chain domain that is at least 98% identical to SEQ ID NO: 341 and a variable light chain domain that is at least 98%, identical to SEQ ID NO: 345.
3. The isolated antibody or antigen-binding fragment according to claim 1, comprising:
- a variable heavy chain domain comprising SEQ ID NO: 41 and a variable light chain domain comprising SEQ ID NO: 45; or
- a variable heavy chain domain comprising SEQ ID NO: 61 and a variable light chain domain comprising SEQ ID NO: 65; or
- a variable heavy chain domain comprising SEQ ID NO: 101 and a variable light chain domain comprising SEQ ID NO: 105; or
- a variable heavy chain domain comprising SEQ ID NO: 121 and a variable light chain domain comprising SEQ ID NO: 125; or
- a variable heavy chain domain comprising SEQ ID NO: 161 and a variable light chain domain comprising SEQ ID NO: 165; or
- a variable heavy chain domain comprising SEQ ID NO: 181 and a variable light chain domain comprising SEQ ID NO: 185; or
- a variable heavy chain domain comprising SEQ ID NO: 341 and a variable light chain domain comprising SEQ ID NO: 345.
4. The isolated antibody according to claim 1, wherein said isolated antibody is an IgG1 or an IgG4 antibody.
5. The isolated antibody according to claim 1, comprising:

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- a heavy chain comprising SEQ ID NO: 57 and a light chain comprising SEQ ID NO: 58; or
- a heavy chain comprising SEQ ID NO: 77 and a light chain comprising SEQ ID NO: 78; or
- a heavy chain comprising SEQ ID NO: 117 and a light chain comprising SEQ ID NO: 118; or
- a heavy chain comprising SEQ ID NO: 137 and a light chain comprising SEQ ID NO: 138; or
- a heavy chain comprising SEQ ID NO: 177 and a light chain comprising SEQ ID NO: 178; or
- a heavy chain comprising SEQ ID NO: 197 and a light chain comprising SEQ ID NO: 198; or
- a heavy chain comprising SEQ ID NO: 357 and a light chain comprising SEQ ID NO: 358.
6. The antigen-binding fragment according to claim 1, which is an scFv, Fab, Fab' fragment or a F(ab')₂ fragment.
7. The isolated antibody or antigen-binding fragment according to claim 1, which is a monoclonal antibody or antigen-binding fragment thereof.
8. The isolated antibody or antigen-binding fragment according to claim 1, which is a human, humanized, or chimeric antibody or antigen-binding fragment thereof.
9. An antibody conjugate, comprising the isolated antibody or antigen binding fragment according to claim 1.
10. A pharmaceutical composition comprising the isolated antibody or antigen-binding fragment according to claim 1 or an antibody conjugate comprising the isolated antibody or antigen binding fragment according to claim 1.
11. A kit comprising the isolated antibody or antigen-binding fragment according to claim 1 or an antibody conjugate comprising the isolated antibody or antigen binding fragment according to claim 1 or a pharmaceutical composition comprising the isolated antibody or antigen-binding fragment of claim 1; and instructions for use.

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